

CHAPTER 1

P. E. 1.1

(a) $\mathbf{A} + \mathbf{B} = (1, 0, 3) + (5, 2, -6) = (6, 2, -3)$

$$|\mathbf{A} + \mathbf{B}| = \sqrt{36 + 4 + 9} = \underline{\underline{7}}$$

(b) $5\mathbf{A} - \mathbf{B} = (5, 0, 15) - (5, 2, -6) = \underline{\underline{(0, -2, 21)}}$

(c) The component of $\mathbf{A}$ along $\mathbf{a}_y$ is $A_y = \underline{\underline{0}}$

(d) $3\mathbf{A} + \mathbf{B} = (3, 0, 9) + (5, 2, -6) = (8, 2, 3)$

A unit vector parallel to this vector is

$$\begin{aligned} \mathbf{a}_{11} &= \frac{(8, 2, 3)}{\sqrt{64 + 4 + 9}} \\ &= \underline{\underline{\pm(0.9117\mathbf{a}_x + 0.2279\mathbf{a}_y + 0.3419\mathbf{a}_z)}} \end{aligned}$$

P. E. 1.2 (a) $\mathbf{r}_p = \underline{\underline{\mathbf{a}_x - 3\mathbf{a}_y + 5\mathbf{a}_z}}$

$$\mathbf{r}_R = \underline{\underline{3\mathbf{a}_y + 8\mathbf{a}_z}}$$

(b) The distance vector is

$$\mathbf{r}_{QR} = \mathbf{r}_R - \mathbf{r}_Q = (0, 3, 8) - (2, 4, 6) = \underline{\underline{-2\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z}}$$

(c) The distance between Q and R is

$$|\mathbf{r}_{QR}| = \sqrt{4 + 1 + 4} = \underline{\underline{3}}$$

P. E. 1.3 Consider the figure shown on the next page:

$$\mathbf{u}_z = \mathbf{u}_p + \mathbf{u}_w = -350\mathbf{a}_x + \frac{40}{\sqrt{2}}(-\mathbf{a}_x + \mathbf{a}_y)$$

$$= -378.28\mathbf{a}_x + 28.28\mathbf{a}_y \text{ km/hr}$$

or

$$\mathbf{u}_z = 379.3 \angle 175.72^\circ \text{ km/hr}$$

Where $\mathbf{u}_p$ = velocity of the airplane in the absence of wind

$\mathbf{u}_w$ = wind velocity

$\mathbf{u}_z$ = observed velocity

CHAPTER 2

P. E. 2.1

(a) At $P(1,3,5)$, $x = 1$, $y = 3$, $z = 5$,
 $\rho = \sqrt{x^2 + y^2} = \sqrt{10}$, $z = 5$, $\phi = \tan^{-1} y/x = \tan^{-1} 3 = 71.6^\circ$
 $P(\rho, \phi, z) = P(\sqrt{10}, \tan^{-1} 3, 5) = \underline{\underline{P(3.162, 71.6^\circ, 5)}}$

Spherical system:

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{35} = 5.916$$

$$\theta = \tan^{-1} \sqrt{x^2 + y^2} / z = \tan^{-1} \sqrt{10} / 5 = \tan^{-1} 0.6325 = 32.31^\circ$$

$$P(r, \theta, \phi) = \underline{\underline{P(5.916, 32.31^\circ, 71.57^\circ)}}$$

At $T(0, -4, 3)$, $x = 0$, $y = -4$, $z = 3$;
 $\rho = \sqrt{x^2 + y^2} = 4$, $z = 3$, $\phi = \tan^{-1} y/x = \tan^{-1} -4/0 = 270^\circ$
 $T(\rho, \phi, z) = \underline{\underline{T(4, 270^\circ, 3)}}$.

Spherical system:

$$r = \sqrt{x^2 + y^2 + z^2} = 5$$
, $\theta = \tan^{-1} \rho / z = \tan^{-1} 4/3 = 53.13^\circ$.
 $T(r, \theta, \phi) = \underline{\underline{T(5, 53.13^\circ, 270^\circ)}}$.

At $S(-3, -4, -10)$, $x = -3$, $y = -4$, $z = -10$;
 $\rho = \sqrt{x^2 + y^2} = 5$, $\phi = \tan^{-1} \left(\frac{-4}{-3} \right) = 233.1^\circ$
 $S(\rho, \phi, z) = \underline{\underline{S(5, 233.1^\circ, -10)}}$.

Spherical system:

$$r = \sqrt{x^2 + y^2 + z^2} = 5\sqrt{5} = 11.18.$$

$$\theta = \tan^{-1} \rho / z = \tan^{-1} \frac{5}{-10} = 153.43^\circ;$$

$$S(r, \theta, \phi) = \underline{\underline{S(11.18, 153.43^\circ, 233.1^\circ)}}$$
.

(b) In Cylindrical system, $\rho = \sqrt{x^2 + y^2}$; $yz = z\rho \sin \phi$,
 $Q_x = \frac{\rho}{\sqrt{\rho^2 + z^2}}$; $Q_y = 0$; $Q_z = -\frac{z\rho \sin \phi}{\sqrt{\rho^2 + z^2}}$;

$$\begin{bmatrix} Q_\rho \\ Q_\phi \\ Q_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} Q_x \\ 0 \\ Q_z \end{bmatrix};$$

$$Q_\rho = Q_x \cos\phi = \frac{\rho \cos\phi}{\sqrt{\rho^2 + z^2}}, \quad Q_\phi = -Q_x \sin\phi = \frac{-\rho \sin\phi}{\sqrt{\rho^2 + z^2}}$$

Hence,

$$\underline{\underline{Q = \frac{\rho}{\sqrt{\rho^2 + z^2}} (\cos\phi \mathbf{a}_\rho - \sin\phi \mathbf{a}_\phi - z \sin\phi \mathbf{a}_z).}}$$

In Spherical coordinates:

$$Q_x = \frac{r \sin\theta}{r} = \sin\theta;$$

$$Q_z = -r \sin\phi \sin\theta r \cos\theta \frac{1}{r} = -r \sin\theta \cos\theta \sin\phi.$$

$$\begin{bmatrix} Q_r \\ Q_\theta \\ Q_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta \cos\phi & \sin\theta \sin\phi & \cos\theta \\ \cos\theta \cos\phi & \cos\theta \sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} Q_x \\ 0 \\ Q_z \end{bmatrix};$$

$$Q_r = Q_x \sin\theta \cos\phi + Q_z \cos\theta = \sin^2\theta \cos\phi - r \sin\theta \cos^2\theta \sin\phi.$$

$$Q_\theta = Q_x \cos\theta \cos\phi - Q_z \sin\theta = \sin\theta \cos\theta \cos\phi + r \sin^2\theta \cos\theta \sin\phi.$$

$$Q_\phi = -Q_x \sin\phi = -\sin\theta \sin\phi.$$

$$\underline{\underline{\therefore Q = \sin\theta (\sin\theta \cos\phi - r \cos^2\theta \sin\phi) \mathbf{a}_r + \sin\theta \cos\theta (\cos\phi + r \sin\theta \sin\phi) \mathbf{a}_\theta - \sin\theta \sin\phi \mathbf{a}_\phi.}}$$

At T :

$$Q(x, y, z) = \frac{4}{5} \mathbf{a}_x + \frac{12}{5} \mathbf{a}_z = 0.8 \mathbf{a}_x + 2.4 \mathbf{a}_z;$$

$$\begin{aligned} Q(\rho, \phi, z) &= \frac{4}{5} (\cos 270^\circ \mathbf{a}_\rho - \sin 270^\circ \mathbf{a}_\phi - 3 \sin 270^\circ \mathbf{a}_z) \\ &= 0.8 \mathbf{a}_\phi + 2.4 \mathbf{a}_z; \end{aligned}$$

$$\begin{aligned} Q(r, \theta, \phi) &= \frac{4}{5} \left(0 - \frac{45}{25} (-1) \right) \mathbf{a}_r + \frac{4}{5} \left(\frac{3}{5} \right) \left(0 + \frac{20}{5} (-1) \right) \mathbf{a}_\theta - \frac{4}{5} (-1) \mathbf{a}_\phi \\ &= \frac{36}{25} \mathbf{a}_r - \frac{48}{25} \mathbf{a}_\theta + \frac{4}{5} \mathbf{a}_\phi = \underline{\underline{1.44 \mathbf{a}_r - 1.92 \mathbf{a}_\theta + 0.8 \mathbf{a}_\phi;}} \end{aligned}$$

Note, that the magnitude of vector $Q = 2.53$ in all 3 cases above.

P.E. 2.2 (a)

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \rho z \sin \phi \\ 3\rho \cos \phi \\ \rho \cos \phi \sin \phi \end{bmatrix}$$

$$\mathbf{A} = (\rho z \cos \phi \sin \phi - 3\rho \cos \phi \sin \phi) \mathbf{a}_x + (\rho z \sin^2 \phi + 3\rho \cos^2 \phi) \mathbf{a}_y + \rho \cos \phi \sin \phi \mathbf{a}_z.$$

$$\text{But } \rho = \sqrt{x^2 + y^2}, \quad \tan \phi = \frac{y}{x}, \quad \cos \phi = \frac{x}{\sqrt{x^2 + y^2}}, \quad \sin \phi = \frac{y}{\sqrt{x^2 + y^2}};$$

Substituting all this yields:

$$\mathbf{A} = \frac{1}{\sqrt{x^2 + y^2}} [(xyz - 3xy) \mathbf{a}_x + (zy^2 + 3x^2) \mathbf{a}_y + xy \mathbf{a}_z].$$

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} r^2 \\ 0 \\ \sin \theta \end{bmatrix}$$

$$\text{Since } r = \sqrt{x^2 + y^2 + z^2}, \quad \tan \theta = \frac{\sqrt{x^2 + y^2}}{z}, \quad \tan \phi = \frac{y}{x};$$

$$\text{and } \sin \theta = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}, \quad \cos \theta = \frac{z}{\sqrt{x^2 + y^2 + z^2}};$$

$$\text{and } \sin \phi = \frac{y}{\sqrt{x^2 + y^2}}, \quad \cos \phi = \frac{x}{\sqrt{x^2 + y^2}};$$

$$B_x = r^2 \sin \theta \cos \phi - \sin \theta \sin \phi = rx - \frac{y}{r} = \frac{1}{r}(r^2 x - y).$$

$$B_y = r^2 \sin \theta \sin \phi + \sin \theta \cos \phi = ry + \frac{x}{r} = \frac{1}{r}(r^2 y + x).$$

$$B_z = r^2 \cos \theta = rz = \frac{1}{r}(r^2 z).$$

Hence,

$$\mathbf{B} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} [\{x(x^2 + y^2 + z^2) - y\} \mathbf{a}_x + \{y(x^2 + y^2 + z^2) + x\} \mathbf{a}_y + z(x^2 + y^2 + z^2) \mathbf{a}_z].$$

P.E.2.3 (a) At:

$$(1, \pi/3, 0), \quad \mathbf{H} = (0, 0.06767, 1)$$

$$\mathbf{a}_x = \cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi = \frac{1}{2}(\mathbf{a}_\rho - \sqrt{3} \mathbf{a}_\phi)$$

$$\mathbf{H} \bullet \mathbf{a}_x = \underline{\underline{-0.0586.}}$$

(b) At:

$$(1, \pi/3, 0), \quad \mathbf{a}_\theta = \cos \theta \mathbf{a}_\rho - \sin \theta \mathbf{a}_z = -\mathbf{a}_z.$$

$$\mathbf{H} \times \mathbf{a}_z = \begin{vmatrix} \mathbf{a}_\rho & \mathbf{a}_\phi & \mathbf{a}_z \\ 0 & 0.06767 & 1 \\ 0 & 0 & 1 \end{vmatrix} = \underline{\underline{-0.06767 \mathbf{a}_\rho.}}$$

$$(c) \quad (\mathbf{H} \bullet \mathbf{a}_\rho) \mathbf{a}_\rho = \underline{\underline{0 \mathbf{a}_\rho.}}$$

$$(d) \quad \mathbf{H} \times \mathbf{a}_z = \begin{vmatrix} \mathbf{a}_\rho & \mathbf{a}_\phi & \mathbf{a}_z \\ 0 & 0.06767 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 0.06767 \mathbf{a}_\rho.$$

$$|\mathbf{H} \times \mathbf{a}_z| = \underline{\underline{0.06767}}$$

P.E. 2.4

(a)

$$\mathbf{A} \bullet \mathbf{B} = (3, 2, -6) \bullet (4, 0, 3) = \underline{\underline{-6.}}$$

$$(b) \quad |\mathbf{A} \times \mathbf{B}| = \begin{vmatrix} 3 & 2 & -6 \\ 4 & 0 & 3 \end{vmatrix} = |6\mathbf{a}_r - 33\mathbf{a}_\theta - 8\mathbf{a}_\phi|.$$

$$\text{Thus the magnitude of } \mathbf{A} \times \mathbf{B} = \underline{\underline{34.48.}}$$

(c)

$$\text{At } (1, \pi/3, 5\pi/4), \quad \theta = \pi/3,$$

$$\mathbf{a}_z = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_\theta = \frac{1}{2} \mathbf{a}_r - \frac{\sqrt{3}}{2} \mathbf{a}_\theta.$$

$$(\mathbf{A} \bullet \mathbf{a}_z) \mathbf{a}_z = \left(\frac{3}{2} - \sqrt{3} \right) \left(\frac{1}{2} \mathbf{a}_r - \frac{\sqrt{3}}{2} \mathbf{a}_\theta \right) = \underline{\underline{-0.116 \mathbf{a}_r + 0.201 \mathbf{a}_\theta}}$$

Prob. 2.1

(a)

$$\rho = \sqrt{x^2 + y^2} = \sqrt{4 + 25} = 5.3852, \quad \phi = \tan^{-1} \frac{y}{x} = \tan^{-1} 2.5 = 68.2^\circ$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{4 + 25 + 1} = 5.477, \quad \theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z} = \tan^{-1} \frac{5.3852}{1} = 79.48^\circ$$

$$P(\rho, \phi, z) = \underline{\underline{P(5.3852, 68.2^\circ, 1)}}, \quad P(r, \theta, \phi) = \underline{\underline{P(5.477, 79.48^\circ, 68.2^\circ)}}$$

(b)

$$\rho = \sqrt{x^2 + y^2} = \sqrt{9 + 16} = 5, \quad \phi = \tan^{-1} \frac{y}{x} = \tan^{-1} \frac{4}{-3} = 360^\circ - 53.123^\circ = 306.88^\circ$$

$$r = \sqrt{x^2 + y^2 + z^2} = 5, \quad \theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z} = \tan^{-1} \infty = 90^\circ$$

$$Q(\rho, \phi, z) = \underline{\underline{Q(5, 306.88^\circ, 0)}}, \quad P(r, \theta, \phi) = \underline{\underline{P(5, 90^\circ, 306.88^\circ)}}$$

(c)

$$\rho = \sqrt{x^2 + y^2} = \sqrt{36 + 4} = 6.325, \quad \phi = \tan^{-1} \frac{y}{x} = \tan^{-1} \frac{2}{6} = 18.43^\circ$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{36 + 4 + 16} = 7.483,$$

$$\theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z} = \tan^{-1} \frac{6.325}{-4} = 180^\circ - 57.69^\circ = 122.31^\circ$$

$$R(\rho, \phi, z) = \underline{\underline{R(6.325, 18.43^\circ, -4)}}, \quad R(r, \theta, \phi) = \underline{\underline{R(7.483, 122.31^\circ, 18.43^\circ)}}$$

Prob. 2.2

(a)

$$x = \rho \cos \phi = 2 \cos 30^\circ = 1.732;$$

$$y = \rho \sin \phi = 2 \sin 30^\circ = 1;$$

$$z = 5;$$

$$P_1(x, y, z) = \underline{\underline{P_1(1.732, 1, 5)}}.$$

(b)

$$x = 1 \cos 90^\circ = 0; \quad y = 1 \sin 90^\circ = 1; \quad z = -3.$$

$$P_2(x, y, z) = \underline{\underline{P_2(0, 1, -3)}}.$$

(c)

$$x = r \sin \theta \cos \phi = 10 \sin(\pi/4) \cos(\pi/3) = 3.535;$$

$$y = r \sin \theta \sin \phi = 10 \sin(\pi/4) \sin(\pi/3) = 6.124;$$

$$z = r \cos \theta = 10 \cos(\pi/4) = 7.0711$$

$$P_3(x, y, z) = \underline{\underline{P_3(3.535, 6.124, 7.0711)}}.$$

(d)

$$x = 4 \sin 30^\circ \cos 60^\circ = 1$$

$$y = 4 \sin 30^\circ \sin 60^\circ = 1.7321$$

$$z = r \cos \theta = 4 \cos 30^\circ = 3.464$$

$$P_4(x, y, z) = \underline{\underline{P_4(1, 1.7321, 3.464)}}.$$

Prob. 2.3

$$\rho = \sqrt{x^2 + y^2} = \sqrt{4 + 36} = 6.324$$

$$(a) \quad \phi = \tan^{-1} \frac{y}{x} = \tan^{-1} \frac{6}{2} = 71.56^\circ$$

$$P \text{ is } \underline{\underline{(6.324, 71.56^\circ, -4)}}$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{4 + 36 + 16} = 7.485$$

$$(b) \quad \theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z} = \tan^{-1} \frac{6.324}{-4} = 90^\circ + \tan^{-1} \frac{4}{6.324} = 122.3^\circ$$

$$P \text{ is } \underline{\underline{(7.483, 122.3^\circ, 71.56^\circ)}}$$

Prob. 2.4

(a)

$$x = \rho \cos \phi = 5 \cos 120^\circ = -2.5$$

$$y = \rho \sin \phi = 5 \sin 120^\circ = 4.33$$

$$z = 1$$

$$\text{Hence } Q = \underline{\underline{(-2.5, 4.33, 1)}}$$

(b)

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{\rho^2 + z^2} = \sqrt{25 + 1} = 5.099$$

$$\theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z} = \tan^{-1} \frac{\rho}{z} = \tan^{-1} \frac{5}{1} = 78.69^\circ$$

$$\phi = 120^\circ$$

$$\text{Hence } Q = \underline{\underline{(5.099, 78.69^\circ, 120^\circ)}}$$

Prob. 2.5

$$T(r, \theta, \phi) \longrightarrow r = 10, \theta = 60^\circ, \phi = 30^\circ$$

$$x = r \sin \theta \cos \phi = 10 \sin 60^\circ \cos 30^\circ = 7.5$$

$$y = r \sin \theta \sin \phi = 10 \sin 60^\circ \sin 30^\circ = 4.33$$

$$z = r \cos \theta = 10 \cos 60^\circ = 5$$

$$T(x, y, z) = \underline{\underline{(7.5, 4.33, 5)}}$$

$$\rho = r \sin \theta = 10 \sin 60^\circ = 8.66$$

$$T(\rho, \phi, z) = \underline{\underline{(8.66, 30^\circ, 5)}}$$

Prob. 2.6

(a)

$$x = \rho \cos \phi, \quad y = \rho \sin \phi,$$

$$V = \underline{\underline{\rho z \cos \phi - \rho^2 \sin \phi \cos \phi + \rho z \sin \phi}}$$

(b)

$$\begin{aligned} U &= x^2 + y^2 + z^2 + y^2 + 2z^2 \\ &= r^2 + r^2 \sin^2 \theta \sin^2 \phi + 2r^2 \cos^2 \theta \\ &= \underline{\underline{r^2 [1 + \sin^2 \theta \sin^2 \phi + 2 \cos^2 \theta]}} \end{aligned}$$

Prob. 2.7

(a)

$$\begin{bmatrix} F_\rho \\ F_\phi \\ F_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{x}{\sqrt{\rho^2 + z^2}} \\ \frac{y}{\sqrt{\rho^2 + z^2}} \\ \frac{4}{\sqrt{\rho^2 + z^2}} \end{bmatrix}$$

$$F_\rho = \frac{1}{\sqrt{\rho^2 + z^2}} [\rho \cos^2 \phi + \rho \sin^2 \phi] = \frac{\rho}{\sqrt{\rho^2 + z^2}};$$

$$F_\phi = \frac{1}{\sqrt{\rho^2 + z^2}} [-\rho \cos \phi \sin \phi + \rho \cos \phi \sin \phi] = 0;$$

$$F_z = \frac{4}{\sqrt{\rho^2 + z^2}};$$

$$\underline{\underline{\bar{F} = \frac{1}{\sqrt{\rho^2 + z^2}} (\rho \mathbf{a}_\rho + 4 \mathbf{a}_z)}}$$

In Spherical:

$$\begin{bmatrix} F_r \\ F_\theta \\ F_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} \frac{x}{r} \\ \frac{y}{r} \\ \frac{4}{r} \end{bmatrix}$$

$$F_r = \frac{r}{r} \sin^2 \theta \cos^2 \phi + \frac{r}{r} \sin^2 \theta \sin^2 \phi + \frac{4}{r} \cos \theta = \sin^2 \theta + \frac{4}{r} \cos \theta;$$

$$F_\theta = \sin \theta \cos \theta \cos^2 \phi + \sin \theta \cos \theta \sin^2 \phi - \frac{4}{r} \sin \theta = \sin \theta \cos \theta - \frac{4}{r} \sin \theta;$$

$$F_\phi = -\sin \theta \cos \phi \sin \phi + \sin \theta \sin \phi \cos \phi = 0;$$

$$\underline{\underline{\therefore \bar{F} = (\sin^2 \theta + \frac{4}{r} \cos \theta) \mathbf{a}_r + \sin \theta (\cos \theta - \frac{4}{r}) \mathbf{a}_\theta}}$$

(b)

$$\begin{bmatrix} G_\rho \\ G_\phi \\ G_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{x\rho^2}{\sqrt{\rho^2 + z^2}} \\ \frac{y\rho^2}{\sqrt{\rho^2 + z^2}} \\ \frac{z\rho^2}{\sqrt{\rho^2 + z^2}} \end{bmatrix}$$

$$G_\rho = \frac{\rho^2}{\sqrt{\rho^2 + z^2}} [\rho \cos^2 \phi + \rho \sin^2 \phi] = \frac{\rho^3}{\sqrt{\rho^2 + z^2}};$$

$$G_\phi = 0;$$

$$G_z = \frac{z\rho^2}{\sqrt{\rho^2 + z^2}};$$

$$\underline{\underline{\mathbf{G} = \frac{\rho^2}{\sqrt{\rho^2 + z^2}} (\rho \mathbf{a}_\rho + z \mathbf{a}_z)}}$$

Spherical :

$$\mathbf{G} = \frac{\rho^2}{r} (x\mathbf{a}_x + y\mathbf{a}_y + z\mathbf{a}_z) = \frac{r^2 \sin^2 \theta}{r} r\mathbf{a}_r = \underline{\underline{r^2 \sin^2 \theta \mathbf{a}_r}}$$

Prob. 2.8

$$\mathbf{B} = \rho \mathbf{a}_x + \frac{y}{\rho} \mathbf{a}_y + z \mathbf{a}_z$$

$$\begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \rho \\ y/\rho \\ z \end{bmatrix}$$

$$B_\rho = \rho \cos \phi + \frac{y}{\rho} \sin \phi$$

$$B_\phi = -\rho \sin \phi + \frac{y}{\rho} \cos \phi$$

$$B_z = z$$

$$\text{But } y = \rho \sin \phi$$

$$B_\rho = \rho \cos \phi + \sin^2 \phi, B_\phi = -\rho \sin \phi + \sin \phi \cos \phi$$

Hence,

$$\underline{\underline{\mathbf{B} = (\rho \cos \phi + \sin^2 \phi) \mathbf{a}_\rho + \sin \phi (\cos \phi - \rho) \mathbf{a}_\phi + z \mathbf{a}_z}}$$

Prob. 2.9

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

At P, $\rho = 2$, $\phi = \pi/2$, $z = -1$

$$A_x = 2 \cos \phi - 3 \sin \phi = 2 \cos 90^\circ - 3 \sin 90^\circ = -3$$

$$A_y = 2 \sin \phi + 3 \cos \phi = 2 \sin 90^\circ + 3 \cos 90^\circ = 2$$

$$A_z = 4$$

Hence, $\underline{\underline{A = -3\mathbf{a}_x + 2\mathbf{a}_y + 4\mathbf{a}_z}}$

Prob. 2.10

(a)

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \rho \sin \phi \\ \rho \cos \phi \\ -2z \end{bmatrix}$$

$$A_x = \rho \sin \phi \cos \phi - \rho \cos \phi \sin \phi = 0$$

$$A_y = \rho \sin^2 \phi + \rho \cos^2 \phi = \rho = \sqrt{x^2 + y^2}$$

$$A_z = -2z$$

Hence,

$$\underline{\underline{A = \sqrt{x^2 + y^2} \mathbf{a}_y - 2z \mathbf{a}_z}}$$

(b)

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} 4r \cos \phi \\ r \\ 0 \end{bmatrix}$$

$$B_x = 4r \sin \theta \cos^2 \phi + r \cos \theta \cos \phi$$

$$B_y = 4r \sin \theta \sin \phi \cos \phi + r \cos \theta \sin \phi$$

$$B_z = 4r \cos \theta \cos \phi - r \sin \theta$$

But $r = \sqrt{x^2 + y^2 + z^2}$, $\sin \theta = \frac{\sqrt{x^2 + y^2}}{r}$, $\cos \theta = \frac{z}{r}$

$$\sin \phi = \frac{y}{\sqrt{x^2 + y^2}}, \quad \cos \phi = \frac{x}{\sqrt{x^2 + y^2}}$$

$$B_x = 4\sqrt{x^2 + y^2} \frac{x^2}{x^2 + y^2} + \frac{zx}{\sqrt{x^2 + y^2}}$$

$$B_y = 4\sqrt{x^2 + y^2} \frac{xy}{x^2 + y^2} + \frac{zy}{\sqrt{x^2 + y^2}}$$

$$B_z = 4z \frac{x}{\sqrt{x^2 + y^2}} - \sqrt{x^2 + y^2}$$

$$\mathbf{B} = \frac{1}{\sqrt{x^2 + y^2}} \left[x(4x + z)\mathbf{a}_x + y(4x + z)\mathbf{a}_y + (4xz - x^2 - y^2)\mathbf{a}_z \right]$$

Prob. 2.11

Method 1:

$$\begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} 4/r^2 \\ 0 \\ 0 \end{bmatrix}$$

$$F_x = \frac{4}{r^2} \sin \theta \cos \phi, \quad F_y = \frac{4}{r^2} \sin \theta \sin \phi, \quad F_z = \frac{4}{r^2} \cos \theta$$

$$r^2 = x^2 + y^2 + z^2, \quad \sin \theta = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}, \quad \cos \theta = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\sin \phi = \frac{y}{\sqrt{x^2 + y^2}}, \quad \cos \phi = \frac{x}{\sqrt{x^2 + y^2}}$$

$$F_x = \frac{4}{x^2 + y^2 + z^2} \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \frac{x}{\sqrt{x^2 + y^2}} = \frac{4x}{(x^2 + y^2 + z^2)^{3/2}}$$

$$F_y = \frac{4}{x^2 + y^2 + z^2} \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \frac{y}{\sqrt{x^2 + y^2}} = \frac{4y}{(x^2 + y^2 + z^2)^{3/2}}$$

$$F_z = \frac{4}{x^2 + y^2 + z^2} \frac{z}{(x^2 + y^2 + z^2)} = \frac{4z}{(x^2 + y^2 + z^2)^{3/2}}$$

Thus,

$$\mathbf{F} = \frac{4}{(x^2 + y^2 + z^2)^{3/2}} \left[x\mathbf{a}_x + y\mathbf{a}_y + z\mathbf{a}_z \right]$$

Method 2:

$$\mathbf{F} = \frac{4\mathbf{a}_r}{r^2} \cdot \frac{\mathbf{r}}{r} = \frac{4r\mathbf{a}_r}{r^3}$$

$$\mathbf{F} = \frac{4}{(x^2 + y^2 + z^2)^{3/2}} [x\mathbf{a}_x + y\mathbf{a}_y + z\mathbf{a}_z]$$

Prob. 2.12

$$r = 2, \quad \theta = \pi/2, \quad \phi = 3\pi/2$$

$$(a) \quad \mathbf{B} = 2 \sin(\pi/2)\mathbf{a}_r - 4 \cos(3\pi/2)\mathbf{a}_\phi = \underline{\underline{2\mathbf{a}_r}}$$

(b)

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \cos \theta \cos \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} r \sin \theta \\ 0 \\ -r^2 \cos \phi \end{bmatrix}$$

$$B_x = r \sin^2 \theta \cos \phi - r^2 \sin \phi \cos \phi, \quad B_y = r \sin \theta \cos \theta \cos \phi - r^2 \cos^2 \phi$$

$$B_z = r \sin \theta \cos \theta$$

$$\text{But } r = \sqrt{x^2 + y^2 + z^2}, \cos \theta = \frac{z}{r}, \sin \theta = \frac{\rho}{r} = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}$$

$$\cos \phi = \frac{x}{\rho} = \frac{x}{\sqrt{x^2 + y^2}}, \quad \sin \phi = \frac{y}{\rho} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\begin{aligned} B_x &= \sqrt{x^2 + y^2 + z^2} \frac{x^2 + y^2}{x^2 + y^2 + z^2} \frac{x}{\sqrt{x^2 + y^2}} - (x^2 + y^2 + z^2) \frac{xy}{x^2 + y^2} \\ &= \frac{x\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} - \frac{xy(x^2 + y^2 + z^2)}{x^2 + y^2} \end{aligned}$$

$$\begin{aligned} B_y &= \sqrt{x^2 + y^2 + z^2} \frac{z\sqrt{x^2 + y^2}}{x^2 + y^2 + z^2} \frac{x}{\sqrt{x^2 + y^2}} - (x^2 + y^2 + z^2) \frac{x^2}{x^2 + y^2} \\ &= \frac{xz}{\sqrt{x^2 + y^2 + z^2}} - \frac{x^2(x^2 + y^2 + z^2)}{x^2 + y^2} \end{aligned}$$

$$B_z = \sqrt{x^2 + y^2 + z^2} \frac{z\sqrt{x^2 + y^2}}{x^2 + y^2 + z^2} = \frac{z\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}$$

$$\mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$$

Prob. 2.13

$$(a) \quad x = \rho \cos \phi$$

$$\underline{\underline{\mathbf{B} = \rho \cos \phi \mathbf{a}_z}}$$

$$x = r \sin \theta \cos \phi$$

$$(b) \quad \mathbf{B} = r \sin \theta \cos \phi \mathbf{a}_z, \quad B_x = 0 = B_y, B_z = r \sin \theta \cos \phi$$

$$\begin{bmatrix} B_r \\ B_\theta \\ B_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ r \sin \theta \cos \phi \end{bmatrix}$$

$$B_r = r \sin \theta \cos \theta \cos \phi = 0.5r \sin(2\theta) \cos \phi$$

$$B_\theta = -r \sin^2 \theta \cos \phi, \quad B_\phi = 0$$

$$\underline{\underline{\mathbf{B} = 0.5r \sin(2\theta) \cos \phi \mathbf{a}_r - r \sin^2 \theta \cos \phi \mathbf{a}_\theta}}$$

Prob. 2.14

(a)

$$\mathbf{a}_x \times \mathbf{a}_\rho = (\cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi) \times \mathbf{a}_\rho = \cos \phi$$

$$\mathbf{a}_x \times \mathbf{a}_\phi = (\cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi) \times \mathbf{a}_\phi = -\sin \phi$$

$$\mathbf{a}_y \times \mathbf{a}_\rho = (\sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi) \times \mathbf{a}_\rho = \sin \phi$$

$$\bar{\mathbf{a}}_y \times \bar{\mathbf{a}}_\phi = (\sin \phi \mathbf{a}_\rho + \sin \phi \mathbf{a}_\phi) \times \mathbf{a}_\phi = \cos \phi$$

(b) and (c)

In spherical system :

$$\mathbf{a}_x = \sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi.$$

$$\mathbf{a}_y = \sin \theta \sin \phi \mathbf{a}_r + \cos \theta \sin \phi \mathbf{a}_\theta - \cos \phi \mathbf{a}_\phi.$$

$$\mathbf{a}_z = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_\theta.$$

Hence,

$$\mathbf{a}_x \times \mathbf{a}_r = \sin \theta \cos \phi;$$

$$\mathbf{a}_x \times \mathbf{a}_\theta = \cos \theta \cos \phi;$$

$$\mathbf{a}_y \times \mathbf{a}_r = \sin \theta \sin \phi;$$

$$\mathbf{a}_y \times \mathbf{a}_\theta = \cos \theta \sin \phi;$$

$$\bar{\mathbf{a}}_z \times \bar{\mathbf{a}}_r = \cos \theta;$$

$$\bar{\mathbf{a}}_z \times \bar{\mathbf{a}}_\theta = -\sin \theta;$$

Prob. 2.15

(a)

$$\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y, \quad \mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$$

$$\mathbf{a}_\rho \times \mathbf{a}_\phi = \begin{vmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \end{vmatrix} = (\cos^2 \phi + \sin^2 \phi) \mathbf{a}_z = \mathbf{a}_z$$

$$\mathbf{a}_z \times \mathbf{a}_\rho = \begin{vmatrix} 0 & 0 & 1 \\ \cos \phi & \sin \phi & 0 \end{vmatrix} = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y = \mathbf{a}_\phi$$

$$\mathbf{a}_\phi \times \mathbf{a}_z = \begin{vmatrix} -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{vmatrix} = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y = \mathbf{a}_\rho$$

(b)

$$\mathbf{a}_r = \sin \theta \cos \phi \mathbf{a}_x + \sin \theta \sin \phi \mathbf{a}_y + \cos \theta \mathbf{a}_z$$

$$\mathbf{a}_\theta = \cos \theta \cos \phi \mathbf{a}_x + \cos \theta \sin \phi \mathbf{a}_y - \sin \theta \mathbf{a}_z$$

$$\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$$

$$\mathbf{a}_r \times \mathbf{a}_\theta = \begin{vmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \end{vmatrix}$$

$$= (-\sin^2 \theta \sin \phi - \cos^2 \theta \sin \phi) \mathbf{a}_x + (\cos^2 \theta \cos \phi + \sin^2 \theta \cos \phi) \mathbf{a}_y$$

$$+ (\sin \theta \cos \theta \sin \phi \cos \phi - \sin \theta \cos \theta \sin \phi \cos \phi) \mathbf{a}_z$$

$$= -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y = \mathbf{a}_\phi$$

$$\begin{aligned}
\mathbf{a}_\phi \times \mathbf{a}_r &= \begin{vmatrix} -\sin \phi & \cos \phi & 0 \\ \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \end{vmatrix} \\
&= \cos \theta \cos \phi \mathbf{a}_x + \cos \theta \sin \phi \mathbf{a}_y + (-\sin \theta \sin^2 \phi - \sin \theta \cos^2 \phi) \mathbf{a}_z \\
&= \cos \theta \cos \phi \mathbf{a}_x + \cos \theta \sin \phi \mathbf{a}_y - \sin \theta \mathbf{a}_z = \mathbf{a}_\theta \\
\mathbf{a}_\theta \times \mathbf{a}_\phi &= \begin{vmatrix} \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{vmatrix} \\
&= \sin \theta \cos \phi \mathbf{a}_x + \sin \theta \sin \phi \mathbf{a}_y + (\cos \theta \cos^2 \phi + \cos \theta \sin^2 \phi) \mathbf{a}_z \\
&= \sin \theta \cos \phi \mathbf{a}_x + \sin \theta \sin \phi \mathbf{a}_y + \cos \theta \mathbf{a}_z = \mathbf{a}_r
\end{aligned}$$

Prob. 2.16

(a)

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{\rho^2 + z^2}.$$

$$\theta = \tan^{-1} \frac{\rho}{z}; \quad \phi = \phi.$$

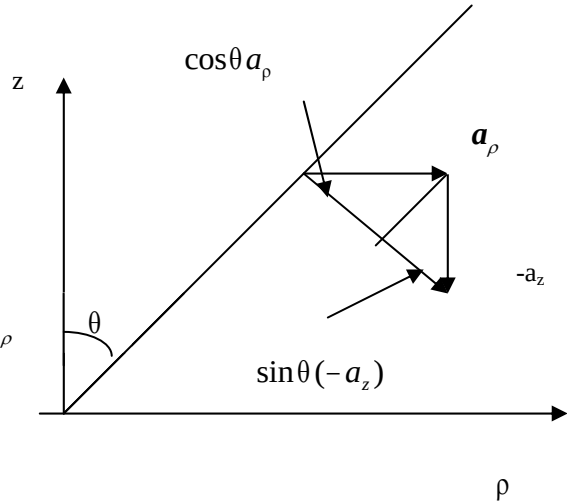
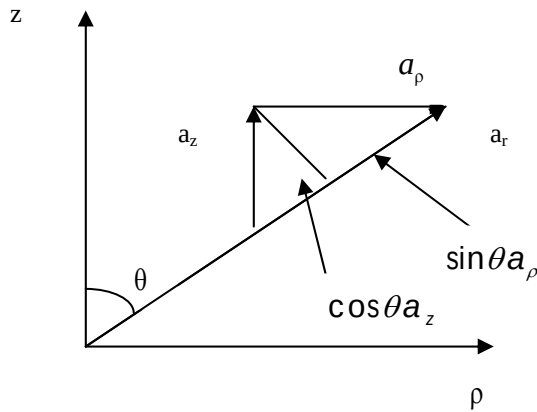
or

$$\rho = \sqrt{x^2 + y^2} = \sqrt{r^2 \sin^2 \theta \cos^2 \phi + r^2 \sin^2 \theta \sin^2 \phi}.$$

$$= r \sin \theta;$$

$$z = r \cos \theta; \quad \phi = \phi.$$

(b) From the figures below,



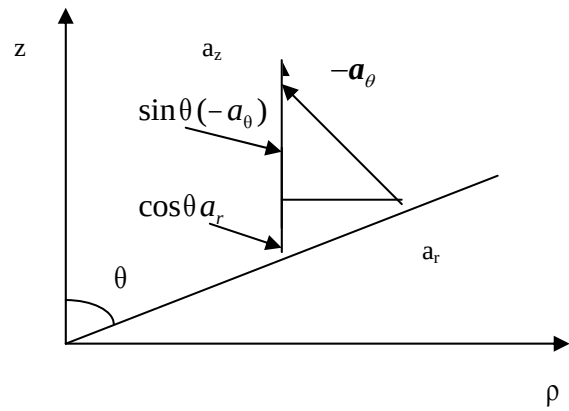
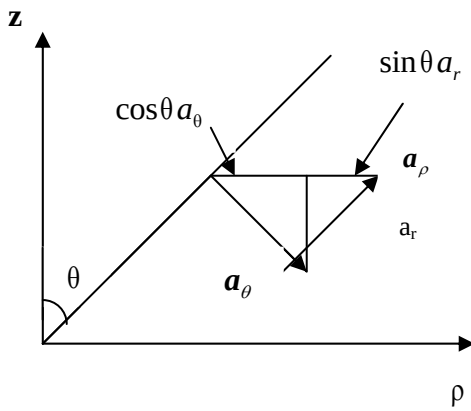
$$\mathbf{a}_r = \sin\theta \mathbf{a}_\rho + \cos\theta \mathbf{a}_z; \quad \mathbf{a}_\theta = \cos\theta \mathbf{a}_\rho - \sin\theta \mathbf{a}_z; \quad \mathbf{a}_\phi = \mathbf{a}_\phi.$$

Hence,

$$\begin{bmatrix} \mathbf{a}_r \\ \mathbf{a}_\theta \\ \mathbf{a}_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta & 0 & \cos\theta \\ \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{a}_\rho \\ \mathbf{a}_\phi \\ \mathbf{a}_z \end{bmatrix}$$

From the figures below,

$$\mathbf{a}_\rho = \cos\theta \mathbf{a}_\theta + \sin\theta \mathbf{a}_r; \quad \mathbf{a}_z = \cos\theta \mathbf{a}_r - \sin\theta \mathbf{a}_\theta; \quad \mathbf{a}_\phi = \mathbf{a}_\phi.$$



$$\begin{bmatrix} \mathbf{a}_\rho \\ \mathbf{a}_\phi \\ \mathbf{a}_z \end{bmatrix} = \begin{bmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \mathbf{a}_r \\ \mathbf{a}_\theta \\ \mathbf{a}_z \end{bmatrix}$$

Prob. 2.17

At $P(2, 0, -1)$, $\phi = 0$, $\theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right) = \cos^{-1}\left(\frac{-1}{\sqrt{5}}\right) = 116.56^\circ$

- (a) $\mathbf{a}_\rho \bullet \mathbf{a}_x = \cos\phi = \underline{\underline{1}}$
- (b) $\mathbf{a}_\phi \bullet \mathbf{a}_y = \cos\phi = \underline{\underline{1}}$
- (c) $\mathbf{a}_r \bullet \mathbf{a}_z = \cos\theta = \underline{\underline{-0.4472}}$

Prob. 2.18

If **A** and **B** are perpendicular to each other, $\mathbf{A} \cdot \mathbf{B} = 0$

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= \rho^2 \sin^2 \phi + \rho^2 \cos^2 \phi - \rho^2 \\ &= \rho^2 (\sin^2 \phi + \cos^2 \phi) - \rho^2 \\ &= \rho^2 - \rho^2 \\ &= 0\end{aligned}$$

As expected.

Prob. 2.19

$$(a) \mathbf{A} + \mathbf{B} = \underline{\underline{8\mathbf{a}_\rho + 2\mathbf{a}_\phi - 7\mathbf{a}_z}}$$

$$(b) \mathbf{A} \cdot \mathbf{B} = 15 + 0 - 8 = \underline{\underline{7}}$$

$$\begin{aligned}(c) \mathbf{A} \times \mathbf{B} &= \begin{vmatrix} 3 & 2 & 1 \\ 5 & 0 & -8 \end{vmatrix} \\ &= -16\mathbf{a}_\rho + (5 + 24)\mathbf{a}_\phi - 10\mathbf{a}_z \\ &= \underline{\underline{-16\mathbf{a}_\rho + 29\mathbf{a}_\phi - 10\mathbf{a}_z}}\end{aligned}$$

$$\begin{aligned}(d) \cos \theta_{AB} &= \frac{\mathbf{A} \cdot \mathbf{B}}{AB} = \frac{7}{\sqrt{9+4+1}\sqrt{25+64}} = \frac{7}{\sqrt{14}\sqrt{89}} \\ &= 0.19831\end{aligned}$$

$$\underline{\underline{\theta_{AB} = 78.56^\circ}}$$

Prob. 2.20

$$\begin{bmatrix} G_x \\ G_y \\ G_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} G_\rho \\ G_\phi \\ G_z \end{bmatrix}$$

$$\begin{aligned}G_x &= G_\rho \cos \phi - G_\phi \sin \phi = 3\rho \cos \phi - \rho \cos \phi \sin \phi \\ &= 3x - x \sin \phi = 3(3) - (3)\sin(306.87^\circ) = 11.4\end{aligned}$$

$$\mathbf{G}_x = G_x \mathbf{a}_x = \underline{\underline{11.4\mathbf{a}_x}}$$

Prob. 2.21

$$\begin{bmatrix} G_\rho \\ G_\phi \\ G_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} yz \\ xz \\ xy \end{bmatrix}$$

$$G_\rho = yz \cos \phi + xz \sin \phi$$

$$x = \rho \cos \phi, y = \rho \sin \phi, \quad yz = \rho z \sin \phi, xz = \rho z \cos \phi$$

$$G_\rho = \rho z \sin \phi \cos \phi + \rho z \cos \phi \sin \phi = 2\rho z \sin \phi \cos \phi = \rho z \sin 2\phi$$

$$G_\phi = -yz \sin \phi + xz \cos \phi = \rho z (\cos^2 \phi - \sin^2 \phi) = \rho z \cos 2\phi$$

$$G_z = xy = \rho^2 \cos \phi \sin \phi = 0.5\rho^2 \sin 2\phi$$

$$\underline{\underline{G = \rho z \sin 2\phi \mathbf{a}_y + \rho z \cos 2\phi \mathbf{a}_\phi + 0.5\rho^2 \sin 2\phi \mathbf{a}_z}}$$

Prob. 2.22

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

$$= \begin{bmatrix} \frac{x}{\sqrt{x^2 + y^2}} & -\frac{y}{\sqrt{x^2 + y^2}} & 0 \\ \frac{y}{\sqrt{x^2 + y^2}} & \frac{x}{\sqrt{x^2 + y^2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

$$= \begin{bmatrix} \frac{x}{\sqrt{x^2 + y^2 + z^2}} & \frac{xz}{\sqrt{x^2 + y^2} \sqrt{x^2 + y^2 + z^2}} & \frac{-y}{\sqrt{x^2 + y^2}} \\ \frac{y}{\sqrt{x^2 + y^2 + z^2}} & \frac{yz}{\sqrt{x^2 + y^2} \sqrt{x^2 + y^2 + z^2}} & \frac{x}{\sqrt{x^2 + y^2}} \\ \frac{z}{\sqrt{x^2 + y^2 + z^2}} & -\frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

Prob. 2.23 (a) Using the results in Prob.2.14,

$$A_\rho = \rho z \sin \phi = r^2 \sin \theta \cos \theta \sin \phi$$

$$A_\phi = 3\rho \cos \phi = 3r \sin \theta \cos \phi$$

$$A_z = \rho \cos \phi \sin \phi = r \sin \theta \cos \phi \sin \phi$$

Hence,

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta & 0 & \cos \theta \\ \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r^2 \sin \theta \cos \theta \sin \phi \\ 3r \sin \theta \cos \phi \\ r \sin \theta \cos \phi \sin \phi \end{bmatrix}$$

$$\underline{\underline{\mathbf{A}(r, \theta, \phi) = r \sin \theta \left[\sin \phi \cos \theta (r \sin \theta + \cos \phi) \mathbf{a}_r + \sin \phi (r \cos^2 \theta - \sin \theta \cos \phi) \mathbf{a}_\theta + 3 \cos \phi \mathbf{a}_\phi \right]}}$$

At $(10, \pi/2, 3\pi/4)$, $r = 10, \theta = \pi/2, \phi = 3\pi/4$

$$\underline{\underline{\mathbf{A} = 10(0\mathbf{a}_r + 0.5\mathbf{a}_\theta - \frac{3}{\sqrt{2}}\mathbf{a}_\phi) = 5\mathbf{a}_\theta - 21.21\mathbf{a}_\phi}}$$

(b) $B_r = r^2 = (\rho^2 + z^2), \quad B_\theta = 0, \quad B_\phi = \sin \theta = \frac{\rho}{\sqrt{\rho^2 + z^2}}$

$$\begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} B_r \\ B_\theta \\ B_\phi \end{bmatrix}$$

$$\underline{\underline{\mathbf{B}(\rho, \phi, z) = \sqrt{\rho^2 + z^2} \left(\rho \mathbf{a}_\rho + \frac{\rho}{\rho^2 + z^2} \mathbf{a}_\phi + z \mathbf{a}_z \right)}}$$

At $(2, \pi/6, 1)$, $\rho = 2, \phi = \pi/6, z = 1$

$$\underline{\underline{\mathbf{B} = \sqrt{5}(2\mathbf{a}_\rho + 0.4\mathbf{a}_\phi + \mathbf{a}_z) = 4.472\mathbf{a}_\rho + 0.8944\mathbf{a}_\phi + 2.236\mathbf{a}_z}}$$

Prob. 2.24

(a) $d = \sqrt{(6-2)^2 + (-1-1)^2 + (2-5)^2} = \sqrt{29} = \underline{\underline{5.385}}$

(b) $d^2 = 3^2 + 5^2 - 2(3)(5) \cos \pi + (-1-5)^2 = 100$
 $d = \sqrt{100} = \underline{\underline{10}}$

(c)

$$\begin{aligned}
 d^2 &= 10^2 + 5^2 - 2(10)(5) \cos \frac{\pi}{4} \cos \frac{\pi}{6} - 2(10)(5) \sin \frac{\pi}{4} \sin \frac{\pi}{6} \cos \left(7\frac{\pi}{4} - \frac{3\pi}{4} \right) \\
 &= 125 - 100(0.7071)(0.866) - 100(0.7071)(0.5)(-0.2334) \\
 &= 125 - 61.23 + 35.33 = 99.118 \\
 d &= \sqrt{99.118} = \underline{\underline{9.956}}.
 \end{aligned}$$

Prob. 2.25

Using eq. (2.33),

$$\begin{aligned}
 d^2 &= r_1^2 + r_2^2 - 2r_1r_2 \cos \theta_1 \cos \theta_2 - 2r_1r_2 \sin \theta_1 \sin \theta_2 \cos(\phi_2 - \phi_1) \\
 &= 16 + 36 - 2(4)(6) \cos 30^\circ \cos 90^\circ - 2(4)(6) \sin 30^\circ \sin 90^\circ \cos(180^\circ) \\
 &= 16 + 36 - 0 - 48(0.5)(1)(-1) = 52 + 24 = 76 \\
 d &= 8.718
 \end{aligned}$$

Prob. 2.26

$$\begin{bmatrix} \mathbf{a}_\rho \\ \mathbf{a}_\phi \\ \mathbf{a}_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{a}_x \\ \mathbf{a}_y \\ \mathbf{a}_z \end{bmatrix}$$

$$\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y, \quad \mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$$

$$\text{At } (0, 4, -1), \quad \phi = 90^\circ$$

$$\mathbf{a}_\rho = \sin 90^\circ \mathbf{a}_y = \underline{\underline{\mathbf{a}_y}}$$

$$\mathbf{a}_\phi = -\sin 90^\circ \mathbf{a}_x = \underline{\underline{-\mathbf{a}_x}}$$

Prob. 2.27

$$\text{At } (1, 60^\circ, -1), \quad \rho = 1, \phi = 60^\circ, z = -1,$$

$$\begin{aligned}
 \text{(a) } \mathbf{A} &= (-2 - \sin 60^\circ) \mathbf{a}_\rho + (4 + 2 \cos 60^\circ) \mathbf{a}_\phi - 3(1)(-1) \mathbf{a}_z \\
 &= -2.866 \mathbf{a}_\rho + 5 \mathbf{a}_\phi + 3 \mathbf{a}_z
 \end{aligned}$$

$$\mathbf{B} = 1 \cos 60^\circ \mathbf{a}_\rho + \sin 60^\circ \mathbf{a}_\phi + \mathbf{a}_z = 0.5 \mathbf{a}_\rho + 0.866 \mathbf{a}_\phi + \mathbf{a}_z$$

$$\mathbf{A} \cdot \mathbf{B} = -1.433 + 4.33 + 3 = 5.897$$

$$AB = \sqrt{2.866^2 + 26 + 9} \sqrt{0.25 + 1 + 0.866^2} = 9.1885$$

$$\cos \theta_{AB} = \frac{\mathbf{A} \cdot \mathbf{B}}{AB} = \frac{5.897}{9.1885} = 0.6419 \quad \longrightarrow \quad \theta_{AB} = \underline{\underline{50.07^\circ}}$$

Let $\mathbf{D} = \mathbf{A} \times \mathbf{B}$. At $(1, 90^\circ, 0)$, $\rho = 1, \phi = 90^\circ, z = 0$

$$(b) \mathbf{A} = -\sin 90^\circ \mathbf{a}_\rho + 4\mathbf{a}_\phi = -\mathbf{a}_\rho + 4\mathbf{a}_\phi$$

$$\mathbf{B} = 1 \cos 90^\circ \mathbf{a}_\rho + \sin 90^\circ \mathbf{a}_\phi + \mathbf{a}_z = \mathbf{a}_\phi + \mathbf{a}_z$$

$$\mathbf{D} = \mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_\rho & \mathbf{a}_\phi & \mathbf{a}_z \\ -1 & 4 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 4\mathbf{a}_\rho + \mathbf{a}_\phi - \mathbf{a}_z$$

$$\mathbf{a}_D = \frac{\mathbf{D}}{D} = \frac{(4, 1, -1)}{\sqrt{16+1+1}} = \underline{\underline{0.9428\mathbf{a}_\rho + 0.2357\mathbf{a}_\phi - 0.2357\mathbf{a}_z}}$$

Prob.2.28

At $P(0, 2, -5)$, $\phi = 90^\circ$;

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -5 \\ 1 \\ -3 \end{bmatrix}$$

$$\mathbf{B} = -\mathbf{a}_x - 5\mathbf{a}_y - 3\mathbf{a}_z$$

$$(a) \mathbf{A} + \mathbf{B} = (2, 4, 10) + (-1, -5, -3)$$

$$= \underline{\underline{\mathbf{a}_x - \mathbf{a}_y + 7\mathbf{a}_z}}$$

$$(b) \cos \theta_{AB} = \frac{\mathbf{A} \cdot \mathbf{B}}{AB} = \frac{-52}{\sqrt{4200}}$$

$$\theta_{AB} = \cos^{-1}\left(\frac{-52}{\sqrt{4200}}\right) = \underline{\underline{143.36^\circ}}$$

$$(c) A_B = \mathbf{A} \cdot \mathbf{a}_B = \frac{\mathbf{A} \cdot \mathbf{B}}{B} = -\frac{52}{\sqrt{35}} = \underline{\underline{-8.789}}$$

Prob. 2.29

$$\mathbf{B} \cdot \mathbf{a}_x = B_x$$

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B_\rho \\ B_\phi \\ B_z \end{bmatrix}$$

$$\begin{aligned} B_x &= B_\rho \cos \phi - B_\phi \sin \phi = \rho^2 \sin \phi \cos \phi - (z-1) \cos \phi \sin \phi \\ &= 16(0.5) - (-2)(0.5) = 8 + 1 = \underline{\underline{9}} \end{aligned}$$

Prob. 2.30

$$\bar{\mathbf{G}} = \cos^2 \phi \mathbf{a}_x + \frac{2r \cos \theta \sin \phi}{r \sin \theta} \bar{\mathbf{a}}_y + (1 - \cos^2 \phi) \bar{\mathbf{a}}_z$$

$$= \cos^2 \phi \bar{\mathbf{a}}_x + 2 \cot \theta \sin \phi \bar{\mathbf{a}}_y + \sin^2 \phi \bar{\mathbf{a}}_z$$

$$\begin{pmatrix} G_r \\ G_\theta \\ G_\phi \end{pmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} \cos^2 \phi \\ 2 \cot \theta \sin \phi \\ \sin^2 \phi \end{bmatrix}$$

$$\begin{aligned} G_r &= \sin \theta \cos^3 \phi + 2 \cos \theta \sin^2 \phi + \cos \theta \sin^2 \phi \\ &= \sin \theta \cos^3 \phi + 3 \cos \theta \sin^2 \phi \end{aligned}$$

$$G_\theta = \cos \theta \cos^3 \phi + 2 \cot \theta \cos \theta \sin^2 \phi - \sin \theta \sin^2 \phi$$

$$G_\phi = -\sin \phi \cos^2 \phi + 2 \cot \theta \sin \phi \cos \phi$$

$$\begin{aligned} \bar{\mathbf{G}} &= [\sin \theta \cos^3 \phi + 3 \cos \theta \sin^2 \phi] \mathbf{a}_r \\ &\quad + [\cos \theta \cos^3 \phi + 2 \cot \theta \cos \theta \sin^2 \phi - \sin \theta \sin^2 \phi] \mathbf{a}_\theta \\ &\quad + \underline{\underline{\sin \phi \cos \phi (2 \cot \theta - \cos \phi) \mathbf{a}_\phi}} \end{aligned}$$

Prob. 2.31

- (a) An infinite line parallel to the z-axis.
- (b) Point (2,-1,10).
- (c) A circle of radius $r \sin \theta = 5$, i.e. the intersection of a cone and a sphere.
- (d) An infinite line parallel to the z-axis.
- (e) A semi-infinite line parallel to the x-y plane.
- (f) A semi-circle of radius 5 in the y-z plane.

Prob. 2.32

$$(a) \mathbf{J}_z = (\mathbf{J} \bullet \mathbf{a}_z) \mathbf{a}_z.$$

$$\text{At } (2, \pi/2, 3\pi/2), \mathbf{a}_z = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_\theta = -\mathbf{a}_\theta.$$

$$\mathbf{J}_z = -\cos 2\theta \sin \phi \mathbf{a}_\theta = -\cos \pi \sin(3\pi/2) \mathbf{a}_\theta = -\mathbf{a}_\theta.$$

$$(b) \mathbf{J}_\phi = \tan \frac{\theta}{2} \ln r \mathbf{a}_\phi = \tan \frac{\pi}{4} \ln 2 \mathbf{a}_\phi = \ln 2 \mathbf{a}_\phi = 0.6931 \mathbf{a}_\phi.$$

$$(c) \mathbf{J}_t = \mathbf{J} - \mathbf{J}_n = \mathbf{J} - \mathbf{J}_r = -\mathbf{a}_\theta + \ln 2 \mathbf{a}_\phi = \underline{\underline{-\mathbf{a}_\theta + 0.6931 \mathbf{a}_\phi}}$$

$$(d) \mathbf{J}_p = (\mathbf{J} \bullet \mathbf{a}_x) \mathbf{a}_x$$

$$\mathbf{a}_x = \sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi = \mathbf{a}_\phi.$$

$$\text{At } (2, \pi/2, 3\pi/2),$$

$$\mathbf{J}_p = \underline{\underline{\ln 2 \mathbf{a}_\phi.}}$$

Prob. 2.33

$$\mathbf{H} \cdot \mathbf{a}_x = H_x$$

$$\begin{bmatrix} H_x \\ H_y \\ H_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \rho^2 \cos \phi \\ -\rho \sin \phi \\ 0 \end{bmatrix}$$

$$H_x = \rho^2 \cos^2 \phi + \rho \sin^2 \phi$$

$$\text{At P, } \rho = 2, \phi = 60^\circ, z = -1$$

$$H_x = 4(1/4) + 2(3/4) = 1 + 1.5 = \underline{\underline{2.5}}$$

Prob. 2.34

$$(a) 5 = \mathbf{r} \cdot \mathbf{a}_x + \mathbf{r} \cdot \mathbf{a}_y = x + y \quad \underline{\underline{\text{a plane}}}$$

$$(b) 10 = |\mathbf{r} \times \mathbf{a}_z| = \begin{vmatrix} x & y & z \\ 0 & 0 & 1 \end{vmatrix} = |y \mathbf{a}_x - x \mathbf{a}_y| = \sqrt{x^2 + y^2} = \rho$$

a cylinder of infinite length

CHAPTER 3

P. E. 3.1

$$(a) \quad DH = \int_{\phi=45^\circ}^{\phi=60^\circ} r \sin \theta d\phi \Big|_{r=3, 90^\circ} = 3(1) \left[\frac{\pi}{3} - \frac{\pi}{4} \right] = \frac{\pi}{4} = \underline{\underline{0.7854.}}$$

$$(b) \quad FG = \int_{\theta=60^\circ}^{\theta=90^\circ} r d\theta \Big|_{r=5} = 5 \left(\frac{\pi}{2} - \frac{\pi}{3} \right) = \frac{5\pi}{6} = \underline{\underline{2.618.}}$$

(c)

$$\begin{aligned} AEHD &= \int_{\theta=60^\circ}^{\theta=90^\circ} \int_{\phi=45^\circ}^{\phi=60^\circ} r^2 \sin \theta d\theta d\phi \Big|_{r=3} = 9(-\cos \theta) \Big|_{\theta=60^\circ}^{\theta=90^\circ} \phi \Big|_{\phi=45^\circ}^{\phi=60^\circ} \\ &= 9 \left(\frac{1}{2} \right) \left(\frac{\pi}{12} \right) = \frac{3\pi}{8} = \underline{\underline{1.178.}} \end{aligned}$$

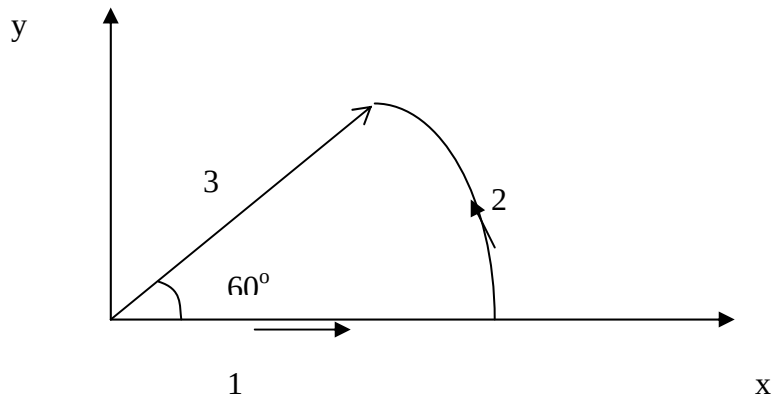
(d)

$$ABCD = \int_{r=3}^{r=5} \int_{\theta=60^\circ}^{\theta=90^\circ} r d\theta dr = \frac{r^2}{2} \Big|_{r=3}^{r=5} \left(\frac{\pi}{2} - \frac{\pi}{3} \right) = \frac{4\pi}{3} = \underline{\underline{4.189.}}$$

(e)

$$\begin{aligned} \text{Volume} &= \int_{r=3}^{r=5} \int_{\phi=45^\circ}^{\phi=60^\circ} \int_{\theta=60^\circ}^{\theta=90^\circ} r^2 \sin \theta dr d\theta d\phi = \frac{r^3}{3} \Big|_{r=3}^{r=5} (-\cos \theta) \Big|_{\theta=60^\circ}^{\theta=90^\circ} \phi \Big|_{\phi=45^\circ}^{\phi=60^\circ} = \frac{1}{3}(98) \left(\frac{1}{2} \right) \frac{\pi}{12} \\ &= \frac{49\pi}{36} = \underline{\underline{4.276.}} \end{aligned}$$

P.E. 3.2



$$\oint_L \mathbf{A} \bullet d\mathbf{l} = \left(\int_1 + \int_2 + \int_3 \right) \mathbf{A} \bullet d\mathbf{l} = C_1 + C_2 + C_3$$

$$\text{Along (1), } C_1 = \int \mathbf{A} \bullet d\mathbf{l} = \int_0^2 \rho \cos \phi d\rho \big|_{\phi=0} = \frac{\rho^2}{2} \bigg|_0^2 = 2.$$

$$\text{Along (2), } d\mathbf{l} = \rho d\phi \mathbf{a}_\phi, \mathbf{A} \bullet d\mathbf{l} = 0, \quad C_2 = 0$$

$$\text{Along (3), } C_3 = \int_2^0 \rho \cos \phi d\rho \big|_{\phi=60^\circ} = -\frac{\rho^2}{2} \bigg|_2^0 \left(\frac{1}{2} \right) = -1$$

$$\oint_L \mathbf{A} \bullet d\mathbf{l} = C_1 + C_2 + C_3 = 2 + 0 - 1 = \underline{\underline{1}}$$

P.E. 3.3

$$(a) \quad \nabla U = \frac{\partial U}{\partial x} \mathbf{a}_x + \frac{\partial U}{\partial y} \mathbf{a}_y + \frac{\partial U}{\partial z} \mathbf{a}_z$$

$$= \underline{\underline{y(2x+z) \mathbf{a}_x + x(x+z) \mathbf{a}_y + xy \mathbf{a}_z}}$$

$$(b) \quad \nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z$$

$$= \underline{\underline{(z \sin \phi + 2\rho) \mathbf{a}_\rho + (z \cos \phi - \frac{z^2}{\rho} \sin 2\phi) \mathbf{a}_\phi + (\rho \sin \phi + 2z \cos^2 \phi) \mathbf{a}_z}}$$

(c)

$$\begin{aligned} \nabla f &= \frac{\partial f}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \mathbf{a}_\phi \\ &= \left(\frac{\cos \theta \sin \phi}{r} + 2r\phi \right) \mathbf{a}_r - \frac{\sin \theta \sin \phi \ln r}{r} \mathbf{a}_\theta + \frac{(\cos \theta \cos \phi \ln r + r^2)}{r \sin \theta} \mathbf{a}_\phi \\ &= \underline{\underline{\left(\frac{\cos \theta \sin \phi}{r} + 2r\phi \right) \mathbf{a}_r - \frac{\sin \theta \sin \phi \ln r}{r} \mathbf{a}_\theta + \left(\frac{\cot \theta \cos \phi \ln r}{r} + r \csc \theta \right) \mathbf{a}_\phi}} \end{aligned}$$

P.E. 3.4

$$\nabla \Phi = (y+z) \mathbf{a}_x + (x+z) \mathbf{a}_y + (x+y) \mathbf{a}_z$$

$$\text{At } (1, 2, 3), \quad \nabla \Phi = \underline{\underline{(5, 4, 3)}}$$

$$\nabla \Phi \bullet \mathbf{a}_1 = (5, 4, 3) \bullet \frac{(2, 2, 1)}{3} = \frac{21}{3} = \underline{\underline{7}},$$

$$\text{where } (2, 2, 1) = (3, 4, 4) - (1, 2, 3)$$

P.E. 3.5

$$\text{Let } f = x^2y + z - 3, \quad g = x \log z - y^2 + 4,$$

$$\nabla f = 2xy \mathbf{a}_x + x^2 \mathbf{a}_y + \mathbf{a}_z,$$

$$\nabla g = \log z \mathbf{a}_x - 2y \mathbf{a}_y + \frac{x}{z} \mathbf{a}_z$$

$$\text{At } P(-1, 2, 1),$$

$$\mathbf{n}_f = \pm \frac{\nabla f}{|\nabla f|} = \pm \frac{(-4\mathbf{a}_x + \mathbf{a}_y + \mathbf{a}_z)}{\sqrt{18}}, \quad \mathbf{n}_g = \pm \frac{\nabla g}{|\nabla g|} = \pm \frac{(-4\mathbf{a}_y - \mathbf{a}_z)}{\sqrt{17}}$$

$$\cos \theta = \mathbf{n}_f \cdot \mathbf{n}_g = \pm \frac{(-5)}{\sqrt{18 \times 17}}$$

Take positive value to get acute angle.

$$\theta = \cos^{-1} \frac{5}{17.493} = \underline{\underline{73.39^\circ}}$$

P.E. 3.6

$$(a) \quad \nabla \bullet \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 0 + 4x + 0 = \underline{\underline{4x.}}$$

$$\text{At } (1, -2, 3), \quad \nabla \bullet \mathbf{A} = \underline{\underline{4.}}$$

(b)

$$\begin{aligned} \nabla \bullet \mathbf{B} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho B_\rho) + \frac{1}{\rho} \frac{\partial B_\phi}{\partial \phi} + \frac{\partial B_z}{\partial z} \\ &= \frac{1}{\rho} 2\rho z \sin \phi - \frac{1}{\rho} 3\rho z^2 \sin \phi = 2z \sin \phi - 3z^2 \sin \phi \\ &= \underline{\underline{(2-3z)z \sin \phi.}} \end{aligned}$$

$$\text{At } (5, \frac{\pi}{2}, 1), \quad \nabla \bullet \mathbf{B} = (2-3)(1) = \underline{\underline{-1.}}$$

(c)

$$\begin{aligned} \nabla \bullet \mathbf{C} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 C_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (C_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial C_\phi}{\partial \phi} \\ &= \frac{1}{r^2} 6r^2 \cos \theta \cos \phi \\ &= \underline{\underline{6 \cos \theta \cos \phi}} \end{aligned}$$

$$\text{At } (1, \frac{\pi}{6}, \frac{\pi}{3}), \quad \nabla \bullet \mathbf{C} = 6 \cos \frac{\pi}{6} \cos \frac{\pi}{3} = \underline{\underline{2.598.}}$$

P.E. 3.7 This is similar to Example 3.7.

$$\Psi = \oint_S \mathbf{D} \bullet d\mathbf{S} = \Psi_t + \Psi_b + \Psi_c$$

$$\Psi_t = 0 = \Psi_b \quad \text{since } \mathbf{D} \text{ has no z-component}$$

$$\begin{aligned} \Psi_c &= \iint \rho^2 \cos^2 \phi \rho d\phi dz = \rho^3 \int_{\phi=0}^{\phi=2\pi} \cos^2 \phi d\phi \int_{z=0}^{z=1} dz \Big|_{\rho=4} \\ &= (4)^3 \pi(1) = 64\pi \\ \Psi &= 0 + 0 + 64\pi = \underline{\underline{64\pi}} \end{aligned}$$

By the divergence theorem,

$$\oint_S \mathbf{D} \bullet d\mathbf{S} = \oint_V \nabla \bullet \mathbf{D} dv$$

$$\begin{aligned} \nabla \bullet \mathbf{D} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho^3 \cos^2 \phi) + \frac{1}{\rho} \frac{\partial}{\partial \phi} z \sin \phi + \frac{\partial A_z}{dz} \\ &= 3\rho \cos^2 \phi + \frac{z}{\rho} \cos \phi. \end{aligned}$$

$$\begin{aligned} \Psi &= \int_V \nabla \bullet \mathbf{D} dv = \int_V (3\rho \cos^2 \phi + \frac{z}{\rho} \cos \phi) \rho d\phi dz d\rho \\ &= 3 \int_0^4 \rho^2 d\rho \int_0^{2\pi} \cos^2 \phi d\phi \int_0^1 dz + \int_0^4 d\rho \int_0^{2\pi} \cos \phi d\phi \int_0^1 z dz \\ &= 3(\frac{4^3}{3})\pi(1) = \underline{\underline{64\pi.}} \end{aligned}$$

P.E. 3.8

(a)

$$\nabla \times \mathbf{A} = \mathbf{a}_x(1-0) + \mathbf{a}_y(y-0) + \mathbf{a}_z(4y-z)$$

$$= \underline{\underline{\mathbf{a}_x + y\mathbf{a}_y + (4y-z)\mathbf{a}_z}}$$

$$\text{At } (1, -2, 3), \nabla \times \mathbf{A} = \underline{\underline{\mathbf{a}_x - 2\mathbf{a}_y - 11\mathbf{a}_z}}$$

(b)

$$\nabla \times \mathbf{B} = \mathbf{a}_\rho(0 - 6\rho z \cos \phi) + \mathbf{a}_\phi(\rho \sin \phi - 0) + \mathbf{a}_z \frac{1}{\rho}(6\rho z^2 \cos \phi - \rho z \cos \phi)$$

$$= \underline{\underline{-6\rho z \cos \phi \mathbf{a}_\rho + \rho \sin \phi \mathbf{a}_\phi + (6z - 1)z \cos \phi \mathbf{a}_z}}$$

$$\text{At } (5, \frac{\pi}{2}, -1), \nabla \times \mathbf{B} = \underline{\underline{5\mathbf{a}_\phi}}$$

(c)

$$\begin{aligned} \nabla \times \mathbf{C} &= \mathbf{a}_r \frac{1}{r \sin \theta} (r^{-1/2} \cos \theta - 0) + \frac{\mathbf{a}_\theta}{r} \left(-\frac{2r \cos \theta \sin \phi}{\sin \theta} - \frac{3}{2} r^{1/2} \right) + \frac{\mathbf{a}_\phi}{r} (0 + 2r \sin \theta \cos \phi) \\ &= \underline{\underline{r^{-1/2} \cot \theta \mathbf{a}_r - (2 \cot \theta \sin \phi + \frac{3}{2} r^{-1/2}) \mathbf{a}_\theta + 2 \sin \theta \cos \phi \mathbf{a}_\phi}} \end{aligned}$$

$$\text{At } (1, \frac{\pi}{6}, \frac{\pi}{3}), \nabla \times \mathbf{C} = \underline{\underline{1.732 \mathbf{a}_r - 4.5 \mathbf{a}_\theta + 0.5 \mathbf{a}_\phi}}$$

P.E. 3.9

$$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

$$\text{But } (\nabla \times \mathbf{A}) = \sin \phi \mathbf{a}_z + \frac{z \cos \phi}{\rho} \mathbf{a}_\rho \quad \text{and} \quad dS = \rho d\phi d\rho \mathbf{a}_z$$

$$\begin{aligned} \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} &= \iint \rho \sin \phi d\phi d\rho \\ &= \frac{\rho^2}{2} \Big|_0^2 (-\cos \phi) \Big|_0^{60^\circ} \\ &= 2 \left(-\frac{1}{2} + 1 \right) = \underline{\underline{1}}. \end{aligned}$$

P.E. 3.10

$$\begin{aligned} \nabla \times \nabla V &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial V}{\partial x} & \frac{\partial V}{\partial y} & \frac{\partial V}{\partial z} \end{vmatrix} = \\ &= \left(\frac{\partial^2 V}{\partial y \partial z} - \frac{\partial^2 V}{\partial y \partial z} \right) \mathbf{a}_x + \left(\frac{\partial^2 V}{\partial x \partial z} - \frac{\partial^2 V}{\partial z \partial x} \right) \mathbf{a}_y + \left(\frac{\partial^2 V}{\partial x \partial y} - \frac{\partial^2 V}{\partial y \partial x} \right) \mathbf{a}_z = 0 \end{aligned}$$

P.E. 3.11

(a)

$$\begin{aligned}\nabla^2 U &= \frac{\partial}{\partial x}(2xy + yz) + \frac{\partial}{\partial y}(x^2 + xz) + \frac{\partial}{\partial z}(xy) \\ &= \underline{\underline{2y.}}\end{aligned}$$

(b)

$$\begin{aligned}\nabla^2 V &= \frac{1}{\rho} \frac{\partial}{\partial \rho} \rho(z \sin \phi + 2\rho) + \frac{1}{\rho^2} (-\rho z \sin \phi - 2z^2 \frac{\partial}{\partial \rho} \sin \phi \cos \phi) + \frac{\partial}{\partial z} (\rho \sin \phi + 2z \cos^2 \phi) \\ &= \frac{1}{\rho} (z \sin \phi + 4\rho) - \frac{1}{\rho^2} (z \rho \sin \phi + 2z^2 \cos 2\phi) + 2 \cos^2 \phi. \\ &= \underline{\underline{4 + 2 \cos^2 \phi - \frac{2z^2}{\rho^2} \cos 2\phi.}}\end{aligned}$$

(c)

$$\begin{aligned}\nabla^2 f &= \frac{1}{r^2} \frac{\partial}{\partial r} [r^2 \frac{1}{r} \cos \theta \sin \phi + 2r^3 \phi] + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} [-\sin^2 \theta \sin \phi \ln r] \\ &\quad + \frac{1}{r^2 \sin^2 \theta} [-\cos \theta \sin \phi \ln r] \\ &= \underline{\underline{\frac{1}{r^2} \cos \theta \sin \phi (1 - 2 \ln r - \csc^2 \theta \ln r) + 6\phi}}\end{aligned}$$

P.E. 3.12

If $\mathbf{B}$ is conservative, $\nabla \times \mathbf{B} = \mathbf{0}$ must be satisfied.

$$\nabla \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y + z \cos xz & x & x \cos xz \end{vmatrix}$$

$$= 0\mathbf{a}_x + (\cos xz - xz \sin xz - \cos xz + xz \sin xz)\mathbf{a}_y + (1 - 1)\mathbf{a}_z = \mathbf{0}$$

Hence $\mathbf{B}$ is a conservative field.

Prob. 3.1

(a)

$$dl = \rho d\phi; \quad \rho = 3$$

$$L = \int dl = 3 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\phi = 3\left(\frac{\pi}{2} - \frac{\pi}{4}\right) = \frac{3\pi}{4} = \underline{\underline{2.356}}$$

(b)

$$dl = r \sin \theta d\phi; \quad r = 1, \quad \theta = 30^\circ;$$

$$L = \int dl = r \sin \theta \int_0^{\frac{\pi}{3}} d\phi = (1) \sin 30^\circ \left[\left(\frac{\pi}{3}\right) - 0\right] = \underline{\underline{0.5236.}}$$

(c)

$$dl = r d\theta$$

$$L = \int dl = r \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} d\theta = 4\left(\frac{\pi}{2} - \frac{\pi}{6}\right) = \frac{4\pi}{3} = \underline{\underline{4.189}}$$

Prob. 3.2

(a)

$$dS = \rho d\phi dz$$

$$S = \int dS = \rho \int \int d\phi dz = 2 \int_0^5 dz \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} d\phi = 2(5) \left[\frac{\pi}{2} - \frac{\pi}{3}\right] = \frac{10\pi}{6} = \underline{\underline{5.236}}$$

(b)

In cylindrical, $dS = \rho d\rho d\phi$

$$S = \int dS = \int_1^3 \rho d\rho \int_0^{\frac{\pi}{4}} d\phi = \frac{\pi}{4} \left(\frac{\rho^2}{2}\right) \Big|_1^3 = \underline{\underline{3.142}}$$

(c) In spherical, $dS = r^2 \sin \theta d\phi d\theta$

$$S = \int dS = 100 \int_{\frac{\pi}{4}}^{\frac{2\pi}{3}} \sin \theta d\theta \int_0^{2\pi} d\phi = 100(2\pi)(-\cos \theta) \Big|_{\frac{\pi}{4}}^{\frac{2\pi}{3}} = 200\pi(0.5 + 0.7071) = \underline{\underline{758.4}}$$

(d)

$$dS = r dr d\theta$$

$$S = \int dS = \int_0^4 r dr \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} d\theta = \frac{r^2}{2} \Big|_0^4 \left(\frac{\pi}{2} - \frac{\pi}{3} \right) = \frac{8\pi}{6} = \underline{\underline{4.189}}$$

Prob.3.3

$$(a) dV = dx dy dz$$

$$V = \int dx dy dz = \int_0^1 dx \int_1^2 dy \int_{-3}^3 dz = (1)(2-1)(3-(-3)) = \underline{\underline{6}}$$

$$(b) dV = \rho d\phi d\rho dz$$

$$V = \int_2^5 \rho d\rho \int_{-1}^4 dz \int_{\frac{\pi}{3}}^{\pi} d\phi = \frac{\rho^2}{2} \Big|_2^5 (4-(-1)) \left(\pi - \frac{\pi}{3} \right) = \frac{1}{2} (25-4)(5) \left(\frac{2\pi}{3} \right) = 35\pi = \underline{\underline{110}}$$

$$(c) dV = r^2 \sin\theta dr d\theta d\phi$$

$$\begin{aligned} V &= \int_1^3 r^2 dr \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \sin\theta d\theta \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} d\phi = \frac{r^3}{3} \Big|_1^3 (-\cos\theta) \Big|_{\pi/2}^{\pi/3} \left(\frac{\pi}{2} - \frac{\pi}{6} \right) \\ &= \frac{1}{3} (27-1) \left(\frac{1}{2} \right) \left(\frac{\pi}{3} \right) = \frac{26\pi}{18} = \underline{\underline{4.538}} \end{aligned}$$

Prob.3.4

$$L = \int_L dl = \int_{\phi=0}^{\pi/6} \rho d\phi \Big|_{\rho=4} = 4(\pi/6) = \underline{\underline{2.094}}$$

Prob. 3.5

$$dS = r^2 \sin\theta d\theta d\phi$$

$$\begin{aligned} S &= r^2 \int_0^{\pi/2} d\phi \int_0^{\pi/4} \sin\theta d\theta \Big|_{r=5} = 25(\pi/2)(-\cos\theta) \Big|_0^{\pi/4} \\ &= \frac{25\pi}{2} (-\cos(\pi/4) + 1) = \underline{\underline{11.502}} \end{aligned}$$

Prob. 3.6

$$dv = \rho d\rho d\phi dz$$

$$V = \int_{z=0}^{10} dz \int_{\phi=0}^{30} d\phi \int_{\rho=2}^5 \rho d\rho = 10(\pi/6) \left(\frac{\rho^2}{2} \Big|_2^5 \right) = \frac{5\pi}{6} (25-4) = \underline{\underline{54.98}}$$

Prob. 3.7

$$dl = dx\mathbf{a}_x + dy\mathbf{a}_y$$

$$I = \int_L \mathbf{H} \bullet d\mathbf{l} = \int (xy^2 dx + x^2 y dy)$$

But on L, $x = y^2 \rightarrow dx = 2y dy$

$$\begin{aligned} I &= \int y^4 (2y dy) + y^3 dy = \int_{y=1}^4 (2y^5 dy + y^3 dy) = \left(\frac{2y^6}{6} + \frac{y^4}{4} \right) \Big|_1^4 \\ &= \frac{1}{3}(4096) + 64 - \frac{2}{6} - \frac{1}{4} = \underline{\underline{1428.75}} \end{aligned}$$

Prob. 3.8

The line joining P and Q is $y = x - 2, dy = dx$

$$\begin{aligned} I &= \int_L (2x^2 - 4xy) dx + (3xy - 2x^2 y) dy = \int_{x=1}^3 [2x^2 - 4x(x-2)] dx + [3x(x-2) - 2x^2(x-2)] dx \\ &= \int_{x=1}^3 [3x^2 - 2x - 2x^3] dx = \left(x^3 + x^2 - \frac{x^4}{4} \right) \Big|_1^3 = 27 + 9 - 81/4 = \underline{\underline{-4.5}} \end{aligned}$$

Prob. 3.9**(a)**

$$\begin{aligned} \int \mathbf{F} \bullet d\mathbf{l} &= \int_{y=0}^1 (x^2 - z^2) dy \Big|_{x=0, z=0} + \int_{x=0}^{x=2} 2xy dx \Big|_{y=1, z=0} + \int_{z=0}^{z=3} (-3xz^2) dz \Big|_{x=2, y=1} \\ &= 0 + 2(1) \frac{x^2}{2} \Big|_0^2 - 3(2) \frac{z^3}{3} \Big|_0^3 \\ &= 0 + 4 - 54 = \underline{\underline{-50}} \end{aligned}$$

(b)

Let $x = 2t, y = t, z = 3t$

$dx = 2dt, dy = dt, dz = 3dt;$

$$\begin{aligned} \int \mathbf{F} \bullet d\mathbf{l} &= \int_0^1 (8t^2 - 5t^2 - 162t^3) dt \\ &= (t^3 - 40.5t^4) \Big|_0^1 = \underline{\underline{-39.5}} \end{aligned}$$

Prob.3.10

$$W = \int_L \mathbf{F} \cdot d\mathbf{l} = \int_{\phi=0}^{\pi/4} z \rho d\phi \Big|_{z=0, \rho=2} + \int_{z=0}^3 \cos \phi dz \Big|_{\phi=\pi/4}$$

$$= 0 + \cos(\pi/4)(3) = 3 \cos 45^\circ = \underline{\underline{2.121 \text{ J}}}$$

Prob. 3.11

$$\int \mathbf{H} \cdot d\mathbf{l} = \int_{x=1}^0 (x-y) dx \Big|_{y=0, z=0} + \int_{z=0}^1 5yz dz \Big|_{x=0, y=0}$$

$$+ \int (x^2 + zy) dy + 5yz dz \Big|_{x=0, z=1-y/2}$$

$$= \int_1^0 x dx + \int_0^2 (y - \frac{y^2}{2}) dy + \int_1^0 (10z - 10z^2) dz$$

$$= \underline{\underline{-1.5}}$$

Prob. 3.12*Method 1:*

$$\oint_L \mathbf{B} \cdot d\mathbf{l} = - \int_{y=0}^1 yz dy \Big|_{z=0} + \int_{z=0}^1 xz dz \Big|_{x=1} + \int (-yz dy + xz dz) \Big|_{x=1}$$

But $z = y \longrightarrow dz = dy$ on the last segment (or integral).

$$\oint_L \mathbf{B} \cdot d\mathbf{l} = 0 + \frac{z^2}{2} \Big|_0^1 + \int_{y=1}^0 (-y^2 + y) dy = \frac{1}{2} + \left(-\frac{y^3}{3} + \frac{y^2}{2}\right) \Big|_1^0$$

$$= \frac{1}{2} + \frac{1}{3} - \frac{1}{2} = \frac{1}{3} = \underline{\underline{0.333}}$$

Method 2:

$$\oint_L \mathbf{B} \cdot d\mathbf{l} = \int_S \nabla \times \mathbf{B} \cdot d\mathbf{S}$$

$$\nabla \times \mathbf{B} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & -yz & xz \end{vmatrix} = y\mathbf{a}_x - z\mathbf{a}_y - x\mathbf{a}_z, \quad d\mathbf{S} = dydz\mathbf{a}_x$$

$$\int_S \nabla \times \mathbf{B} \cdot d\mathbf{S} = \int_{y=0}^1 \int_{z=0}^y y dz dy = \int_0^1 y^2 dy = \frac{y^3}{3} \Big|_0^1 = \frac{1}{3} = \underline{\underline{0.333}}$$

Prob. 3.13

$$\psi = \int_S \mathbf{A} \cdot d\mathbf{S} = \int \int z dx dz = \int_0^1 dx \int_0^2 z dz = (1) \frac{z^2}{2} \Big|_0^2 = 2$$

Prob. 3.14

$$\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{D} dv$$

$$\nabla \cdot \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = 2xz + 3y^2 + 2yz$$

$$\begin{aligned} \Psi &= \int_V \nabla \cdot \mathbf{D} dv = \iiint (2xz + 3y^2 + 2yz) dx dy dz \\ &= 2 \int_{-1}^1 x dx \int_0^4 dy \int_1^3 z dz + 3 \int_{-1}^1 dx \int_0^4 y^2 dy \int_1^3 dz + 2 \int_{-1}^1 dx \int_0^4 y dy \int_1^3 z dz \\ &= 0 + 3(2) \left(\frac{y^3}{3} \Big|_0^4 \right) (3) + 2(2) \left(\frac{y^2}{2} \Big|_0^4 \right) \left(\frac{z^2}{2} \Big|_1^3 \right) = 6(64) + 16(9-1) = 384 + 128 \end{aligned}$$

$$\underline{\underline{\Psi = 512}}$$

Prob. 3.15

$$\Psi = \oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{A} dv$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (-3 \sin \theta) + \frac{1}{r \sin \theta} (5) = 3 - \frac{3 \cos \theta}{r \sin \theta} + \frac{5}{r \sin \theta}$$

$$dv = r^2 \sin \theta d\theta dr d\phi$$

$$\begin{aligned} \Psi &= 3 \iiint r^2 \sin \theta d\theta dr d\phi - 3 \iiint r \cos \theta d\theta d\phi dr + \iiint 5 d\theta d\phi dr \\ &= 3 \int_0^4 r^2 dr \int_0^{\pi/2} \sin \theta d\theta \int_0^{3\pi/2} d\phi - 3 \int_0^4 r dr \int_0^{\pi/2} \cos \theta d\theta \int_0^{3\pi/2} d\phi + 5 \int_0^4 dr \int_0^{\pi/2} d\theta \int_0^{3\pi/2} d\phi \\ &= 3 \left(\frac{r^3}{3} \Big|_0^4 \right) (-\cos \theta) \Big|_0^{\pi/2} (3\pi/2) - 3 \left(\frac{r^2}{2} \Big|_0^4 \right) (\sin \theta) \Big|_0^{\pi/2} (3\pi/2) + 5(4)(\pi/2)(3\pi/2) \\ &= 96\pi - 36\pi + 15\pi^2 = \underline{\underline{336.54}} \end{aligned}$$

Prob. 3.16

(a) $dv = dx dy dz$

$$\int_v xy dv = \int_{z=0}^2 \int_{y=0}^1 \int_{x=0}^1 xy dx dy dz = \int_0^1 x dx \int_0^1 y dy \int_0^2 dz$$

$$= \frac{x^2}{2} \Big|_0^1 \frac{y^2}{2} \Big|_0^1 z \Big|_0^2 = (1/2)(1/2)(2) = \underline{\underline{0.5}}$$

(b)

$$dv = \rho d\rho d\phi dz$$

$$\int_v \rho z dv = \int_{\phi=0}^{\pi} \int_{z=0}^2 \int_{\rho=1}^3 \rho z \rho d\rho d\phi dz = \int_1^3 \rho^2 d\rho \int_0^2 z dz \int_0^{\pi} d\phi$$

$$= \frac{\rho^3}{3} \Big|_1^3 \frac{z^2}{2} \Big|_0^2 (\pi) = (9 - \frac{1}{3})(2\pi) = \underline{\underline{54.45}}$$

Prob. 3.17

$$(a) \quad \nabla V_1 = \frac{\partial V_1}{\partial x} \mathbf{a}_x + \frac{\partial V_1}{\partial y} \mathbf{a}_y + \frac{\partial V_1}{\partial z} \mathbf{a}_z$$

$$= \underline{\underline{(6y - 2z)\mathbf{a}_x + 6x\mathbf{a}_y + (1 - 2x)\mathbf{a}_z}}$$

$$(b) \quad \nabla V_2 = \frac{\partial V_2}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V_2}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V_2}{\partial z} \mathbf{a}_z$$

$$= \underline{\underline{(10 \cos \phi - z)\mathbf{a}_\rho - 10 \sin \phi \mathbf{a}_\phi - \rho \mathbf{a}_z}}$$

$$\nabla V_3 = \frac{\partial V_3}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V_3}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V_3}{\partial \phi} \mathbf{a}_\phi$$

$$(c) \quad = -\frac{2}{r^2} \cos \phi \mathbf{a}_r + 0 + \frac{1}{r \sin \theta} \left(-\frac{2}{r} \sin \phi \right) \mathbf{a}_\phi$$

$$= \underline{\underline{-\frac{2}{r^2} \cos \phi \mathbf{a}_r - \frac{2 \sin \phi}{r^2 \sin \theta} \mathbf{a}_\phi}}$$

Prob. 3.18

(a)

$$\nabla V = (10yz - 4xz)\mathbf{a}_x + 10xz\mathbf{a}_y + (10xy - 2x^2)\mathbf{a}_z$$

At P, $x = -1, y = 4, z = 3$. Hence,

$$\nabla V = (120 + 12)\mathbf{a}_x - 30\mathbf{a}_y + (-40 - 2)\mathbf{a}_z = \underline{\underline{132\mathbf{a}_x - 30\mathbf{a}_y - 42\mathbf{a}_z}}$$

(b)

$$\nabla U = (2 \sin \phi + z) \mathbf{a}_\rho + 2 \cos \phi \mathbf{a}_\phi + \rho \mathbf{a}_z$$

$$\text{At } Q, \rho = 2, \phi = 90^\circ, z = -1$$

$$\nabla U = (2 - 1) \mathbf{a}_\rho + 0 \mathbf{a}_\phi + 2 \mathbf{a}_z = \mathbf{a}_\rho + 2 \mathbf{a}_z$$

(c)

$$\nabla W = -\frac{8}{r^3} \sin \theta \cos \phi \mathbf{a}_r + \frac{1}{r} \frac{4 \cos \theta \cos \phi}{r^2} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \left(\frac{-4 \sin \theta \sin \phi}{r^2} \right) \mathbf{a}_\phi$$

$$\text{At } R, r = 1, \theta = \pi/6, \phi = \pi/2$$

$$\nabla W = -\frac{4}{(1)^3} \mathbf{a}_\phi = \underline{\underline{-4 \mathbf{a}_\phi}}$$

Prob. 3.19

$$r = \sqrt{x^2 + y^2 + z^2}, \quad r^n = (x^2 + y^2 + z^2)^{n/2}$$

Method 1:

$$\begin{aligned} \nabla r^n &= \frac{\partial r^n}{\partial x} \mathbf{a}_x + \frac{\partial r^n}{\partial y} \mathbf{a}_y + \frac{\partial r^n}{\partial z} \mathbf{a}_z = \frac{n}{2} (x^2 + y^2 + z^2)^{n/2-1} (2x) \mathbf{a}_x + \cdots \\ &= n(x^2 + y^2 + z^2)^{n/2-1} (x \mathbf{a}_x + y \mathbf{a}_y + z \mathbf{a}_z) = \underline{\underline{nr^{n-2} \mathbf{r}}} \end{aligned}$$

Method 2:

$$\nabla r^n = \frac{\partial r^n}{\partial r} \mathbf{a}_r = nr^{n-1} \frac{\mathbf{r}}{r} = nr^{n-2} \mathbf{r}$$

Prob. 3.20

$$\nabla T = 2x \mathbf{a}_x + 2y \mathbf{a}_y - \mathbf{a}_z$$

At $(1, 1, 2)$, $\nabla T = (2, 2, -1)$. The mosquito should move in the direction of

$$\underline{\underline{2 \mathbf{a}_x + 2 \mathbf{a}_y - \mathbf{a}_z}}$$

Prob. 3.21

$$\nabla F = \mathbf{a}_x - 2 \mathbf{a}_y + \mathbf{a}_z$$

$$a_n = \frac{\nabla F}{|\nabla F|} = \frac{\mathbf{a}_x - 2 \mathbf{a}_y + \mathbf{a}_z}{\sqrt{1+4+1}} = \underline{\underline{0.4082 \mathbf{a}_x - 0.8165 \mathbf{a}_y + 0.4082 \mathbf{a}_z}}$$

Prob. 3.22

Method 1:

$$\begin{aligned}\nabla T &= \frac{\partial T}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial T}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \mathbf{a}_\phi \\ &= \sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi\end{aligned}$$

At P, $r = 2, \theta = 60^\circ, \phi = 30^\circ$

$$\begin{aligned}\nabla T &= \sin 60^\circ \cos 30^\circ \mathbf{a}_r + \cos 60^\circ \cos 30^\circ \mathbf{a}_\theta - \sin 30^\circ \mathbf{a}_\phi \\ &= 0.75 \mathbf{a}_r - 0.433 \mathbf{a}_\theta - 0.5 \mathbf{a}_\phi\end{aligned}$$

$$|\nabla T| = \sqrt{0.75^2 + 0.433^2 + 0.5^2} = 1$$

The magnitude of T is 1 and its direction is along ∇T .

Method 2:

$$T = r \sin \theta \cos \phi = x$$

$$\nabla T = \mathbf{a}_x$$

$$|\nabla T| = 1$$

Prob. 3.23

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{a}_x + \frac{\partial f}{\partial y} \mathbf{a}_y + \frac{\partial f}{\partial z} \mathbf{a}_z = (2xy - 2y^2) \mathbf{a}_x + (x^2 - 4xy) \mathbf{a}_y + 3z^2 \mathbf{a}_z$$

At point (2,4,-3), $x = 2, y = 4, z = -3$

$$\nabla f = (16 - 32) \mathbf{a}_x + (4 - 32) \mathbf{a}_y + 27 \mathbf{a}_z = -16 \mathbf{a}_x - 28 \mathbf{a}_y + 27 \mathbf{a}_z$$

$$\mathbf{a} = \frac{\mathbf{a}_x + 2\mathbf{a}_y - \mathbf{a}_z}{\sqrt{1+4+1}} = \frac{1}{\sqrt{6}}(1, 2, -1)$$

The directional derivative is

$$\nabla f \cdot \mathbf{a} = (-16, -28, 27) \cdot \frac{1}{\sqrt{6}}(1, 2, -1) = -\frac{99}{\sqrt{6}} = \underline{\underline{-40.42}}$$

Prob. 3.24

(a) Let $f = ax + by + cz - d = 0$

$$\nabla f = a\mathbf{a}_x + b\mathbf{a}_y + c\mathbf{a}_z$$

$$\mathbf{a}_{n1} = \frac{\nabla f}{|\nabla f|} = \frac{a\mathbf{a}_x + b\mathbf{a}_y + c\mathbf{a}_z}{\sqrt{a^2 + b^2 + c^2}}$$

$$\text{Let } g = \alpha x + \beta y + \gamma z - \delta$$

$$\mathbf{a}_{n2} = \frac{\nabla g}{|\nabla g|} = \frac{\alpha\mathbf{a}_x + \beta\mathbf{a}_y + \gamma\mathbf{a}_z}{\sqrt{\alpha^2 + \beta^2 + \gamma^2}}$$

$$\cos \theta = \mathbf{a}_{n1} \cdot \mathbf{a}_{n2} = \frac{a\alpha + b\beta + c\gamma}{\sqrt{a^2 + b^2 + c^2} \sqrt{\alpha^2 + \beta^2 + \gamma^2}}$$

$$\theta = \cos^{-1} \frac{a\alpha + b\beta + c\gamma}{\sqrt{(a^2 + b^2 + c^2)(\alpha^2 + \beta^2 + \gamma^2)}}$$

$$a = 1, b = 2, c = 3$$

$$(b) \quad \alpha = 1, \beta = 1, \gamma = 0$$

$$\theta = \cos^{-1} \frac{1 + 2 + 0}{\sqrt{(1^2 + 2^2 + 3^2)(1^2 + 1^2 + 0^2)}} = \cos^{-1} \frac{3}{\sqrt{28}} = \cos^{-1} 0.5669 = \underline{\underline{55.46^\circ}}$$

Prob. 3.25

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z = 4ye^z \mathbf{a}_x + 4xe^z \mathbf{a}_y + 4xye^z \mathbf{a}_z$$

$$\text{At } (3, 1, -2), \quad x = 3, y = 1, z = -2$$

$$\nabla V = 4e^{-2} \mathbf{a}_x + 12e^{-2} \mathbf{a}_y + 12e^{-2} \mathbf{a}_z = \underline{\underline{0.5413\mathbf{a}_x + 1.624\mathbf{a}_y + 1.624\mathbf{a}_z}}$$

This is the direction. The maximum rate of change is

$$|\nabla V| = e^{-2} \sqrt{4^2 + 12^2 + 12^2} = 0.1353 \times 17.44 = \underline{\underline{2.36}}$$

Prob. 3.26

(a)

$$\begin{aligned} \nabla UV &= \frac{\partial}{\partial x}(UV) \mathbf{a}_x + \frac{\partial}{\partial y}(UV) \mathbf{a}_y + \frac{\partial}{\partial z}(UV) \mathbf{a}_z \\ &= \left(U \frac{\partial V}{\partial x} + V \frac{\partial U}{\partial x} \right) \mathbf{a}_x + \left(U \frac{\partial V}{\partial y} + V \frac{\partial U}{\partial y} \right) \mathbf{a}_y + \left(U \frac{\partial V}{\partial z} + V \frac{\partial U}{\partial z} \right) \mathbf{a}_z \\ &= U \nabla V + V \nabla U \end{aligned}$$

(b)

$$\nabla V = 10xy\mathbf{a}_x + (5x^2 + 2z)\mathbf{a}_y + 2y\mathbf{a}_z$$

$$U\nabla V = 30x^2y^2z\mathbf{a}_x + (15x^3yz + 6xyz^2)\mathbf{a}_y + 6xy^2z\mathbf{a}_z$$

$$\nabla U = 3yz\mathbf{a}_x + 3xz\mathbf{a}_y + 3xy\mathbf{a}_z$$

$$V\nabla U = (15x^2y^2z + 6y^2z^2)\mathbf{a}_x + (15x^3yz + 6xyz^2)\mathbf{a}_y + (15x^3y + 6xy^2z)\mathbf{a}_z$$

$$U\nabla V + V\nabla U = (45x^2y^2z + 6y^2z^2)\mathbf{a}_x + (30x^3yz + 12xyz^2)\mathbf{a}_y + (15x^3y + 12xy^2z)\mathbf{a}_z - (1)$$

$$UV = 15x^3y^2z + 6xy^2z^2$$

$$\nabla(UV) = (45x^2y^2z + 6y^2z^2)\mathbf{a}_x + (30x^3yz + 12xyz^2)\mathbf{a}_y + (15x^3y + 12xy^2z)\mathbf{a}_z - (2)$$

From (1) and (2), the formula is verified.

Prob. 3.27

$$(a) \nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = \underline{\underline{3y - x}}$$

$$(b) \nabla \cdot \mathbf{B} = \frac{1}{\rho} 2\rho z^2 + \frac{1}{\rho} \rho 2 \sin \phi \cos \phi + 2\rho \sin^2 \phi$$

$$= \underline{\underline{2z^2 + \sin 2\phi + 2\rho \sin^2 \phi}}$$

$$(c) \nabla \cdot \mathbf{C} = \frac{1}{r^2} 3r^2 + 0 = \underline{\underline{3}}$$

Prob. 3.28

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 2xy + 0 + 2y = 2y(1 + x)$$

$$(a) \text{ At } (-3, 4, 2), x = -3, y = 4$$

$$\nabla \cdot \mathbf{A} = 2(4)(1 - 3) = \underline{\underline{-16}}$$

$$\nabla \cdot \mathbf{B} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho B_\rho) + \frac{1}{\rho} \frac{\partial B_\phi}{\partial \phi} + \frac{\partial B_z}{\partial z} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (3\rho^2 \sin \phi) + 0 + 8z \cos^2 \phi$$

$$(b) = 6 \sin \phi + 8z \cos^2 \phi$$

$$\text{At } (5, 30^\circ, 1), z = 1, \phi = 30^\circ$$

$$\nabla \cdot \mathbf{B} = 6 \sin 30^\circ + 8(1) \cos^2 30^\circ = 3 + 6 = \underline{\underline{9}}$$

$$\nabla C = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 C_r) + 0 + \frac{1}{r \sin \theta} \frac{\partial C_\phi}{\partial \phi} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^4 \cos \theta) + 0 = 4r \cos \theta$$

(c) At $(2, \pi/3, \pi/2), r = 2, \theta = \pi/3$

$$\nabla C = 4(2) \cos(\pi/3) = \underline{\underline{4}}$$

Prob. 3.29

$$\nabla \cdot \mathbf{H} = k \nabla \cdot \nabla T = k \nabla^2 T$$

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 50 \sin \frac{\pi x}{2} \cos h \frac{\pi y}{2} \left(-\frac{\pi^2}{4} + \frac{\pi^2}{4} \right) = 0$$

$$\text{Hence, } \nabla \cdot \mathbf{H} = 0$$

Prob. 3.30

(a)

$$\begin{aligned} \nabla \cdot (V \mathbf{A}) &= \frac{\partial}{\partial x} (V A_x) + \frac{\partial}{\partial y} (V A_y) + \frac{\partial}{\partial z} (V A_z) \\ &= \left(A_x \frac{\partial V}{\partial x} + V \frac{\partial A_x}{\partial x} \right) + \left(A_y \frac{\partial V}{\partial y} + V \frac{\partial A_y}{\partial y} \right) + \left(A_z \frac{\partial V}{\partial z} + V \frac{\partial A_z}{\partial z} \right) \\ &= V \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) + A_x \frac{\partial V}{\partial x} + A_y \frac{\partial V}{\partial y} + A_z \frac{\partial V}{\partial z} \\ &= \underline{\underline{V \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla V}} \end{aligned}$$

(b)

$$\nabla \cdot \mathbf{A} = 2 + 3 - 4 = 1; \quad \nabla V = yz \mathbf{a}_x + xz \mathbf{a}_y + xy \mathbf{a}_z$$

$$\begin{aligned} \nabla \cdot (V \mathbf{A}) &= V \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla V \\ &= xyz + 2xyz + 3xyz - 4xyz = \underline{\underline{2xyz}} \end{aligned}$$

Prob. 3.31

(a)

$$(\nabla \cdot \mathbf{r}) \mathbf{T} = 3\mathbf{T} = \underline{\underline{6yz \mathbf{a}_y + 3xy^2 \mathbf{a}_y + 3x^2 yz \mathbf{a}_z}}$$

(b)

$$\begin{aligned}
x \frac{\partial T}{\partial x} + y \frac{\partial T}{\partial y} + z \frac{\partial T}{\partial z} &= x (y^2 \mathbf{a}_y + 2xyz \mathbf{a}_z) + y(2z \mathbf{a}_x + 2xy \mathbf{a}_y + x^2 z \mathbf{a}_z) \\
&\quad + z(2y \mathbf{a}_x + x^2 y \mathbf{a}_z) \\
&= \underline{\underline{4yz \mathbf{a}_x + 3xy^2 \mathbf{a}_y + 4x^2 yz \mathbf{a}_z}}
\end{aligned}$$

(c)

$$\begin{aligned}
\nabla \bullet \mathbf{r}(\mathbf{r} \bullet \mathbf{T}) &= 3 (2xyz + xy^3 + x^2 yz^2) \\
&= \underline{\underline{6xyz + 3xy^3 + 3x^2 yz^2}}
\end{aligned}$$

(d)

$$\begin{aligned}
(\mathbf{r} \bullet \nabla) r^2 &= (x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z})(x^2 + y^2 + z^2) \\
&= x(2x) + y(2y) + z(2z) \\
&= \underline{\underline{2(x^2 + y^2 + z^2) = 2r^2}}
\end{aligned}$$

Prob. 3.32

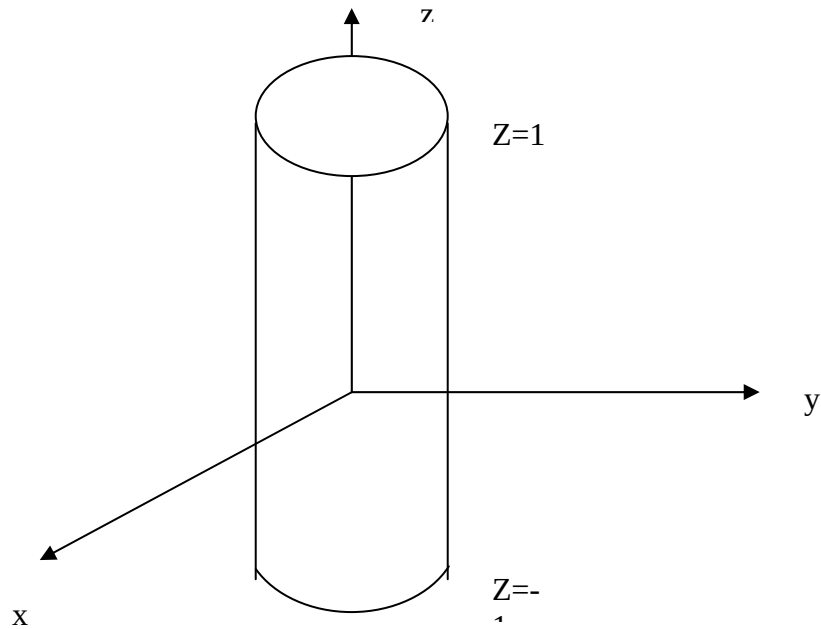
We convert A to cylindrical coordinates; only the ρ -component is needed.

$$A_\rho = A_x \cos \phi + A_y \sin \phi = 2x \cos \phi - z^2 \sin \phi$$

But $x = \rho \cos \phi$,

$$A_\rho = 2\rho \cos^2 \phi - z^2 \sin \phi$$

$$\begin{aligned}
\Psi &= \int_s \mathbf{A} \cdot d\mathbf{S} = \iint A_\rho \rho d\phi dz = \iint [2\rho^2 \cos^2 \phi - \rho z^2 \sin \phi] d\phi dz \\
&= 2(2)^2 \int_0^{\pi/2} \frac{1}{2} (1 + \cos 2\phi) d\phi \int_0^1 dz - 2 \int_0^1 z^2 dz \int_0^{\pi/2} \sin \phi d\phi \\
&= 4\left(\phi + \frac{1}{2} \sin 2\phi\right) \Big|_0^{\pi/2} - 2 \frac{z^3}{3} \Big|_0^1 (-\cos \phi) \Big|_0^{\pi/2} = 2\pi - 2/3 = \underline{\underline{5.6165}}
\end{aligned}$$

Prob. 3.33**(a)**

$$\oint \mathbf{D} \cdot d\mathbf{S} = \left[\iint_{z=-1} + \iint_{z=1} + \iint_{\rho=5} \right] \mathbf{D} \cdot d\mathbf{S}$$

$$= - \iint \rho^2 \cos^2 \phi d\phi d\rho + \iint \rho^2 \cos^2 \phi d\phi d\rho + \iint 2\rho^2 z^2 d\phi dz \Big|_{\rho=5}$$

$$= 2(5)^2 \int_0^{2\pi} d\phi \int_{-1}^1 z^2 dz = +50(2\pi) \left(\frac{z^3}{3} \Big|_{-1}^1 \right)$$

$$= \frac{200\pi}{3} = \underline{\underline{209.44}}$$

$$(b) \nabla \cdot \mathbf{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (2\rho^2 z^2) = 4z^2$$

$$\int \nabla \cdot \mathbf{D} dv = \iiint 4z^2 \rho d\rho d\phi dz = 4 \int_{-1}^1 z^2 dz \int_0^5 \rho d\rho \int_0^{2\pi} d\phi$$

$$= 4x \frac{z^3}{3} \Big|_{-1}^1 \frac{\rho^2}{2} \Big|_0^5 (2\pi) = \frac{200\pi}{3} = \underline{\underline{209.44}}$$

Prob. 3.34

$$\begin{aligned}
\int_S \mathbf{H} \cdot d\mathbf{S} &= \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} 10 \cos \theta r^2 \sin \theta d\theta d\phi \Big|_{r=1} \\
&= 10(1)^2 \int_0^{2\pi} d\phi \int_0^{\pi/2} \sin \theta \cos \theta d\theta = 10(2\pi) \int_0^{\pi/2} \sin \theta d(\sin \theta) \\
&= 20\pi \left(\frac{\sin^2 \theta}{2} \right) \Big|_0^{\pi/2} = 10\pi = \underline{\underline{31.416}}
\end{aligned}$$

Prob. 3.35

$$\oint_S \mathbf{H} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{H} dv$$

$$\begin{aligned}
\oint_S \mathbf{H} \cdot d\mathbf{S} &= -\iint_{x=0} 2xy dy dz \Big|_{x=0} + \iint_{x=1} 2xy dy dz \Big|_{x=1} - \iint_{y=1} (x^2 + z^2) dx dz \Big|_{y=1} \\
&\quad + \iint_{y=2} (x^2 + z^2) dx dz \Big|_{y=2} - \iint_{z=-1} 2yz dx dy \Big|_{z=-1} + \iint_{z=3} 2yz dx dy \Big|_{z=3} \\
&= 0 + 2 \int_1^2 y dy \int_{-1}^3 dz + 2 \int_0^1 dx \int_1^2 y dy + 6 \int_0^1 dx \int_1^2 y dy \\
&= 12 + 3 + 9 = \underline{\underline{24}}
\end{aligned}$$

$$\nabla \cdot \mathbf{H} = \frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z} = 2y + 0 + 2y = 4y$$

$$\begin{aligned}
\int_V \nabla \cdot \mathbf{H} dv &= \iiint 4y dx dy dz = 4 \int_0^1 dx \int_1^2 y dy \int_{-1}^3 dz \\
&= 4(1) \frac{y^2}{2} \Big|_1^2 (3+1) = \underline{\underline{24}}
\end{aligned}$$

Prob. 36

$$\Psi = \oint_S \mathbf{H} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{H} dv$$

To find $\oint_S \mathbf{H} \cdot d\mathbf{S}$, let

$$\Psi = \Psi_t + \Psi_b + \Psi_s$$

where Ψ_t , Ψ_b , and Ψ_s are the fluxes from the top, bottom, and side of the cylinder.

For Ψ_t , $dS = \rho d\rho d\phi \mathbf{a}_z$,

$$\Psi_t = -2z \iint \rho d\rho d\phi \Big|_{z=3} = -6 \int_0^{10} \rho d\rho \int_0^{2\pi} d\phi = -12\pi \frac{\rho^2}{2} \Big|_0^{10} = -600\pi$$

For Ψ_r , $dS = \rho d\rho d\phi (-\mathbf{a}_z)$,

$$\Psi_b = 2z \iint \rho d\rho d\phi \Big|_{z=0} = 0$$

For Ψ_s , $dS = \rho d\phi dz \mathbf{a}_\rho$,

$$\Psi_t = 4\rho^3 \iint d\phi dz \Big|_{\rho=10} = 4000 \int_0^3 dz \int_0^{2\pi} d\phi = 4000(3)(2\pi) = 24000\pi$$

$$\Psi = -600\pi + 0 + 24000\pi = 23400\pi = \underline{\underline{73,513.27}}$$

To get $\int_v \nabla \cdot \mathbf{H} dv$, $dv = \rho d\phi d\rho dz$

$$\nabla \cdot \mathbf{H} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (4\rho^3) - 2 = 12\rho - 2$$

$$\begin{aligned} \Psi &= \iiint (12\rho - 2) \rho d\phi d\rho dz = 12 \int_0^{10} \rho^2 d\rho \int_0^{2\pi} d\phi \int_0^3 dz - 2 \int_0^{10} \rho d\rho \int_0^{2\pi} d\phi \int_0^3 dz \\ &= 12 \frac{(1000)}{3} (2\pi)(3) - 2 \frac{(100)}{2} (2\pi)(3) = 24000\pi - 600\pi = \underline{\underline{73,513.27}} \end{aligned}$$

Prob. 3.37

$$\text{Let } \psi = \oiint_S \mathbf{A} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{A} dv$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 2(x + y + z)$$

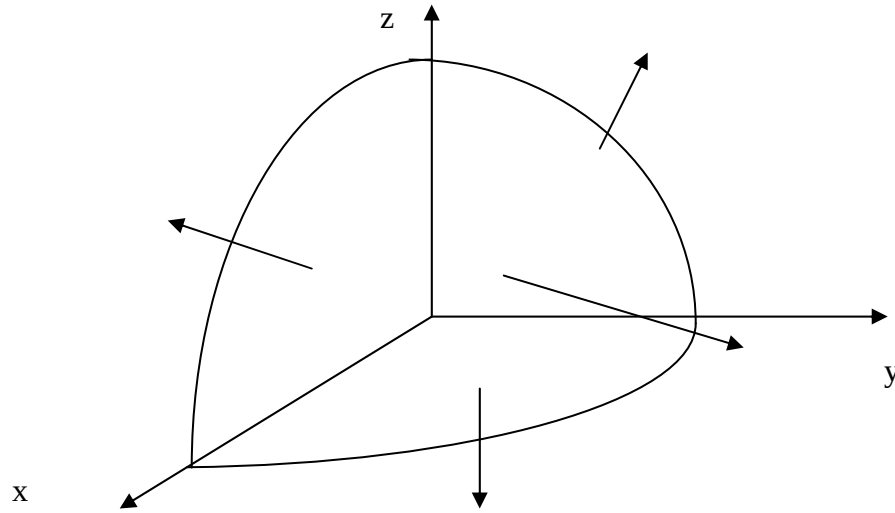
$$\nabla \cdot \mathbf{A} = 2(\rho \cos \phi + \rho \sin \phi + z), \quad dv = \rho d\rho d\phi dz$$

$$\begin{aligned} \psi &= \iiint 2(\rho \cos \phi + \rho \sin \phi + z) \rho d\rho d\phi dz \\ &= 0 + 0 + 2 \int_0^1 \rho d\rho \int_2^4 z dz \int_0^{2\pi} d\phi = \frac{2\rho^2}{2} \Big|_0^1 \frac{z^2}{2} \Big|_2^4 (2\pi) = \frac{1}{2} (16 - 4)(2\pi) \\ &= 12\pi = \underline{\underline{37.7}} \end{aligned}$$

Prob. 3.38

$$\begin{aligned} \nabla \cdot \mathbf{A} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^4) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r \sin^2 \theta \cos \phi) \\ &= 4r + 2 \cos \theta \cos \phi \end{aligned}$$

$$\begin{aligned}
\int \nabla \cdot \mathbf{A} dv &= \iiint 4r^3 \sin \theta d\theta d\phi dr + \iiint 2r^2 \sin \theta \cos \theta \cos \phi d\theta d\phi dr \\
&= 4 \frac{r^4}{4} \Big|_0^3 (-\cos \theta) \Big|_0^{\pi/2} \left(\frac{\pi}{2}\right) + \frac{2r^3}{3} \Big|_0^3 \left(-\frac{\cos^2 \theta}{2}\right) \Big|_0^{\pi/2} \sin \phi \Big|_0^{\pi/2} \\
&= 81(1)\left(\frac{\pi}{2}\right) + 18\left(0 + \frac{1}{2}\right)(1-0) \\
&= \frac{81\pi}{2} + 9 = \underline{\underline{136.23}}
\end{aligned}$$



$$\int \mathbf{A} \cdot d\mathbf{S} = \left[\iint_{\phi=0} + \iint_{\phi=\pi/2} + \iint_{r=3} + \iint_{\theta=\pi/2} \right] \mathbf{A} \cdot d\mathbf{S}$$

Since $\mathbf{A}$ has no ϕ -component, the first two integrals on the right hand side vanish.

$$\begin{aligned}
\int \mathbf{A} \cdot d\mathbf{S} &= \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{\pi/2} r^4 \sin \theta d\theta d\phi \Big|_{r=3} + \int_{r=0}^3 \int_{\phi=0}^{\pi/2} r^2 \sin^2 \theta \cos \phi dr d\phi \Big|_{\theta=\pi/2} \\
&= 81 \left(\frac{\pi}{2}\right) (-\cos \theta) \Big|_0^{\pi/2} + 9(1) \sin \phi \Big|_0^{\pi/2} \\
&= \frac{81\pi}{2} + 9 = \underline{\underline{136.23}}
\end{aligned}$$

Prob. 3.39

$$\text{Let } \psi = \oiint \mathbf{F} \cdot d\mathbf{S} = \psi_t + \psi_b + \psi_o + \psi_i$$

where $\psi_t, \psi_b, \psi_o, \psi_i$ are the fluxes through the top surface, bottom surface, outer surface ($\rho = 3$), and inner surface respectively.

For the top surface, $d\mathbf{S} = \rho d\phi d\rho \mathbf{a}_z$, $z = 5$;

$$\mathbf{F} \cdot d\mathbf{S} = \rho^2 z d\phi dz. \text{ Hence:}$$

$$\psi_t = \int_{\rho=2}^3 \int_{\phi=0}^{2\pi} \rho^2 z d\phi dz \Big|_{z=5} = \frac{190}{3} \pi = 198.97$$

For the bottom surface, $z = 0$, $d\mathbf{S} = \rho d\phi d\rho (-\mathbf{a}_z)$

$$\mathbf{F} \cdot d\mathbf{S} = -\rho^2 z d\phi d\rho = 0. \text{ Hence, } \psi_b = 0.$$

For the outer curved surface, $\rho = 3$, $d\mathbf{S} = \rho d\phi dz \mathbf{a}_\rho$

$$\mathbf{F} \cdot d\mathbf{S} = \rho^2 \sin \phi d\phi dz. \text{ Hence,}$$

$$\psi_o = \int_{z=0}^5 dz \rho^3 \int_{\phi=0}^{2\pi} \sin \phi d\phi \Big|_{\rho=3} = 0$$

For the inner curved surface, $\rho = 2$, $d\mathbf{S} = \rho d\phi dz (-\mathbf{a}_\rho)$

$$\mathbf{F} \cdot d\mathbf{S} = -\rho^3 \sin \phi d\phi dz. \text{ Hence,}$$

$$\psi_i = - \int_{z=0}^5 dz \rho^3 \int_{\phi=0}^{2\pi} \sin \phi d\phi \Big|_{\rho=2} = 0$$

$$\psi = \frac{190\pi}{3} + 0 + 0 + 0 = \frac{190}{3} \pi = \underline{\underline{198.97}}$$

$$\psi = \oiint \mathbf{F} \cdot d\mathbf{S} = \int \nabla \cdot \mathbf{F} dV$$

$$\nabla \cdot \mathbf{F} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho^3 \sin \phi) + \frac{1}{\rho} \frac{\partial}{\partial \phi} (z \cos \phi) + \rho$$

$$= 3\rho \sin \phi - \frac{z}{\rho} \sin \phi + \rho$$

$$\begin{aligned}
 \int_V \nabla \cdot \mathbf{F} dv &= \iiint (3\rho \sin \phi - \frac{z}{\rho} \sin \phi + \rho) \rho d\phi d\rho dz \\
 &= 0 + 0 + \int_0^5 dz \int_0^{2\pi} d\phi \int_2^3 \rho^2 d\rho \\
 &= \frac{190 \pi}{3} = 198.97
 \end{aligned}$$

Prob. 3.40

$$(a) \quad \nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & y^2 & -xz \end{vmatrix} = z\mathbf{a}_y - x\mathbf{a}_z$$

$$\begin{aligned}
 \nabla \times \mathbf{B} &= \left(\frac{1}{\rho} 2\rho z 2 \sin \phi \cos \phi - 0 \right) \mathbf{a}_\rho + (2\rho z - 2z \sin^2 \phi) \mathbf{a}_\phi + \frac{1}{\rho} (2\rho \sin^2 \phi - 0) \mathbf{a}_z \\
 (b) \quad &= 4z \sin \phi \cos \phi \mathbf{a}_\rho + 2(\rho z - z \sin^2 \phi) \mathbf{a}_\phi + 2 \sin^2 \phi \mathbf{a}_z \\
 &= \underline{\underline{2z \sin 2\phi \mathbf{a}_\rho + 2z(\rho - \sin^2 \phi) \mathbf{a}_\phi + 2 \sin^2 \phi \mathbf{a}_z}}
 \end{aligned}$$

$$\begin{aligned}
 \nabla \times \mathbf{C} &= \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (r \cos^2 \theta \sin \theta) \right] \mathbf{a}_r - \frac{1}{r} \left[\frac{\partial}{\partial r} (r^2 \cos^2 \theta) \right] \mathbf{a}_\theta \\
 (c) \quad &= \frac{r}{r \sin \theta} \left[(2 \cos \theta)(-\sin \theta) \sin \theta + \cos \theta (\cos^2 \theta) \right] \mathbf{a}_r - \frac{\cos^2 \theta}{r} (2r) \mathbf{a}_\theta \\
 &= \underline{\underline{\frac{(\cos^3 \theta - 2 \sin^2 \theta \cos \theta)}{\sin \theta} \mathbf{a}_r - 2 \cos^2 \theta \mathbf{a}_\theta}}
 \end{aligned}$$

Prob. 3.41

$$\begin{aligned}
 (a) \quad \nabla \times \mathbf{A} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y & y^2 z & -2xz \end{vmatrix} = \underline{\underline{-y^2 \mathbf{a}_x + 2z \mathbf{a}_y - x^2 \mathbf{a}_z}} \\
 \nabla \cdot \nabla \times \mathbf{A} &= \underline{\underline{0}}
 \end{aligned}$$

(b)

$$\begin{aligned}
\nabla \times \mathbf{A} &= \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \mathbf{a}_\rho + \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \mathbf{a}_\phi + \frac{1}{\rho} \left(\frac{\partial(\rho A_\rho)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi} \right) \mathbf{a}_z \\
&= (0 - 0) \mathbf{a}_\rho + (\rho^2 - 3z^2) \mathbf{a}_\phi + \frac{1}{\rho} (4\rho^3 - 0) \mathbf{a}_z \\
&= \underline{\underline{(\rho^2 - 3z^2) \mathbf{a}_\phi + 4\rho^2 \mathbf{a}_z}}
\end{aligned}$$

$$\nabla \bullet \nabla \times \mathbf{A} = \underline{\underline{0}}$$

$$\begin{aligned}
\nabla \times \mathbf{A} &= \frac{1}{r \sin \theta} \left[0 - \frac{\sin \phi}{r^2} \right] \mathbf{a}_r + \frac{1}{r} \left[\frac{-1}{\sin \theta} \frac{\cos \phi}{r^2} - 0 \right] \mathbf{a}_\theta + \frac{1}{r} \left[\frac{\partial}{\partial r} \left(\frac{\cos \phi}{r} \right) - 0 \right] \mathbf{a}_\phi \\
(c) \quad &= \underline{\underline{-\frac{\sin \phi}{r^3 \sin \theta} \mathbf{a}_r + \frac{\cos \phi}{r^3 \sin \theta} \mathbf{a}_\theta + \frac{\cos \phi}{r^3} \mathbf{a}_\phi}}
\end{aligned}$$

$$\nabla \bullet \nabla \times \mathbf{A} = \frac{-\sin \phi}{r^4 \sin \theta} + 0 + \frac{\sin \phi}{r^4 \sin \theta} = 0$$

$$\nabla \bullet \nabla \times \mathbf{A} = \underline{\underline{0}}$$

Prob. 3.42

$$\nabla \times \mathbf{H} = 0 \mathbf{a}_\rho + 1 \mathbf{a}_\phi + \frac{1}{\rho} (2\rho \cos \phi - \rho \cos \phi) \mathbf{a}_z = \underline{\underline{\mathbf{a}_\phi + \cos \phi \mathbf{a}_z}}$$

$$\nabla \times \nabla \times \mathbf{H} = \left(-\frac{1}{\rho} \sin \phi - 0 \right) \mathbf{a}_\rho + 0 \mathbf{a}_\phi + \frac{1}{\rho} (1 - 0) \mathbf{a}_z = \underline{\underline{-\frac{1}{\rho} \sin \phi \mathbf{a}_\rho + \frac{1}{\rho} \mathbf{a}_z}}$$

Prob. 3.43

Method 1: We can express $\mathbf{A}$ in spherical coordinates.

$$\mathbf{A} = \frac{r}{r^3} \mathbf{a}_r = \frac{\mathbf{a}_r}{r^2},$$

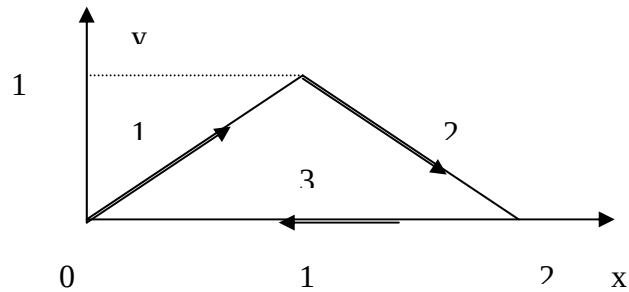
$$\nabla \times \mathbf{A} = \nabla \times \left(\frac{\mathbf{a}_r}{r^2} \right) = \nabla \left(\frac{1}{r^2} \right) \times \mathbf{a}_r = \frac{-2}{r^3} \mathbf{a}_r \times \mathbf{a}_r = \mathbf{0}$$

Method 2:

$$\mathbf{A} = \frac{x}{r^3} \mathbf{a}_x + \frac{y}{r^3} \mathbf{a}_y + \frac{z}{r^3} \mathbf{a}_z$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{x}{r^3} & \frac{y}{r^3} & \frac{z}{r^3} \end{vmatrix} = \left\{ -\frac{3}{2} z (x^2 + y^2 + z^2)^{-5/2} (2y) - \frac{3}{2} y (x^2 + y^2 + z^2)^{-5/2} (2z) \right\} \mathbf{a}_x + \dots$$

$$= \mathbf{0}$$

Prob. 3.44

(a)

$$\oint_L \mathbf{F} \bullet d\mathbf{l} = \left(\int_1 + \int_2 + \int_3 \right) \mathbf{F} \bullet d\mathbf{l}$$

For 1, $y = x$ $dy = dx$, $\bar{dl} = dx \bar{a}_x + dy \bar{a}_y$,

$$\int_1 \mathbf{F} \bullet d\mathbf{l} = \int_0^1 x^3 dx - x dx = -\frac{1}{4}$$

For 2, $y = -x + 2$, $dy = -dx$, $\bar{dl} = dx \bar{a}_x + dy \bar{a}_y$,

$$\int_2 \mathbf{F} \bullet d\mathbf{l} = \int_1^2 (-x^3 + 2x^2 - x + 2) dx = \frac{17}{12}$$

For 3,

$$\int_3 \mathbf{F} \bullet d\mathbf{l} = \int_2^0 x^2 y dx \Big|_{y=0} = 0$$

$$\oint_L \mathbf{F} \bullet d\mathbf{l} = -\frac{1}{4} + \frac{17}{12} + 0 = \underline{\underline{\frac{7}{6}}}$$

(b)

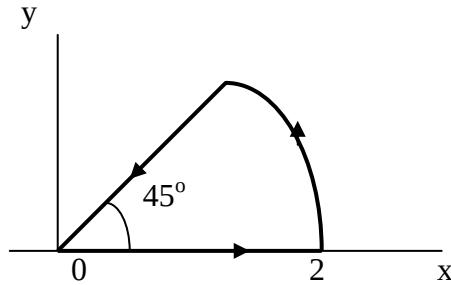
$$\nabla \times \mathbf{F} = -x^2 \mathbf{a}_z ; \quad d\mathbf{S} = dx dy (-\mathbf{a}_z)$$

$$\begin{aligned} \int (\nabla \times \mathbf{F}) \bullet d\mathbf{S} &= - \iint (-x^2) dx dy = \int_0^1 \int_0^x x^2 dy dx + \int_1^2 \int_{y=0}^{-x+2} x^2 dy dx \\ &= \int_0^1 x^2 y \Big|_0^x dx + \int_1^2 x^2 y \Big|_0^{-x+2} dx = \frac{x}{4} \Big|_0^1 + \int_1^2 x^2 (-x+2) dx = \underline{\underline{\frac{7}{6}}} \end{aligned}$$

(c) Yes

Prob. 3.45

$$\begin{aligned} \oint \mathbf{A} \bullet d\mathbf{l} &= \int_{\rho=2}^1 \rho \sin \phi d\rho \Big|_{\phi=0} + \int_{\phi=0}^{\pi/2} \rho^2 \rho d\phi \Big|_{\rho=1} + \int_{\rho=1}^2 \rho \sin \phi d\rho \Big|_{\phi=90^\circ} + \int_{\phi=\pi/2}^0 \rho^3 d\phi \Big|_{\rho=2} \\ &= \frac{\pi}{2} + \frac{1}{2}(4-1) + 8\left(-\frac{\pi}{2}\right) = \underline{\underline{-9.4956}} \end{aligned}$$

Prob. 3.46

$$\oint \mathbf{F} \cdot d\mathbf{l} = \int_0^2 2\rho z d\rho \Big|_{z=1} + \int_0^{\pi/4} 3z \sin \phi \rho d\phi \Big|_{\rho=2, z=1} + \int_2^0 2\rho z d\rho \Big|_{z=1}$$

$$= \rho^2 \Big|_0^2 + (-6 \cos \phi) \Big|_0^{\pi/4} + \rho^2 \Big|_2^0 = (4 - 0) + 6(-\cos \pi/4 + 1) + (0 - 4) = \underline{\underline{1.757}}$$

$$\nabla \times \mathbf{F} = \frac{1}{\rho} [3z \sin \phi - 0] \mathbf{a}_z + \dots$$

$$\int (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\rho=0}^2 \int_{\phi=0}^{\pi/4} \frac{3z}{\rho} \sin \phi \rho d\phi d\rho \Big|_{z=1} = 3(2)(-\cos \phi) \Big|_0^{\pi/4}$$

$$= 6(-\cos \pi + 1) = \underline{\underline{1.757}}$$

Prob. 3.47

$$\nabla \cdot \mathbf{A} = 8xe^{-y} + 8xe^{-y} = 16xe^{-y}$$

$$\nabla(\nabla \cdot \mathbf{A}) = 16e^{-y} \mathbf{a}_x - 16xe^{-y} \mathbf{a}_y$$

$$\nabla \times \nabla(\nabla \cdot \mathbf{A}) = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 16e^{-y} & -16xe^{-y} & 0 \end{vmatrix} = (-16e^{-y} + 16e^{-y}) \mathbf{a}_z = \underline{\underline{0}}$$

Should be expected since $\nabla \times \nabla V = 0$.

Prob. 3.48

$$(a) \nabla V = -\frac{\sin \theta \cos \phi}{r^2} \mathbf{a}_r + \frac{\cos \theta \cos \phi}{r^2} \mathbf{a}_\theta - \frac{\sin \phi}{r^2} \mathbf{a}_\phi$$

(b) $\nabla \times \nabla V = \underline{\underline{0}}$

(c)

$$\begin{aligned}\nabla \cdot \nabla V &= \nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} (-\sin \theta \cos \phi) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\cos \theta \cos \phi}{r} \right) + \frac{1}{r^2 \sin^2 \theta} \left(-\frac{\sin \theta \cos \phi}{r} \right) \\ &= 0 + \frac{\cos \phi}{r^3 \sin \theta} (1 - 2 \sin^2 \theta) - \frac{\cos \phi}{r^3 \sin \theta} \\ &= -\frac{2 \sin \theta \cos \phi}{r^3}\end{aligned}$$

Prob.3.49

$$\begin{aligned}\mathbf{Q} &= \frac{r}{r \sin \theta} r \sin \theta [(\cos \phi - \sin \phi) \mathbf{a}_x + (\cos \phi + \sin \phi) \mathbf{a}_y] \\ &= r(\cos \phi - \sin \phi) \mathbf{a}_x + r(\cos \phi + \sin \phi) \mathbf{a}_y\end{aligned}$$

$$\begin{bmatrix} Q_r \\ Q_\theta \\ Q_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} Q_x \\ Q_y \\ Q_z \end{bmatrix}$$

$$\mathbf{Q} = r \sin \theta \mathbf{a}_r + r \cos \theta \mathbf{a}_\theta + r \mathbf{a}_\phi$$

(a)

$$d\mathbf{l} = \rho d\phi \mathbf{a}_\phi, \quad \rho = r \sin 30^\circ = 2\left(\frac{1}{2}\right) = 1$$

$$z = r \cos 30^\circ = \sqrt{3}$$

$$Q_\phi = r = \sqrt{\rho^2 + z^2}$$

$$\oint \mathbf{Q} \cdot d\mathbf{l} = \int_0^{2\pi} \sqrt{\rho^2 + z^2} \rho d\phi = 2(1)(2\pi) = \underline{\underline{4\pi}}$$

(b)

$$\nabla \times \mathbf{Q} = \cot \theta \mathbf{a}_r - 2 \mathbf{a}_\theta + \cos \theta \mathbf{a}_\phi$$

$$\text{For } S_1, \quad d\mathbf{S} = r^2 \sin \theta d\theta d\phi \mathbf{a}_r$$

$$\int_{S_1} (\nabla \times \mathbf{Q}) \cdot d\mathbf{S} = \int r^2 \sin \theta \cot \theta d\theta d\phi \Big|_{r=2}$$

$$= 4 \int_0^{2\pi} d\phi \int_0^{30^\circ} \cos \theta d\theta = \underline{\underline{4\pi}}$$

(c)

For S_2 , $d\mathbf{S} = r \sin \theta d\theta dr \mathbf{a}_\theta$

$$\begin{aligned} \int_{S_2} (\nabla \times \mathbf{Q}) \cdot d\mathbf{S} &= -2 \int r \sin \theta d\phi dr \Big|_{\theta=30^\circ} \\ &= -2 \sin 30^\circ \int_0^2 r dr \int_0^{2\pi} d\phi \\ &= \underline{\underline{-4\pi}} \end{aligned}$$

(d)

For S_1 , $d\mathbf{S} = r^2 \sin \theta d\phi d\theta \mathbf{a}_r$

$$\begin{aligned} \int_{S_1} \mathbf{Q} \cdot d\mathbf{S} &= r^3 \int \sin^2 \theta d\theta d\phi \Big|_{r=2} \\ &= 8 \int_0^{2\pi} d\phi \int_0^{30^\circ} \sin^2 \theta d\theta \\ &= 4\pi \left[\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right] = \underline{\underline{2.2767}} \end{aligned}$$

(e)

For S_2 , $d\mathbf{S} = r \sin \theta d\phi dr \mathbf{a}_\theta$

$$\begin{aligned} \int_{S_2} \mathbf{Q} \cdot d\mathbf{S} &= \int r^2 \sin \theta \cos \theta d\phi dr \Big|_{\theta=30^\circ} \\ &= \frac{4\pi\sqrt{3}}{3} = \underline{\underline{7.2552}} \end{aligned}$$

(f)

$$\begin{aligned} \nabla \cdot \mathbf{Q} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^3 \sin \theta) + \frac{r}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \cos \theta) + 0 \\ &= 2 \sin \theta + \cos \theta \cot \theta \end{aligned}$$

$$\begin{aligned} \int \nabla \cdot \mathbf{Q} dv &= \int (2 \sin \theta + \cos \theta \cot \theta) r^2 \sin \theta d\theta d\phi dr \\ &= \frac{r^3}{3} \Big|_0^2 (2\pi) \int_0^{30^\circ} (1 + \sin^2 \theta) d\theta \\ &= \frac{4\pi}{3} \left(\pi - \frac{\sqrt{3}}{2} \right) = \underline{\underline{9.532}} \end{aligned}$$

$$\begin{aligned}
 \text{Check: } \int \nabla \cdot \mathbf{Q} dv &= \left(\int_{S_1} + \int_{S_2} \right) \mathbf{Q} \cdot d\mathbf{s} \\
 &= 4\pi \left[\frac{\pi}{3} - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{3} \right] \\
 &= \frac{4\pi}{3} \left[\pi - \frac{\sqrt{3}}{2} \right] \quad (\text{It checks!})
 \end{aligned}$$

Prob. 3.50

Since $\mathbf{u} = \boldsymbol{\omega} \times \mathbf{r}$, $\nabla \times \mathbf{u} = \nabla \times (\boldsymbol{\omega} \times \mathbf{r})$. From Appendix A.10,

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B}$$

$$\nabla \times \mathbf{u} = \nabla \times (\boldsymbol{\omega} \times \mathbf{r})$$

$$\begin{aligned}
 \nabla \times (\boldsymbol{\omega} \times \mathbf{r}) &= \boldsymbol{\omega}(\nabla \cdot \mathbf{r}) - \mathbf{r}(\nabla \cdot \boldsymbol{\omega}) + (\mathbf{r} \cdot \nabla) \boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \nabla) \mathbf{r} \\
 &= \boldsymbol{\omega}(3) - \boldsymbol{\omega} = 2\boldsymbol{\omega}
 \end{aligned}$$

$$\text{or } \boldsymbol{\omega} = \frac{1}{2} \nabla \times \mathbf{u}.$$

Alternatively, let $x = r \cos \omega t$, $y = r \sin \omega t$

$$\mathbf{u} = \frac{\partial x}{\partial t} \mathbf{a}_x + \frac{\partial y}{\partial t} \mathbf{a}_y$$

$$= -\omega r \sin \omega t \mathbf{a}_x + \omega r \cos \omega t \mathbf{a}_y$$

$$= -\omega y \mathbf{a}_x + \omega x \mathbf{a}_y$$

$$\nabla \times \mathbf{u} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -\omega y & \omega x & 0 \end{vmatrix} = 2\omega \mathbf{a}_z = 2\boldsymbol{\omega}$$

$$\text{i.e., } \boldsymbol{\omega} = \frac{1}{2} \nabla \times \mathbf{u}$$

Note that we have used the fact that $\nabla \cdot \boldsymbol{\omega} = 0$, $(\mathbf{r} \cdot \nabla) \boldsymbol{\omega} = 0$, $(\boldsymbol{\omega} \cdot \nabla) \mathbf{r} = \boldsymbol{\omega}$

Prob. 3.51

$$(a) \quad \nabla \cdot \mathbf{H} = \frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y} + \frac{\partial H_z}{\partial z} = \underline{\underline{2z + 5x + 8}}$$

$$(b) \quad \nabla \times \mathbf{H} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & 5xy & 8(y+z) \end{vmatrix} = \underline{\underline{8\mathbf{a}_x + 2x\mathbf{a}_y + 5y\mathbf{a}_z}}$$

Prob. 3.52

$$\begin{aligned} \nabla \cdot \mathbf{B} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 B_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (B_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial B_\phi}{\partial \phi} \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^4) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (4r \sin \theta \cos 2\theta) + 0 \\ &= 4r + \frac{1}{\sin \theta} [4r \cos \theta \cos 2\theta + 4r \sin \theta (-2 \sin 2\theta)] \end{aligned}$$

$$\underline{\underline{\nabla \cdot \mathbf{B} = 4r + 4r \cot \theta \cos 2\theta - 8r \sin 2\theta}}$$

$$\nabla \times \mathbf{B} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (B_\phi \sin \theta) - \frac{\partial B_\phi}{\partial \phi} \right] \mathbf{a}_r + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial B_r}{\partial \phi} - \frac{\partial}{\partial r} (r B_\phi) \right] \mathbf{a}_\theta + \frac{1}{r} \left[\frac{\partial}{\partial r} (r B_\theta) - \frac{\partial B_r}{\partial \theta} \right] \mathbf{a}_\phi$$

All terms are zero except one.

$$\nabla \times \mathbf{B} = 0\mathbf{a}_r + 0\mathbf{a}_\theta + \frac{1}{r} \left[\frac{\partial}{\partial r} (r B_\theta) - 0 \right] \mathbf{a}_\phi = \frac{1}{r} \frac{\partial}{\partial r} (4r^2 \cos 2\theta) \mathbf{a}_\phi = \underline{\underline{8 \cos 2\theta \mathbf{a}_\phi}}$$

Prob. 3.53

$$\begin{aligned} (a) \quad \nabla \cdot (\nabla V) &= \nabla \cdot \left(\mathbf{V} \frac{\partial V}{\partial x} \mathbf{a}_x + \mathbf{V} \frac{\partial V}{\partial y} \mathbf{a}_y + \mathbf{V} \frac{\partial V}{\partial z} \mathbf{a}_z \right) \\ &= \frac{\partial}{\partial x} \left(\mathbf{V} \frac{\partial V}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mathbf{V} \frac{\partial V}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mathbf{V} \frac{\partial V}{\partial z} \right) \\ &= V \frac{\partial^2 V}{\partial x^2} + V \frac{\partial^2 V}{\partial y^2} + V \frac{\partial^2 V}{\partial z^2} + \left(\frac{\partial V}{\partial x} \right)^2 + \left(\frac{\partial V}{\partial y} \right)^2 + \left(\frac{\partial V}{\partial z} \right)^2 \\ &= V \nabla^2 V + |\nabla V|^2 \end{aligned}$$

(b)

$$\begin{aligned}
\nabla \times V\mathbf{A} &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ VA_x & VA_y & VA_z \end{vmatrix} \\
&= \left[\frac{\partial}{\partial y}(VA_z) - \frac{\partial}{\partial z}(VA_y) \right] \mathbf{a}_x + \left[\frac{\partial}{\partial z}(VA_x) - \frac{\partial}{\partial x}(VA_z) \right] \mathbf{a}_y + \left[\frac{\partial}{\partial x}(VA_y) - \frac{\partial}{\partial y}(VA_x) \right] \mathbf{a}_z \\
&= \left[A_z \frac{\partial V}{\partial y} + V \frac{\partial A_z}{\partial y} - A_y \frac{\partial V}{\partial z} - V \frac{\partial A_y}{\partial z} \right] \mathbf{a}_x \\
&\quad + \left[A_x \frac{\partial V}{\partial z} + V \frac{\partial A_x}{\partial z} - A_z \frac{\partial V}{\partial x} - V \frac{\partial A_z}{\partial x} \right] \mathbf{a}_y \\
&\quad + \left[A_y \frac{\partial V}{\partial x} + V \frac{\partial A_y}{\partial x} - A_x \frac{\partial V}{\partial y} - V \frac{\partial A_x}{\partial y} \right] \mathbf{a}_z
\end{aligned}$$

$$\begin{aligned}
\nabla \times V\mathbf{A} &= V \left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \mathbf{a}_x + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \mathbf{a}_y + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \mathbf{a}_z \right] \\
&\quad + \left(A_z \frac{\partial V}{\partial y} - A_y \frac{\partial V}{\partial z} \right) \mathbf{a}_x + \left(A_x \frac{\partial V}{\partial z} - A_z \frac{\partial V}{\partial x} \right) \mathbf{a}_y + \left(A_y \frac{\partial V}{\partial x} - A_x \frac{\partial V}{\partial y} \right) \mathbf{a}_z \\
&= V \nabla \times \mathbf{A} + \nabla V \times \mathbf{A}
\end{aligned}$$

Prob. 3.54

(a)

$$\nabla \cdot \mathbf{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 2xy + 1 + 1 = \underline{\underline{2 + 2xy}}$$

(b)

$$\begin{aligned}
\nabla \times \mathbf{B} &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & (2x^2 + y) & (z - y) \end{vmatrix} = (-1 + 0)\mathbf{a}_x + (0 - 0)\mathbf{a}_y + (4x + x^2)\mathbf{a}_z \\
&= \underline{\underline{-\mathbf{a}_x + x(4 + x)\mathbf{a}_z}}
\end{aligned}$$

(c)

$$\nabla(\nabla \cdot \mathbf{B}) = \nabla(2 + 2xy) = \underline{\underline{2y\mathbf{a}_x + 2x\mathbf{a}_y}}$$

(d)

$$\begin{aligned}\nabla \times \nabla \times \mathbf{B} &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -1 & 0 & (4x - x^2) \end{vmatrix} = 0\mathbf{a}_x - (4 - 2x)\mathbf{a}_y + 0\mathbf{a}_z \\ &= \underline{\underline{2(x - 2)\mathbf{a}_y}}\end{aligned}$$

Prob. 3.55

(a)

$$V_1 = x^3 + y^3 + z^3$$

$$\begin{aligned}\nabla^2 V_1 &= \frac{\partial^2 V_1}{\partial x^2} + \frac{\partial^2 V_1}{\partial y^2} + \frac{\partial^2 V_1}{\partial z^2} \\ &= \frac{\partial}{\partial x}(3x^2) + \frac{\partial}{\partial y}(3y^2) + \frac{\partial}{\partial z}(3z^2) \\ &= 6x + 6y + 6z = \underline{\underline{6(x + y + z)}}$$

(b)

$$V_2 = \rho z^2 \sin 2\phi$$

$$\begin{aligned}\nabla^2 V_2 &= \frac{1}{\rho} \frac{\partial}{\partial \rho}(\rho z^2 \sin 2\phi) - \frac{4z^2}{\rho} \sin 2\phi + \frac{\partial}{\partial z}(2\rho z \sin 2\phi) \\ &= \frac{z^2}{\rho} \sin 2\phi - \frac{4z^2}{\rho} \sin 2\phi + 2\rho \sin 2\phi \\ &= \underline{\underline{\left(\frac{-3z^2}{\rho} + 2\rho\right) \sin 2\phi}}\end{aligned}$$

(c)

$$V_3 = r^2(1 + \cos \theta \sin \phi)$$

$$\begin{aligned}\nabla^2 V_3 &= \frac{1}{r^2} \frac{\partial}{\partial r}[2r^3(1 + \cos \theta \sin \phi)] \\ &\quad + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta}(-\sin^2 \theta \sin \phi)r^2 + \frac{1}{r^2 \sin^2 \theta} r^2 \left(-\cos \theta \sin \phi\right) \\ &= 6(1 + \cos \theta \sin \phi) - \frac{2 \sin \theta}{\sin \theta} \cos \theta \sin \phi - \frac{\cos \theta}{\sin \theta} \frac{\sin \phi}{\theta} \\ &= \underline{\underline{6 + 4 \cos \theta \sin \phi - \frac{\cos \theta \sin \phi}{\sin^2 \theta}}}\end{aligned}$$

Prob. 3.56

(a)

$$U = x^3 y^2 e^{xz}$$

$$\begin{aligned}\nabla^2 U &= \frac{\partial}{\partial x}(3x^2 y^2 e^{xz} + x^3 y^2 z e^{xz}) + \frac{\partial}{\partial y}(2x^3 y e^{xz}) + \frac{\partial}{\partial z}(x^4 y^2 e^{xz}) \\ &= 6xy^2 e^{xz} + 3x^2 y z e^{xz} + 3x^2 y^2 z e^{xz} + x^3 y^2 z^2 e^{xz} + 2x^3 e^{xz} + x^5 y^2 e^{xz} \\ &= \underline{\underline{e^{xz}(6xy^2 + 3x^2 y^2 z + 3x^2 y^2 z + x^3 y^2 z^2 + 2x^3 + x^5 y^2)}}$$

At (1, -1, 1),

$$\nabla^2 U = e^1 (6 + 3 + 3 + 1 + 2 + 1) = 16e = \underline{\underline{43.493}}$$

(b)

$$V = \rho^2 z (\cos \phi + \sin \phi)$$

$$\begin{aligned}\nabla^2 V &= \frac{1}{\rho} \frac{\partial}{\partial \rho} [2\rho^2 z (\cos \phi + \sin \phi)] - z (\cos \phi + \sin \phi) + 0 \\ &= 4z (\cos \phi + \sin \phi) - z (\cos \phi + \sin \phi) \\ &= \underline{\underline{3z (\cos \phi + \sin \phi)}}$$

$$\text{At } (5, \frac{\pi}{6}, -2), \quad \nabla^2 V = -6(0.866 + 0.5) = \underline{\underline{-8.196}}$$

(c)

$$W = e^{-r} \sin \theta \cos \phi$$

$$\begin{aligned}\nabla^2 W &= \frac{1}{r^2} \frac{\partial}{\partial r} (-r^2 e^{-r} \sin \theta \cos \phi) + \frac{e^{-r}}{r^2 \sin \theta} \cos \phi \frac{\partial}{\partial \theta} (\sin \theta \cos \theta) \\ &\quad - \frac{e^{-r} \sin \theta \cos \phi}{r^2 \sin^2 \theta}\end{aligned}$$

$$= \frac{1}{r^2} (-2r e^{-r} \sin \theta \cos \phi) + e^{-r} \sin \theta \cos \phi$$

$$+ \frac{e^{-r} \cos \phi}{r^2 \sin \theta} (1 - 2 \sin^2 \theta) - \frac{e^{-r} \cos \phi}{r^2 \sin \theta}$$

$$\underline{\underline{\nabla^2 W = e^{-r} \sin \theta \cos \phi (1 - \frac{2}{r} - \frac{2}{r^2})}}$$

At (1, 60°, 30°),

$$\nabla^2 W = e^{-1} \sin 60 \cos 30 (1 - 2 - 2) = -2.25e^{-1} = \underline{\underline{-0.8277}}$$

Prob. 3.57

(a) Let $V = 1nr = 1n\sqrt{x^2 + y^2 + z^2}$

$$\frac{\partial V}{\partial x} = \frac{1}{r} \frac{1}{2} (2x) (x^2 + y^2 + z^2)^{-1/2} = \frac{x}{r^2}$$

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z = \frac{x\mathbf{a}_x + y\mathbf{a}_y + z\mathbf{a}_z}{r^2} = \underline{\underline{\frac{\mathbf{r}}{r^2}}}$$

(b) Let $\nabla V = \mathbf{A} = \frac{\mathbf{r}}{r^2} = \frac{1}{r} \mathbf{a}_r$ in spherical coordinates.

$$\begin{aligned} \nabla^2(1nr) &= \nabla \cdot \nabla (1nr) = \nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) = \frac{1}{r^2} \frac{\partial}{\partial r} (r) \\ &= \underline{\underline{\frac{1}{r^2}}} \end{aligned}$$

Prob. 3.58

(a)

$$\nabla U = \frac{\partial U}{\partial x} \mathbf{a}_x + \frac{\partial U}{\partial y} \mathbf{a}_y + \frac{\partial U}{\partial z} \mathbf{a}_z = y^2 z^3 \mathbf{a}_x + 2xyz^3 \mathbf{a}_y + \underline{\underline{3xy^2 z^2 \frac{\partial U}{\partial z} \mathbf{a}_z}}$$

$$\nabla^2 U = \frac{\partial}{\partial x} \left(\frac{\partial U}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial U}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial U}{\partial z} \right) = 0 + 2xz^3 + 6xy^2 z = \underline{\underline{2xz^3 + 6xy^2 z}}$$

(b)

$$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z = -\frac{\sin \phi}{\rho^2} \mathbf{a}_\rho + \frac{\cos \phi}{\rho^2} \mathbf{a}_\phi + 0 = \underline{\underline{\frac{1}{\rho^2} [-\sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi]}}$$

$$\begin{aligned} \nabla^2 V &= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (-\rho^{-1} \sin \theta) + \frac{1}{\rho^2} \left(-\frac{\sin \phi}{\rho} \right) + 0 \\ &= \frac{\sin \phi}{\rho^3} - \frac{1}{\rho^3} \sin \phi = \underline{\underline{0}} \end{aligned}$$

(c)
$$\begin{aligned} \nabla W &= \frac{\partial W}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial W}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial W}{\partial \phi} \mathbf{a}_\phi \\ &= \underline{\underline{2r \sin \theta \cos \phi \mathbf{a}_r + r \cos \theta \cos \phi \mathbf{a}_\theta - r \sin \phi \mathbf{a}_\phi}} \end{aligned}$$

$$\begin{aligned}
\nabla^2 W &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial W}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial W}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 W}{\partial \phi^2} \\
&= \frac{1}{r^2} \frac{\partial}{\partial r} (2r^3 \sin \theta \cos \phi) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (r^2 \sin \theta \cos \theta \cos \phi) + \frac{1}{r^2 \sin^2 \theta} (-r^2 \sin \theta \cos \phi) \\
&= 6 \sin \theta \cos \phi + \frac{1}{\sin \theta} (\cos^2 \theta \cos \phi - \sin^2 \theta \cos \phi) - \frac{\cos \phi}{\sin \theta} \\
&= 6 \sin \theta \cos \phi + \frac{\cos \phi}{\sin \theta} - 2 \sin \theta \cos \phi - \frac{\cos \phi}{\sin \theta} = \underline{\underline{4 \sin \theta \cos \phi}}
\end{aligned}$$

Prob. 3.59

$$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z = \underline{\underline{2\rho z \cos \phi \mathbf{a}_\rho - \rho z \sin \phi \mathbf{a}_\phi + \rho^2 \cos \phi \mathbf{a}_z}}$$

$$\begin{aligned}
\nabla^2 V &= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (2\rho^2 z \cos \phi) - \frac{1}{\rho^2} \rho^2 z \cos \phi + 0 \\
&= (4-1) z \cos \phi = \underline{\underline{3z \cos \phi}}
\end{aligned}$$

Prob. 3.60

$$\nabla V = \frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi$$

$$(a) \quad \underline{\underline{-\frac{10}{r^3} \cos \phi \mathbf{a}_r - \frac{5 \sin \phi}{r^3 \sin \theta} \mathbf{a}_\phi}}$$

$$\nabla \nabla V = \nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial V}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial V}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

$$(b) \quad = \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \left(-\frac{10 \cos \phi}{r^3} \right) + 0 + \frac{1}{r^2 \sin^2 \theta} \left(-\frac{5 \cos \phi}{r^2} \right)$$

$$\underline{\underline{\nabla \nabla V = \frac{10 \cos \phi}{r^4} - \frac{5 \cos \phi}{r^4 \sin^2 \theta}}}$$

(c) $\nabla \times \nabla V = 0$, see Example 3.10.

Prob. 3.61

$$\nabla U = \frac{\partial U}{\partial x} \mathbf{a}_x + \frac{\partial U}{\partial y} \mathbf{a}_y + \frac{\partial U}{\partial z} \mathbf{a}_z = 4yz^2 \mathbf{a}_x + (4xz^2 + 10z) \mathbf{a}_y + (8xyz + 10y) \mathbf{a}_z$$

$$\nabla \square \nabla U = \frac{\partial}{\partial x} (\nabla U_x) + \frac{\partial}{\partial y} (\nabla U_y) + \frac{\partial}{\partial z} (\nabla U_z) = 0 + 0 + 8xy = 8xy$$

$$\nabla^2 U = \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = 0 + 0 + 8xy = 8xy$$

Hence, $\nabla^2 U = \nabla \square \nabla U$

Prob. 3.62Method 1

$$\nabla^2 \mathbf{G} \Big|_{\rho} = \nabla^2 G_{\rho} - \frac{2}{\rho^2} \frac{\partial G_{\phi}}{\partial \phi} - \frac{G_{\rho}}{\rho^2}$$

$$\begin{aligned} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (2\rho \sin \phi) - \frac{2\rho \sin \phi}{\rho^2} + 0 + \frac{8\rho \sin \phi}{\rho^2} - \frac{2\rho \sin \phi}{\rho^2} \\ &= \frac{2 \sin \phi}{\rho} - \frac{2 \sin \phi}{\rho} + \frac{8 \sin \phi}{\rho} - \frac{2 \sin \phi}{\rho} = \frac{6 \sin \phi}{\rho} \end{aligned}$$

$$\nabla^2 \mathbf{G} \Big|_{\phi} = \nabla^2 G_{\phi} + \frac{2}{\rho^2} \frac{\partial G_{\rho}}{\partial \phi} - \frac{G_{\phi}}{\rho^2}$$

$$\begin{aligned} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (4\rho \cos \phi) - \frac{1}{\rho} 4\rho \cos \phi + 0 + \frac{4\rho \cos \phi}{\rho^2} - \frac{4\rho \cos \phi}{\rho^2} \\ &= \frac{4 \cos \phi}{\rho} - \frac{4 \cos \phi}{\rho} + \frac{4 \cos \phi}{\rho} - \frac{4 \cos \phi}{\rho} = 0 \end{aligned}$$

$$\begin{aligned} \nabla^2 \mathbf{G} \Big|_z &= \nabla^2 G_z = \frac{1}{\rho} \frac{\partial}{\partial \rho} [\rho(z^2 + 1)] + 0 + \frac{\partial}{\partial z} (2z\rho) \\ &= \frac{1}{\rho} (z^2 + 1) + 2\rho \end{aligned}$$

Adding the components together gives

$$\underline{\underline{\nabla^2 \mathbf{G} = \frac{6 \sin \phi}{\rho} \mathbf{a}_{\rho} + \left[2\rho + \frac{1}{\rho} (z^2 + 1) \right] \mathbf{a}_z}}$$

Method 2:

$$\nabla^2 \mathbf{G} = \nabla(\nabla \cdot \mathbf{G}) - \nabla \times (\nabla \times \mathbf{G})$$

$$\text{Let } \mathbf{V} = \nabla \cdot \mathbf{G} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (2\rho^2 \sin \phi) + \frac{1}{\rho} (-4\rho \sin \phi) + 2z\rho = 2z\rho$$

$$\nabla(\nabla \cdot \mathbf{G}) = \nabla V = 2z\mathbf{a}_\rho + 2\rho\mathbf{a}_z$$

$$\begin{aligned} \text{Let } \mathbf{A} = \nabla \times \mathbf{G} &= \left[\frac{1}{\rho} 0 - 0 \right] \mathbf{a}_\rho + \left[0 - (z^2 + 1) \right] \mathbf{a}_\phi + \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (4\rho^2 \cos \phi) - 2\rho \cos \phi \right] \mathbf{a}_z \\ &= -(z^2 + 1)\mathbf{a}_\phi + 6 \cos \phi \mathbf{a}_z \end{aligned}$$

$$\begin{aligned} \nabla \times \nabla \times \mathbf{G} = \nabla \times \mathbf{A} &= \left[-\frac{6}{\rho} \sin \phi + 2z \right] \mathbf{a}_\rho + (0 - 0)\mathbf{a}_\phi + \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (-\rho(z^2 + 1)) - 0 \right] \mathbf{a}_z \\ &= \left[2z - \frac{6}{\rho} \sin \phi \right] \mathbf{a}_\rho - \frac{1}{\rho} (z^2 + 1) \mathbf{a}_z \end{aligned}$$

$$\nabla^2 \mathbf{G} = \nabla V - \nabla \times \mathbf{A}$$

$$\begin{aligned} &= 2z\mathbf{a}_\rho + 2\rho\mathbf{a}_z - \left[2z - \frac{6}{\rho} \sin \phi \right] \mathbf{a}_\rho + \frac{1}{\rho} (z^2 + 1) \mathbf{a}_z \\ &= \frac{6}{\rho} \sin \phi \mathbf{a}_\rho + \left[2\rho + \frac{1}{\rho} (z^2 + 1) \right] \mathbf{a}_z \end{aligned}$$

Prob. 3.63

$$\nabla \cdot \mathbf{A} = \frac{\partial}{\partial x} (xz) + \frac{\partial}{\partial y} (z^2) + \frac{\partial}{\partial z} (yz) = z + y$$

$$\nabla(\nabla \cdot \mathbf{A}) = \mathbf{a}_y + \mathbf{a}_z$$

$$\nabla^2 \mathbf{A} = \nabla^2 A_x \mathbf{a}_x + \nabla^2 A_y \mathbf{a}_y + \nabla^2 A_z \mathbf{a}_z = 0 + 2\mathbf{a}_y + 0 = 2\mathbf{a}_y$$

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = -\mathbf{a}_y + \mathbf{a}_z \quad (1)$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & z^2 & yz \end{vmatrix} = -z\mathbf{a}_x + x\mathbf{a}_y$$

$$\nabla \times \nabla \times \mathbf{A} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -z & x & 0 \end{vmatrix} = -\mathbf{a}_y + \mathbf{a}_z \quad (2)$$

From (1) and (2),

$$\nabla \times \nabla \times \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

Prob. 3.64

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 1 + 1 + 1 = 3 \neq 0$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \mathbf{0}$$

$$\nabla \cdot \mathbf{B} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho B_\rho) + \frac{1}{\rho} \frac{\partial B_\phi}{\partial \phi} + \frac{\partial B_z}{\partial z} = 4 \cos \phi - 4 \cos \phi = 0$$

$$\begin{aligned} \nabla \times \mathbf{B} &= \left[\frac{1}{\rho} \frac{\partial B_z}{\partial \phi} - \frac{\partial B_\phi}{\partial z} \right] \mathbf{a}_\rho + \left[\frac{\partial B_\rho}{\partial z} - \frac{\partial B_z}{\partial \rho} \right] \mathbf{a}_\phi + \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (\rho B_\rho) - \frac{\partial B_\rho}{\partial \phi} \right] \mathbf{a}_z \\ &= 0 \mathbf{a}_\rho + 0 \mathbf{a}_\phi + \frac{1}{\rho} [-8\rho \sin \phi + 2\rho \sin \phi] \mathbf{a}_z = -6 \sin \phi \mathbf{a}_z \neq \mathbf{0} \end{aligned}$$

$$\nabla \cdot \mathbf{C} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \sin \theta) + 0 + 0 = \frac{2 \sin \theta}{r} \neq 0$$

$$\begin{aligned} \nabla \times \mathbf{C} &= \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (r \sin^2 \theta) - 0 \right] \mathbf{a}_r + \frac{1}{r} \left[0 - \frac{\partial}{\partial r} (r^2 \sin \theta) \right] \mathbf{a}_\theta \\ &\quad + \frac{1}{r} [0 - \cos \theta] \mathbf{a}_\phi \\ &= 2 \cos \theta \mathbf{a}_r - 2 \sin \theta \mathbf{a}_\theta - \frac{\cos \theta}{r} \mathbf{a}_\phi \neq \mathbf{0} \end{aligned}$$

(a) $\mathbf{B}$ is solenoidal.

(b) $\mathbf{A}$ is irrotational.

Prob. 3.65 (a)

$$\nabla \times \mathbf{G} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 16xy - z & 8x^2 & -x \end{vmatrix}$$

$$= 0 \mathbf{a}_x + (-1 + 1) \mathbf{a}_y + (16x - 16x) \mathbf{a}_z = \mathbf{0}$$

Thus, $\mathbf{G}$ is irrotational.

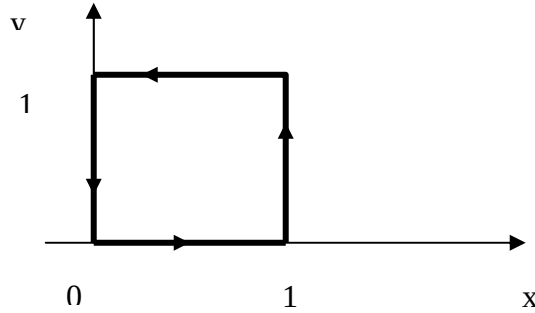
(b) Assume that ψ represents the net flux.

$$\psi = \oint \mathbf{G} \cdot d\mathbf{S} = \int \nabla \cdot \mathbf{G} dv$$

$$\nabla \cdot \mathbf{G} = 16y + 0 + 0 = 16y$$

$$\psi = \iiint 16y dx dy dz = 16 \int_0^1 dx \int_0^1 dz \int_0^1 y dy = 16(1)(1) \left(\frac{y^2}{2} \right) \Big|_0^1 = \underline{\underline{8}}$$

(c)



$$\begin{aligned} \oint_L \mathbf{G} \cdot d\mathbf{l} &= \int_{x=0}^{x=1} (16xy - z) dx \Big|_{z=0}^{y=0} + \int_{y=0}^{y=1} 8x^2 dy \Big|_{x=1}^{z=0} + \int_{x=1}^{x=0} (16xy - z) dx \Big|_{z=0}^{y=1} + \int_{y=1}^{y=0} 8x^2 dy \Big|_{x=0}^{z=0} \\ &= 0 + 8(1)y \Big|_0^1 + 16(1) \frac{x^2}{2} \Big|_1^0 + 0 \\ &= 8 - 8 = \underline{\underline{0}} \end{aligned}$$

This is expected since $\mathbf{G}$ is irrotational, i.e.

$$\oint \mathbf{G} \cdot d\mathbf{l} = \int (\nabla \times \mathbf{G}) \cdot d\mathbf{S} = 0$$

$$\nabla \cdot \mathbf{T} = -6 + 0 = \underline{\underline{-6}}$$

Prob. 3.66

$$\nabla \cdot \mathbf{E} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho E_\rho) = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\frac{\lambda}{2\pi\epsilon} \right) = 0$$

Hence, $\mathbf{E}$ is solenoidal.

$$\nabla \times \mathbf{E} = \frac{\partial E_\rho}{\partial z} \mathbf{a}_\phi - \frac{\partial E_\rho}{\partial \phi} \mathbf{a}_z = \mathbf{0}$$

showing that $\mathbf{E}$ is conservative.

Prob. 3.67

$$\nabla \times \mathbf{H} = \mathbf{0}$$

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = 0$$

Prob. 3.68

$$\nabla \times \mathbf{B} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2z + y^2 & 2xy & x^3 \end{vmatrix} = (0 - 0)\mathbf{a}_x + (3x^2 - 3x^2)\mathbf{a}_y + (2y - 2y)\mathbf{a}_z = \mathbf{0}$$

showing that $\mathbf{B}$ is conservative.

Prob. 3.69

$$\nabla \cdot \mathbf{D} = \frac{1}{\rho}(\rho 0) + \frac{1}{\rho}(0) + 0 = 0$$

We conclude that $\mathbf{D}$ is solenoidal.

Prob. 3.70

$$\begin{aligned} \nabla \times \mathbf{E} &= \frac{1}{r \sin \theta} (0 - 0)\mathbf{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} 0 - 0 \right) \mathbf{a}_\theta + \frac{1}{r} \left[\frac{\partial}{\partial r} (kr^{-2} \sin \theta) - \frac{2k(-\sin \theta)}{r^3} \right] \mathbf{a}_\phi \\ &= 0 + \frac{1}{r} \left[-\frac{2k \sin \theta}{r^3} + \frac{2k \sin \theta}{r^3} \right] \mathbf{a}_\phi = \mathbf{0} \end{aligned}$$

showing that $\mathbf{E}$ is conservative.

CHAPTER 4

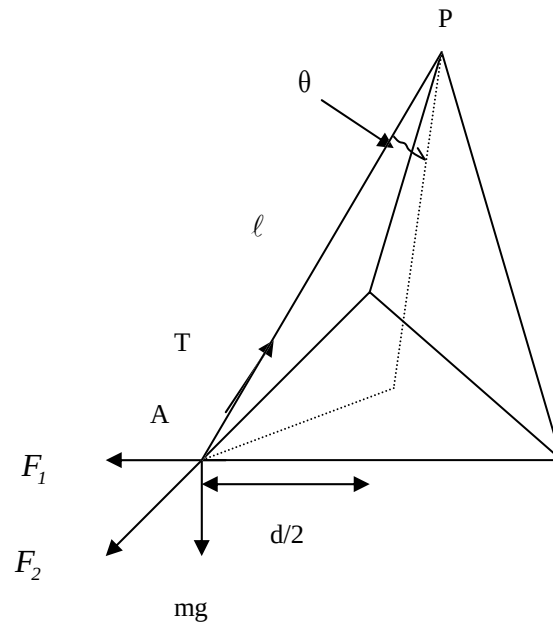
P. E. 4.1

$$\begin{aligned}
 (a) \quad \mathbf{F} &= \frac{1 \times 10^{-9}}{4\pi \left(\frac{10^{-9}}{36\pi} \right)} \left[\frac{5 \times 10^{-9} [(1, -3, 7) - (2, 0, 4)]}{[(1, -3, 7) - (2, 0, 4)]^3} + \frac{(-2 \times 10^{-9}) [(1, -3, 7) - (-3, 0, 5)]}{[(1, -3, 7) - (-3, 0, 5)]^3} \right] \\
 &= \left[\frac{45(-1, -3, 3)}{19^{3/2}} - \frac{18(4, -3, 2)}{29^{3/2}} \right] \text{ nN} \\
 &= \underline{\underline{-1.004\mathbf{a}_x - 1.284\mathbf{a}_y + 1.4\mathbf{a}_z \text{ nN}}}
 \end{aligned}$$

$$(b) \quad \mathbf{E} = \frac{\mathbf{F}}{Q} = \underline{\underline{-1.004\mathbf{a}_x - 1.284\mathbf{a}_y + 1.4\mathbf{a}_z \text{ V/m}}}$$

P. E. 4.2

Let q be the charge on each sphere, i.e. $q=Q/3$. The free body diagram below helps us to establish the relationship between various forces.



At point A,

$$\begin{aligned}
 T \sin \theta \cos 30^\circ &= F_1 + F_2 \cos 60^\circ \\
 &= \frac{q^2}{4\pi\epsilon_0 d^2} + \frac{q^2}{4\pi\epsilon_0 d^2} \left(\frac{1}{2}\right) \\
 &= \frac{3q^2}{8\pi\epsilon_0 d^2}
 \end{aligned}$$

$$T \cos \theta = mg$$

$$\text{Hence, } \tan \theta \cos 30^\circ = \frac{3q^2}{8\pi\epsilon_0 d^2 mg}$$

$$\text{But } \sin \theta = \frac{h}{l} = \frac{d}{\sqrt{3} l} \tan \theta = \frac{\frac{d}{\sqrt{3}}}{\sqrt{l^2 - \frac{d^2}{3}}}$$

$$\text{Thus, } \frac{\frac{d}{\sqrt{3}} \left(\frac{\sqrt{3}}{2}\right)}{\sqrt{l^2 - \frac{d^2}{3}}} = \frac{3q^2}{8\pi\epsilon_0 d^2 mg}$$

$$\text{or } q^2 = \frac{4\pi\epsilon_0 d^3 mg}{3\sqrt{l^2 - \frac{d^2}{3}}}$$

$$\text{but } q = \frac{Q}{3} \longrightarrow q^2 = \frac{Q^2}{9}. \text{ Hence,}$$

$$Q^2 = \frac{12\pi\epsilon_0 d^3 mg}{\sqrt{l^2 - \frac{d^2}{3}}}$$

P.E. 4.3

$$e\bar{E} = m \frac{d^2 \bar{l}}{dt^2}$$

$$eE_0(-2\bar{a}_x + \bar{a}_y) = m\left(\frac{d^2 x}{dt^2}\bar{a}_x + \frac{d^2 y}{dt^2}\bar{a}_y + \frac{d^2 z}{dt^2}\bar{a}_z\right)$$

where $E_0 = 200 \text{ kV/m}$

$$\frac{d^2 z}{dt^2} = 0 \quad \longrightarrow \quad z = ct + c_2$$

$$m \frac{d^2 x}{dt^2} = -2eE_0 \quad \longrightarrow \quad x = \frac{-2eE_0 t^2}{2m} + c_3 t + c_4$$

$$m \frac{d^2 y}{dt^2} = eE_0 \quad \longrightarrow \quad y = \frac{eE_0 t^2}{2m} + c_5 t + c_6$$

At $t = 0$, $(x, y, z) = (0, 0, 0)$ $c_1 = 0 = c_4 = c_6$

Also, $\left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right) = (0, 0, 0)$

At $t = 0 \quad \longrightarrow \quad c_1 = 0 = c_3 = c_5$

Hence, $(x, y) = \frac{eE_0 t^2}{2m} (-2, 1)$

i.e. $2|y| = |x|$

Thus the largest value of is

$$80 \text{ cm} = \underline{\underline{0.8 \text{ m}}}$$

P.E. 4.4

(a)

Consider an element of area dS of the disk.

The contribution due to $dS = \rho d\phi d\rho$ is

$$dE = \frac{\rho_s dS}{4\pi\epsilon_0 r^2} = \frac{\rho_s dS}{4\pi\epsilon_0 (\rho^2 + h^2)}$$

The sum of the contribution along ρ gives zero.

$$\begin{aligned} E_z &= \frac{\rho_s}{4\pi\epsilon_0} \int_{\rho=0}^a \int_{\phi=0}^{2\pi} \frac{h\rho d\rho d\phi}{(\rho^2 + h^2)^{3/2}} = \frac{h\rho_s}{2\epsilon_0} \int_{\rho=0}^a \frac{\rho d\rho}{(\rho^2 + h^2)^{3/2}} \\ &= \frac{h\rho_s}{4\epsilon_0} \int_0^a (\rho^2 + h^2)^{-3/2} d(\rho^2) = \frac{h\rho_s}{4\epsilon_0} (-2(\rho^2 + h^2)^{-1/2}) \Big|_0^a \\ &= \frac{\rho_s}{2\epsilon_0} \left[1 - \frac{h}{(h^2 + a^2)^{1/2}} \right] \end{aligned}$$

(b)

As $a \longrightarrow \infty$,

$$\underline{\underline{E = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_z}}}$$

(c) Let us recall that if $a/h \ll 1$ then $(1+a/h)^n$ can be approximated by $(1+na/h)$. Thus the expression for E_z from (a) can be modified for $a \ll h$ as follows.

$$\begin{aligned} E_z &= \frac{\rho_s}{2\epsilon_0} \left[1 - \frac{1}{\sqrt{1 + \frac{a^2}{h^2}}} \right] = \frac{\rho_s}{2\epsilon_0} \left[1 - \left(1 + \frac{a^2}{h^2} \right)^{-\frac{1}{2}} \right] \xrightarrow{a \rightarrow 0, \text{ but } \rho_s \pi a^2 = Q} \frac{\rho_s}{2\epsilon_0} \left[\frac{a^2}{2h^2} \right] \\ &= \frac{\rho_s}{2\epsilon_0} \left[\frac{\pi a^2}{2\pi h^2} \right] = \frac{Q}{4\pi\epsilon_0 h^2} \end{aligned}$$

This is in keeping with original Coulomb's law.

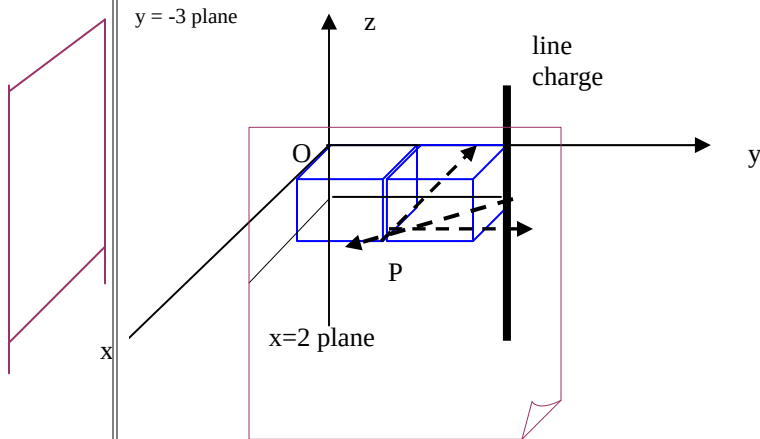
P. E. 4.5

$$\begin{aligned} Q_s &= \int \rho_s dS = \int_{-2}^2 \int_{-2}^2 12|y| dx dy \\ &= 12(4) \int_0^2 2y dy = \underline{\underline{192 \text{ mC}}} \end{aligned}$$

$$\mathbf{E} = \int \frac{\rho_s dS}{4\pi\epsilon r^2} \mathbf{a}_r = \int \frac{\rho_s dS}{4\pi\epsilon_0} \frac{|\mathbf{r} - \mathbf{r}'|}{|\mathbf{r} - \mathbf{r}'|^3}$$

where $\mathbf{r} - \mathbf{r}' = (0, 0, 10) - (x, y, z) = (-x, -y, 10)$.

$$\begin{aligned}
 \mathbf{E} &= \int_{x=-2}^2 \int_{y=-2}^2 \frac{12 |y| 10^{-3} (-x, -y, 10)}{4\pi \left(\frac{10^{-9}}{36\pi}\right) (x^2 + y^2 + 100)^{3/2}} \\
 &= 108(10^6) \left[\int_{-2}^2 |y| \int_{-2}^2 \frac{-x dx dy \mathbf{a}_x}{(x^2 + y^2 + 100)^{3/2}} + \int_{-2}^2 \int_{-2}^2 \frac{-y |y| dy dx \mathbf{a}_y}{(x^2 + y^2 + 100)^{3/2}} \right. \\
 &\quad \left. + 10 \mathbf{a}_z \int_{-2}^2 \int_{-2}^2 \frac{|y| dx dy}{(x^2 + y^2 + 100)^{3/2}} \right] \\
 \mathbf{E} &= 108(10^7) \mathbf{a}_z \int_{-2}^2 \left[2 \int_0^2 \frac{\frac{1}{2} d(y^2)}{(x^2 + y^2 + 100)^{3/2}} \right] dx \\
 &= -216(10^7) \mathbf{a}_z \int_{-2}^2 \left[\frac{1}{(x^2 + 104)^{1/2}} - \frac{1}{(x^2 + 100)^{1/2}} \right] dx \\
 &= -216(10^7) \mathbf{a}_z \ln \left| \frac{x + \sqrt{x^2 + 104}}{x + \sqrt{x^2 + 100}} \right|_{-2}^2 \\
 &= -216(10^7) \mathbf{a}_z \left(\ln \left(\frac{2 + \sqrt{108}}{2 + \sqrt{104}} \right) - \ln \left(\frac{-2 + \sqrt{108}}{-2 + \sqrt{104}} \right) \right) \\
 &= -216(10^7) \mathbf{a}_z (-7.6202 (10^{-3})) \\
 \mathbf{E} &= \underline{\underline{16.46 \mathbf{a}_z \text{ MV/m}}}
 \end{aligned}$$

P.E. 4.6

E_1 and E_2 remain the same as in Example 4.6.

$$\mathbf{E}_3 = \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho$$

This expression, which represents the field due to a line charge,

is modified as follows. To get $\mathbf{a}_\rho$, consider the $z = -1$ plane. $\rho = \sqrt{2}$

$$\mathbf{a}_\rho = \mathbf{a}_x \cos 45^\circ - \mathbf{a}_y \sin 45^\circ$$

$$= \frac{1}{\sqrt{2}}(\mathbf{a}_x - \mathbf{a}_y)$$

$$\mathbf{E}_3 = \frac{10(10^{-9})}{2\pi\left(\frac{10^{-9}}{36\pi}\right)} \frac{1}{2}(\mathbf{a}_x - \mathbf{a}_y)$$

$$= 90\pi(\mathbf{a}_x - \mathbf{a}_y). \quad \text{Hence,}$$

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 \\ &= -180\pi\mathbf{a}_x + 270\pi\mathbf{a}_y + 90\pi\mathbf{a}_x - 90\pi\mathbf{a}_y \\ &= \underline{\underline{-282.7\mathbf{a}_x + 565.5\mathbf{a}_y \text{ V/m}}} \end{aligned}$$

P.E. 4.7

$$\begin{aligned} \mathbf{D} &= \mathbf{D}_Q + \mathbf{D}_\rho = \frac{Q}{4\pi r^2} \mathbf{a}_r + \frac{\rho_s}{2} \mathbf{a}_n \\ &= \frac{30 \times 10^{-9}}{4\pi(5)^2} \frac{[(0, 4, 3) - (0, 0, 0)]}{5} + \frac{10 \times 10^{-9}}{2} \mathbf{a}_y \\ &= \frac{30}{500\pi} (0, 4, 3) + 5\mathbf{a}_y \text{ nC/m}^2 \\ &= \underline{\underline{5.076\mathbf{a}_y + 0.0573\mathbf{a}_z \text{ nC/m}^2}} \end{aligned}$$

P.E. 4.8

$$(a) \quad \rho_v = \nabla \cdot \mathbf{D} = 4x$$

$$\rho_v(-1, 0, 3) = \underline{\underline{-4 \text{ C/m}^3}}$$

$$\begin{aligned} (b) \quad \Psi &= \iiint \mathbf{D} \cdot d\mathbf{S} = \iint \mathbf{D} \cdot d\mathbf{S} \Big|_{x=0} + \iint \mathbf{D} \cdot d\mathbf{S} \Big|_{x=1} + \iint \mathbf{D} \cdot d\mathbf{S} \Big|_{y=0} + \iint \mathbf{D} \cdot d\mathbf{S} \Big|_{y=1} + \iint \mathbf{D} \cdot d\mathbf{S} \Big|_{z=0} + \iint \mathbf{D} \cdot d\mathbf{S} \Big|_{z=1} \\ &= - \int_0^1 \int_0^1 (2y^2 + z) dy dz + \int_0^1 \int_0^1 (2y^2 + z) dy dz + \int_0^1 \int_0^1 4x(1) dx dz - \int_0^1 \int_0^1 (x) dx dz + \int_0^1 \int_0^1 (x) dx dz \\ &= 4 \int_0^1 \int_0^1 x dx dz = 4(1/2)(1) = 2C \end{aligned}$$

$$\begin{aligned} (c) \quad \Psi = Q &= \int \rho_v dv = \int_0^1 \int_0^1 \int_0^1 4x dx dy dz \\ &= 4(1)(1)(1/2) = \underline{\underline{2C}} \end{aligned}$$

$$Q = \Psi = \underline{\underline{2C}}$$

P.E. 4.9

$$Q = \int \rho_v dv = \Psi = \iiint \mathbf{D} \cdot d\mathbf{S}$$

For $0 \leq r \leq 10$,

$$D_r(4\pi r^2) = \iiint 2r(r^2) \sin \theta d\theta dr d\phi$$

$$D_r(4\pi r^2) = 4\pi \left(\frac{2r^4}{4} \Big|_0^r \right) = 2\pi r^4$$

$$D_r = \frac{r^2}{2} \quad \mathbf{E} = \frac{r^2}{2\epsilon_0} \mathbf{a}_r \text{ nV/m}$$

$$\mathbf{E}(r=2) = \frac{4(10^{-9})}{2(\frac{10^{-9}}{36\pi})} \mathbf{a}_r = 72\pi \mathbf{a}_r = \underline{\underline{226 \mathbf{a}_r \text{ V/m}}}$$

For $r \geq 10$,

$$D_r(4\pi r^2) = 2\pi r_0^4, \quad r_0 = 10\text{m}$$

$$D_r = \frac{r_0^4}{2r^2} \longrightarrow \mathbf{E} = \frac{r_0^4}{2\epsilon_0 r^2} \mathbf{a}_r \text{ nV/m}$$

$$\begin{aligned} \mathbf{E}(r=12) &= \frac{10^4(10^{-9})}{2(\frac{10^{-9}}{36\pi})(144)} \mathbf{a}_r = 1250\pi \mathbf{a}_r \\ &= \underline{\underline{3.927 \mathbf{a}_r \text{ kV/m}}} \end{aligned}$$

P. E. 4.10

$$V(\mathbf{r}) = \sum_{k=1}^3 \frac{Q_k}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_k|} + C$$

$$\text{At } V(\infty) = 0, \quad C = 0$$

$$|\mathbf{r} - \mathbf{r}_1| = |(-1, 5, 2) - (2, -1, 3)| = \sqrt{46}$$

$$|\mathbf{r} - \mathbf{r}_2| = |(-1, 5, 2) - (0, 4, -2)| = \sqrt{18}$$

$$|\mathbf{r} - \mathbf{r}_3| = |(-1, 5, 2) - (0, 0, 0)| = \sqrt{30}$$

$$\begin{aligned} V(-1, 5, 2) &= \frac{10^{-6}}{4\pi\left(\frac{10^{-9}}{36\pi}\right)} \left[\frac{-4}{\sqrt{46}} + \frac{5}{\sqrt{18}} + \frac{3}{\sqrt{30}} \right] \\ &= \underline{\underline{10.23 \text{ kV}}} \end{aligned}$$

P.E. 4.11

$$V = \frac{Q}{4\pi\epsilon_0 r} + C$$

$$\text{If } V(0, 6, -8) = V(r = 10) = 2;$$

$$2 = \frac{5(10^{-9})}{4\pi\left(\frac{10^{-9}}{36\pi}\right)(10)} + C \quad \longrightarrow \quad C = -2.5$$

(a)

$$\begin{aligned} V_A &= \frac{5(10^{-9})}{4\pi\left(\frac{10^{-9}}{36\pi}\right)|(-3, 2, 6) - (0, 0, 0)|} - 2.5 \\ &= \underline{\underline{3.929 \text{ V}}} \end{aligned}$$

(b)

$$V_B = \frac{45}{\sqrt{7^2 + 1^2 + 5^2}} - 2.5 = \underline{\underline{2.696 \text{ V}}}$$

$$(c) \quad V_{AB} = V_B - V_A = 2.696 - 3.929 = - \underline{\underline{1.233 \text{ V}}}$$

P.E. 4.12

(a)

$$\begin{aligned}
 -\frac{W}{Q} &= \int \mathbf{E} \cdot d\mathbf{l} = \int (3x^2 + y)dx + xdy \\
 &= \int_0^2 (3x^2 + y)dx \Big|_{y=5} + \int_5^{-1} xdy \Big|_{x=2} \\
 &= 18 - 12 = 6 \text{ kV} \\
 W &= -6Q = \underline{\underline{12 \text{ mJ}}}
 \end{aligned}$$

(b)

$$\begin{aligned}
 dy &= -3dx \\
 -\frac{W}{Q} &= \int \mathbf{E} \cdot d\mathbf{l} = \int_0^2 (3x^2 + 5 - 3x)dx + x(-3)dx \\
 &= \int_0^2 (3x^2 - 6x + 5)dx = 8 - 12 + 10 = 6 \\
 W &= \underline{\underline{12 \text{ mJ}}}
 \end{aligned}$$

P.E. 4.13

(a)

$$\begin{aligned}
 (0, 0, 10) &\longrightarrow (r = 10, \theta = 0, \phi = 0) \\
 V &= \frac{100 \cos 0}{4\pi\epsilon_0(10^2)}(10^{-12}) = \frac{10^{-12}}{4\pi(\frac{10^{-9}}{36\pi})} = \underline{\underline{9 \text{ mV}}} \\
 \mathbf{E} &= \frac{100(10^{-12})}{4\pi(\frac{10^{-9}}{36\pi})10^3} [2 \cos 0 \mathbf{a}_r + \sin 0 \mathbf{a}_\theta] \\
 &= \underline{\underline{1.8 \mathbf{a}_r \text{ mV/m}}}
 \end{aligned}$$

(b)

$$\begin{aligned}
 \text{At } (1, \frac{\pi}{3}, \frac{\pi}{2}), \\
 V &= \frac{100 \cos \frac{\pi}{3} (10^{-12})}{4\pi(\frac{10^{-9}}{36\pi})(1)^2} = \underline{\underline{0.45 \text{ V}}} \\
 \mathbf{E} &= \frac{100(10^{-12})}{4\pi(\frac{10^{-9}}{36\pi})(1)^2} (2 \cos \frac{\pi}{3} \mathbf{a}_r + \sin \frac{\pi}{3} \mathbf{a}_\theta) \\
 &= \underline{\underline{0.9 \mathbf{a}_r + 0.7794 \mathbf{a}_\theta \text{ V/m}}}
 \end{aligned}$$

P.E. 4.14

After Q_1 , $W_1 = 0$

$$\begin{aligned}\text{After } Q_2, W_2 &= Q_2 V_{21} = \frac{Q_2 Q_1}{4\pi\epsilon_0 |(1,0,0) - (0,0,0)|} \\ &= \frac{1(-2)(10^{-18})}{4\pi(10^{-9}) \frac{1}{36\pi}} = \underline{\underline{-18 \text{ nJ}}}\end{aligned}$$

After Q_3 ,

$$\begin{aligned}W_3 &= Q_3(V_{31} + V_{32}) + Q_2 V_{21} \\ &= 3(9)(10^{-9}) \left\{ \frac{1}{|(0,0,-1) - (0,0,0)|} + \frac{-2}{|(0,0,-1) - (1,0,0)|} \right\} - 18 \text{ nJ} \\ &= 27(1 - \frac{2}{\sqrt{2}}) - 18 \\ &= \underline{\underline{-29.18 \text{ nJ}}}\end{aligned}$$

After Q_4 ,

$$\begin{aligned}W_4 &= Q_4(V_{41} + V_{42} + V_{43}) + Q_3(V_{31} + V_{32}) + Q_2 V_{21} \\ &= -4(9)(10^{-9}) \left\{ \frac{1}{|(0,0,1) - (0,0,0)|} + \frac{-2}{|(0,0,1) - (1,0,0)|} + \frac{3}{|(0,0,1) - (0,0,-1)|} \right\} + W_3 \\ &= -36(1 - \frac{2}{\sqrt{2}} + \frac{3}{2}) + W_3 \\ &= -39.09 - 29.18 \text{ nJ} = \underline{\underline{-68.27 \text{ nJ}}}\end{aligned}$$

P.E. 4.15

$$\mathbf{E} = -\nabla V = -(y+1)\mathbf{a}_x + (1-x)\mathbf{a}_y - 2\mathbf{a}_z$$

At (1,2,3), $\mathbf{E} = \underline{\underline{-3\mathbf{a}_x - 2\mathbf{a}_z \text{ V/m}}}$

$$\begin{aligned}W &= \frac{1}{2} \epsilon_o \int \mathbf{E} \bullet \mathbf{E} dv = \frac{1}{2} \epsilon_o \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (x^2 + y^2 - 2x + 2y + 6) dx dy dz \\ &= \frac{1}{2} \epsilon_o \left[\int_{-1}^1 x^2 dx \iint dy dz + \int_{-1}^1 y^2 dy \iint dx dz - 2 \int_{-1}^1 x dx \iint dy dz + 2 \int_{-1}^1 y dy \iint dx dz + 6(2)(2)(2) \right] \\ &= \frac{1}{2} \epsilon_o \left[2 \frac{x^3}{3} \Big|_{-1}^1 (2)(2) + 0 + 0 + 6(8) \right] = \frac{80\epsilon_o}{3} \\ &= \underline{\underline{0.2358 \text{ nJ}}}\end{aligned}$$

Prob. 4.1

$$\begin{aligned}
 \mathbf{F}_{Q_1} &= \frac{Q_1 Q_2 (\mathbf{r}_{Q_1} - \mathbf{r}_{Q_2})}{4\pi \varepsilon_0 |\mathbf{r}_{Q_1} - \mathbf{r}_{Q_2}|^3} = \frac{-20(10^{-12})[(3, 2, 1) - (-4, 0, 6)]}{4\pi \frac{10^{-9}}{36\pi} |(3, 2, 1) - (-4, 0, 6)|^3} = -180 \frac{(7, 2, -5)}{688.88} \times 10^{-3} \\
 &= \underline{\underline{-1.8291 \mathbf{a}_x - 0.5226 \mathbf{a}_y + 1.3065 \mathbf{a}_z \text{ mN}}}
 \end{aligned}$$

Prob. 4.2 (a)

$$\begin{aligned}
 E(5, 0, 6) &= \frac{Q_1}{4\pi \varepsilon_0} \frac{[(5, 0, 6) - (4, 0, -3)]}{|(5, 0, 6) - (4, 0, -3)|^3} + \frac{Q_2}{4\pi \varepsilon_0} \frac{[(5, 0, 6) - (2, 0, 1)]}{|(5, 0, 6) - (2, 0, 1)|^3} \\
 &= \frac{Q_1}{4\pi \varepsilon_0} \frac{(1, 0, 9)}{(\sqrt{82})^3} + \frac{Q_2}{4\pi \varepsilon_0} \frac{(3, 0, 5)}{(34)^{3/2}}
 \end{aligned}$$

If $E_z = 0$, then

$$\begin{aligned}
 \frac{9Q_1}{4\pi \varepsilon_0 (82)^{3/2}} + \frac{5Q_2}{4\pi \varepsilon_0 (34)^{3/2}} &= 0 \\
 Q_1 &= -\frac{5}{9} Q_2 \left(\frac{82}{34}\right)^{3/2} = -\frac{5}{9} 4 \left(\frac{82}{34}\right)^{3/2} \text{ nC} \\
 &= \underline{\underline{-8.3232 \text{ nC}}}
 \end{aligned}$$

(b)

$$\mathbf{F}(5, 0, 6) = q\mathbf{E}(5, 0, 6)$$

If $F_x = 0$, then

$$\begin{aligned}
 \frac{qQ_1}{4\pi \varepsilon_0 (82)^{3/2}} + \frac{3qQ_2}{4\pi \varepsilon_0 (34)^{3/2}} &= 0 \\
 Q_1 &= -3Q_2 \left(\frac{82}{34}\right)^{3/2} = -12 \left(\frac{82}{34}\right)^{3/2} \text{ nC} \\
 Q_1 &= \underline{\underline{-44.945 \text{ nC}}}
 \end{aligned}$$

Prob. 4.3

$$\begin{aligned}
 \text{(a)} \quad E &= \sum_{k=1}^2 \frac{Q(\mathbf{r} - \mathbf{r}_k')}{4\pi\epsilon_o |\mathbf{r} - \mathbf{r}_k'|^3} = \frac{Q[(0,0,0) - (a,0,0)]}{4\pi\epsilon_o |(0,0,0) - (a,0,0)|^3} - \frac{Q[(0,0,0) - (-a,0,0)]}{4\pi\epsilon_o |(0,0,0) - (-a,0,0)|^3} \\
 &= \frac{Q(-a,0,0)}{4\pi\epsilon_o a^3} - \frac{Q(a,0,0)}{4\pi\epsilon_o a^3} = \underline{\underline{\frac{Q}{2\pi\epsilon_o a^2} \mathbf{a}_x}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad E &= \frac{Q[(0,a,0) - (a,0,0)]}{4\pi\epsilon_o |(0,a,0) - (a,0,0)|^3} - \frac{Q[(0,a,0) - (-a,0,0)]}{4\pi\epsilon_o |(0,a,0) - (-a,0,0)|^3} \\
 &= \frac{Q(-a,a,0)}{4\pi\epsilon_o (2a^2)^{3/2}} - \frac{Q(a,a,0)}{4\pi\epsilon_o (2a^2)^{3/2}} = \underline{\underline{\frac{-Q}{4\sqrt{2}\pi\epsilon_o a^2} \mathbf{a}_x}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad E &= \frac{Q[(a,0,a) - (a,0,0)]}{4\pi\epsilon_o |(a,0,a) - (a,0,0)|^3} - \frac{Q[(a,0,a) - (-a,0,0)]}{4\pi\epsilon_o |(a,0,a) - (-a,0,0)|^3} \\
 &= \frac{Q(0,0,a)}{4\pi\epsilon_o a^3} - \frac{Q(2a,0,a)}{4\pi\epsilon_o (5a^2)^{3/2}} = \underline{\underline{\frac{-Q}{10\sqrt{5}\pi\epsilon_o a^2} \mathbf{a}_x + \frac{Q}{4\pi\epsilon_o a^2} \left[1 - \frac{1}{5\sqrt{5}}\right] \mathbf{a}_z}}
 \end{aligned}$$

Prob. 4.4

$$F = qE = mg \quad \longrightarrow \quad E = \frac{mg}{q} = \frac{2 \times 9.8}{4 \times 10^{-3}} = \underline{\underline{4.9 \text{ kV/m}}}$$

Prob. 4.5

$$\text{(a)} \quad Q = \int \rho_L dl = \int_0^5 12x^2 dx = 4x^3 \Big|_0^5 = \underline{\underline{0.5 C}}$$

$$\begin{aligned}
 \text{(b)} \quad Q &= \int \rho_S dS = \int_{z=0}^4 \int_{\phi=0}^{2\pi} \rho z^2 \rho d\phi dz \Big|_{\rho=3} = 9(2\pi) \frac{z^3}{3} \Big|_0^4 \text{ nC} \\
 &= \underline{\underline{1.206 \mu C}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad Q &= \int \rho_V dV = \iiint \frac{10}{r \sin \theta} r^2 \sin \theta d\theta d\phi dr \\
 &= 10 \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \int_0^4 r dr = 10(2\pi)(\pi) \frac{4^2}{2} \\
 &= \underline{\underline{1579.1 C}}
 \end{aligned}$$

Prob. 4.6

$$Q = \int_V \rho_v dv = \int_0^a \int_0^a \int_0^a \frac{\rho_o x}{a} dx dy dz = (a)(a)\rho_o \left(\frac{x^2}{2a} \Big|_0^a \right) = \underline{\underline{\frac{a^3 \rho_o}{2}}}$$

Prob. 4.7

$$Q = \int_V \rho_v dv = \int_{\rho=0}^2 \int_{z=0}^1 \int_{\phi=\pi/6}^{\pi/2} 5\rho^2 z \rho d\phi d\rho dz \text{ mC}$$

$$= 5 \frac{\rho^4}{4} \Big|_0^2 \frac{z^2}{2} \Big|_0^1 \phi \Big|_{\pi/6}^{\pi/2} = \frac{5}{8} (16)(1)(\pi/2 - \pi/6) = \frac{10\pi}{3}$$

$$Q = \underline{\underline{10.472 \text{ mC}}}$$

Prob. 4.8

$$Q = \int \rho_s dS = \int 6xy dx dy$$

$$= \int_{x=0}^2 \int_{y=0}^x 6xy dx dy + \int_{x=2}^4 \int_{y=2}^{-x+4} 6xy dx dy$$

$$= 6 \int_{x=0}^2 x \frac{y^2}{2} \Big|_0^x dx + \int_{x=2}^4 6x \frac{y^2}{2} \Big|_0^{-x+4} dx$$

$$= 6 \int_{x=0}^2 x \left(\frac{x^2}{2} - 0 \right) dx + 3 \int_{x=2}^4 x \left[(4-x)^2 - 0 \right] dx$$

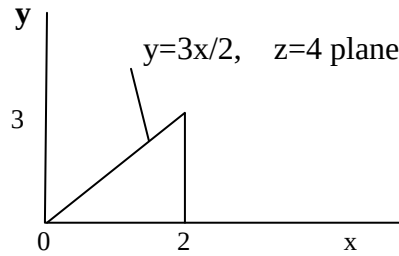
$$= 6 \int_0^2 \frac{x^3}{2} dx + 3 \int_2^4 (16x - 8x^2 + x^3) dx$$

$$= 3 \frac{x^4}{4} \Big|_0^2 + 6 \left(8x^2 - \frac{8x^3}{3} + \frac{x^4}{4} \right) \Big|_2^4$$

$$= 12 + 3 \left(128 - 32 - \frac{512}{3} + \frac{64}{3} + 64 - 4 \right)$$

$$= 12 + 3 \left(96 - \frac{448}{3} + 60 \right)$$

$$Q = \underline{\underline{32 \text{ C}}}$$

Prob. 4.9

$$\begin{aligned}
 Q &= \int_S \rho_s dS = \iint 10x^2 yz dx dy \Big|_{z=4} \text{ mC} = 10(4) \int_{x=0}^2 \int_{y=0}^{3x/2} x^2 y dy dx \\
 &= 40 \int_{x=0}^2 x^2 \frac{y^2}{2} \Big|_0^{3x/2} dx = 20 \int_{x=0}^2 x^2 \left(\frac{9}{4} x^2 \right) dx = 20(9/4) \frac{x^5}{5} \Big|_0^2 = 9(32) = \underline{\underline{288 \text{ mC}}}
 \end{aligned}$$

Prob. 4.10

$$\begin{aligned}
 Q &= \int_V \rho_v dv = \iiint 4\rho^2 z \cos \phi \rho d\rho d\phi dz \quad \text{nC} \\
 &= 4 \int_0^2 \rho^3 d\rho \int_0^1 z dz \int_0^{\pi/4} \cos \phi d\phi = \rho^4 \Big|_0^2 \frac{z^2}{2} \Big|_0^1 (\sin \phi) \Big|_0^{\pi/4} \\
 &= (16)(0.5)(\sin \pi/4) = \underline{\underline{5.657 \text{ nC}}}
 \end{aligned}$$

Prob. 4.11

$$(a) \quad Q = \int_L \rho_L dl = \int_0^1 12x^2 dx = \frac{12x^3}{3} \Big|_0^1 = \underline{\underline{4 \text{ nC}}}$$

(b) Method 1:

$$E = \int \frac{\rho_L dl}{4\pi\epsilon r^3} \mathbf{r}$$

$$\mathbf{r} = (0, 0, h) - (x, 0, 0) = (-x, 0, h), \quad r = |\mathbf{r}| = \sqrt{h^2 + x^2}$$

$$E = \frac{1}{4\pi\epsilon} \int \frac{12x^2(-x, 0, h)dx}{(h^2 + x^2)^{3/2}}$$

Since $x \ll h$, we can ignore the x-component.

$$E = \frac{h}{4\pi\epsilon} \mathbf{a}_z \int \frac{12x^2 dx}{h^3} = \frac{Q}{4\pi\epsilon h^2} \mathbf{a}_z$$

Method 2:

The line charge can be regarded as a point charge because it is very far from the observation point.

$$\mathbf{E} = \frac{Q}{4\pi\epsilon r^2} \mathbf{a}_r = \frac{4 \times 10^{-9}}{4\pi \times \frac{10^{-9}}{36\pi} (1000)^2} \mathbf{a}_z = \frac{4(9)}{10^6} \mathbf{a}_z = \underline{\underline{3.6 \times 10^{-5} \mathbf{a}_z \text{ V/m}}}$$

Prob. 4.12

$$\mathbf{E} = \int_L \frac{\rho_L dl}{4\pi\epsilon_o R^2} \mathbf{a}_R$$

$$\mathbf{R} = -a\mathbf{a}_\rho + h\mathbf{a}_z, R = |\mathbf{R}| = \sqrt{a^2 + h^2}, dl = a d\phi$$

$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_o} \int \frac{(-a\mathbf{a}_\rho + h\mathbf{a}_z)}{(a^2 + h^2)^{3/2}} a d\phi$$

Due to symmetry, the ρ -component cancels

$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_o} \int_0^{2\pi} \frac{ha\mathbf{a}_z d\phi}{(a^2 + h^2)^{3/2}} = \frac{\rho_L ha\mathbf{a}_z}{4\pi\epsilon_o (a^2 + h^2)^{3/2}} (2\pi)$$

out.

$$\mathbf{F} = Q\mathbf{E} = \frac{4 \times 10^{-3} \times 12 \times 10^{-6} \times 4 \times 3\mathbf{a}_z}{4\pi \times \frac{10^{-9}}{36\pi} \times 5^3} (2\pi) = \underline{\underline{260.58\mathbf{a}_z \text{ N}}}$$

Prob. 4.13

$$\mathbf{E} = \int_S \frac{\rho_s dS}{4\pi\epsilon_o R^3} \mathbf{R}, \quad \mathbf{R} = \rho(-\mathbf{a}_\rho) + h\mathbf{a}_z, dS = \rho d\phi d\rho$$

$$\mathbf{E} = \frac{\rho_s}{4\pi\epsilon_o} \int_S \frac{\rho d\phi d\rho}{(\rho^2 + h^2)^{3/2}} (-\rho\mathbf{a}_\rho + h\mathbf{a}_z)$$

Due to symmetry, the ρ -component vanishes.

$$\mathbf{E} = \frac{\rho_s h\mathbf{a}_z}{4\pi\epsilon_o} \int_a^b \rho(\rho^2 + h^2)^{-3/2} d\rho \int_0^{2\pi} d\phi$$

$$\text{Let } u = \rho^2 + h^2, du = 2\rho d\rho$$

$$\mathbf{E} = \frac{\rho_s h\mathbf{a}_z}{4\pi\epsilon_o} (2\pi) \int \frac{1}{2} u^{-3/2} du = \frac{\rho_s h\mathbf{a}_z}{2\epsilon_o} \frac{1/2 u^{-1/2}}{-1/2} = \frac{\rho_s h\mathbf{a}_z}{2\epsilon_o} \left(-\frac{1}{\sqrt{\rho^2 + h^2}} \Big|_a^b \right)$$

$$\mathbf{E} = \underline{\underline{\frac{\rho_s h}{4\pi\epsilon_o} \left[\frac{1}{\sqrt{a^2 + h^2}} - \frac{1}{\sqrt{b^2 + h^2}} \right] \mathbf{a}_z}}$$

Prob. 4.14

(a) From eq. (4.26),

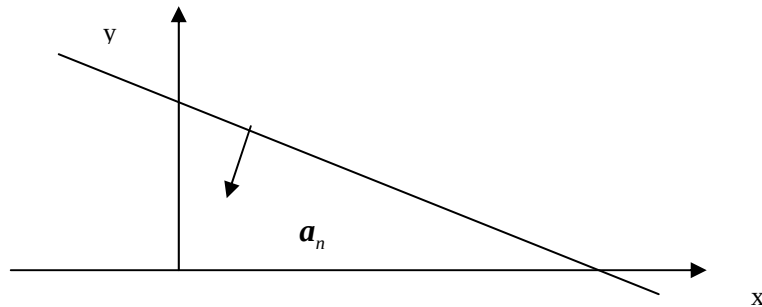
$$\mathbf{E} = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_n = \frac{12 \times 10^{-9}}{2 \times \frac{10^{-9}}{36\pi}} \mathbf{a}_n = 18\pi(12) \mathbf{a}_n = \begin{cases} 678.58 \mathbf{a}_z \text{ V/m, } z > 0 \\ -678.58 \mathbf{a}_z \text{ V/m, } z < 0 \end{cases}$$

(b) Similarly,

$$\mathbf{E} = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_n = \frac{12 \times 10^{-9}}{2 \times \frac{10^{-9}}{36\pi}} \mathbf{a}_n = 18\pi(12) \mathbf{a}_n = \begin{cases} -678.58 \mathbf{a}_z \text{ V/m, } z > 4 \\ 678.58 \mathbf{a}_z \text{ V/m, } z < 4 \end{cases}$$

For $z > 4$ and $z < 0$, the fields cancel out. For $0 < z < 4$, they add up. Thus

$$\mathbf{E} = 2 \times 678.58 \mathbf{a}_z \text{ V/m} = \underline{\underline{1.357 \mathbf{a}_z \text{ kV/m}}}$$

Prob. 4.15

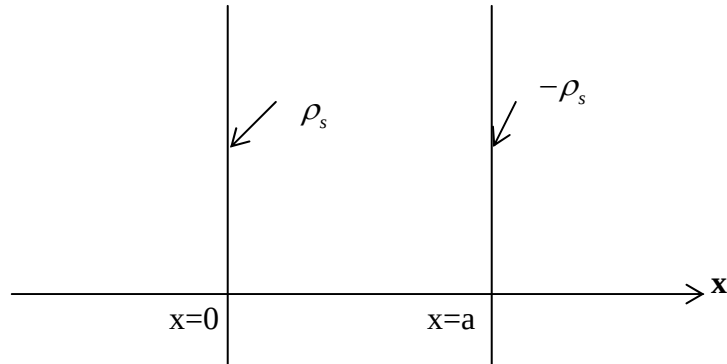
Let $f(x, y) = x + 2y - 5$; $\nabla f = \mathbf{a}_x + 2\mathbf{a}_y$

$$\mathbf{a}_n = \pm \frac{\nabla f}{|\nabla f|} = \pm \frac{(\mathbf{a}_x + 2\mathbf{a}_y)}{\sqrt{5}}$$

Since point $(-1, 0, 1)$ is below the plane,

$$\mathbf{a}_n = -\frac{(\mathbf{a}_x + 2\mathbf{a}_y)}{\sqrt{5}}.$$

$$\begin{aligned} \mathbf{E} &= \frac{\rho_s}{2\epsilon_0} \mathbf{a}_n = \frac{6(10^{-9})}{2(10^{-9}/36\pi)} \left(-\frac{(\mathbf{a}_x + 2\mathbf{a}_y)}{\sqrt{5}} \right) \\ &= \underline{\underline{-151.7 \mathbf{a}_x - 303.5 \mathbf{a}_y \text{ V/m}}} \end{aligned}$$

Prob. 4. 16

(a) For $x < 0$,

$$\underline{\underline{E = E_1 + E_2 = \frac{\rho_s}{2\epsilon_0}(-a_x) + \frac{(-\rho_s)}{2\epsilon_0}(-a_x) = 0}}$$

(b) For $0 < x < a$,

$$\underline{\underline{E = \frac{\rho_s}{2\epsilon_0}a_x + \frac{(-\rho_s)}{2\epsilon_0}(-a_x) = \frac{\rho_s}{\epsilon_0}a_x}}$$

(c) For $x > a$,

$$\underline{\underline{E = \frac{\rho_s}{2\epsilon_0}a_x + \frac{(-\rho_s)}{2\epsilon_0}(a_x) = 0}}$$

Prob. 4.17

(a) At P(5,-1,4),

$$\begin{aligned} E &= \sum_{k=1}^3 \frac{\rho_{sk}}{2\epsilon_0} a_{nk} = \frac{10 \times 10^{-6}}{2 \times \frac{10^{-9}}{36\pi}} (a_x) + \frac{-20 \times 10^{-6}}{2 \times \frac{10^{-9}}{36\pi}} (a_y) + \frac{30 \times 10^{-6}}{2 \times \frac{10^{-9}}{36\pi}} (-a_z) \\ &= 36\pi(5, -10, -15) \times 10^3 = \underline{\underline{565.5a_x - 1131a_y - 1696.5a_z \text{ kV/m}}} \end{aligned}$$

(b) At R(0,-2,1)

$$\underline{\underline{E = 36\pi[5(-a_x) - 10(a_y) + 15(-a_z)] \times 10^3 = -565.5a_x - 1131a_y - 1696.5a_z \text{ kV/m}}}$$

(c) At Q(3,-4,10),

$$\underline{\underline{E = 36\pi[5a_x - 10(-a_y) + 15a_z] \times 10^3 = 565.5a_x + 1131a_y + 1696.5a_z \text{ kV/m}}}$$

Prob. 4.18

$$\mathbf{F}_e = \frac{e^2}{4\pi\epsilon_0 r^2} \mathbf{a}_r$$

$$\begin{aligned} \frac{F_e}{F_g} &= \frac{e^2}{4\pi\epsilon_0} \frac{1}{Gm^2} = \frac{1}{4\pi \times 10^{-9} / 36\pi \times 6.67 \times 10^{-11}} \left(\frac{1.6 \times 10^{-19}}{9.1 \times 10^{-31}} \right)^2 \\ &= \underline{\underline{4.17 \times 10^{42}}} \end{aligned}$$

Prob. 4.19

Let Q_1 be located at the origin. At the spherical surface of radius r ,

$$Q_1 = \oint \mathbf{D} \cdot d\mathbf{S} = \epsilon E_r (4\pi r^2)$$

Or

$$\mathbf{E} = \frac{Q_1}{4\pi\epsilon r^2} \mathbf{a}_r \quad \text{by Gauss's law}$$

If a second charge Q_2 is placed on the spherical surface, Q_2 experiences a force

$$\mathbf{F} = Q_2 \mathbf{E} = \frac{Q_1 Q_2}{4\pi\epsilon r^2} \mathbf{a}_r$$

which is Coulomb's law.

Prob. 4.20

For a point charge, $\mathbf{D} = \frac{Q}{4\pi R^3} \mathbf{R}$

For the given three point charges,

$$\mathbf{D} = \frac{1}{4\pi} \left[\frac{Q\mathbf{R}_1}{R_1^3} + \frac{Q\mathbf{R}_2}{R_2^3} - \frac{2Q\mathbf{R}_3}{R_3^3} \right]$$

$$\mathbf{R}_1 = (0, 0) - (-1, 0) = (1, 0), R_1 = 1$$

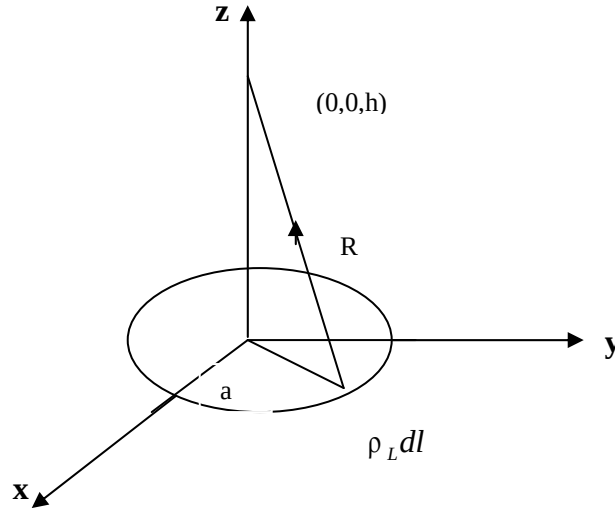
$$\mathbf{R}_2 = (0, 0) - (1, 0) = (-1, 0), R_2 = 1$$

$$\mathbf{R}_3 = (0, 0) - (0, 1) = (0, -1), R_3 = 1$$

$$\mathbf{D} = \frac{Q}{4\pi} [(1, 0) + (-1, 0) - 2(0, -1)] = \frac{Q}{4\pi} (0, 2) = \underline{\underline{\frac{Q}{2\pi} \mathbf{a}_y}}$$

Prob. 4.21

(a) Assume for now that the ring is placed on the $z=0$ plane.



$$\mathbf{D} = \int \frac{\rho_L dl \mathbf{R}}{4\pi R^3}, \mathbf{R} = -a\mathbf{a}_\rho + h\mathbf{a}_z$$

$$\mathbf{D} = \frac{\rho_L}{4\pi} \int_{\phi=0}^{\phi=2\pi} \frac{ad\phi(-a\mathbf{a}_\rho + h\mathbf{a}_z)}{(a^2 + h^2)^{3/2}}$$

Due to symmetry, the ρ component vanishes.

$$\mathbf{D} = \frac{\rho_L a(2\pi h)\mathbf{a}_z}{4\pi(a^2 + h^2)^{3/2}} = \frac{\rho_L a h \mathbf{a}_z}{2(a^2 + h^2)^{3/2}}$$

$$a = 2, \quad h = 3, \quad \rho_L = 5 \mu\text{C/m}$$

Since the ring is actually placed in $x = 0$, $\mathbf{a}_z$ becomes $\mathbf{a}_x$.

$$\mathbf{D} = \frac{(6)(5)\mathbf{a}_x}{2(4+9)^{3/2}} = \underline{\underline{0.32\mathbf{a}_x \mu\text{C/m}^2}}$$

(b)

$$\begin{aligned}\mathbf{D}_Q &= \frac{Q}{4\pi} \frac{[(3,0,0)-(0,-3,0)]}{|(3,0,0)-(0,-3,0)|^3} + \frac{Q}{4\pi} \frac{[(3,0,0)-(0,3,0)]}{|(3,0,0)-(0,3,0)|^3} \\ &= \frac{Q(3,3,0)}{4\pi(18)^{3/2}} + \frac{Q(3,-3,0)}{4\pi(18)^{3/2}} = \frac{6Q(1,0,0)}{4\pi(18)^{3/2}}\end{aligned}$$

$$\mathbf{D} = \mathbf{D}_R + \mathbf{D}_Q = 0$$

$$0.32(10^{-6}) + \frac{6Q}{4\pi(18)^{3/2}} = 0$$

$$\therefore Q = -0.32(4\pi)(18^{3/2})10^{-6} \frac{1}{6} = \underline{\underline{-51.182 \mu C}}$$

Prob. 4.22

$$(a) \quad \rho_v = \nabla \cdot \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = 0 + 2x - 4 = \underline{\underline{2x - 4 \text{ nC/m}^3}}$$

(b)

$$\Psi = \int_S \mathbf{D} \cdot d\mathbf{S} = \iint y^2 dy dz \Big|_{x=3} = \int_0^5 dz \int_0^6 y^2 dy = (5) \frac{y^3}{3} \Big|_0^6 = \frac{5}{3} (6)^3 = \underline{\underline{360 \text{ nC}}}$$

Prob. 4.23

$$\Psi = \int_S \mathbf{D} \cdot d\mathbf{S} = \epsilon_o \int_S \mathbf{E} \cdot d\mathbf{S}, \quad d\mathbf{S} = d\rho dz \mathbf{a}_\phi$$

$$\begin{aligned}\frac{\Psi}{\epsilon_o} &= \int_S \mathbf{E} \cdot d\mathbf{S} = - \iint 6\rho z \sin \phi d\rho dz \Big|_{\phi=90^\circ} \\ &= -6 \sin 90^\circ \int_0^2 \rho d\rho \int_0^5 z dz = -6 \left(\frac{\rho^2}{2} \Big|_0^2 \right) \left(\frac{z^2}{2} \Big|_0^5 \right) = -3(4-0) \frac{1}{2} (25-0) = -150\end{aligned}$$

$$\Psi = -150\epsilon_o = -150 \times \frac{10^{-9}}{36\pi} = \underline{\underline{-1.326 \text{ nC}}}$$

Prob. 4.24

(a)

$$\begin{aligned}\rho_v &= \nabla \cdot \mathbf{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \sin \theta \sin \phi) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \cos \theta \sin \phi) + \frac{1}{r \sin \theta} (-\sin \phi) \\ &= \frac{2}{r} \sin \theta \sin \phi + \frac{\sin \phi}{\sin \theta} (\cos^2 \theta - \sin^2 \theta) - \frac{\sin \phi}{r \sin \theta}\end{aligned}$$

$$\text{But } \cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta$$

$$\rho_v = \frac{2}{r} \sin \theta \sin \phi + \frac{\sin \phi}{\sin \theta} - \frac{2 \sin \theta \sin \phi}{r} - \frac{\sin \phi}{\sin \theta} = 0$$

$$\text{At } (2, 30^\circ, 60^\circ), \quad r = 2, \theta = 30^\circ, \phi = 60^\circ$$

$$\rho_v = \underline{\underline{0}}$$

$$\Psi = \int_S \mathbf{D} \cdot d\mathbf{S}, \quad d\mathbf{S} = r^2 \sin \theta d\theta d\phi \mathbf{a}_r$$

$$\begin{aligned}\Psi &= \int_{\phi=0}^{60^\circ} \int_{\theta=0}^{30^\circ} r^2 \sin^2 \theta \sin \phi d\theta d\phi \Big|_{r=2} \\ &= 4 \int_0^{60^\circ} \sin \phi d\phi \int_0^{30^\circ} \sin^2 \theta d\theta\end{aligned}$$

(b)

$$\text{But } \sin^2 \theta = \frac{1}{2} (1 - \cos 2\theta).$$

$$\begin{aligned}\Psi &= 4 \left(-\cos \phi \Big|_0^{60^\circ} \right) \frac{1}{2} \int_0^{30^\circ} (1 - \cos 2\theta) d\theta = 2(-\cos 60^\circ + 1) \left(\theta - \frac{1}{2} \sin 2\theta \right) \Big|_0^{\pi/6} \\ &= 2(-1/2 + 1) \left(\pi/6 - \frac{1}{2} \sin(\pi/3) - 0 \right) = 0.5236 - 0.433 = 0.0906 \text{ nC} = \underline{\underline{90.6 \text{ pC}}}\end{aligned}$$

Prob. 4.25

$$(a) \quad \rho_v = \nabla \cdot \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \underline{\underline{8y \text{ C/m}^2}}$$

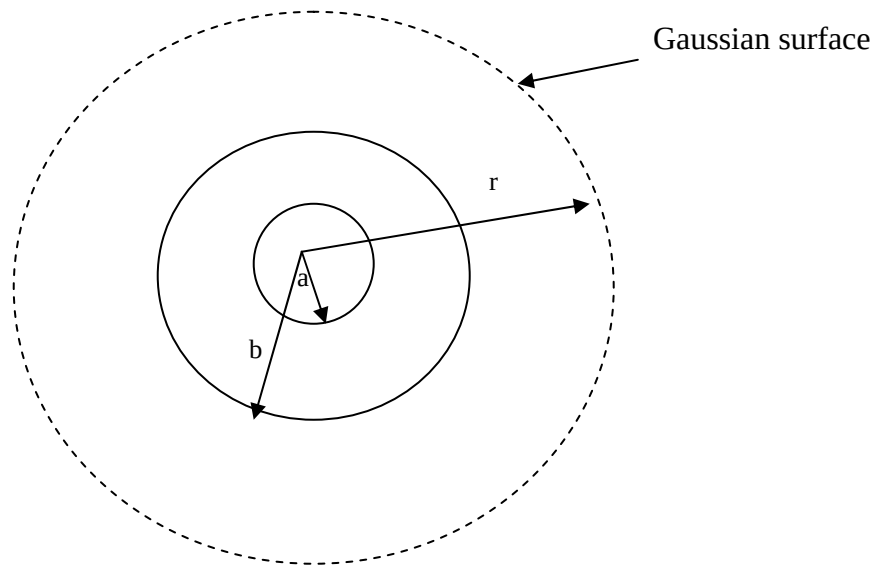
$$\begin{aligned}(b) \quad \rho_v &= \nabla \cdot \mathbf{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z} \\ &= 8 \sin \phi - 2 \sin \phi + 4z \\ &= \underline{\underline{6 \sin \phi + 4z \text{ C/m}^3}}\end{aligned}$$

$$\begin{aligned}
 (c) \quad \rho_v = \nabla \cdot \mathbf{D} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (D_\theta \sin \theta) + 0 \\
 &= -\frac{2}{r^4} \cos \theta + \frac{2 \cos \theta}{r^4} \\
 &= \underline{\underline{0}}
 \end{aligned}$$

Prob. 4.26

$$(a) \quad Q = \int_v \rho_v dv = \int_0^2 \int_0^2 \int_0^2 12xyz dx dy dz = 12 \left(\frac{x^2}{2} \Big|_0^2 \right) \left(\frac{y^2}{2} \Big|_0^2 \right) \left(\frac{z^2}{2} \Big|_0^2 \right) = 12(2)(2)(2) = \underline{\underline{96 \text{ mC}}}$$

$$(b) \quad \Psi = Q = \underline{\underline{96 \text{ mC}}}$$

Prob. 4.27

Apply Gauss's law,

$$\oint \mathbf{D} \cdot d\mathbf{S} = Q_{enc}$$

$$D_r 4\pi r^2 = \rho_{s1} 4\pi a^2 + \rho_{s2} 4\pi b^2 = 8 \times 10^{-9} \times 4\pi (1)^2 + (-6 \times 10^{-3}) \times 4\pi (2)^2 = -0.3016$$

$$D_r = \frac{-0.3016}{4\pi r^2} = \frac{0.3016}{4\pi (3)^2} = -0.0027$$

$$\underline{\underline{\mathbf{D} = -2.7 \mathbf{a}_r \text{ mC/m}^2}}$$

Prob. 4.28

For $r < a$,

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{enc} = \int_V \rho_v dv$$

$$D_r(4\pi r^2) = \iiint 5r^{1/2} r^2 \sin \theta d\theta d\phi dr = 5 \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \int_0^r r^{5/2} dr$$

$$= 5(2)(2\pi) \frac{r^{7/2}}{7/2} \Big|_0^r = \frac{40\pi r^{7/2}}{7}$$

$$D_r = \varepsilon_o E_r = \frac{\frac{40\pi r^{7/2}}{7}}{4\pi r^2} = \frac{10}{7} r^{3/2}$$

$$E_r = \frac{10}{7\varepsilon_o} r^{3/2}, 0 < r < a$$

For $r > a$,

$$D_r(4\pi r^2) = \frac{40}{7} \pi a^{7/2}$$

$$D_r = \varepsilon_o E_r = \frac{\frac{40}{7} \pi a^{7/2}}{4\pi r^2}$$

$$E_r = \frac{10a^{7/2}}{7\varepsilon_o r^2}, r > a$$

Thus,

$$\mathbf{E} = \begin{cases} \frac{10}{7\varepsilon_o} r^{3/2} \mathbf{a}_r, 0 < r < a \\ \frac{10a^{7/2}}{7\varepsilon_o r^2} \mathbf{a}_r, r > a \end{cases}$$

Prob. 4.29

$$(a) \quad \rho_v = \nabla \cdot \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \underline{\underline{2y \text{ C/m}^3}}$$

$$(b) \quad \Psi = \int \mathbf{D} \cdot d\mathbf{S} = \iint x^2 dx dz \Big|_{y=1} = \int_0^1 x^2 dx \int_0^1 dz = \underline{\underline{\frac{1}{3} \text{ C}}}$$

$$(c) \quad Q = \int_V \rho_v dv = \iiint 2y dx dy dz = 2 \int_0^1 dx \int_0^1 y dy \int_0^1 dz = \underline{\underline{1 \text{ C}}}$$

Prob. 4.30

(a)

$$\rho_V = \nabla \cdot \mathbf{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

$$\rho_V = 4(z+1) \cos \phi - (z+1) \cos \phi + 0$$

$$\rho_V = \underline{\underline{3(z+1) \cos \phi \quad \mu\text{C}/\text{m}^3}}$$

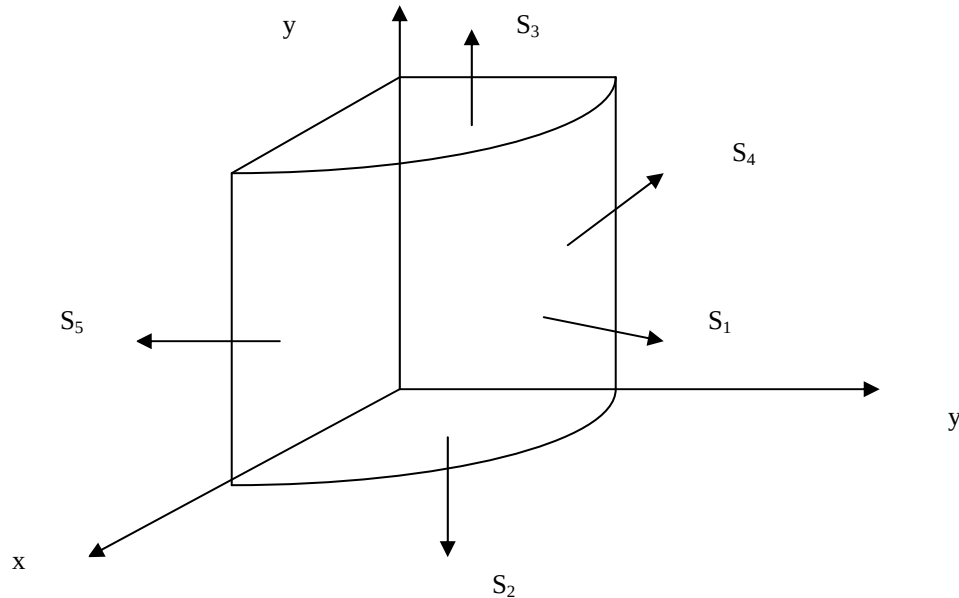
(b)

$$\begin{aligned} Q_{enc} &= \int \rho_V dv = \iiint 3(z+1) \cos \phi \, \rho d\phi \, d\rho \, dz \\ &= 3 \int_0^2 \rho d\rho \int_0^4 (z+1) dz \int_0^{\pi/2} \cos \phi d\phi = 3 \left(\frac{1}{2} \rho^2 \Big|_0^2 \right) \left(\frac{z^2}{2} + z \right) \Big|_0^4 (\sin \phi \Big|_0^{\pi/2}) \\ &= 3(2)(8+4)(1-0) = \underline{\underline{72 \mu\text{C}}} \end{aligned}$$

(c)

$$\text{Let } \psi = \psi_1 + \psi_2 + \psi_3 + \psi_4 + \psi_5 = \oiint \mathbf{D} \cdot d\mathbf{S}$$

where $\psi_1, \psi_2, \psi_3, \psi_4, \psi_5$ respectively correspond with surfaces S_1, S_2, S_3, S_4, S_5 (in the figure below) respectively.



For S_1 $\rho = 2, d\mathbf{S} = \rho d\phi dz \mathbf{a}_\rho$

$$\psi_1 = \iint 2\rho(z+1) \cos \phi dS \Big|_{\rho=2} = 2(2)^2 \int_0^4 (z+1) dz \int_0^{\pi/2} \cos \phi d\phi$$

$$= 8(12)(1) = 96$$

For S_2 , $z = 0, d\mathbf{S} = \rho d\phi d\rho (-\mathbf{a}_z)$

$$\psi_2 = -\iint \rho^2 \cos \phi \rho d\phi d\rho = -\int_0^2 \rho^3 d\rho \int_0^{\pi/2} \cos \phi d\phi$$

$$= -\frac{\rho^4}{4} \Big|_0^2 (1) = -4$$

For S_3 , $z = 4, d\mathbf{S} = \rho d\phi d\rho \mathbf{a}_z$, $\psi_3 = +4$

For S_4 , $\phi = \pi/2, d\mathbf{S} = d\rho dz \mathbf{a}_\phi$

$$\psi_4 = -\iint \rho(z+1) \sin \phi d\rho dz \Big|_{\phi=\pi/2} = -\left(\sin \frac{\pi}{2}\right) \int_0^2 \rho d\rho \int_0^4 (z+1) dz$$

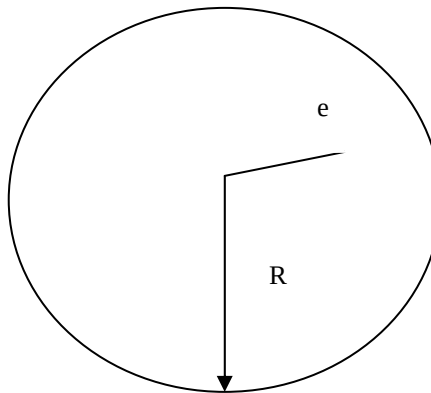
$$= -\frac{\rho^2}{2} \Big|_0^2 (12) = -(2)(12) = -24$$

For S_5 , $\phi = 0, d\mathbf{S} = d\rho dz (-\mathbf{a}_\phi)$, $\psi_5 = \iint \rho(z+1) \sin \phi d\rho dz \Big|_{\phi=0} = 0$

$$\psi = 96 - 4 + 4 - 24 + 0 = \underline{\underline{72\mu C}}$$

This is exactly the answer obtained in part (b).

Prob. 4.31



$$F = eE$$

$$\rho_0 = \frac{e}{4\pi \frac{R^3}{3}} = \frac{3e}{4\pi R^3}$$

$$\rho_V = \begin{cases} \rho_0, & 0 < r < R \\ 0, & \text{elsewhere} \end{cases}$$

$$\oint \mathbf{D} \cdot d\mathbf{S} = Q_{enc} = \int \rho_V dV = \frac{3e}{4\pi R^3} \frac{4\pi r^3}{3} = D_r(4\pi r^2)$$

$$E_r = \frac{3er}{12\pi\epsilon_0 R^3}$$

$$F = eE = \frac{e^2 r}{\underline{\underline{4\pi\epsilon_0 R^3}}}$$

Prob. 4.32

Using Gauss' law, when $\rho < 0$, $\mathbf{E} = \mathbf{0}$

$$\text{For } a < \rho < b, \quad Q_{enc} = \oint_S \mathbf{D} \cdot d\mathbf{S} = \epsilon_o \oint_S \mathbf{E} \cdot d\mathbf{S} = \epsilon_o E_\rho (2\pi\rho L)$$

where L is the length of the cable. But

$$Q_{enc} = \int_S \rho_s dS = \int_{\phi=0}^{2\pi} \int_{z=0}^L \frac{\rho_o}{\rho} \rho d\phi dz = \rho_o 2\pi L$$

$$\rho_o 2\pi L = \epsilon_o E_\rho (2\pi\rho L) \quad \rightarrow \quad E_\rho = \frac{\rho_o}{\epsilon_o \rho}$$

For $\rho > b$, $Q_{enc} = 0$. Thus,

$$\underline{\underline{\mathbf{E} = \begin{cases} \frac{\rho_o}{\epsilon_o \rho} \mathbf{a}_\rho, & a < \rho < b \\ 0, & \text{otherwise} \end{cases}}}$$

Prob. 4.33**(a)**

$$\psi = Q_{enc} \quad \text{at } r = 2$$

$$Q_{enc} = \int \rho_V dV = \iiint \frac{10}{r^2} r^2 \sin \theta d\theta dr d\phi$$

$$= 10 \int_{r=1}^2 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \sin \theta d\theta dr d\phi$$

$$= 10(1)(2\pi)(2) = (40\pi) \text{ mC}$$

$$\text{Thus, } \psi = \underline{\underline{125.7 \text{ mC}}}$$

At $r = 6$;

$$Q_{enc.} = 10 \int_{r=1}^4 dr \int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi} \sin \theta d\theta$$

$$= 10(3)(2\pi)(2) = 120\pi \text{ mC}$$

$$\psi = \underline{\underline{377 \text{ mC}}}$$

(b)

$$\psi = Q_{enc}$$

$$\text{But } \psi = \oint \mathbf{D} \cdot d\mathbf{S} = D_r \oint dS = D_r(4\pi r^2)$$

At $r = 1$,

$$Q_{enc} = 0 \quad \longrightarrow \quad \underline{\underline{\mathbf{D} = 0}}$$

At $r = 5$, $Q_{enc} = 120\pi$

$$D_r = \frac{Q_{enc}}{4\pi r^2} = \frac{120\pi}{4\pi(5)^2} = 1.2$$

$$\mathbf{D} = \underline{\underline{1.2\mathbf{a}_r \text{ mC/m}^2}}$$

Prob. 4.34

$$\rho_v = \frac{Q}{\text{volume}} = \frac{Q}{4\pi a^3/3} = \frac{3Q}{4\pi a^3}$$

$$\text{For } r < a, \quad \oint \mathbf{D} \cdot d\mathbf{S} = Q_{enc} = \int \rho_v dv$$

$$D_r 4\pi r^2 = \frac{3Q}{4\pi a^3} \frac{4\pi r^3}{3} = \frac{Qr^3}{a^3} \quad \longrightarrow \quad D_r = \frac{Qr}{4\pi a^3}$$

For $r > a$, $\oint \mathbf{D} \cdot d\mathbf{S} = Q$

$$D_r 4\pi r^2 = Q \longrightarrow D_r = \frac{Q}{4\pi r^2}$$

Hence,

$$\underline{\underline{\mathbf{D} = \begin{cases} \frac{Qr}{4\pi a^3} \mathbf{a}_r, & r < a \\ \frac{Q}{4\pi r^2} \mathbf{a}_r, & r > a \end{cases}}}$$

Prob. 4.35

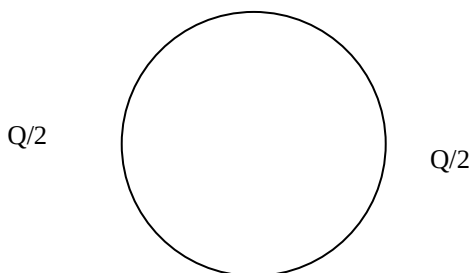
$$\begin{aligned} V_p &= \frac{Q_1}{4\pi\epsilon_0 r_1} + \frac{Q_2}{4\pi\epsilon_0 r_2} = \frac{10^{-9}}{4\pi \times \frac{10^{-9}}{36\pi}} \left[\frac{2}{|(1, -2, 3) - (1, 0, 3)|} - \frac{4}{|(1, -2, 3) - (-2, 1, 5)|} \right] \\ &= 9 \left[\frac{2}{2} - \frac{4}{\sqrt{9+9+4}} \right] = \underline{\underline{1.325 \text{ V}}} \end{aligned}$$

Prob. 4.36

$$V = 4 \frac{Q}{4\pi\epsilon_0 r}, \quad r = \sqrt{a^2 + a^2 + h^2} = \sqrt{2^2 + 2^2 + 3^2} = \sqrt{17} \text{ cm}$$

$$V = \frac{4 \times 8 \times 10^{-9}}{4\pi \times \frac{10^{-9}}{36\pi} \times \sqrt{17} \times 10^{-2}} = \underline{\underline{6.985 \text{ kV}}}$$

Prob. 4.37 (a)



$$\begin{aligned} V &= \frac{2 \frac{Q}{2}}{4\pi\epsilon_0 r} = \frac{Q}{4\pi\epsilon_0 r} \\ &= \frac{60(10^{-6})}{4\pi \times \frac{10^{-9}}{36\pi} \times 4} = \underline{\underline{135 \text{ kV}}} \end{aligned}$$

(b)

$$V = \frac{3 \left(\frac{Q}{3} \right)}{4 \pi \epsilon_0 r} = \underline{\underline{135 \text{ kV}}}$$

(c)

$$V = \int \frac{\rho_L dl}{4 \pi \epsilon_0 r} = \frac{\frac{Q}{8\pi} 2\pi(4)}{4 \pi \epsilon_0 r} = \underline{\underline{135 \text{ kV}}}$$

Prob. 4.38

(a)

$$V_p = \sum \frac{Q_k}{4 \pi |\mathbf{r}_p - \mathbf{r}_k|}$$

$$4 \pi \epsilon_0 V_p = \frac{10^{-3}}{|(-1, 1, 2) - (0, 0, 4)|} + \frac{-2(10^{-3})}{|(-1, 1, 2) - (-2, 5, 1)|} + \frac{3(10^{-3})}{|(-1, 1, 2) - (3, -4, 6)|}$$

$$4 \pi \epsilon_0 (10^3) V_p = \frac{1}{|(-1, 1, -2)|} - \frac{2}{|(1, -4, 1)|} + \frac{3}{|(-4, 5, -4)|} = \frac{1}{\sqrt{6}} - \frac{2}{\sqrt{18}} + \frac{3}{\sqrt{57}}$$

$$4 \pi \frac{10^{-9}}{36 \pi} (10^3) V_p = 0.3542$$

$$\therefore V_p = \underline{\underline{3.008 \times 10^6 \text{ V}}}$$

(b)

$$V_Q = \sum \frac{Q_k}{4 \pi \epsilon_0 |\mathbf{r}_p - \mathbf{r}_k|}$$

$$4 \pi \epsilon_0 V_Q = \frac{10^{-3}}{|(1, 2, 3) - (0, 0, 4)|} + \frac{-2(10^{-3})}{|(1, 2, 3) - (-2, 5, 1)|} + \frac{3(10^{-3})}{|(1, 2, 3) - (3, -4, 6)|}$$

$$4 \pi \epsilon_0 (10^3) V_p = \frac{1}{|(1, 2, -1)|} - \frac{2}{|(3, -3, 2)|} + \frac{3}{|(-2, 6, -3)|} = \frac{1}{\sqrt{6}} - \frac{2}{\sqrt{22}} + \frac{3}{7}$$

$$4 \pi \frac{10^{-9}}{36 \pi} (10^3) V_p = 0.410$$

$$V_Q = \underline{\underline{3.694 (10^6) \text{ V}}}$$

$$\therefore V_{PQ} = V_Q - V_p = 0.686(10^6) = \underline{\underline{686 \text{ kV}}}$$

Prob. 4.39

$$\begin{aligned}
 -\mathbf{E} &= \nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z \\
 &= 2\rho e^{-z} \sin \phi \mathbf{a}_\rho + \rho e^{-z} \cos \phi \mathbf{a}_\phi - \rho^2 e^{-z} \sin \phi \mathbf{a}_z
 \end{aligned}$$

At $(4, \pi/4, -1)$, $\rho=4, \phi=\pi/4, z=-1$

$$\begin{aligned}
 -\mathbf{E} &= 2(4)e^1 \sin(\pi/4) \mathbf{a}_\rho + 4e^1 \cos(\pi/4) \mathbf{a}_\phi - 16e^1 \sin(\pi/4) \mathbf{a}_z \\
 &= 2(4)(2.7183)(0.7071) \mathbf{a}_\rho + 4(2.7183)(0.7071) \mathbf{a}_\phi - 16(2.7183)(0.7071) \mathbf{a}_z
 \end{aligned}$$

$$\mathbf{E} = \underline{\underline{-15.38 \mathbf{a}_\rho - 7.688 \mathbf{a}_\phi + 30.75 \mathbf{a}_z \text{ V/m}}}$$

Prob. 4.40

(a)

$$\begin{aligned}
 \mathbf{E} &= -\left(\frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z\right) \\
 &= -2xy(z+3) \mathbf{a}_x - x^2(z+3) \mathbf{a}_y - x^2y \mathbf{a}_z
 \end{aligned}$$

At $(3, 4, -6)$, $x=3, y=4, z=-6$,

$$\begin{aligned}
 \mathbf{E} &= -2(3)(4)(-3) \mathbf{a}_x - 9(-3) \mathbf{a}_y - 9(4) \mathbf{a}_z \\
 &= \underline{\underline{72 \mathbf{a}_x + 27 \mathbf{a}_y - 36 \mathbf{a}_z \text{ V/m}}}
 \end{aligned}$$

(b)

$$\rho_V = \nabla \cdot \mathbf{D} = \varepsilon_0 \nabla \cdot \mathbf{E} = -\varepsilon_0 (2y)(z+3)$$

$$\begin{aligned}
 \psi &= Q_{enc} = \int \rho_V dV = -2\varepsilon_0 \iiint y(z+3) dx dy dz \\
 &= -2\varepsilon_0 \int_0^1 dx \int_0^1 y dy \int_0^1 (z+3) dz = -2\varepsilon_0 (1)(1/2) \left(\frac{z^2}{2} + 3z\right) \Big|_0^1 \\
 &= -\varepsilon_0 \left(\frac{1}{2} + 3\right) = \frac{-7}{2} \left(\frac{10^{-9}}{36\pi}\right)
 \end{aligned}$$

$$Q_{enc} = \underline{\underline{-30.95 \text{ pC}}}$$

Prob. 4.41

(a)

$$\begin{aligned}
 Q &= \int_V \rho_v dv = \iiint \rho_o \left(1 - \frac{r^2}{a^2}\right) r^2 \sin \theta d\theta d\phi dr \\
 &= \rho_o \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta \int_0^a \left(r^2 - \frac{r^4}{a^2}\right) dr = \rho_o (2\pi)(2) \left(\frac{a^3}{3} - \frac{a^3}{5}\right) = \underline{\underline{\frac{8\pi}{15} a^3 \rho_o}}
 \end{aligned}$$

(b) Outside the nucleus, $r > a$,

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{enc} \longrightarrow \mathbf{E} = \frac{Q_{enc}}{4\pi\epsilon_0 r^2} \mathbf{a}_r$$

$$\mathbf{E} = \frac{\frac{8\pi a^3 \rho_o}{15}}{4\pi\epsilon_0 r^2} \mathbf{a}_r = \frac{2a^3 \rho_o}{15\epsilon_0 r^2} \mathbf{a}_r$$

$$V = -\int \mathbf{E} \cdot d\mathbf{l} = -\int E_r dr = \frac{2a^3 \rho_o}{15\epsilon_0 r} + C_1$$

Since $V(\infty) = 0, C_1 = 0$.

$$V = \frac{2a^3 \rho_o}{15\epsilon_0 r}$$

(c) Inside the nucleus, $r < a$

$$Q_{enc} = 4\pi\rho_o \left(\frac{r^3}{3} - \frac{r^5}{5a^2} \right)$$

$$\mathbf{E} = \frac{Q_{enc}}{4\pi\epsilon_0 r^2} \mathbf{a}_r = \frac{\rho_o}{\epsilon_0} \left(\frac{r}{3} - \frac{r^3}{5a^2} \right) \mathbf{a}_r$$

$$V = -\int E_r dr = -\frac{\rho_o}{\epsilon_0} \left(\frac{r^2}{6} - \frac{r^4}{20a^2} \right) + C_2$$

$$V(r=a) = \frac{2a^2 \rho_o}{15\epsilon_0} = \frac{\rho_o}{\epsilon_0} \left(\frac{a^2}{20} - \frac{a^2}{6} \right) + C_2$$

$$C_2 = \frac{2a^2 \rho_o}{15\epsilon_0} + \frac{7a^2 \rho_o}{60\epsilon_0} = \frac{a^2 \rho_o}{4\epsilon_0}$$

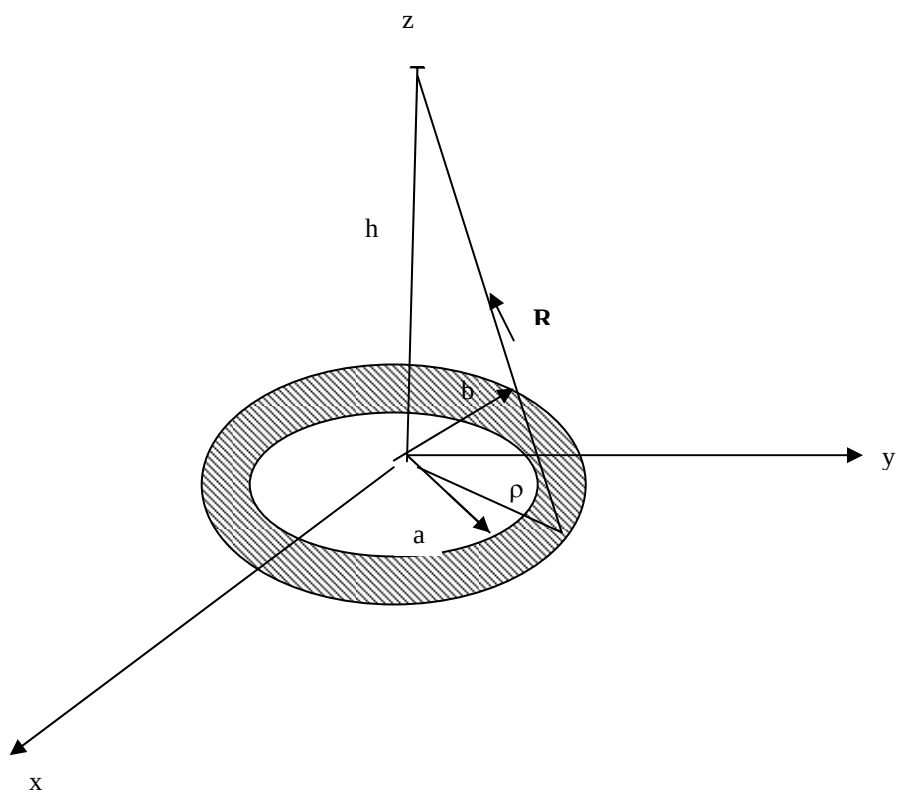
$$V = \frac{\rho_o}{\epsilon_0} \left(\frac{r^4}{20a^2} - \frac{r^2}{6} \right) + \frac{a^2 \rho_o}{4\epsilon_0}$$

(d) E is maximum when

$$\frac{dE}{dr} = 0 = \frac{\rho_o}{\epsilon_0} \left(\frac{1}{3} - \frac{3r^2}{5a^2} \right) \longrightarrow 9r^2 = 5a^2$$

$$r = \frac{\sqrt{5}}{3} a = \underline{\underline{0.7454a}}$$

We are able to say maximum because $\frac{d^2E}{dr^2} = -\frac{6r}{5a^2} < 0$.

Prob. 4.42

$$\mathbf{D} = \int \frac{\rho_s dS \mathbf{R}}{4\pi R^3}, \quad \mathbf{R} = -\rho \mathbf{a}_\rho + h \mathbf{a}_z, \quad R = |\mathbf{R}| = \sqrt{\rho^2 + h^2}, \quad dS = \rho d\phi d\rho$$

$$\rho_s = \frac{Q}{S} = \frac{Q}{\pi(b^2 - a^2)}$$

$$\mathbf{D} = \frac{\rho_s}{4\pi} \int \frac{\rho d\phi d\rho \rho(-\rho \mathbf{a}_\rho + h \mathbf{a}_z)}{(\rho^2 + h^2)^{3/2}}$$

Due to symmetry, the component along $\mathbf{a}_\rho$ vanishes.

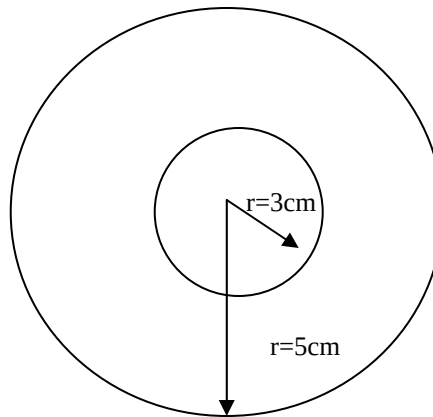
$$\begin{aligned}
 D_z &= \frac{\rho_s h}{4\pi} \int_{\rho=a}^b \int_{\phi=0}^{2\pi} \frac{\rho d\phi d\rho}{(\rho^2 + h^2)^{3/2}} = \frac{\rho_s h}{4\pi} (2\pi) \int_a^b (\rho^2 + h^2)^{-3/2} \rho d\rho \\
 &= \frac{\rho_s h}{2} \left[\frac{-1}{\sqrt{\rho^2 + h^2}} \right]_a^b = \frac{\rho_s h}{2} \left[\frac{1}{\sqrt{a^2 + h^2}} - \frac{1}{\sqrt{b^2 + h^2}} \right] \\
 \underline{\underline{\mathbf{D} = \frac{Qh}{2\pi(b^2 - a^2)} \left[\frac{1}{\sqrt{a^2 + h^2}} - \frac{1}{\sqrt{b^2 + h^2}} \right] \mathbf{a}_z}}}
 \end{aligned}$$

Prob. 4.43

$$\rho_v = \nabla \cdot \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = 4 - 20y + 2z$$

At P(1,2,3), $x=1$, $y=2$, $z=3$

$$\rho_v = 4 - 20(2) + 2(3) = \underline{\underline{-30 \text{ C/m}^3}}$$

Prob. 4.44

For $r < 3\text{cm}$, $Q_{\text{enc}} = 0 \longrightarrow \mathbf{D} = 0$

For $3 < r < 5\text{cm}$,

$$\oint \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}} = 10 \text{ nC}$$

$$D_r 4\pi r^2 = 10 \text{ nC} \longrightarrow D_r = \frac{10}{4\pi r^2} \text{ nC/m}^2$$

For $r > 5 \text{ cm}$,

$$\oint \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}} = 10 - 5 = 5\text{nC} \longrightarrow D_r = \frac{5}{4\pi r^2} \text{ nC/m}^2$$

Thus,

$$\mathbf{D} = \begin{cases} 0, & r < 3 \text{ cm} \\ \frac{10}{4\pi r^2} \mathbf{a}_r \text{ nC/m}^2, & 3 < r < 5 \text{ cm} \\ \frac{5}{4\pi r^2} \mathbf{a}_r \text{ nC/m}^2, & r > 5 \text{ cm} \end{cases}$$

Prob. 4.45

$$\rho_v = \nabla \cdot \mathbf{D} = \nabla \cdot \epsilon_o \mathbf{E} = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\frac{\epsilon_o E_o \rho^2}{a} \right) = \frac{2\epsilon_o E_o}{a}, 0 < \rho < a$$

Prob. 4.46

Let us choose the following path of two segments.

$$(2, 1, -1) \rightarrow (5, 1, -1) \rightarrow (5, 1, 2)$$

$$W = -q \int \mathbf{E} \cdot d\mathbf{l}$$

$$\begin{aligned} -\frac{W}{q} &= \int \mathbf{E} \cdot d\mathbf{l} = \int_{x=2}^5 2xyz dx \Big|_{z=-1, y=1} + \int_{z=-1}^2 x^2 y dz \Big|_{x=5, y=1} \\ &= 2(1)(-1) \frac{x^2}{2} \Big|_2^5 + (5)^2 (1) z \Big|_{-1}^2 = -21 + 75 = 54 \end{aligned}$$

$$W = -54q = \underline{\underline{-108 \mu\text{J}}}$$

Prob. 4.47

(a)

$$\rho_v = \nabla \cdot \mathbf{D} = \nabla \cdot \epsilon \mathbf{E}$$

$$\begin{aligned} \frac{\rho_v}{\epsilon} &= \nabla \cdot \mathbf{E} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (12\rho^2 z \cos \phi) + \frac{1}{\rho} (-6\rho z \cos \phi) + 0 \\ &= 24z \cos \phi - 6z \cos \phi = 18z \cos \phi \end{aligned}$$

$$\text{At } A(2, 180^\circ, -1), \quad \rho = 2, \phi = 180^\circ, z = -1$$

$$\rho_v = 18\epsilon z \cos \phi = 18(-1) \cos(180^\circ) \times \frac{10^{-9}}{36\pi} = \frac{10^{-9}}{2\pi} = \underline{\underline{0.1592 \text{ nC/m}^3}}$$

(b)

$$W = -Q \int_L \mathbf{E} \cdot d\mathbf{l}, \quad d\mathbf{l} = \rho d\phi \mathbf{a}_\phi$$

$$W = Q \int_{\phi=180^\circ}^{0^\circ} 6\rho z \sin \phi \rho d\phi \Big|_{\rho=2, z=-1} = Q6(2)^2(-1) \left(-\cos \phi \Big|_{180^\circ}^{0^\circ} \right) = -24Q(-1-1) \\ = 48Q = 48 \times 10 \times 10^{-6} = \underline{\underline{480 \mu\text{J}}}$$

Prob. 4.48

(a)

From A to B, $d\mathbf{l} = r d\theta \mathbf{a}_\theta$,

$$W_{AB} = -Q \int_{\theta=30^\circ}^{90^\circ} 10r \cos \theta r d\theta \Big|_{r=5} = -Q(10)(5)^2 (\sin \theta) \Big|_{30^\circ}^{90^\circ} = \underline{\underline{-1250 \text{ nJ}}}$$

(b)

From A to C, $d\mathbf{l} = dr \mathbf{a}_r$,

$$W_{AC} = -Q \int_{r=5}^{10} 20r \sin \theta dr \Big|_{\theta=30^\circ} = -Q(20)(\sin 30^\circ) \left(\frac{1}{2} r^2 \right) \Big|_5^{10} = \underline{\underline{-3750 \text{ nJ}}}$$

(c)

From A to D, $d\mathbf{l} = r \sin \theta d\phi \mathbf{a}_\phi$,

$$W_{AD} = -Q \int 0(r \sin \theta) d\phi = \underline{\underline{0 \text{ J}}}$$

(d)

$$W_{AE} = W_{AD} + W_{DF} + W_{FE}$$

where F is $(10, 30^\circ, 60^\circ)$. Hence,

$$W_{AE} = -Q \left\{ \int_{r=5}^{10} 20r \sin \theta dr \Big|_{\theta=30^\circ} + \int_{\theta=30^\circ}^{90^\circ} 10r \cos \theta r d\theta \Big|_{r=10} \right\} \\ = -100 \left[\frac{75}{2} + \frac{100}{2} \right] \text{ nJ} = \underline{\underline{-8750 \text{ nJ}}}$$

Prob. 4.49

$$V_{AB} = - \int_A^B \mathbf{E} \cdot d\mathbf{l} = - \int_1^5 \frac{10}{r^2} dr \\ = \frac{10}{r} \Big|_1^5 = 10 \left(\frac{1}{5} - 1 \right) = \underline{\underline{-8V}}$$

Prob. 4.50*Method 1:*

$$W = -Q \int_L \mathbf{E} \cdot d\mathbf{l}, \quad d\mathbf{l} = \rho d\phi \mathbf{a}_\phi$$

$$\begin{bmatrix} E_\rho \\ E_\phi \\ E_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}$$

$$E_\phi = -E_x \sin \phi + E_y \cos \phi = -20x \sin \phi + 40y \cos \phi$$

$$x = \rho \cos \phi, y = \rho \sin \phi$$

$$E_\phi = -20\rho \cos \phi \sin \phi + 40\rho \sin \phi \cos \phi = 20\rho \cos \phi \sin \phi$$

$$\begin{aligned} W &= -Q \int_L \mathbf{E} \cdot d\mathbf{l} = -2 \times 10^{-3} \int 20\rho \cos \phi \sin \phi \rho d\phi \Big|_{\rho=2} \\ &= -2(20)(2)^2 \int_0^{\pi/2} \sin \phi d(\sin \phi) \text{ mJ} = 160 \frac{\sin^2 \phi}{2} \Big|_0^{\pi/2} = \underline{\underline{-80 \text{ mJ}}} \end{aligned}$$

Method 2:

$$-\frac{W}{Q} = \int_L \mathbf{E} \cdot d\mathbf{l} = \int 20x dx + 40y dy$$

$$y = 2 - x, dy = -dx$$

$$\begin{aligned} -\frac{W}{Q} &= \int 20x dx + 40(2 - x)(-dx) = \int_{x=2}^0 (60x - 80) dx \\ &= \frac{60x^2}{2} - 80x \Big|_2^0 = 40 \end{aligned}$$

$$W = -40Q = \underline{\underline{-80 \text{ mJ}}}$$

Method 3:

$$\nabla \times \mathbf{E} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 20x & 40y & -10z \end{vmatrix} = 0$$

$$V = -\int_L \mathbf{E} \cdot d\mathbf{l} = -10x^2 - 20y^2 + 5z^2 + C$$

$$W = Q(V_2 - V_1) = Q(-20 \times 4 + 10 \times 4) = -40Q$$

$$W = -40Q = \underline{\underline{-80 \text{ mJ}}}$$

Prob. 4.51

$$W = -Q \int_L \mathbf{E} \cdot d\mathbf{l}$$

$$\mathbf{E} = \frac{\rho_s}{2\epsilon_o} \mathbf{a}_n = \frac{\rho_s}{2\epsilon_o} \mathbf{a}_x, \quad d\mathbf{l} = dx \mathbf{a}_x$$

$$\begin{aligned} W &= -\frac{Q\rho_s}{2\epsilon_o} \int_3^1 dx = -\frac{Q\rho_s}{2\epsilon_o} (-2) = \frac{Q\rho_s}{\epsilon_o} \\ &= \frac{10 \times 10^{-6} \times 40 \times 10^{-9}}{\frac{10^{-9}}{36\pi}} = 400 \times 36\pi \times 10^{-6} = \underline{\underline{45.24 \text{ mJ}}} \end{aligned}$$

Prob. 4.52

$$(a) \quad \mathbf{E} = -\nabla V = -\frac{\partial V}{\partial x} \mathbf{a}_x - \frac{\partial V}{\partial y} \mathbf{a}_y - \frac{\partial V}{\partial z} \mathbf{a}_z = \underline{\underline{-4x\mathbf{a}_x - 8y\mathbf{a}_y}}$$

$$\rho_v = \nabla \cdot \mathbf{D} = \epsilon_o \nabla \cdot \mathbf{E} = \epsilon_o (-4 - 8) = -12\epsilon_o = \underline{\underline{-106.25 \text{ pC/m}^3}}$$

$$\begin{aligned} (b) \quad \mathbf{E} &= -\nabla V = -\frac{\partial V}{\partial \rho} \mathbf{a}_\rho - \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi - \frac{\partial V}{\partial z} \mathbf{a}_z \\ &= \underline{\underline{-(20\rho \sin \phi + 6z)\mathbf{a}_\rho - 10\rho \cos \phi \mathbf{a}_\phi - 6\rho \mathbf{a}_z}} \end{aligned}$$

$$\begin{aligned} \rho_v &= \nabla \cdot \mathbf{D} = \epsilon_o \nabla \cdot \mathbf{E} = \epsilon_o \left(-\frac{1}{\rho} [40\rho \sin \phi + 6z] + 10 \sin \phi \right) \\ &= \underline{\underline{-\left(30 \sin \phi + \frac{6z}{\rho} \right) \epsilon_o \text{ C/m}^3}} \end{aligned}$$

$$\begin{aligned} (c) \quad \mathbf{E} &= -\nabla V = -\frac{\partial V}{\partial r} \mathbf{a}_r - \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta - \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \\ &= \underline{\underline{-10r \cos \theta \sin \phi \mathbf{a}_r + 5r \sin \theta \sin \phi \mathbf{a}_\theta - 5r \cot \theta \cos \phi \mathbf{a}_\phi}} \end{aligned}$$

$$\rho_v = \nabla \cdot \mathbf{D} = \epsilon_o \nabla \cdot \mathbf{E}$$

$$\begin{aligned} \frac{\rho_v}{\epsilon_o} &= \frac{1}{r^2} (-30r^2 \cos \theta \sin \phi) + \frac{5r \sin \phi}{r \sin \theta} 2 \sin \theta \cos \theta + \frac{5r \cot \theta \sin \phi}{r \sin \theta} \\ \rho_v &= \epsilon_o (5 \sin \phi \csc^2 \theta \cos \theta - 20 \cos \theta \sin \phi) \text{ C/m}^3 \end{aligned}$$

Prob. 4.53

(a)

$$-\mathbf{E} = \nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z = e^{-z} \sin \phi \mathbf{a}_\rho + e^{-z} \cos \phi \mathbf{a}_\phi - \rho e^{-z} \sin \phi \mathbf{a}_z$$

$$\mathbf{E} = -e^{-z} \sin \phi \mathbf{a}_\rho - e^{-z} \cos \phi \mathbf{a}_\phi + \rho e^{-z} \sin \phi \mathbf{a}_z$$

(b)

$$\begin{aligned} \nabla \times \mathbf{E} &= \left[\frac{1}{\rho} \frac{\partial E_z}{\partial \phi} - \frac{\partial E_\phi}{\partial z} \right] \mathbf{a}_\rho + \left[\frac{\partial E_\rho}{\partial z} - \frac{\partial E_z}{\partial \rho} \right] \mathbf{a}_\phi + \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (\rho E_\phi) - \frac{\partial E_\rho}{\partial \phi} \right] \mathbf{a}_z \\ &= \left[\frac{1}{\rho} \rho e^{-z} \cos \phi - e^{-z} \cos \phi \right] \mathbf{a}_\rho + \left[e^{-z} \sin \phi - e^{-z} \sin \phi \right] \mathbf{a}_\phi + \frac{1}{\rho} \left[-e^{-z} \cos \phi + e^{-z} \cos \phi \right] \mathbf{a}_z = \mathbf{0} \end{aligned}$$

since each component is zero. Alternatively,

$$\nabla \times \mathbf{E} = \nabla \times \nabla V = \mathbf{0}.$$

Prob. 4.54

$$V = -r^{-3} \sin \theta \cos \phi$$

$$\begin{aligned} -\mathbf{E} = \nabla V &= \frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \\ &= -3r^{-4} \sin \theta \cos \phi \mathbf{a}_r + r^{-4} \cos \theta \cos \phi \mathbf{a}_\theta + \frac{r^{-4} \sin \theta}{\sin \theta} (-\sin \phi) \mathbf{a}_\phi \end{aligned}$$

$$\mathbf{E} = \frac{3}{r^4} \sin \theta \cos \phi \mathbf{a}_r - \frac{1}{r^4} \cos \theta \cos \phi \mathbf{a}_\theta + \frac{\sin \phi}{r^4} \mathbf{a}_\phi$$

At $(1, 30^\circ, 60^\circ)$, $r = 1$, $\theta = 30^\circ$, $\phi = 60^\circ$

$$\begin{aligned} \mathbf{E} &= 3 \sin 30^\circ \cos 60^\circ \mathbf{a}_r - \cos 30^\circ \cos 60^\circ \mathbf{a}_\theta + \sin 60^\circ \mathbf{a}_\phi \\ &= 0.75 \mathbf{a}_r - 0.433 \mathbf{a}_\theta + 0.866 \mathbf{a}_\phi \end{aligned}$$

$$\begin{aligned} \mathbf{D} = \epsilon_0 \mathbf{E} &= \frac{10^{-9}}{36\pi} (0.75 \mathbf{a}_r - 0.433 \mathbf{a}_\theta + 0.866 \mathbf{a}_\phi) \\ &= \underline{\underline{6.635 \mathbf{a}_r - 3.829 \mathbf{a}_\theta + 7.657 \mathbf{a}_\phi \text{ pC/m}^2}} \end{aligned}$$

Prob. 4.55

For $a < r < b$, we apply Gauss's law.

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{enc} = Q$$

$$D_r(4\pi r^2) = Q \longrightarrow E_r = \frac{D_r}{\epsilon_o} = \frac{Q}{4\pi\epsilon_o r^2}$$

$$V_{ab} = -\int_a^b \mathbf{E} \cdot d\mathbf{l} = -\frac{Q}{4\pi\epsilon_o} \int_a^b \frac{1}{r^2} dr = \frac{Q}{4\pi\epsilon_o} \frac{1}{r} \bigg|_a^b = \underline{\underline{-\frac{Q}{4\pi\epsilon_o} \left(\frac{1}{a} - \frac{1}{b} \right)}}$$

Prob. 4.56

$$\begin{aligned} V &= \int_S \frac{\rho_s dS}{4\pi\epsilon_o r} = \frac{\rho_s}{4\pi\epsilon_o} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} \frac{r^2 \sin\theta d\theta d\phi}{r} = \frac{\rho_s}{4\pi\epsilon_o r} (2\pi) \int_0^{\pi/2} \sin\theta d\theta \\ &= \frac{\rho_s}{2\epsilon_o r} \left(-\cos\theta \bigg|_0^{\pi/2} \right) \bigg|_{r=a} \\ V &= \underline{\underline{\frac{\rho_s}{2\epsilon_o a}}} \end{aligned}$$

Prob. 4.57

$$\nabla \times \mathbf{E} = 0 \longrightarrow \nabla \times \mathbf{D} = 0$$

$$\begin{aligned} \nabla \times \mathbf{D} &= \left[\frac{1}{\rho} \frac{\partial D_z}{\partial \phi} - \frac{\partial D_\phi}{\partial z} \right] \mathbf{a}_\rho + \left[\frac{\partial D_\rho}{\partial z} - \frac{\partial D_z}{\partial \rho} \right] \mathbf{a}_\phi + \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (\rho D_\phi) - \frac{\partial D_\rho}{\partial \phi} \right] \mathbf{a}_z \\ &= 0 \mathbf{a}_\rho - 0 \mathbf{a}_\phi - \frac{1}{\rho} 2\rho \cos\phi \mathbf{a}_z \neq 0 \end{aligned}$$

Hence **D** is not a genuine EM field.

$$\begin{aligned} \psi &= \int_S \mathbf{D} \cdot d\mathbf{S} = \int_{\phi=0}^{\pi/4} \int_{z=0}^1 2\rho \sin\phi \rho d\phi dz = 2 \int_0^{\pi/4} \sin\phi d\phi \int_0^1 dz \rho^2 \bigg|_{\rho=1} \\ &= -2 \cos\phi \bigg|_0^{\pi/4} (1)(1)^2 = -2(\cos\pi/4 - 1) = \underline{\underline{0.5858 \text{ C}}} \end{aligned}$$

Prob. 4.58

(a)

$m \frac{d^2 y}{dt^2} = eE$; divide by m , and integrate once, one obtains :

$$u = \frac{dy}{dt} = \frac{eEt}{m} + c_0$$

$$y = \frac{eEt^2}{2m} + c_0 t + c_1 \quad (1)$$

"From rest" implies $c_1 = 0 = c_0$

At $t = t_0$, $y = d$, $E = \frac{V}{d}$ or $V = Ed$.

Substituting this in (1) yields :

$$t^2 = \frac{2md}{eE}$$

Hence :

$$u = \frac{eE}{m} \sqrt{\frac{2md}{eE}} = \sqrt{\frac{2eEd}{m}} = \sqrt{\frac{2eV}{m}}$$

that is, $u \propto \sqrt{V}$

or $u = k \sqrt{V}$

(b)

$$k = \sqrt{\frac{2e}{m}} = \sqrt{\frac{2(1.603) 10^{-19}}{9.1066 (10^{-31})}}$$

$$= \underline{\underline{5.933 \times 10^5}}$$

(c)

$$V = \frac{u^2 m}{2e} = \frac{9(10^{16}) \frac{1}{100}}{2(1.76) (10^{11})} = \underline{\underline{2.557 \text{ kV}}}$$

Prob. 4.59

(a)

This is similar to Example 4.3.

$$u_y = \frac{eEt}{m}, \quad u_x = u_0$$

$$y = \frac{eEt^2}{2m}, \quad x = u_0 t$$

$$t = \frac{x}{u_0} = \frac{10(10^{-2})}{10^7} = 10 \text{ ns}$$

Since $x = 10 \text{ cm}$ when $y = 1 \text{ cm}$,

$$E = \frac{2my}{et^2} = \frac{2(10^{-2})}{1.76(10^{11})(10^{-16})} = 1.136 \text{ kV/m}$$

$$\underline{\underline{\mathbf{E} = -1.136 \mathbf{a}_y \text{ kV/m}}}$$

(b)

$$u_x = u_0 = 10^7,$$

$$u_y = \frac{2000}{1.76}(1.76)10^{11}(10^{-8}) = 2(10^6)$$

$$\underline{\underline{\mathbf{u} = (\mathbf{a}_x + 0.2\mathbf{a}_y)(10^7) \text{ m/s}}}$$

Prob. 4.60

$$V = \frac{p \cos \theta}{4\pi \epsilon_0 r^2} = \frac{k \cos \theta}{r^2}$$

At $(0, 1 \text{ nm})$, $\theta = 0$, $r = 1 \text{ nm}$, $V = 9$;

$$\text{that is, } 9 = \frac{k(1)}{1(10^{-18})}, \quad \therefore k = 9(10^{-18})$$

$$V = 9(10^{-18}) \frac{\cos \theta}{r^2}$$

At $(1, 1) \text{ nm}$, $r = \sqrt{2} \text{ nm}$, $\theta = 45^\circ$,

$$V = \frac{9(10^{-18}) \cos 45^\circ}{10^{-18} (\sqrt{2})^2} = \frac{9}{2\sqrt{2}} = \underline{\underline{3.182 \text{ V}}}$$

Prob. 4.61

The dipole is oriented along y – axis.

$$V = \frac{\mathbf{p} \bullet \mathbf{r}}{4\pi\epsilon_0 r^2}; \mathbf{p} \bullet \mathbf{r} = Qd \mathbf{a}_y \bullet \mathbf{a}_r = Qd \sin \theta \sin \phi$$

$$V = \frac{Qd \sin \theta \sin \phi}{4\pi\epsilon_0 r^2}$$

$$\begin{aligned} \mathbf{E} = -\nabla V &= -\frac{\partial V}{\partial r} \mathbf{a}_r - \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta - \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \\ &= \frac{Qd}{4\pi\epsilon_0} \left\{ \frac{2 \sin \theta \sin \phi}{r^3} \mathbf{a}_r - \frac{\cos \theta \sin \phi}{r^3} \mathbf{a}_\theta - \frac{\cos \phi}{r^3} \mathbf{a}_\phi \right\} \end{aligned}$$

$$\mathbf{E} = \frac{Qd}{4\pi\epsilon_0 r^3} (2 \sin \theta \sin \phi \mathbf{a}_r - \cos \theta \sin \phi \mathbf{a}_\theta - \cos \phi \mathbf{a}_\phi)$$

Prob. 4.62

Using eq. (4.81),

$$V = \frac{\mathbf{p} \bullet (\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3}$$

$$\mathbf{r} - \mathbf{r}' = (4, 0, 1) - (2, 3, -1) = (2, -3, 2)$$

$$|\mathbf{r} - \mathbf{r}'| = \sqrt{4 + 9 + 4} = \sqrt{17}$$

$$\mathbf{p} \bullet (\mathbf{r} - \mathbf{r}') = (2, 6, -4) \bullet (2, -3, 2) = 4 - 18 - 8 = -22$$

$$V = \frac{-22 \times 10^{-6}}{4\pi \times \frac{10^{-9}}{36\pi} \times 17^{3/2}} = -\frac{198}{70.093} \text{ kV} = \underline{\underline{-2.825 \text{ kV}}}$$

Prob. 4.63

$$\mathbf{E} = k(2 \cos \theta \mathbf{a}_r + \sin \theta \mathbf{a}_\theta)$$

$$\begin{bmatrix} E_\rho \\ E_\phi \\ E_z \end{bmatrix} = \begin{bmatrix} \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} 2k \cos \theta \\ k \sin \theta \\ 0 \end{bmatrix}$$

$$E_z = 2k \cos^2 \theta - k \sin^2 \theta = 2k \cos^2 \theta - k(1 - \cos^2 \theta) = 3k \cos^2 \theta - k$$

Setting this to zero gives

$$3 \cos^2 \theta = 1 \quad \rightarrow \quad \cos \theta = \pm \frac{1}{\sqrt{3}} = \pm 0.5773 \quad \rightarrow \quad \underline{\underline{\theta = 54.74^\circ, 125.26^\circ}}$$

Prob. 4.64

$$\begin{aligned}
 W &= W_1 + W_2 = 0 + Q_2 V_{21} = Q_2 \frac{Q_1}{4\pi\epsilon_o |(2, 0, 0) - (0, 0, 1)|} \\
 &= \frac{40 \times 10^{-9} \times (-50) \times 10^{-9}}{4\pi \times \frac{10^{-9}}{36\pi} |(2, 0, -1)|} = \frac{40 \times 9 \times (-50) \times 10^{-9}}{\sqrt{4+1}} \\
 &= \underline{\underline{-8.05 \mu\text{J}}}
 \end{aligned}$$

Prob. 4.65

$$\begin{aligned}
 \mathbf{E} &= -\nabla V = -\frac{\partial V}{\partial x} \mathbf{a}_x - \frac{\partial V}{\partial y} \mathbf{a}_y = -4x\mathbf{a}_x - 12y\mathbf{a}_y \text{ V/m} \\
 W &= \frac{1}{2} \epsilon_o \int_v |\mathbf{E}|^2 dv = \frac{1}{2} \epsilon_o \int_{z=-1}^1 \int_{y=-1}^1 \int_{x=-1}^1 (16x^2 + 144y^2) dx dy dz \\
 &= \frac{1}{2} \epsilon_o \left[16(4) \frac{x^3}{3} \Big|_{-1}^1 + 144(4) \frac{y^3}{3} \Big|_{-1}^1 \right] = \frac{1}{2} \frac{10^{-9}}{36\pi} (160)(4) \frac{1}{3} (1+1) \\
 &= \underline{\underline{1.886 \text{ nJ}}}
 \end{aligned}$$

Prob. 4.66

Given that $\mathbf{E} = 2r \sin \theta \cos \phi \mathbf{a}_r + r \cos \theta \cos \phi \mathbf{a}_\theta - r \sin \theta \mathbf{a}_\phi$

$$\begin{aligned}
 E^2 &= 4r^2 \sin^2 \theta \cos^2 \phi + r^2 \cos^2 \theta \cos^2 \phi + r^2 \sin^2 \phi \\
 &= r^2 \cos^2 \phi (4 \sin^2 \theta + \cos^2 \theta) + r^2 \sin^2 \phi \\
 &= r^2 \cos^2 \phi + 3r^2 \cos^2 \phi \sin^2 \theta + r^2 \sin^2 \phi \\
 &= r^2 (1 + 3 \cos^2 \phi \sin^2 \theta)
 \end{aligned}$$

$$\begin{aligned}
 W &= \frac{\epsilon}{2} \iiint E^2 r^2 \sin \theta dr d\theta d\phi \\
 &= \frac{\epsilon}{2} \int_0^2 r^4 dr \int_{\theta=0}^{\pi} \int_{\phi}^{\pi} (1 + 3 \cos^2 \phi \sin^2 \theta) \sin \theta d\theta d\phi \\
 &= \frac{16\epsilon}{5} \int_0^{\pi} (\pi \sin \theta + \frac{3\pi}{2} \sin^2 \theta) d\theta \\
 &= \frac{16}{5} \times \frac{10^{-9}}{36\pi} (4\pi) = \frac{16}{45} \text{ nJ} = \underline{\underline{0.36 \text{ nJ}}}
 \end{aligned}$$

Prob. 4.67*Method 1:*

$$W = \frac{1}{2} \int_S \rho_s V dS = \frac{V}{2} \int_S \rho_s dS = \frac{1}{2} QV$$

$$\text{But } V = \frac{Q}{4\pi\epsilon_o a}$$

$$W = \frac{Q^2}{\underline{\underline{8\pi\epsilon_o a}}}$$

Method 2:

$$\begin{aligned} W &= \frac{1}{2} \int_V \mathbf{D} \cdot \mathbf{E} dv = \frac{1}{2} \epsilon_o \int_V E^2 dv \\ &= \frac{1}{2} \epsilon_o \iiint \left(\frac{Q}{4\pi\epsilon_o r^2} \right)^2 r^2 \sin\theta d\theta dr d\phi \\ &= \frac{1}{2} \epsilon_o \int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta \int_{r=a}^\infty \frac{Q^2}{16\pi^2 \epsilon_o^2 r^2} dr = \frac{\epsilon_o}{2} (2\pi)(2) \frac{Q^2}{16\pi^2 \epsilon_o^2 a} \\ W &= \frac{Q^2}{\underline{\underline{8\pi\epsilon_o a}}} \end{aligned}$$

Prob. 68

$$\begin{aligned} W &= \frac{1}{2} \int_V \mathbf{D} \cdot \mathbf{E} dv = \frac{1}{2} \epsilon_o \int_V |\mathbf{E}|^2 dv = \frac{1}{2} \epsilon_o \int_V (y^4 + 4x^2 y^2 + 16z^2) dx dy dz \\ \frac{2W}{\epsilon_o} &= \int_0^2 dx \int_0^4 dz \int_{-1}^1 y^4 dy + 4 \int_0^2 dx \int_0^4 dz \int_{-1}^1 y^2 dy + 16 \int_0^2 dx \int_0^4 dz \int_{-1}^1 z^2 dz \\ &= 2(4) \left(2 \frac{y^5}{5} \Big|_0^1 \right) + 4(4) \left(\frac{x^3}{3} \Big|_0^2 \right) \left(2 \frac{y^3}{3} \Big|_0^1 \right) + 16(2)(2) \left(\frac{z^3}{3} \Big|_0^4 \right) \\ &= \frac{16}{5} + \frac{256}{9} + \frac{64 \times 64}{3} = 1396.98 \\ W &= \frac{1}{2} \times \frac{10^{-9}}{36\pi} \times 1396.98 = \underline{\underline{6.176 \text{ nJ}}} \end{aligned}$$

Prob. 4.69

$$(a) \quad \mathbf{E} = -\nabla V = -\left(\frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z \right)$$

$$= -e^{-z} \sin \phi \mathbf{a}_\rho - e^{-z} \cos \phi \mathbf{a}_\phi + \rho e^{-z} \sin \phi \mathbf{a}_z$$

$$|\mathbf{E}|^2 = e^{-2z} \sin^2 \phi + e^{-2z} \cos^2 \phi + \rho^2 e^{-2z} \sin^2 \phi = e^{-2z} + \rho^2 e^{-2z} \sin^2 \phi$$

$$W = \frac{1}{2} \epsilon_o \int_v |\mathbf{E}|^2 dv$$

$$\frac{2W}{\epsilon_o} = \int_v |\mathbf{E}|^2 dv = \iiint (e^{-2z} + \rho^2 e^{-2z} \sin^2 \phi) \rho d\rho d\phi dz$$

$$(b) \quad = \int_0^1 \rho d\rho \int_0^{2\pi} d\phi \int_0^2 e^{-2z} dz + \int_0^1 \rho^3 d\rho \int_0^{2\pi} \sin^2 \phi d\phi \int_0^2 e^{-2z} dz$$

$$= \frac{\rho^2}{2} \Big|_0^1 (2\pi) \left(\frac{e^{-2z}}{-2} \Big|_0^2 \right) + \frac{\rho^4}{4} \Big|_0^1 \left(\frac{\phi}{2} - \frac{\sin 2\phi}{4} \right) \Big|_0^{2\pi} \left(\frac{e^{-2z}}{-2} \Big|_0^2 \right)$$

$$= \pi(-1/2)(e^{-4} - 1) + \frac{1}{4}(\pi - 0)(-1/2)(e^{-4} - 1) = \frac{5\pi}{8}(1 - e^{-4}) = 1.9275$$

$$W = \frac{1}{2} \epsilon_o (1.9275) = \frac{1}{2} \times \frac{10^{-9}}{36\pi} (1.9275) = \underline{\underline{8.512 \text{ pJ}}}$$

CHAPTER 5

P. E. 5.1 $d\mathbf{S} = \rho d\phi dz \mathbf{a}_\rho$

$$I = \int \mathbf{J} \cdot d\mathbf{S} = \int_{\phi=0}^{2\pi} \int_{z=1}^5 10z \sin^2 \phi \rho dz d\phi \Big|_{\rho=2} =$$

$$10(2) \frac{z^2}{2} \Big|_1^5 \int_0^{2\pi} \frac{1}{2} (1 - \cos 2\phi) d\phi = 10(5^2 - 1^2) \frac{1}{2} \left(\phi - \frac{1}{2} \sin 2\phi \right) \Big|_0^{2\pi} = 10(24)\pi = 240\pi$$

$I = 754 \text{ A}$

P. E. 5.2

$$I = \rho_s w u = 0.5 \times 10^{-6} \times 0.1 \times 10 = 0.5 \mu \text{ A}$$

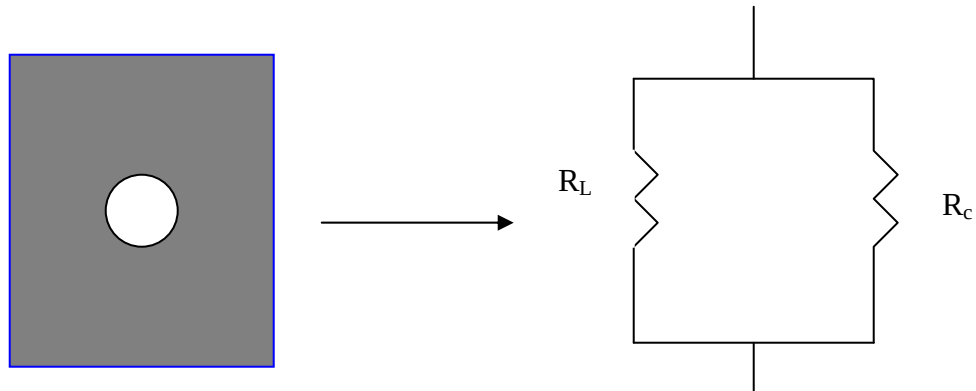
$$V = IR = 0.5 \times 10^{-6} \times 10^{14} = \underline{\underline{50 \text{ MV}}}$$

P. E. 5.3 $\sigma = 5.8 \times 10^7 \text{ S/m}$

$$J = \sigma E \longrightarrow E = \frac{J}{\sigma} = \frac{8 \times 10^6}{5.8 \times 10^7} = \underline{\underline{0.138 \text{ V/m}}}$$

$$J = \rho_v u \longrightarrow u = \frac{J}{\rho_v} = \frac{8 \times 10^6}{1.81 \times 10^{10}} = \underline{\underline{4.42 \times 10^{-4} \text{ m/s}}}$$

P. E. 5.4 The composite bar can be modeled as a parallel combination of resistors as shown below.



$$R_L = \frac{l}{\sigma_L S_L}, \quad l = 4\text{m}, S_L = d^2 - \pi r^2 = 9 - \frac{\pi}{4} \text{ cm}^2, \sigma_L = 5 \times 10^6 \text{ S/m}$$

For the lead,

$$R_L = \frac{4}{(5 \times 10^6) \left(9 - \frac{\pi}{4}\right) \times 10^{-4}} = 973.8 \mu\Omega$$

For copper, $R_c = \frac{l}{\sigma_c S_c}, \quad S_c = \pi r^2 = \frac{\pi}{4}, \sigma_c = 5.8 \times 10^7 \text{ S/m cm}^2$

$$R_c = \frac{4}{5.8 \times 10^7 \times \frac{\pi}{4} \times 10^{-4}} = 878.5 \mu\Omega$$

$$R = \frac{R_L R_c}{R_L + R_c} = \frac{973.8 \times 878.5}{973.8 + 878.5} = \underline{\underline{461.8 \mu\Omega}}$$

P. E. 5.5 $\rho_{ps} = \mathbf{P} \cdot \mathbf{a}_s = ax^2 + b$

$$\rho_{ps} \Big|_{x=0} = \mathbf{P} \cdot (-\mathbf{a}_x) \Big|_{x=0} = -(ax^2 + b) \Big|_{x=0} = b$$

$$\rho_{ps} \Big|_{x=L} = \mathbf{P} \cdot \mathbf{a}_x \Big|_{x=L} = (ax^2 + b) \Big|_{x=L} = \underline{\underline{aL^2 + b}}$$

$$Q_s = \int \rho_{ps} dS = -bA + (aL^2 + b)A = AaL^2$$

$$\rho_{pv} = -\nabla \cdot \mathbf{P} = -\frac{d}{dx}(ax^2 + b) = -2ax$$

$$\rho_{pv} \Big|_{x=0} = \underline{\underline{0}}, \quad \rho_{pv} \Big|_{x=L} = \underline{\underline{-2aL}}$$

$$Q_v = \int \rho_{pv} dv = \int_0^L (-2ax) A dx = -AaL^2$$

Hence,

$$Q_T = Q_v + Q_s = -AaL^2 + AaL^2 = 0$$

P. E. 5.6

$$\mathbf{E} = \frac{V}{d} \mathbf{a}_x = \frac{10^3}{2 \times 10^{-3}} \mathbf{a}_x = 500 \mathbf{a}_x \text{ kV/m}$$

$$\mathbf{P} = \chi_e \epsilon_o \mathbf{E} = (2.55 - 1) \times \frac{10^{-9}}{36\pi} \times 0.5 \times 10^6 \mathbf{a}_x = \underline{\underline{6.853 \mathbf{a}_x \mu\text{C/m}^2}}$$

$$\rho_{ps} = \mathbf{P} \cdot \mathbf{a}_x = \underline{\underline{6.853 \mu\text{C/m}^2}}$$

P. E. 5.7 (a) Since $P = \epsilon_o \chi_e E$, $P_x = \epsilon_o \chi_e E_x$

$$\chi_e = \frac{P_x}{\epsilon_o E_x} = \frac{3 \times 10^{-9}}{10\pi} \frac{1}{5} \times 36\pi \times 10^9 = \underline{\underline{2.16}}$$

$$(b) \quad E = \frac{P}{\chi_e \epsilon_o} = \frac{36\pi \times 10^9}{2.16} \frac{1}{10\pi} (3, -1, 4) 10^{-9} = \underline{\underline{5\mathbf{a}_x - 1.67\mathbf{a}_y + 6.67\mathbf{a}_z \text{ V/m}}}$$

(c)

$$D = \epsilon_o \epsilon_r E = \frac{\epsilon_r P}{\chi_e} = \frac{3.16}{2.16} \left(\frac{1}{10\pi} \right) (3, -1, 4) \text{ nC/m}^2 = \underline{\underline{139.7\mathbf{a}_x - 46.6\mathbf{a}_y + 186.3\mathbf{a}_z \text{ pC/m}^2}}$$

P. E. 5.8 From Example 5.8,

$$F = \frac{\rho_s^2 S}{2\epsilon_o} \longrightarrow \rho_s^2 = \frac{2\epsilon_o F}{S}$$

But $\rho_s = \epsilon_o E = \epsilon_o \frac{V_d}{d}$. Hence

$$\rho_s^2 = \frac{2\epsilon_o F}{S} = \frac{\epsilon_o^2 V_d^2}{d^2} \longrightarrow V_d^2 = \frac{2Fd^2}{\epsilon_o S}$$

i.e.

$$V_d = V_1 - V_2 = \sqrt{\frac{2Fd^2}{\epsilon_o S}}$$

as required.

P. E. 5.9 (a) Since $a_n = a_x$,

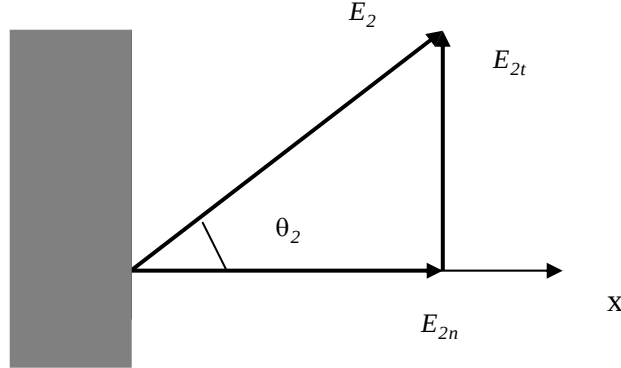
$$D_{1n} = 12\mathbf{a}_x, \quad D_{1t} = -10\mathbf{a}_x + 4\mathbf{a}_z, \quad D_{2n} = D_{1n} = 12\mathbf{a}_x$$

$$E_{2t} = E_{1t} \longrightarrow D_{2t} = \frac{\epsilon_2 D_{1t}}{\epsilon_1} = \frac{1}{2.5} (-10\mathbf{a}_y + 4\mathbf{a}_z) = -4\mathbf{a}_y + 1.6\mathbf{a}_z$$

$$D_2 = D_{2n} + D_{2t} = \underline{\underline{12\mathbf{a}_x - 4\mathbf{a}_y + 1.6\mathbf{a}_z \text{ nC/m}^2}}.$$

$$\tan \theta_2 = \frac{D_{2t}}{D_{2n}} = \frac{\sqrt{(-4)^2 + (1.6)^2}}{12} = 0.359 \quad \longrightarrow \quad \underline{\underline{\theta_2 = 19.75^\circ}}$$

(b) $E_{1t} = E_{2t} = E_2 \sin \theta_2 = 12 \sin 60^\circ = 10.392$



$$E_{1n} = \frac{\epsilon_{r2}}{\epsilon_{r1}} E_{2n} = \frac{1}{2.5} 12 \cos 60^\circ = 2.4$$

$$E_1 = \sqrt{E_{1t}^2 + E_{1n}^2} = \underline{\underline{10.67}}$$

$$\tan \theta_1 = \frac{E_{1t}}{E_{1n}} = \frac{E_{2t}}{(\epsilon_{r1} / \epsilon_{r2}) E_{2n}} = \frac{\epsilon_{r1}}{\epsilon_{r2}} \frac{E_{2t}}{E_{2n}} = \frac{\epsilon_{r1}}{\epsilon_{r2}} \tan \theta_2 = \frac{2.5}{1} \tan 60^\circ = 4.33 \quad \longrightarrow \quad \underline{\underline{\theta_1 = 77^\circ}}$$

Note that $\theta_1 > \theta_2$.

P. E. 5.10

$$D = \epsilon_o E = \frac{10^{-9}}{36\pi} (60, 20, -30) \times 10^{-3} = \underline{\underline{0.531a_x + 0.177a_y - 0.265a_z}} \text{ pC/m}^2$$

$$\rho_s = D_n = |D| = \frac{10^{-9}}{36\pi} (10) \sqrt{36 + 4 + 9} (10^{-3}) = \underline{\underline{0.619}} \text{ pC/m}^2$$

Prob. 5.1

$$I = \int \mathbf{J} \cdot d\mathbf{S}, \quad d\mathbf{S} = dydz\mathbf{a}_x$$

$$\begin{aligned} I &= \iint e^{-x} \cos(4y) dydz \Big|_{x=2} = e^{-2} \int_0^{\pi/3} \cos(4y) dy \int_0^4 dz \\ &= 4e^{-2} \left(\frac{\sin 4y}{4} \Big|_0^{\pi/3} \right) = e^{-2} \left(\sin\left(\frac{4\pi}{3}\right) - 0 \right) = \underline{\underline{-0.1172 \text{ A}}} \end{aligned}$$

Prob. 5.2

Method 1:

$$\begin{aligned} I &= \int \mathbf{J} \cdot d\mathbf{S} = \iint \frac{10}{r} e^{-10^3 t} r^2 \sin \theta d\theta d\phi \Big|_{t=2\text{ms}, r=4\text{m}} \\ &= 10(4)e^{-10^3 \times 2 \times 10^{-3}} \int_{\theta=0}^{\pi} \sin \theta d\theta \int_{\phi=0}^{2\pi} d\phi = 40e^{-2}(2)(2\pi) = 160\pi e^{-2} \\ &= \underline{\underline{68.03 \text{ A}}} \end{aligned}$$

Method 2:

$$I = \int \mathbf{J} \cdot d\mathbf{S} = \iint \frac{10}{r} e^{-10^3 t} dS = \frac{10}{r} e^{-10^3 t} (4\pi r^2)$$

since r is constant on the surface.

$$I = 40\pi r e^{-2} = 160\pi e^{-2} = \underline{\underline{68.03 \text{ A}}}$$

Prob. 5.3

$$\begin{aligned} I &= \int \mathbf{J} \cdot d\mathbf{S} = \iint \frac{10}{\rho} \sin \phi \rho d\phi dz = 10 \int_0^5 dz \int_0^{\pi} \sin \phi d\phi \\ &= 10(5)(-\cos \phi) \Big|_0^{\pi} = \underline{\underline{100 \text{ A}}} \end{aligned}$$

Prob. 5.4

$$\begin{aligned} I &= \int \mathbf{J} \cdot d\mathbf{S} = 5 \int_{\rho=0}^a \int_{\phi=0}^{2\pi} e^{-10\rho} \rho d\phi d\rho = 5 \int_0^{2\pi} d\phi \int_{\rho=0}^a \rho e^{-10\rho} d\rho \\ &= 5(2\pi) \left(\frac{e^{-10\rho}}{100} (-10\rho - 1) \right) \Big|_0^a = \frac{10\pi}{100} [e^{-10a} (-10a - 1) - 1(-0 - 1)], \quad a = 0.004 \\ &= \frac{\pi}{10} [e^{-0.04} (-0.04 - 1) + 1] = \frac{\pi}{10} (0.00078) = \underline{\underline{244.7 \mu\text{A}}} \end{aligned}$$

Prob. 5.5

$$I = \int_S \mathbf{J} \cdot d\mathbf{S}, \quad d\mathbf{S} = r^2 \sin \theta d\theta d\phi \mathbf{a}_r$$

$$I = \int_{\phi=0}^{2\pi} \int_{\theta=\pi/4}^{\pi/2} \frac{20 \cos \theta}{r+3} r^2 \sin \theta d\theta d\phi \Big|_{r=3} = \frac{20(9)}{6} \int_0^{2\pi} d\phi \int_{\pi/4}^{\pi/2} \cos \theta \sin \theta d\theta$$

$$= 30(2\pi) \int_{\pi/4}^{\pi/2} \cos \theta d(-\cos \theta) = -60\pi \frac{\cos^2 \theta}{2} \Big|_{\pi/4}^{\pi/2} = -60\pi(0 - \cos^2(\pi/4)) = -60\pi(-1/2) = 30\pi$$

$$\underline{\underline{I = 94.2 \text{ A}}}$$

Prob. 5.6

$$R = \frac{l}{\sigma S} \longrightarrow \sigma = \frac{l}{RS} = \frac{2 \times 10^{-2}}{10^6(\pi)(4 \times 10^{-3})^2} = \underline{\underline{3.978 \times 10^{-4} \text{ S/m}}}$$

$$\text{Prob. 5.7 (a)} \quad R = \frac{l}{\sigma S} = \frac{8 \times 10^{-2}}{3 \times 10^4 \pi (25) 10^{-6}} = \frac{8}{75\pi} = \underline{\underline{33.95 \text{ m}\Omega}}$$

$$\text{(b)} \quad I = V/R = 9 \times \frac{75\pi}{8} = \underline{\underline{265.1 \text{ A}}}$$

$$\text{(c)} \quad P = IV = \underline{\underline{2.386 \text{ kW}}}$$

Prob. 5.8

$$\text{(a)} \quad E = \frac{V}{l} = \frac{9}{100} = \underline{\underline{90 \text{ mV/m}}}$$

$$\text{(b)} \quad R = \frac{V}{I} = \frac{9}{0.3} = 30 \text{ }\Omega$$

$$R = \frac{l}{\sigma S} \longrightarrow \sigma = \frac{l}{RS} = \frac{100}{30(\pi \times 2^2) 10^{-6}} = \underline{\underline{2.653 \times 10^5 \text{ S/m}}}$$

Prob. 5.9

If R and S are the same,

$$R_1 = \frac{\ell_1}{\sigma_1 S} = R_2 = \frac{\ell_2}{\sigma_2 S} \longrightarrow \ell_1 = \ell_2 \frac{\sigma_1}{\sigma_2}$$

If 1 corresponds to copper and 2 to silver,

$$\sigma_1 = 5.8 \times 10^7 \text{ S/m}, \quad \sigma_2 = 6.1 \times 10^7 \text{ S/m}$$

$$\ell_1 = \ell_2 \frac{5.8}{6.1} = 0.951 \ell_2$$

That is, the copper wire is shorter than silver wire or the silver wire is longer.

Prob. 5.10

$$R = \frac{\ell}{\sigma S} = \frac{V}{I} \rightarrow \sigma = \frac{I \ell}{SV}, \quad S = \pi r^2$$

$$\sigma = \frac{2(5)}{\pi(2^2 \times 10^{-6})(12)} = \frac{10^7}{48\pi} = \underline{\underline{6.635 \times 10^4 \text{ S/m}}}$$

$$\text{Prob. 5.11 (a)} \quad S_i = \pi r_i^2 = \pi(1.5)^2 \times 10^{-4} = 7.068 \times 10^{-4}$$

$$S_o = \pi(r_o^2 - r_i^2) = \pi(4 - 2.25) \times 10^{-4} = 5.498 \times 10^{-4}$$

$$R_i = \frac{\rho l}{S_i} = \frac{11.8 \times 10^{-8} \times 10}{7.068 \times 10^{-4}} = 16.69 \times 10^{-4}$$

$$R_o = \frac{\rho_o l}{S_o} = \frac{1.77 \times 10^{-8} \times 10}{5.498 \times 10^{-4}} = 3.219 \times 10^{-4}$$

$$R = R_i // R_o = \frac{R_i R_o}{R_i + R_o} = \frac{16.69 \times 3.219 \times 10^{-4}}{16.69 + 3.219} = \underline{\underline{0.27 \text{ m}\Omega}}$$

$$(b) \quad V = I_i R_i = I_o R_o \longrightarrow \frac{I_i}{I_o} = \frac{R_o}{R_i} = \frac{0.3219}{1.669} = 0.1929$$

$$I_i + I_o = 1.1929 I_o = 60 \text{ A}$$

$$I_o = \underline{\underline{50.3 \text{ A}}} \quad (\text{copper}), \quad I_i = \underline{\underline{9.7 \text{ A}}} \quad (\text{steel})$$

Alternatively, using the principle of current division,

$$I_o = 60 \frac{R_i}{R_i + R_o} = 50.3 \text{ A}$$

$$I_i = 60 \frac{R_o}{R_i + R_o} = 9.7 \text{ A}$$

$$(c) \quad R = \frac{10 \times 1.77 \times 10^{-8}}{\pi(2^2) \times 10^{-4}} = \underline{\underline{0.141 \text{ m}\Omega}}$$

Prob. 5.12

From eq. (5.16),

$$R_1 = \frac{\rho_1 \ell}{S_1} = \frac{\rho_1 \ell}{ab}, \quad R_2 = \frac{\rho_2 \ell}{S_2} = \frac{\rho_2 \ell}{ac}$$

$$R = R_1 \square R_2 = \frac{R_1 R_2}{R_1 + R_2} = \frac{\frac{\rho_1 \ell}{ab} \frac{\rho_2 \ell}{ac}}{\frac{\rho_1 \ell}{ab} + \frac{\rho_2 \ell}{ac}} = \frac{\rho_1 \rho_2 \ell^2}{ac \rho_1 \ell + ab \rho_2 \ell}$$

$$R = \underline{\underline{\frac{\rho_1 \rho_2 \ell}{a(c \rho_1 + b \rho_2)}}}$$

Prob. 5.13

$$R = \frac{\ell}{\sigma S} = \frac{\ell}{\pi r^2 \sigma} = \frac{V}{I}$$

$$I = V \frac{\pi r^2 \sigma}{\ell} = \frac{12\pi(0.84 \times 10^{-3})^2 \times 6.1 \times 10^7}{12.4} = \underline{\underline{130.86 \text{ A}}}$$

Prob. 5.14

$$|\mathbf{P}| = n |\mathbf{p}| = nQd = 2ned = \chi_e \varepsilon_o E \quad (Q = 2e)$$

$$\chi_e = \frac{2ned}{\varepsilon_o E} = \frac{2 \times 5 \times 10^{25} \times 1.602 \times 10^{-19} \times 10^{-18}}{\frac{10^{-9}}{36\pi} \times 10^4} = 0.000182$$

$$\varepsilon_r = 1 + \chi_e = \underline{\underline{1.000182}}$$

Prob. 5.15

$$\mathbf{P} = \frac{\sum_{i=1}^N q_i \mathbf{d}_i}{V} = \frac{\sum_{i=1}^N \mathbf{p}_i}{V}$$

$$|\mathbf{P}| = \frac{N}{V} |\mathbf{p}| = 2 \times 10^{19} \times 1.8 \times 10^{-27} = 3.6 \times 10^{-8}$$

$$\mathbf{P} = |\mathbf{P}| \mathbf{a}_x = \underline{\underline{3.6 \times 10^{-8} \mathbf{a}_x \text{ C/m}^2}}$$

But $P = \chi_e \varepsilon_o E$ or $\chi_e = \frac{P}{\varepsilon_o E} = \frac{3.6 \times 10^{-8}}{(10^{-9} / 36\pi) 10^5} = 0.0407$

$$\varepsilon_r = 1 + \chi_e = \underline{\underline{1.0407}}$$

Prob. 5.16

$$\mathbf{E} = \frac{Q}{4\pi\varepsilon_o\varepsilon_r r^2} \mathbf{a}_r$$

$$\mathbf{P} = \chi_e \varepsilon_o \mathbf{E} = \frac{\chi_e Q}{4\pi\varepsilon_r r^2} \mathbf{a}_r = \frac{3(10)10^{-3}}{4\pi(4)1^2} \mathbf{a}_r = \underline{\underline{596.8 \mathbf{a}_r \text{ } \mu\text{C/m}^2}}$$

Prob. 5.17

$$\mathbf{P} = \chi_e \varepsilon_o \mathbf{E} \longrightarrow \mathbf{E} = \frac{\mathbf{P}}{\chi_e \varepsilon_o} = \frac{100 \times 10^{-9}}{2.5 \frac{10^{-9}}{36\pi} (2)} \mathbf{a}_\rho = \underline{\underline{2.261 \mathbf{a}_\rho \text{ kV/m}}}$$

$$\mathbf{D} = \varepsilon_o \varepsilon_r \mathbf{E} = 3.5 \times \frac{10^{-9}}{36\pi} 2.261 \times 10^3 \mathbf{a}_\rho = \underline{\underline{70 \mathbf{a}_\rho \text{ nC/m}^2}}$$

Prob. 5.18

$$\rho_{pv} = -\nabla \cdot \mathbf{P} = -\frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho_o \rho^2) = \underline{\underline{-2\rho_o}}$$

The surface polarization charge is

$$\rho_{ps} = \mathbf{P} \cdot \mathbf{a}_\rho \Big|_{\rho=a} = \underline{\underline{\rho_o a}}$$

Prob. 5.19

(a)

$$\begin{aligned}
Q_{s1} &= \int_S \mathbf{P} \cdot d\mathbf{S}, d\mathbf{S} = r^2 \sin \theta d\theta d\phi (-\mathbf{a}_r) \\
&= -\iint 4r r^2 \sin \theta d\theta d\phi \Big|_{r=1.2\text{cm}} \\
&= -4(1.2)^3 (10^{-6}) \int_0^{2\pi} d\phi \int_0^{\pi} \sin \theta d\theta (10^{-12}) \\
&= -6.912(2\pi)(2) \times 10^{-18} \\
&= \underline{\underline{-86.86 \times 10^{-18} \text{ C}}}
\end{aligned}$$

(b)

$$\begin{aligned}
Q_{s2} &= \int_S \mathbf{P} \cdot d\mathbf{S}, d\mathbf{S} = r^2 \sin \theta d\theta d\phi (\mathbf{a}_r) \\
&= \iint 4r r^2 \sin \theta d\theta d\phi \Big|_{r=2.6\text{cm}} \\
&= 4(2.6)^3 (10^{-6}) \int_0^{2\pi} d\phi \int_0^{\pi} \sin \theta d\theta (10^{-12}) \\
&= 4(2.6)^3 (2\pi)(2) \times 10^{-18} = \underline{\underline{883.5 \times 10^{-18} \text{ C}}}
\end{aligned}$$

(c)

$$\begin{aligned}
\rho_{pv} &= -\nabla \cdot \mathbf{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} (4r^3) \text{ pC/m}^3 = -12 \text{ pC/m}^3 \\
Q_v &= \int_v \rho_{pv} dv = -12 \iiint dv = -12 \int_0^{\pi} \sin \theta d\theta \int_0^{2\pi} d\phi \int_{1.2}^{2.6} r^2 dr (10^{-18}) \\
&= -12(2)(2\pi) \frac{r^3}{3} \Big|_{1.2}^{2.6} (10^{-18}) = -16\pi(2.6^3 - 1.2^3)(10^{-18}) \\
&= \underline{\underline{-796.61 \times 10^{-18} \text{ C}}}
\end{aligned}$$

Prob. 5.20

$$\begin{aligned}
\mathbf{D} &= \varepsilon_o \varepsilon_r \mathbf{E} = 2.1 \times \frac{10^{-9}}{36\pi} (6, 12, -20) = \underline{\underline{0.1114\mathbf{a}_x + 0.2228\mathbf{a}_y - 0.3714\mathbf{a}_z \text{ nC/m}^2}} \\
\mathbf{P} &= \chi_e \varepsilon_o \mathbf{E} = 1.1 \times \frac{10^{-9}}{36\pi} (6, 12, -20) = \underline{\underline{0.0584\mathbf{a}_x + 0.1167\mathbf{a}_y - 0.1945\mathbf{a}_z \text{ nC/m}^2}}
\end{aligned}$$

Prob. 5.21

$$\text{At } P(-2, 5, 3), \quad x = -2, y = 5, z = 3$$

$$V = 4(4)(5)(27) = \underline{\underline{2.16 \text{ kV}}}$$

$$\mathbf{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z\right) = -(8xyz^3\mathbf{a}_x + 4x^2z^3\mathbf{a}_y + 12x^2yz^2\mathbf{a}_z)$$

At P,

$$\mathbf{E} = -(-16)(5)(27)\mathbf{a}_x - 4(4)(27)\mathbf{a}_y - 12(4)(5)(9)\mathbf{a}_z = \underline{\underline{2.16\mathbf{a}_x - 0.432\mathbf{a}_y - 2.16\mathbf{a}_z \text{ kV/m}}}$$

$$\mathbf{P} = \chi_e \varepsilon_o \mathbf{E} = \frac{7 \times 10^{-9}}{36\pi} (2160, -432, -2160) = \underline{\underline{133.69\mathbf{a}_x - 26.74\mathbf{a}_y - 133.69\mathbf{a}_z \text{ nC/m}^2}}$$

Prob. 5.22

$$(a) \quad \mathbf{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z\right) = \underline{\underline{-20xyz\mathbf{a}_x - 10x^2z\mathbf{a}_y - 10(x^2y - z)\mathbf{a}_z \text{ V/m}}}$$

$$(b) \quad \mathbf{D} = \varepsilon \mathbf{E} = 5\varepsilon_o \mathbf{E} = \underline{\underline{-0.8842xyz\mathbf{a}_x - 0.4421x^2z\mathbf{a}_y - 0.4421(x^2y - z)\mathbf{a}_z \text{ nC/m}^2}}$$

$$(c) \quad \mathbf{P} = \chi_e \varepsilon_o \mathbf{E} = 4\varepsilon_o \mathbf{E} = \underline{\underline{-0.7073xyz\mathbf{a}_x - 0.3537x^2z\mathbf{a}_y - 0.3537(x^2y - z)\mathbf{a}_z \text{ nC/m}^2}}$$

$$(d) \quad \rho_v = -\varepsilon \nabla^2 V$$

$$\nabla^2 V = \frac{\partial}{\partial x}(20xyz) + \frac{\partial}{\partial y}(10x^2z) + \frac{\partial}{\partial z}(10x^2y - 10z) = 20yz - 10$$

$$\rho_v = -5\varepsilon_o 10(2yz - 1) = \underline{\underline{-0.8854yz + 0.4427 \text{ nC/m}^3}}$$

Prob. 5.23

Using Gauss' law,

$$Q_{enc} = \oint \mathbf{D} \cdot d\mathbf{S}$$

$$\text{For } r < a, \quad Q_{enc} = 0 \quad \rightarrow \quad \mathbf{E} = \mathbf{0} = \mathbf{D} = \mathbf{P}$$

$$\text{For } a < r < b, \quad Q_{enc} = 4C$$

$$4 = D_r(4\pi r^2) \quad \rightarrow \quad \mathbf{D} = \frac{4}{4\pi r^2} \mathbf{a}_r = \frac{\mathbf{a}_r}{\pi r^2}$$

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon} = \frac{\mathbf{D}}{2\varepsilon_0} = \frac{\mathbf{a}_r}{2\varepsilon_0\pi r^2}$$

$$\mathbf{P} = \chi_e \varepsilon_0 \mathbf{E} = (1)\varepsilon_0 \mathbf{E} = \frac{\mathbf{a}_r}{2\pi r^2}$$

$$\text{For } b < r < c, \quad Q_{enc} = 4 - 6 = -2C$$

$$-2 = D_r(4\pi r^2) \quad \rightarrow \quad \mathbf{D} = \frac{-2}{4\pi r^2} \mathbf{a}_r = -\frac{\mathbf{a}_r}{2\pi r^2}$$

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon} = \frac{\mathbf{D}}{5\varepsilon_0} = -\frac{\mathbf{a}_r}{10\varepsilon_0\pi r^2}$$

$$\mathbf{P} = \chi_e \varepsilon_0 \mathbf{E} = (4)\varepsilon_0 \mathbf{E} = -\frac{4\mathbf{a}_r}{10\pi r^2} = -\frac{2\mathbf{a}_r}{5\pi r^2}$$

$$\text{For } r > c, \quad Q_{enc} = 4 - 6 + 10 = 8C$$

$$8 = D_r(4\pi r^2) \quad \rightarrow \quad \mathbf{D} = \frac{8}{4\pi r^2} \mathbf{a}_r = \frac{2\mathbf{a}_r}{\pi r^2}$$

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon_0} = \frac{2\mathbf{a}_r}{\varepsilon_0\pi r^2}$$

$$\mathbf{P} = \chi_e \varepsilon_0 \mathbf{E} = (0)\varepsilon_0 \mathbf{E} = \mathbf{0}$$

Prob. 5.24 (a) Applying Coulomb's law, we obtain the electric field intensity due to a point charge as

$$E_r = \begin{cases} \frac{D_r}{\varepsilon_0} = \frac{Q}{4\pi\varepsilon_0 r^2}, & r > b \\ \frac{D_r}{\varepsilon} = \frac{Q}{4\pi\varepsilon r^2}, & a < r < b \end{cases}$$

$$P = \frac{\varepsilon_r - 1}{\varepsilon_r} D \quad (= D - \varepsilon_0 E)$$

Hence

$$P_r = \frac{\varepsilon_r - 1}{\varepsilon_r} \cdot \frac{Q}{4\pi r^2}, \quad a < r < b$$

$$(b) \quad \rho_{pv} = -\nabla \cdot \mathbf{P} = -\frac{1}{r^2} \frac{d}{dr} (r^2 P_r) = \underline{\underline{0}}$$

(c)

$$\rho_{ps} = \mathbf{P} \cdot (-\mathbf{a}_r) = -\frac{Q}{4\pi a^2} \left(\frac{\varepsilon_r - 1}{\varepsilon_r} \right), \quad r = a$$

$$\rho_{ps} = \mathbf{P} \cdot (\mathbf{a}_r) = -\frac{Q}{4\pi b^2} \left(\frac{\varepsilon_r - 1}{\varepsilon_r} \right), \quad r = b$$

Prob. 5.25

$$F_1 = \frac{Q_1 Q_2}{4\pi \varepsilon_o d^2} = 2.6 \text{ nN}, \quad F_2 = \frac{Q_1 Q_2}{4\pi \varepsilon_o \varepsilon_r d^2} = 1.5 \text{ nN}$$

$$\frac{F_1}{F_2} = \frac{2.6}{1.5} = \varepsilon_r = \underline{\underline{1.733}}$$

Prob. 5.26

(a) By Gauss's law,

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{enc} \quad \longrightarrow \quad D_r = \frac{Q}{4\pi r^2}$$

$$E_r = \frac{D_r}{\varepsilon} = \frac{Q}{4\pi \varepsilon r^2}$$

$$W = \int_v \frac{1}{2} \varepsilon |\mathbf{E}|^2 dv, \quad dv = r^2 \sin \theta dr d\phi d\theta$$

$$W = \frac{1}{2} \varepsilon \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=a}^{\infty} \frac{Q^2}{16\pi^2 \varepsilon^2 r^4} r^2 \sin \theta dr d\phi d\theta = \underline{\underline{\frac{Q^2}{8\pi \varepsilon a}}}$$

(b) D_r remains the same but

$$E_r = \frac{D_r}{\varepsilon} = \frac{Q}{4\pi r^2 \varepsilon_o \left(1 + \frac{a}{r}\right)^2} = \frac{Q}{4\pi \varepsilon_o (r+a)^2}$$

$$\begin{aligned} W &= \int_v \frac{1}{2} \varepsilon |E|^2 dv = \frac{1}{2} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=a}^{\infty} \frac{Q^2 r^2 \sin \theta dr d\theta d\phi}{16\pi^2 \varepsilon^2 (r+a)^4} \varepsilon_o \left(\frac{r+a}{r}\right)^2 \\ &= \frac{Q^2}{32\pi^2 \varepsilon_o} (4\pi) \int_a^{\infty} \frac{dr}{(r+a)^2} = \frac{Q^2}{8\pi \varepsilon_o} \left(-\frac{1}{r+a} \Big|_a^{\infty} \right) = \frac{Q^2}{8\pi \varepsilon_o} \frac{1}{2a} \end{aligned}$$

$$W = \frac{Q^2}{16a\pi \varepsilon_o}$$

Prob. 5.27

(a)

$$\rho_v = \begin{cases} \rho_o, & 0 < r < a \\ 0, & r > a \end{cases}$$

$$\text{For } r < a, \quad \varepsilon E_r (4\pi r^2) = \rho_o \frac{4\pi r^3}{3} \quad \longrightarrow \quad E_r = \frac{\rho_o r}{3\varepsilon}$$

$$V = -\int \mathbf{E} \cdot d\mathbf{l} = -\frac{\rho_o r^2}{6\varepsilon} + c_1$$

$$\text{For } r > a, \quad \varepsilon_o E_r (4\pi r^2) = \rho_o \frac{4\pi a^3}{3} \quad \longrightarrow \quad E_r = \frac{\rho_o a^3}{3\varepsilon_o r^2}$$

$$V = -\int \mathbf{E} \cdot d\mathbf{l} = \frac{\rho_o a^3}{3\varepsilon_o r} + c_2$$

As $r \longrightarrow \infty$, $V = 0$ and $c_2 = 0$

At $r = a$, $V(a^+) = V(a^-)$

$$-\frac{\rho_o a^2}{6\varepsilon_o \varepsilon_r} + c_1 = \frac{\rho_o a^2}{3\varepsilon_o} \quad \longrightarrow \quad c_1 = \frac{\rho_o a^2}{6\varepsilon_o \varepsilon_r} (2\varepsilon_r + 1)$$

$$V(r=0) = c_1 = \frac{\rho_o a^2 (2\varepsilon_r + 1)}{6\varepsilon_o \varepsilon_r}$$

$$(b) \quad V(r=a) = \frac{\rho_o a^2}{3\varepsilon_o}$$

Prob. 5.28

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \epsilon_o E_o \begin{bmatrix} 4 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$D_x = \epsilon_o E_o (4+1-1) = 4\epsilon_o E_o$$

$$D_y = \epsilon_o E_o (1+3-1) = 3\epsilon_o E_o$$

$$D_z = \epsilon_o E_o (1+1-2) = 0$$

$$\underline{\underline{D = \epsilon_o E_o (4\mathbf{a}_x + 3\mathbf{a}_y) \text{ C/m}^2}}$$

Prob. 5.29

Since $\frac{\partial \rho_v}{\partial t} = 0$, $\nabla \cdot \mathbf{J} = 0$ must hold.

(a) $\nabla \cdot \mathbf{J} = 6x^2y + 0 - 6x^2y = 0 \longrightarrow$ This is possible.

(b) $\nabla \cdot \mathbf{J} = y + (z+1) \neq 0 \longrightarrow$ This is not possible.

(c) $\nabla \cdot \mathbf{J} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (z^2) + \cos \phi \neq 0 \longrightarrow$ This is not possible.

(d) $\nabla \cdot \mathbf{J} = \frac{1}{r^2} \frac{\partial}{\partial r} (\sin \theta) = 0 \longrightarrow$ This is possible.

Prob. 5.30

$$\nabla \cdot \mathbf{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} = 2e^{-2y} \cos 2x - 2e^{-2y} \cos 2x + 1 = 1 = -\frac{\partial \rho_v}{\partial t}$$

Hence, $\frac{\partial \rho_v}{\partial t} = \underline{\underline{-1 \text{ C/m}^3 \text{ s}}}$

Prob. 5.31

(a) $\nabla \cdot \mathbf{J} = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\frac{100}{\rho} \right) = -\frac{100}{\rho^3}$

$$-\frac{\partial \rho_v}{\partial t} = \nabla \cdot \mathbf{J} = -\frac{100}{\rho^3} \longrightarrow \underline{\underline{\frac{\partial \rho_v}{\partial t} = \frac{100}{\rho^3} \text{ C/m}^3 \cdot \text{s}}}$$

$$(b) \quad I = \int \mathbf{J} \cdot d\mathbf{S} = \iint \frac{100}{\rho^2} \rho d\phi dz \Big|_{\rho=2} = \frac{100}{2} \int_0^{2\pi} d\phi \int_0^1 dz = 100\pi = \underline{\underline{314.16 \text{ A}}}$$

Prob. 5.32

$$\rho = \rho_{vo} e^{-t/T_r}$$

where ρ_{vo} is the initial value ($t=0$). When $t = 80 \mu s$, $\rho = \frac{1}{2} \rho_{vo}$. Then,

$$\frac{1}{2} \rho_{vo} = \rho_{vo} e^{-(80 \times 10^{-6})/T_r} \rightarrow (80 \times 10^{-6})/T_r = \ln 2$$

$$T_r = \frac{80 \times 10^{-6}}{\ln 2} = \underline{\underline{115.42 \mu s}}$$

$$T_r = \frac{\varepsilon}{\sigma} \rightarrow \sigma = \frac{\varepsilon}{T_r} = \frac{7.5 \times \frac{10^{-9}}{36\pi} \ln 2}{80 \times 10^{-6}} = \underline{\underline{5.746 \times 10^{-7} \text{ S/m}}}$$

Prob. 5.33

From the continuity equation,

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \quad (1)$$

$$\text{But } \mathbf{J} = \rho_v \mathbf{u}$$

Applying the vector identity

$$\nabla \cdot (\mathbf{V}\mathbf{A}) = \mathbf{V} \cdot \nabla \mathbf{A} + \mathbf{A} \cdot \nabla \mathbf{V}$$

$$\nabla \cdot \mathbf{J} = \rho_v (\nabla \cdot \mathbf{u}) + \mathbf{u} \cdot \nabla \rho_v \quad (2)$$

Substituting (2) into (1) gives

$$(\mathbf{u} \cdot \nabla) \rho_v + \rho_v (\nabla \cdot \mathbf{u}) + \frac{\partial \rho_v}{\partial t} = 0$$

as required.

Prob. 5.34

$$-\frac{\partial \rho_v}{\partial t} = \nabla \cdot \mathbf{J} = \frac{\partial J_x}{\partial x} = 0.5\pi \cos \pi x$$

At P(2,4,-3), $x=2$

$$\frac{\partial \rho_v}{\partial t} = -0.5\pi \cos(2\pi) = -0.5\pi = \underline{\underline{-1.571 \text{ C/m}^2\text{s}}}$$

Prob. 5.35

(a)

$$\frac{\varepsilon}{\sigma} = \frac{3.1 \times \frac{10^{-9}}{36\pi}}{10^{-15}} = \underline{\underline{2.741 \times 10^4 \text{ s}}}$$

$$(b) \quad \frac{\varepsilon}{\sigma} = \frac{6 \times \frac{10^{-9}}{36\pi}}{10^{-15}} = \underline{\underline{5.305 \times 10^4 \text{ s}}}$$

$$(c) \quad \frac{\varepsilon}{\sigma} = \frac{80 \times \frac{10^{-9}}{36\pi}}{10^{-4}} = \underline{\underline{7.07 \mu\text{s}}}$$

Prob. 5.36

$$T_r = \frac{\varepsilon}{\sigma} = \frac{2.5 \times 10^{-9}}{5 \times 10^{-6} \times 36\pi} = 4.42 \mu\text{s}$$

$$\rho_{vo} = \frac{Q}{V} = \frac{1}{\frac{4\pi}{3} \times 10^{-6} \times 8} = \underline{\underline{29.84 \text{ kC/m}^3}}$$

$$\rho_v = \rho_{vo} e^{-t/T_r} = 29.84 e^{-2/4.42} = \underline{\underline{18.98 \text{ kC/m}^3}}$$

Prob. 5.37

The normal component of a solenoidal vector field is continuous across an interface.

Hence, since $\nabla \cdot \mathbf{J} = 0$,

$$J_{1n} = J_{2n}$$

Similarly, the tangential component of a curl-free vector field is continuous.

Hence, since $\nabla \times (\mathbf{J}/\sigma) = 0$,

$$\frac{J_{1t}}{J_{2t}} = \frac{\sigma_1}{\sigma_2}$$

Alternatively,

$$E_{1n} = E_{2n} \quad \rightarrow \quad \frac{J_{1t}}{\sigma_1} = \frac{J_{2t}}{\sigma_2} \quad \text{or} \quad \frac{J_{1t}}{J_{2t}} = \frac{\sigma_1}{\sigma_2}$$

as required.

Prob. 5.38

(a)

$$\mathbf{P}_1 = \chi_{e1} \varepsilon_o \mathbf{E}_1 = 3 \times \frac{10^{-9}}{36\pi} (60, -100, 40) = \underline{\underline{1.591\mathbf{a}_x - 2.6526\mathbf{a}_y + 1.061\mathbf{a}_z \text{ nC/m}^2}}$$

(b)

$$\mathbf{E}_{2t} = \mathbf{E}_{1t} = 60\mathbf{a}_x - 100\mathbf{a}_y$$

$$\mathbf{D}_{2n} = \mathbf{D}_{1n} \rightarrow \varepsilon_2 \mathbf{E}_{2n} = \varepsilon_1 \mathbf{E}_{1n}$$

$$\mathbf{E}_{2n} = \frac{\varepsilon_1}{\varepsilon_2} \mathbf{E}_{1n} = \frac{4}{7.5} (40\mathbf{a}_z) = 21.33\mathbf{a}_z$$

$$\mathbf{E}_2 = 60\mathbf{a}_x - 100\mathbf{a}_y + 21.33\mathbf{a}_z$$

$$\mathbf{D}_2 = \varepsilon_o \varepsilon_{r2} \mathbf{E}_2 = 7.5 \times \frac{10^{-9}}{36\pi} (60, -100, 21.33) = \underline{\underline{3.979\mathbf{a}_x - 6.631\mathbf{a}_y + 1.414\mathbf{a}_z \text{ nC/m}^2}}$$

Prob. 5.39

(a)

$$\mathbf{E}_{2t} = \mathbf{E}_{1t} = -10\mathbf{a}_y + 8\mathbf{a}_z$$

$$\mathbf{D}_{2n} = \mathbf{D}_{1n} \rightarrow \varepsilon_2 \mathbf{E}_{2n} = \varepsilon_1 \mathbf{E}_{1n}$$

$$\mathbf{E}_{1n} = \frac{\varepsilon_2}{\varepsilon_1} \mathbf{E}_{2n} = \frac{2\varepsilon_o}{4\varepsilon_o} (6\mathbf{a}_x) = 3\mathbf{a}_x$$

$$\mathbf{E}_1 = 3\mathbf{a}_x - 10\mathbf{a}_y + 8\mathbf{a}_z$$

$$\mathbf{P}_1 = \chi_{e1} \varepsilon_o \mathbf{E}_1 = 3 \times \frac{10^{-9}}{36\pi} (3, -10, 8) = \underline{\underline{79.6\mathbf{a}_x - 265.3\mathbf{a}_y + 212.2\mathbf{a}_z \text{ nC/m}^2}}$$

$$\mathbf{P}_2 = \chi_{e2} \varepsilon_o \mathbf{E}_2 = 1 \times \frac{10^{-9}}{36\pi} (6, -10, 8) = \underline{\underline{53.05\mathbf{a}_x - 88.42\mathbf{a}_y + 70.74\mathbf{a}_z \text{ nC/m}^2}}$$

(b)

$$w_1 = \frac{1}{2} \varepsilon_1 E_1^2 = \frac{1}{2} (4\varepsilon_o)(9 + 100 + 64) = 346\varepsilon_o = \underline{\underline{3.0593 \text{ nJ/m}^2}}$$

$$w_2 = \frac{1}{2} \varepsilon_2 E_2^2 = \frac{1}{2} (2\varepsilon_o)(36 + 100 + 64) = 200\varepsilon_o = \underline{\underline{1.7684 \text{ nJ/m}^2}}$$

Prob. 5.40

$$f(x, y) = 4x + 3y - 10 = 0$$

$$\nabla f = 4\mathbf{a}_x + 3\mathbf{a}_y \longrightarrow \mathbf{a}_n = -\frac{\nabla f}{|\nabla f|} = \frac{-(4\mathbf{a}_x + 3\mathbf{a}_y)}{5} = -0.8\mathbf{a}_x - 0.6\mathbf{a}_y$$

The minus sign is chosen for $\mathbf{a}_n$ because it is directed toward the origin.

$$\mathbf{D}_{1n} = (\mathbf{D}_1 \cdot \mathbf{a}_n) \mathbf{a}_n = (1.6 - 2.4) \mathbf{a}_n = -0.64\mathbf{a}_x - 0.48\mathbf{a}_y$$

$$\mathbf{D}_{1t} = \mathbf{D}_1 - \mathbf{D}_{1n} = 2.64\mathbf{a}_x - 3.52\mathbf{a}_y + 6.5\mathbf{a}_z$$

$$\mathbf{D}_{2n} = \mathbf{D}_{1n} = -0.64\mathbf{a}_x - 0.48\mathbf{a}_y$$

$$\mathbf{E}_{2t} = \mathbf{E}_{1t} \longrightarrow \frac{\mathbf{D}_{2t}}{\varepsilon_2} = \frac{\mathbf{D}_{1t}}{\varepsilon_2}$$

$$\mathbf{D}_{2t} = \frac{\varepsilon_2}{\varepsilon_1} \mathbf{D}_{1t} = \frac{2.5}{1} (2.64, -3.52, 6.5) = (6.6, -8.8, 16.25)$$

$$\mathbf{D}_2 = \mathbf{D}_{2n} + \mathbf{D}_{2t} = \underline{\underline{5.96\mathbf{a}_x - 9.28\mathbf{a}_y + 16.25\mathbf{a}_z \text{ nC/m}^2}}$$

$$\theta_2 = \cos^{-1} \frac{\mathbf{D}_2 \cdot \mathbf{a}_n}{|\mathbf{D}_2|} = \underline{\underline{87.66^\circ}}$$

Prob. 5.41

(a)

$$\text{Let } f(x, y, z) = x + 2y + z - 1 = 0$$

$$\nabla f = \mathbf{a}_x + 2\mathbf{a}_y + \mathbf{a}_z \quad \rightarrow \quad \mathbf{a}_n = \frac{\nabla f}{|\nabla f|} = \frac{1}{\sqrt{6}} (\mathbf{a}_x + 2\mathbf{a}_y + \mathbf{a}_z)$$

$$\mathbf{E}_{1n} = (\mathbf{E}_1 \cdot \mathbf{a}_n) \mathbf{a}_n = \frac{1}{\sqrt{6}} (20 - 20 + 40) \frac{1}{\sqrt{6}} (\mathbf{a}_x + 2\mathbf{a}_y + \mathbf{a}_z) = \underline{\underline{6.667\mathbf{a}_x + 13.33\mathbf{a}_y + 6.667\mathbf{a}_z \text{ V/m}}}$$

$$\mathbf{E}_{1t} = \mathbf{E}_1 - \mathbf{E}_{1n} = \underline{\underline{13.3\mathbf{a}_x - 23.3\mathbf{a}_y + 33.3\mathbf{a}_z \text{ V/m}}}$$

(b)

$$\mathbf{E}_{2t} = \mathbf{E}_{1t} = 13.3\mathbf{a}_x - 23.3\mathbf{a}_y + 33.3\mathbf{a}_z$$

$$\mathbf{D}_{2n} = \mathbf{D}_{1n} \quad \rightarrow \quad \varepsilon_2 \mathbf{E}_{2n} = \varepsilon_1 \mathbf{E}_{1n}$$

$$\mathbf{E}_{2n} = \frac{\varepsilon_1}{\varepsilon_2} \mathbf{E}_{1n} = \frac{2}{5} (6.667, 13.33, 6.667) = 2.7\mathbf{a}_x + 5.3\mathbf{a}_y + 3.3\mathbf{a}_z$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n} = \underline{\underline{16\mathbf{a}_x - 18\mathbf{a}_y + 36\mathbf{a}_z \text{ V/m}}}$$

Prob. 5.42

$$(a) \quad \mathbf{P}_1 = \varepsilon_o \chi_{e1} \mathbf{E}_1 = 2 \times \frac{10^{-9}}{36\pi} (10, -6, 12) = \underline{\underline{0.1768\mathbf{a}_x - 0.1061\mathbf{a}_y + 0.2122\mathbf{a}_z \text{ nC/m}^2}}$$

$$(b) \quad \mathbf{E}_{1n} = -6\mathbf{a}_y, \quad \mathbf{E}_{2t} = \mathbf{E}_{1t} = 10\mathbf{a}_x + 12\mathbf{a}_z$$

$$\mathbf{D}_{2n} = \mathbf{D}_{1n} \quad \longrightarrow \quad \varepsilon_2 \mathbf{E}_{2n} = \varepsilon_1 \mathbf{E}_{1n}$$

$$\text{or} \quad \mathbf{E}_{2n} = \frac{\varepsilon_1}{\varepsilon_2} \mathbf{E}_{1n} = \frac{3\varepsilon_o}{4.5\varepsilon_o} (-6\mathbf{a}_y) = -4\mathbf{a}_y$$

$$\mathbf{E}_2 = \underline{\underline{10\mathbf{a}_x - 4\mathbf{a}_y + 12\mathbf{a}_z \text{ V/m}}}$$

$$\tan \theta_2 = \frac{E_{2t}}{E_{2n}} = \frac{\sqrt{10^2 + 12^2}}{4} = 3.905 \quad \longrightarrow \quad \underline{\underline{\theta_2 = 75.64^\circ}}$$

$$(c) \quad w_E = \frac{1}{2} \mathbf{D} \bullet \mathbf{E} = \frac{1}{2} \varepsilon |\mathbf{E}|^2$$

$$w_{E1} = \frac{1}{2} \varepsilon_1 |\mathbf{E}_1|^2 = \frac{1}{2} \times 3 \times \frac{10^{-9}}{36\pi} (10^2 + 6^2 + 12^2) = \underline{\underline{3.7136 \text{ nJ/m}^3}}$$

$$w_{E2} = \frac{1}{2} \varepsilon_2 |\mathbf{E}_2|^2 = \frac{1}{2} \times 4.5 \times \frac{10^{-9}}{36\pi} (10^2 + 4^2 + 12^2) = \underline{\underline{5.1725 \text{ nJ/m}^3}}$$

Prob. 5.43 (a) $\mathbf{D}_{2n} = 12\mathbf{a}_\rho = \mathbf{D}_{1n}, \quad \mathbf{D}_{2t} = -6\mathbf{a}_\phi + 9\mathbf{a}_z$

$$\mathbf{E}_{2t} = \mathbf{E}_{2t} \longrightarrow \frac{\mathbf{D}_{1t}}{\varepsilon_1} = \frac{\mathbf{D}_{2t}}{\varepsilon_2}$$

$$\mathbf{D}_{1t} = \frac{\varepsilon_1}{\varepsilon_2} \mathbf{D}_{2t} = \frac{3.5\varepsilon_o}{1.5\varepsilon_o} (-6\mathbf{a}_\phi + 9\mathbf{a}_z) = -14\mathbf{a}_\phi + 21\mathbf{a}_z$$

$$\underline{\underline{\mathbf{D}_1 = 12\mathbf{a}_\rho - 14\mathbf{a}_\phi + 21\mathbf{a}_z \text{ nC/m}^2}}$$

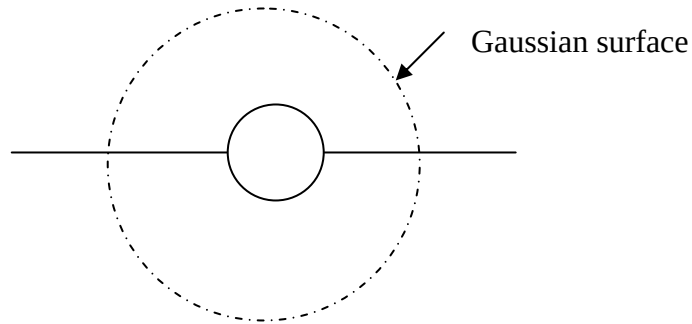
$$\mathbf{E}_1 = \mathbf{D}_1 / \varepsilon_1 = \frac{(12, -14, 21) \times 10^{-9}}{3.5 \times \frac{10^{-9}}{36\pi}} = 387.8\mathbf{a}_\rho - 452.4\mathbf{a}_\phi + 678.6\mathbf{a}_z$$

$$(b) \quad \mathbf{P}_2 = \varepsilon_o \chi_{e2} \mathbf{E}_2 = 0.5\varepsilon_o \frac{\mathbf{D}_2}{\varepsilon_2} = \frac{0.5\varepsilon_o}{1.5\varepsilon_o} (12, -6, 9) = \underline{\underline{4\mathbf{a}_\rho - 2\mathbf{a}_\phi + 3\mathbf{a}_z \text{ nC/m}^2}}$$

$$\rho_{v2} = \nabla \bullet \mathbf{P}_2 = 0$$

$$(c) \quad w_{E1} = \frac{1}{2} \mathbf{D}_1 \bullet \mathbf{E}_1 = \frac{1}{2} \frac{\mathbf{D}_1 \bullet \mathbf{D}_1}{\varepsilon_o \varepsilon_{r1}} = \frac{1}{2} \frac{(12^2 + 14^2 + 21^2) \times 10^{-18}}{3.5 \times \frac{10^{-9}}{36\pi}} = \underline{\underline{12.62 \text{ } \mu\text{J/m}^2}}$$

$$w_{E2} = \frac{1}{2} \frac{\mathbf{D}_2 \bullet \mathbf{D}_2}{\varepsilon_o \varepsilon_{r2}} = \frac{1}{2} \frac{(12^2 + 6^2 + 9^2) \times 10^{-18}}{1.5 \times \frac{10^{-9}}{36\pi}} = \underline{\underline{9.839 \text{ } \mu\text{J/m}^2}}$$

Prob. 5.44

$$Q = \int \mathbf{D} \cdot d\mathbf{S} = \varepsilon_1 E_r \frac{4\pi r^2}{2} + \varepsilon_2 E_r \frac{4\pi r^2}{2} = 2\pi r^2 (\varepsilon_1 + \varepsilon_2) E_r$$

$$E_r = \begin{cases} \frac{Q}{2\pi(\varepsilon_1 + \varepsilon_2)r^2}, & r > a \\ 0, & r < a \end{cases}$$

Prob. 5.45

$$E_{1t} = E_{2t} = 5 \cos \theta \mathbf{a}_\theta$$

$$D_{1n} = D_{2n} \rightarrow \varepsilon_1 E_{1n} = \varepsilon_2 E_{2n}$$

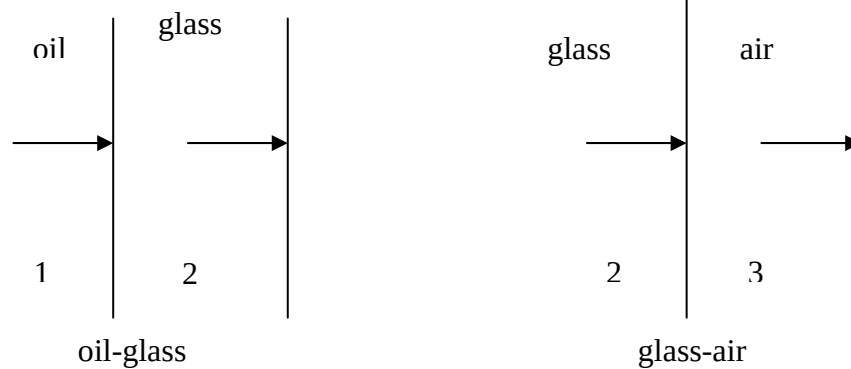
$$E_{1n} = \frac{\varepsilon_2}{\varepsilon_1} E_{2n} = \frac{6\varepsilon_o}{2\varepsilon_o} (10 \sin \theta) \mathbf{a}_r = 30 \sin \theta \mathbf{a}_r$$

$$E_{1t} = E_{1t} + E_{1n} = \underline{30 \sin \theta \mathbf{a}_r + 5 \cos \theta \mathbf{a}_\theta}$$

$$D_1 = \varepsilon_1 E_1 = 2\varepsilon_o E_1 = \underline{\underline{\varepsilon_o (60 \sin \theta \mathbf{a}_r + 10 \cos \theta \mathbf{a}_\theta)}}$$

Prob. 5.46

(a) The two interfaces are shown below



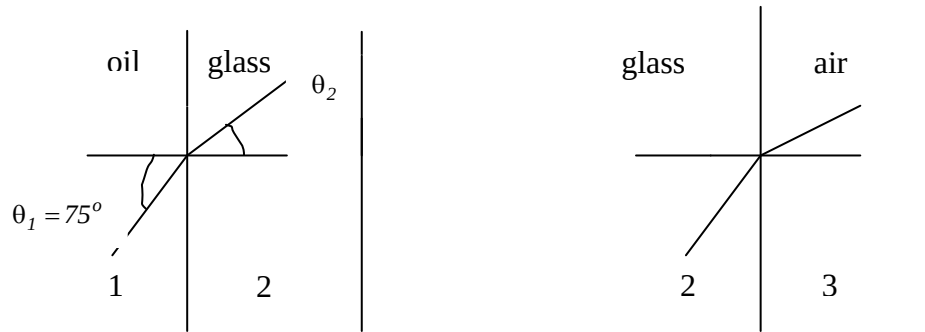
$$E_{1n} = 2000, \quad E_{1t} = 0 = E_{2t} = E_{3t}$$

$$D_{1n} = D_{2n} = D_{3n} \longrightarrow \epsilon_1 E_{1n} = \epsilon_2 E_{2n} = \epsilon_3 E_{3n}$$

$$E_{2n} = \frac{\epsilon_1}{\epsilon_2} E_{1n} = \frac{3.0}{8.5} (2000) = \underline{\underline{705.9 \text{ V/m}}}, \quad \theta_2 = 0^\circ$$

$$E_{3n} = \frac{\epsilon_1}{\epsilon_3} E_{1n} = \frac{3.0}{1.0} (2000) = \underline{\underline{6000 \text{ V/m}}}, \quad \theta_3 = 0^\circ$$

(b)



$$E_{1n} = 2000 \cos 75^\circ = 517.63, \quad E_{1t} = 2000 \sin 75^\circ = E_{2t} = E_{3t} = 1931.85$$

$$E_{2n} = \frac{\epsilon_1}{\epsilon_2} E_{1n} = \frac{3}{8.5} (517.63) = 182.7, \quad E_{3n} = \frac{\epsilon_1}{\epsilon_3} E_{1n} = \frac{3}{1} (517.63) = 1552.89$$

$$E_2 = \sqrt{E_{2n}^2 + E_{2t}^2} = 1940.5, \quad \theta_2 = \tan^{-1} \frac{E_{2t}}{E_{2n}} = \underline{\underline{84.6^\circ}},$$

$$E_3 = \sqrt{E_{3n}^2 + E_{3t}^2} = 2478.6, \quad \theta_3 = \tan^{-1} \frac{E_{3t}}{E_{3n}} = \underline{\underline{51.2^\circ}}$$

Prob. 5.47

$$\begin{aligned} \rho_s = D_n = \epsilon_o E &= \frac{10^{-9}}{36\pi} \sqrt{30^2 + 40^2 + 20^2} \times 10^{-3} = \frac{\sqrt{2900}}{36\pi} \text{ pC/m}^2 \\ &= \underline{\underline{0.476 \text{ pC/m}^2}} \end{aligned}$$

$$\text{Prob. 5.48 (a)} \quad \rho_s = D_n = \epsilon_o E_n = \frac{10^{-9}}{36\pi} \sqrt{15^2 + 8^2} = \underline{\underline{0.1503 \text{ nC/m}^2}}$$

$$(b) \quad D_n = \rho_s = -20 \text{ nC/m}^2$$

$$D = D_n a_n = (-20 \text{ nC})(-a_y) = \underline{\underline{20 a_y \text{ nC} / \text{m}^2}}$$

Prob. 5.49

At the interface between ϵ_o and $2\epsilon_o$,

$$E_{1n} = E_o \cos 30^\circ, \quad E_{1t} = E_o \sin 30^\circ$$

$$E_{2t} = E_{1t} = 0.5 E_o$$

$$D_{2n} = D_{1n} \longrightarrow \epsilon_1 E_{1n} = \epsilon_2 E_{2n}$$

$$E_{2n} = \frac{\epsilon_1}{\epsilon_2} E_{1n} = \frac{\epsilon_o}{2\epsilon_o} (0.866 E_o) = 0.433 E_o$$

The angle **E** makes with the z-axis is

$$\theta_1 = \tan^{-1} \frac{E_{2t}}{E_{2n}} = \tan^{-1} \frac{0.5}{0.433} = \underline{\underline{49.11^\circ}}$$

At the interface between $2\epsilon_o$ and $3\epsilon_o$,

$$E_{3t} = E_{2t} = 0.5 E_o$$

$$D_{3n} = D_{2n} \longrightarrow E_{3n} = \frac{\epsilon_2}{\epsilon_3} E_{2n} = \frac{2\epsilon_o}{3\epsilon_o} (0.433 E_o) = 0.2887 E_o$$

The angle **E** makes with the z-axis is

$$\theta_2 = \tan^{-1} \frac{E_{3t}}{E_{3n}} = \tan^{-1} \frac{0.5}{0.2887} = \underline{\underline{60^\circ}}$$

At the interface between $3\epsilon_o$ and ϵ_o ,

$$E_{4t} = E_{3t} = 0.5 E_o$$

$$D_{4n} = D_{3n} \longrightarrow E_{4n} = \frac{\epsilon_3}{\epsilon_4} E_{3n} = \frac{3\epsilon_o}{\epsilon_o} (0.2887 E_o) = 0.866 E_o$$

The angle **E** makes with the z-axis is

$$\theta_3 = \tan^{-1} \frac{E_{4t}}{E_{4n}} = \tan^{-1} \frac{0.5}{0.866} = \underline{\underline{30^\circ}}$$

CHAPTER 6

P. E. 6.1

$$\nabla^2 V = -\frac{\rho}{\epsilon} \longrightarrow \frac{d^2 V}{dx^2} = -\frac{\rho_o x}{\epsilon a}$$

$$V = -\frac{\rho_o x^3}{6\epsilon a} + Ax + B$$

$$\mathbf{E} = -\frac{dV}{dx} \mathbf{a}_x = \left(\frac{\rho_o x^2}{2\epsilon a} - A \right) \mathbf{a}_x$$

If $\mathbf{E} = 0$ at $x=0$, then

$$0 = 0 - A \longrightarrow A = 0$$

If $V = 0$ at $x=a$, then

$$0 = -\frac{\rho_o a^3}{6\epsilon a} + B \longrightarrow B = \frac{\rho_o a^2}{6\epsilon}$$

Thus

$$\underline{\underline{V = \frac{\rho_o}{6\epsilon a}(a^3 - x^3)}}, \quad \underline{\underline{\mathbf{E} = \frac{\rho_o x^2}{2\epsilon a} \mathbf{a}_x}}$$

$$\mathbf{P. E. 6.2} \quad V_1 = A_1 x + B_1, \quad V_2 = A_2 x + B_2$$

$$V_1(x=d) = V_o = A_1 d + B_1 \longrightarrow B_1 = V_o - A_1 d$$

$$V_1(x=0) = 0 = 0 + B_2 \longrightarrow B_2 = 0$$

$$V_1(x=a) = V_2(x=a) \longrightarrow aA_1 + B_1 = A_2 a$$

$$D_{1n} = D_{2n} \longrightarrow \epsilon_1 A_1 = \epsilon_2 A_2 \longrightarrow A_2 = \frac{\epsilon_1}{\epsilon_2} A_1$$

$$A_1 a + V_o - A_1 d = \frac{\epsilon_1}{\epsilon_2} a A_1 \longrightarrow V_o = A_1 \left(-a + d + \frac{\epsilon_1}{\epsilon_2} a \right)$$

or

$$A_1 = \frac{V_o}{d - a + \epsilon_1 a / \epsilon_2}, \quad A_2 = \frac{\epsilon_1}{\epsilon_2} A_1 \frac{\epsilon_1 V_o}{\epsilon_2 d - \epsilon_2 a + \epsilon_1 a}$$

Hence

$$\underline{\underline{\mathbf{E}_1 = -A_1 \mathbf{a}_x = \frac{-V_o \mathbf{a}_x}{d - a + \epsilon_1 a / \epsilon_2}}}, \quad \underline{\underline{\mathbf{E}_2 = -A_2 \mathbf{a}_x = \frac{-V_o \mathbf{a}_x}{a + \epsilon_2 d / \epsilon_1 - \epsilon_2 a / \epsilon_1}}}$$

P. E. 6.3 From Example 6.3,

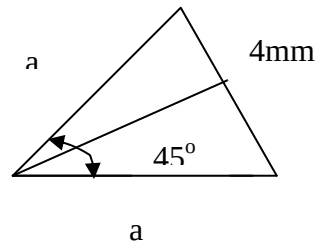
$$\mathbf{E} = -\frac{V_o}{\rho\phi_o} \mathbf{a}_\phi, \quad D = \epsilon_o E$$

$$\rho_s = D_n(\phi = 0) = -\frac{V_o \epsilon_o}{\rho\phi_o}$$

The charge on the plate $\phi = 0$ is

$$Q = \int \rho_s dS = -\frac{V_o \epsilon_o}{\phi_o} \int_{z=0}^L \int_{\rho=a}^b \frac{1}{\rho} dz d\rho = -\frac{V_o \epsilon_o}{\phi_o} L \ln(b/a)$$

$$C = \frac{|Q|}{V_o} = \frac{\epsilon_o L}{\phi_o} \ln \frac{b}{a}$$



$$a \sin \frac{45^\circ}{2} = 2 \quad \longrightarrow \quad a = \frac{2}{\sin 22.5^\circ} = 5.226 \text{ mm}$$

$$C = \frac{1.5 \times \frac{10^{-9}}{36\pi}}{\frac{\pi}{4}} 5 \ln \frac{1000}{5.226} = 444 \text{ pF}$$

$$Q = CV_o = 444 \times 10^{-12} \times 50 \text{ C} = \underline{\underline{22.2 \text{ nC}}}$$

P. E. 6.4 From Example 6.4,

$$V_o = 50, \quad \theta_2 = 45^\circ, \quad \theta_1 = 90^\circ, \quad r = \sqrt{3^2 + 4^2 + 2^2} = \sqrt{29}, \quad \theta = \tan^{-1} \frac{\rho}{z} =$$

$$\tan^{-1} \frac{5}{2} \quad \longrightarrow \quad \theta = 68.2^\circ; \quad \tan 45^\circ = 1$$

$$V = \frac{50 \ln(\tan 34.1^\circ)}{\ln(\tan 22.5^\circ)} = \underline{\underline{22.125 \text{ V}}},$$

$$\mathbf{E} = \frac{50 \mathbf{a}_\theta}{\sqrt{29} \sin 68.2^\circ \ln(\tan 22.5^\circ)} = \underline{\underline{11.35 \mathbf{a}_\theta \text{ V/m}}}$$

P. E. 6.5

From Example 6.5,

$$V(x,y) = \frac{4V_o}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{\sin(n\pi x / b) \sinh[n\pi y / b]}{n \sinh(n\pi a / b)}$$

$$\begin{aligned} \mathbf{E} &= -\nabla V = -\frac{\partial V}{\partial x} \mathbf{a}_x - \frac{\partial V}{\partial y} \mathbf{a}_y \\ &= -\frac{4V_o}{b} \sum_{n=\text{odd}}^{\infty} \frac{1}{\sinh n\pi a/b} \left[\cos(n\pi x/b) \sinh(n\pi y/b) \mathbf{a}_x + \sin(n\pi x/b) \cosh(n\pi y/b) \mathbf{a}_y \right] \end{aligned}$$

(a) At $(x,y) = (a, a/2)$,

$$V = \frac{400}{\pi} (0.3775 - 0.0313 + 0.00394 - 0.000585 + \dots) = \underline{\underline{44.51 \text{ V}}}$$

$$\begin{aligned} \mathbf{E} &= 0 \mathbf{a}_x + (-115.12 + 19.127 - 3.9411 + 0.8192 - 0.1703 + 0.035 - 0.0074 + \dots) \mathbf{a}_y \\ &= \underline{\underline{-99.25 \mathbf{a}_y \text{ V/m}}} \end{aligned}$$

(b) At $(x,y) = (3a/2, a/4)$,

$$V = \frac{400}{\pi} (0.1238 + 0.006226 - 0.00383 + 0.0000264 + \dots) = \underline{\underline{16.50 \text{ V}}}$$

$$\begin{aligned} \mathbf{E} &= (24.757 - 3.7358 - 0.3834 + 0.0369 + 0.00351 - 0.00033 + \dots) \mathbf{a}_x \\ &\quad + (-66.25 - 4.518 + 0.3988 + 0.03722 - 0.00352 - 0.000333 + \dots) \mathbf{a}_y \\ &= \underline{\underline{20.68 \mathbf{a}_x - 70.34 \mathbf{a}_y \text{ V/m}}} \end{aligned}$$

P. E. 6.6

$$V(y=a) = V_o \sin(7\pi x / b) = \sum_{n=1}^{\infty} c_n \sin(n\pi x / b) \sinh(n\pi a / b)$$

By equating coefficients, we notice that $c_n = 0$ for $n \neq 7$. For $n=7$,

$$V_o \sin(7\pi x / b) = c_7 \sin(7\pi x / b) \sinh(7\pi a / b) \longrightarrow c_7 = \frac{V_o}{\sinh(7\pi a / b)}$$

Hence

$$V(x,y) = \underline{\underline{\frac{V_o}{\sinh(7\pi a / b)} \sin(7\pi x / b) \sinh(7\pi y / b)}}$$

P. E. 6.7 Let $V(r, \theta, \phi) = R(r)F(\theta)\Phi(\phi)$.

Substituting this in Laplace's equation in spherical coordinates gives

$$\frac{\Phi F}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{R\Phi}{r^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dF}{d\theta} \right) + \frac{RF}{r^2 \sin^2 \theta} \frac{d^2 \Phi}{d\phi^2} = 0$$

Dividing by $RF\Phi / r^2 \sin^2 \theta$ gives

$$\frac{\sin^2 \theta}{R} \frac{d}{dr} (r^2 R') + \frac{\sin \theta}{F} \frac{d}{d\theta} (\sin \theta F') = - \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = \lambda^2$$

$$\underline{\underline{\Phi'' + \lambda^2 \Phi = 0}}$$

$$\frac{1}{R} \frac{d}{dr} (r^2 R') + \frac{1}{F \sin \theta} \frac{d}{d\theta} (\sin \theta F') = \lambda^2 / \sin^2 \theta$$

$$\frac{1}{R} \frac{d}{dr} (r^2 R') = \frac{\lambda^2}{\sin^2 \theta} - \frac{1}{F \sin \theta} \frac{d}{d\theta} (\sin \theta F') = \mu^2$$

$$2rR' + r^2 R'' = \mu^2 R$$

or

$$\underline{\underline{R'' + \frac{2}{r} R' - \frac{\mu^2}{r^2} R = 0}}$$

$$\frac{\sin \theta}{F} \frac{d}{d\theta} (\sin \theta F') - \lambda^2 + \mu^2 \sin^2 \theta = 0$$

or

$$\underline{\underline{F'' + \cos \theta F' + (\mu^2 \sin \theta - \lambda^2 \csc \theta) F = 0}}$$

P. E. 6.8 (a) This is similar to Example 6.8(a) except that here $0 < \phi < 2\pi$ instead of $0 < \phi < \pi/2$. Hence

$$I = \frac{2\pi t V_o \sigma}{\ln(b/a)} \quad \text{and} \quad R = \frac{V_o}{I} = \frac{\ln \frac{b}{a}}{\underline{\underline{2\pi t \sigma}}}$$

(b) This is similar to Example 6.8(b) except that here $0 < \phi < 2\pi$. Hence

$$I = \frac{V_o \sigma}{t} \int_a^b \int_0^{2\pi} \rho d\rho d\phi = \frac{V_o \sigma \pi (b^2 - a^2)}{t}$$

and $R = \frac{V_o}{I} = \frac{t}{\sigma \pi (b^2 - a^2)}$

P. E. 6.9 From Example 6.9,

$$J_1 = \frac{\sigma_1 V_o}{\rho \ln \frac{b}{a}}, \quad J_2 = \frac{\sigma_2 V_o}{\rho \ln \frac{b}{a}}$$

$$I = \int J \cdot dS = \int_{z=0}^L \left[\int_{\phi=0}^{\pi} J_1 \rho d\phi + \int_{\phi=\pi}^{2\pi} J_2 \rho d\phi \right] dz = \frac{V_o l}{\ln \frac{b}{a}} [\pi \sigma_1 + \pi \sigma_2]$$

$$R = \frac{V_o}{I} = \frac{\ln \frac{b}{a}}{\pi l [\sigma_1 + \sigma_2]}$$

P. E. 6.10 (a) $C = \frac{4\pi\epsilon}{\frac{1}{a} - \frac{1}{b}}$, C_1 and C_2 are in series.

$$C_1 = 4\pi \times \frac{10^{-9}}{36\pi} \left(\frac{2.5}{\frac{10^3}{2} - \frac{10^3}{3}} \right) = 5/3 \text{ pF}, \quad C_2 = 4\pi \times \frac{10^{-9}}{36\pi} \left(\frac{3.5}{\frac{10^3}{1} - \frac{10^3}{2}} \right) = 7/9 \text{ pF}$$

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{(5/3)(7/9)}{(5/3) + (7/9)} = \underline{\underline{0.53 \text{ pF}}}$$

(b) $C = \frac{2\pi\epsilon}{\frac{1}{a} - \frac{1}{b}}$, C_1 and C_2 are in parallel.

$$C_1 = 2\pi \times \frac{10^{-9}}{36\pi} \left(\frac{2.5}{\frac{10^3}{1} - \frac{10^3}{3}} \right) = 5/24 \text{ pF}, \quad C_2 = 2\pi \times \frac{10^{-9}}{36\pi} \left(\frac{3.5}{\frac{10^3}{1} - \frac{10^3}{3}} \right) = 7/24 \text{ pF}$$

$$C = C_1 + C_2 = \underline{\underline{0.5 \text{ pF}}}$$

P. E. 6.11 As in Example 6.8, the solution of Laplace's equation yields
 $V(\rho) = A \ln \rho + B$

Using the boundary conditions $V(\rho = a) = 0$, $V(\rho = b) = V_o$,

$$0 = A \ln a + B$$

$$V_o = A \ln b + B$$

Solving this yields

$$V = V_o \frac{\ln \rho / a}{\ln b / a}, \quad \mathbf{E} = -\nabla V = -\frac{V_o}{\rho \ln b / a} \mathbf{a}_\rho$$

$$Q = \int \epsilon \mathbf{E} \cdot d\mathbf{S} = \frac{V_o \epsilon}{\ln b / a} \int_{z=0}^L \int_{\phi=0}^{2\pi} \frac{1}{\rho} dz \rho d\phi = \frac{V_o 2\pi \epsilon L}{\ln b / a}$$

$$C = \frac{Q}{V_o} = \frac{2\pi \epsilon L}{\ln b / a}$$

P. E. 6.12

(a) Let C_1 and C_2 be capacitances per unit length of each section and C_T be the total capacitance of 10m length. C_1 and C_2 are in series.

$$C_1 = \frac{2\pi \epsilon_{r1} \epsilon_o}{\ln b / c} = \frac{2\pi \times 2.5 \times 10^{-9}}{\ln 3 / 2 \times 36\pi} = 342.54 \text{ pF/m},$$

$$C_2 = \frac{2\pi \epsilon_{r2} \epsilon_o}{\ln c / a} = \frac{2\pi \times 3.5 \times 10^{-9}}{\ln 2 \times 36\pi} = 280.52 \text{ pF/m}$$

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{342.54 \times 280.52}{342.54 + 280.52} = 154.22 \text{ pF}$$

$$C_T = Cl = \underline{\underline{1.54}} \text{ nF}$$

(b) C_1 and C_2 are in parallel.

$$C = C_1 + C_2 = \frac{\pi \epsilon_{r1} \epsilon_o}{\ln b / a} + \frac{\pi \epsilon_{r2} \epsilon_o}{\ln b / a} = \frac{\pi (\epsilon_{r1} + \epsilon_{r2}) \epsilon_o}{\ln b / a} = \frac{6\pi \times 10^{-9}}{\ln 3 \times 36\pi} = 151.7 \text{ pF/m}$$

$$C_T = Cl = \underline{\underline{1.52}} \text{ nF}$$

P. E. 6.13 Instead of Eq. (6.31), we now have

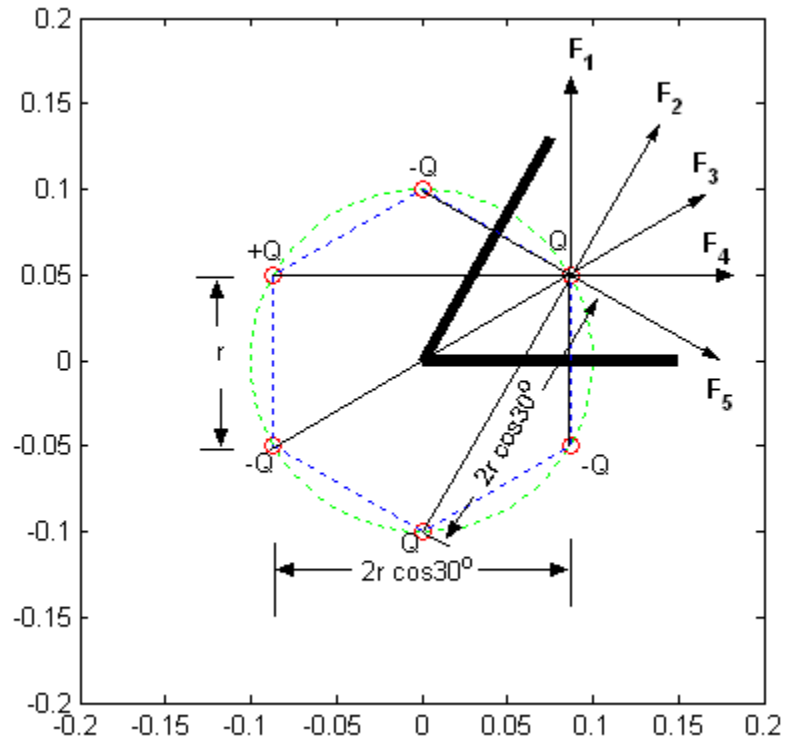
$$V = -\int_b^a \frac{Q dr}{4\pi \epsilon r^2} = -\int_b^a \frac{Q dr}{4\pi \frac{10\epsilon_o}{r} r^2} = -\frac{Q}{40\pi \epsilon_o} \ln b / a$$

$$C = \frac{Q}{|V|} = \frac{40\pi}{\ln 4 / 1.5} \frac{10^{-9}}{36\pi} = \underline{\underline{1.13 \text{ nF}}}$$

P. E. 6.14 Let

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \mathbf{F}_4 + \mathbf{F}_5$$

where $\mathbf{F}_i$, $i = 1, 2, \dots, 5$ are shown on in the figure below.



$$\begin{aligned} \mathbf{F} &= -\frac{Q^2}{4\pi\epsilon_0 r^2} \mathbf{a}_y + \frac{Q^2(\mathbf{a}_x \sin 30^\circ + \mathbf{a}_y \cos 30^\circ)}{4\pi\epsilon_0 (2r \cos 30^\circ)^2} - \frac{Q^2(\mathbf{a}_x \cos 30^\circ + \mathbf{a}_y \sin 30^\circ)}{4\pi\epsilon_0 (2r)^2} \\ &\quad + \frac{Q^2 \mathbf{a}_x}{4\pi\epsilon_0 (2r \cos 30^\circ)^2} - \frac{Q^2(\mathbf{a}_x \cos 30^\circ - \mathbf{a}_y \sin 30^\circ)}{4\pi\epsilon_0 r^2} \\ &= \frac{Q^2}{4\pi\epsilon_0 r^2} \left[-\mathbf{a}_y + \frac{1}{3} \left(\frac{\mathbf{a}_x}{2} + \frac{\sqrt{3}\mathbf{a}_y}{2} \right) - \frac{1}{4} \left(\frac{\sqrt{3}\mathbf{a}_x}{2} + \frac{\mathbf{a}_y}{2} \right) + \frac{1}{3} \mathbf{a}_x - \frac{\sqrt{3}\mathbf{a}_x}{2} + \frac{\mathbf{a}_y}{2} \right] \end{aligned}$$

$$= 9 \times 10^{-5} \left[\mathbf{a}_x \left(\frac{1}{2} - \frac{5\sqrt{3}}{8} \right) + \mathbf{a}_y \left(\frac{-5}{8} + \frac{\sqrt{3}}{6} \right) \right] = -52.4279 \mathbf{a}_x - 30.27 \mathbf{a}_y \text{ } \mu\text{N}$$

$$\underline{|\mathbf{F}| = 60.54 \text{ } \mu\text{N}}$$

Note that the force tends to pull Q toward the origin.

Prob. 6.1

(a)

$$\begin{aligned} \mathbf{E} &= -\nabla V = - \left(\frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \right) \\ &= -(15x^2 y^2 z \mathbf{a}_x + 10x^3 y z \mathbf{a}_y + 5x^3 y^2 \mathbf{a}_z) \end{aligned}$$

At P, $x=-3$, $y=1$, $z=2$,

$$\mathbf{E} = -15(9)(1)(2) \mathbf{a}_x + 10(-27)(1)(2) \mathbf{a}_y - 5(-27)(1) \mathbf{a}_z = \underline{\underline{-270 \mathbf{a}_x + 540 \mathbf{a}_y + 135 \mathbf{a}_z \text{ V/m}}}$$

(b) $\rho_v = \nabla \cdot \mathbf{D} \quad \text{or} \quad \rho_v = -\epsilon \nabla^2 V$

$$\begin{aligned} \nabla^2 V &= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = \frac{\partial}{\partial x} (15x^2 y^2 z) + \frac{\partial}{\partial y} (10x^3 y z) + \frac{\partial}{\partial z} (5x^3 y^2) \\ &= 30xy^2 z + 10x^3 z \end{aligned}$$

At P,

$$\rho_v = -\epsilon \nabla^2 V = -2.25 \times \frac{10^{-9}}{36\pi} [30(-3)(1)(2) + 10(-27)(2)] = \underline{\underline{14.324 \text{ nC/m}^3}}$$

Prob. 6.2

(a)

$$-\mathbf{E} = \nabla V = -\frac{20}{r^3} \cos \theta \sin \phi \mathbf{a}_r - \frac{10}{r^3} \sin \theta \sin \phi \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{10}{r^2} \cos \theta \cos \phi \mathbf{a}_\phi$$

At P(1, 60°, 30°), $r=1$, $\theta=60^\circ$, $\phi=30^\circ$

$$\begin{aligned} \mathbf{E} &= \frac{20}{1^3} \cos 60^\circ \sin 30^\circ \mathbf{a}_r + \frac{10}{1^3} \sin 60^\circ \sin 30^\circ \mathbf{a}_\theta - \frac{1}{\sin 60^\circ} \frac{10 \cos 60^\circ \cos 30^\circ}{1^3} \mathbf{a}_\phi \\ &= \underline{\underline{5 \mathbf{a}_r + 4.33 \mathbf{a}_\theta - 5 \mathbf{a}_\phi \text{ V/m}}} \end{aligned}$$

(b)

$$\begin{aligned}
 \nabla^2 V &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(\frac{-20 \cos \theta \sin \phi}{r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\frac{-10 \sin^2 \theta \sin \phi}{r^2} \right) \\
 &\quad - \frac{1}{r^2 \sin^2 \theta} \frac{10 \cos \theta \sin \phi}{r^2} \\
 &= \frac{20 \cos \theta \sin \phi}{r^4} - \frac{20 \sin \theta \cos \theta \sin \phi}{r^4 \sin \theta} - \frac{10 \cos \theta \sin \phi}{r^4 \sin^2 \theta} \\
 &= -\frac{10 \cos \theta \sin \phi}{r^4 \sin^2 \theta}
 \end{aligned}$$

$$\nabla^2 V = -\frac{\rho_v}{\epsilon} \longrightarrow \rho_v = -\epsilon \nabla^2 V = \frac{10 \epsilon_o \cos \theta \sin \phi}{r^4 \sin^2 \theta}$$

At P, $r=1$, $\theta=60^\circ$, $\phi=30^\circ$

$$\rho_v = 10 \times \frac{10^{-9} \cos 60^\circ \sin 30^\circ}{36\pi \cdot 1^4 \sin^2 60^\circ} = \underline{\underline{29.47 \text{ pC/m}^3}}$$

Prob. 6.3

$$\nabla^2 V = -\frac{\rho_v}{\epsilon}$$

$$\frac{d^2 V}{dy^2} = -\frac{y}{4\pi} \cdot \frac{10^{-9}}{4\epsilon_o} = -\frac{y}{4\pi} \cdot \frac{10^{-9}}{4 \frac{10^{-9}}{36\pi}} = -2.25y$$

$$\frac{dV}{dy} = -2.25 \frac{y^2}{2} + B$$

$$V = -0.375y^3 + By + C$$

$$V(1) = 0 = -0.375 + B + C \quad (1)$$

$$V(3) = 50 = -10.125 + 3B + C \quad (2)$$

From (1) and (2), $B=29.875$ and $C=-29.5$

$$V = -0.375y^3 + 29.875y - 29.5$$

$$V(2) = \underline{\underline{27.25 \text{ V}}}$$

Prob. 6.4

$$\mathbf{E} = -\nabla V = -\frac{dV}{d\rho} \mathbf{a}_\rho = -(0.8)(10)\rho^{-0.2} \mathbf{a}_\rho = -8(0.6)^{-0.2} \mathbf{a}_\rho = \underline{\underline{-8.861 \mathbf{a}_\rho}}$$

$$\nabla^2 V = \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dV}{d\rho} \right) = -\frac{\rho_v}{\epsilon_o} = \frac{1}{\rho} \frac{d}{d\rho} (8\rho^{0.8}) = \frac{1}{\rho} (8)0.8\rho^{-0.2} = 6.4\rho^{-1.2}$$

$$\rho_v = -\epsilon_o \nabla^2 V = -6.4 \times \frac{10^{-9}}{36\pi} \rho^{-1.2} = -\frac{6.4(0.6)^{-1.2}}{36\pi} \text{ nC/m}^3 = \underline{\underline{-0.1044 \text{ nC/m}^3}}$$

Prob. 6.5

$$\nabla^2 V = \frac{d^2 V}{dx^2} = -\frac{\rho_v}{\epsilon} = -\frac{50(1-y^2) \times 10^{-6}}{\epsilon} = -k(1-y^2)$$

where $k = \frac{50 \times 10^{-6}}{3 \times \frac{10^{-9}}{36\pi}} = 600\pi \times 10^3$

$$\frac{dV}{dy} = -k(y - y^3/3) + A$$

$$V = -k \left(\frac{y^2}{2} - \frac{y^4}{12} \right) + Ay + B = 50\pi \cdot 10^3 y^4 - 300\pi \cdot 10^3 y^2 + Ay + B$$

When $y=2\text{cm}$, $V=30 \times 10^3$,

$$30 \times 10^3 = 50\pi \times 10^3 \times 16 \times 10^{-6} - 300\pi \times 10^3 \times 4 \times 10^{-4} + Ay + B$$

or

$$30,376.77 = 0.02A + B \quad (1)$$

When $y=-2\text{cm}$, $V=30 \times 10^3$,

$$30,376.77 = -0.02A + B \quad (2)$$

From (1) and (2), $A=0$, $B=30,376.77$. Thus,

$$\underline{\underline{V = 157.08y^4 - 942.5y^2 + 30.377 \text{ kV}}}$$

Prob. 6.6

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon} \quad \longrightarrow \quad \frac{d^2 V}{dz^2} = -\frac{\rho_o z}{\varepsilon d}$$

$$\frac{dV}{dz} = -\frac{\rho_o z^2}{2\varepsilon d} + A$$

$$V = -\frac{\rho_o z^3}{6\varepsilon d} + Az + B$$

$$z = 0, V = 0 \quad \longrightarrow \quad 0 = 0 + B, \quad \text{i.e. } B = 0$$

$$z = d, V = V_o \quad \longrightarrow \quad V_o = -\frac{\rho_o d^2}{6\varepsilon} + Ad$$

$$A = \frac{V_o}{d} + \frac{\rho_o d}{6\varepsilon}$$

Hence,

$$V = \underline{\underline{-\frac{\rho_o z^3}{6\varepsilon d} + \left(\frac{V_o}{d} + \frac{\rho_o d}{6\varepsilon}\right)z}}$$

Prob. 6.7

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon} = \frac{-\frac{10}{\rho} \times 10^{-12}}{3.6 \times \frac{10^{-9}}{36\pi}} = -\frac{0.1\pi}{\rho}$$

Let $\alpha = 0.1\pi$.

$$\nabla^2 V = -\frac{\alpha}{\rho} = \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dV}{d\rho} \right)$$

$$-\alpha = \frac{d}{d\rho} \left(\rho \frac{dV}{d\rho} \right)$$

$$\rho \frac{dV}{d\rho} = -\alpha\rho + A$$

$$\frac{dV}{d\rho} = -\alpha + \frac{A}{\rho}$$

$$V = -\alpha\rho + A \ln \rho + B$$

$$\text{At } \rho=2, V=0 \longrightarrow 0 = -2\alpha + A \ln 2 + B \quad (1)$$

$$\text{At } \rho=5, V=60 \longrightarrow 60 = -5\alpha + A \ln 5 + B \quad (2)$$

Subtracting (1) from (2),

$$60 = -3\alpha + A \ln 5/2 \longrightarrow A = \frac{60 + 3\alpha}{\ln 2.5} = 66.51$$

From (1),

$$B = 2\alpha - A \ln 2 = -45.473$$

$$E = -\frac{dV}{d\rho} \mathbf{a}_\rho = \left(\alpha - \frac{A}{\rho} \right) \mathbf{a}_\rho = \left(0.3142 - \frac{66.51}{\rho} \right) \mathbf{a}_\rho$$

Prob. 6.8

$$\frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dV}{d\rho} \right) = -\frac{\rho_o}{\epsilon}$$

$$\frac{d}{d\rho} \left(\rho \frac{dV}{d\rho} \right) = -\frac{\rho_o \rho}{\epsilon}$$

Integrating gives

$$\rho \frac{dV}{d\rho} = -\frac{\rho_o}{\epsilon} \frac{\rho^2}{2} + A \quad \rightarrow \quad \frac{dV}{d\rho} = -\frac{\rho_o}{2\epsilon} \rho + \frac{A}{\rho}$$

Integrating again,

$$V = -\frac{\rho_o \rho^2}{4\epsilon} + A \ln \rho + B$$

where A and B are integration constants.

Prob. 6.9

$$\nabla^2 V = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = -\frac{\rho_v}{\epsilon} = -\frac{10 \times 10^{-9}}{6r \times \frac{10^{-9}}{36\pi}} = -\frac{60\pi}{r}$$

$$\frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = -60\pi r \quad \rightarrow \quad r^2 \frac{dV}{dr} = -30\pi r^2 + A$$

$$\frac{dV}{dr} = -30\pi + \frac{A}{r^2} \quad \rightarrow \quad V = -30\pi r - \frac{A}{r} + B$$

$$V(r=1) = 0 \quad \rightarrow \quad 0 = -30\pi - A + B \quad (1)$$

$$V(r=4) = 50 \quad \rightarrow \quad 50 = -120\pi - A/4 + B \quad (2)$$

Solving (1) and (2) yields $A = 443.66$, $B = 537.91$

Thus,

$$V = -30\pi r - \frac{443.66}{r} + 537.91$$

$$V(r=2) = -60\pi - \frac{443.66}{2} + 537.91 = \underline{\underline{127.58 \text{ V}}}$$

Prob. 6.10

(a)

$$\nabla^2 V_1 = \frac{\partial^2 V_1}{\partial x^2} + \frac{\partial^2 V_1}{\partial y^2} + \frac{\partial^2 V_1}{\partial z^2} = 0 + 0 - 2 \neq 0$$

It does not satisfy Laplace's equation.

(b)

$$\nabla^2 V_2 = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left[\rho (-\rho^{-2} 10 \sin \phi) \right] + \frac{1}{\rho^2} \left(-\frac{10 \sin \phi}{\rho} \right) = \frac{10}{\rho} (\rho^{-2}) \sin \phi - \frac{10 \sin \phi}{\rho^3} = 0$$

It does satisfy Laplace's equation.

(c)

$$\begin{aligned} \nabla^2 V_3 &= \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 (-r^{-2} 5 \sin \theta) \right] + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left[\sin \theta \frac{(5 \cos \theta)}{r^2} \right] \\ &= 0 + \frac{5}{r^3 \sin \theta} (1 - 2 \sin^2 \theta) \neq 0 \end{aligned}$$

It does not satisfy Laplace's equation.**Prob. 6.11**

$$\nabla^2 U = \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = 6xy + 0 + 2c = 0$$

$$\underline{\underline{c = -3xy}}$$

Prob. 6.12

$$(a) \quad \frac{\partial V}{\partial x} = 4xyz, \quad \frac{\partial^2 V}{\partial x^2} = 4yz$$

$$\frac{\partial V}{\partial y} = 2x^2 z - 3y^2 z, \quad \frac{\partial^2 V}{\partial y^2} = -6yz$$

$$\frac{\partial V}{\partial z} = 2x^2 y - y^3, \quad \frac{\partial^2 V}{\partial z^2} = 0$$

$$\nabla^2 V = 4yz - 6yz + 0 = -2yz$$

 $\nabla^2 V \neq 0$, V does not satisfy Laplace's equation.

(b)

$$\nabla^2 V = -\frac{\rho_v}{\epsilon} = -2yz \quad \longrightarrow \quad \rho_v = 2yz\epsilon$$

$$Q = \int \rho_v dv = \int_0^1 \int_0^1 \int_0^1 (2yz\epsilon) dx dy dz = 2\epsilon \left(1 \right) \frac{y^2}{2} \left| \frac{1}{0} \frac{z^2}{2} \right| \frac{1}{0} = \epsilon / 2 = 2\epsilon_o / 2 = \epsilon_o$$

$$\underline{\underline{Q = 8.854 \text{ pC}}}$$

Prob. 6.13

$$\nabla^2 V = \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dV}{d\rho} \right) = 0 \quad \rightarrow \quad V = A \ln \rho + B$$

Let $a = 1 \text{ cm}$, $b = 1.5 \text{ cm}$, $V_o = 50 \text{ V}$

$$V(\rho = b) = 0 \quad \rightarrow \quad 0 = A \ln b + B \quad \text{or} \quad B = -A \ln b$$

$$V(\rho = a) = V_o \quad \rightarrow \quad V_o = A \ln a - A \ln b = A \ln \frac{a}{b} \quad \text{or} \quad A = \frac{V_o}{\ln \frac{a}{b}}$$

$$V = A \ln \rho - A \ln b = A \ln \frac{\rho}{b}$$

$$\mathbf{E} = -\nabla V = -\frac{dV}{d\rho} \mathbf{a}_\rho = -\frac{A}{\rho} \mathbf{a}_\rho = -\frac{V_o}{\rho \ln \frac{a}{b}} \mathbf{a}_\rho$$

$$\rho_s = D_n = \varepsilon E_n = \frac{\varepsilon_o \varepsilon_r V_o}{a \ln \frac{b}{a}} = \frac{50(4) \frac{10^{-9}}{36\pi}}{10^{-2} \ln 1.5} = \frac{400(50)}{36\pi \ln 1.5} \text{ nC/m}^2 = \underline{\underline{436.14 \text{ nC/m}^2}}$$

Prob. 6.14

$$\nabla^2 V = \frac{d^2 V}{dz^2} = 0 \quad \longrightarrow \quad V = Az + B$$

When $z=0$, $V = 0 \quad \longrightarrow \quad B=0$

When $z=d$, $V = V_o \quad \longrightarrow \quad V_o = Ad \quad \text{or} \quad A = V_o/d$

Hence,

$$V = \frac{V_o z}{d}$$

$$\mathbf{E} = -\nabla V = -\frac{dV}{dz} \mathbf{a}_z = -\frac{V_o}{d} \mathbf{a}_z$$

$$\mathbf{D} = \varepsilon \mathbf{E} = -\varepsilon_o \varepsilon_r \frac{V_o}{d} \mathbf{a}_z$$

Since $V_o = 50 \text{ V}$ and $d = 2 \text{ mm}$,

$$\underline{V = 25z \text{ kV}}, \quad \underline{\underline{\mathbf{E} = -25 \mathbf{a}_z \text{ kV/m}}}$$

$$\mathbf{D} = -\frac{10^{-9}}{36\pi} (1.5) 25 \times 10^3 \mathbf{a}_z = \underline{\underline{-332 \mathbf{a}_z \text{ nC/m}^2}}$$

$$\rho_s = D_n = \underline{\underline{\pm 332 \text{ nC/m}^2}}$$

The surface charge density is positive on the plate at $z=d$ and negative on the plate at $z=0$.

Prob. 6.15 From Example 6.8, solving $\nabla^2 V = 0$ when $V = V(\rho)$ leads to

$$V = \frac{V_o \ln \rho / a}{\ln b / a} = V_o \frac{\ln(a / \rho)}{\ln(a / b)}$$

$$\mathbf{E} = -\nabla V = -\frac{V_o}{\rho \ln b / a} \mathbf{a}_\rho = \frac{V_o}{\rho \ln a / b} \mathbf{a}_\rho, \quad \mathbf{D} = \epsilon \mathbf{E} = -\frac{\epsilon_o \epsilon_r V_o}{\rho \ln b / a} \mathbf{a}_\rho$$

$$\rho_s = D_n = \pm \frac{\epsilon_o \epsilon_r V_o}{\rho \ln b / a} \Big|_{\rho=a,b}$$

In this case, $V_o = 100 \text{ V}$, $b = 5 \text{ mm}$, $a = 15 \text{ mm}$, $\epsilon_r = 2$. Hence at $\rho = 10 \text{ mm}$,

$$V = \frac{100 \ln(10/15)}{\ln(5/15)} = \underline{\underline{36.91 \text{ V}}}$$

$$\mathbf{E} = \frac{100}{10 \times 10^{-3} \ln 3} \mathbf{a}_\rho = \underline{\underline{9.102 \mathbf{a}_\rho \text{ kV/m}}}$$

$$\mathbf{D} = 9.102 \times 10^3 \times \frac{10^{-9}}{36\pi} 2 \mathbf{a}_\rho = \underline{\underline{161 \mathbf{a}_\rho \text{ nC/m}^2}}$$

$$\rho_s(\rho = 5 \text{ mm}) = \frac{10^{-9}}{36\pi} (2) \frac{10^5}{5 \ln 3} = \underline{\underline{322 \text{ nC/m}^2}}$$

$$\rho_s(\rho = 15 \text{ mm}) = -\frac{10^{-9}}{36\pi} (2) \frac{10^5}{15 \ln 3} = \underline{\underline{-107.3 \text{ nC/m}^2}}$$

Prob. 6.16

$$\frac{1}{\rho} \frac{d^2 V}{d\phi^2} = 0 \quad \longrightarrow \quad \frac{d^2 V}{d\phi^2} = 0 \quad \longrightarrow \quad \frac{dV}{d\phi} = A$$

$$V = A\phi + B$$

$$0 = 0 + B \quad \longrightarrow \quad B = 0$$

$$50 = A\pi/2 \quad \longrightarrow \quad A = \frac{100}{\pi}$$

$$\mathbf{E} = -\nabla V = -\frac{1}{\rho} \frac{dV}{d\phi} \mathbf{a}_\phi = -\frac{A}{\rho} \mathbf{a}_\phi = \underline{\underline{-\frac{100}{\pi\rho} \mathbf{a}_\phi}}$$

Prob. 6.17

(a)

$$\frac{\partial V}{\partial \rho} = V_o \left(1 + \frac{a^2}{\rho^2}\right) \sin \phi$$

$$\rho \frac{\partial V}{\partial \rho} = V_o \left(\rho + \frac{a^2}{\rho}\right) \sin \phi$$

$$\frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = V_o \left(1 - \frac{a^2}{\rho^2}\right) \sin \phi$$

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = V_o \left(\frac{1}{\rho} - \frac{a^2}{\rho^3}\right) \sin \phi$$

$$\frac{\partial^2 V}{\partial \phi^2} = -V_o \left(\rho - \frac{a^2}{\rho}\right) \sin \phi$$

$$\frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} = -V_o \left(\frac{1}{\rho} - \frac{a^2}{\rho^3}\right) \sin \phi$$

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{\partial^2 V}{\partial \phi^2} = 0$$

(b)

If $\rho^2 \gg a^2$, then $\frac{a^2}{\rho^2} \ll 1$ and $V \approx V_o \rho \sin \phi$

$$\mathbf{E} = -\nabla V = -\frac{\partial V}{\partial \rho} \mathbf{a}_\rho - \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi = \underline{\underline{-V_o \sin \phi \mathbf{a}_\rho - V_o \cos \phi \mathbf{a}_\phi}}$$

Prob. 6.18

$$\nabla^2 V = \frac{d^2 V}{dx^2} = 0 \quad \longrightarrow \quad V = Ax + B$$

At $x = 20 \text{ mm} = 0.02 \text{ m}$, $V = 0$

$$0 = 0.02A + B \quad \longrightarrow (1)$$

$$\mathbf{E} = -100 \mathbf{a}_x - \frac{dV}{dx} \mathbf{a}_x \quad \longrightarrow A = 110 \longrightarrow (2)$$

From (1) $B = -0.02A = -2.2$

Then $V = 110x - 2.2$

At $x = 0$ $V = -2.2 \text{ V}$

At $x = 50 \text{ mm} = 0.05 \text{ m}$,

$$V = 110 \times 0.05 - 2.2 = \underline{\underline{3.3 \text{ V}}}$$

Prob. 6.19

$$\nabla^2 V = 0 \longrightarrow V = -A/r + B$$

$$\text{At } r=0.5, V=-50 \longrightarrow -50 = -A/0.5 + B$$

Or

$$-50 = -2A + B \quad (1)$$

$$\text{At } r = 1, V = 50 \longrightarrow 50 = -A + B \quad (2)$$

From (1) and (2), $A = 100$, $B = 150$, and

$$\underline{\underline{V = -\frac{100}{r} + 150}}$$

$$\underline{\underline{E = -\nabla V = -\frac{A}{r^2} \mathbf{a}_r = -\frac{100}{r^2} \mathbf{a}_r \text{ V/m}}}$$

Prob. 6.20 From Example 6.4,

$$V = \frac{V_o \ln\left(\frac{\tan \theta / 2}{\tan \theta_1 / 2}\right)}{\ln\left(\frac{\tan \theta_2 / 2}{\tan \theta_1 / 2}\right)}$$

$$V_o = 100, \quad \theta_1 = 30^\circ, \quad \theta_2 = 120^\circ, \quad r = \sqrt{3^2 + 0^2 + 4^2} = 5, \quad \theta = \tan^{-1} \rho / z = \tan^{-1} 3 / 4 = 36.87^\circ$$

$$V = 100 \frac{\ln\left(\frac{\tan 18.435^\circ}{\tan 15^\circ}\right)}{\ln\left(\frac{\tan 60^\circ}{\tan 15^\circ}\right)} = \underline{\underline{117 \text{ V}}}$$

$$\mathbf{E} = -\frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta = \frac{-V_o \mathbf{a}_\theta}{r \ln\left(\frac{\tan \theta_2}{\tan \theta_1}\right)} \frac{\frac{\sec^2 \theta / 2}{\tan \theta_1 / 2}}{\frac{\tan \theta / 2}{\tan \theta_1 / 2}} = \frac{-V_o \mathbf{a}_\theta}{r \ln\left(\frac{\tan \theta_2}{\tan \theta_1}\right) 2 \sin(\theta / 2) \cos(\theta / 2)}$$

$$\mathbf{E} = \frac{-V_o \mathbf{a}_\theta}{r \sin \theta \ln \left(\frac{\tan \theta_2 / 2}{\tan \theta_1 / 2} \right)} = \frac{-100 \mathbf{a}_\theta}{5 \sin 36.87^\circ \ln 6.464} = \underline{\underline{-17.86 \mathbf{a}_\theta \text{ V/m}}}$$

Prob. 6.21

(a)

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = 0 \quad \longrightarrow \quad V = A \ln \rho + B$$

$$V(\rho = b) = 0 \quad \longrightarrow \quad 0 = A \ln b + B \quad \longrightarrow \quad B = -A \ln b$$

$$V(\rho = a) = V_o \quad \longrightarrow \quad V_o = A \ln a / b \quad \longrightarrow \quad A = -\frac{V_o}{\ln b / a}$$

$$V = -\frac{V_o}{\ln b / a} \ln \rho / b = \frac{V_o \ln b / \rho}{\ln b / a}$$

$$V(\rho = 15 \text{ mm}) = 70 \frac{\ln 2}{\ln 50} = \underline{\underline{12.4 \text{ V}}}$$

(b) As the electron decelerates, potential energy gained = K.E. loss

$$e[70 - 12.4] = \frac{1}{2} m [(10^7)^2 - u^2] \quad \longrightarrow \quad 10^{14} - u^2 = \frac{2e}{m} \times 57.6$$

$$u^2 = 10^{14} - \frac{2 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}} \times 57.6 = 10^{12} (100 - 20.25)$$

$$\underline{\underline{u = 8.93 \times 10^6 \text{ m/s}}}$$

Prob. 6.22 This is similar to case 1 of Example 6.5.

$$X = c_1 x + c_2, \quad Y = c_3 y + c_4$$

$$\text{But } X(0) = 0 \quad \longrightarrow \quad 0 = c_2, \quad Y(0) = 0 \quad \longrightarrow \quad 0 = c_4$$

Hence,

$$V(x, y) = XY = a_o xy, \quad a_o = c_1 c_3$$

$$\text{Also, } V(xy = 4) = 20 \quad \longrightarrow \quad 20 = 4a_o \quad \longrightarrow \quad a_o = 5$$

Thus,

$$V(x, y) = 5xy \text{ and } \mathbf{E} = -\nabla V = -5y \mathbf{a}_x - 5x \mathbf{a}_y$$

At $(x,y) = (1,2)$,

$$\underline{\underline{V = 10 \text{ V}, \quad \mathbf{E} = -10\mathbf{a}_x - 5\mathbf{a}_y \text{ V/m}}}$$

Prob. 6.23 (a) As in Example 6.5, $X(x) = A \sin(n\pi x / b)$

For Y,

$$Y(y) = c_1 \cosh(n\pi y / b) + c_2 \sinh(n\pi y / b)$$

$$Y(a) = 0 \longrightarrow 0 = c_1 \cosh(n\pi a / b) + c_2 \sinh(n\pi a / b) \longrightarrow c_1 = -c_2 \tanh(n\pi a / b)$$

$$V = \sum_{n=1}^{\infty} a_n \sin(n\pi x / b) [\sinh(n\pi y / b) - \tanh(n\pi a / b) \cosh(n\pi y / b)]$$

$$V(x, y = 0) = V_o = -\sum_{n=1}^{\infty} a_n \tanh(n\pi a / b) \sin(n\pi x / b)$$

$$-a_n \tanh(n\pi a / b) = \frac{2}{b} \int_0^b V_o \sin(n\pi x / b) dx = \begin{cases} \frac{4V_o}{n\pi}, & n = \text{odd} \\ 0, & n = \text{even} \end{cases}$$

Hence,

$$\begin{aligned} V &= -\frac{4V_o}{\pi} \sum_{n=\text{odd}} \sin(n\pi x / b) \left[\frac{\sinh(n\pi y / b)}{n \tanh(n\pi a / b)} - \frac{\cosh(n\pi y / b)}{n} \right] \\ &= -\frac{4V_o}{\pi} \sum_{n=\text{odd}} \frac{\sin(n\pi x / b)}{n \sinh(n\pi a / b)} [\sinh(n\pi y / b) \cosh(n\pi a / b) - \cosh(n\pi y / b) \sinh(n\pi a / b)] \\ &= \frac{4V_o}{\pi} \sum_{n=\text{odd}} \frac{\sin(n\pi x / b) \sinh[n\pi(a - y) / b]}{n \sinh(n\pi a / b)} \end{aligned}$$

Alternatively, for Y

$$Y(y) = c_1 \sinh n\pi(y - c_2) / b$$

$$Y(a) = 0 \longrightarrow 0 = c_1 \sinh[n\pi(a - c_2) / b] \longrightarrow c_2 = a$$

$$V = \sum_{n=1}^{\infty} b_n \sin(n\pi x / b) \sinh[n\pi(y - a) / b]$$

where

$$b_n = \begin{cases} -\frac{4V_o}{n\pi \sinh(n\pi a / b)}, & n = \text{odd} \\ 0, & n = \text{even} \end{cases}$$

(b) This is the same as Example 6.5 except that we exchange y and x. Hence

$$\underline{\underline{V(x,y) = \frac{4V_o}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{\sin(n\pi y / a) \sinh(n\pi x / a)}{n \sinh(n\pi b / a)}}}$$

(c) This is the same as part (a) except that we must exchange x and y. Hence

$$\underline{\underline{V(x,y) = \frac{4V_o}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{\sin(n\pi y / a) \sinh[n\pi(b-x) / a]}{n \sinh(n\pi b / a)}}}$$

Prob. 6.24 (a) $X(x)$ is the same as in Example 6.5. Hence

$$V(x,y) = \sum_{n=1}^{\infty} \sin(n\pi x / b) [a_n \sinh(n\pi y / b) + b_n \cosh(n\pi y / b)]$$

At $y=0$, $V = V_1$

$$V_1 = \sum_{n=1}^{\infty} b_n \sin(n\pi x / b) \longrightarrow b_n = \begin{cases} \frac{4V_1}{n\pi}, & n = \text{odd} \\ 0, & n = \text{even} \end{cases}$$

At $y=a$, $V = V_2$

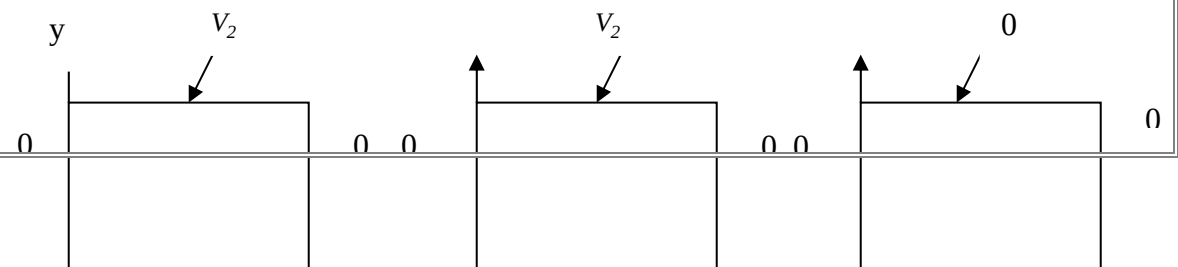
$$V_2 = \sum_{n=1}^{\infty} \sin(n\pi x / b) [a_n \sinh(n\pi a / b) + b_n \cosh(n\pi a / b)]$$

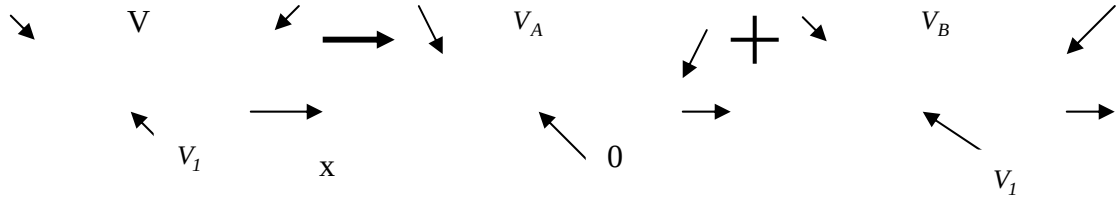
$$a_n \sinh(n\pi a / b) + b_n \cosh(n\pi a / b) = \begin{cases} \frac{4V_2}{n\pi}, & n = \text{odd} \\ 0, & n = \text{even} \end{cases}$$

or

$$a_n = \begin{cases} \frac{4V_2}{n\pi \sinh(n\pi a / b)} (V_2 - V_1 \cosh(n\pi a / b)), & n = \text{odd} \\ 0, & n = \text{even} \end{cases}$$

Alternatively, we may apply superposition principle.





i.e. $V = V_A + V_B$

V_A is exactly the same as Example 6.5 with $V_o = V_2$, while V_B is exactly the same as Prob. 6.19(a). Hence

$$V = \frac{4}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{\sin(n\pi x / b)}{n \sinh(n\pi a / b)} [V_1 \sinh[n\pi(a - y) / b] + V_2 \sinh(n\pi y / b)]$$

(b)

$$V(x, y) = (a_1 e^{-\alpha x} + a_2 e^{+\alpha x})(a_3 \sin \alpha y + a_4 \cos \alpha y)$$

$$\lim_{x \rightarrow \infty} V(x, y) = 0 \quad \longrightarrow \quad a_2 = 0$$

$$V(x, y = 0) = 0 \quad \longrightarrow \quad a_4 = 0$$

$$V(x, y = a) = 0 \quad \longrightarrow \quad \alpha = n\pi / a, \quad n = 1, 2, 3, \dots$$

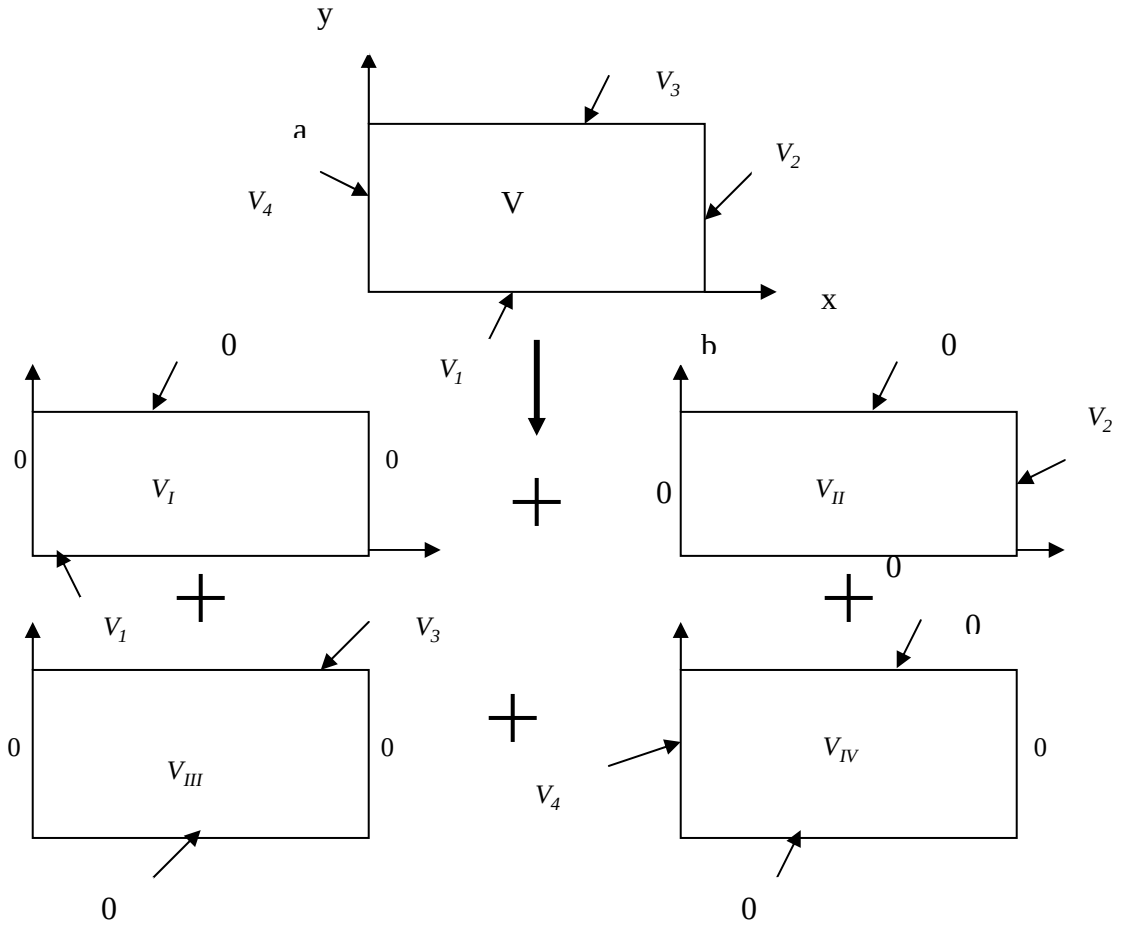
Hence,

$$V(x, y) = \sum_{n=1}^{\infty} a_n e^{-n\pi x / a} \sin(n\pi y / a)$$

$$V(x = 0, y) = V_o = \sum_{n=1}^{\infty} a_n \sin(n\pi y / a) \quad \longrightarrow \quad a_n = \begin{cases} \frac{4V_o}{n\pi}, & n = \text{odd} \\ 0, & n = \text{even} \end{cases}$$

$$V(x, y) = \frac{4V_o}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{\sin(n\pi y / a)}{n} \exp(-n\pi x / a)$$

(c) The problem is easily solved using superposition theorem, as illustrated below.



Therefore,

$$V = V_I + V_{II} + V_{III} + V_{IV}$$

$$= \frac{4}{\pi} \sum_{n=\text{odd}} \frac{1}{n} \left\{ \frac{\sin(n\pi x/b)}{\sinh(n\pi a/b)} [V_1 \sinh(n\pi(a-y)/b) + V_3 \sinh(n\pi y/b)] \right. \\ \left. + \frac{\sin(n\pi x/a)}{\sinh(n\pi b/a)} [V_2 \sinh(n\pi y/a) + V_4 \sinh(n\pi(b-x)/a)] \right\}$$

where

$$V_I = \frac{4V_1}{\pi} \sum_{n=\text{odd}} \frac{\sin(n\pi x/b) \sinh[n\pi(a-y)/b]}{n \sinh(n\pi a/b)}$$

$$V_{II} = \frac{4V_2}{\pi} \sum_{n=\text{odd}} \frac{\sin(n\pi x/a) \sinh(n\pi y/a)}{n \sinh(n\pi b/a)}$$

$$V_{III} = \frac{4V_3}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{\sin(n\pi x / b) \sinh(n\pi y / b)}{n \sinh(n\pi a / b)}$$

$$V_{IV} = \frac{4V_4}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{\sin(n\pi y / a) \sinh[n\pi(b-x) / a]}{n \sinh(n\pi b / a)}$$

Prob. 6.25

$$\mathbf{E} = -\nabla V = -\frac{\partial V}{\partial x} \mathbf{a}_x - \frac{\partial V}{\partial y} \mathbf{a}_y$$

$$E_x = \frac{4V_o}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{n\pi}{a} \frac{\sin(n\pi y / a)}{n} \exp(-n\pi x / a)$$

$$E_y = -\frac{4V_o}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{n\pi}{a} \frac{\cos(n\pi y / a)}{n} \exp(-n\pi x / a)$$

$$\mathbf{E} = \frac{4V_o}{a} \sum_{n=\text{odd}}^{\infty} \exp(-n\pi x / a) \left[\sin(n\pi y / a) \mathbf{a}_x - \cos(n\pi y / a) \mathbf{a}_y \right]$$

Prob. 6.26

This is similar to Example 6.5 except that we must exchange x and y. Going through the same arguments, we have

$$V(x, y) = \sum c_n \sinh\left(\frac{n\pi x}{b}\right) \sin\left(\frac{n\pi y}{b}\right)$$

Applying the condition at x=a, we get

$$V_o \sin\left(\frac{\pi y}{b}\right) = \sum c_n \sinh\left(\frac{n\pi a}{b}\right) \sin\left(\frac{n\pi y}{b}\right)$$

This yields

$$c_n \sinh\left(\frac{n\pi a}{b}\right) = \begin{cases} V_o, & n=1 \\ 0, & n \neq 1 \end{cases}$$

Hence,

$$V(x, y) = V_o \frac{\sinh\left(\frac{\pi x}{b}\right) \sin\left(\frac{\pi y}{b}\right)}{\sinh\left(\frac{\pi a}{b}\right)}$$

Prob. 6.27

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} = 0$$

If we let $V(\rho, \phi) = R(\rho)\Phi(\phi)$,

$$\frac{\Phi}{\rho} \frac{\partial}{\partial \rho}(\rho R') + \frac{1}{\rho^2} R \Phi'' = 0$$

or

$$\frac{\rho}{R} \frac{\partial}{\partial \rho}(\rho R') = -\frac{\Phi''}{\Phi} = \lambda$$

Hence

$$\underline{\underline{\Phi'' + \lambda \Phi = 0}}$$

and

$$\frac{\partial}{\partial \rho}(\rho R') - \frac{\lambda R}{\rho} = 0$$

or

$$\underline{\underline{R'' + \frac{R'}{\rho} - \frac{\lambda R}{\rho^2} = 0}}$$

Prob. 6.28

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial V}{\partial \theta}) = 0$$

If $V(r, \theta) = R(r)F(\theta)$, $r \neq 0$,

$$F \frac{d}{dr} (r^2 R') + \frac{R}{\sin \theta} \frac{d}{d\theta} (\sin \theta F') = 0$$

Dividing through by RF gives

$$\frac{1}{R} \frac{d}{dr} (r^2 R') = -\frac{1}{F \sin \theta} \frac{d}{d\theta} (\sin \theta F') = \lambda$$

Hence,

$$\sin \theta F'' + \cos \theta F' + \lambda F \sin \theta = 0$$

or

$$\underline{\underline{F'' + \cot \theta F' + \lambda F = 0}}$$

Also,

$$\frac{d}{dr} (r^2 R') - \lambda R = 0$$

or

$$\underline{\underline{R'' + \frac{2R'}{r} - \frac{\lambda}{r^2} R = 0}}$$

Prob. 6.29 If the centers at $\phi = 0$ and $\phi = \pi/2$ are maintained at a potential difference of V_o , from Example 6.3,

$$E_\phi = \frac{2V_o}{\pi\rho}, \quad J = \sigma E$$

Hence,

$$I = \int J \cdot dS = \frac{2V_o\sigma}{\pi} \int_{\rho=a}^b \int_{z=0}^t \frac{1}{\rho} d\rho dz = \frac{2V_o\sigma t}{\pi} \ln(b/a)$$

and

$$R = \frac{V_o}{I} = \frac{\pi}{2\sigma t \ln(b/a)}$$

Prob. 6.30 If $V(r=a)=0$, $V(r=b)=V_o$, from Example 6.9,

$$E = \frac{V_o}{r^2(1/a - 1/b)}, \quad J = \sigma E$$

Hence,

$$I = \int J \cdot dS = \frac{V_o\sigma}{1/a - 1/b} \int_{\theta=0}^{\alpha} \int_{\phi=0}^{2\pi} \frac{1}{r^2} r^2 \sin\theta d\theta d\phi = \frac{2\pi V_o\sigma}{1/a - 1/b} (-\cos\theta)|_0^\alpha$$

$$R = \frac{V_o}{I} = \frac{\frac{1}{a} - \frac{1}{b}}{2\pi\sigma(1 - \cos\alpha)}$$

Prob. 6.31

This is the same as Problem 6.30 except that $\alpha = \pi$. Hence,

$$R = \frac{1}{2\pi\sigma(1 - \cos\pi)} \left(\frac{1}{a} - \frac{1}{b} \right) = \underline{\underline{\frac{1}{4\pi\sigma} \left(\frac{1}{a} - \frac{1}{b} \right)}}$$

Prob. 6.32 For a spherical capacitor, from Eq. (6.38),

$$R = \frac{\frac{1}{a} - \frac{1}{b}}{4\pi\sigma}$$

For the hemisphere, $R' = 2R$ since the sphere consists of two hemispheres in parallel. As $b \longrightarrow \infty$,

$$R' = \lim_{b \longrightarrow \infty} \frac{2 \left[\frac{1}{a} - \frac{1}{b} \right]}{4\pi\sigma} = \frac{1}{2\pi a\sigma}$$

$$G = 1/R' = 2\pi a\sigma$$

Alternatively, for an isolated sphere, $C = 4\pi\epsilon a$. But

$$RC = \frac{\epsilon}{\sigma} \longrightarrow R = \frac{1}{4\pi a\sigma}$$

$$R' = 2R = \frac{1}{2\pi a\sigma} \quad \text{or} \quad G = 2\pi a\sigma$$

Prob. 6.33

(a) For the parallel-plate capacitor,

$$\mathbf{E} = -\frac{V_o}{d} \mathbf{a}_x$$

From Example 6.11,

$$C = \frac{1}{V_o^2} \int \epsilon |\mathbf{E}|^2 dv = \frac{1}{V_o^2} \int \epsilon \frac{V_o^2}{d^2} dv = \frac{\epsilon}{d^2} Sd = \frac{\epsilon S}{d}$$

(b) For the cylindrical capacitor,

$$\mathbf{E} = -\frac{V_o}{\rho \ln b/a} \mathbf{a}_\rho$$

From Example 6.8,

$$C = \frac{1}{V_o^2} \iiint \frac{\epsilon V_o^2}{(\rho \ln b/a)^2} \rho d\rho d\phi dz = \frac{2\pi\epsilon L}{(\ln b/a)^2} \int_a^b \frac{d\rho}{\rho} = \frac{2\pi\epsilon L}{\ln b/a}$$

(c) For the spherical capacitor,

$$\mathbf{E} = \frac{V_o}{r^2(1/a - 1/b)} \mathbf{a}_r$$

From Example 6.10,

$$C = \frac{1}{V_o^2} \iiint \frac{\epsilon V_o^2}{r^4(1/a - 1/b)^2} r^2 \sin\theta d\theta dr d\phi = \frac{\epsilon}{(1/a - 1/b)^2} 4\pi \int_a^b \frac{dr}{r^2} = \frac{4\pi\epsilon}{\frac{1}{a} - \frac{1}{b}}$$

Prob. 6.34

Assume $V(\rho=a) = 0$ and $V(\rho=b)=V_o$. Following Example 6.8,

$$V = A \ln \rho + B = \frac{V_o}{\ln \frac{b}{a}} \ln \frac{\rho}{a}$$

$$\mathbf{J} = \sigma \mathbf{E} = -\sigma \nabla V, \quad d\mathbf{S} = \rho d\phi dz \mathbf{a}_\rho$$

$$I = \int_S \mathbf{J} \cdot d\mathbf{S} = \int_{\phi=0}^{2\pi} \int_{z=0}^L \frac{\sigma V_o}{\rho \ln \frac{b}{a}} \rho d\phi dz = \frac{\sigma V_o}{\ln \frac{b}{a}} (2\pi L)$$

$$R = \frac{V_o}{I} = \frac{\ln \frac{b}{a}}{2\pi\sigma L}$$

$$G = \frac{I}{V_o} = \frac{2\pi\sigma L}{\ln \frac{b}{a}}$$

The conductance per unit length is

$$G' = \frac{G}{L} = \frac{2\pi\sigma}{\ln \frac{b}{a}}$$

Prob. 6.35

From eq. (6.37) or from previous problem

$$R = \frac{\ln \frac{b}{a}}{2\pi\sigma L}$$

$$P = VI = \frac{V^2}{R} = \frac{2\pi\sigma L V^2}{\ln \frac{b}{a}}$$

The power loss per unit length is

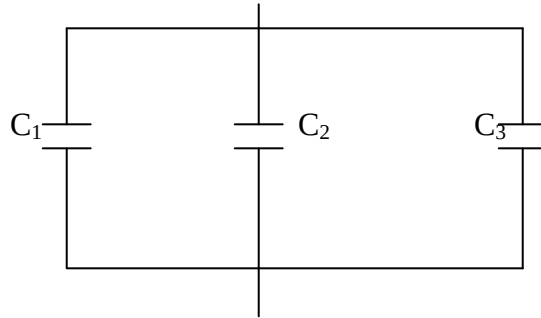
$$P' = \frac{P}{L} = \frac{2\pi\sigma V^2}{\ln \frac{b}{a}}$$

Prob. 6.36

$$C = \frac{\epsilon S}{d} \longrightarrow S = \frac{Cd}{\epsilon_o \epsilon_r} = \frac{2 \times 10^{-9} \times 10^{-6}}{4 \times 10^{-9} / 36\pi} \text{ m}^2 = \underline{\underline{0.5655 \text{ cm}^2}}$$

Prob. 6.37

This can be regarded as three capacitors in parallel.

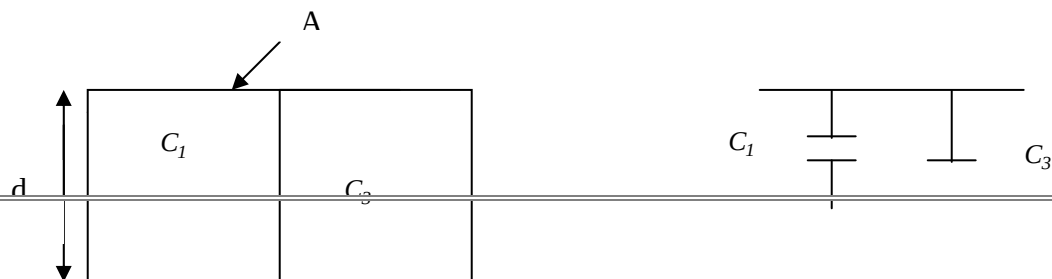


$$\begin{aligned} C &= C_1 + C_2 + C_3 = \sum \frac{\epsilon_o \epsilon_{rk} S_k}{d_k} \\ &= \frac{\epsilon_o}{2 \times 10^{-3}} \left[3 \times 15 \times 10^{-2} \times 20 \times 10^{-2} + 5 \times 15 \times 10^{-2} \times 20 \times 10^{-2} + 8 \times 15 \times 10^{-2} \times 20 \times 10^{-2} \right] \\ &= \frac{10^{-9}}{36\pi} \times \frac{15 \times 10^{-2} \times 20 \times 10^{-2}}{2 \times 10^{-3}} [3 + 5 + 8] = \underline{\underline{2.122 \text{ nF}}} \end{aligned}$$

Prob. 6.38

The structure may be treated as consisting of three capacitors in series.

$$\begin{aligned} C_1 &= \frac{\epsilon_o A}{a}, \quad C_2 = \frac{\epsilon_r \epsilon_o A}{a}, \quad C_3 = \frac{\epsilon_o A}{a} \\ \frac{1}{C} &= \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} = \frac{a}{\epsilon_o A} + \frac{a}{\epsilon_o \epsilon_r A} + \frac{a}{\epsilon_o A} \\ \frac{A}{aC} &= \frac{2}{\epsilon_o} + \frac{1}{\epsilon_o \epsilon_r} = \frac{2\epsilon_r + 1}{\epsilon_o \epsilon_r} \\ C &= \underline{\underline{\frac{\epsilon_o \epsilon_r A}{a(1 + 2\epsilon_r)}}} \end{aligned}$$

Prob. 6.39



From the figure above,

$$C = \frac{C_1 C_2}{C_1 + C_2} + C_3$$

here

$$C_1 = \frac{\epsilon_o A / 2}{d / 2} = \frac{\epsilon_o A}{d}, \quad C_2 = \frac{\epsilon_o \epsilon_r A}{d}, \quad C_3 = \frac{\epsilon_o A}{2d}$$

$$C = \frac{\epsilon_o^2 \epsilon_r A^2 / d^2}{\epsilon_o (\epsilon_r + 1) A / d} + \frac{\epsilon_o A}{2d} = \frac{\epsilon_o A}{d} \left(\frac{1}{2} + \frac{\epsilon_r}{\epsilon_r + 1} \right) = \frac{10^{-9}}{36\pi} \frac{10 \times 10^{-4}}{2 \times 10^{-3}} \left(\frac{1}{2} + \frac{6}{7} \right) \cong \underline{\underline{6 \text{ pF}}}$$

Prob. 6.40

$$C = \frac{\epsilon_o S}{d} \longrightarrow S = \frac{Cd}{\epsilon_o}$$

$$S = \frac{1 \times 10^{-3}}{10^{-9} / 36\pi} = 36\pi \times 10^6$$

$$\underline{\underline{S = 1.131 \times 10^8 \text{ m}^2}}$$

Prob. 6.41

$$F dx = dW_E \longrightarrow F = \frac{dW_E}{dx}$$

$$W_E = \int \frac{1}{2} \epsilon |E|^2 dv = \frac{1}{2} \epsilon_o \epsilon_r E^2 x a d + \frac{1}{2} \epsilon_o E^2 d a (1 - x)$$

where $E = V_o / d$.

$$\frac{dW_E}{dx} = \frac{1}{2} \epsilon_o \frac{V_o^2}{d^2} (\epsilon_r - 1) d a \longrightarrow F = \frac{\epsilon_o (\epsilon_r - 1) V_o^2 a}{2d}$$

Alternatively, $W_E = \frac{1}{2} C V_o^2$, where

$$C = C_1 + C_2 = \frac{\epsilon_o \epsilon_r a x}{d} + \frac{\epsilon_o \epsilon_r (L - x)}{d}$$

$$\frac{dW_E}{dx} = \frac{1}{2} \epsilon_o \frac{V_o^2 a}{d} (\epsilon_r - 1)$$

$$F = \frac{\epsilon_o(\epsilon_r - 1)V_o^2 a}{2d}$$

Prob. 6.42

(a)

$$C = \frac{\epsilon_o S}{d} = \frac{10^{-9}}{36\pi} \frac{200 \times 10^{-4}}{3 \times 10^{-3}} = \underline{\underline{59 \text{ pF}}}$$

(b) $\rho_s = D_n = 10^{-6} \text{ nC/m}^2$. But

$$D_n = \epsilon E_n = \frac{\epsilon_o V_o}{d} = \rho_s$$

or

$$V_o = \frac{\rho_s d}{\epsilon_o} = 10^{-6} \times 3 \times 10^{-3} \times 36\pi \times 10^9 = \underline{\underline{339.3 \text{ V}}}$$

(c)

$$F = \frac{Q^2}{2S\epsilon_o} = \frac{\rho_s^2 S}{2\epsilon_o} = \frac{10^{-12} \times 200 \times 10^{-4} \times 36\pi \times 10^9}{2} = \underline{\underline{1.131 \text{ mN}}}$$

Prob. 6.43

$$C_1 = \frac{\epsilon_o \epsilon_r S}{d}, \quad C_2 = \frac{\epsilon_o S}{d}$$

$$\frac{C_1}{C_2} = \epsilon_r \longrightarrow \epsilon_r = \frac{56 \mu\text{F}}{32 \mu\text{F}} = \underline{\underline{1.75}}$$

Prob. 6.44

$$(a) \quad C = \frac{\epsilon S}{d} = \frac{6.8 \times \frac{10^{-9}}{36\pi} \times 0.5}{4 \times 10^{-3}} = \underline{\underline{7.515 \text{ nF}}}$$

$$\rho_s = \frac{Q}{S}, \quad C = \frac{Q}{V} \longrightarrow Q = CV$$

(b)

$$\rho_s = \pm \frac{CV}{S} = \pm \frac{7.515 \times 10^{-9} \times 9}{0.5} = \underline{\underline{\pm 135.27 \text{ nC/m}^2}}$$

Prob. 6.45

$$C_o = \frac{\epsilon S}{d}, \quad C = \frac{\epsilon S}{3d} = \underline{\underline{C_o / 3}}$$

$$C_o = \frac{Q_o}{V} \rightarrow Q_o = C_o V$$

$$Q = CV = (C_o / 3)V = \underline{\underline{Q_o / 3}}$$

$$E_o = \frac{V}{d}, \quad E = \frac{V}{3d} = \underline{\underline{\frac{E_o}{3}}}$$

$$W = \frac{1}{2}CV^2 = \frac{1}{2}(C_o / 3)V^2 = \underline{\underline{W_o / 3}}$$

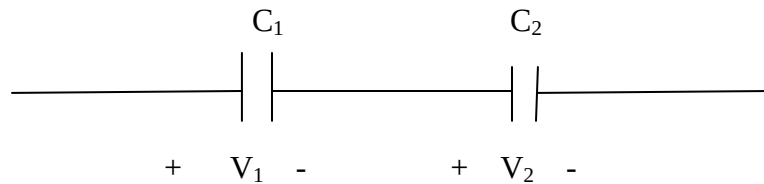
This indicates that two-thirds of the energy stored is lost in the connecting wires and source resistance.

Prob. 6.46

$$(a) \quad C_1 = \frac{\epsilon_o \epsilon_{r1} S}{d}, \quad C_2 = \frac{\epsilon_o \epsilon_{r2} S}{d}$$

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{\frac{\epsilon_o \epsilon_{r1} S}{d} \frac{\epsilon_o \epsilon_{r2} S}{d}}{\frac{\epsilon_o \epsilon_{r1} S}{d} + \frac{\epsilon_o \epsilon_{r2} S}{d}} = \frac{\epsilon_o \epsilon_{r1} \epsilon_{r2} S}{\epsilon_{r1} + \epsilon_{r2}} = \frac{10^{-9}}{36\pi} \frac{40 \times 10^{-4}}{2 \times 10^{-3}} \frac{4 \times 6}{4 + 6} = \frac{4.8}{36\pi} \text{ nF} = \underline{\underline{42.44 \text{ pF}}}$$

$$(b) \quad C = \frac{Q}{V} \rightarrow Q = CV = \underline{\underline{509.3 \text{ pC}}}$$



(c) C_1 and C_2 are in series as shown above

$$Q = C_1 V_1 + C_2 V_2, \quad V_1 + V_2 = V = 12$$

Solving these gives

$$V_1 = \frac{C_2}{C_1 + C_2} V = \frac{6}{10}(12) = 7.2$$

$$V_2 = \frac{C_1}{C_1 + C_2} V = \frac{4}{10}(12) = 4.8$$

$$E_1 = \frac{V_1}{d} = \underline{\underline{3.6 \text{ kV/m}}}$$

$$E_2 = \frac{V_2}{d} = \underline{\underline{2.4 \text{ kV/m}}}$$

$$D = D_1 = D_2 = \epsilon_1 E_1 = \underline{\underline{1.2732 \times 10^{-7} \text{ C/m}^2}}$$

$$P_1 = \chi_{e1} \epsilon_o E_1 = \underline{\underline{9.549 \times 10^{-8} \text{ C/m}^2}}$$

$$P_2 = \chi_{e2} \epsilon_o E_2 = \underline{\underline{1.061 \times 10^{-8} \text{ C/m}^2}}$$

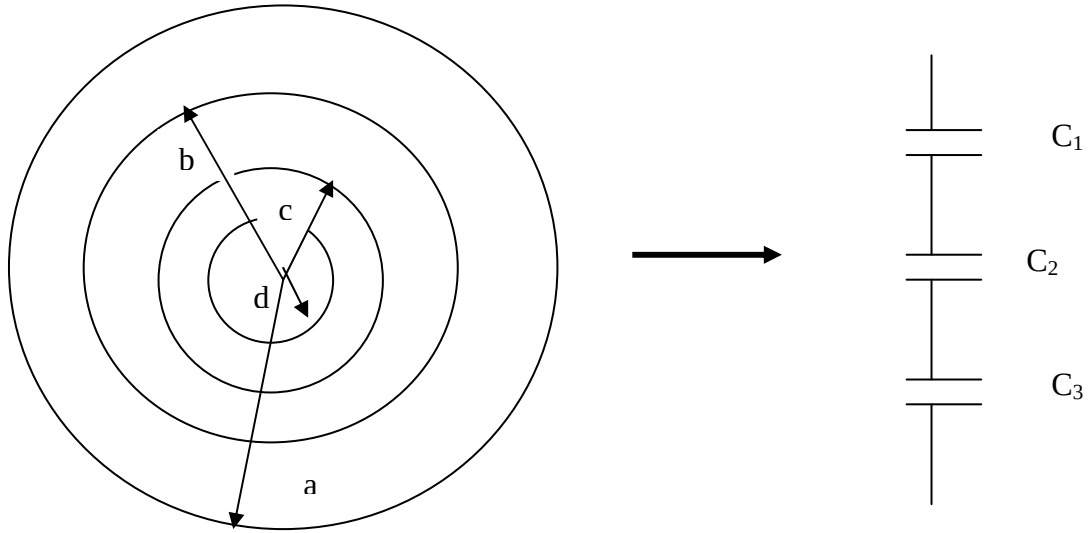
Prob.6.47

(a)

$$C = \frac{4\pi\epsilon}{\frac{1}{a} - \frac{1}{b}} = \frac{4\pi \times 2.25 \times \frac{10^{-9}}{36\pi}}{\frac{1}{5 \times 10^{-2}} - \frac{1}{10 \times 10^{-2}}} = \underline{\underline{25 \text{ pF}}}$$

(b) $Q = C V_o = 25 \times 80 \text{ pC}$

$$\rho_s = \frac{Q}{4\pi r^2} = \frac{25 \times 80}{4\pi \times 25 \times 10^{-4}} \text{ pC/m}^2 = \underline{\underline{63.66 \text{ nC/m}^2}}$$

Prob. 6.48

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

where $C_1 = \frac{4\pi\epsilon_3}{\frac{1}{b} - \frac{1}{a}}$, $C_2 = \frac{4\pi\epsilon_2}{\frac{1}{c} - \frac{1}{b}}$, $C_3 = \frac{4\pi\epsilon_1}{\frac{1}{d} - \frac{1}{c}}$,

$$\frac{4\pi}{C} = \frac{1/b - 1/a}{\epsilon_3} + \frac{1/c - 1/b}{\epsilon_2} + \frac{1/d - 1/c}{\epsilon_1}$$

$$C = \frac{4\pi}{\frac{\epsilon_1}{\frac{1}{d} - \frac{1}{c}} + \frac{\epsilon_2}{\frac{1}{c} - \frac{1}{b}} + \frac{\epsilon_3}{\frac{1}{b} - \frac{1}{a}}}$$

Prob. 6.49

We may place a charge Q on the inner conductor. The negative charge $-Q$ is on the outer surface of the shell. Within the shell, $\mathbf{E} = 0$, i.e. between $r=c$ and $r=b$. Otherwise,

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r$$

The potential at $r=a$ is

$$\begin{aligned} V_a &= -\int_{-\infty}^a \mathbf{E} \cdot d\mathbf{l} = -\int_{-\infty}^c E_r dr - \int_c^b E_r dr - \int_b^a E_r dr \\ &= -\frac{Q}{4\pi\epsilon_0} \int_{-\infty}^c \frac{dr}{r^2} - 0 - \frac{Q}{4\pi\epsilon_0} \int_b^a \frac{dr}{r^2} = \frac{Q}{4\pi\epsilon_0 c} + \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right) \\ C &= \frac{Q}{V_a} = \frac{1}{\frac{1}{4\pi\epsilon_0 c} + \frac{1}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)} \end{aligned}$$

Prob. 6.50

We can regard this as having two cylindrical capacitors in series.

$$\begin{aligned} C_1 &= \frac{2\pi\epsilon_0\epsilon_{r1}L}{\ln \frac{c}{a}}, \quad C_2 = \frac{2\pi\epsilon_0\epsilon_{r2}L}{\ln \frac{b}{c}} \\ C &= \frac{C_1 C_2}{C_1 + C_2} = \frac{\frac{2\pi\epsilon_0\epsilon_{r1}L}{\ln \frac{c}{a}} \frac{2\pi\epsilon_0\epsilon_{r2}L}{\ln \frac{b}{c}}}{\frac{2\pi\epsilon_0\epsilon_{r1}L}{\ln \frac{c}{a}} + \frac{2\pi\epsilon_0\epsilon_{r2}L}{\ln \frac{b}{c}}} = \frac{2\pi\epsilon_0\epsilon_{r1}\epsilon_{r2}L}{\epsilon_{r1} \ln \frac{b}{c} + \epsilon_{r2} \ln \frac{c}{a}} \end{aligned}$$

Prob. 6.51

$$C = \frac{2\pi\epsilon L}{\ln(b/a)} = \frac{2\pi \times 2.5 \times \frac{10^{-9}}{36\pi} \times 3 \times 10^3}{\ln(8/5)} = \underline{\underline{0.8665 \mu\text{F}}}$$

Prob. 6.52

Let the plate at $\phi=0$ be 0, i.e. $V(0)=0$ and let the plate at $\phi=\pi/4$ be V_0 , i.e. $V(\pi/4)=V_0$.

$$\nabla^2 V = \frac{1}{\rho^2} \frac{d^2 V}{d\phi^2} = 0 \quad \longrightarrow \quad \frac{dV}{d\phi} = A \quad \longrightarrow \quad V = A\phi + B$$

$$V(0) = 0 \quad \longrightarrow \quad 0 = 0 + B \quad \longrightarrow \quad B = 0$$

$$V(\pi/4) = V_o \quad \longrightarrow \quad V_o = A\pi/4 \quad \longrightarrow \quad A = \frac{4V_o}{\pi}$$

$$\mathbf{E} = -\nabla V = -\frac{1}{\rho} \frac{dV}{d\phi} \mathbf{a}_\phi = -\frac{A}{\rho} \mathbf{a}_\phi = -\frac{4V_o}{\pi\rho} \mathbf{a}_\phi$$

$$\mathbf{D} = \epsilon \mathbf{E} = -\frac{4\epsilon V_o}{\pi\rho} \mathbf{a}_\phi$$

$$\rho_s = D_n = -\frac{4\epsilon V_o}{\pi\rho}$$

$$Q = \int \rho_s dS = - \int_{\rho=a}^b \int_{z=0}^L \frac{4\epsilon V_o}{\pi\rho} d\rho dz = -\frac{4\epsilon V_o}{\pi} L \ln(b/a)$$

$$C = \frac{|Q|}{V_o} = \frac{4\epsilon L}{\pi} \ln(b/a)$$

Prob. 6.53

Since $V = V(\phi)$,

$$\nabla^2 V = 0 \quad \rightarrow \quad \frac{1}{\rho^2} \frac{d^2 V}{d\phi^2} = 0$$

$$\text{But } \rho \neq 0, \quad \frac{d^2 V}{d\phi^2} = 0 \quad \rightarrow \quad \frac{dV}{d\phi} = A \quad \rightarrow \quad V = A\phi + B$$

$$\text{For } \phi=0, V=0 \quad \rightarrow \quad 0=0+B \quad \rightarrow \quad B=0$$

$$\text{For } \phi=\alpha, V=V_o \quad \rightarrow \quad V_o = A\alpha \quad \rightarrow \quad A = \frac{V_o}{\alpha}$$

$$\mathbf{E} = -\nabla V = -\frac{1}{\rho} \frac{dV}{d\phi} \mathbf{a}_\phi = -\frac{A}{\rho} \mathbf{a}_\phi$$

$$\rho_s = D_n = \epsilon E_\phi = \frac{A}{\rho} \epsilon, \quad dS = d\rho dz$$

$$Q = \int_S \rho_s dS = \int_{\rho=\rho_1}^{\rho_2} \int_{z=0}^L \frac{A}{\rho} \epsilon d\rho dz = A\epsilon L \ln \frac{\rho_2}{\rho_1} = \frac{V_o}{\alpha} \epsilon L \ln \frac{\rho_2}{\rho_1}$$

$$C = \frac{Q}{V_o} = \frac{\epsilon L}{\alpha} \ln \frac{\rho_2}{\rho_1}$$

Prob. 6.54

$$C = \frac{2\pi\epsilon_o L}{\ln(b/a)} = \frac{2\pi \times \frac{10^{-9}}{36\pi} \times 100 \times 10^{-6}}{\ln(600/20)} = 1.633 \times 10^{-15} \text{ F}$$

$$V = Q/C = \frac{50 \times 10^{-15}}{1.633 \times 10^{-15}} = \underline{\underline{30.62 \text{ V}}}$$

Prob. 6.55

$$C_1 = \frac{2\pi\epsilon_1}{\ln(b/a)}, \quad C_2 = \frac{2\pi\epsilon_2}{\ln(c/b)}$$

Since the capacitance are in series, the total capacitance per unit length is

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{2\pi\epsilon_1\epsilon_2}{\epsilon_2 \ln(b/a) + \epsilon_1 \ln(c/b)}$$

Prob. 6.56

(a) This is similar to Example 6.10.

$$\nabla^2 V = 0 \quad \rightarrow \quad V = \frac{V_o \left(\frac{1}{r} - \frac{1}{b} \right)}{\frac{1}{a} - \frac{1}{b}}$$

$$\mathbf{E} = -\nabla V = -\frac{dV}{dr} \mathbf{a}_r = \frac{-V_o}{r^2 \left(\frac{1}{a} - \frac{1}{b} \right)} \mathbf{a}_r$$

$$\text{At } r=a, \quad \rho_s = D_n = \epsilon E_r = \frac{\epsilon_o \epsilon_r V_o}{a^2 \left(\frac{1}{a} - \frac{1}{b} \right)}$$

$$\epsilon_r = \frac{\rho_s a^2 \left(\frac{1}{a} - \frac{1}{b} \right)}{\epsilon_o V} = \frac{400 \times 10^{-9} (4 \times 10^{-4}) \left(\frac{1}{2} - \frac{1}{4} \right) \frac{1}{10^{-2}}}{\frac{10^{-9}}{36\pi} (100)} = 16(36\pi)10^{-2} (1/4) = \underline{\underline{4.524}}$$

(b)

$$C = \frac{4\pi\epsilon_o \epsilon_r}{\frac{1}{a} - \frac{1}{b}} = \frac{4\pi \times \frac{10^{-9}}{36\pi} 4.524}{\left(\frac{1}{2} - \frac{1}{4} \right) \frac{1}{10^{-2}}} = \frac{452.4}{9(1/4)} = \underline{\underline{201.1 \text{ nF}}}$$

Prob. 6.57

Each half has capacitance given by

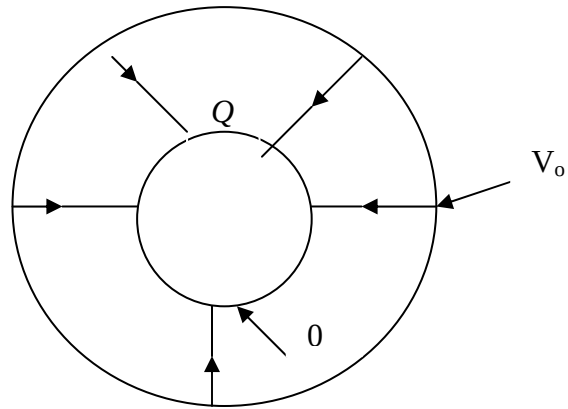
$$C = \frac{2\pi\epsilon}{\frac{1}{a} - \frac{1}{b}} = \frac{2\pi\epsilon ab}{b-a}$$

The two halves may be regarded as capacitors in parallel. Hence,

$$C = C_1 + C_2 = \frac{2\pi\epsilon_1 ab}{b-a} + \frac{2\pi\epsilon_2 ab}{b-a} = \frac{2\pi(\epsilon_1 + \epsilon_2)ab}{b-a}$$

Prob. 6.58

$$\mathbf{E} = \frac{Q}{4\pi\epsilon r^2} \mathbf{a}_r$$



$$W = \frac{1}{2} \int \epsilon |\mathbf{E}|^2 dv = \iiint \frac{Q^2}{32\pi^2 \epsilon^2 r^4} \epsilon r^2 \sin\theta d\theta d\phi dr$$

$$= \frac{Q^2}{32\pi^2 \epsilon} (2\pi)(2) \int_c^b \frac{dr}{r^2} = \frac{Q^2}{8\pi\epsilon} \left(\frac{1}{c} - \frac{1}{b} \right)$$

$$W = \frac{Q^2(b-c)}{8\pi\epsilon bc}$$

Prob. 6.59

(a) Method 1: $\mathbf{E} = \frac{\rho_s}{\epsilon} (-\mathbf{a}_x)$, where ρ_s is to be determined.

$$V_o = -\int \mathbf{E} \cdot d\mathbf{l} = -\int \frac{-\rho_s}{\epsilon} dx = \rho_s \int_0^d \frac{1}{\epsilon_0} \frac{d}{d+x} dx = \frac{\rho_s}{\epsilon} d \ln(x+d) \Big|_0^d$$

$$V_o = \rho_s d \ln \frac{2d}{d} \longrightarrow \rho_s = \frac{V_o \epsilon_o}{d \ln 2}$$

$$\mathbf{E} = -\frac{\rho_s}{\epsilon} \mathbf{a}_x = -\frac{V_o}{(x+d) \ln 2} \mathbf{a}_x$$

Method 2: We solve Laplace's equation

$$\nabla \cdot (\epsilon \nabla V) = \frac{d}{dx} \left(\epsilon \frac{dV}{dx} \right) = 0 \longrightarrow \epsilon \frac{dV}{dx} = A$$

$$\frac{dV}{dx} = \frac{A}{\epsilon} = \frac{Ad}{\epsilon_o(x+d)} = \frac{c_1}{x+d}$$

$$V = c_1 \ln(x+d) + c_2$$

$$V(x=0) = 0 \longrightarrow 0 = c_1 \ln d + c_2 \longrightarrow c_2 = -c_1 \ln d$$

$$V(x=d) = V_o \longrightarrow V_o = c_1 \ln 2d - c_1 \ln d = c_1 \ln 2$$

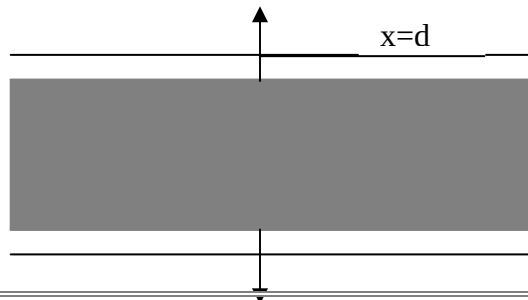
$$c_1 = \frac{V_o}{\ln 2}$$

$$V = c_1 \ln \frac{x+d}{d} = \frac{V_o}{\ln 2} \ln \frac{x+d}{d}$$

$$\mathbf{E} = -\frac{dV}{dx} \mathbf{a}_x = -\frac{V_o}{(x+d) \ln 2} \mathbf{a}_x$$

$$(b) \quad \mathbf{P} = (\epsilon_r - 1) \epsilon_o \mathbf{E} = -\left(\frac{x+d}{d} - 1 \right) \frac{\epsilon_o V_o}{(x+d) \ln 2} \mathbf{a}_x = -\frac{\epsilon_o x V_o}{d(x+d) \ln 2} \mathbf{a}_x$$

(c)



$$x=0$$

$$\rho_{ps}|_{x=0} = \mathbf{P} \cdot (-\mathbf{a}_x)|_{x=0} = \underline{\underline{0}}$$

$$\rho_{ps}|_{x=d} = \mathbf{P} \cdot \mathbf{a}_x|_{x=d} = -\frac{\epsilon_o V_o}{\underline{\underline{2d \ln 2}}}$$

$$(d) \quad E = \frac{\rho_s}{\epsilon} \mathbf{a}_x = \frac{Q}{\epsilon S} \mathbf{a}_x = \frac{Q}{\epsilon_o (1 + \frac{x}{d}) S} \mathbf{a}_x$$

$$V = -\int \mathbf{E} \cdot d\mathbf{l} = -\frac{Q}{\epsilon_o S} \int_a^d \frac{dx}{(1 + \frac{x}{d})} = \frac{Q}{\epsilon_o S} d \ln 2$$

$$C = \frac{Q}{V} = \frac{\epsilon_o S}{\underline{\underline{d \ln 2}}}$$

Prob. 6.60

We solve Laplace's equation for an inhomogeneous medium.

$$\nabla \cdot (\epsilon \nabla V) = \frac{d}{dx} \left(\epsilon \frac{dV}{dx} \right) = 0 \quad \longrightarrow \quad \epsilon \frac{dV}{dx} = A$$

$$\frac{dV}{dx} = \frac{A}{\epsilon} = \frac{A}{2\epsilon_o} \left[1 + \left(\frac{x}{d} \right)^2 \right]$$

$$V = \frac{A}{2\epsilon_o} \left(x + \frac{x^3}{3d^2} \right) + B$$

When $x=d$, $V=V_o$,

$$V_o = \frac{A}{2\epsilon_o} \left(d + \frac{d}{3} \right) + B \quad \longrightarrow \quad V_o = \frac{2Ad}{3\epsilon_o} + B \quad (1)$$

When $x = -d$, $V=0$,

$$0 = \frac{A}{2\epsilon_o} \left(-d - \frac{d}{3} \right) + B \quad \longrightarrow \quad 0 = -\frac{2Ad}{3\epsilon_o} + B \quad (2)$$

$$\text{Adding (1) and (2), } V_o = 2B \quad \longrightarrow \quad B = V_o / 2$$

From (2),

$$B = \frac{2Ad}{3\epsilon_o} = \frac{V_o}{2} \quad \longrightarrow \quad A = \frac{3\epsilon_o V_o}{4d}$$

$$\mathbf{E} = -\nabla V = -\frac{dV}{dx} \mathbf{a}_x = -\frac{A}{\epsilon} \mathbf{a}_x = -\frac{3\epsilon_o V_o}{4d} \frac{\left[1 + \left(\frac{x}{d} \right)^2 \right]}{2\epsilon_o} \mathbf{a}_x = -\frac{3V_o}{8d} \left[1 + \left(\frac{x}{d} \right)^2 \right] \mathbf{a}_x$$

$$\rho_s = D \cdot \mathbf{a}_n = \epsilon E \cdot \mathbf{a}_x \Big|_{x=d} = -A = -\frac{3\epsilon_o V_o}{4d}$$

$$Q = \int_s \rho_s dS = \rho_s S = -\frac{3S\epsilon_o V_o}{4d}$$

$$C = \frac{|Q|}{V_o} = \underline{\underline{\frac{3\epsilon_o S}{4d}}}$$

Prob. 6.61

Method 1: Using Gauss's law,

$$Q = \int \mathbf{D} \cdot d\mathbf{S} = 4\pi r^2 D_r \quad \longrightarrow \quad \mathbf{D} = \frac{Q}{4\pi r^2} \mathbf{a}_r, \quad \epsilon = \frac{\epsilon_o k}{r^2}$$

$$\mathbf{E} = \mathbf{D} / \epsilon = \frac{Q}{4\pi \epsilon_o k} \mathbf{a}_r$$

$$V = -\int \mathbf{E} \cdot d\mathbf{l} = -\frac{Q}{4\pi\epsilon_o k} \int_b^a dr = -\frac{Q}{4\pi\epsilon_o k} (b-a)$$

$$C = \frac{Q}{|V|} = \frac{4\pi\epsilon_o k}{\underline{\underline{b-a}}}$$

Method 2: Using the inhomogeneous Laplace's equation,

$$\nabla \cdot (\epsilon \nabla V) = 0 \quad \longrightarrow \quad \frac{1}{r^2} \frac{d}{dr} \left(\frac{\epsilon_o k}{r^2} r^2 \frac{dV}{dr} \right) = 0$$

$$\epsilon_o k \frac{dV}{dr} = A' \quad \longrightarrow \quad \frac{dV}{dr} = A \text{ or } V = Ar + B$$

$$V(r=a) = 0 \quad \longrightarrow \quad 0 = Aa + B \quad \longrightarrow \quad B = -Aa$$

$$V(r=b) = V_o \quad \longrightarrow \quad V_o = Ab + B = A(b-a) \quad \longrightarrow \quad A = \frac{V_o}{b-a}$$

$$\mathbf{E} = -\frac{dV}{dr} \mathbf{a}_r = -A \mathbf{a}_r = -\frac{V_o}{b-a} \mathbf{a}_r$$

$$\rho_s = D_n = -\frac{V_o}{b-a} \frac{\epsilon_o k}{r^2} \Big|_{r=a,b}$$

$$Q = \int \rho_s dS = -\frac{V_o \epsilon_o k}{b-a} \iint \frac{1}{r^2} r^2 \sin \theta d\theta d\phi = -\frac{V_o \epsilon_o k}{b-a} 4\pi$$

$$C = \frac{|Q|}{V_o} = \frac{4\pi\epsilon_o k}{\underline{\underline{b-a}}}$$

Prob. 6.62

$$C = 4\pi\epsilon_o a = 4\pi \times \frac{10^{-9}}{36\pi} \times 6.37 \times 10^6 = \underline{\underline{0.708 \text{ mF}}}$$

Prob.6.63

$$C = \frac{Q}{V}$$

$$D = \frac{Q}{2\pi\rho L} a_\rho$$

$$E = \frac{D}{\varepsilon} = \frac{Q}{2\pi\rho L\varepsilon_0(3)(1+\rho)}$$

$$V = -\int \mathbf{E} \cdot d\mathbf{l} = \frac{Q}{6\pi L\varepsilon_0} \int_a^b \frac{d\rho}{\rho(1+\rho)}$$

$$\text{Let } \frac{1}{\rho(1+\rho)} = \frac{A}{\rho} + \frac{B}{1+\rho}$$

Using partial fractions

$$A=1, B=-1$$

$$V = \frac{Q}{6\pi\varepsilon_0 L} \left[\int_a^b \frac{d\rho}{\rho} - \int_a^b \frac{d\rho}{1+\rho} \right]$$

$$= \frac{Q}{6\pi\varepsilon_0 L} \left[\ln \rho - \ln(1+\rho) \right] \Big|_a^b$$

$$= \frac{Q}{6\pi\varepsilon_0 L} \left[\ln \frac{b}{1+b} - \ln \frac{a}{1+a} \right]$$

If $a=1$ mm, and $b=5$ mm

$$C = \frac{Q}{|V|}$$

$$= \frac{6\pi\varepsilon_0}{\ln \frac{b}{1+b} - \ln \frac{a}{1+a}}$$

$$= \frac{6\pi \times \frac{10^{-9}}{36\pi}}{\ln \frac{5}{6} - \ln \frac{1}{2}}$$

$$= \frac{\frac{1}{6} \times 10^{-9}}{\ln 0.8333 - \ln 0.5} = \frac{1}{6} \times 1.9591 \text{ nF}$$

$$C = \underline{\underline{0.326 \text{ nF}}}$$

Prob. 6.64

$$\frac{D-a}{a} = \frac{D}{a} - 1 = 11$$

$$C \square \frac{\pi \times \frac{10^{-9}}{36\pi} (4)}{\ln 11} = \frac{4}{36 \ln 11} \text{ nF/m} = \underline{\underline{46.34 \text{ pF/m}}}$$

Prob. 6.65

(a) From eq. (6.46),

$$\rho_s = \frac{-Qh}{2\pi[x^2 + y^2 + h^2]^{3/2}} = \frac{-10 \times 10^{-9}(10)}{2\pi[4 + 16 + 100]^{3/2}} = \frac{-10^{-7}}{2\pi(120)^{3/2}} = \underline{\underline{-12.107 \text{ pF/m}^2}}$$

(b) $Q_{in} = -Q = \underline{\underline{-10 \text{ nC}}}$

Prob. 6.66

4nC	-3nC		3nC	-4nC
-2	-1	0	1	2
4	3		2	1

(a) $Q_i = -(3\text{nC} - 4\text{nC}) = \underline{\underline{1\text{nC}}}$

(b) The force of attraction between the charges and the plates is

$$\mathbf{F} = \mathbf{F}_{13} + \mathbf{F}_{14} + \mathbf{F}_{23} + \mathbf{F}_{24}$$

$$|F| = \frac{10^{-18}}{4\pi \times 10^{-9} / 36\pi} \left[\frac{9}{2^2} - \frac{2(12)}{3^2} + \frac{16}{4^2} \right] = \underline{\underline{5.25 \text{ nN}}}$$

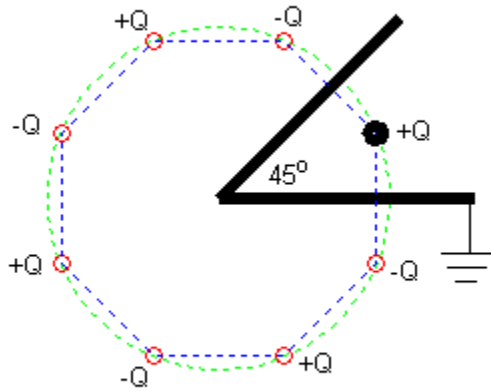
Prob. 6.67

We have 7 images as follows: -Q at (-1,1,1), -Q at (1,-1,1), -Q at (1,1,-1),

-Q at (-1,-1,-1), Q at (1,-1,-1), Q at (-1,-1,1), and Q at (-1,1,-1). Hence,

$$\mathbf{F} = \frac{Q^2}{4\pi\epsilon_0} \left[-\frac{2}{2^3} \mathbf{a}_x - \frac{2}{2^3} \mathbf{a}_y - \frac{2}{2^3} \mathbf{a}_z - \frac{(2\mathbf{a}_x + 2\mathbf{a}_y + 2\mathbf{a}_z)}{12^{3/2}} + \frac{(2\mathbf{a}_y + 2\mathbf{a}_z)}{8^{3/2}} \right. \\ \left. + \frac{(2\mathbf{a}_x + 2\mathbf{a}_y)}{8^{3/2}} + \frac{(2\mathbf{a}_x + 2\mathbf{a}_z)}{8^{3/2}} \right]$$

$$= 0.9(\mathbf{a}_x + \mathbf{a}_y + \mathbf{a}_z) \left(-\frac{1}{4} - \frac{1}{12\sqrt{3}} + \frac{1}{4\sqrt{2}} \right) = \underline{\underline{-0.1092(\mathbf{a}_x + \mathbf{a}_y + \mathbf{a}_z) \text{ N}}}$$

Prob. 6.68

$$N = \left(\frac{360^\circ}{45^\circ} - 1 \right) = \underline{\underline{7}}$$

Prob. 6.69

(a)

$$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_- = \frac{\rho_L}{2\pi\epsilon_0} \left(\frac{\mathbf{a}_{\rho 1}}{\rho_1} - \frac{\mathbf{a}_{\rho 2}}{\rho_2} \right) = \frac{16 \times 10^{-9}}{2\pi \times \frac{10^{-9}}{36\pi}} \left[\frac{(2, -2, 3) - (3, -2, 4)}{|(2, -2, 3) - (3, -2, 4)|^2} - \frac{(2, -2, 3) - (3, -2, -4)}{|(2, -2, 3) - (3, -2, -4)|^2} \right]$$

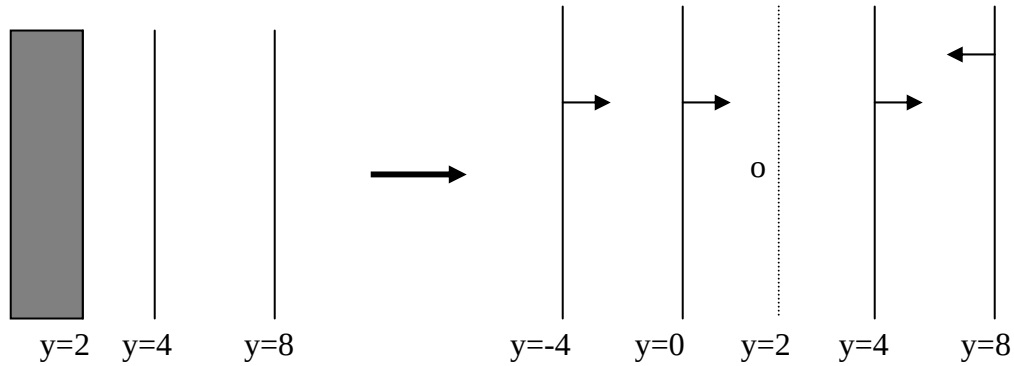
$$= 18 \times 16 \left[\frac{(-1, 0, -1)}{2} - \frac{(-1, 0, 7)}{50} \right] = \underline{\underline{-138.2\mathbf{a}_x - 184.3\mathbf{a}_y \text{ V/m}}}$$

(b) $\rho_s = D_n$

$$\mathbf{D} = \mathbf{D}_+ + \mathbf{D}_- = \frac{\rho_L}{2\pi} \left(\frac{\mathbf{a}_{\rho 1}}{\rho_1} - \frac{\mathbf{a}_{\rho 2}}{\rho_2} \right) = \frac{16 \times 10^{-9}}{2\pi} \left[\frac{(5, -6, 0) - (3, -6, 4)}{|(5, -6, 0) - (3, -6, 4)|^2} - \frac{(5, -6, 0) - (3, -6, -4)}{|(5, -6, 0) - (3, -6, -4)|^2} \right]$$

$$= \frac{8}{\pi} \left[\frac{(2, 0, -4)}{20} - \frac{(2, 0, 4)}{20} \right] \text{ nC/m}^2 = \underline{\underline{-1.018\mathbf{a}_z \text{ nC/m}^2}}$$

$$\underline{\underline{\rho_s = -1.018 \text{ nC/m}^2}}$$

Prob. 6.70


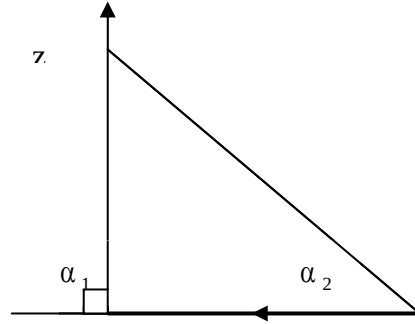
At P(0,0,0), $\mathbf{E}=0$ since $\mathbf{E}$ does not exist for $y < 2$.

At Q(-4,6,2), $y=6$ and

$$\begin{aligned} \mathbf{E} &= \sum \frac{\rho_s}{2\epsilon_0} \mathbf{a}_n = \frac{10^{-9}}{2 \times 10^{-9} / 36\pi} (-30\mathbf{a}_y + 20\mathbf{a}_y - 20\mathbf{a}_y - 30\mathbf{a}_y) = 18\pi(-60)\mathbf{a}_y \\ &= \underline{\underline{-3.4\mathbf{a}_y \text{ kV/m}}} \end{aligned}$$

CHAPTER 7

P.E. 7.1



$$\rho = 5, \quad \cos \alpha_1 = 0, \quad \cos \alpha_2 = \sqrt{\frac{2}{27}}$$

$$\mathbf{a}_\phi = \mathbf{a}_l \times \mathbf{a}_\rho = \left(\frac{-\mathbf{a}_x - \mathbf{a}_y}{\sqrt{2}} \right) \times \mathbf{a}_z = \frac{-\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}}$$

$$\mathbf{H}_3 = \frac{10}{4\pi(5)} \left(\sqrt{\frac{2}{27}} - 0 \right) \left(\frac{-\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}} \right) = \underline{\underline{-30.63\mathbf{a}_x + 30.63\mathbf{a}_y}} \text{ mA/m}$$

P.E. 7.2

$$(a) \quad \mathbf{H} = \frac{2}{4\pi(2)} \left(1 + \frac{3}{\sqrt{13}} \right) \mathbf{a}_z = \underline{\underline{0.1458\mathbf{a}_z}} \text{ A/m}$$

$$(b) \quad \rho = \sqrt{3^2 + 4^2} = 5, \quad \alpha_2 = 0, \quad \cos \alpha_1 = -\frac{12}{13},$$

$$\mathbf{a}_\phi = -\mathbf{a}_y \times \left(\frac{3\mathbf{a}_x - 4\mathbf{a}_z}{5} \right) = \frac{4\mathbf{a}_x + 3\mathbf{a}_z}{5}$$

$$\begin{aligned} \mathbf{H} &= \frac{2}{4\pi(5)} \left(1 + \frac{12}{13} \right) \left(\frac{4\mathbf{a}_x + 3\mathbf{a}_z}{5} \right) = \frac{1}{26\pi} (4\mathbf{a}_x + 3\mathbf{a}_z) \\ &= \underline{\underline{48.97\mathbf{a}_x + 36.73\mathbf{a}_z}} \text{ mA/m} \end{aligned}$$

P.E. 7.3

(a) From Example 7.3,

$$\mathbf{H} = \frac{Ia^2}{2(a^2 + z^2)^{3/2}} \mathbf{a}_z$$

At (0,0,-1cm), $z = 2\text{cm}$,

$$\mathbf{H} = \frac{50 \times 10^{-3} \times 25 \times 10^{-4}}{2(5^2 + 2^2)^{3/2} \times 10^{-6}} \mathbf{a}_z = \underline{\underline{400.2\mathbf{a}_z}} \text{ mA/m}$$

(b) At (0,0,10cm), $z = 9\text{cm}$,

$$\mathbf{H} = \frac{50 \times 10^{-3} \times 25 \times 10^{-4}}{2(5^2 + 9^2)^{3/2} \times 10^{-6}} \mathbf{a}_z = \underline{\underline{57.3 \mathbf{a}_z \text{ mA/m}}}$$

P.E. 7.4

$$\begin{aligned} \mathbf{H} &= \frac{NI}{2L} (\cos \theta_2 - \cos \theta_1) \mathbf{a}_z = \frac{2 \times 10^3 \times 50 \times 10^{-3} (\cos \theta_2 - \cos \theta_1) \mathbf{a}_z}{2 \times 0.75} \\ &= \frac{100}{1.5} (\cos \theta_2 - \cos \theta_1) \mathbf{a}_z \end{aligned}$$

$$\begin{aligned} \text{(a) At } (0,0,0), \theta &= 90^\circ, \cos \theta_2 = \frac{0.75}{\sqrt{0.75^2 + 0.05^2}} \\ &= 0.9978 \end{aligned}$$

$$\mathbf{H} = \frac{100}{1.5} (0.9978 - 0) \mathbf{a}_z$$

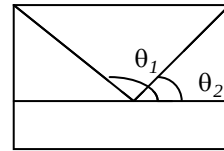
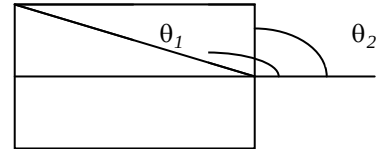
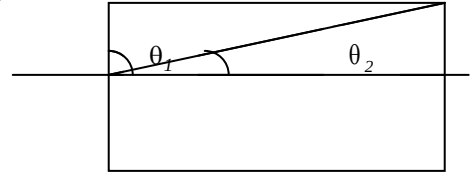
$$= \underline{\underline{66.52 \mathbf{a}_z \text{ A/m}}}$$

$$\text{(b) At } (0,0,0.75), \theta_2 = 90^\circ, \cos \theta_1 = -0.9978$$

$$\begin{aligned} \mathbf{H} &= \frac{100}{1.5} (0 + 0.9978) \mathbf{a}_z \\ &= \underline{\underline{66.52 \mathbf{a}_z \text{ A/m}}} \end{aligned}$$

$$\begin{aligned} \text{(c) At } (0,0,0.5), \cos \theta_1 &= \frac{-0.5}{\sqrt{0.5^2 + 0.05^2}} = -0.995 \\ \cos \theta_2 &= \frac{0.25}{\sqrt{0.25^2 + 0.05^2}} = 0.9806 \end{aligned}$$

$$\begin{aligned} \mathbf{H} &= \frac{100}{1.5} (0.9806 + 0.995) \mathbf{a}_z \\ &= \underline{\underline{131.7 \mathbf{a}_z \text{ A/m}}} \end{aligned}$$

**P.E. 7.5**

$$\mathbf{H} = \frac{1}{2} \mathbf{K} \times \mathbf{a}_n$$

$$\text{(a) } \mathbf{H}(0,0,0) = \frac{1}{2} 50 \mathbf{a}_z \times (-\mathbf{a}_y) = \underline{\underline{25 \mathbf{a}_x \text{ mA/m}}}$$

$$\text{(b) } \mathbf{H}(1,5,-3) = \frac{1}{2} 50 \mathbf{a}_z \times \mathbf{a}_y = \underline{\underline{-25 \mathbf{a}_x \text{ mA/m}}}$$

P.E. 7.6

$$|\mathbf{H}| = \begin{cases} \frac{NI}{2\pi\rho}, & \rho - a < \rho < \rho + a, \quad 9 < \rho < 11 \\ 0, & \text{otherwise} \end{cases}$$

(a) At $(3, -4, 0)$, $\rho = \sqrt{3^2 + 4^2} = 5\text{cm} < 9\text{cm}$

$$|\mathbf{H}| = \underline{\underline{0}}$$

(b) At $(6, 9, 0)$, $\rho = \sqrt{6^2 + 9^2} = \sqrt{117} < 11$

$$|\mathbf{H}| = \frac{10^3 \times 100 \times 10^{-3}}{2\pi \sqrt{117} \times 10^2} = \underline{\underline{147.1 \text{ A/m}}}$$

P.E. 7.7

(a) $\mathbf{B} = \nabla \times \mathbf{A} = (-4xz - 0)\mathbf{a}_x + (0 + 4yz)\mathbf{a}_y + (y^2 - x^2)\mathbf{a}_z$

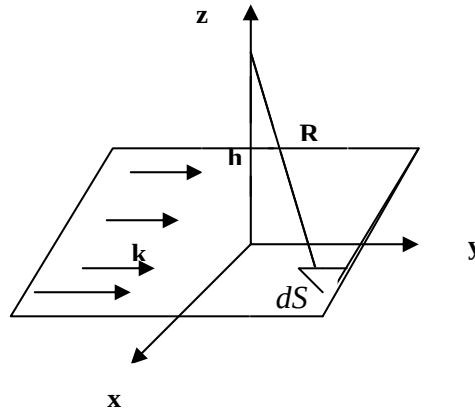
$$\mathbf{B}(-1, 2, 5) = \underline{\underline{20\mathbf{a}_x + 40\mathbf{a}_y + 3\mathbf{a}_z \text{ Wb/m}^2}}$$

$$\begin{aligned} \text{(b)} \quad \psi &= \int \mathbf{B} \cdot d\mathbf{S} = \int_{y=-1}^4 \int_{x=0}^1 (y^2 - x^2) dx dy = \int_{-1}^4 y^2 dy - 5 \int_0^1 x^2 dx \\ &= \frac{1}{3}(64 + 1) - \frac{5}{3} = \underline{\underline{20 \text{ Wb}}} \end{aligned}$$

Alternatively,

$$\begin{aligned} \psi &= \int \mathbf{A} \cdot d\mathbf{l} = \int_0^1 x^2(-1)dx + \int_{-1}^4 y^2(1)dy + \int_1^0 x^2(4)dx + 0 \\ &= -\frac{5}{3} + \frac{65}{3} = \underline{\underline{20 \text{ Wb}}} \end{aligned}$$

P.E. 7.8



$$\mathbf{H} = \int \frac{\mathbf{k} dS \times \mathbf{R}}{4\pi R^3},$$

$$dS = dx dy, \mathbf{k} = k_y \mathbf{a}_y,$$

$$\mathbf{R} = (-x, -y, h),$$

$$\mathbf{k} \times \mathbf{R} = (h\mathbf{a}_x + x\mathbf{a}_z)k_y,$$

$$\begin{aligned}\mathbf{H} &= \int \frac{k_y(h\mathbf{a}_x + x\mathbf{a}_z)dxdy}{4\pi(x^2 + y^2 + h^2)^{3/2}} \\ &= \frac{k_y h \mathbf{a}_x}{4\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dxdy}{(x^2 + y^2 + h^2)^{3/2}} + \frac{k_y \mathbf{a}_z}{4\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{xdxdy}{(x^2 + y^2 + h^2)^{3/2}}\end{aligned}$$

The integrand in the last term is zero because it is an odd function of x.

$$\begin{aligned}\mathbf{H} &= \frac{k_y h \mathbf{a}_x}{4\pi} \int_{\phi=0}^{2\pi} \int_{\rho=0}^{\infty} \frac{\rho d\phi d\rho}{(\rho^2 + h^2)^{3/2}} = \frac{k_y h 2\pi \mathbf{a}_x}{4\pi} \int_0^{\infty} (\rho^2 + h^2)^{-3/2} \frac{d(\rho^2)}{2} \\ &= \frac{k_y h}{2} \mathbf{a}_x \left(\frac{-1}{(\rho^2 + h^2)^{1/2}} \right) \Big|_0^{\infty} = \frac{k_y}{2} \mathbf{a}_x\end{aligned}$$

Similarly, for point (0,0,-h), $\mathbf{H} = -\frac{1}{2} k_y \mathbf{a}_x$

Hence,

$$\mathbf{H} = \begin{cases} \frac{1}{2} k_y \mathbf{a}_x, & z > 0 \\ -\frac{1}{2} k_y \mathbf{a}_x, & z < 0 \end{cases}$$

Prob. 7.1

(a) See text

(b) Let $\mathbf{H} = \mathbf{H}_y + \mathbf{H}_z$

$$\text{For } \mathbf{H}_z = \frac{I}{2\pi\rho} \mathbf{a}_{\phi} \quad \rho = \sqrt{(-3)^2 + 4^2} = 5$$

$$\mathbf{a}_{\phi} = -\mathbf{a}_z \times \frac{(-3\mathbf{a}_x + 4\mathbf{a}_y)}{5} = \frac{(3\mathbf{a}_y - 4\mathbf{a}_x)}{5}$$

$$\mathbf{H}_z = \frac{20}{2\pi(25)} (4\mathbf{a}_x + 3\mathbf{a}_y) = 0.5093\mathbf{a}_x + 0.382\mathbf{a}_y$$

$$\text{For } \mathbf{H}_y = \frac{I}{2\pi\rho} \mathbf{a}_{\phi}, \quad \rho = \sqrt{(-3)^2 + 5^2} = \sqrt{34}$$

$$\mathbf{a}_{\phi} = \mathbf{a}_y \times \frac{(-3\mathbf{a}_x + 5\mathbf{a}_z)}{\sqrt{34}} = \frac{3\mathbf{a}_z + 5\mathbf{a}_x}{\sqrt{34}}$$

$$\mathbf{H}_y = \frac{10}{2\pi(34)}(5\mathbf{a}_x + 3\mathbf{a}_z) = 0.234\mathbf{a}_x + 0.1404\mathbf{a}_z$$

$$\mathbf{H} = \mathbf{H}_y + \mathbf{H}_z$$

$$= \underline{\underline{0.7433\mathbf{a}_x + 0.382\mathbf{a}_y + 0.1404\mathbf{a}_z \text{ A/m}}}$$

Prob. 7.2

$$H = \frac{I}{2\pi\rho} \rightarrow \rho = \frac{I}{2\pi H} = \frac{2}{2\pi(10 \times 10^{-3})} = \frac{100}{\pi} = \underline{\underline{31.83 \text{ m}}}$$

Prob. 7.3

$$\text{Let } \mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2$$

where $\mathbf{H}_1$ and $\mathbf{H}_2$ are respectively due to the lines located at (0,0) and (0,5).

$$\mathbf{H}_1 = \frac{I}{2\pi\rho}\mathbf{a}_\phi, \quad \rho = 5, \quad \mathbf{a}_\phi = \mathbf{a}_\ell \times \mathbf{a}_\rho = \mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y$$

$$\mathbf{H}_1 = \frac{10}{2\pi(5)}\mathbf{a}_y = \frac{\mathbf{a}_y}{\pi}$$

$$\mathbf{H}_2 = \frac{I}{2\pi\rho}\mathbf{a}_\phi, \quad \rho = 5\sqrt{2}, \quad \mathbf{a}_\phi = \mathbf{a}_\ell \times \mathbf{a}_\rho, \mathbf{a}_\ell = -\mathbf{a}_z$$

$$\mathbf{a}_\rho = \frac{5\mathbf{a}_x - 5\mathbf{a}_y}{5\sqrt{2}} = \frac{\mathbf{a}_x - \mathbf{a}_y}{\sqrt{2}}$$

$$\mathbf{a}_\phi = -\mathbf{a}_z \times \left(\frac{\mathbf{a}_x - \mathbf{a}_y}{\sqrt{2}} \right) = \frac{-\mathbf{a}_x - \mathbf{a}_y}{\sqrt{2}}$$

$$\mathbf{H}_2 = \frac{10}{2\pi 5\sqrt{2}} \left(\frac{-\mathbf{a}_x - \mathbf{a}_y}{\sqrt{2}} \right) = \frac{1}{2\pi} (-\mathbf{a}_x - \mathbf{a}_y)$$

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 = \frac{\mathbf{a}_y}{\pi} + \frac{1}{2\pi} (-\mathbf{a}_x - \mathbf{a}_y) = \underline{\underline{-0.1592\mathbf{a}_x + 0.1592\mathbf{a}_y}}$$

Prob. 7.4

$$\mathbf{H} = d\mathbf{H} = \frac{Id\mathbf{l} \times \mathbf{R}}{4\pi R^3} = \mathbf{H}_1 + \mathbf{H}_2$$

$$\text{For } \mathbf{H}_1, \quad \mathbf{R} = (3, 1, -2) - (0, 0, 0) = (3, 1, -2), \quad R = \sqrt{9+1+4} = \sqrt{14}$$

$$Id\mathbf{l} \times \mathbf{R} = 4 \times 10^{-5} \begin{vmatrix} 1 & 0 & 0 \\ 3 & 1 & -2 \end{vmatrix} = 4 \times 10^{-5} (0, 2, 1)$$

$$\mathbf{H}_1 = \frac{4 \times 10^{-5} (0, 2, 1)}{4\pi(14)^{3/2}} = (0, 0.01215, 0.006076)10^{-5}$$

$$\text{For } \mathbf{H}_2, \quad \mathbf{R} = (3, 1, -2) - (0, 0, 1) = (3, 1, -3), \quad R = \sqrt{9+1+9} = \sqrt{19}$$

$$Id\mathbf{l} \times \mathbf{R} = 6 \times 10^{-5} \begin{vmatrix} 0 & 1 & 0 \\ 3 & 1 & -3 \end{vmatrix} = 6 \times 10^{-5} (-3, 0, -3)$$

$$\mathbf{H}_2 = \frac{6 \times 10^{-5} (-3, 0, -3)}{4\pi(19)^{3/2}} = (-0.017, 0, -0.017)10^{-5}$$

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 = (-0.0173, 0.01215, -0.01122)10^{-5} = \underline{\underline{(-0.173\mathbf{a}_x + 1.215\mathbf{a}_y - 0.1122\mathbf{a}_z) \mu\text{A/m}}}$$

Prob. 7.5

$$\text{Let } \mathbf{H} = \mathbf{H}_y + \mathbf{H}_z$$

$$\text{For } \mathbf{H}_z, \quad \mathbf{H}_z = \frac{I}{2\pi\rho} \mathbf{a}_\phi$$

$$\text{where } I = 20, \quad \rho = -3\mathbf{a}_x + 4\mathbf{a}_y, \quad \rho = \sqrt{3^2 + 4^2} = 5$$

$$\mathbf{a}_\phi = \mathbf{a}_\ell \times \mathbf{a}_\rho = -\mathbf{a}_z \times \left(\frac{-3\mathbf{a}_x + 4\mathbf{a}_y}{5} \right) = \begin{vmatrix} 0 & 0 & -1 \\ -0.6 & 0.8 & 0 \end{vmatrix} = 0.8\mathbf{a}_x + 0.6\mathbf{a}_y$$

$$\mathbf{H}_z = \frac{20}{2\pi(5)} (0.8\mathbf{a}_x + 0.6\mathbf{a}_y) = 0.5093\mathbf{a}_x + 0.382\mathbf{a}_y$$

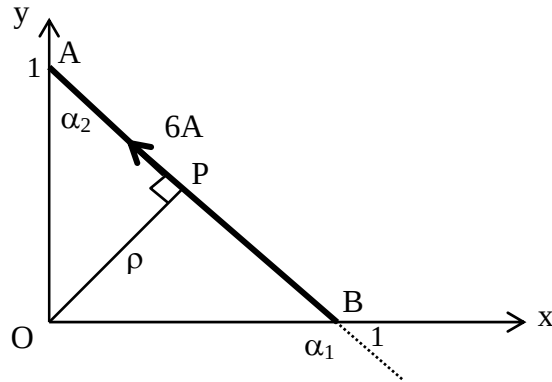
$$\text{For } \mathbf{H}_y, \quad \mathbf{H}_y = \frac{I}{2\pi\rho} \mathbf{a}_\phi$$

$$\text{where } I = 10, \quad \rho = -3\mathbf{a}_x + 5\mathbf{a}_z, \quad \rho = \sqrt{3^2 + 5^2} = \sqrt{34}$$

$$\mathbf{a}_\phi = \mathbf{a}_\ell \times \mathbf{a}_\rho = \frac{1}{\sqrt{34}} \begin{vmatrix} 0 & 1 & 0 \\ -3 & 0 & 5 \end{vmatrix} = \frac{1}{\sqrt{34}} (5\mathbf{a}_x + 3\mathbf{a}_z)$$

$$\mathbf{H}_y = \frac{10}{2\pi(34)} (5\mathbf{a}_x + 3\mathbf{a}_z) = 0.234\mathbf{a}_x + 0.1404\mathbf{a}_z$$

$$\mathbf{H} = \mathbf{H}_y + \mathbf{H}_z = 0.7432\mathbf{a}_x + 0.382\mathbf{a}_y + 0.1404\mathbf{a}_z \text{ A/m}$$

Prob. 7.6

$$\mathbf{H} = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\mathbf{a}_\phi$$

$$\alpha_1 = 135^\circ, \quad \alpha_2 = 45^\circ, \quad \rho = \frac{1}{2}\sqrt{2} = \frac{\sqrt{2}}{2}$$

$$\mathbf{a}_\phi = \mathbf{a}_l \times \mathbf{a}_\rho = \left(\frac{-\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}} \right) \times \left(\frac{-\mathbf{a}_x - \mathbf{a}_y}{\sqrt{2}} \right) = \frac{1}{2} \begin{vmatrix} -1 & 1 & 0 \\ -1 & -1 & 0 \end{vmatrix} = \mathbf{a}_z$$

$$\mathbf{H} = \frac{6}{4\pi \frac{\sqrt{2}}{2}} (\cos 45^\circ - \cos 135^\circ) \mathbf{a}_z = \frac{3}{\pi} \mathbf{a}_z$$

$$\mathbf{H} (0, 0, 0) = \underline{\underline{0.954\mathbf{a}_z \text{ A/m}}}$$

Prob. 7.7

(a) At (5,0,0), $\rho=5$, $\mathbf{a}_\phi = \mathbf{a}_y$, $\cos\alpha_1=0$, $\cos\alpha_2 = \frac{10}{\sqrt{125}}$

$$\mathbf{H} = \frac{2}{4\pi(5)} \left(\frac{10}{\sqrt{125}} \right) \mathbf{a}_y = \underline{\underline{28.471\mathbf{a}_y \text{ mA/m}}}$$

(b) At (5,5,0), $\rho=5\sqrt{2}$, $\cos\alpha_1=0$, $\cos\alpha_2 = \frac{10}{\sqrt{150}}$

$$\mathbf{a}_\phi = \mathbf{a}_z \times \left(\frac{\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}} \right) = \frac{-\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}}$$

$$\mathbf{H} = \frac{2}{4\pi(5\sqrt{2})} \left(\frac{10}{\sqrt{150}} \right) \left(\frac{-\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}} \right) = \underline{\underline{13(-\mathbf{a}_x + \mathbf{a}_y) \text{ mA/m}}}$$

(c) At (5,15,0), $\rho = \sqrt{250} = 5\sqrt{10}$, $\cos \alpha_1 = 0$, $\cos \alpha_2 = \frac{10}{\sqrt{350}}$

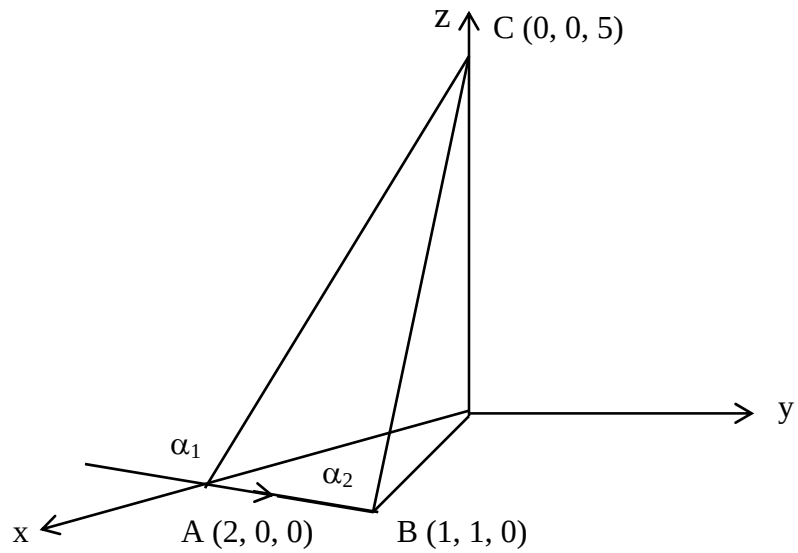
$$\mathbf{a}_\phi = \mathbf{a}_z \times \left(\frac{5\mathbf{a}_x + 15\mathbf{a}_y}{5\sqrt{10}} \right) = \frac{5\mathbf{a}_y - 15\mathbf{a}_x}{5\sqrt{10}}$$

$$\mathbf{H} = \frac{2}{4\pi(5\sqrt{10})} \left(\frac{10}{\sqrt{350}} \right) \left(\frac{-15\mathbf{a}_x + 5\mathbf{a}_y}{5\sqrt{10}} \right) = \underline{\underline{-5.1\mathbf{a}_x + 1.7\mathbf{a}_y \text{ mA/m}}}$$

d) At (5,-15,0), by symmetry,

$$\underline{\underline{\mathbf{H} = 5.1\mathbf{a}_x + 1.7\mathbf{a}_y \text{ mA/m}}}$$

Prob. 7.8



(a) Consider the figure above.

$$\mathbf{AB} = (1, 1, 0) - (2, 0, 0) = (-1, 1, 0)$$

$$\mathbf{AC} = (0, 0, 5) - (2, 0, 0) = (-2, 0, 5)$$

$\mathbf{AB} \cdot \mathbf{AC} = 2$, i.e $\mathbf{AB}$ and $\mathbf{AC}$ are not perpendicular.

$$\cos(180^\circ - \alpha_1) = \frac{\mathbf{AB} \cdot \mathbf{AC}}{|\mathbf{AB}||\mathbf{AC}|} = \frac{2}{\sqrt{2}\sqrt{29}} \rightarrow \cos \alpha_1 = -\sqrt{\frac{2}{29}}$$

$$\mathbf{BC} = (0, 0, 5) - (1, 1, 0) = (-1, -1, 5)$$

$$\mathbf{BA} = (1, -1, 0)$$

$$\cos \alpha_2 = \frac{\mathbf{BC} \cdot \mathbf{BA}}{|\mathbf{BC}||\mathbf{BA}|} = \frac{-1+1}{|\mathbf{BC}||\mathbf{BA}|} = 0$$

i.e. $\mathbf{BC} = \boldsymbol{\rho} = (-1, -1, 5), \quad \rho = \sqrt{27}$

$$\mathbf{a}_\phi = \mathbf{a}_l \times \mathbf{a}_\rho = \frac{(-1, 1, 0)}{\sqrt{2}} \times \frac{(-1, -1, 5)}{\sqrt{27}} = \frac{(5, 5, 2)}{\sqrt{54}}$$

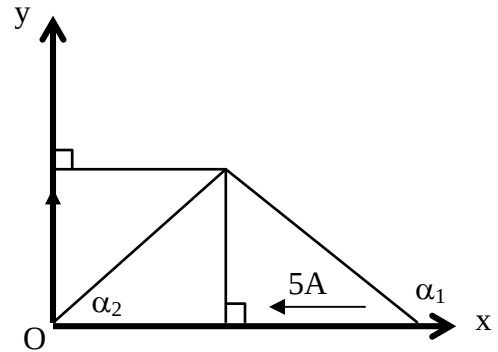
$$\begin{aligned} \mathbf{H}_2 &= \frac{10}{4\pi\sqrt{27}} \left(0 + \sqrt{\frac{2}{29}} \right) \frac{(5, 5, 2)}{\sqrt{2}\sqrt{27}} = \frac{5}{2\pi\sqrt{29}} \cdot \frac{(5, 5, 2)}{27} \text{ A/m} \\ &= \underline{\underline{27.37 \mathbf{a}_x + 27.37 \mathbf{a}_y + 10.95 \mathbf{a}_z \text{ mA/m}}} \end{aligned}$$

(b) $\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 + \mathbf{H}_3 = (0, -59.1, 0) + (27.37, 27.37, 10.95)$
 $+ (-30.63, 30.63, 0)$
 $= \underline{\underline{-3.26 \mathbf{a}_x - 1.1 \mathbf{a}_y + 10.95 \mathbf{a}_z \text{ mA/m}}}$

Prob. 7.9

(a) Let $\mathbf{H} = \mathbf{H}_x + \mathbf{H}_y = 2\mathbf{H}_x$

$$\mathbf{H}_x = \frac{I}{4\pi\rho} (\cos \alpha_2 - \cos \alpha_1) \mathbf{a}_\phi$$



where $\mathbf{a}_\phi = -\mathbf{a}_x \times \mathbf{a}_y = -\mathbf{a}_z$, $\alpha_1 = 180^\circ$, $\alpha_2 = 45^\circ$

$$\mathbf{H}_x = \frac{5}{4\pi(2)} (\cos 45^\circ - \cos 180^\circ) (-\mathbf{a}_z) = -0.3397\mathbf{a}_z$$

$$\mathbf{H} = 2\mathbf{H}_x = \underline{\underline{-0.6792 \mathbf{a}_z \text{ A/m}}}$$

(b) $\mathbf{H} = \mathbf{H}_x + \mathbf{H}_y$

where $\mathbf{H}_x = \frac{5}{4\pi(2)} (1-0) \mathbf{a}_\phi$, $\mathbf{a}_\phi = -\mathbf{a}_x \times -\mathbf{a}_y = \mathbf{a}_z$
 $= 198.9\mathbf{a}_z \text{ mA/m}$

$\mathbf{H}_y = 0$ since $\alpha_1 = \alpha_2 = 0$

$$\mathbf{H} = \underline{\underline{0.1989 \mathbf{a}_z \text{ A/m}}}$$

(c) $\mathbf{H} = \mathbf{H}_x + \mathbf{H}_y$

where $\mathbf{H}_x = \frac{5}{4\pi(2)} (1-0) (-\mathbf{a}_x \times \mathbf{a}_z) = 198.9 \mathbf{a}_y \text{ mA/m}$

$$\mathbf{H}_y = \frac{5}{4\pi(2)} (1-0) (\mathbf{a}_y \times \mathbf{a}_z) = 198.9 \mathbf{a}_x \text{ mA/m}$$

$$\mathbf{H} = \underline{\underline{0.1989 \mathbf{a}_x + 0.1989 \mathbf{a}_y \text{ A/m.}}}$$

Prob. 7.10

For the side of the loop along y-axis,

$$\mathbf{H}_1 = \frac{I}{4\pi\rho} (\cos \alpha_2 - \cos \alpha_1) \mathbf{a}_\phi$$

where $\mathbf{a}_\phi = -\mathbf{a}_x$, $\rho = 2 \tan 30^\circ = \frac{2}{\sqrt{3}}$, $\alpha_2 = 30^\circ$, $\alpha_1 = 150^\circ$

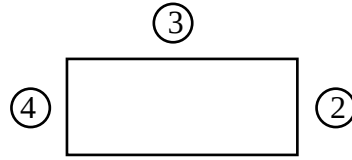
$$\mathbf{H}_1 = \frac{5}{4\pi} \frac{\sqrt{3}}{2} (\cos 30^\circ - \cos 150^\circ) (-\mathbf{a}_x) = -\frac{15}{8\pi} \mathbf{a}_x$$

$$\mathbf{H} = 3\mathbf{H}_1 = -1.79\mathbf{a}_x \text{ A/m}$$

Prob. 7.11

Let $\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 + \mathbf{H}_3 + \mathbf{H}_4$

where $\bar{H}_n$ is the contribution by side n.



(a) $\mathbf{H} = 2\mathbf{H}_1 + \mathbf{H}_2 + \mathbf{H}_4$ since $\mathbf{H}_1 = \mathbf{H}_3$

$$\mathbf{H}_1 = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\mathbf{a}_\phi = \frac{10}{4\pi(2)}\left(\frac{6}{\sqrt{40}} + \frac{1}{\sqrt{2}}\right)\mathbf{a}_z$$

$$\mathbf{H}_2 = \frac{10}{4\pi(6)}\left(2 \times \frac{2}{\sqrt{40}}\right)\mathbf{a}_z, \quad \mathbf{H}_4 = \frac{10}{4\pi(2)}\left(2 \cdot \frac{1}{\sqrt{2}}\right)\mathbf{a}_z$$

$$\mathbf{H} = \left[\frac{5}{2\pi}\left(\frac{3}{\sqrt{10}} + \frac{1}{\sqrt{2}}\right) + \frac{5}{6\pi\sqrt{10}} + \frac{5}{2\pi\sqrt{2}} \right] \mathbf{a}_z = \underline{\underline{1.964\mathbf{a}_z \text{ A/m}}}$$

(b) At $(4, 2, 0)$, $\mathbf{H} = 2(\mathbf{H}_1 + \mathbf{H}_4)$

$$\mathbf{H}_1 = \frac{10}{4\pi(2)} \frac{8}{\sqrt{20}} \mathbf{a}_z, \quad \mathbf{H}_4 = \frac{10}{4\pi(4)} \frac{4}{\sqrt{20}} \mathbf{a}_z$$

$$\mathbf{H} = \frac{2\sqrt{5}}{\pi} \left(1 + \frac{1}{4}\right) \mathbf{a}_z = \underline{\underline{1.78\mathbf{a}_z \text{ A/m}}}$$

(c) At $(4, 8, 0)$, $\mathbf{H} = \mathbf{H}_1 + 2\mathbf{H}_2 + \mathbf{H}_3$

$$\mathbf{H}_1 = \frac{10}{4\pi(8)} \left(2 \cdot \frac{4}{4\sqrt{5}}\right) \mathbf{a}_z, \quad \mathbf{H}_2 = \frac{10}{4\pi(4)} \left(\frac{8}{4\sqrt{5}} - \frac{1}{\sqrt{2}}\right) \mathbf{a}_z$$

$$\mathbf{H}_3 = \frac{10}{4\pi(4)} \left(\frac{2}{\sqrt{2}}\right) (-\mathbf{a}_z)$$

$$\mathbf{H} = \frac{5}{8\pi} (\mathbf{a}_z) \left(\frac{1}{\sqrt{5}} + \frac{4}{\sqrt{5}} - \frac{4}{\sqrt{2}}\right) = \underline{\underline{-0.1178\mathbf{a}_z \text{ A/m}}}$$

(d) At $(0, 0, 2)$,

$$\mathbf{H}_1 = \frac{10}{4\pi(2)} \left(\frac{8}{\sqrt{68}} - 0\right) (\mathbf{a}_x \times \mathbf{a}_z) = -\frac{10}{\pi\sqrt{68}} \mathbf{a}_y$$

$$\mathbf{H}_2 = \frac{10}{4\pi\sqrt{68}} \left(\frac{4}{\sqrt{84}} - 0\right) \mathbf{a}_y \times \left(\frac{2\mathbf{a}_z - 8\mathbf{a}_x}{\sqrt{68}}\right) = \frac{5(\mathbf{a}_x + 4\mathbf{a}_z)}{17\pi\sqrt{84}}$$

$$\begin{aligned}
 \mathbf{H}_3 &= \frac{10}{4\pi\sqrt{20}} \left(-\frac{8}{\sqrt{84}} - 0 \right) \mathbf{a}_x \times \left(\frac{2\mathbf{a}_x - 4\mathbf{a}_y}{\sqrt{20}} \right) = \frac{\mathbf{a}_y + 2\mathbf{a}_z}{\pi\sqrt{21}} \\
 \mathbf{H}_4 &= \frac{10}{4\pi\sqrt{2}} \left(0 + \frac{4}{\sqrt{20}} \right) (-\mathbf{a}_y \times \mathbf{a}_z) = \frac{-5\mathbf{a}_x}{\pi\sqrt{20}} \\
 \mathbf{H} &= \left(\frac{5}{34\pi\sqrt{21}} - \frac{5}{\pi\sqrt{20}} \right) \mathbf{a}_x + \left(\frac{1}{\pi\sqrt{21}} - \frac{10}{\pi\sqrt{68}} \right) \mathbf{a}_y + \left(\frac{20}{34\pi\sqrt{21}} + \frac{2}{\pi\sqrt{21}} \right) \mathbf{a}_z \\
 &= \underline{\underline{-0.3457 \bar{a}_x - 0.3165 \bar{a}_y + 0.1798 \bar{a}_z \text{ A/m}}}
 \end{aligned}$$

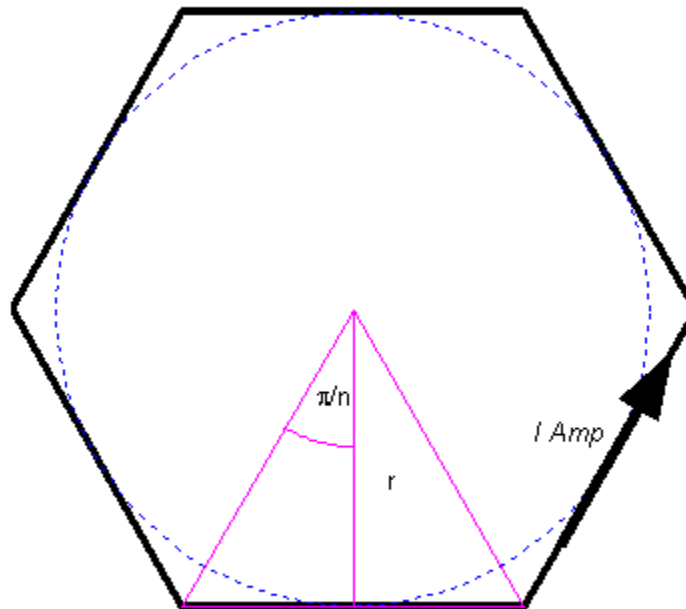
Prob. 7.12

$$\mathbf{H} = 4\mathbf{H}_1 = 4 \frac{I}{4\pi\rho} (\cos\alpha_2 - \cos\alpha_1) \mathbf{a}_\phi$$

$$\rho = a = 2\text{cm}, I = 5\text{mA}, \alpha_2 = 45^\circ, \alpha_1 = 90^\circ + 45^\circ = 135^\circ$$

$$\mathbf{a}_\phi = \mathbf{a}_\ell \times \mathbf{a}_\rho = \mathbf{a}_y \times (-\mathbf{a}_x) = \mathbf{a}_z$$

$$\mathbf{H} = \frac{I}{\pi a} \left(\frac{1}{\sqrt{2}} - -\frac{1}{\sqrt{2}} \right) \mathbf{a}_z = \frac{\sqrt{2}I}{\pi a} \mathbf{a}_z = \frac{\sqrt{2} \times 5 \times 10^{-3}}{\pi \times 2 \times 10^{-2}} \mathbf{a}_z = \underline{\underline{0.1125 \mathbf{a}_z}}$$

Prob. 7.13

- (a) Consider one side of the polygon as shown. The angle subtended by the Side At the center of the circle

$$\frac{360^\circ}{n} = \frac{2\pi}{n}$$

The field due to this side is

$$H_1 = \frac{I}{4\pi\rho} (\cos \alpha_2 - \cos \alpha_1)$$

where $\rho = r$, $\cos \alpha_2 = \cos(90 - \frac{\pi}{n}) = \sin \frac{\pi}{n}$

$$\cos \alpha_1 = -\sin \frac{\pi}{n}$$

$$H_1 = \frac{I}{4\pi r} 2 \sin \frac{\pi}{n}$$

$$H = nH_1 = \frac{nI}{2\pi r} \sin \frac{\pi}{n}$$

(b) For $n=3$, $H = \frac{3I}{2\pi r} \sin \frac{\pi}{3}$

$$r \cot 30^\circ = 2 \rightarrow r = \frac{2}{\sqrt{3}}$$

$$H = \frac{3 \times 5}{2\pi \cdot \frac{2}{\sqrt{3}}} \cdot \frac{\sqrt{3}}{2} = \frac{45}{8\pi} = 1.79 \text{ A/m.}$$

For $n=4$, $H = \frac{4I}{2\pi r} \sin \frac{\pi}{4} = \frac{4 \times 5}{2\pi(2)} \cdot \frac{1}{\sqrt{2}}$
 $= 1.128 \text{ A/m.}$

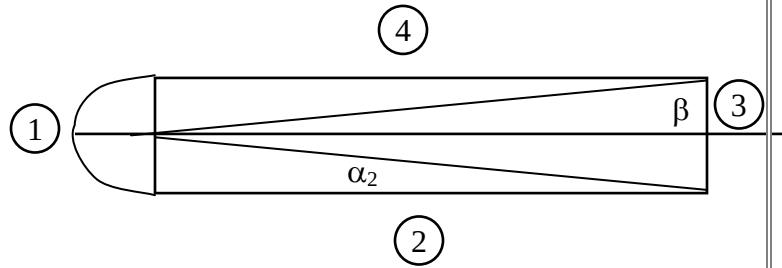
- (c) As $n \rightarrow \infty$,

$$H = \lim_{n \rightarrow \infty} \frac{nI}{2\pi r} \sin \frac{\pi}{n} = \frac{nI}{2\pi r} \cdot \frac{\pi}{n} = \frac{I}{2r}$$

From Example 7.3, when $h=0$,

$$H = \frac{I}{2r}$$

which agrees.

Prob. 7.14

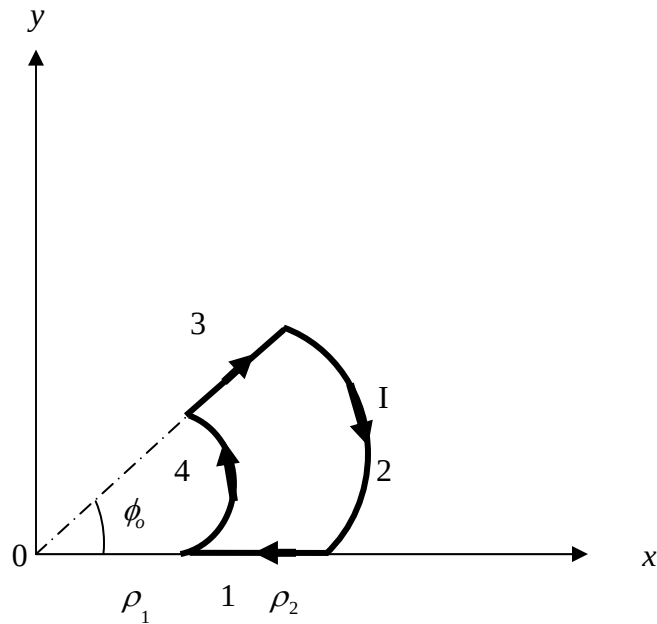
Let $\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 + \mathbf{H}_3 + \mathbf{H}_4$

$$\mathbf{H}_1 = \frac{I}{4a} \mathbf{a}_z = \frac{10}{4 \times 4 \times 10^{-2}} \mathbf{a}_z = 62.5 \mathbf{a}_z$$

$$\begin{aligned} \mathbf{H}_2 = \mathbf{H}_4 &= \frac{I}{4\pi \times 4 \times 10^{-2}} (\cos \alpha_2 - \cos 90^\circ) \mathbf{a}_z, \quad \alpha_2 = \tan^{-1} \frac{4}{100} = 2.29^\circ \\ &= 19.88 \mathbf{a}_z \end{aligned}$$

$$\begin{aligned} \mathbf{H}_3 &= \frac{I}{4\pi(1)} 2 \cos \beta \mathbf{a}_z, \quad \beta = \tan^{-1} \frac{100}{4} = 87.7^\circ \\ &= \frac{10}{4\pi} 2 \cos 87.7^\circ \mathbf{a}_z = 0.06361 \mathbf{a}_z \end{aligned}$$

$$\begin{aligned} \mathbf{H} &= (62.5 + 2 \times 19.88 + 0.06361) \mathbf{a}_z \\ &= 102.32 \mathbf{a}_z \text{ A/m.} \end{aligned}$$

Prob. 7.15

Let $\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 + \mathbf{H}_3 + \mathbf{H}_4$

which correspond with sides 1, 2, 3, and 4 as shown in the figure above.

$\mathbf{H}_1$ and $\mathbf{H}_3$ can be found using eq. (7.12). It can be shown that

$$\mathbf{H}_3 = -\mathbf{H}_1$$

For $\mathbf{H}_4$,
$$d\mathbf{H}_4 = \frac{Id\mathbf{l} \times \mathbf{R}}{4\pi R^3}, \quad d\mathbf{l} = \rho_1 d\phi \mathbf{a}_\phi, \quad \mathbf{R} = -\rho_1 \mathbf{a}_\rho$$

$$d\mathbf{H}_4 = \frac{I\rho_1^2 d\phi \mathbf{a}_\phi \times (-\mathbf{a}_\rho)}{4\pi\rho_1^3} = \frac{Id\phi(\mathbf{a}_z)}{4\pi\rho_1}$$

$$\mathbf{H}_4 = \frac{I(\mathbf{a}_z)}{4\pi\rho_1} \int_0^{\phi_0} d\phi = \frac{I\phi_0(\mathbf{a}_z)}{4\pi\rho_1}$$

Similarly, for $\mathbf{H}_2$,

$$\mathbf{H}_2 = \frac{I\phi_0(-\mathbf{a}_z)}{4\pi\rho_2}$$

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 + \mathbf{H}_3 + \mathbf{H}_4 = \underline{\underline{\frac{I\phi_0}{4\pi} \left(\frac{1}{\rho_1} - \frac{1}{\rho_2} \right) \mathbf{a}_z}}$$

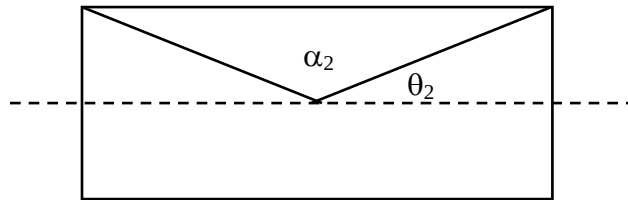
Prob. 7.16

From Example 7.3,

$$\mathbf{H}_1 = \frac{I}{2a} \mathbf{a}_z, \quad \mathbf{H}_2 = \frac{Ia^2}{2[a^2 + d^2]^{3/2}} \mathbf{a}_z$$

$$\begin{aligned} \mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 &= \frac{I}{2} \left[\frac{1}{a} + \frac{a^2}{[a^2 + d^2]^{3/2}} \right] \mathbf{a}_z = \frac{10}{2} \left[\frac{1}{3 \times 10^{-2}} + \frac{3^2}{[3^2 + 4^2]^{3/2} (10^{-2})} \right] \mathbf{a}_z \\ &= 500 \left[\frac{1}{3} + \frac{9}{125} \right] \mathbf{a}_z = \underline{\underline{202.67 \mathbf{a}_z \text{ A/m}}} \end{aligned}$$

Prob. 7.17

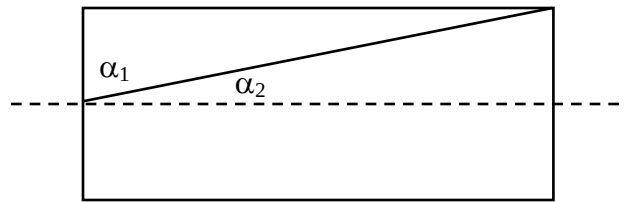


$$|\mathbf{H}| = \frac{nI}{2} (\cos \theta_2 - \cos \theta_1)$$

$$\cos \theta_2 = -\cos \theta_1 = \frac{\ell/2}{\left(a^2 + \ell^2/4\right)^{1/2}}$$

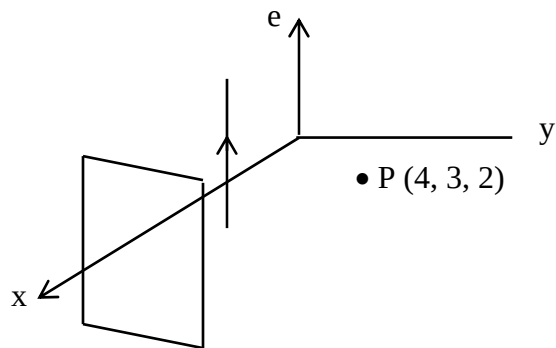
$$|\mathbf{H}| = \frac{nI\ell}{2\left(a^2 + \ell^2/4\right)^{1/2}} = \frac{0.5 \times 150 \times 2 \times 10^{-2}}{2 \times 10^{-3} \times \sqrt{4^2 + 10^2}} = \underline{\underline{69.63 \text{ A/m}}}$$

(b)



$$\alpha_1 = 90^\circ, \tan \theta_2 = \frac{a}{b} = \frac{4}{20} = 0.2 \rightarrow \theta_2 = 11.31^\circ$$

$$|\mathbf{H}| = \frac{nI}{2} \cos \theta_2 = \frac{150 \times 0.5}{2} \cos 11.31^\circ = \underline{\underline{36.77 \text{ A/m}}}$$

Prob. 7.18

Let $\mathbf{H} = \mathbf{H}_l + \mathbf{H}_p$

$$\mathbf{H}_l = \frac{1}{2\pi\rho} \mathbf{a}_\phi$$

$$\boldsymbol{\rho} = (4, 3, 2) - (1, -2, 2) = (3, 5, 0), \quad \rho = |\boldsymbol{\rho}| = \sqrt{34}$$

$$\mathbf{a}_\rho = \frac{3\mathbf{a}_x + 5\mathbf{a}_y}{\sqrt{34}}, \quad \mathbf{a}_l = \mathbf{a}_z$$

$$\mathbf{a}_\phi = \mathbf{a}_l \times \mathbf{a}_\rho = \mathbf{a}_z \times \left(\frac{3\mathbf{a}_x + 5\mathbf{a}_y}{\sqrt{34}} \right) = \frac{3\mathbf{a}_y - 5\mathbf{a}_x}{\sqrt{34}}$$

$$\mathbf{H}_l = \frac{20\pi}{2\pi} \left(\frac{-5\mathbf{a}_x + 3\mathbf{a}_y}{34} \right) \times 10^{-3} = (-1.47\mathbf{a}_y + 0.88\mathbf{a}_x) \text{ mA/m}$$

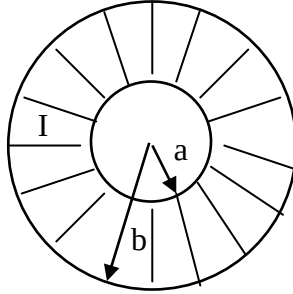
$$\mathbf{H}_p = \frac{1}{2} \mathbf{K} \times \mathbf{a}_n = \frac{1}{2} (100 \times 10^{-3}) \mathbf{a}_z \times (-\mathbf{a}_x) = -0.05\mathbf{a}_y \text{ A/m}$$

$$\mathbf{H} = \mathbf{H}_l + \mathbf{H}_p = \underline{\underline{-1.47\mathbf{a}_x - 49.12 \mathbf{a}_y \text{ mA/m}}}$$

Prob. 7.19

(a) See text

(b)



$$\text{For } \rho < a, \quad \oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} = 0 \rightarrow \mathbf{H} = 0$$

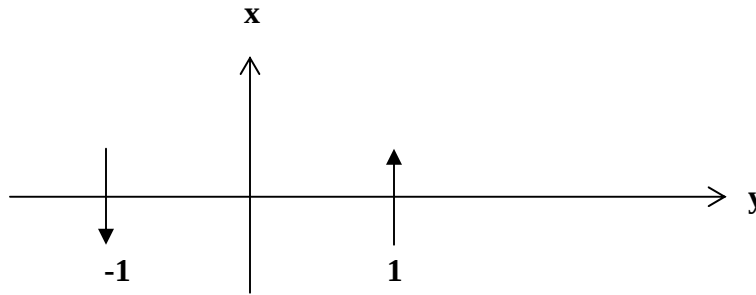
$$\text{For } a < \rho < b, \quad H_{\phi} \cdot 2\pi\rho = \frac{I\pi(\rho^2 - a^2)}{\pi(b^2 - a^2)}$$

$$H_{\phi} = \frac{I}{2\pi\rho} \left(\frac{\rho^2 - a^2}{b^2 - a^2} \right)$$

$$\text{For } \rho > b, \quad H_{\phi} \cdot 2\pi\rho = I \rightarrow H_{\phi} = \frac{I}{2\pi\rho}$$

Thus,

$$H_{\phi} = \begin{cases} 0, & \rho < a \\ \frac{I}{2\pi\rho} \left(\frac{\rho^2 - a^2}{b^2 - a^2} \right), & a < \rho < b \\ \frac{I}{2\pi\rho}, & \rho > b \end{cases}$$

Prob.7.20

$$\begin{aligned}
 \mathbf{H} &= \sum \frac{1}{2} \mathbf{K} \times \mathbf{a}_n \\
 &= \frac{1}{2} (20\mathbf{a}_x) \times (-\mathbf{a}_y) + \frac{1}{2} (-20\mathbf{a}_x) \times \mathbf{a}_y \\
 &= 10(-\mathbf{a}_z) - 10(\mathbf{a}_z) \\
 &= \underline{\underline{-20\mathbf{a}_z \text{ A/m}}}
 \end{aligned}$$

Prob. 7.21

$$\mathbf{H}_p = \frac{1}{2} \mathbf{k} \times \mathbf{a}_n = \frac{1}{2} 10\mathbf{a}_x \times \mathbf{a}_z = -5\mathbf{a}_y$$

$$\mathbf{H}_L = \frac{I}{2\pi\rho} \mathbf{a}_\phi = \frac{I}{2\pi(3)} (\mathbf{a}_x \times -\mathbf{a}_z) = \frac{I}{6\pi} \mathbf{a}_y$$

$$\mathbf{H}_p + \mathbf{H}_L = -5\mathbf{a}_y + \frac{I}{6\pi} \mathbf{a}_y = 0 \quad \longrightarrow \quad I = 30\pi = \underline{\underline{94.25 \text{ A}}}$$

Prob. 7.22

(a)

From eq. (7.29),

$$\mathbf{H} = \begin{cases} \frac{I\rho}{2\pi a^2} \mathbf{a}_\phi, & 0 < \rho < a \\ \frac{I}{2\pi\rho} \mathbf{a}_\phi, & \rho > a \end{cases}$$

(b)

$$\mathbf{J} = \nabla \times \mathbf{H} = \frac{1}{\rho} \frac{d}{d\rho} (\rho H_\phi) \mathbf{a}_z = \begin{cases} \frac{I}{\pi a^2} \mathbf{a}_z, & \rho < a \\ 0\mathbf{a}_z, & \rho > a \end{cases}$$

Prob. 7.23

For $0 < \rho < a$

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = I_{enc} = \int \mathbf{J} \cdot d\mathbf{S}$$

$$H_\phi 2\pi\rho = \int_{\phi=0}^{2\pi} \int_{\rho=0}^{\rho} \frac{J_o}{\rho} \rho d\phi d\rho$$

$$= J_o 2\pi\rho$$

$$H_\phi = J_o$$

For $\rho > a$

$$\int \mathbf{H} \cdot d\mathbf{l} = \int \mathbf{J} \cdot d\mathbf{S} = \int_{\phi=0}^{2\pi} \int_{\rho=0}^a \frac{J_o}{\rho} \rho d\phi d\rho$$

$$H_\rho 2\pi\rho = J_o 2\pi a$$

$$H_\phi = \frac{J_o a}{\rho}$$

$$\text{Hence } H_\phi = \begin{cases} J_o, & 0 < \rho < a \\ \frac{J_o a}{\rho}, & \rho > a \end{cases}$$

Prob. 7.24

$$(a) \quad \mathbf{J} = \nabla \times \mathbf{H} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x^2 & 0 \end{vmatrix} = \underline{\underline{2(x-y)\mathbf{a}_z}}$$

$$(b) \quad d\mathbf{S} = dx dy \mathbf{a}_z,$$

$$\begin{aligned} I &= \int_S \mathbf{J} \cdot d\mathbf{S} = \iint (2x - 2y) dx dy = 2 \int_1^5 dy \int_0^2 x dx - 2 \int_0^2 dx \int_1^5 y dy = 2(4) \left(\frac{x^2}{2} \Big|_0^2 \right) - 2(2) \left(\frac{y^2}{2} \Big|_1^5 \right) \\ &= 4(4 - 0) - 2(25 - 1) = 16 - 48 = \underline{\underline{-32 \text{ A}}} \end{aligned}$$

Prob. 7.25

$$(a) \quad \mathbf{J} = \nabla \times \mathbf{H} = \frac{1}{\rho} \frac{d}{d\rho} (\rho H_\phi) \mathbf{a}_z = \frac{1}{\rho} \frac{d}{d\rho} (k_o \frac{\rho^2}{a}) \mathbf{a}_z = \underline{\underline{\frac{2k_o}{a} \mathbf{a}_z}}$$

(b) For $\rho > a$,

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{enc} = \int \mathbf{J} \cdot d\mathbf{S} = \int_{\rho=0}^a \int_{\phi=0}^{2\pi} \frac{2k_o}{a} \rho d\rho d\phi = \frac{2k_o}{a} (2\pi) \frac{\rho^2}{2} \bigg|_0^a$$

$$H_\phi 2\pi\rho = 2\pi k_o a \longrightarrow H_\phi = \frac{k_o a}{\rho}$$

$$\underline{\underline{\mathbf{H} = k_o \left(\frac{a}{\rho} \right) \mathbf{a}_\phi, \quad \rho > a}}$$

Prob. 7.26

$$\mathbf{J} = \nabla \times \mathbf{H} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x^2 & 0 \end{vmatrix} = (2x - 2y)\mathbf{a}_z$$

At (1, -4, 7), x = 1, y = -4, z = 7,

$$\mathbf{J} = [2(1) - 2(-4)]\mathbf{a}_z = \underline{\underline{10\mathbf{a}_z \text{ A/m}^2}}$$

Prob. 7.27

(a)

$$\begin{aligned} \mathbf{J} = \nabla \times \mathbf{H} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho H_\phi) \mathbf{a}_z = \frac{1}{\rho} \frac{\partial}{\partial \rho} (10^3 \rho^3) \mathbf{a}_z \\ &= \underline{\underline{3\rho \times 10^3 \mathbf{a}_z \text{ A/m}^2}} \end{aligned}$$

(b)

Method 1:

$$\begin{aligned} I = \int_S \mathbf{J} \cdot d\mathbf{S} &= \iint 3\rho \rho d\phi d\rho 10^3 = 3 \times 10^3 \int_0^2 \rho^2 d\rho \int_0^{2\pi} d\phi \\ &= 3 \times 10^3 (2\pi) \frac{\rho^3}{3} \bigg|_0^2 = 16\pi \times 10^3 \text{ A} = \underline{\underline{50.265 \text{ kA}}} \end{aligned}$$

Method 2:

$$I = \oint_L \mathbf{H} \cdot d\mathbf{l} = 10^3 \int_0^{2\pi} \rho^2 \rho d\phi = 10^3 (8)(2\pi) = \underline{\underline{50.265 \text{ kA}}}$$

Prob. 7.28

$$\mathbf{J} = \nabla \times \mathbf{H} = \frac{1}{\rho} \frac{d}{d\rho} (\rho H_\phi) \mathbf{a}_z = \frac{1}{\rho} \frac{d}{d\rho} (4\rho^2) \mathbf{a}_z = 8\mathbf{a}_z$$

$$I = \int_S \mathbf{J} \cdot d\mathbf{S} = JS = 8(\pi a^2) = 8\pi \times 10^{-4} = \underline{\underline{2.513 \text{ mA}}}$$

Prob. 7.29

$$(a) \quad \mathbf{B} = \frac{\mu_o I}{2\pi\rho} \mathbf{a}_\phi$$

At (-3,4,5), $\rho=5$.

$$B = \frac{4\pi \times 10^{-7} \times 2}{2\pi(5)} a_\phi = \underline{\underline{80 a_\phi \text{ nWb/m}^2}}$$

$$(b) \quad \Psi = \int \mathbf{B} \cdot d\mathbf{S} = \frac{\mu_o I}{2\pi} \iint \frac{d\rho dz}{\rho} = \frac{4\pi \times 10^{-7} \times 2}{2\pi} \ln \rho \bigg|_2^6 \bigg|_0^4$$

$$= 16 \times 10^{-7} \ln 3 = \underline{\underline{1.756 \mu\text{Wb}}}$$

Prob. 7.30

Let $\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2$

where $\mathbf{H}_1$ and $\mathbf{H}_2$ are due to the wires centered at $x = 0$ and $x = 10\text{cm}$ respectively.

$$(a) \quad \text{For } \mathbf{H}_1, \rho = 50 \text{ cm}, \mathbf{a}_\phi = \mathbf{a}_l \times \mathbf{a}_\rho = \mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y$$

$$\mathbf{H}_1 = \frac{5}{2\pi(5 \times 10^{-2})} \mathbf{a}_y = \frac{50}{\pi} \mathbf{a}_y$$

$$\text{For } \mathbf{H}_2, \rho = 5 \text{ cm}, \mathbf{a}_\phi = -\mathbf{a}_z \times -\mathbf{a}_x = \mathbf{a}_y, \mathbf{H}_2 = \mathbf{H}_1$$

$$\mathbf{H} = 2\mathbf{H}_1 = \frac{100}{\pi} \mathbf{a}_y$$

$$= \underline{\underline{31.83 \mathbf{a}_y \text{ A/m}}}$$

$$(b) \quad \text{For } \mathbf{H}_1, \mathbf{a}_\phi = \mathbf{a}_z \times \left(\frac{2\mathbf{a}_x + \mathbf{a}_y}{\sqrt{5}} \right) = \frac{2\mathbf{a}_y - \mathbf{a}_x}{\sqrt{5}}$$

$$\mathbf{H}_1 = \frac{5}{2\pi 5\sqrt{5} \times 10^{-2}} \left(\frac{-\mathbf{a}_x + 2\mathbf{a}_y}{\sqrt{5}} \right) = -3.183\mathbf{a}_x + 6.366\mathbf{a}_y$$

$$\text{For } \mathbf{H}_2, \mathbf{a}_\rho = -\mathbf{a}_z \times \mathbf{a}_y = \mathbf{a}_x$$

$$\mathbf{H}_2 = \frac{5}{2\pi(5)} \mathbf{a}_x = 15.915\mathbf{a}_x$$

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2$$

$$= \underline{\underline{12.3 \mathbf{a}_x + 6.366\mathbf{a}_y \text{ A/m}}}$$

Prob. 7.31

$$(a) I = \int J \, dS$$

$$\begin{aligned}
 &= \int_{\phi=0}^{2\pi} \int_{\rho=0}^a J_o \left(1 - \frac{\rho^2}{a^2}\right) \rho \, d\rho \, d\phi = J_o \int_0^{2\pi} d\phi \int_0^a \left(\rho - \frac{\rho^3}{a^2}\right) d\rho \\
 &= 2\pi J_o \left(\frac{\rho^2}{2} - \frac{\rho^4}{4a^2} \right) \Big|_0^a = \frac{2\pi}{2} J_o \left(a^2 - \frac{a^2}{2} \right) \\
 &= \frac{1}{2} \pi a^2 J_o
 \end{aligned}$$

$$(b) \oint \mathbf{H} \, dl = I_{enc} = \int J \, dS$$

For $\rho < a$,

$$\begin{aligned}
 H_\phi 2\pi\rho &= \int J \, dS \\
 &= 2\pi J_o \left(\frac{\rho^2}{2} - \frac{\rho^4}{4a^2} \right) \\
 H_\rho \cancel{2\pi\rho} &= 2\pi J_o \frac{\rho^2}{4} \left(2 - \frac{\rho^2}{a^2} \right) \\
 H_\rho &= \frac{J_o \rho}{4} \left(2 - \frac{\rho^2}{a^2} \right)
 \end{aligned}$$

For $\rho > a$,

$$\oint \mathbf{H} \, dl = \int J_o \, dS = I$$

$$H_\phi \cancel{2\pi\rho} = \frac{1}{2} \pi a^2 J_o$$

$$H_\phi = \frac{a^2 J_o}{4\rho}$$

$$\text{Hence } H_\phi = \begin{cases} \frac{J_o \rho}{4} \left(2 - \frac{\rho^2}{a^2} \right), & \rho < a \\ \frac{a J_o}{4\rho}, & \rho > a \end{cases}$$

Prob. 7.32

$$\mathbf{B} = \frac{\mu_0 I}{2\pi\rho} \mathbf{a}_\phi$$

$$\begin{aligned} \Psi &= \int \mathbf{B} \cdot d\mathbf{S} = \int_{\rho=d}^{d+a} \int_{z=0}^b \frac{\mu_0 I}{2\pi\rho} d\rho dz \\ &= \frac{\mu_0 I b}{2\pi} \ln \frac{d+a}{d} \end{aligned}$$

Prob. 7.33

For a whole circular loop of radius a , Example 7.3 gives

$$\mathbf{H} = \frac{I a^2 \mathbf{a}_z}{2[a^2 + h^2]^{3/2}}$$

Let $h \rightarrow 0$

$$\mathbf{H} = \frac{I}{2a} \mathbf{a}_z$$

For a semicircular loop, H is halved

$$\mathbf{H} = \frac{I}{4a} \mathbf{a}_z$$

$$\mathbf{B} = \mu_0 \mathbf{H} = \frac{\mu_0 I}{4a} \mathbf{a}_z$$

Prob. 7.34

$$(a) \quad \nabla \cdot \mathbf{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0$$

showing that $\mathbf{B}$ satisfies Maxwell's equation.

$$(b) \quad d\mathbf{S} = dydz \mathbf{a}_x$$

$$\Psi = \int \mathbf{B} \cdot d\mathbf{S} = \int_{z=1}^4 \int_{y=0}^1 y^2 dy dz = \frac{y^3}{3} \bigg|_0^1 (z) \bigg|_1^4 = \underline{\underline{1 \text{ Wb}}}$$

$$(c) \nabla \times \mathbf{H} = \mathbf{J} \longrightarrow \mathbf{J} = \nabla \times \frac{\mathbf{B}}{\mu_o}$$

$$\nabla \times \mathbf{B} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & z^2 & x^2 \end{vmatrix} = -2z\mathbf{a}_x - 2x\mathbf{a}_y - 2y\mathbf{a}_z$$

$$\mathbf{J} = -\frac{2}{\mu_o}(z\mathbf{a}_x + x\mathbf{a}_y + y\mathbf{a}_z) \text{ A/m}^2$$

(d)

Since $\nabla \cdot \mathbf{B} = 0$,

$$\Psi = \oint_S \mathbf{B} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{B} dv = 0$$

Prob. 7.35

On the slant side of the ring, $z = \frac{h}{6}(\rho - a)$

where $\mathbf{H}_1$ and $\mathbf{H}_2$ are due to the wires centered at $x = 0$ and $x = 10\text{cm}$ respectively.

$$\begin{aligned} \psi &= \int \mathbf{B} \cdot d\mathbf{S} = \int \frac{\mu_o I}{2\pi\rho} d\rho dz \\ &= \frac{\mu_o I}{2\pi} \int_{\rho=a}^{a+b} \int_{z=0}^{\frac{h}{6}(\rho-a)} \frac{dz d\rho}{\rho} = \frac{\mu_o I h}{2\pi b} \int_{\rho=a}^{a+b} \left(1 - \frac{a}{\rho}\right) d\rho \\ &= \frac{\mu_o I h}{2\pi b} \left(b - a \ln \frac{a+b}{a} \right) \text{ as required.} \end{aligned}$$

If $a = 30\text{ cm}$, $b = 10\text{ cm}$, $h = 5\text{ cm}$, $I = 10\text{ A}$,

$$\begin{aligned} \psi &= \frac{4\pi \times 10^{-7} \times 10 \times 0.05}{2\pi(10 \times 10^{-2})} \left(0.1 - 0.3 \ln \frac{4}{3} \right) \\ &= \underline{\underline{1.37 \times 10^{-8} \text{ Wb}}} \end{aligned}$$

Prob. 7.36

$$\psi = \int \mathbf{B} \cdot d\mathbf{S} = \mu_o \int_{z=0}^{0.2} \int_{\phi=0}^{50^\circ} \frac{10^6}{\rho} \sin 2\phi \rho d\phi dz$$

$$\psi = 4\pi \times 10^{-7} \times 10^6 (0.2) \left(-\frac{\cos 2\phi}{2} \right) \bigg|_0^{50^\circ}$$

$$= 0.04\pi (1 - \cos 100^\circ)$$

$$= \underline{\underline{0.1475 \text{ Wb}}}$$

Prob. 7.37

$$\begin{aligned}
 \psi &= \int_S \mathbf{B} \cdot d\mathbf{S} = \int_{\phi=0}^{\pi/4} \int_{\rho=1}^2 \frac{20}{\rho} \sin^2 \phi \, \rho d\rho d\phi = 20 \int_1^2 d\rho \int_0^{\pi/4} \sin^2 \phi d\phi \\
 &= 20(1) \int_0^{\pi/4} \frac{1}{2} (1 - \cos 2\phi) d\phi = 10 \left(\phi - \frac{1}{2} \sin 2\phi \right) \Big|_0^{\pi/4} \\
 &= 10 \left(\frac{\pi}{4} - \frac{1}{2} \right) = \underline{\underline{2.854 \text{ Wb}}}
 \end{aligned}$$

Prob. 7.38

$$\begin{aligned}
 \psi &= \int_S \mathbf{B} \cdot d\mathbf{S}, \quad d\mathbf{S} = r^2 \sin \theta d\theta d\phi \mathbf{a}_r \\
 \psi &= \iint \frac{2}{r^3} \cos \theta \, r^2 \sin \theta d\theta d\phi \Big|_{r=1} = 2 \int_0^{2\pi} d\phi \int_0^{\pi/3} \cos \theta \sin \theta d\theta \\
 &= 2(2\pi) \int_0^{\pi/3} \sin \theta d(\sin \theta) = 4\pi \frac{\sin^2 \theta}{2} \Big|_0^{\pi/3} = 2\pi \sin^2(\pi/3) \\
 &= \underline{\underline{4.7123 \text{ Wb}}}
 \end{aligned}$$

Prob. 7.39

$$\mathbf{B} = \mu_o \mathbf{H} = \frac{\mu_o}{4\pi} \int_v \frac{\mathbf{J} \times \mathbf{R}}{R^3} dv$$

Since current is the flow of charge, we can express this in terms of a charge moving with velocity $\mathbf{u}$. $\mathbf{J}dv = dq\mathbf{u}$.

$$\mathbf{B} = \frac{\mu_o}{4\pi} \left[\frac{q\mathbf{u} \times \mathbf{R}}{R^3} \right]$$

In our case, $\mathbf{u}$ and $\mathbf{R}$ are perpendicular. Hence,

$$\begin{aligned}
 B &= \frac{\mu_o}{4\pi} \frac{qu}{R^2} = \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{1.6 \times 10^{-19} \times 2.2 \times 10^6}{(5.3 \times 10^{-11})^2} = \frac{1.6 \times 10^{-20}}{(5.3)^2 \times 10^{-22}} \\
 &= \underline{\underline{12.53 \text{ Wb/m}^2}}
 \end{aligned}$$

Prob. 7.40

$$(a) \quad \nabla \cdot \mathbf{A} = -ya \sin ax \neq 0$$

$$\begin{aligned} \nabla \times \mathbf{A} &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y \cos ax & 0 & y + e^{-x} \end{vmatrix} \\ &= \mathbf{a}_x + e^{-x} \mathbf{a}_y - \cos ax \mathbf{a}_z \neq \mathbf{0} \end{aligned}$$

$\mathbf{A}$ is neither electrostatic nor magnetostatic field

$$(b) \quad \nabla \cdot \mathbf{B} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho B_\rho) = \frac{1}{\rho} \frac{\partial}{\partial \rho} (20) = 0$$

$$\nabla \times \mathbf{B} = \mathbf{0}$$

$\mathbf{B}$ can be $\mathbf{E}$ -field in a charge-free region.

$$(c) \quad \nabla \cdot \mathbf{C} = \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (r^2 \sin \theta) = 0$$

$$\nabla \times \mathbf{C} = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r^2 \sin^2 \theta) \mathbf{a}_r - \frac{1}{r} \frac{\partial}{\partial r} (r^3 \sin \theta) \mathbf{a}_\theta \neq \mathbf{0}$$

$\mathbf{C}$ is possibly $\mathbf{H}$ field.

Prob. 7.41

$$(a) \quad \nabla \cdot \mathbf{D} = 0$$

$$\begin{aligned} \nabla \times \mathbf{D} &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 z & 2(x+1)yz & -(x+1)z^2 \end{vmatrix} \\ &= 2(x+1)y \mathbf{a}_x + \dots \neq \mathbf{0} \end{aligned}$$

$\mathbf{D}$ is possibly a magnetostatic field.

$$(b) \quad \nabla \cdot \mathbf{E} = \frac{1}{\rho} \frac{\partial}{\partial \rho} ((z+1) \cos \phi) + \frac{\partial}{\partial z} \left(\frac{\sin \phi}{\rho} \right) = 0$$

$$\nabla \times \mathbf{E} = \frac{1}{\rho^2} \cos \theta \mathbf{a}_\rho + \dots \neq \mathbf{0}$$

$\mathbf{E}$ could be a magnetostatic field.

$$(c) \quad \nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (2 \cos \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left(\frac{\sin \theta}{r^2} \right) \neq 0$$

$$\nabla \times \mathbf{F} = \frac{1}{r} \left[\frac{\partial}{\partial r} (r^{-1} \sin \theta) + \frac{2 \sin \theta}{r^2} \right] \mathbf{a}_\theta \neq \mathbf{0}$$

$\mathbf{F}$ can be neither electrostatic nor magnetostatic field.

Prob. 7.42

$$\mathbf{A} = \int \frac{\mu_o I d\mathbf{l}}{4\pi r} = \frac{\mu_o IL \mathbf{a}_z}{4\pi r}$$

This requires no integration since $L \ll r$.

$$\mathbf{B} = \nabla \times \mathbf{A} = \frac{1}{\rho} \frac{\partial A_z}{\partial \phi} \mathbf{a}_\rho - \frac{\partial A_z}{\partial \rho} \mathbf{a}_\phi$$

$$\text{But } r = \sqrt{\rho^2 + z^2}$$

$$\mathbf{A} = \frac{\mu_o IL \mathbf{a}_z}{4\pi(\rho^2 + z^2)^{1/2}}$$

$$\frac{\partial A_z}{\partial \rho} = \frac{\mu_o IL}{4\pi} \frac{\partial}{\partial \rho} (\rho^2 + z^2)^{1/2} = \frac{\mu_o IL}{4\pi} \left(-\frac{1}{2}\right) (\rho^2 + z^2)^{-3/2} (2\rho)$$

$$\mathbf{B} = \frac{\mu_o IL \rho \mathbf{a}_\phi}{4\pi(\rho^2 + z^2)^{3/2}} = \frac{\mu_o IL \rho \mathbf{a}_\phi}{4\pi r^3}$$

Prob. 7.43

$$\mathbf{B} = \mu_o \mathbf{H} = \nabla \times \mathbf{A} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 10 \sin \pi y & 0 & 4 + \cos \pi x \end{vmatrix} = \pi \sin \pi x \mathbf{a}_y - 10\pi \cos \pi y \mathbf{a}_z$$

$$\mathbf{H} = \frac{\pi}{\mu_o} \left(\sin \pi x \mathbf{a}_y - 10 \cos \pi y \mathbf{a}_z \right)$$

$$\mathbf{J} = \nabla \times \mathbf{H} = \frac{\pi}{\mu_o} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & \sin \pi x & -10 \cos \pi y \end{vmatrix} = \frac{\pi}{\mu_o} \left(10\pi \sin \pi y \mathbf{a}_x + \pi \cos \pi x \mathbf{a}_z \right)$$

$$\mathbf{J} = \frac{\pi^2}{\mu_o} (10 \sin \pi y \mathbf{a}_x + \cos \pi x \mathbf{a}_z)$$

Prob. 7.44

(a)

$$\begin{aligned} \nabla \cdot \mathbf{A} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + 0 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(\frac{2 \cos \theta}{r} \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left(\frac{\sin^2 \theta}{r^3} \right) \\ &= -\frac{1}{r^4} 2 \cos \theta + \frac{1}{r^4 \sin \theta} 2 \sin \theta \cos \theta = 0 \end{aligned}$$

(b)

$$\begin{aligned}
\mathbf{B} = \nabla \times \mathbf{A} &= \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] \mathbf{a}_r + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \mathbf{a}_\theta \\
&\quad + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \mathbf{a}_\phi \\
&= 0 \mathbf{a}_r + 0 \mathbf{a}_\theta + \frac{1}{r} \left[\frac{\partial}{\partial r} \left(\frac{\sin \theta}{r^2} \right) + \frac{2 \sin \theta}{r^3} \right] \mathbf{a}_\phi = \frac{1}{r} \left[-\frac{2 \sin \theta}{r^3} + \frac{2 \sin \theta}{r^3} \right] \mathbf{a}_\phi = \mathbf{0}
\end{aligned}$$

Prob. 7.45

$$\begin{aligned}
(a) \quad \mathbf{B} = \nabla \times \mathbf{A} &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x^2y + yz & xy^2 - xz^3 & -6xy + 2z^2y^2 \end{vmatrix} \\
\mathbf{B} &= \underline{\underline{(-6xz + 4x^2y + 3xz^2)\mathbf{a}_x + (y + 6yz - 4xy^2)\mathbf{a}_y + (y^2 - z^3 - 2x^2 - z)\mathbf{a}_z \text{ Wb/m}^2}}
\end{aligned}$$

(b)

$$\begin{aligned}
\psi &= \int_{z=0}^2 \int_{y=0}^2 (-6xz + 4x^2y + 3xz^2) dy dz \Big|_{x=1} \\
&= \int \int_0^2 (-6xz) dy dz + 4 \int \int_0^2 x^2y dy dz + 3 \int \int_0^2 xz^2 dy dz \\
&= -6 \int_0^2 dz \int_0^2 dy + 4 \int_0^2 dz \int_0^2 y dy + 3 \int_0^2 dy \int_0^2 z^2 dz \\
&= -6(2)(2) + 4(2) \left(\frac{y^2}{2} \Big|_0^2 \right) + 3(2) \left(\frac{z^3}{3} \Big|_0^2 \right) = -24 + 16 + 16 \\
\psi &= \underline{\underline{8 \text{ Wb}}}
\end{aligned}$$

$$(c) \quad \nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 4xy + 2xy - 6xy = 0$$

$$\nabla \cdot \mathbf{B} = -6z + 8xy + 3z^3 + 6z - 8xy + 1 - 3z^3 - 1 = 0$$

As a matter of mathematical necessity,

$$\nabla \bullet \mathbf{B} = \nabla \bullet (\nabla \times \mathbf{A}) = 0$$

Prob. 7.46

$$\begin{aligned}
\mathbf{B} &= \nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] \mathbf{a}_r + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \mathbf{a}_\theta \\
&= \frac{1}{r \sin \theta} \frac{k}{r^2} \frac{\partial}{\partial \theta} (\sin^2 \theta) \mathbf{a}_r - \frac{1}{r} \frac{\partial}{\partial r} (k r^{-1}) \sin \theta \mathbf{a}_\theta \\
&= \frac{k}{r^3 \sin \theta} 2 \sin \theta \cos \theta \mathbf{a}_r + \frac{k \sin \theta}{r^3} \mathbf{a}_\theta = \underline{\underline{\frac{2k \cos \theta}{r^3} \mathbf{a}_r + \frac{k \sin \theta}{r^3} \mathbf{a}_\theta}}
\end{aligned}$$

Prob. 7.47

$$\begin{aligned}
\mathbf{B} &= \nabla \times \mathbf{A} = \frac{1}{\rho} \frac{\partial A_z}{\partial \phi} \mathbf{a}_\rho - \frac{\partial A_\rho}{\partial \phi} \mathbf{a}_\phi \\
&= \frac{15}{\rho} e^{-\rho} \cos \phi \mathbf{a}_\rho + 15 e^{-\rho} \sin \phi \mathbf{a}_\phi \\
\mathbf{B} \left(3, \frac{\pi}{4}, -10 \right) &= 5 e^{-3} \frac{1}{\sqrt{2}} \mathbf{a}_\rho + 15 e^{-3} \frac{1}{\sqrt{2}} \mathbf{a}_\phi \\
\mathbf{H} &= \frac{\mathbf{B}}{\mu_0} = \frac{10^7}{4\pi} \frac{15}{\sqrt{2}} e^{-3} \left(\frac{1}{3} \mathbf{a}_\rho + \mathbf{a}_\phi \right) \\
\mathbf{H} &= (14 \mathbf{a}_\rho + 42 \mathbf{a}_\phi) \cdot 10^4 \text{ A/m} \\
\psi &= \int \mathbf{B} \cdot d\mathbf{S} = \iint \frac{15}{\rho} e^{-\rho} \cos \phi \rho d\phi dz \\
&= 15 z \Big|_0^{10} (\sin \phi) \Big|_0^{\pi/2} e^{-5} = 150 e^{-5} \Rightarrow \psi = \underline{\underline{1.011 \text{ Wb}}}
\end{aligned}$$

Prob. 7.48

$$\begin{aligned}
\mathbf{B} &= \nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] \mathbf{a}_r + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \mathbf{a}_\theta \\
&\quad + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \mathbf{a}_\phi \\
&= \frac{1}{r \sin \theta} \frac{10}{r} 2 \sin \theta \cos \theta \mathbf{a}_r - \frac{1}{r} \frac{\partial}{\partial r} (10) \sin \theta \mathbf{a}_\theta + 0 \mathbf{a}_\phi \\
\mathbf{B} &= \frac{20}{r^2} \cos \theta \mathbf{a}_r
\end{aligned}$$

At $(4, 60^\circ, 30^\circ)$, $r = 4$, $\theta = 60^\circ$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} = \frac{1}{4\pi \times 10^{-7}} \left[\frac{20}{4^2} \cos 60^\circ \mathbf{a}_r \right] = \underline{\underline{4.974 \times 10^5 \mathbf{a}_r \text{ A/m}}}$$

Prob. 7.49

Applying Ampere's law gives

$$H_\phi \cdot 2\pi\rho = J_o \cdot \pi\rho^2$$

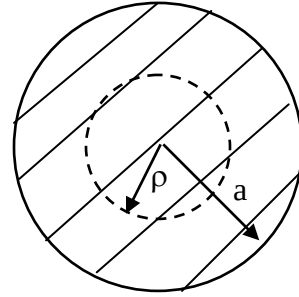
$$H_\phi = \frac{J_o}{2} \rho$$

$$B_\phi = \mu_o H_\phi = \mu_o \frac{J_o \rho}{2}$$

$$\text{But } \mathbf{B} = \nabla \times \mathbf{A} = -\frac{\partial A_z}{\partial \rho} \bar{\mathbf{a}}_\phi + \dots$$

$$-\frac{\partial A_z}{\partial \rho} = \frac{1}{2} \mu_o J_o \rho \longrightarrow A_z = -\mu_o \frac{J_o \rho^2}{4}$$

$$\text{or } \mathbf{A} = \underline{\underline{-\frac{1}{4} \mu_o J_o \rho^2 \mathbf{a}_z}}$$

**Prob. 7.50**

$$\begin{aligned} \mathbf{B} = \nabla \times \mathbf{A} &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & A_z(x, y) \end{vmatrix} = \frac{\partial A_z}{\partial y} \mathbf{a}_x - \frac{\partial A_z}{\partial x} \mathbf{a}_y \\ &= \underline{\underline{-\frac{\pi}{2} \sin \frac{\pi x}{2} \sin \frac{\pi y}{2} \mathbf{a}_x - \frac{\pi}{2} \cos \frac{\pi x}{2} \cos \frac{\pi y}{2} \mathbf{a}_y}} \end{aligned}$$

Prob. 7.51

$$\begin{aligned} \mathbf{B} = \nabla \times \mathbf{A} &= \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\phi \sin \theta) \mathbf{a}_r - \frac{1}{r} \frac{\partial}{\partial r} (r A_\phi) \mathbf{a}_\theta \\ &= \frac{1}{r \sin \theta} \frac{A_o}{r^2} (2 \sin \theta \cos \theta) \mathbf{a}_r - \frac{A_o}{r} \sin \theta (-r^{-2}) \mathbf{a}_\theta \\ &= \underline{\underline{\frac{A_o}{r^3} (2 \cos \theta \mathbf{a}_r + \sin \theta \mathbf{a}_\theta)}} \end{aligned}$$

Prob. 7.52

$$(a) \quad \mathbf{J} = \nabla \times \mathbf{H} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & x^2z & -y^2z \end{vmatrix} = (-2yz - x^2)\mathbf{a}_x + (2xz - 2xy)\mathbf{a}_z$$

At (2,-1,3), $x=2, y=-1, z=3$.

$$\mathbf{J} = \underline{\underline{2\mathbf{a}_x + 16\mathbf{a}_z \text{ A/m}^2}}$$

$$(b) \quad -\frac{\partial \rho_v}{\partial t} = \nabla \cdot \mathbf{J} = 0 - 2x + 2x = 0$$

At (2,-1,3),

$$\frac{\partial \rho_v}{\partial t} = \underline{\underline{0 \text{ C/m}^3\text{s}}}$$

Prob. 7.53

$$(a) \quad \mathbf{B} = \nabla \times \mathbf{A}$$

$$\begin{aligned} &= \left[\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \mathbf{a}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right] \mathbf{a}_\phi + \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (\rho A_\phi) - \frac{\partial A_\rho}{\partial \phi} \right] \mathbf{a}_z \\ &= -\frac{\partial A_z}{\partial \rho} \mathbf{a}_\phi = 20\rho \mathbf{a}_\phi \text{ } \mu\text{Wb/m}^2 \end{aligned}$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_o} = \underline{\underline{\frac{-20\rho}{\mu_o} \mathbf{a}_\phi \text{ } \mu\text{A/m}}}$$

$$\begin{aligned} \mathbf{J} = \nabla \times \mathbf{H} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\phi) \mathbf{a}_z \\ &= \frac{1}{\mu_o \rho} (-40\rho) \mathbf{a}_z = \underline{\underline{\frac{-40}{\mu_o} \mathbf{a}_z \text{ } \mu\text{A/m}^2}} \end{aligned}$$

$$\begin{aligned} (b) \quad I &= \int \mathbf{J} \cdot d\mathbf{S} = \frac{-40}{\mu_o} \int_{\rho=0}^2 \int_{\phi=0}^{2\pi} \rho d\phi d\rho, \quad d\mathbf{S} = \rho d\phi d\rho \mathbf{a}_z \\ &= \frac{-40}{\mu_o} \int_0^2 \rho d\rho \int_0^{2\pi} d\phi = \frac{-40}{\mu_o} \frac{\rho^2}{2} \Big|_0^2 (2\pi) \\ &= \frac{-80\pi \times 2 \times 10^{-6}}{4\pi \times 10^{-7}} = \underline{\underline{-400 \text{ A}}} \end{aligned}$$

Prob. 7.54

$$\mathbf{H} = -\nabla V_m \rightarrow V_m = -\int \mathbf{H} \cdot d\mathbf{l} = -\text{mmf}$$

From Example 7.3, $\mathbf{H} = \frac{Ia^2}{2(z^2 + a^2)^{3/2}} \mathbf{a}_z$

$$V_m = -\frac{Ia^2}{2} \int (z^2 + a^2)^{-3/2} dz = \frac{-Iz}{2(z^2 + a^2)^{1/2}} + c$$

As $z \rightarrow \infty$, $V_m = 0$, i.e.

$$0 = -\frac{I}{2} + c \rightarrow c = \frac{I}{2}$$

Hence,

$$V_m = \frac{I}{2} \left[1 - \frac{z}{\sqrt{z^2 + a^2}} \right]$$

Prob. 7.55

$$\mathbf{H} = \frac{I}{2\pi\rho} \mathbf{a}_\phi$$

But $\mathbf{H} = -\nabla V_m \quad (J = 0)$

$$\frac{I}{2\pi\rho} \mathbf{a}_\phi = -\frac{1}{\rho} \frac{\partial V_m}{\partial \phi} \mathbf{a}_\phi \rightarrow V_m = -\frac{I}{2\pi} \phi + C$$

At $(10, 60^\circ, 7)$, $\phi = 60^\circ = \frac{\pi}{3}$, $V_m = 0 \rightarrow 0 = -\frac{I}{2\pi} \cdot \frac{\pi}{3} + C$

or $C = \frac{I}{6}$

$$V_m = -\frac{I}{2\pi} \phi + \frac{I}{6}$$

At $(4, 30^\circ, -2)$, $\phi = 30^\circ = \frac{\pi}{6}$,

$$V_m = -\frac{I}{2\pi} \cdot \frac{\pi}{6} + \frac{I}{6} = \frac{I}{12} = \frac{12}{12}$$

$$\underline{\underline{V_m = 1 \text{ A}}}$$

Prob. 7.56

For an infinite current sheet,

$$\bar{H} = \frac{1}{2} \bar{K} \times \bar{a}_n = \frac{1}{2} 50 \bar{a}_y \times \bar{a}_n = 25 \bar{a}_x$$

But $\bar{H} = -\nabla V_m \quad (\bar{J} = 0)$

$$25 \bar{a}_x = -\frac{\partial V_m}{\partial x} \bar{a}_n \rightarrow V_m = -25x + c$$

At the origin, $x=0$, $V_m=0$, $c=0$, i.e.

$$V_m = -25x$$

(a) At $(-2, 0, 5)$, $\underline{V_m = 50A}$.

(b) At $(10, 3, 1)$, $\underline{V_m = -250A}$.

Prob. 7.57

$$\begin{aligned} \text{(a)} \quad \nabla \times \nabla V &= \nabla \times \left(\frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z \right) \\ &= \left(\frac{1}{\rho} \frac{\partial^2 V}{\partial \phi \partial z} - \frac{1}{\rho} \frac{\partial^2 V}{\partial z \partial \phi} \right) \mathbf{a}_\rho + \left(\frac{\partial^2 V}{\partial z \partial \rho} - \frac{\partial^2 V}{\partial \rho \partial z} \right) \mathbf{a}_\phi \\ &\quad + \frac{1}{\rho} \left(\frac{\partial^2 V}{\partial \rho \partial \phi} - \frac{\partial^2 V}{\partial \phi \partial \rho} \right) \mathbf{a}_z = 0 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \nabla \cdot (\nabla \times \mathbf{A}) &= \nabla \cdot \left[\left(\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \mathbf{a}_\rho \right. \\ &\quad \left. + \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \mathbf{a}_\phi + \frac{1}{\rho} \left(\frac{\partial}{\partial \rho} (\rho A_\phi) - \frac{\partial A_\rho}{\partial \phi} \right) \mathbf{a}_z \right] \\ &= \frac{1}{\rho} \frac{\partial^2 A_z}{\partial \rho \partial \phi} - \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial A_\phi}{\partial z} \right) + \frac{1}{\rho} \frac{\partial^2 A_\rho}{\partial \phi \partial z} - \frac{1}{\rho} \frac{\partial^2 A_z}{\partial \phi \partial \rho} \\ &\quad + \frac{\partial}{\partial z} \left(\frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\phi) \right) - \frac{\partial}{\partial z} \left(\frac{1}{\rho} \frac{\partial A_\rho}{\partial \phi} \right) \\ &= -\frac{\partial^2 A_\phi}{\partial \rho \partial z} - \frac{1}{\rho} \frac{\partial A_\phi}{\partial z} + \frac{\partial^2 A_\phi}{\partial z \partial \rho} + \frac{1}{\rho} \frac{\partial A_\phi}{\partial z} = 0 \end{aligned}$$

Prob. 7.58

$$\begin{aligned}
\nabla' \frac{1}{R} &= \left(\frac{\partial}{\partial x'} \mathbf{a}_x + \frac{\partial}{\partial y'} \mathbf{a}_y + \frac{\partial}{\partial z'} \mathbf{a}_z \right) \left[(x-x')^2 + (y-y')^2 + (z-z')^2 \right]^{-1/2} \\
&= \left(-\frac{1}{2} \right) (-2) (x-x') \mathbf{a}_x \left[(x-x')^2 + (y-y')^2 + (z-z')^2 \right]^{-3/2} + \mathbf{a}_y \text{ and } \mathbf{a}_z \text{ terms} \\
&= \frac{\mathbf{R}}{R^3}
\end{aligned}$$

$$R = |\mathbf{r} - \mathbf{r}'| = \left[(x-x')^2 + (y-y')^2 + (z-z')^2 \right]^{1/2}$$

$$\begin{aligned}
\nabla \frac{1}{R} &= \left(\frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z \right) \left[(x-x')^2 + (y-y')^2 + (z-z')^2 \right]^{-1/2} \\
&= -\frac{1}{2} 2 (x-x') \mathbf{a}_x \left[(x-x')^2 + (y-y')^2 + (z-z')^2 \right]^{-3/2} + \mathbf{a}_y \text{ and } \mathbf{a}_z \text{ terms} \\
&= - \left[(x-x') \mathbf{a}_x + (y-y') \mathbf{a}_y + (z-z') \mathbf{a}_z \right] / R^3 = -\frac{\bar{\mathbf{R}}}{R^3}
\end{aligned}$$

CHAPTER 8

P.E. 8.1

$$(a) \quad \mathbf{F} = m \frac{\partial \mathbf{u}}{\partial t} = Q\mathbf{E} = \underline{\underline{6\mathbf{a}_z N}}$$

$$(b) \quad \frac{\partial \mathbf{u}}{\partial t} = 6\mathbf{a}_z = \frac{\partial}{\partial t}(u_x, u_y, u_z) \Rightarrow$$

$$\frac{\partial u_x}{\partial t} = 0 \rightarrow u_x = A$$

$$\frac{\partial u_y}{\partial t} = 0 \rightarrow u_y = B$$

$$\frac{\partial u_z}{\partial t} = 6 \rightarrow u_z = 6t + C$$

$$\text{Since } \mathbf{u}(t=0) = 0, \quad A = B = C = 0$$

$$u_x = 0 = u_y, \quad u_z = 6t$$

$$u_x = \frac{\partial x}{\partial t} = 0 \rightarrow x = A$$

$$u_y = \frac{\partial y}{\partial t} = 0 \rightarrow y = B$$

$$u_z = \frac{\partial z}{\partial t} = 6t \rightarrow z = 3t^2 + C_1$$

$$\text{At } t = 0, (x, y, z) = (0, 0, 0) \rightarrow A_1 = 0 = B_1 = C_1$$

$$\text{Hence, } (x, y, z) = (0, 0, 3t^2),$$

$$\mathbf{u} = 6t\mathbf{a}_z \text{ at any time. At } P(0, 0, 12), z = 12 = 3t^2 \rightarrow t = 2\text{s}$$

$$\underline{\underline{t = 2\text{s}}}$$

$$(c) \quad \mathbf{u} = 6t\mathbf{a}_z = 12\mathbf{a}_z \text{ m/s.}$$

$$\mathbf{a} = \frac{\partial \mathbf{u}}{\partial t} = \underline{\underline{6\mathbf{a}_z \text{ m/s}^2}}$$

$$(d) \quad K.E = \frac{1}{2} m |\mathbf{u}|^2 = \frac{1}{2} (1)(144) = \underline{\underline{72J}}$$

P.E. 8.2

$$(a) \quad m\mathbf{a} = e\mathbf{u} \times \mathbf{B} = (eB_0 u_y, -eB_0 u_x, 0)$$

$$\frac{d^2 x}{dt^2} = \frac{eB_0}{m} \frac{dy}{dt} = \omega \frac{dy}{dt} \quad (1)$$

$$\frac{d^2 y}{dt^2} = -\frac{eB_0}{m} \frac{dx}{dt} = -\omega \frac{dx}{dt} \quad (2)$$

$$\frac{d^2 z}{dt^2} = 0; \Rightarrow \frac{dz}{dt} = C_1 \quad (3)$$

From (1) and (2),

$$\frac{d^3 x}{dt^3} = \omega \frac{d^2 y}{dt^2} = -\omega^2 \frac{dx}{dt}$$

$$(D^2 + \omega^2 D)x = 0 \rightarrow Dx = (0, \pm j\omega)x$$

$$x = c_2 + c_3 \cos \omega t + c_4 \sin \omega t$$

$$\frac{dy}{dt} = \frac{1}{\omega} \frac{d^2 x}{dt^2} = -c_3 \omega \cos \omega t - c_4 \omega \sin \omega t$$

At $t = 0$, $\mathbf{u} = (\alpha, 0, \beta)$. Hence,

$$c_1 = \beta, c_3 = 0, c_4 = \frac{\alpha}{\omega}$$

$$\underline{\underline{\frac{dx}{dt} = \alpha \cos \omega t, \frac{dy}{dt} = -\alpha \sin \omega t, \frac{dz}{dt} = \beta}}}$$

(b) Solving these yields

$$x = \frac{\alpha}{\omega} \sin \omega t, y = \frac{\alpha}{\omega} \cos \omega t, z = \beta t$$

The starting point of the particle is $(0, \frac{\alpha}{\omega}, 0)$

$$(c) \quad x^2 + y^2 = \frac{\alpha^2}{\omega^2}, z = \beta t$$

showing that the particles move along a helix of radius $\frac{\alpha}{\omega}$ placed along the z-axis.

P.E. 8.3

(a) From Example 8.3, $QuB = QE$ regardless of the sign of the charge.

$$E = uB = 8 \times 10^6 \times 0.5 \times 10^{-3} = \underline{4 \text{ kV/m}}$$

(b) Yes, since $QuB = QE$ holds for any Q and m .

P.E. 8.4

By Newton's 3rd law, $\mathbf{F}_{12} = \mathbf{F}_{21}$, the force on the infinitely long wire is:

$$\begin{aligned}\mathbf{F}_l = -\mathbf{F} &= \frac{\mu_o I_1 I_2 b}{2\pi} \left(\frac{1}{\rho_o} - \frac{1}{\rho_o + a} \right) \mathbf{a}_\rho \\ &= \frac{4\pi \times 10^{-7} \times 50 \times 3}{2\pi} \left(\frac{1}{2} - \frac{1}{3} \right) \mathbf{a}_\rho = \underline{\underline{5\mathbf{a}_\rho \text{ } \mu\text{N}}}\end{aligned}$$

P.E. 8.5

$$\begin{aligned}\mathbf{m} &= IS\mathbf{a}_n = 10 \times 10^{-4} \times 50 \frac{(2, 6, -3)}{7} \\ &= 7.143 \times 10^{-3} (2, 6, -3) \\ &= \underline{\underline{(1.429\mathbf{a}_x + 4.286\mathbf{a}_y - 2.143\mathbf{a}_z) \times 10^{-2} \text{ A}\cdot\text{m}^2}}\end{aligned}$$

P.E. 8.6

$$\begin{aligned}\text{(a)} \quad \mathbf{T} = \mathbf{m} \times \mathbf{B} &= \frac{10 \times 10^{-4} \times 50}{7 \times 10} \begin{vmatrix} 2 & 6 & -3 \\ 6 & 4 & 5 \end{vmatrix} \\ &= \underline{\underline{0.03\mathbf{a}_x - 0.02\mathbf{a}_y - 0.02\mathbf{a}_z \text{ N}\cdot\text{m}}}\end{aligned}$$

$$\text{(b)} \quad |\mathbf{T}| = ISB \sin \theta \rightarrow |\mathbf{T}|_{\max} = ISB$$

$$|\mathbf{T}|_{\max} = \frac{50 \times 10^{-3}}{10} |6\mathbf{a}_x + 4\mathbf{a}_y + 5\mathbf{a}_z| = \underline{\underline{0.04387 \text{ Nm}}}$$

P.E. 8.7

$$\text{(a)} \quad \mu_r = \frac{\mu}{\mu_o} = 4.6, \chi_m = \mu_r - 1 = \underline{\underline{3.6}}$$

$$\text{(b)} \quad \mathbf{H} = \frac{\mathbf{B}}{\mu} = \frac{10 \times 10^{-3} e^{-y}}{4\pi \times 10^{-7} \times 4.6} \mathbf{a}_z \text{ A/m} = \underline{\underline{1730e^{-y}\mathbf{a}_z \text{ A/m}}}$$

$$\text{(c)} \quad \mathbf{M} = \chi_m \mathbf{H} = \underline{\underline{6228e^{-y}\mathbf{a}_z \text{ A/m}}}$$

P.E. 8.8

$$\mathbf{a}_n = \frac{3\mathbf{a}_x + 4\mathbf{a}_y}{5} = \frac{6\mathbf{a}_x + 8\mathbf{a}_y}{10}$$

$$\mathbf{B}_{1n} = (\mathbf{B}_1 \bullet \mathbf{a}_n) \mathbf{a}_n = \frac{(6 + 32)(6\mathbf{a}_x + 8\mathbf{a}_y)}{1000}$$

$$= 0.228\mathbf{a}_x + 0.304\mathbf{a}_y = \mathbf{B}_{2n}$$

$$\mathbf{B}_{1t} = \mathbf{B}_1 - \mathbf{B}_{1n} = -0.128\mathbf{a}_x + 0.096\mathbf{a}_y + 0.2\mathbf{a}_z$$

$$\mathbf{B}_{2t} = \frac{\mu_2}{\mu_1} \mathbf{B}_{1t} = 10\mathbf{B}_{1t} = -1.28\mathbf{a}_x + 0.96\mathbf{a}_y + 2\mathbf{a}_z$$

$$\mathbf{B}_2 = \mathbf{B}_{2n} + \mathbf{B}_{2t} = \underline{\underline{-1.052\mathbf{a}_x + 1.264\mathbf{a}_y + 2\mathbf{a}_z \text{ Wb/m}^2}}$$

P.E. 8.9

$$(a) \quad \mathbf{B}_{1n} = \mathbf{B}_{2n} \rightarrow \mu_1 \mathbf{H}_{1n} = \mu_2 \mathbf{H}_{2n}$$

$$\text{or } \mu_1 \mathbf{H}_1 \bullet \mathbf{a}_{n21} = \mu_2 \mathbf{H}_2 \bullet \mathbf{a}_{n21}$$

$$\mu_o \frac{(60+2-36)}{7} = 2\mu_o \frac{(6H_{2x}-10-12)}{7}$$

$$35 = 6H_{2x}$$

$$\underline{\underline{H_{2x} = 5.833 \text{ A/m}}}$$

$$(b) \quad \mathbf{K} = (\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{n12} = \mathbf{a}_{n21} \times (\mathbf{H}_1 - \mathbf{H}_2)$$

$$= \mathbf{a}_{n21} \times \left[(10, 1, 12) - \left(\frac{35}{6}, -5, 4 \right) \right]$$

$$= \frac{1}{7} \begin{vmatrix} 6 & 2 & -3 \\ 25 & 6 & 8 \\ 6 & 8 & 8 \end{vmatrix}$$

$$\underline{\underline{\mathbf{K} = 4.86\mathbf{a}_x - 8.64\mathbf{a}_y + 3.95\mathbf{a}_z \text{ A/m}}}$$

(c) Since $\mathbf{B} = \mu\mathbf{H}$, $\mathbf{B}_1$ and $\mathbf{H}_1$ are parallel, i.e. they make the same angle with the normal to the interface.

$$\cos \theta_1 = \frac{\mathbf{H}_1 \bullet \mathbf{a}_{n21}}{|\mathbf{H}_1|} = \frac{26}{7\sqrt{100+1+144}} = 0.2373$$

$$\underline{\underline{\theta_1 = 76.27^\circ}}$$

$$\cos \theta_2 = \frac{\mathbf{H}_2 \bullet \mathbf{a}_{n21}}{|\mathbf{H}_2|} = \frac{13}{7\sqrt{(5.833)^2 + 25 + 16}} = 0.2144$$

$$\underline{\underline{\theta_2 = 77.62^\circ}}$$

P.E. 8.10

$$(a) \quad L' = \mu_o \mu_r n^2 S = 4\pi \times 10^{-7} \times 1000 \times 16 \times 10^6 \times 4 \times 10^{-4}$$

$$= \underline{\underline{8.042 \text{ H/m}}}$$

$$(b) \quad W_m' = \frac{1}{2} L' I^2 = \frac{1}{2} (8.042)(0.5^2) = \underline{\underline{1.005 \text{ J/m}}}$$

P.E. 8.11 From Example 8.11,

$$L_{\text{in}} = \frac{\mu_o l}{8\pi}$$

$$L_{\text{ext}} = \frac{2W_m}{I^2} = \frac{1}{I^2} \iiint \frac{\mu I^2}{4\pi^2 \rho^2} \rho d\rho d\phi dz$$

$$= \frac{1}{4\pi^2} \int_0^l dz \int_0^{2\pi} d\phi \int_a^b \frac{2\mu_o}{(1+\rho)\rho} d\rho$$

$$= \frac{2\mu_o}{4\pi^2} \bullet 2\pi l \int_a^b \left[\frac{1}{\rho} - \frac{1}{(1+\rho)} \right] d\rho$$

$$= \frac{\mu_o l}{\pi} \left[\ln \frac{b}{a} - \ln \frac{1+b}{1+a} \right]$$

$$L = L_{\text{in}} + L_{\text{ext}} = \frac{\mu_o l}{8\pi} + \frac{\mu_o l}{\pi} \left[\ln \frac{b}{a} - \ln \frac{1+b}{1+a} \right]$$

P.E. 8.12

$$(a) \quad L'_{\text{in}} = \frac{\mu_o}{8\pi} = \frac{4\pi \times 10^{-7}}{8\pi} = \underline{\underline{0.05 \text{ } \mu\text{H/m}}}$$

$$L'_{\text{ext}} = L' - L'_{\text{in}} = 1.2 - 0.05 = \underline{\underline{1.15 \text{ } \mu\text{H/m}}}$$

$$(b) \quad L' = \frac{\mu_o}{2\pi} \left[\frac{1}{4} + \ln \frac{d-a}{a} \right]$$

$$\ln \frac{d-a}{a} = \frac{2\pi L'}{\mu_o} - 0.25 = \frac{2\pi \times 1.2 \times 10^{-6}}{4\pi \times 10^{-7}} - 0.25$$

$$= 6 - 0.25 = 5.75$$

$$\frac{d-a}{a} = e^{5.75} = 314.19$$

$$d-a = 314.19a = 314.19 \times \frac{2.588 \times 10^{-3}}{2} = 406.6 \text{ mm}$$

$$d = 407.9 \text{ mm} = \underline{\underline{40.79 \text{ cm}}}$$

P.E. 8.13

This is similar to Example 8.13. In this case, however, $h=0$ so that

$$\begin{aligned}
 A_1 &= \frac{\mu_o I_1 a^2 b}{4b^3} \mathbf{a}_\phi \\
 \phi_{12} &= \frac{\mu_o I_1 a^2}{4b^2} \bullet 2\pi b = \frac{\mu_o \pi I_1 a^2}{2b} \\
 m_{12} &= \frac{\phi_{12}}{I_1} = \frac{\mu_o \pi a^2}{2b} = \frac{4\pi \times 10^{-7} \times \pi \times 4}{2 \times 3} \\
 &= \underline{2.632 \mu\text{H}}
 \end{aligned}$$

P.E. 8.14

$$\begin{aligned}
 L_{\text{in}} &= \frac{\mu_o}{8\pi} l = \frac{\mu_o 2\pi \rho_o}{8\pi} = \frac{4\pi \times 10^{-7} \times 10 \times 10^{-2}}{4} \\
 &= \underline{31.42 \text{ nH}}
 \end{aligned}$$

P.E. 8.15

(a) From Example 7.6,

$$\begin{aligned}
 B_{\text{ave}} &= \frac{\mu_o NI}{l} = \frac{\mu_o NI}{2\pi \rho_o} \\
 \phi &= B_{\text{ave}} \bullet S = \frac{\mu_o NI}{2\pi \rho_o} \bullet \pi a^2 \\
 \text{or } I &= \frac{2\rho_o \phi}{\mu a^2 N} = \frac{2 \times 10 \times 10^{-2} \times 0.5 \times 10^{-3}}{4\pi \times 10^{-7} \times 10^{-4} \times 10^3} \\
 &= \underline{795.77\text{A}}
 \end{aligned}$$

Alternatively, using circuit approach

$$\begin{aligned}
 R &= \frac{l}{\mu S} = \frac{2\pi \rho_o}{\mu_o S} = \frac{2\pi \rho_o}{\mu_o \pi a^2} \\
 \Im &= NI = \frac{\phi \Re}{N} = \frac{2\rho_o \phi}{\mu a^2 N}, \quad \text{as obtained before.} \\
 \Re &= \frac{2\rho_o}{\mu a^2} = \frac{2 \times 10 \times 10^{-2}}{4\pi \times 10^{-7} \times 10^{-4}} = 1.591 \times 10^9 \\
 \Im &= \phi \Re = 0.5 \times 10^{-3} \times 1.591 \times 10^9 = 7.9577 \times 10^5 \\
 I &= \frac{\Im}{N} = 795.77 \text{ A as obtained before.}
 \end{aligned}$$

(b) If $\mu=500\mu_o$,

$$I = \frac{795.77}{500} = \underline{\underline{1.592 \text{ A}}}$$

P.E. 8.16

$$\mathfrak{S} = \frac{B_a^2 S}{2\mu_o} = \frac{(1.5)^2 \times 10 \times 10^{-4}}{2 \times 4\pi \times 10^{-7}} = \frac{22500}{8\pi} = \underline{\underline{895.25 \text{ N}}}$$

Prob. 8.1

At P, $x = 2$, $y = 5$, $z = -3$

$$\mathbf{E} = 2(2)(5)(-3)\mathbf{a}_x + (2)^2(-3)\mathbf{a}_y + (2)^2(5)\mathbf{a}_z = -60\mathbf{a}_x - 12\mathbf{a}_y + 20\mathbf{a}_z$$

$$\mathbf{B} = (5)^2\mathbf{a}_x + (-3)^2\mathbf{a}_y + 2^2\mathbf{a}_z = 25\mathbf{a}_x + 9\mathbf{a}_y + 4\mathbf{a}_z$$

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

$$\mathbf{u} \times \mathbf{B} = \begin{vmatrix} 1.4 & -3.2 & -1 \\ 25 & 9 & 4 \end{vmatrix} = -3.8\mathbf{a}_x - 30.6\mathbf{a}_y + 92.6\mathbf{a}_z$$

$$\mathbf{E} + \mathbf{u} \times \mathbf{B} = (-60, -12, 20) + (-3.8, -30.6, 92.6) = (-63.8, -42.6, 112.6)$$

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{u} \times \mathbf{B}) = 4(\mathbf{E} + \mathbf{u} \times \mathbf{B}) \text{ mN}$$

$$= \underline{\underline{-255.2\mathbf{a}_x - 170.4\mathbf{a}_y + 450.4\mathbf{a}_z \text{ mN}}}$$

Prob. 8.2

$$F = m\omega^2 r = 9.11 \times 10^{-31} \times (2 \times 10^{16})^2 (0.4 \times 10^{-10}) = \underline{\underline{14.576 \text{ nN}}}$$

Prob. 8.3

(a)

$$\mathbf{F} = Q(\mathbf{u} \times \mathbf{B}) = 10^{-3} \begin{vmatrix} 10 & -2 & 6 \\ 0 & 0 & 25 \end{vmatrix} = 10^{-3}(-50\mathbf{a}_x - 250\mathbf{a}_y)$$

$$= \underline{\underline{-0.05\mathbf{a}_x - 0.25\mathbf{a}_y \text{ N}}}$$

(b) Constant velocity implies that acceleration $\mathbf{a} = \mathbf{0}$.

$$\mathbf{F} = m\mathbf{a} = \mathbf{0} = Q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

$$\mathbf{E} = -\mathbf{u} \times \mathbf{B} = \underline{\underline{50\mathbf{a}_x + 250\mathbf{a}_y \text{ V/m}}}$$

Prob. 8.4

$$\mathbf{F}_e = q\mathbf{E}, \quad \mathbf{F}_m = q\mathbf{u} \times \mathbf{B}$$

$$\frac{F_e}{F_m} = \frac{E}{uB} = \frac{20 \times 10^3}{0.5 \times 10^8 \times 5 \times 10^{-6}} = \underline{\underline{80}}$$

Prob. 8.5

$$m\mathbf{a} = Q\mathbf{u} \times \mathbf{B}$$

$$10^{-3}\mathbf{a} = -2 \times 10^{-3} \begin{vmatrix} u_x & u_y & u_z \\ 0 & 6 & 0 \end{vmatrix}$$

$$\frac{d}{dt}(u_x, u_y, u_z) = (12u_z, 0, -12u_x)$$

$$\text{i.e. } \frac{du_x}{dt} = 12u_z \quad (1)$$

$$\frac{du_y}{dt} = 0 \rightarrow u_y = A_1 \quad (2)$$

$$\frac{du_z}{dt} = -12u_x \quad (3)$$

From (1) and (3),

$$\ddot{u}_x = 12\dot{u}_z = -144u_x$$

or

$$\ddot{u}_x + 144u_x = 0 \rightarrow u_x = c_1 \cos 12t + c_2 \sin 12t$$

From (1), $u_z = -c_1 \sin 12t + c_2 \cos 12t$

At $t=0$,

$$u_x=5, u_y=0, u_z=0 \rightarrow A_1=0=c_2, c_1=5$$

Hence,

$$\mathbf{u} = (5 \cos 12t, 0, -5 \sin 12t)$$

$$\mathbf{u}(t=10s) = (5 \cos 120, 0, -5 \sin 120) = \underline{\underline{4.071\mathbf{a}_x - 2.903\mathbf{a}_z}} \text{ m/s}$$

$$u_x = \frac{dx}{dt} = 5 \cos 12t \rightarrow x = \frac{5}{12} \sin 12t + B_1$$

$$u_y = \frac{dy}{dt} = 0 \rightarrow y = B_2$$

$$u_z = \frac{dz}{dt} = -5 \sin 12t \rightarrow z = \frac{5}{12} \cos 12t + B_3$$

At $t=0$, $(x, y, z) = (0, 1, 2) \rightarrow B_1=0, B_2=1, B_3=\frac{19}{12}$

$$(x, y, z) = \left(\frac{5}{12} \sin 12t, 1, \frac{5}{12} \cos 12t + \frac{19}{12} \right) \quad (4)$$

At $t=10s$,

$$(x, y, z) = \left(\frac{5}{12} \sin 120, 1, \frac{5}{12} \cos 120 + \frac{19}{12} \right) = (0.2419, 1, 1.923)$$

By eliminating t from (4),

$x^2 + (z - \frac{19}{12})^2 = (\frac{5}{12})^2$, $y=1$ which is a circle in the $y=1$ plane with center at $(0, 1, 19/12)$. The particle gyrates.

Prob. 8.6

(a) $m\mathbf{a} = -e(\mathbf{u} \times \mathbf{B})$

$$-\frac{m}{e} \frac{d}{dt} (u_x, u_y, u_z) = \begin{vmatrix} u_x & u_y & u_z \\ 0 & 0 & B_o \end{vmatrix} = u_y B_o \vec{a}_x - B_o u_x \vec{a}_y$$

$$\frac{du_z}{dt} = 0 \rightarrow u_z = c = 0$$

$$\frac{du_x}{dt} = -u_y \frac{B_o e}{m} = -u_y w, \text{ where } w = \frac{B_o e}{m}$$

$$\frac{du_y}{dt} = u_x w$$

Hence,

$$\ddot{u}_x = -w \dot{u}_y = -w^2 u_x$$

$$\text{or } \ddot{u}_x + w^2 u_x = 0 \rightarrow u_x = A \cos wt + B \sin wt$$

$$u_y = -\frac{\dot{u}_x}{w} = A \sin wt - B \cos wt$$

At $t=0$, $u_x = u_o$, $u_y = 0 \rightarrow A = u_o$, $B=0$

Hence,

$$u_x = u_o \cos wt = \frac{dx}{dt} \rightarrow x = \frac{u_o}{w} \sin wt + c_1$$

$$u_y = u_o \sin wt = \frac{dy}{dt} \rightarrow y = -\frac{u_o}{w} \cos wt + c_2$$

At $t=0$, $x = 0 = y \rightarrow c_1=0$, $c_2=\frac{u_o}{w}$. Hence,

$$x = \frac{u_o}{w} \sin wt, y = \frac{u_o}{w} (1 - \cos wt)$$

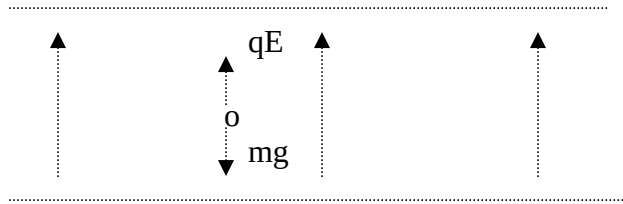
$$\frac{u_o^2}{w^2} (\cos^2 wt + \sin^2 wt) = \left(\frac{u_o}{w} \right)^2 = x^2 + \left(y - \frac{u_o}{w} \right)^2$$

showing that the electron would move in a circle centered at $(0, \frac{u_o}{w})$. But since the field does not exist throughout the circular region, the electron passes through a semi-circle and leaves the field horizontally.

(b) $d =$ twice the radius of the semi-circle

$$= \frac{2u_o}{w} = \frac{2u_o m}{B_o e}$$

Prob. 8.7



$$mg = qE \quad \longrightarrow \quad q = \frac{mg}{E} = \frac{0.4 \times 10^{-3} \times 9.81}{1.5 \times 10^5} = \underline{\underline{26.67 \text{ nC}}}$$

Prob. 8.8

$$\mathbf{F} = \int I d\mathbf{l} \times \mathbf{B} = I \mathbf{L} \times \mathbf{B}$$

$$= 4.5(0.2)\mathbf{a}_x \times (2.5)(\mathbf{a}_y + \mathbf{a}_z)10^{-3} = 4.5(0.5) \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} \text{ mN}$$

$$\underline{\underline{\mathbf{F} = 2.25(-\mathbf{a}_y + \mathbf{a}_z) \text{ mN}}}$$

Prob. 8.9

$$qE = quB \quad \rightarrow \quad B = \frac{E}{u} = \frac{12 \times 10^3}{140} = \frac{1200}{14} = \underline{\underline{85.714 \text{ Wb/m}^2}}$$

Prob. 8.10

$$\mathcal{J} = IL \times \mathbf{B} \rightarrow \mathcal{J} = \frac{\mathbf{F}}{L} = I_1 \mathbf{a}_l \times \mathbf{B}_2 = \frac{\mu_o I_1 I_2 \mathbf{a}_l \times \mathbf{a}_\phi}{2\pi\rho}$$

$$(a) \quad \mathbf{F}_{21} = \frac{\mathbf{a}_z \times (-\mathbf{a}_y) 4\pi \times 10^{-7} (-100)(200)}{2\pi} = \underline{\underline{4\mathbf{a}_x \text{ mN/m}}} \text{ (repulsive)}$$

$$(b) \quad \mathbf{F}_{12} = -\mathbf{F}_{21} = -4\mathbf{a}_x \text{ mN/m (repulsive)}$$

$$(c) \quad \mathbf{a}_l \times \mathbf{a}_\phi = \mathbf{a}_z \times \left(-\frac{4}{5}\mathbf{a}_x + \frac{3}{5}\mathbf{a}_y\right) = -\frac{3}{5}\mathbf{a}_x - \frac{4}{5}\mathbf{a}_y, \rho = 5$$

$$\begin{aligned} \mathbf{F}_{31} &= \frac{4\pi \times 10^{-7} (-3 \times 10^4)}{2\pi(5)} \left(-\frac{3}{5}\mathbf{a}_x - \frac{4}{5}\mathbf{a}_y\right) \\ &= \underline{\underline{0.72\mathbf{a}_x + 0.96\mathbf{a}_y \text{ mN/m}}} \text{ (attractive)} \end{aligned}$$

$$(d) \quad \mathbf{F}_3 = \mathbf{F}_{31} + \mathbf{F}_{32}$$

$$\mathbf{F}_{32} = \frac{4\pi \times 10^{-7} \times 6 \times 10^4}{2\pi(3)} (\mathbf{a}_z \times \mathbf{a}_y) = -4\mathbf{a}_x \text{ mN/m (attractive)}$$

$$\mathbf{F}_3 = \underline{\underline{-3.28\mathbf{a}_x + 0.96\mathbf{a}_y \text{ mN/m}}}$$

(attractive due to L_2 and repulsive due to L_1)

Prob. 8.11

$$F = \frac{\mu_o I_1 I_2}{2\pi\rho} = \frac{4\pi \times 10^{-7} (10)10}{2\pi(20 \times 10^{-2})} = \underline{\underline{100 \mu\text{N}}}$$

Prob. 8.12

$$W = -\int \mathbf{F} \cdot d\mathbf{l}, \quad \mathbf{F} = \int L d\mathbf{l} \times \mathbf{B} = 3(2\mathbf{a}_z) \times \cos \frac{\phi}{3} \mathbf{a}_\phi$$

$$F = 6 \cos \frac{\phi}{3} \bar{a}_\phi \text{ N}$$

$$W = -\int_0^{2\pi} 6 \cos \frac{\phi}{3} \rho_o d\phi = -6\rho_o \times 3 \sin \frac{\phi}{3} \Big|_0^{2\pi} \text{ J}$$

$$= -1.8 \sin \frac{2\pi}{3} = \underline{\underline{-1.559 \text{ J}}}$$

Prob. 8.13

$$\begin{aligned} (a) \quad \mathbf{F}_1 &= \int_{\rho=2}^6 \frac{\mu_o I_1 I_2}{2\pi\rho} d\rho \mathbf{a}_\rho \times \mathbf{a}_\phi = \frac{4\pi \times 10^{-7}}{2\pi} (2)(5) \ln \frac{6}{2} \mathbf{a}_z \\ &= 2 \ln 3 \mathbf{a}_z \mu\text{N} = \underline{\underline{2.197\mathbf{a}_z \mu\text{N}}} \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \mathbf{F}_2 &= \int I_2 d\mathbf{l}_2 \times \mathbf{B}_1 \\
 &= \frac{\mu_0 I_1 I_2}{2\pi} \int \frac{1}{\rho} [d\rho \mathbf{a}_\rho + dz \mathbf{a}_z] \times \mathbf{a}_\phi \\
 &= \frac{\mu_0 I_1 I_2}{2\pi} \int \frac{1}{\rho} [d\rho \mathbf{a}_z - dz \mathbf{a}_\rho]
 \end{aligned}$$

But $\rho = z+2$, $dz=d\rho$

$$\begin{aligned}
 \mathbf{F}_2 &= \frac{4\pi \times 10^{-7}}{2\pi} (5)(2) \int_{\rho=4}^2 \frac{1}{\rho} [d\rho \mathbf{a}_z - dz \mathbf{a}_\rho] \\
 2 \ln \frac{2}{4} (\mathbf{a}_z - \mathbf{a}_\rho) \mu N &= 1.386 \mathbf{a}_\rho - 1.386 \mathbf{a}_z \mu N \\
 \mathbf{F}_3 &= \frac{\mu_0 I_1 I_2}{2\pi} \int \frac{1}{\rho} [d\rho \mathbf{a}_z - dz \mathbf{a}_\rho]
 \end{aligned}$$

But $z = -\rho + 6$, $dz = -d\rho$

$$\begin{aligned}
 \mathbf{F}_3 &= \frac{4\pi \times 10^{-7}}{2\pi} (5)(2) \int_{\rho=6}^4 \frac{1}{\rho} [d\rho \mathbf{a}_z - dz \mathbf{a}_\rho] \\
 2 \ln \frac{4}{6} (\mathbf{a}_z + \mathbf{a}_\rho) \mu N &= -0.8109 \mathbf{a}_\rho - 0.8109 \mathbf{a}_z \mu N \\
 \mathbf{F} &= \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 \\
 &= \mathbf{a}_\rho (\ln 4 + \ln 4 - \ln 9) + \mathbf{a}_z (\ln 9 - \ln 4 + \ln 4 - \ln 9) \\
 &= \underline{\underline{0.575 \mathbf{a}_\rho \mu N}}
 \end{aligned}$$

Prob. 8.14

From Prob. 8.7,

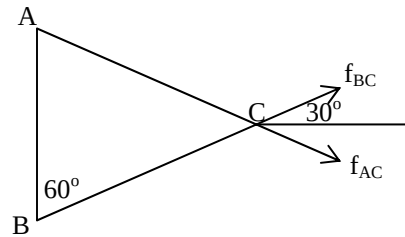
$$\mathbf{f} = \frac{\mu_0 I_1 I_2}{2\pi \rho} \mathbf{a}_\rho$$

$$\mathbf{f} = \mathbf{f}_{AC} + \mathbf{f}_{BC}$$

$$|\mathbf{f}_{AC}| = |\mathbf{f}_{BC}| = \frac{4\pi \times 10^{-7} \times 75 \times 150}{2\pi \times 2} = 1.125 \times 10^{-3}$$

$$\mathbf{f} = 2 \times 1.125 \cos 30^\circ \mathbf{a}_x \text{ mN/m}$$

$$= \underline{\underline{1.949 \mathbf{a}_x \text{ mN/m}}}$$



Prob. 8.15

The field due to the current sheet is

$$\mathbf{B} = \frac{\mu}{2} \mathbf{K} \times \mathbf{a}_n = \frac{\mu_o}{2} 10 \mathbf{a}_x \times (-\mathbf{a}_z) = 5\mu_o \mathbf{a}_y$$

$$\mathbf{F} = I_2 \int d\mathbf{l}_2 \times \mathbf{B} = 2.5 \int_0^L dx \mathbf{a}_x \times (5\mu_o \mathbf{a}_y) = 2.5L \times 5\mu_o (\mathbf{a}_z)$$

$$\frac{\mathbf{F}}{L} = 12.5 \times 4\pi \times 10^{-7} (\mathbf{a}_z) = \underline{\underline{15.71 \mathbf{a}_z \text{ } \mu\text{N/m}}}$$

Prob. 8.16

$$\mathbf{F} = \int Id\mathbf{l} \times \mathbf{B} = IL \times \mathbf{B} = 5(2\mathbf{a}_z) \times 40\mathbf{a}_x 10^{-3} = \underline{\underline{0.4 \mathbf{a}_y \text{ } \text{N}}}$$

Prob. 8.17

$$\mathbf{m} = IS\mathbf{a}_n = 10(2 \times 6)(\mathbf{a}_x) = 120\mathbf{a}_x$$

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} = 120\mathbf{a}_x \times 4.5(\mathbf{a}_y - \mathbf{a}_z) = 540 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{vmatrix} = \underline{\underline{540(\mathbf{a}_y + \mathbf{a}_z) \text{ } \text{N.m}}}$$

Prob. 8.18

$$f(x, y, z) = x + 2y - 5z - 12 = 0 \quad \longrightarrow \quad \nabla f = \mathbf{a}_x + 2\mathbf{a}_y - 5\mathbf{a}_z$$

$$\mathbf{a}_n = \frac{\nabla f}{|\nabla f|} = \frac{\mathbf{a}_x + 2\mathbf{a}_y - 5\mathbf{a}_z}{\sqrt{30}}$$

$$\mathbf{m} = NIS\mathbf{a}_n = 2 \times 60 \times 8 \times 10^{-4} \frac{(\mathbf{a}_x + 2\mathbf{a}_y - 5\mathbf{a}_z)}{\sqrt{30}} = \underline{\underline{17.53\mathbf{a}_x + 35.05\mathbf{a}_y - 87.64\mathbf{a}_z \text{ } \text{mAm}}}$$

Prob. 8.19

$$\mathbf{m} = IS \quad \longrightarrow \quad I = \frac{\mathbf{m}}{S} = \frac{\mathbf{m}}{\pi r^2}$$

$$I = \frac{8 \times 10^{22}}{\pi (6370 \times 10^3)^2} = 6.275 \times 10^8 = \underline{\underline{627.5 \text{ } \text{MA}}}$$

Prob. 8.20

Let $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$

$$\mathbf{F}_1 = \int Id\mathbf{l} \times \mathbf{B} = \int_5^0 2dx\mathbf{a}_x \times 30\mathbf{a}_z \text{ mN}$$

$$= -60\mathbf{a}_y x \Big|_5^0 = 300\mathbf{a}_y \text{ mN}$$

$$\mathbf{F}_2 = \int_0^5 2dy\mathbf{a}_y \times 30\mathbf{a}_z \text{ mN}$$

$$= 60\mathbf{a}_x y \Big|_0^5 = 300\mathbf{a}_x \text{ mN}$$

$$\mathbf{F}_3 = \int_0^5 2(dx\mathbf{a}_x + dz\mathbf{a}_z) \times 30\mathbf{a}_z \text{ mN}$$

$$= 60(-\mathbf{a}_y)x \Big|_0^5 = -300\mathbf{a}_y \text{ mN}$$

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = 300\mathbf{a}_y + 300\mathbf{a}_x - 300\mathbf{a}_y \text{ mN} = \underline{\underline{300\mathbf{a}_x \text{ mN}}}$$

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} = IS\mathbf{a}_n \times \mathbf{B} = 2\left(\frac{1}{2}\right)(5)(5)\mathbf{a}_y \times 30\mathbf{a}_z 10^{-3} = \underline{\underline{0.75\mathbf{a}_x \text{ N.m}}}$$

Prob. 8.21

For each turn, $\mathbf{T} = \mathbf{m} \times \mathbf{B}$, $\mathbf{m} = IS\mathbf{a}_n$

For N turns,

$$T = NISB = 50 \times 4 \times 12 \times 10^{-4} \times 100 \times 10^{-3} = \underline{\underline{24 \text{ mNm}}}$$

Prob. 8.22

$$\mathbf{F} = \int Id\mathbf{l} \times \mathbf{B} \longrightarrow F = IB\ell = 520 \times 0.4 \times 10^{-3} \times 30 \times 10^{-3}$$

$$F = \underline{\underline{6.24 \text{ mN}}}$$

Prob. 8.23

$$M = \chi_m H = \chi_m \frac{B}{\mu_o \mu_r} = \frac{\chi_m B}{\mu_o (1 + \chi_m)}$$

Prob. 8.24

$$(a) \quad \mathbf{M} = \chi_m \mathbf{H} = \chi_m \frac{\mathbf{B}}{\mu_0 \mu}$$

$$M = \frac{4999}{5000} \times \frac{1.5}{4\pi \times 10^{-7}} = \underline{\underline{1.193 \times 10^6 \text{ A/m}}}$$

$$(b) \quad \mathbf{M} = \frac{\sum_{k=1}^N \mathbf{m}_k}{\Delta v}$$

If we assume that all $\mathbf{m}_k$ align with the applied $\mathbf{B}$ field,

$$M = \frac{Nm_k}{\Delta v} \rightarrow m_k = \frac{M}{N/\Delta v} = \frac{1.193 \times 10^6}{8.5 \times 10^{28}}$$

$$m_k = \underline{\underline{1.404 \times 10^{-23} \text{ A} \cdot \text{m}^2}}$$

Prob. 8.25

$$\mu_r = \chi_m + 1 = 6.5 + 1 = \underline{\underline{7.5}}$$

$$\mathbf{M} = \chi_m \mathbf{H} \rightarrow \mathbf{H} = \frac{\mathbf{M}}{\chi_m} = \frac{24y^2}{6.5} \mathbf{a}_z$$

At $y = 2\text{cm}$,

$$\mathbf{H} = \frac{24 \times 4 \times 10^{-4}}{6.5} \mathbf{a}_z = \underline{\underline{1.477 \mathbf{a}_z \text{ mA/m}}}$$

$$\mathbf{J} = \nabla \times \mathbf{H} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & \frac{24y^2}{6.5} \end{vmatrix} = \frac{48y}{6.5} \mathbf{a}_x$$

At $y=2\text{cm}$,

$$\mathbf{J} = \frac{48 \times 2 \times 10^{-2}}{6.5} \mathbf{a}_x = \underline{\underline{0.1477 \mathbf{a}_x \text{ A/m}^2}}$$

Prob. 8.26

$$(a) \quad \mu = 80\mu_0 \rightarrow \mu_r = 80$$

$$(b) \quad \chi_m = \mu_r - 1 = 79$$

$$(c) \quad \mathbf{H} = \frac{\mathbf{B}}{\mu} = \frac{20x\mathbf{a}_y 10^{-3}}{80(4\pi \times 10^{-7})} = 198.9x\mathbf{a}_y \text{ A/m}$$

$$(d) \quad \mathbf{M} = \chi_m \mathbf{H} = 15.713x \mathbf{a}_y \text{ kA/m}$$

$$(e) \quad \mathbf{J}_b = \nabla \times \mathbf{M} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 15.713x & 0 \end{vmatrix} = 15.713 \mathbf{a}_z \text{ kA/m}$$

Prob. 8.27

When $H = 250$,

$$B = \frac{2H}{100 + H} = \frac{2(250)}{100 + 250} = 1.4286 \text{ mWb/m}^2$$

But $B = \mu_o \mu_r H$

$$\mu_r = \frac{B}{\mu_o H} = \frac{1.4286 \times 10^{-3}}{4\pi \times 10^{-7} \times 250} = \underline{\underline{4.54}}$$

Prob. 8.28

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

$$H_\phi \cdot 2\pi\rho = \frac{\pi\rho^2}{\pi a^2} \cdot I \rightarrow H_\phi = \frac{I\rho}{2\pi a^2}$$

$$\mathbf{M} = \chi_m \mathbf{H} = (\mu_r - 1) \frac{I\rho}{2\pi a^2} \mathbf{a}_\phi$$

$$\mathbf{J}_b = \nabla \times \mathbf{M} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho M_\phi) \mathbf{a}_z = (\mu_r - 1) \frac{I}{\pi a^2} \mathbf{a}_z$$

Prob. 8.29

(a) From $H_{1t} - H_{2t} = K$ and $M = \chi_m H$, we obtain:

$$\frac{M_{1t}}{\chi_{m1}} - \frac{M_{2t}}{\chi_{m2}} = K$$

Also from $B_{1n} - B_{2n} = 0$ and $B = \mu H = (\mu/\chi_m)M$, we get:

$$\frac{\mu_1 M_{1n}}{\chi_{m1}} = \frac{\mu_2 M_{2n}}{\chi_{m2}}$$

(b) From $B_1 \cos \theta_1 = B_{1n} = B_{2n} = B_2 \cos \theta_2$ (1)

$$\text{and } \frac{B_1 \sin \theta_1}{\mu_1} = H_{1t} = K + H_{2t} = K + \frac{B_2 \sin \theta_2}{\mu_2} \quad (2)$$

Dividing (2) by (1) gives

$$\frac{\tan \theta_1}{\mu_1} = \frac{k}{B_2 \cos \theta_2} + \frac{\tan \theta_2}{\mu_2} = \frac{\tan \theta_2}{\mu_2} \left(1 + \frac{k\mu_2}{B_2 \sin \theta_2} \right)$$

$$\text{i.e. } \underline{\underline{\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2} \left(1 + \frac{k\mu_2}{B_2 \sin \theta_2} \right)}}$$

Prob. 8.30

$$\mathbf{B}_{2n} = \mathbf{B}_{1n} = 1.8\mathbf{a}_z$$

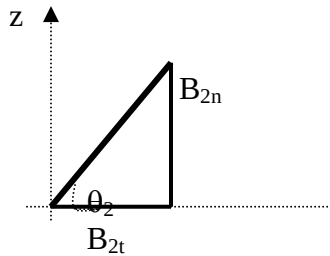
$$\mathbf{H}_{2t} = \mathbf{H}_{1t} \longrightarrow \frac{\mathbf{B}_{2t}}{\mu_2} = \frac{\mathbf{B}_{1t}}{\mu_1}$$

$$\mathbf{B}_{2t} = \frac{\mu_2}{\mu_1} \mathbf{B}_{1t} = \frac{4\mu_o}{2.5\mu_o} (6\mathbf{a}_x - 4.2\mathbf{a}_y) = 9.6\mathbf{a}_x - 6.72\mathbf{a}_y$$

$$\mathbf{B}_2 = \mathbf{B}_{2n} + \mathbf{B}_{2t} = 9.6\mathbf{a}_x - 6.72\mathbf{a}_y + 1.8\mathbf{a}_z \text{ mWb/m}^2$$

$$\mathbf{H}_2 = \frac{\mathbf{B}_2}{\mu_2} = \frac{10^{-3}(9.6, -6.72, 1.8)}{4 \times 4\pi \times 10^{-7}}$$

$$= 1,909.86\mathbf{a}_x - 1,336.9\mathbf{a}_y + 358.1\mathbf{a}_z \text{ A/m}$$



$$\tan \theta_2 = \frac{B_{2n}}{B_{2t}} = \frac{1.8}{\sqrt{9.6^2 + 6.72^2}} = 0.1536$$

$$\theta_2 = \underline{\underline{8.73^\circ}}$$

Prob. 8.31

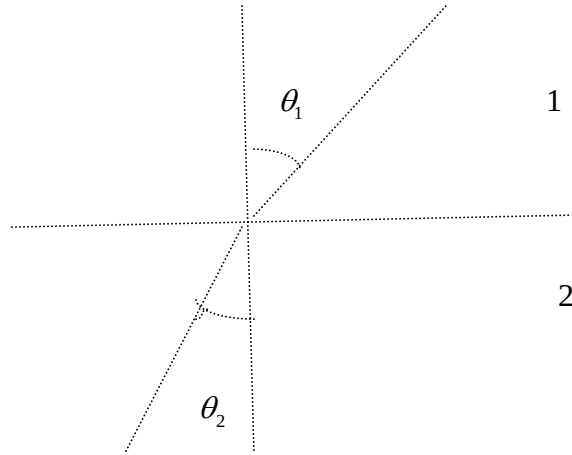
$$\mathbf{B}_{2n} = \mathbf{B}_{1n} = 12\mathbf{a}_z$$

$$\mathbf{H}_{2t} = \mathbf{H}_{1t} \longrightarrow \frac{\mathbf{B}_{2t}}{\mu_2} = \frac{\mathbf{B}_{1t}}{\mu_1}$$

$$\mathbf{B}_{2t} = \frac{\mu_2}{\mu_1} \mathbf{B}_{1t} = \frac{2\mu_0}{5\mu_0} (4\mathbf{a}_x - 10\mathbf{a}_y) = 1.6\mathbf{a}_x - 4\mathbf{a}_y$$

$$\mathbf{B}_2 = \mathbf{B}_{2n} + \mathbf{B}_{2t} = 1.6\mathbf{a}_x - 4\mathbf{a}_y + 12\mathbf{a}_z \text{ mWb/m}^2$$

$$w_2 = \frac{1}{2} \mu_2 H_2^3 = \frac{B_2^3}{2\mu_2} = \frac{(1.6^2 + 4^2 + 12^2) \times 10^{-6}}{2(2)(4\pi \times 10^{-7})} = \underline{\underline{32.34 \text{ J/m}^3}}$$

Prob. 8.32

$$\tan \theta_1 = \frac{H_{1t}}{H_{1n}}, \quad \tan \theta_2 = \frac{H_{2t}}{H_{2n}}$$

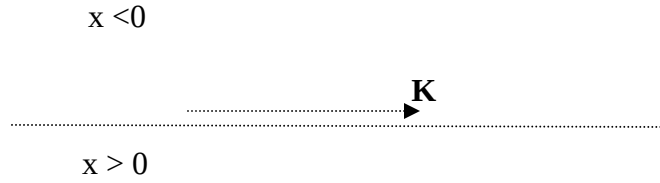
$$\text{But } H_{1t} = H_{2t}$$

$$B_{1n} = B_{2t} \rightarrow \mu_1 H_{1n} = \mu_2 H_{2n} \rightarrow H_{2n} = \frac{\mu_1}{\mu_2} H_{1n}$$

$$\frac{H_{2t}}{H_{2n}} = \frac{H_{1t}}{\frac{1}{6.5} H_{1n}} = \frac{6.5 H_{1t}}{H_{1n}}$$

$$\text{If } \theta_1 = 42^\circ, \quad \tan 42^\circ = \frac{H_{1t}}{H_{1n}} \rightarrow \frac{H_{1t}}{H_{1n}} = 0.9004$$

$$\tan \theta_2 = 6.5(0.9004) = 5.832 \rightarrow \underline{\underline{\theta_2 = 80.3^\circ}}$$

Prob. 8.33

$$(\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{n12} = \mathbf{K} \quad \rightarrow \quad \mathbf{H}_1 \times \mathbf{a}_x = \mathbf{K} + \mathbf{H}_2 \times \mathbf{a}_x$$

$$\mathbf{H}_1 \times \mathbf{a}_x = (10\mathbf{a}_x + 6\mathbf{a}_z) \times \mathbf{a}_x = 6\mathbf{a}_y$$

$$\mathbf{H}_2 \times \mathbf{a}_x = (H_{2x}, H_{2y}, H_{2z}) \times \mathbf{a}_x = H_{2z}\mathbf{a}_y - H_{2y}\mathbf{a}_z$$

$$6\mathbf{a}_y = 12\mathbf{a}_y + H_{2z}\mathbf{a}_y - H_{2y}\mathbf{a}_z$$

Equating components,

$$6 = 12 + H_{2z} \quad \rightarrow \quad H_{2z} = -6,$$

$$H_{2y} = 0$$

$$\text{Also, } \mathbf{B}_{1n} = \mathbf{B}_{2n} \quad \rightarrow \quad \mu_1 \mathbf{H}_{1n} = \mu_2 \mathbf{H}_{2n}$$

$$\mathbf{H}_{2n} = \frac{\mu_1}{\mu_2} \mathbf{H}_{1n} = \frac{2\mu_o}{4\mu_o} (10\mathbf{a}_x) = 5\mathbf{a}_x$$

$$\underline{\underline{\mathbf{H}_2 = 5\mathbf{a}_x - 6\mathbf{a}_z \text{ A/m}}}$$

Prob. 8.34

$$\mathbf{H}_{2t} = \mathbf{H}_{1t} = \alpha\mathbf{a}_x + \delta\mathbf{a}_z$$

$$\mathbf{B}_{2n} = \mathbf{B}_{1n} \quad \longrightarrow \quad \mu_2 \mathbf{H}_{2n} = \mu_1 \mathbf{H}_{1n}$$

$$\mathbf{H}_{2n} = \frac{\mu_1}{\mu_2} \mathbf{H}_{1n} = \frac{\mu_{r1}}{\mu_{r2}} \beta\mathbf{a}_y$$

$$\underline{\underline{\mathbf{H} = \alpha\mathbf{a}_x + \frac{\mu_{r1}}{\mu_{r2}} \beta\mathbf{a}_y + \delta\mathbf{a}_z}}}$$

Prob. 8.35

$$(a) \quad \mathbf{B}_{1n} = \mathbf{B}_{2n} = 15\mathbf{a}_\phi$$

$$\mathbf{H}_{1t} = \mathbf{H}_{2t} \rightarrow \frac{\mathbf{B}_{1t}}{\mu_1} = \frac{\mathbf{B}_{2t}}{\mu_2}$$

$$\mathbf{B}_{1t} = \frac{\mu_1}{\mu_2} \mathbf{B}_{2t} = \frac{2}{5} (10\mathbf{a}_\rho - 20\mathbf{a}_z) = 4\mathbf{a}_\rho - 8\mathbf{a}_z$$

Hence,

$$\mathbf{B}_1 = \frac{4\mathbf{a}_\rho + 15\mathbf{a}_\phi - 8\mathbf{a}_z \text{ mWb/m}^2}{}$$

$$(b) \quad w_{m1} = \frac{1}{2} \mathbf{B}_1 \cdot \mathbf{H}_1 = \frac{B_1^2}{2\mu_1} = \frac{(4^2 + 15^2 + 8^2) \times 10^{-6}}{2 \times 2 \times 4\pi \times 10^{-7}}$$

$$w_{m1} = \frac{60.68 \text{ J/m}^3}{}$$

$$w_{m2} = \frac{B_2^2}{2\mu_2} = \frac{(10^2 + 15^2 + 20^2) \times 10^{-6}}{2 \times 5 \times 4\pi \times 10^{-7}} = \frac{57.7 \text{ J/m}^3}{}$$

Prob. 8.36

$$f(x, y, z) = x - y + 2z$$

$$\nabla f = \mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z$$

$$\mathbf{a}_n = \frac{\nabla f}{|\nabla f|} = \frac{1}{\sqrt{6}} (\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z)$$

(a)

$$\begin{aligned} \mathbf{H}_{1n} &= (\mathbf{H}_1 \cdot \mathbf{a}_n) \mathbf{a}_n = (40 - 20 - 60) \frac{(\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z)}{6} \\ &= \underline{\underline{-6.667\mathbf{a}_x + 6.667\mathbf{a}_y - 13.333\mathbf{a}_z \text{ A/m}}} \end{aligned}$$

(b)

$$\mathbf{H}_2 = \mathbf{H}_{2n} + \mathbf{H}_{2t}$$

$$\text{But } \mathbf{B}_{2n} = \mathbf{B}_{1n} \longrightarrow \mu_2 \mathbf{H}_{2n} = \mu_1 \mathbf{H}_{1n}$$

$$\begin{aligned} \mathbf{B}_2 &= \mu_2 \mathbf{H}_2 = \mu_2 \mathbf{H}_{2n} + \mu_2 \mathbf{H}_{2t} = \mu_1 \mathbf{H}_{1n} + \mu_2 \mathbf{H}_{2t} = \mu_o (2\mathbf{H}_{1n} + 5\mathbf{H}_{2t}) \\ &= 4\pi \times 10^{-7} [(-13.333, 13.333, -26.667) + (233.333, 66.666, -83.333)] \\ &= 4\pi \times 10^{-7} (220, 80, -110) \\ &= \underline{\underline{276.5\mathbf{a}_x + 100.5\mathbf{a}_y - 138.2\mathbf{a}_z \text{ } \mu\text{Wb/m}^2}} \end{aligned}$$

Prob. 8.37

$$\mathbf{a}_n = \mathbf{a}_\rho$$

$$\mathbf{B}_{2n} = \mathbf{B}_{1n} = 22\mu_o \mathbf{a}_\rho$$

$$\mathbf{H}_{2t} = \mathbf{H}_{1t} \longrightarrow \frac{\mathbf{B}_{2t}}{\mu_2} = \frac{\mathbf{B}_{1t}}{\mu_1}$$

$$\mathbf{B}_{2t} = \frac{\mu_2}{\mu_1} \mathbf{B}_{1t} = \frac{\mu_o}{800\mu_o} (45\mu_o \mathbf{a}_\phi) = 0.05625\mu_o \mathbf{a}_\phi$$

$$\mathbf{B}_2 = \underline{\underline{\mu_o(22\mathbf{a}_\rho + 0.05625\mathbf{a}_\phi) \text{ Wb/m}^2}}$$

Prob. 8.38

$$\mathbf{H}_{1n} = -3\mathbf{a}_z, \quad \mathbf{H}_{1t} = 10\mathbf{a}_x + 15\mathbf{a}_y$$

$$\mathbf{H}_{2t} = \mathbf{H}_{1t} = 10\mathbf{a}_x + 15\mathbf{a}_y$$

$$\mathbf{H}_{2n} = \frac{\mu_1}{\mu_2} \mathbf{H}_{1n} = \frac{1}{200} (-3\mathbf{a}_z) = -0.015\mathbf{a}_z$$

$$\mathbf{H}_2 = 10\mathbf{a}_x + 15\mathbf{a}_y - 0.015\mathbf{a}_z$$

$$\mathbf{B}_2 = \mu_2 \mathbf{H}_2 = 200 \times 4\pi \times 10^{-7} (10, 15, -0.015)$$

$$\mathbf{B}_2 = \underline{\underline{2.51\mathbf{a}_x + 3.77\mathbf{a}_y - 0.0037\mathbf{a}_z \text{ mWb/m}^2}}$$

$$\tan \alpha = \frac{B_{2n}}{B_{2t}}$$

$$\text{or } \alpha = \tan^{-1} \frac{0.0037}{\sqrt{2.51^2 + 3.77^2}} = \underline{\underline{0.047^\circ}}$$

Prob. 8.39

$$(a) \quad \mathbf{H} = \frac{1}{2} \mathbf{K} \times \mathbf{a}_n = \frac{1}{2} (30 - 40)\mathbf{a}_x \times (-\mathbf{a}_z) = \underline{\underline{-5\mathbf{a}_y \text{ A/m}}}$$

$$\mathbf{B} = \mu_o \mathbf{H} = 4\pi \times 10^{-7} (-5\mathbf{a}_y) = \underline{\underline{-6.28\mathbf{a}_y \mu \text{ Wb/m}^2}}$$

$$(b) \quad \mathbf{H} = \frac{1}{2} (-30 - 40)\mathbf{a}_y = \underline{\underline{-35\mathbf{a}_y \text{ A/m}}}$$

$$\mathbf{B} = \mu_o \mu_r \mathbf{H} = 4\pi \times 10^{-7} (2.5)(-35\mathbf{a}_y) = \underline{\underline{-110\mathbf{a}_y \mu \text{ Wb/m}^2}}$$

$$(c) \quad \mathbf{H} = \frac{1}{2}(-30 + 40)\mathbf{a}_y = \underline{\underline{5\mathbf{a}_y}}$$

$$\mathbf{B} = \mu_o \mathbf{H} = \underline{\underline{6.283\mathbf{a}_y \mu \text{ Wb/m}^2}}$$

Prob. 8.40

$r = a$ is the interface between the two media.

$$\begin{aligned} \mathbf{B}_{2n} = \mathbf{B}_{1n} &\longrightarrow B_{o1}(1+1.6)\cos\theta\mathbf{a}_r = B_{o2}\cos\theta\mathbf{a}_r \\ 2.6B_{o1} = B_{o2} &\quad (1) \end{aligned}$$

$$\mathbf{H}_{2t} = \mathbf{H}_{1t} \longrightarrow \frac{\mathbf{B}_{2t}}{\mu_2} = \frac{\mathbf{B}_{1t}}{\mu_1}$$

$$\mu_2 \mathbf{B}_{1t} = \mu_1 \mathbf{B}_{2t}$$

$$\mu_2 B_{o1}(-0.2)\sin\theta\mathbf{a}_\theta = \mu_o B_{o2}(-\sin\theta)\mathbf{a}_\theta$$

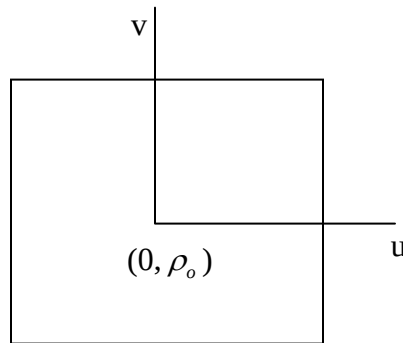
$$\mu_2 = \frac{\mu_o B_{o2}}{0.2 B_{o1}} \quad (2)$$

Substituting (1) into (2) gives

$$\mu_2 = \frac{\mu_o}{0.2}(2.6) = \underline{\underline{13\mu_o}}$$

Prob. 8.41

- (a) The square cross-section of the toroid is shown below. Let (u,v) be the local coordinates and ρ_o = mean radius. Using Ampere's law around a circle passing through P, we get



$$H(2\pi)(\rho_o + v) = NI \longrightarrow H = \frac{NI}{2\pi(\rho_o + v)}$$

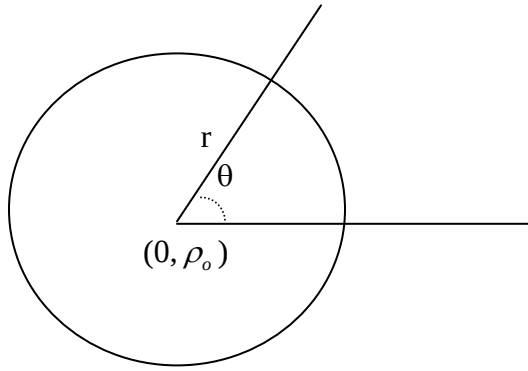
The flux per turn is

$$\Psi = \int_{u=-a/2}^{a/2} \int_{v=-a/2}^{a/2} B du dv = \frac{\mu_o N I a}{2\pi} \ln \left(\frac{\rho_o + a/2}{\rho_o - a/2} \right)$$

$$L = \frac{N\Psi}{I} = \frac{\mu_o N^2 a}{2\pi} \ln \left(\frac{2\rho_o + a}{2\rho_o - a} \right)$$

- (b) The circular cross-section of the toroid is shown below. Let (r, θ) be the local coordinates. Consider a point $P(r \cos \theta, \rho_o + r \sin \theta)$ and apply Ampere's law around a circle that passes through P.

$$H(2\pi)(\rho_o + r \sin \theta) = NI \quad \longrightarrow \quad H = \frac{NI}{2\pi(\rho_o + r \sin \theta)} \approx \frac{NI}{2\pi\rho_o} \left(1 - \frac{r \sin \theta}{\rho_o} \right)$$



$$\text{Flux per turn } \Psi = \int_{r=0}^a \int_{\theta}^{2\pi} \frac{\mu NI}{2\pi\rho_o} \left(1 - \frac{r \sin \theta}{\rho_o} \right) r dr d\theta = \frac{\mu NI}{2\pi\rho_o} \frac{a^2}{2} (2\pi)$$

$$L = \frac{N\Psi}{I} = \frac{\mu N^2 a^2}{2\rho_o}$$

Or from Example 8.10,

$$L = L' l = \frac{\mu_o N^2 l S}{l^2} = \frac{\mu_o N^2 \pi a^2}{2\pi\rho_o} = \frac{\mu_o N^2 a^2}{2\rho_o}$$

Prob. 8.42

$$\rho_o = \frac{1}{2}(3+5) = 4\text{cm}$$

$$a = 2\text{ cm}$$

$$L = \frac{\mu_o N^2 a}{2\pi} \ln \left[\frac{2\rho_o + a}{2\rho_o - a} \right]$$

$$N^2 = \frac{2\pi L}{\mu_o a \ln \left[\frac{2\rho_o + a}{2\rho_o - a} \right]} = \frac{2\pi(45 \times 10^{-6})}{4\pi \times 10^{-7} (2 \times 10^{-2}) \ln \left(\frac{8+2}{8-2} \right)} = 22,023.17$$

$$\underline{\underline{N = 148.4 \text{ or } 148}}$$

Prob. 8.43

$$L = \frac{\mu_o \ell}{8\pi} = \frac{4\pi \times 10^{-7} (40)}{8\pi} = 20 \times 10^{-7} = \underline{\underline{2 \mu\text{H}}}$$

Prob. 8.44

$$L_{\text{in}} = \frac{\mu_o \ell}{8\pi}, \quad L_{\text{ext}} = \frac{\mu_o \ell}{2\pi} \ln(b/a)$$

$$\text{If } L_{\text{in}} = 2L_{\text{ext}} \longrightarrow \frac{\mu_o \ell}{8\pi} = \frac{\mu_o \ell}{\pi} \ln(b/a)$$

$$\ln(b/a) = \frac{1}{8} \quad \frac{b}{a} = e^{1/8} = 1.1331$$

$$b = 1.1331a = \underline{\underline{7.365 \text{ mm}}}$$

Prob. 8.45

$$L = \frac{\mu_o \ell}{2\pi} \left[\ln \frac{2\ell}{a} - 1 \right] = \frac{4\pi \times 10^{-7} (10)}{2\pi} \left[\ln \frac{2 \times 10}{2 \times 10^{-2}} - 1 \right] = 2 \times 10^{-6} (\ln 1000 - 1)$$

$$= 2(5.908) \mu\text{H} = \underline{\underline{11.82 \mu\text{H}}}$$

Prob. 8.46

$$\psi_{12} = \int \mathbf{B}_1 \cdot d\mathbf{S} = \int_{\rho=\rho_o}^{\rho_o+a} \int_{z=0}^b \frac{\mu_o I}{2\pi\rho} dz d\rho = \frac{\mu_o I b}{2\pi} \ln \frac{a + \rho_o}{\rho_o}$$

$$M_{12} = \frac{N\psi_{12}}{I} = \frac{N\mu_o b}{2\pi} \ln \frac{a + \rho_o}{\rho_o}$$

For $N = 1$,

$$\begin{aligned} M_{12} &= \frac{\psi_{12}}{I_1} = \frac{\mu_o b}{2\pi} \ln \frac{a + \rho_o}{\rho_o} \\ &= \frac{4\pi \times 10^{-7}}{2\pi} (1) \ln 2 = \underline{\underline{0.1386 \mu \text{ H}}} \end{aligned}$$

Prob. 8.47

We may approximate the longer solenoid as infinite so that $B_1 = \frac{\mu_o N_1 I_1}{l_1}$. The flux linking the second solenoid is:

$$\begin{aligned} \psi_2 &= N_2 B_1 S_1 = \frac{\mu_o N_1 I_1}{l_1} \bullet \pi r_1^2 \square N_2 \\ M &= \frac{\psi_2}{I_1} = \frac{\mu_o N_1 N_2}{l_1} \bullet \pi r_1^2 \end{aligned}$$

Here we assume air-core solenoids.

Prob. 8.48

For a straight infinitely long conductor,

$$\begin{aligned} B_\phi &= \frac{\mu_o I}{2\pi\rho} \\ \Psi &= \int_S \mathbf{B} \bullet d\mathbf{S} = \frac{\mu_o I}{2\pi} \left[\int_{z=0}^h \int_{\rho=b}^{b+w} \frac{1}{\rho} dz d\rho - \int_{z=0}^h \int_{\rho=a+b}^{a+b+w} \frac{1}{\rho} dz d\rho \right] \\ &= \frac{\mu_o I h}{2\pi} \left[\ln \rho \Big|_b^{b+w} - \ln \rho \Big|_{a+b}^{a+b+w} \right] = \frac{\mu_o I h}{2\pi} \left[\ln \frac{b+w}{b} - \ln \frac{a+b+w}{a+b} \right] \\ \underline{\underline{\Psi}} &= \frac{\mu_o I h}{2\pi} \ln \left[\frac{(a+b)(b+w)}{b(a+b+w)} \right] \end{aligned}$$

Prob. 8.49

$$\begin{aligned} \mathbf{H} &= \frac{I}{2\pi\rho} \mathbf{a}_\rho \\ w_m &= \frac{1}{2} \mu |\mathbf{H}|^2 = \frac{1}{2} \mu \frac{I^2}{4\pi^2 \rho^2} \\ W &= \int w_m dv = \iiint \frac{1}{2} \mu \frac{I^2}{4\pi^2 \rho^2} \rho d\phi d\rho dz = \frac{1}{4\pi} \mu I^2 L \ln(b/a) \end{aligned}$$

$$= \frac{1}{4\pi} \times 4 \times 4\pi \times 10^{-7} (625 \times 10^{-6}) 3 \ln(18/12) = \underline{\underline{304.1 \text{ pJ}}}$$

Alternatively,

$$W = \frac{1}{2} LI^2 = \frac{1}{2} \frac{\mu L}{2\pi} \ln \frac{b}{a} \times I^2 = \frac{\mu I^2 L}{4\pi} \ln \frac{b}{a}$$

Prob. 8.50

$$\begin{aligned} \mu_r &= \chi_m + 1 = 20 \\ w_m &= \frac{1}{2} \mathbf{B}_1 \cdot \mathbf{H}_1 = \frac{1}{2} \mu \mathbf{H} \cdot \mathbf{H} \\ &= \frac{1}{2} \mu (25x^4 y^2 z^2 + 100x^2 y^4 z^2 + 225x^2 y^2 z^4) \\ W_m &= \int w_m dv \\ &= \frac{1}{2} \mu \left[25 \int_0^1 x^4 dx \int_0^2 y^2 dy \int_{-1}^2 z^2 dz + 100 \int_0^1 x^2 dx \int_0^2 y^4 dy \int_{-1}^2 z^2 dz \right. \\ &\quad \left. + 225 \int_0^1 x^2 dx \int_0^2 y^2 dy \int_{-1}^2 z^2 dz \right] \\ &= \frac{25\mu}{2} \left[\frac{x^5}{5} \Big|_0^1 \frac{y^3}{3} \Big|_0^2 \frac{z^3}{3} \Big|_{-1}^2 + 4 \frac{x^3}{3} \Big|_0^1 \frac{y^5}{5} \Big|_0^2 \frac{z^3}{3} \Big|_{-1}^2 \right. \\ &\quad \left. + 9 \frac{x^3}{3} \Big|_0^1 \frac{y^3}{3} \Big|_0^2 \frac{z^5}{5} \Big|_{-1}^2 \right] \\ &= \frac{25\mu}{2} \left(\frac{1}{5} \cdot \frac{8}{3} \cdot \frac{9}{3} + \frac{4}{3} \cdot \frac{32}{3} \cdot \frac{9}{3} + \frac{9}{3} \cdot \frac{8}{3} \cdot \frac{33}{5} \right) \\ &= \frac{25}{2} \times 4\pi \times 10^{-7} \times 20 \times \frac{3600}{45} \end{aligned}$$

$$W_m = \underline{\underline{25.13 \text{ mJ}}}$$

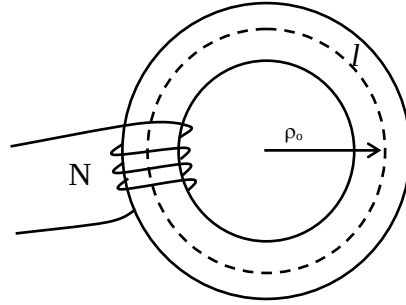
Prob. 8.51

$$\begin{aligned} W &= \frac{1}{2} \int_v \mathbf{B} \cdot \mathbf{H} dv = \int_v \frac{|\mathbf{B}|^2}{2\mu} dv = \frac{1}{2(15)4\pi \times 10^{-7}} \int_{z=0}^4 \int_{y=0}^3 \int_{x=0}^2 (4^2 + 12^2) 10^{-6} dx dy dz \\ &= \frac{10^{-6}}{(30)4\pi \times 10^{-7}} (16 + 144)(2)(3)(4) = \frac{320}{\pi} = \underline{\underline{101.86 \text{ J}}} \end{aligned}$$

Prob. 8.52

$$NI = Hl = \frac{Bl}{\mu}$$

$$N = \frac{Bl}{\mu_o \mu_r I} = \frac{1.5 \times 0.6\pi}{4\pi \times 10^{-7} \times 600 \times 12} = \underline{\underline{313 \text{ turns}}}$$

**Prob. 8.53**

$$F = NI = 400 \times 0.5 = 200 \text{ A.t}$$

$$R_a = \frac{100}{4\pi} \text{ MA.t/Wb}, \quad R_1 = R_2 = \frac{6}{4\pi} \text{ MA.t/Wb}, \quad R_3 = \frac{1.8}{4\pi} \text{ MA.t/Wb}$$

$$F_a = \frac{R_a F}{R_a + R_3 + R_1 // R_2} = \underline{\underline{190.8 \text{ A.t}}}$$

$$H_a = \frac{F_a}{l_a} = \frac{190.8}{1 \times 10^{-2}} = \underline{\underline{19080 \text{ A/m}}}$$

Prob. 8.54

$$\text{Total } F = NI = 2000 \times 10 = 20,000 \text{ A.t}$$

$$R_c = \frac{l_c}{\mu_o \mu_r S} = \frac{(24 + 20 - 0.6) \times 10^{-2}}{4\pi \times 10^{-7} \times 1500 \times 2 \times 10^{-4}} = \underline{\underline{0.115 \times 10^7 \text{ A.t/m}}}$$

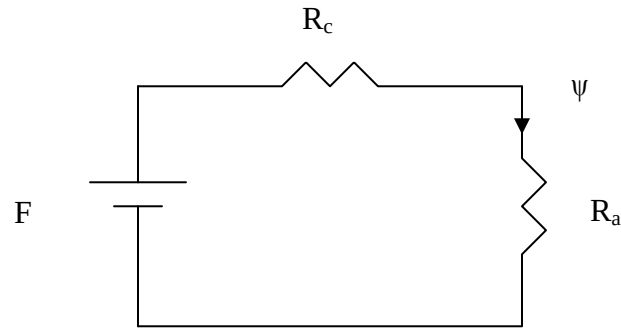
$$R_a = \frac{l_a}{\mu_o \mu_r S} = \frac{0.6 \times 10^{-2}}{4\pi \times 10^{-7} (1) \times 2 \times 10^{-4}} = \underline{\underline{2.387 \times 10^7 \text{ A.t/m}}}$$

$$R = R_a + R_c = 2.502 \times 10^7 \text{ A.t/m}$$

$$\psi = \frac{\mathfrak{F}}{R} = \psi_a = \psi_c = \frac{20,000}{2.502 \times 10^7} = \underline{\underline{8 \times 10^{-4} \text{ Wb/m}^2}}$$

$$\mathfrak{F}_a = \frac{R_a}{R_a + R_c} \mathfrak{F} = \frac{2.387 \times 20,000}{2.502} = \underline{\underline{19,081 \text{ A.t}}}$$

$$\mathfrak{F}_c = \frac{R_c}{R_a + R_c} \mathfrak{F} = \frac{0.115 \times 20,000}{2.502} = \underline{\underline{919 \text{ A.t}}}$$

Prob. 8.55

$$F = NI = 500 \times 0.2 = 100 \text{ A.t}$$

$$R_c = \frac{l_c}{\mu S} = \frac{42 \times 10^{-2}}{4\pi \times 10^{-7} \times 10^3 \times 4 \times 10^{-4}} = \frac{42 \times 10^6}{16\pi}$$

$$R_a = \frac{l_a}{\mu_o S} = \frac{10^{-3}}{4\pi \times 10^{-7} \times 4 \times 10^{-4}} = \frac{10^8}{16\pi}$$

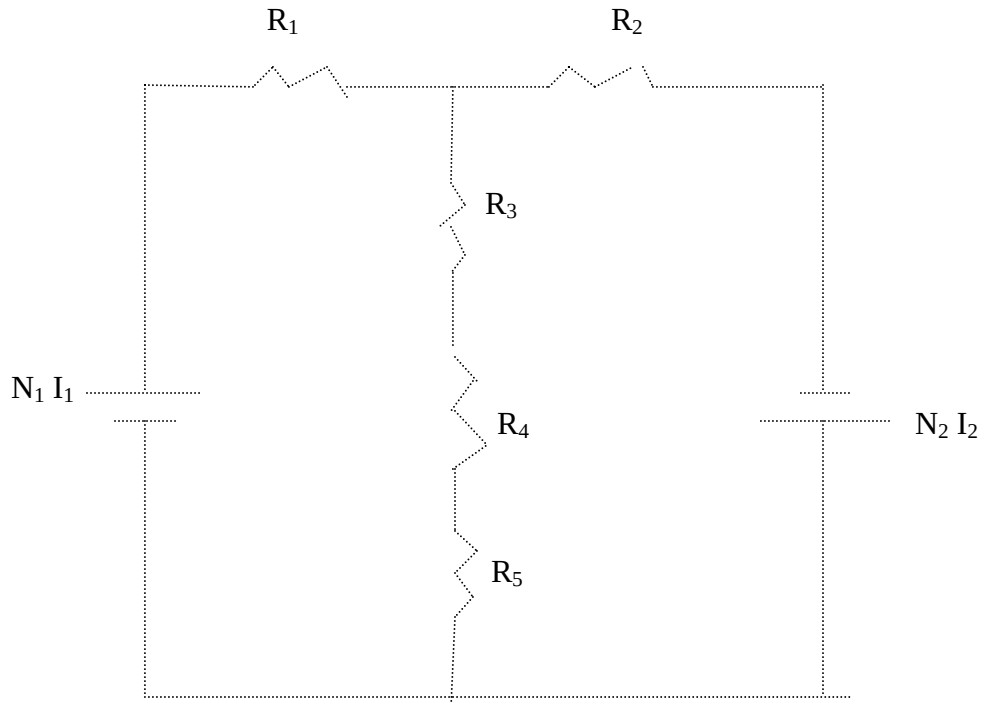
$$R_a + R_c = \frac{1.42 \times 10^8}{16\pi}$$

$$\psi = \frac{F}{R_a + R_c} = \frac{16\pi \times 100}{1.42 \times 10^8} = \frac{16\pi}{1.42} \mu\text{Wb}$$

$$B_a = \frac{\psi}{S} = \frac{16\pi \times 10^{-6}}{1.42 \times 4 \times 10^{-4}} = \underline{\underline{88.5 \text{ mWb/m}^2}}$$

Prob. 8.56

The equivalent circuit is shown below.

**Prob. 8.57**

$$R = \frac{\ell}{\mu_o S} = \frac{4.4 \times 10^{-3}}{4\pi \times 10^{-7} (4.82 \times 10^{-2})} = \frac{0.2282 \times 10^6}{\pi} = \underline{\underline{7.2643 \times 10^4 \text{ A.t/Wb}}}$$

Prob. 8.58

$$F = \frac{B^2 S}{2\mu_o} = \frac{\psi^2}{2\mu_o S} = \frac{4 \times 10^{-6}}{2 \times 4\pi \times 10^{-7} \times 0.3 \times 10^{-4}} = \underline{\underline{53.05 \text{ kN}}}$$

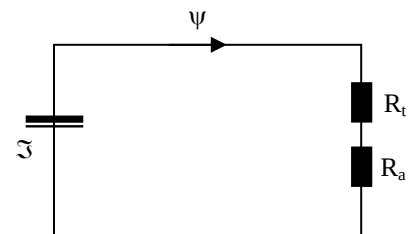
Prob. 8.59

(a) $F = NI = 200 \times 10^{-3} \times 750 = 150 \text{ A.t.}$

$$R_a = \frac{l_a}{\mu_o S} = \frac{10^{-3}}{25 \times 10^{-6} \mu_o} = 3.183 \times 10^7$$

$$R_t = \frac{l_t}{\mu_o \mu_r S} = \frac{2\pi \times 0.1}{\mu_o \times 300 \times 25 \times 10^{-6}} = 6.7 \times 10^7$$

$$\psi = \frac{\mathfrak{F}}{R_a + R_t} = \frac{150}{10^7 (3.183 + 20/3)} = 15.23 \times 10^{-7}$$



$$F = \frac{B^2 S}{2\mu_o} = \frac{\psi^2}{2\mu_o S} = \frac{2.32 \times 10^{-12}}{2 \times 4\pi \times 10^{-7} \times 25 \times 10^{-6}}$$

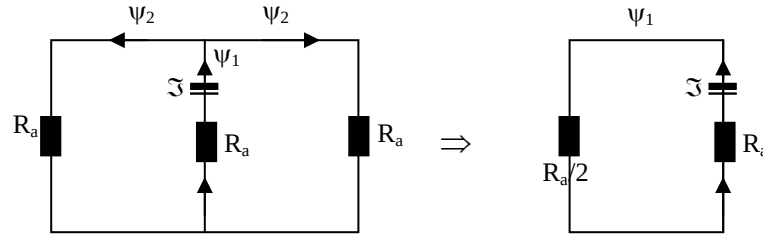
$$= \underline{\underline{37 \text{ mN}}}$$

(b) If $\mu_t \rightarrow \infty$, $R_t = 0$, $\psi = \frac{\mathfrak{F}}{R_a} = \frac{150}{3.183 \times 10^7}$

$$F_2 = I_2 dl_2 \bullet B_1 = I_2 dl_2 \frac{\psi_1}{S} = \frac{2 \times 10^{-3} \times 5 \times 10^{-3} \times 150}{3.183 \times 10^7 \times 25 \times 10^{-6}}$$

$$F_2 = \underline{\underline{1.885 \mu\text{N}}}$$

Prob. 8.60



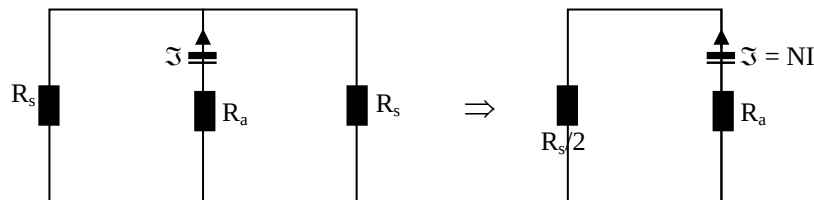
$$\psi_1 = 2\psi_2, \psi_1 = \frac{\mathfrak{F}}{3/2 R_a} = \frac{2\mathfrak{F}}{3R_a} \rightarrow \psi_2 = \frac{\mathfrak{F}}{3R_a}$$

$$\mathfrak{F} = 2 \left(\frac{\psi_2^2}{2\mu_o S} \right) + \frac{\psi_1}{2\mu_o S} = \frac{3\psi_1^2}{4\mu_o S} = \frac{\mathfrak{F}^2}{3R_a^2 \mu_o S}$$

$$= \frac{\mu_o S \mathfrak{F}^2}{3l_a^2} = \frac{4\pi \times 10^{-7} \times 200 \times 10^{-4} \times 9 \times 10^6}{3 \times 10^{-6}}$$

$$= 24\pi \times 10^3 = mg \rightarrow m = \frac{24\pi \times 10^3}{9.8} = \underline{\underline{7694 \text{ kg}}}$$

Prob. 8.61



Since $\mu \rightarrow \infty$ for the core (see Figure), $R_c = 0$.

$$\mathfrak{F} = NI = \psi \left(R_a + \frac{R_s}{2} \right) = \frac{\psi(a/2 + x)}{\mu_o S}$$

$$= \frac{\psi(2x+a)}{2\mu_o S}$$

$$\mathfrak{J} = \frac{B^2 S}{2\mu_o} = \psi^2 \frac{1}{2\mu_o S} = \frac{1}{2\mu_o S} \bullet \frac{N^2 I^2 4\mu_o^2 S^2}{(a+2x)^2}$$

$$= \frac{2N^2 I^2 \mu_o S}{(a+2x)^2}$$

$\mathbf{F} = -F\mathbf{a}_x$ since the force is attractive, i.e.

$$\mathbf{F} = \frac{-2N^2 I^2 \mu_o S \mathbf{a}_x}{(a+2x)^2}$$

CHAPTER 9

P.E. 9.1

$$(a) \quad V_{emf} = \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} = uBl = 8(0.5)(0.1) = \underline{\underline{0.4}} \text{ V}$$

$$(b) \quad I = \frac{V_{emf}}{R} = \frac{0.4}{20} = \underline{\underline{20}} \text{ mA}$$

$$(c) \quad \mathbf{F}_m = I\mathbf{l} \times \mathbf{B} = 0.02(-0.1\mathbf{a}_y \times 0.5\mathbf{a}_z) = \underline{\underline{-\mathbf{a}_x}} \text{ mN}$$

$$(d) \quad P = FU = I^2 R = 8 \text{ mW}$$

$$\text{or} \quad P = \frac{V_{emf}^2}{R} = \frac{(0.4)^2}{20} = \underline{\underline{8}} \text{ mW}$$

P.E. 9.2

$$(a) \quad V_{emf} = \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

$$\text{where } \mathbf{B} = B_o \mathbf{a}_y = B_o (\sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi), \quad B_o = 0.05 \text{ Wb/m}^2$$

$$(\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} = -\rho \omega B_o \sin \phi dz = -0.2\pi \sin\left(\omega t + \frac{\pi}{2}\right) dz$$

$$V_{emf} = \int_0^{0.03} (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} = -6\pi \cos(100\pi t) \text{ mV}$$

At $t = 1\text{ms}$,

$$V_{emf} = -6\pi \cos 0.1\pi = \underline{\underline{-17.93}} \text{ mV}$$

$$i = \frac{V_{emf}}{R} = -60\pi \cos(100\pi t) \text{ mA}$$

$$\text{At } t = 3\text{ms}, \quad i = -60\pi \cos 0.3\pi = \underline{\underline{-110.8}} \text{ mA}$$

(b) Method 1:

$$\Psi = \int \mathbf{B} \cdot d\mathbf{S} = \int B_o t (\cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi) \cdot d\rho dz \mathbf{a}_\phi = - \int_0^{\rho_o} \int_0^{z_o} B_o t \sin \phi d\rho dz = -B_o \rho_o z_o t \sin \phi$$

where $B_o = 0.02$, $\rho_o = 0.04$, $z_o = 0.03$

$$\phi = \omega t + \frac{\pi}{2}$$

$$\Psi = -B_o \rho_o z_o t \cos \omega t$$

$$V_{emf} = -\frac{\partial \Psi}{\partial t} = B_o \rho_o z_o \cos \omega t - B_o \rho_o z_o t \omega \sin \omega t$$

$$= (0.02)(0.04)(0.03)[\cos \omega t - \omega t \sin \omega t]$$

$$= 24[\cos \omega t - \omega t \sin \omega t] \mu V$$

Method 2:

$$V_{emf} = - \int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} + \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

$$\mathbf{B} = B_o t \mathbf{a}_x = B_o t (\cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi), \phi = \omega t + \pi/2$$

$$\frac{\partial \mathbf{B}}{\partial t} = B_o (\cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi)$$

Note that only explicit dependence of $\mathbf{B}$ on time is accounted for, i.e. we make ϕ

= constant because it is transformer (stationary) emf. Thus,

$$V_{emf} = -B_o \int_0^{\rho_o} \int_0^{z_o} (\cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi) \cdot d\rho d\mathbf{z} \mathbf{a}_\phi + \int_{z_o}^0 -\rho_o \omega B_o t \cos \phi dz$$

$$= B_o \rho_o z_o (\sin \phi - \omega t \cos \phi), \phi = \omega t + \pi/2$$

$$= B_o \rho_o z_o (\cos \omega t - \omega t \sin \omega t) \text{ as obtained earlier.}$$

At $t = 1\text{ms}$,

$$V_{emf} = 24[\cos 18^\circ - 100\pi \times 10^{-3} \sin 18^\circ] \mu V$$

$$= \underline{\underline{20.5 \mu V}}$$

At $t = 3\text{ms}$,

$$i = 240[\cos 54^\circ - .03\pi \sin 54^\circ] \text{mA}$$

$$= \underline{\underline{-41.93 \text{ mA}}}$$

P.E. 9.3

$$V_1 = -N_1 \frac{d\psi}{dt}, V_2 = -N_2 \frac{d\psi}{dt}$$

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} \rightarrow V_2 = \frac{N_2}{N_1} V_1 = \frac{300 \times 120}{500} = \underline{\underline{72V}}$$

P.E. 9.4

$$(a) \quad \mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} = \underline{\underline{-20\omega \epsilon_o \sin(\omega t - 50x) \mathbf{a}_y \text{ A/m}^2}}$$

$$(b) \quad \nabla \times \mathbf{H} = \mathbf{J}_d \rightarrow -\frac{\partial H_z}{\partial x} \mathbf{a}_y = -20\omega\epsilon_o \sin(\omega t - 50x) \mathbf{a}_y$$

$$\text{or } \mathbf{H} = \frac{20\omega\epsilon_o}{50} \cos(\omega t - 50x) \mathbf{a}_z$$

$$= \underline{\underline{0.4\omega\epsilon_o \cos(\omega t - 50x) \mathbf{a}_z}} \text{ A/m}$$

$$(c) \quad \nabla \times \mathbf{E} = -\mu_o \frac{\partial \mathbf{H}}{\partial t} \rightarrow \frac{\partial E_y}{\partial x} \mathbf{a}_z = 0.4\mu_o \omega^2 \epsilon_o \sin(\omega t - 50x) \mathbf{a}_z$$

$$1000 = 0.4\mu_o \epsilon_o \omega^2 = 0.4 \frac{\omega^2}{c^2}$$

$$\text{or } \omega = \underline{\underline{1.5 \times 10^{10} \text{ rad/s}}}$$

P.E. 9.5

$$(a) \quad j^3 \left(\frac{1+j}{2-j} \right)^2 = -j \left[\frac{\sqrt{2} \angle 45^\circ}{\sqrt{5} \angle -26.56^\circ} \right]^2 = -j \left(\frac{2}{5} \angle 143.13^\circ \right)$$

$$= \underline{\underline{0.24 + j0.32}}$$

$$(b) \quad 6 \angle 30^\circ + j5 - 3 + e^{j45^\circ} = 5.196 + j3 + j5 - 3 + 0.7071(1 + j)$$

$$= \underline{\underline{2.903 + j8.707}}$$

P.E. 9.6

$$\mathbf{P} = 2 \sin(10t + x - \pi/4) \mathbf{a}_y = 2 \cos(10t + x - \pi/4 - \pi/2) \mathbf{a}_y, w = 10$$

$$= R_e \left(2e^{j(x-3\pi/4)} \mathbf{a}_y e^{j\omega t} \right) = R_e \left(\mathbf{P}_s e^{j\omega t} \right)$$

$$\text{i.e. } \underline{\underline{\mathbf{P}_s = 2e^{j(x-3\pi/4)} \mathbf{a}_y}}$$

$$\mathbf{Q} = R_e \left(\mathbf{Q}_s e^{j\omega t} \right) = R_e \left(e^{j(x+\omega t)} (\mathbf{a}_x - \mathbf{a}_z) \right) \sin \pi y$$

$$= \underline{\underline{\sin \pi y \cos(\omega t + x) (\mathbf{a}_x - \mathbf{a}_z)}}$$

P.E. 9.7

$$-\mu \frac{\partial \mathbf{H}}{\partial t} = \nabla \times \mathbf{E} = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (E_\phi \sin \theta) \mathbf{a}_r - \frac{1}{r} \frac{\partial}{\partial r} (r E_\phi) \mathbf{a}_\theta$$

$$\begin{aligned}
&= \frac{2 \cos \theta}{r^2} \cos(\omega t - \beta r) \mathbf{a}_r - \frac{\beta}{r} \sin \theta \sin(\omega t - \beta r) \mathbf{a}_\theta \\
\mathbf{H} &= -\frac{2 \cos \theta}{\mu \omega r^2} \sin(\omega t - \beta r) \mathbf{a}_r - \frac{\beta}{\mu \omega r} \sin \theta \cos(\omega t - \beta r) \mathbf{a}_\theta \\
\beta &= \frac{\omega}{c} = \frac{6 \times 10^7}{3 \times 10^8} = \underline{\underline{0.2}} \text{ rad/m} \\
\mathbf{H} &= -\frac{1}{12\pi r^2} \cos \theta \sin(6 \times 10^7 - 0.2r) \mathbf{a}_r - \frac{1}{120\pi r} \sin \theta \cos(6 \times 10^7 - 0.2r) \mathbf{a}_\theta
\end{aligned}$$

P.E. 9.8

$$\begin{aligned}
\omega &= \frac{3}{\sqrt{\mu \epsilon}} = \frac{3c}{\sqrt{\mu_r \epsilon_r}} = \frac{9 \times 10^8}{\sqrt{10}} = \underline{\underline{2.846 \times 10^8}} \text{ rad/s} \\
\mathbf{E} &= \frac{1}{\epsilon} \int \nabla \times \mathbf{H} dt = -\frac{6}{\omega \epsilon} \cos(\omega t - 3y) \mathbf{a}_x \\
&= \frac{-6}{\frac{9 \times 10^8}{\sqrt{10}} \cdot \frac{10^{-9}}{36\pi}} \cos(\omega t - 3y) \mathbf{a}_x \\
\mathbf{E} &= \underline{\underline{-476.86 \cos(2.846 \times 10^8 t - 3y) \mathbf{a}_x \text{ V/m}}}
\end{aligned}$$

Prob. 9.1

$$\begin{aligned}
V &= -\frac{\partial \psi}{\partial t} = -\frac{\partial}{\partial t} \int \mathbf{B} \cdot d\mathbf{S} = -\frac{\partial \mathbf{B}}{\partial t} \cdot \mathbf{S} \\
&= 3770 \sin 377t \times \pi (0.2)^2 \times 10^{-3} \\
&= \underline{\underline{0.4738 \sin 377t \text{ V}}}
\end{aligned}$$

Prob. 9.2

$$V_{emf} = -\frac{\partial \psi}{\partial t}, \quad \psi = \iint \mathbf{B} \cdot d\mathbf{S} = BS$$

$$V_{emf} = -\frac{\partial B}{\partial t} S$$

$$i(t) = \frac{V_{emf}}{R} = \frac{(4 \times 20 \sin 20t)(2 \times 10^{-4})}{20 + 30} = \frac{160}{50} \sin 20t(10^{-4})$$

$$\underline{\underline{i(t) = 0.32 \sin 20t \text{ mA}}}$$

Prob.9.3

$$\psi = \mathbf{B} \cdot \mathbf{S} = (0.2)^2 \pi (40 \times 10^{-3} \sin 10^4 t)$$

$$V = -\frac{\partial \psi}{\partial t} = -16\pi \cos 10^4 t$$

$$i = \frac{V}{R} = \frac{16\pi}{4} \cos 10^4 t$$

$$\underline{\underline{= -12.57 \cos 10^4 t \text{ A}}}$$

Prob.9.4

Measuring the induced emf in the clockwise direction,

$$V_{emf} = \oint (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

$$= \int_0^{1.2} (5\mathbf{a}_x \times 0.2\mathbf{a}_z) \cdot dy\mathbf{a}_y + \int_{1.2}^0 (15\mathbf{a}_x \times 0.2\mathbf{a}_z) \cdot dy\mathbf{a}_x$$

$$= -\int_0^{1.2} (1) dy - \int_{1.2}^0 (3) dy$$

$$= -1.2 + 1.2 \times 3 = -1.2 + 3.6$$

$$\underline{\underline{= 2.4 \text{ V}}}$$

Prob. 9.5

$$V_{emf} = \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} = \int_{y=0}^{1.6} (2\mathbf{a}_x \times 10 \cos \beta y \mathbf{a}_z) \cdot dy\mathbf{a}_y = - \int_{y=0}^{1.6} 20 \cos \beta y dy$$

$$= - \left. \frac{20 \sin \beta y}{\beta} \right|_0^{1.6} = - \frac{20 \sin 1.6\beta}{\beta}$$

$$\underline{\underline{\hspace{1.5cm}}}$$

Prob. 9.6

$$\mathbf{B} = \frac{\mu_o I}{2\pi y} (-\mathbf{a}_x)$$

$$\psi = \int \mathbf{B} \cdot d\mathbf{S} = \frac{\mu_o I}{2\pi} \int_{z=0}^a \int_{y=\rho}^{\rho+a} \frac{dz dy}{y} = \frac{\mu_o I a}{2\pi} \ln \frac{\rho+a}{\rho}$$

$$\begin{aligned} V_{emf} &= -\frac{\partial \psi}{\partial t} = -\frac{\partial \psi}{\partial \rho} \bullet \frac{\partial \rho}{\partial t} = -\frac{\mu_o I a}{2\pi} u_o \frac{d}{d\rho} [\ln(\rho+a) - \ln \rho] \\ &= -\frac{\mu_o I a}{2\pi} u_o \left[\frac{1}{\rho+a} - \frac{1}{\rho} \right] = \frac{\mu_o a^2 I u_o}{2\pi \rho(\rho+a)} \end{aligned}$$

where $\rho = \rho_o + u_o t$

Prob. 9.7

$$\begin{aligned} V_{emf} &= \int_{\rho}^{\rho+a} 3\mathbf{a}_z \times \frac{\mu_o I}{2\pi \rho} \mathbf{a}_{\phi} \bullet d\rho \mathbf{a}_{\rho} = -\frac{3\mu_o I}{2\pi} \ln \frac{\rho+a}{\rho} \\ &= -\frac{4\pi \times 10^{-7}}{2\pi} \times 15 \times 3 \ln \frac{60}{20} = -9.888 \mu V \end{aligned}$$

Thus the induced emf = 9.888 μV, point A at higher potential.

Prob. 9.8

$$V = \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

$$\mathbf{u} = \rho \omega \mathbf{a}_{\phi}, \quad \mathbf{B} = B_o \mathbf{a}_z$$

$$\begin{aligned} V &= \int_{\rho=0}^{\ell} \rho \omega B_o d\rho = \frac{1}{2} \omega B_o \rho^2 \Big|_0^{\ell} = \frac{\omega B_o \ell^2}{2} \\ &= \frac{30}{2} \times 60 \times 10^{-3} (8 \times 10^{-2})^2 = \underline{\underline{5.76 \text{ mV}}} \end{aligned}$$

Prob. 9.9

$$\begin{aligned} V_{emf} &= -N \frac{\partial \psi}{\partial t} = -N \frac{\partial}{\partial t} \int \mathbf{B} \cdot d\mathbf{S} = -NB \frac{dS}{dt} \\ &= -NB \ell \frac{d}{dt} (\rho \phi) = -NB \ell \rho \frac{d\phi}{dt} = -NB \ell \rho \omega \\ &= -50(0.2)(30 \times 10^{-4})(60) = \underline{\underline{-1.8 \text{ V}}} \end{aligned}$$

Prob. 9.10Method 1:We assume that the sliding rode is on $-\ell < y < \ell$

$$\ell = x / \sqrt{3} = 5t / \sqrt{3}$$

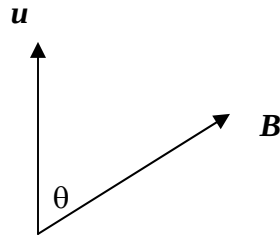
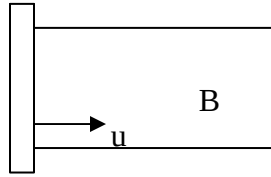
$$V_{emf} = \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} = \int 5\mathbf{a}_x \times 0.6\mathbf{a}_z \cdot dy\mathbf{a}_y = -3x \int_{-\ell}^{\ell} dy = -3x(2\ell) = -6 \times \left(\frac{25t^2}{\sqrt{3}} \right) = \underline{\underline{-86.6025t^2}}$$

Method 2:

The flux linkage is given by

$$\psi = \int_{x=0}^{5t} \int_{y=-x/\sqrt{3}}^{x/\sqrt{3}} 0.6x dx dy = 0.6 \times \frac{2}{\sqrt{3}} \times 125t^3 / 3 = 28,8675t^3$$

$$V_{emf} = -\frac{d\psi}{dt} = \underline{\underline{-86.602t^2}}$$

Prob. 9.11

$$\begin{aligned} V_{emf} &= \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} = uBl \cos \theta \\ &= \left(\frac{120 \times 10^3}{3600} \text{ m/s} \right) (4.3 \times 10^{-5}) (1.6) \cos 65^\circ \\ &= 2.293 \cos 65^\circ = \underline{\underline{0.97 \text{ mV}}} \end{aligned}$$

Prob. 9.12

$$V_{emf} = uB\ell = 410 \times 0.4 \times 10^{-6} \times 36 = \underline{\underline{5.904 \text{ mV}}}$$

Prob. 9.13

$$d\psi = 0.64 - 0.45 = 0.19, dt = 0.02$$

$$V_{emf} = N \frac{d\psi}{dt} = 10 \left(\frac{0.19}{0.02} \right) = 95V$$

$$I = \frac{V_{emf}}{R} = \left(\frac{95}{15} \right) = \underline{\underline{6.33 \text{ A}}}$$

Using Lenz's law, the direction of the induced current is counterclockwise.

Prob. 9.14

$$V = \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}, \text{ where } \mathbf{u} = \rho\omega\mathbf{a}_\phi, \quad \mathbf{B} = B_o\mathbf{a}_z$$

$$V = \int_{\rho_1}^{\rho_2} \rho\omega B_o d\rho = \frac{\omega B_o}{2} (\rho_2^2 - \rho_1^2)$$

$$V = \frac{60 \times 15}{2} \bullet 10^{-3} (100 - 4) \bullet 10^{-4} = \underline{\underline{4.32 \text{ mV}}}$$

Prob. 9.15

$$J_{ds} = j\omega D_s \rightarrow |J_{ds}|_{\max} = \omega \epsilon E_s = \omega \epsilon \frac{V_s}{d}$$

$$= \frac{10^{-9}}{36\pi} \times \frac{2\pi \times 20 \times 10^6 \times 50}{0.2 \times 10^{-3}}$$

$$= \underline{\underline{277.8 \text{ A/m}^2}}$$

$$I_{ds} = J_{ds} \bullet S = \frac{1000}{3.6} \times 2.8 \times 10^{-4} = \underline{\underline{77.78 \text{ mA}}}$$

Prob. 9.16

$$J_c = \sigma E, \quad J_d = \frac{\partial D}{\partial t} = \epsilon \frac{\partial E}{\partial t}$$

$$|J_c| = \sigma |E|, \quad |J_d| = \epsilon \omega |E|$$

$$\text{If } I_c = I_d, \text{ then } |J_c| = |J_d| \longrightarrow \sigma = \epsilon \omega$$

$$\omega = 2\pi f = \frac{\sigma}{\epsilon}$$

$$f = \frac{\sigma}{2\pi\epsilon} = \frac{4}{2\pi \times 9 \times \frac{10^{-9}}{36\pi}} = \underline{\underline{8 \text{ GHz}}}$$

Prob. 9.17 $\frac{J_c}{J_d} = \frac{\sigma E}{\omega \varepsilon E} = \frac{\sigma}{\omega \varepsilon}$

(a) $\frac{\sigma}{\omega \varepsilon} = \frac{2 \times 10^{-3}}{2\pi \times 10^9 \times 81 \times \frac{10^{-9}}{36\pi}} = \underline{\underline{0.444 \times 10^{-3}}}$

(b) $\frac{\sigma}{\omega \varepsilon} = \frac{25}{2\pi \times 10^9 \times 81 \times \frac{10^{-9}}{36\pi}} = \underline{\underline{5.555}}$

(c) $\frac{\sigma}{\omega \varepsilon} = \frac{2 \times 10^{-4}}{2\pi \times 10^9 \times 5 \times \frac{10^{-9}}{36\pi}} = \underline{\underline{7.2 \times 10^{-4}}}$

Prob. 9.18

$$J_c = \sigma E \quad \rightarrow \quad J_{cs} = \sigma E_s$$

$$J_d = \varepsilon \frac{\partial E}{\partial t} \quad \rightarrow \quad J_{ds} = j\omega \varepsilon E_s$$

$$\frac{|J_{cs}|}{|J_{ds}|} = \frac{\sigma}{\omega \varepsilon} = \frac{4}{2\pi \times 10^7 \times 81 \times \frac{10^{-9}}{36\pi}} = \frac{400(18)}{81} = \underline{\underline{88.89}}$$

Prob. 9.19

$$\frac{J_d}{J} = \frac{\omega \varepsilon E}{\sigma E} = \frac{\omega \varepsilon}{\sigma} = 1 \quad \longrightarrow \quad \omega = \frac{\sigma}{\varepsilon} = \frac{10^{-4}}{3 \times \frac{10^{-9}}{36\pi}} = 12\pi \times 10^5$$

$$2\pi f = 12\pi \times 10^5 \quad \longrightarrow \quad f = \underline{\underline{600 \text{ kHz}}}$$

Prob. 9.20

$$J_c = \sigma E = 0.4 \cos(2\pi \times 10^8 t)$$

$$E = \frac{0.4}{\sigma} \cos(2\pi \times 10^8 t)$$

$$\begin{aligned} J_d &= \varepsilon \frac{\partial E}{\partial t} = -\frac{0.4\varepsilon}{\sigma} (2\pi \times 10^8) \sin(2\pi \times 10^8 t) \\ &= -\frac{0.4 \times 4.5 \times 10^{-9}}{10^{-4}} (2\pi \times 10^8) \sin(2\pi \times 10^8 t) \\ &= -100 \sin(2\pi \times 10^8 t) \text{ A/m}^2 \end{aligned}$$

Prob. 9.21

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \mathbf{J} + \mathbf{J}_d$$

Since the region is source-free, $\mathbf{J} = \mathbf{0}$.

$$\begin{aligned} \mathbf{J}_d &= \nabla \times \mathbf{H} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x(z) & 0 & 0 \end{vmatrix} = \frac{\partial H_x}{\partial z} \mathbf{a}_y = -H_o(-\beta) \sin(\omega t - \beta z) \mathbf{a}_y \\ &= \underline{\underline{\beta H_o \sin(\omega t - \beta z) \mathbf{a}_y}} \end{aligned}$$

Prob. 9.22

$$J_d = \frac{\partial D}{\partial t} = \varepsilon \frac{\partial E}{\partial t}$$

$$I = \int_s \mathbf{J}_d \cdot d\mathbf{S} = J_d S = \varepsilon \frac{\partial E}{\partial t} S = \frac{10^{-9}}{36\pi} 25 \times 10^3 \cos 10^3 t (2 \times 5) = 2.2105 \times 10^{-8} \cos 10^3 t$$

$$\underline{\underline{I = 22.1 \cos 10^3 t \text{ nA}}}$$

Prob. 9.23

$$(a) \quad \nabla \cdot \mathbf{E}_s = \rho_s / \varepsilon, \quad \nabla \cdot \mathbf{H}_s = 0$$

$$\underline{\underline{\nabla \times \mathbf{E}_s = j\omega\mu\mathbf{H}_s, \quad \nabla \times \mathbf{H}_s = (\sigma - j\omega\varepsilon)\mathbf{E}_s}}$$

$$(b) \quad \nabla \cdot \mathbf{D} = \rho_v \rightarrow \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \rho_v \quad (1)$$

$$\nabla \cdot \mathbf{B} = 0 \rightarrow \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0 \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \rightarrow \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = -\frac{\partial B_x}{\partial t} \quad (3)$$

$$\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = -\frac{\partial B_y}{\partial t} \quad (4)$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -\frac{\partial B_z}{\partial t} \quad (5)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \rightarrow \frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = J_x + \frac{\partial D_x}{\partial t} \quad (6)$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = J_y + \frac{\partial D_y}{\partial t} \quad (7)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = J_z + \frac{\partial D_z}{\partial t} \quad (8)$$

Prob. 9.24

$$\text{If } \mathbf{J} = 0 = \rho_v, \text{ then } \nabla \cdot \mathbf{B} = 0 \quad (1)$$

$$\nabla \cdot \mathbf{D} = \rho_v \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (3)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \quad (4)$$

Since $\nabla \cdot \nabla \times \mathbf{A} = 0$ for any vector field $\mathbf{A}$,

$$\nabla \cdot \nabla \times \mathbf{E} = -\frac{\partial}{\partial t} \nabla \cdot \mathbf{B} = 0$$

$$\nabla \cdot \nabla \times \mathbf{H} = -\frac{\partial}{\partial t} \nabla \cdot \mathbf{D} = 0$$

showing that (1) and (2) are incorporated in (3) and (4). Thus Maxwell's equations can be reduced to (3) and (4), i.e.

$$\underline{\underline{\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}}}$$

Prob. 9.25

$$\nabla \cdot \mathbf{E} = 0 \longrightarrow (1)$$

$$\nabla \cdot \mathbf{H} = 0 \longrightarrow (2)$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \longrightarrow (3)$$

$$\begin{aligned} \nabla \times \mathbf{E} &= \begin{bmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_y(x, t) & 0 \end{bmatrix} \\ &= \frac{\partial E_y}{\partial x} \mathbf{a}_z = -E_o \sin x \cos t \mathbf{a}_z \end{aligned}$$

$$H = -\frac{1}{\mu} \int \nabla \times \mathbf{E} dt = \frac{E_o}{\mu_o} \sin x \sin t \mathbf{a}_z$$

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} \longrightarrow (4)$$

$$\begin{aligned} \nabla \times \mathbf{H} &= \begin{bmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & H_z(x, t) \end{bmatrix} \\ &= -\frac{\partial H_z}{\partial x} \mathbf{a}_y = -\frac{E_o}{\mu_o} \cos x \sin t \mathbf{a}_y \end{aligned}$$

$$\mathbf{E} = \frac{1}{\epsilon} \int \nabla \times \mathbf{H} dt = \frac{E_o}{\mu_o \epsilon} \cos x \cos t \mathbf{a}_y$$

which is off the given $\mathbf{E}$ by a factor. Thus, Maxwell's equations (1) to (3) are satisfied, but (4) is not. The only way (4) is satisfied is for $\mu_o \epsilon = 1$ which is not true.

Prob. 9.26

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \nabla \times \mathbf{E} = -\frac{\partial}{\partial t} \nabla \times \mathbf{B} = -\mu \frac{\partial}{\partial t} \nabla \times \mathbf{H} = -\mu \frac{\partial \mathbf{J}}{\partial t} - \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

But

$$\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$$

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu \frac{\partial \mathbf{J}}{\partial t} - \mu \varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad \mathbf{J} = \sigma \mathbf{E}$$

In a source-free region, $\nabla \cdot \mathbf{E} = \rho_v / \varepsilon = 0$. Thus,

$$\underline{\underline{\nabla^2 \mathbf{E} = \mu \sigma \frac{\partial \mathbf{E}}{\partial t} + \mu \varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}}}$$

Prob. 9.27

$$\nabla \cdot \mathbf{J} = (0 + 0 + 3z^2) \sin 10^4 t = -\frac{\partial \rho_v}{\partial t}$$

$$\rho_v = -\int \nabla \cdot \mathbf{J} dt = -\int 3z^2 \sin 10^4 t dt = \frac{3z^2}{10^4} \cos 10^4 t + C_o$$

If $\rho_v|_{z=0} = 0$, then $C_o = 0$ and

$$\underline{\underline{\rho_v = 0.3z^2 \cos 10^4 t \text{ mC/m}^3}}$$

Prob. 9.28

$$\nabla \cdot \mathbf{D} = \varepsilon \nabla \cdot \mathbf{E} = \rho_v$$

$$\rho_v = \varepsilon \nabla \cdot \mathbf{E} = \varepsilon_o \frac{\partial E_z}{\partial z} = \underline{\underline{-\varepsilon_o E_o \sin z \cos t}}$$

Prob. 9.29

Let $\omega = 10^8$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \mathbf{0} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \rightarrow \mathbf{E} = \frac{1}{\varepsilon} \int \nabla \times \mathbf{H} dt$$

$$\nabla \times \mathbf{H} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 40 \sin(\omega t + \beta x) & 0 \end{vmatrix} = 40\beta \cos(\omega t + \beta x) \mathbf{a}_z$$

$$\mathbf{E} = \frac{1}{\varepsilon} \int \nabla \times \mathbf{H} dt = \underline{\underline{\frac{40\beta}{\omega \varepsilon_o} \sin(\omega t + \beta x) \mathbf{a}_z}}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

$$\nabla \times \mathbf{E} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & \frac{40\beta}{\omega\epsilon_o} \sin(\omega t + \beta x) \end{vmatrix} = -\frac{40\beta^2}{\omega\epsilon_o} \cos(\omega t + \beta x) \mathbf{a}_y \quad (1)$$

$$-\mu_o \frac{\partial \mathbf{H}}{\partial t} = -40\mu_o \omega \cos(\omega t + \beta x) \mathbf{a}_y \quad (2)$$

Equating (1) and (2) gives

$$40\mu_o \omega = \frac{40\beta^2}{\omega\epsilon_o} \rightarrow \beta^2 = \mu_o \epsilon_o \omega^2$$

$$\beta = \omega \sqrt{\epsilon_o \mu_o} = 10^8 \sqrt{4\pi \times 10^{-7} \times \frac{10^{-9}}{36\pi}} = \frac{1}{3} = \underline{\underline{0.333 \text{ rad/m}}}$$

Prob. 9.30

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} = \epsilon_o \frac{\partial \mathbf{E}}{\partial t} = \frac{50\epsilon_o}{\rho} (-10^8) \sin(10^8 t - kz) \mathbf{a}_\rho = -\frac{4.421 \times 10^{-2}}{\rho} \sin(10^8 t - kz) \mathbf{a}_\rho \text{ A/m}^2$$

$$\nabla \times \mathbf{E} = -\mu_o \frac{\partial \mathbf{H}}{\partial t}$$

$$\nabla \times \mathbf{E} = \frac{\partial E_\rho}{\partial z} \mathbf{a}_\phi = \frac{50k}{\rho} \sin(10^8 t - kz) \mathbf{a}_\phi$$

$$\mathbf{H} = -\frac{1}{\mu_o} \int \nabla \times \mathbf{E} dt = \frac{1}{4\pi \times 10^{-7}} \frac{50k}{10^8 \rho} \cos(10^8 t - kz) \mathbf{a}_\phi$$

$$\mathbf{H} = \frac{2.5k}{2\pi\rho} \cos(10^8 t - kz) \mathbf{a}_\phi \text{ A/m}$$

$$\nabla \times \mathbf{H} = -\frac{\partial H_\phi}{\partial z} \mathbf{a}_\rho = -\frac{2.5k^2}{2\pi\rho} \sin(10^8 t - kz) \mathbf{a}_\rho$$

$$\nabla \times \mathbf{H} = \mathbf{J}_d \rightarrow -\frac{4.421 \times 10^{-2}}{\rho} \sin(10^8 t - kz) \mathbf{a}_\rho = \frac{-2.5k^2}{2\pi\rho} \sin(10^8 t - kz) \mathbf{a}_\rho$$

$$k^2 = \frac{2\pi}{2.5} \times 4.421 \times 10^{-2} \rightarrow \underline{\underline{k = 0.333}}$$

Prob. 9.31

$$\nabla \times \mathbf{E} = -\mu_o \frac{\partial \mathbf{H}}{\partial t} \longrightarrow \mathbf{H} = -\frac{1}{\mu_o} \int \nabla \times \mathbf{E} dt$$

$$\begin{aligned} \nabla \times \mathbf{E} &= \frac{1}{r} \frac{\partial}{\partial r} (r E_\theta) \mathbf{a}_\phi = \frac{1}{r} \frac{\partial}{\partial r} [10 \sin \theta \cos(\omega t - \beta r)] \mathbf{a}_\phi \\ &= \frac{10\beta}{r} \sin \theta \cos(\omega t - \beta r) \mathbf{a}_\phi \end{aligned}$$

$$\begin{aligned} \mathbf{H} &= -\frac{10\beta}{\mu r} \sin \theta \int \sin(\omega t - \beta r) dt \mathbf{a}_\phi \\ &= \frac{10\beta}{\omega \mu_o r} \sin \theta \cos(\omega t - \beta r) \mathbf{a}_\phi \end{aligned}$$

Prob. 9.32

$$(a) \mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} \longrightarrow \mathbf{D} = \int \mathbf{J}_d dt$$

$$\mathbf{D} = \frac{-60 \times 10^{-3}}{109} \cos(10^9 t - \beta z) \mathbf{a}_x = \underline{\underline{-60 \times 10^{-12} \cos(10^9 t - \beta z) \mathbf{a}_x \text{ C/m}^2}}$$

$$\nabla \times \mathbf{E} = \mu \frac{\partial \mathbf{H}}{\partial t} \longrightarrow \nabla \times \frac{\mathbf{D}}{\epsilon} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

$$\begin{aligned} \nabla \times \frac{\mathbf{D}}{\epsilon} &= \frac{1}{\epsilon} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ D_x & 0 & 0 \end{vmatrix} = \frac{1}{\epsilon} (-60)(-1) \times 10^{-12} \sin(10^9 t - \beta z) \mathbf{a}_x \\ &= \frac{60\beta}{\epsilon} \times 10^{-12} \sin(10^9 t - \beta z) \mathbf{a}_y \end{aligned}$$

$$\begin{aligned} \mathbf{H} &= -\frac{1}{\mu} \int \nabla \times \frac{\mathbf{D}}{\epsilon} dt = -\frac{1}{\mu} (-1) \frac{60\beta}{\epsilon} \times \frac{10^{-12}}{10^9} \cos(10^9 t - \beta z) \mathbf{a}_y \\ &= \underline{\underline{\frac{60\beta}{\mu \epsilon} \times 10^{-21} \cos(10^9 t - \beta z) \mathbf{a}_y \text{ A/m}}} \end{aligned}$$

$$(b) \quad \nabla \times \mathbf{H} = \mathbf{J} + \mathbf{J}_\phi = 0 + \mathbf{J}_d$$

$$\mathbf{J}_d = \nabla \times \mathbf{H} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & H_y & 0 \end{vmatrix} = \frac{(-\beta)(-1)60\beta}{\mu\epsilon} \times (10^{-21}) \sin(10^9 t - \beta z) \mathbf{a}_x$$

Equating this with the given $\mathbf{J}_d$

$$60 \times 10^{-3} = \frac{60\beta^2}{\mu\epsilon} \times 10^{-21}$$

$$\beta^2 = \mu\epsilon \times 10^{18} = 2 \times 4\pi \times 10^{-7} \times 10 \times \frac{10^{-9}}{36\pi} = \frac{2000}{9}$$

$$\beta = \underline{\underline{14.907 \text{ rad/m}}}$$

Prob. 9.33

$$\nabla \cdot \mathbf{D} = \rho_v, \quad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla \cdot \nabla \times \mathbf{H} = \nabla \cdot \mathbf{J} + \frac{\partial \nabla \cdot \mathbf{D}}{\partial t} = 0 \quad \rightarrow \quad \nabla \cdot \mathbf{J} + \frac{\partial \rho_v}{\partial t} = 0$$

$$\text{Or} \quad \underline{\underline{\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}}}$$

Prob. 9.34

From Maxwell's equations,

$$\nabla \cdot \mathbf{D} = \rho_v = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla \times \nabla \times \mathbf{E} = -\mu \frac{\partial}{\partial t} \nabla \times \mathbf{H} \quad \rightarrow \quad \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu \frac{\partial}{\partial t} \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right)$$

$$\text{But} \quad \nabla \cdot \mathbf{E} = 0, \quad \mathbf{J} = \sigma \mathbf{E}, \quad \mathbf{D} = \epsilon \mathbf{E}$$

$$-\nabla^2 \mathbf{E} = -\mu \frac{\partial}{\partial t} \left(\sigma \mathbf{E} + \frac{\partial \epsilon \mathbf{E}}{\partial t} \right) = -\mu \sigma \frac{\partial \mathbf{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\underline{\underline{\nabla^2 \mathbf{E} - \mu \sigma \frac{\partial \mathbf{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0}}$$

Prob. 9.35

(a) $\nabla \cdot \mathbf{A} = 0$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & E_z(x, t) \end{vmatrix} = -\frac{\partial E_z(x, t)}{\partial x} \mathbf{a}_y \neq 0$$

Yes, $\mathbf{A}$ is a possible EM field.

(b) $\nabla \cdot \mathbf{B} = 0$

$$\nabla \times \mathbf{B} = \frac{1}{\rho} \frac{\partial}{\partial \rho} [10 \cos(\omega t - 2\rho)] \mathbf{a}_z \neq 0$$

Yes, $\mathbf{B}$ is a possible EM field.

(c) $\nabla \cdot \mathbf{C} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (3\rho^3 \cot \phi \sin \omega t) - \frac{\sin \phi \sin \omega t}{\rho^2} \neq 0$

$$\nabla \times \mathbf{C} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\cos \phi \sin \omega t) \mathbf{a}_z - 3\rho^2 \frac{\partial}{\partial \phi} (\cot \phi \sin \omega t) \mathbf{a}_z \neq 0$$

No, $\mathbf{C}$ cannot be an EM field.

(d) $\nabla \cdot \mathbf{D} = \frac{1}{r^2 \sin \theta} \sin(\omega t - 5r) \frac{\partial}{\partial \theta} (\sin^2 \theta) \neq 0$

$$\nabla \times \mathbf{D} = -\frac{\partial D_\theta}{\partial \phi} \mathbf{a}_r + \frac{1}{r} \frac{\partial}{\partial r} (r D_\theta) \mathbf{a}_\phi = \frac{1}{r} \sin \theta (-5) \sin(\omega t - 5r) \mathbf{a}_\phi \neq 0$$

No, $\mathbf{D}$ cannot be an EM field.

Prob. 9.36

From Maxwell's equations,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (1)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \quad (2)$$

Dotting both sides of (2) with $\vec{E}$ gives:

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) = \mathbf{E} \cdot \mathbf{J} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} \quad (3)$$

But for any arbitrary vectors $\vec{A}$ and $\vec{B}$,

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

Applying this on the left-hand side of (3) by letting $\mathbf{A} \equiv \mathbf{H}$ and $\mathbf{B} \equiv \mathbf{E}$, we get

$$\mathbf{H} \cdot (\nabla \times \mathbf{E}) + \nabla \cdot (\mathbf{H} \times \mathbf{E}) = \mathbf{E} \cdot \mathbf{J} + \frac{1}{2} \frac{\partial}{\partial t} (\mathbf{D} \cdot \mathbf{E}) \quad (4)$$

From (1),

$$\mathbf{H} \cdot (\nabla \times \mathbf{E}) = \mathbf{H} \cdot \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = \frac{1}{2} \frac{\partial}{\partial t} (\mathbf{B} \cdot \mathbf{H})$$

Substituting this in (4) gives:

$$-\frac{1}{2} \frac{\partial}{\partial t} (\mathbf{B} \cdot \mathbf{H}) - \nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{J} \cdot \mathbf{E} + \frac{1}{2} \frac{\partial}{\partial t} (\mathbf{D} \cdot \mathbf{E})$$

Rearranging terms and then taking the volume integral of both sides:

$$\int_v \nabla \cdot (\mathbf{E} \times \mathbf{H}) dv = -\frac{\partial}{\partial t} \frac{1}{2} \int_v (\mathbf{E} \cdot \mathbf{D} + \mathbf{H} \cdot \mathbf{B}) dv - \int_v \mathbf{J} \cdot \mathbf{E} dv$$

Using the divergence theorem, we get

$$\oint_s (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} = -\frac{\partial W}{\partial t} - \int_v \mathbf{J} \cdot \mathbf{E} dv$$

$$\text{or } \frac{\partial W}{\partial t} = -\oint_s (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} - \int_v \mathbf{E} \cdot \mathbf{J} dv \text{ as required.}$$

Prob. 9.37

$$\nabla \times \mathbf{E} = -\mu_o \frac{\partial \mathbf{H}}{\partial t} \quad \rightarrow \quad \mathbf{H} = -\frac{1}{\mu_o} \int \nabla \times \mathbf{E} dt$$

$$\begin{aligned} \nabla \times \mathbf{E} &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & E_z(x, y) \end{vmatrix} = \frac{\partial E_z}{\partial y} \mathbf{a}_x - \frac{\partial E_z}{\partial x} \mathbf{a}_y \\ &= -\alpha \cos(12\pi x) \cos(10^{11}t - \alpha y) \mathbf{a}_x + 12\pi \sin(12\pi x) \sin(10^{11}t - \alpha y) \mathbf{a}_y \end{aligned}$$

$$\begin{aligned} \mathbf{H} &= -\frac{1}{\mu_o} \int \nabla \times \mathbf{E} dt \\ &= -\frac{1}{\mu_o} \left[-\alpha \cos(12\pi x) \int \cos(10^{11}t - \alpha y) dt \mathbf{a}_x + 12\pi \sin(12\pi x) \int \sin(10^{11}t - \alpha y) dt \mathbf{a}_y \right] \\ &= \frac{\alpha}{\mu_o 10^{11}} \cos(12\pi x) \sin(10^{11}t - \alpha y) \mathbf{a}_x + \frac{12\pi}{\mu_o 10^{10}} \sin(12\pi x) \cos(10^{11}t - \alpha y) \mathbf{a}_y \end{aligned}$$

$$\text{But } \nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} = \epsilon_o \frac{\partial \mathbf{E}}{\partial t}$$

$$\epsilon_o \frac{\partial \mathbf{E}}{\partial t} = \epsilon_o (10^{11}) \cos(12\pi x) \cos(10^{11}t - \alpha t) \mathbf{a}_z \quad (1)$$

$$\begin{aligned} \nabla \times \mathbf{H} &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x(x, y) & H_y(x, y) & 0 \end{vmatrix} = \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \mathbf{a}_z \\ &= \left[\frac{(12\pi)^2}{\mu_o 10^{11}} \cos(12\pi x) \cos(10^{11}t - \alpha y) + \frac{\alpha^2}{\mu_o 10^{11}} \cos(12\pi x) \cos(10^{11}t - \alpha y) \right] \mathbf{a}_z \quad (2) \end{aligned}$$

Equating (1) and (2),

$$\epsilon_o (10^{11}) = \frac{(12\pi)^2}{\mu_o 10^{11}} + \frac{\alpha^2}{\mu_o 10^{11}} \quad \rightarrow \quad \mu_o \epsilon_o (10^{22}) = (12\pi)^2 + \alpha^2$$

$$4\pi \times 10^{-7} \times \frac{10^{-9}}{36\pi} (10^{22}) = \frac{10^6}{9} = 144\pi^2 + \alpha^2 \quad \rightarrow \quad \underline{\underline{\alpha = 331.2}}$$

$$\frac{\alpha}{\mu_o 10^{11}} = \frac{331.2 \times 10^{-11}}{4\pi \times 10^{-7}} = 2.636 \times 10^{-3}, \quad \frac{12\pi}{\mu_o 10^{11}} = \frac{12\pi \times 10^{-11}}{4\pi \times 10^{-7}} = 3 \times 10^{-3}$$

Thus,

$$\underline{\underline{\mathbf{H} = 2.636 \cos(12\pi x) \sin(10^{11}t - \alpha y) \mathbf{a}_x + 3 \sin(12\pi x) \cos(10^{11}t - \alpha y) \mathbf{a}_y \text{ mA/m}}}$$

Prob. 9.38

$$\nabla \times \mathbf{E} = -\mu_o \frac{\partial \mathbf{H}}{\partial t} \quad \rightarrow \quad \mathbf{H} = -\frac{1}{\mu_o} \int \nabla \times \mathbf{E} dt$$

$$\nabla \times \mathbf{E} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x(z) & 0 & 0 \end{vmatrix} = \frac{\partial E_x}{\partial z} \mathbf{a}_y = -\beta E_o \cos(1200\pi t - \beta z) \mathbf{a}_y \quad (1)$$

$$-\mu_o \frac{\partial \mathbf{H}}{\partial t} = -\frac{\mu_o E_o}{\eta} (1200\pi) \cos(1200\pi t - \beta z) \mathbf{a}_y \quad (2)$$

Setting (1) and (2) equal,

$$\beta E_o = \frac{\mu_o E_o}{\eta} (1200\pi) \quad \rightarrow \quad \beta = \frac{1200\pi \mu_o}{\eta} \quad (3)$$

$$\text{But } \nabla \times \mathbf{H} = \varepsilon_o \frac{\partial \mathbf{E}}{\partial t}$$

$$\nabla \times \mathbf{H} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & H_y(z) & 0 \end{vmatrix} = -\frac{\partial H_y}{\partial z} \mathbf{a}_x = \frac{\beta E_o}{\eta} \cos(1200\pi t - \beta z) \mathbf{a}_x \quad (4)$$

$$\varepsilon_o \frac{\partial \mathbf{E}}{\partial t} = \varepsilon_o (1200\pi) E_o \cos(1200\pi t - \beta z) \mathbf{a}_x \quad (5)$$

Setting (4) and (5) equal,

$$\beta = \varepsilon_o (1200\pi) \eta \quad (6)$$

From (3) and (6),

$$\frac{1200\pi \mu_o}{\eta} = \varepsilon_o (1200\pi) \eta \quad \rightarrow \quad \eta^2 = \frac{\mu_o}{\varepsilon_o}$$

$$\eta = \sqrt{\frac{\mu_o}{\varepsilon_o}} = \sqrt{\frac{4\pi \times 10^{-7}}{\frac{10^{-9}}{36\pi}}} = 120\pi = \underline{\underline{377}}$$

From (3),

$$\beta = \frac{1200\pi \mu_o}{\eta} = \frac{1200\pi \times 4\pi \times 10^{-7}}{120\pi} = 40\pi \times 10^{-7} = \underline{\underline{1.257 \times 10^{-5} \text{ rad/m}}}$$

Prob. 9.39 Using Maxwell's equations,

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \quad (\sigma = 0) \quad \longrightarrow \quad \mathbf{E} = \frac{1}{\varepsilon} \int \nabla \times \mathbf{H} dt$$

But

$$\nabla \times \mathbf{H} = -\frac{1}{r \sin \theta} \frac{\partial H_\theta}{\partial \phi} \mathbf{a}_r + \frac{1}{r} \frac{\partial}{\partial r} (r H_\theta) \mathbf{a}_\phi = \frac{12 \sin \theta}{r} \beta \sin(2\pi \times 10^8 t - \beta r) \mathbf{a}_\phi$$

$$\begin{aligned} \mathbf{E} &= \frac{12 \sin \theta}{\epsilon_o} \beta \int \sin(2\pi \times 10^8 t - \beta r) dt \mathbf{a}_\phi \\ &= -\frac{12 \sin \theta}{\omega \epsilon_o r} \beta \cos(\omega t - \beta r) \mathbf{a}_\phi, \quad \omega = 2\pi \times 10^8 \end{aligned}$$

Prob. 9.40

With the given $\mathbf{A}$, we need to prove that

$$\nabla^2 \mathbf{A} = \mu \epsilon \frac{\partial^2 \mathbf{A}}{\partial t^2}$$

$$\nabla^2 \mathbf{A} = \mu \epsilon (j\omega)(j\omega) \mathbf{A} = -\omega^2 \mu \epsilon \mathbf{A}$$

Let $\beta^2 = \omega^2 \mu \epsilon$, then $\nabla^2 \mathbf{A} = -\beta^2 \mathbf{A}$ is to be proved. We recognize that

$$\mathbf{A} = \frac{\mu_o}{4\pi r} e^{j\omega t} e^{-j\beta r} \mathbf{a}_z$$

$$\text{Assume } \varphi = \frac{e^{-j\beta r}}{r}, \quad \mathbf{A} = \frac{\mu_o}{4\pi} e^{j\omega t} \varphi \mathbf{a}_z$$

$$\begin{aligned} \nabla^2 \varphi &= \frac{1}{r^2 \sin \theta} \left[\frac{\partial}{\partial r} (r^2 \sin \theta \frac{\partial \varphi}{\partial r}) \right] = \frac{1}{r^2} \left[\frac{\partial}{\partial r} (r^2) \left(\frac{-j\beta}{r} - \frac{1}{r^2} \right) e^{-j\beta r} \right] \\ &= \frac{1}{r^2} (-\beta^2 r + j\beta - j\beta) e^{-j\beta r} = -\beta^2 \frac{e^{-j\beta r}}{r} = -\beta^2 \varphi \end{aligned}$$

Therefore, $\nabla^2 \mathbf{A} = -\beta^2 \mathbf{A}$

We can find V using Lorentz gauge.

$$\begin{aligned} V &= \frac{-1}{\mu_o \epsilon_o} \int \nabla \cdot \mathbf{A} dt = \frac{-1}{j\omega \mu_o \epsilon_o} \nabla \cdot \mathbf{A} \\ &= \frac{-1}{j\omega \mu_o \epsilon_o} \frac{\partial}{\partial r} \left(\frac{\mu_o}{4\pi r} e^{-j\beta r} e^{j\omega t} \right) = \frac{-1}{j\omega \epsilon_o (4\pi)} \left(\frac{-j\beta}{r} - \frac{1}{r^2} \right) e^{-j\beta r} e^{j\omega t} \cos \theta \\ V &= \frac{\cos \theta}{j4\pi \omega \epsilon_o r} \left(j\beta + \frac{1}{r} \right) e^{j(\omega t - \beta r)} \end{aligned}$$

Prob. 9.41

Take the curl of both sides of the equation.

$$\nabla \times \mathbf{E} = -\nabla \times \nabla V - \frac{\partial}{\partial t} \nabla \times \mathbf{A}$$

But $\nabla \times \nabla V = \mathbf{0}$ and $\mathbf{B} = \nabla \times \mathbf{A}$. Hence,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

which is Faraday's law.

Prob. 9.42

$$(a) \quad \nabla \cdot \mathbf{A} = \frac{\partial A_z}{\partial z} = \frac{x}{c}, \quad \frac{\partial V}{\partial t} = -xc, \quad -\mu_o \epsilon_o \frac{\partial V}{\partial t} = \frac{x}{c^2} c = \frac{x}{c}$$

Hence, $\nabla \cdot \mathbf{A} = -\mu_o \epsilon_o \frac{\partial V}{\partial t}$

$$(b) \quad \mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} = -\left(\frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial z} \mathbf{a}_z \right) + x \mathbf{a}_z = -(z \mathbf{a}_x + x \mathbf{a}_z) + x \mathbf{a}_z$$

$$\underline{\underline{\mathbf{E} = -z \mathbf{a}_x}}$$

Prob. 9.43

$$\nabla^2 \mathbf{A} = \mathbf{0} = -\mu \epsilon \frac{\partial V}{\partial t} \longrightarrow \underline{\underline{V = \text{constant}}}$$

$$(a) \quad \mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} = \mathbf{0} - A_o \omega \cos(\omega t - \beta z) \mathbf{a}_x$$

$$\underline{\underline{= -A_o \omega \cos(\omega t - \beta z) \mathbf{a}_x}}$$

(b) Using Maxwell's equations, we can show that

$$\underline{\underline{\beta = \omega \sqrt{\mu_o \epsilon_o}}}$$

Prob. 9.44

$$(a) \quad z = 4 \angle 30^\circ - 10 \angle 50^\circ = 3.464 + 2j - 6.427 - j7.66 = -2.963 - j5.66$$

$$= 6.389 \angle -117.64^\circ$$

$$z^{1/2} = \underline{\underline{2.5277 \angle -58.82^\circ}}$$

$$(b) \quad \frac{1 + j2}{6 - j8 - 7 \angle 15^\circ} = \frac{2.236 \angle 63.43^\circ}{6 - j8 - 7.761 - j1.812} = \frac{2.236 \angle 63.43^\circ}{9.841 \angle 265.57^\circ}$$

$$= \underline{\underline{0.2272 \angle -202.1^\circ}}$$

$$(c) \quad z = \frac{(5 \angle 53.13^\circ)^2}{12 - j7 - 6 - j10} = \frac{25 \angle 106.26^\circ}{18.028 \angle -70.56^\circ}$$

$$= \underline{\underline{1.387 \angle 176.8^\circ}}$$

$$(d) \quad \frac{1.897 \angle -100^\circ}{(576 \angle 90^\circ)(9.434 \angle -122^\circ)} = \underline{\underline{0.0349 \angle -68^\circ}}$$

Prob. 9.45

$$(a) \quad \mathbf{H} = \text{Re}[\mathbf{H}_s e^{j\omega t}], \quad \omega = 10^6$$

$$\mathbf{H} = \text{Re}[-10e^{j(\omega t + \pi/3)} \mathbf{a}_x] \quad \rightarrow \quad \underline{\underline{\mathbf{H}_s = -10e^{j\pi/3} \mathbf{a}_x}}$$

$$(b) \quad \underline{\underline{\mathbf{E}_s = 4 \cos(4y) e^{-j2x} \mathbf{a}_z}}$$

$$(c) \quad \sin A = \cos(A - 90^\circ)$$

$$\mathbf{D} = 5 \cos(\omega t - \pi/2 + \pi/3) \mathbf{a}_x - 8 \cos(\omega t - \pi/4) \mathbf{a}_y$$

$$\underline{\underline{\mathbf{D}_s = 5e^{-j\pi/6} \mathbf{a}_x - 8e^{-j\pi/4} \mathbf{a}_y}}}$$

Prob. 9.46

$$(a) \quad \mathbf{A}_s = 10e^{j\pi/2} \mathbf{a}_x + 20e^{-j\pi/2} \mathbf{a}_y$$

$$\mathbf{A} = \text{Re}[\mathbf{A}_s e^{j\omega t}] = \text{Re}[10e^{j(\omega t + \pi/2)} \mathbf{a}_x + 20e^{j(\omega t - \pi/2)} \mathbf{a}_y]$$

$$= 10 \cos(\omega t + \pi/2) \mathbf{a}_x + 20 \cos(\omega t - \pi/2) \mathbf{a}_y$$

$$\underline{\underline{\mathbf{A} = -10 \sin \omega t \mathbf{a}_x + 20 \sin \omega t \mathbf{a}_y}}}$$

$$(b)$$

$$\mathbf{B} = \text{Re}[\mathbf{B}_s e^{j\omega t}] = \text{Re}[4e^{j(\omega t - 2x + \pi/2)} \mathbf{a}_x + 6e^{j(\omega t + 2x)} \mathbf{a}_y] = 4 \cos(\omega t - 2x + \pi/2) \mathbf{a}_x + 6 \cos(\omega t + 2x) \mathbf{a}_y$$

$$= -4 \sin(\omega t - 2x) \mathbf{a}_x + 6 \sin(\omega t + 2x) \mathbf{a}_y$$

$$(c) \quad \mathbf{C}_s = 2e^{j\pi/2} e^{-10z} e^{-j\pi/4} \mathbf{a}_z$$

$$\mathbf{C} = \text{Re}[\mathbf{C}_s e^{j\omega t}] = 2 \text{Re}[e^{j(\omega t + \pi/2 - \pi/4)} e^{-20z} \mathbf{a}_z] = 2e^{-20z} \cos(\omega t - \pi/4 + \pi/2) \mathbf{a}_z$$

$$\underline{\underline{= -2e^{-20z} \sin(\omega t - \pi/4) \mathbf{a}_z}}}$$

Prob. 9.47

$$(a) \quad \underline{\underline{\mathbf{H}_s = \frac{1}{\rho} e^{-j3z} \mathbf{a}_\phi}}$$

$$(b) \quad \nabla \times \mathbf{H} = \epsilon_o \frac{\partial \mathbf{E}}{\partial t} \rightarrow \mathbf{E} = \frac{1}{\epsilon_o} \int \nabla \times \mathbf{H} dt$$

$$\text{But } \nabla \times \mathbf{H} = -\frac{\partial H_\phi}{\partial z} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho H_\phi) \mathbf{a}_z = -\frac{3}{\rho} \sin(\omega t - 3z) \mathbf{a}_\rho + \mathbf{0}$$

$$\mathbf{E} = -\frac{1}{\epsilon_o} \int \frac{3}{\rho} \sin(\omega t - 3z) \mathbf{a}_\rho dt = \underline{\underline{\frac{3}{\rho \epsilon_o \omega} \cos(\omega t - 3z) \mathbf{a}_\rho}} \text{ V/m}$$

$$(c) \quad \nabla \times \mathbf{E} = \frac{\partial E_\rho}{\partial z} \mathbf{a}_\phi = \frac{9}{\rho \epsilon_o \omega} \sin(\omega t - 3z) \mathbf{a}_\phi$$

$$-\frac{\partial \mathbf{B}}{\partial t} = -\mu_o \frac{\partial \mathbf{H}}{\partial t} = \frac{\mu_o \omega}{\rho} \sin(\omega t - 3z) \mathbf{a}_\phi$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \rightarrow \frac{9}{\rho \epsilon_o \omega} = \frac{\mu_o \omega}{\rho}$$

$$\omega^2 = \frac{9}{\mu_o \epsilon_o} \rightarrow \omega = \frac{3}{\sqrt{\mu_o \epsilon_o}} = 3 \times 3 \times 10^8 = \underline{\underline{9 \times 10^8 \text{ rad/s}}}$$

Prob. 9.48

We can use Maxwell's equations or borrow ideas from chapter 10.

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \eta_o \sqrt{\frac{1}{\epsilon_r}} = \frac{120\pi}{9}$$

$$H_o = \frac{E_o}{\eta} = \frac{10 \times 9}{120\pi} = \underline{\underline{0.2387}}$$

$$\beta = \omega \sqrt{\mu \epsilon} = \frac{\omega}{c} \sqrt{\epsilon_r} = \frac{2\pi \times 10^9}{3 \times 10^8} \sqrt{81} = 60\pi = \underline{\underline{188.5 \text{ rad/m}}}$$

Prob. 9.49

$$\nabla \times \mathbf{H}_s = j\omega\epsilon_o \mathbf{E}_s$$

$$\nabla \times \mathbf{H}_s = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & 12e^{j\alpha x} \end{vmatrix} = -j12\alpha e^{j\alpha x} \mathbf{a}_y$$

$$\mathbf{E}_s = \frac{\nabla \times \mathbf{H}_s}{j\omega\epsilon_o} = -\frac{12\alpha}{\omega\epsilon_o} e^{j\alpha x} \mathbf{a}_y$$

But $\nabla \times \mathbf{E}_s = -j\omega\mu_o \mathbf{H}_s$

$$\nabla \times \mathbf{E}_s = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_{sy}(x) & 0 \end{vmatrix} = \frac{\partial E_{sy}}{\partial x} \mathbf{a}_z = -j \frac{12\alpha^2}{\omega\epsilon_o} e^{j\alpha x} \mathbf{a}_z$$

$$\mathbf{H}_s = -\frac{\nabla \times \mathbf{E}_s}{j\omega\mu_o} = \frac{12\alpha^2}{\omega^2\mu_o\epsilon_o} e^{j\alpha x} \mathbf{a}_z$$

Equating this with the given $\mathbf{H}$,

$$12 = \frac{12\alpha^2}{\omega^2\mu_o\epsilon_o} \rightarrow \alpha^2 = \omega^2\mu_o\epsilon_o \rightarrow \alpha^2 = (10^9)^2 \times 4\pi \times 10^{-7} \times \frac{10^{-9}}{36\pi} = (10^9)^2 \times \frac{10^{-16}}{9}$$

$$\alpha = \frac{10}{3} = \underline{\underline{3.333}}$$

$$\mathbf{E}_s = -\frac{12\alpha}{\omega\epsilon_o} e^{j\alpha x} \mathbf{a}_y = -\frac{12(10/3)}{10^9 \times \frac{10^{-9}}{36\pi}} e^{j10x/3} \mathbf{a}_y = -40(36\pi) e^{j10x/3} \mathbf{a}_y = \underline{\underline{-4.533e^{j3.33x} \text{ kV/m}}}$$

Prob. 9.50

$$(j\omega)^2 Y + 4j\omega Y + Y = 2\angle 0^\circ, \quad \omega = 3$$

$$Y(-\omega^2 + 4j\omega + 1) = 2$$

$$Y = \frac{2}{-\omega^2 + 4j\omega + 1} = \frac{2}{-9 + j12 + 1} = \frac{2}{-8 + j12} = -0.0769 - j0.1154$$

$$= 0.1387 \angle -123.7^\circ$$

$$y(t) = \text{Re}(Y e^{j\omega t}) = \underline{\underline{0.1387 \cos(3t - 123.7^\circ)}}$$

Prob. 9.51 We begin with Maxwell's equations:

$$\nabla \cdot \mathbf{D} = \rho_v / \epsilon = 0, \quad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

We write these in phasor form and in terms of $\mathbf{E}_s$ and $\mathbf{H}_s$ only.

$$\nabla \cdot \mathbf{E}_s = 0 \quad (1)$$

$$\nabla \cdot \mathbf{H}_s = 0 \quad (2)$$

$$\nabla \times \mathbf{E}_s = -j\omega\mu\mathbf{H}_s \quad (3)$$

$$\nabla \times \mathbf{H}_s = (\sigma + j\omega\varepsilon)\mathbf{E}_s \quad (4)$$

Taking the curl of (3),

$$\nabla \times \nabla \times \mathbf{E}_s = -j\omega\mu\nabla \times \mathbf{H}_s$$

$$\nabla(\nabla \cdot \mathbf{E}_s) - \nabla^2 \mathbf{E}_s = -j\omega\mu(\sigma + j\omega\varepsilon)\mathbf{E}_s$$

$$\nabla^2 \mathbf{E}_s + (\omega^2\mu\varepsilon - j\omega\mu\sigma)\mathbf{E}_s = 0 \quad \longrightarrow \quad \underline{\underline{\nabla^2 \mathbf{E}_s + \gamma^2 \mathbf{E}_s = 0}}$$

Similarly, by taking the curl of (4),

$$\nabla \times \nabla \times \mathbf{H}_s = (\sigma + j\omega\varepsilon)\nabla \times \mathbf{E}_s$$

$$\nabla(\nabla \cdot \mathbf{H}_s) - \nabla^2 \mathbf{H}_s = -j\omega\mu(\sigma + j\omega\varepsilon)\mathbf{H}_s$$

$$\nabla^2 \mathbf{H}_s + (\omega^2\mu\varepsilon - j\omega\mu\sigma)\mathbf{H}_s = 0 \quad \longrightarrow \quad \underline{\underline{\nabla^2 \mathbf{H}_s + \gamma^2 \mathbf{H}_s = 0}}$$

CHAPTER 10

P. E. 10.1 (a)

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2 \times 10^8} = \underline{31.42 \text{ ns}},$$

$$\lambda = uT = 3 \times 10^8 \times 31.42 \times 10^{-9} = \underline{9.425 \text{ m}}$$

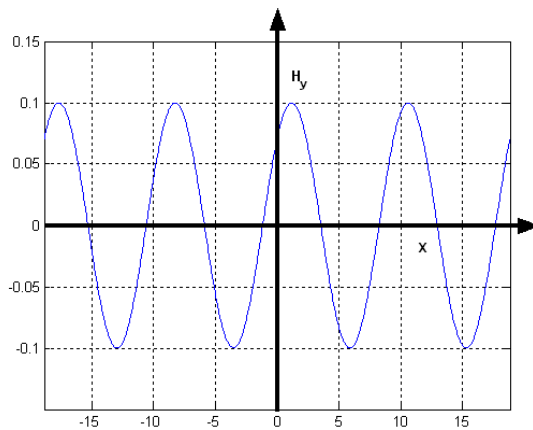
$$k = \beta = 2\pi / \lambda = \underline{0.6667 \text{ rad/m}}$$

$$(b) \quad t_1 = T/8 = \underline{3.927 \text{ ns}}$$

(c)

$$\mathbf{H}(t = t_1) = 0.1 \cos(2 \times 10^8 \frac{\pi}{8 \times 10^8} - 2x/3) \mathbf{a}_y = 0.1 \cos(2x/3 - \pi/4) \mathbf{a}_y$$

as sketched below.

P. E. 10.2 Let $x_o = \sqrt{1 + (\sigma / \omega \epsilon)^2}$, then

$$\alpha = \omega \sqrt{\frac{\mu_o \epsilon_o}{2} \mu_r \epsilon_r (x_o - 1)} = \frac{\omega}{c} \sqrt{\frac{16}{2}} \sqrt{x_o - 1}$$

$$\text{or} \quad \sqrt{x_o - 1} = \frac{\alpha c}{\omega \sqrt{8}} = \frac{1/3 \times 3 \times 10^8}{10^8 \sqrt{8}} = \frac{1}{\sqrt{8}} \longrightarrow x_o = 9/8$$

$$x_o^2 = \frac{81}{64} = 1 + (\sigma / \omega \epsilon)^2 \longrightarrow \frac{\sigma}{\omega \epsilon} = 0.5154$$

$$\tan 2\theta_{\eta} = 0.5154 \longrightarrow \theta_{\eta} = 13.63^{\circ}$$

$$\frac{\beta}{\alpha} = \sqrt{\frac{x_o + 1}{x_o - 1}} = \sqrt{17}$$

$$(a) \quad \beta = \alpha \sqrt{17} = \frac{\sqrt{17}}{3} = \underline{1.374 \text{ rad/m}}$$

$$(b) \quad \frac{\sigma}{\omega \epsilon} = \underline{0.5154}$$

$$(c) \quad |\eta| = \frac{\sqrt{\mu / \epsilon}}{\sqrt{x_o}} = \frac{120\pi \sqrt{2/8}}{\sqrt{9/8}} = 177.72$$

$$\eta = \underline{177.72 \angle 13.63^{\circ} \Omega}$$

$$(d) \quad u = \frac{\omega}{\beta} = \frac{10^8}{1.374} = \underline{7.278 \times 10^7 \text{ m/s}}$$

$$(e) \quad \mathbf{a}_H = \mathbf{a}_k \times \mathbf{a}_E \longrightarrow \mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_H \longrightarrow \mathbf{a}_H = \mathbf{a}_y$$

$$\mathbf{H} = \frac{0.5}{177.5} e^{-z/3} \sin(10^8 t - \beta z - 13.63^{\circ}) \mathbf{a}_y = \underline{2.817 e^{-z/3} \sin(10^8 t - \beta z - 13.63^{\circ}) \mathbf{a}_y \text{ mA/m}}$$

P. E. 10.3 (a) Along -z direction

$$(b) \quad \lambda = \frac{2\pi}{\beta} = 2\pi / 2 = \underline{3.142 \text{ m}}$$

$$f = \frac{\omega}{2\pi} = \frac{10^8}{2\pi} = \underline{15.92 \text{ MHz}}$$

$$\beta = \omega \sqrt{\mu \epsilon} = \omega \sqrt{\mu_o \epsilon_o} \sqrt{\mu_r \epsilon_r} = \frac{\omega}{c} \sqrt{(1) \epsilon_r}$$

$$\text{or } \sqrt{\epsilon_r} = \beta c / \omega = \frac{3 \times 10^8 \times 2}{10^8} = 6 \longrightarrow \underline{\underline{\epsilon_r = 36}}$$

$$(c) \quad \theta_{\eta} = 0, |\eta| = \sqrt{\mu / \epsilon} = \sqrt{\mu_o / \epsilon_o} \sqrt{1 / \epsilon_r} = \frac{120\pi}{6} = 20\pi$$

$$\mathbf{a}_k = \mathbf{a}_E \times \mathbf{a}_H \longrightarrow -\mathbf{a}_z = \mathbf{a}_y \times \mathbf{a}_H \longrightarrow \mathbf{a}_H = \mathbf{a}_x$$

$$\mathbf{H} = \frac{50}{20\pi} \sin(\omega t + \beta z) \mathbf{a}_x = \underline{\underline{795.8 \sin(10^8 t + 2z) \mathbf{a}_x}} \text{ mA/m}$$

P. E. 10.4 (a)

$$\frac{\sigma}{\omega \epsilon} = \frac{10^{-2}}{10^9 \pi \times 4 \times \frac{10^{-9}}{36\pi}} = 0.09$$

$$\alpha \cong \omega \sqrt{\frac{\mu \epsilon}{2} \left[1 + \frac{1}{2} \left(\frac{\sigma}{\omega \epsilon} \right)^2 - 1 \right]} = \frac{\omega}{2c} \sqrt{\mu_r \epsilon_r} \frac{\sigma}{\omega \epsilon} = \frac{10^9 \pi}{2 \times 3 \times 10^8} (2)(0.09) = 0.9425 \text{ Np/m}$$

$$\beta \cong \omega \sqrt{\frac{\mu \epsilon}{2} \left[1 + \frac{1}{2} \left(\frac{\sigma}{\omega \epsilon} \right)^2 + 1 \right]} = \frac{10^9 \pi}{3 \times 10^8} \sqrt{2[2 + 0.5(0.09)^2]} = 20.965 \text{ rad/m}$$

$$\mathbf{E} = 30e^{-0.9425y} \cos(10^9 \pi t - 20.96y + \pi/4) \mathbf{a}_z$$

At $t = 2\text{ns}$, $y = 1\text{m}$,

$$\mathbf{E} = 30e^{-0.9425} \cos(2\pi - 20.96 + \pi/4) \mathbf{a}_z = \underline{\underline{2.844 \mathbf{a}_z}} \text{ V/m}$$

$$(b) \quad \beta y = 10^\circ = \frac{10\pi}{180} \text{ rad}$$

or

$$y = \frac{\pi}{18} \frac{1}{\beta} = \frac{\pi}{18 \times 20.965} = \underline{\underline{8.325 \text{ mm}}}$$

$$(c) \quad 30(0.6) = 30 e^{-\alpha y}$$

$$y = \frac{1}{\alpha} \ln(1/0.6) = \frac{1}{0.9425} \ln \frac{1}{0.6} = \underline{\underline{542 \text{ mm}}}$$

(d)

$$|\eta| \cong \frac{\sqrt{\mu/\epsilon}}{[1 + \frac{1}{4}(0.09)^2]} = \frac{60\pi}{1.002} = 188.11 \Omega$$

$$2\theta_{\eta} = \tan^{-1} 0.09 \longrightarrow \theta_{\eta} = 2.571^{\circ}$$

$$\mathbf{a}_H = \mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_y \times \mathbf{a}_z = \mathbf{a}_x$$

$$\mathbf{H} = \frac{30}{188.11} e^{-0.9425y} \cos(10^9 \pi t - 20.96y + \pi/4 - 2.571^{\circ}) \mathbf{a}_x$$

At $y = 2\text{m}$, $t = 2\text{ns}$,

$$\mathbf{H} = (0.1595)(0.1518) \cos(-34.8963\text{rad}) \mathbf{a}_x = \underline{\underline{-22.83 \mathbf{a}_x \text{ mA/m}}}$$

P. E. 10.5

$$I_s = \int_0^w \int_0^{\infty} J_{xs} dy dz = J_{xs}(0) \int_0^w dy \int_0^{\infty} e^{-z(1+j)/\delta} dz = \frac{J_{xs}(0) w \delta}{1+j}$$

$$\underline{\underline{|I_s| = \frac{J_{xs}(0) w \delta}{\sqrt{2}}}}$$

P. E. 10.6 (a)

$$\frac{R_{ac}}{R_{dc}} = \frac{a}{2\delta} = \frac{a}{2} \sqrt{\pi f \mu \sigma} = \frac{1.3 \times 10^{-3}}{2} \sqrt{\pi \times 10^7 \times 4\pi \times 10^{-7} \times 3.5 \times 10^7} = \underline{\underline{24.16}}$$

(b)

$$\frac{R_{ac}}{R_{dc}} = \frac{1.3 \times 10^{-3}}{2} \sqrt{\pi \times 2 \times 10^9 \times 4\pi \times 10^{-7} \times 3.5 \times 10^7} = \underline{\underline{341.7}}$$

P. E. 10.7

$$E = \text{Re}[E_s e^{j\omega t}] = \text{Re}[E_o e^{j\omega t} e^{-j\beta z} \mathbf{a}_x + E_o e^{-j\pi/2} e^{j\omega t} e^{-j\beta z} \mathbf{a}_y]$$

$$= E_o \cos(\omega t - \beta z) \mathbf{a}_x + E_o \cos(\omega t - \beta z - \pi/2) \mathbf{a}_y$$

$$= E_o \cos(\omega t - \beta z) \mathbf{a}_x + E_o \sin(\omega t - \beta z) \mathbf{a}_y$$

At $z = 0$, $E_x = E_o \cos \omega t$, $E_y = E_o \sin \omega t$

$$\cos^2 \omega t + \sin^2 \omega t = 1 \longrightarrow \left(\frac{E_x}{E_o} \right)^2 + \left(\frac{E_y}{E_o} \right)^2 = 1$$

which describes a circle. Hence the polarization is circular.

P. E. 10.8

$$\mathcal{P}_{\text{ave}} = \frac{1}{2} \eta H_o^2 \mathbf{a}_x$$

(a) Let $f(x, y) = x + y - 1 = 0$

$$\mathbf{a}_n = \frac{\nabla f}{|\nabla f|} = \frac{\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}}, \quad d\mathbf{S} = dS \mathbf{a}_n$$

$$P_t = \int \mathbf{P} \cdot d\mathbf{S} = \mathbf{P} \cdot \mathbf{S} \mathbf{a}_n = \frac{1}{2} \eta H_o^2 \mathbf{a}_x \cdot \frac{\mathbf{a}_x + \mathbf{a}_y}{\sqrt{2}}$$

$$= \frac{1}{2\sqrt{2}} (120\pi)(0.2)^2 (0.1)^2 = \underline{53.31 \text{ mW}}$$

(b) $d\mathbf{S} = dydz \mathbf{a}_x$, $P_t = \int \mathcal{P} \cdot d\mathbf{S} = \frac{1}{2} \eta H_o^2 S$

$$P_t = \frac{1}{2} (120\pi)(0.2)^2 \pi (0.05)^2 = \underline{59.22 \text{ mW}}$$

P. E. 10.9 $\eta_1 = \eta_o = 120\pi, \eta_2 = \sqrt{\frac{\mu}{\epsilon}} = \frac{\eta_o}{2}$

$$\tau = \frac{2\eta_2}{\eta_2 + \eta_1} = 2/3, \Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = -1/3$$

$$E_{ro} = \Gamma E_{io} = -\frac{10}{3}$$

$$\underline{\underline{\mathbf{E}_{rs} = -\frac{10}{3} e^{j\beta_1 z} \mathbf{a}_x \text{ V/m}}}$$

where $\beta_1 = \omega / c = 100\pi / 3$.

$$E_{to} = \tau E_{io} = \frac{20}{3}$$

$$\underline{\underline{\mathbf{E}_{ts} = \frac{20}{3} e^{-j\beta_2 z} \mathbf{a}_x \text{ V/m}}}$$

where $\beta_2 = \omega \sqrt{\epsilon_r} / c = 2\beta_1 = 200\pi / 3$.

P. E. 10.10

$$\alpha_1 = 0, \quad \beta_1 = \frac{\omega}{c} \sqrt{\mu_r \varepsilon_r} = \frac{2\omega}{c} = 5 \longrightarrow \omega = 5c/2 = 7.5 \times 10^8$$

$$\frac{\sigma_2}{\omega \varepsilon_2} = \frac{0.1}{7.5 \times 10^8 \times 4 \times \frac{10^{-9}}{36\pi}} = 1.2\pi$$

$$\alpha_2 = \frac{\omega}{c} \sqrt{\frac{4}{2} \left[\sqrt{1 + 1.44\pi^2} - 1 \right]} = 6.021$$

$$\beta_2 = \frac{\omega}{c} \sqrt{\frac{4}{2} \left[\sqrt{1 + 1.44\pi^2} + 1 \right]} = 7.826$$

$$|\eta_2| = \frac{60\pi}{\sqrt[4]{1 + 1.44\pi^2}} = 95.445, \quad \eta_1 = 120\pi \sqrt{\varepsilon_{r1}} = 754$$

$$\tan 2\theta_{\eta_2} = 1.2\pi \longrightarrow \theta_{\eta_2} = 37.57^\circ$$

$$\eta_2 = 95.445 \angle 37.57^\circ$$

(a)

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{95.445 \angle 37.57^\circ - 754}{95.445 \angle 37.57^\circ + 754} = \underline{\underline{0.8186 \angle 171.08^\circ}}$$

$$\tau = 1 + \Gamma = \underline{\underline{0.2295 \angle 33.56^\circ}}$$

$$s = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.8186}{1 - 0.8186} = \underline{\underline{10.025}}$$

$$(b) \quad \mathbf{E}_i = 50 \sin(\omega t - 5x) \mathbf{a}_y = \text{Im}(\mathbf{E}_{is} e^{j\omega t}), \text{ where } \mathbf{E}_{is} = 50 e^{-j5x} \mathbf{a}_y.$$

$$\mathbf{E}_{ro} = \Gamma \mathbf{E}_{io} = 0.8186 e^{j171.08^\circ} (50) = 40.93 e^{j171.08^\circ}$$

$$\mathbf{E}_{rs} = 40.93 e^{j5x + j171.08^\circ} \mathbf{a}_y$$

$$\mathbf{E}_r = \text{Im}(\mathbf{E}_{rs} e^{j\omega t}) = \underline{\underline{40.93 \sin(\omega t + 5x + 171.1^\circ) \mathbf{a}_y}} \text{ V/m}$$

$$\mathbf{a}_H = \mathbf{a}_k \times \mathbf{a}_E = -\mathbf{a}_x \times \mathbf{a}_y = -\mathbf{a}_z$$

$$\mathbf{H}_r = -\frac{40.93}{754} \sin(\omega t + 5x + 171.1^\circ) \mathbf{a}_z = \underline{\underline{-0.0543 \sin(\omega t + 5x + 171.1^\circ) \mathbf{a}_z}} \text{ A/m}$$

(c)

$$E_{to} = \tau E_{io} = 0.229 e^{j33.56^\circ} (50) = 11.475 e^{j33.56^\circ}$$

$$\mathbf{E}_{ts} = 11.475 e^{-j\beta_2 x + j33.56^\circ} e^{-\alpha_2 x} \mathbf{a}_y$$

$$\mathbf{E}_t = \text{Im}(\mathbf{E}_{ts} e^{j\omega t}) = \underline{\underline{11.475 e^{-6.021x} \sin(\omega t - 7.826x + 33.56^\circ) \mathbf{a}_y}} \text{ V/m}$$

$$\mathbf{a}_H = \mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_x \times \mathbf{a}_y = \mathbf{a}_z$$

$$\begin{aligned} \mathbf{H}_t &= \frac{11.495}{95.445} e^{-6.021x} \sin(\omega t - 7.826x + 33.56^\circ - 37.57^\circ) \mathbf{a}_z \\ &= \underline{\underline{0.1202 e^{-6.021x} \sin(\omega t - 7.826x - 4.01^\circ) \mathbf{a}_z}} \text{ A/m} \end{aligned}$$

(d)

$$\mathcal{P}_{\text{ave}} = \frac{E_{io}^2}{2\eta_1} \mathbf{a}_x + \frac{E_{ro}^2}{2\eta_1} (-\mathbf{a}_x) = \frac{1}{2(240\pi)} [50^2 \mathbf{a}_x - 40.93^2 \mathbf{a}_x] = \underline{\underline{0.5469 \mathbf{a}_x}} \text{ W/m}^2$$

$$\mathbf{P}_{2\text{ave}} = \frac{E_{to}^2}{2|\eta_2|} e^{-2\alpha_2 x} \cos \theta_{\eta_2} \mathbf{a}_x = \frac{(11.475)^2}{2(95.445)} \cos 37.57^\circ e^{-2(6.021)x} \mathbf{a}_x = \underline{\underline{0.5469 e^{-12.04x} \mathbf{a}_x}} \text{ W/m}^2$$

P. E. 10.11 (a)

$$\mathbf{k} = -2\mathbf{a}_y + 4\mathbf{a}_z \longrightarrow k = \sqrt{2^2 + 4^2} = \sqrt{20}$$

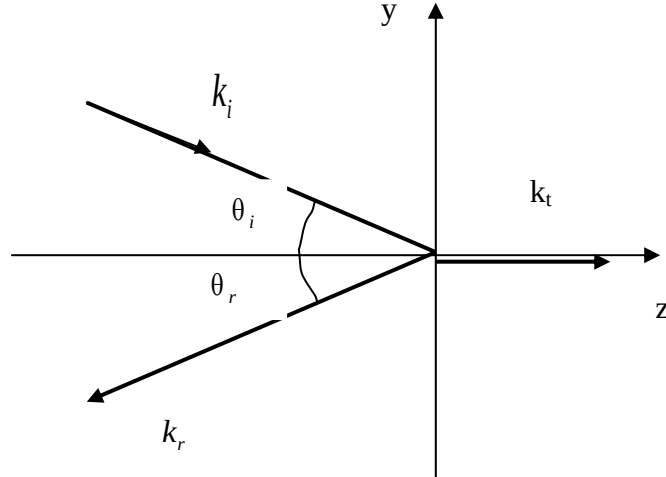
$$\omega = kc = 3 \times 10^8 \sqrt{20} = \underline{\underline{1.342 \times 10^9 \text{ rad/s}}},$$

$$\lambda = 2\pi / k = \underline{\underline{1.405 \text{ m}}}$$

$$\begin{aligned} \text{(b) } \mathbf{H} &= \frac{\mathbf{a}_k \times \mathbf{E}}{\eta_o} = \frac{(-2\mathbf{a}_y + 4\mathbf{a}_z)}{\sqrt{20}(120\pi)} \times (10\mathbf{a}_y + 5\mathbf{a}_z) \cos(\omega t - \mathbf{k} \cdot \mathbf{r}) \\ &= \underline{\underline{-29.66 \cos(1.342 \times 10^9 t + 2y - 4z) \mathbf{a}_x}} \text{ mA/m} \end{aligned}$$

$$(c) \quad \mathcal{P}_{\text{ave}} = \frac{|E_o|^2}{2\eta_o} \mathbf{a}_k = \frac{125}{2(120\pi)} \frac{(-2\mathbf{a}_y + 4\mathbf{a}_z)}{\sqrt{20}} = \underline{\underline{-74.15\mathbf{a}_y + 148.9\mathbf{a}_z}} \quad \text{mW/m}^2$$

P. E. 10.12 (a)



$$\tan \theta_i = \frac{k_{iy}}{k_{iz}} = \frac{2}{4} \longrightarrow \underline{\underline{\theta_i = 26.56 = \theta_r}}$$

$$\sin \theta_t = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} \sin \theta_i = \frac{1}{2} \sin 26.56^\circ \longrightarrow \underline{\underline{\theta_t = 12.92^\circ}}$$

(b) $\eta_1 = \eta_o, \eta_2 = \eta_o / 2$ $\mathbf{E}$ is parallel to the plane of incidence. Since $\mu_1 = \mu_2 = \mu_o$, we may use the result of Prob. 10.42, i.e.

$$\Gamma_{\parallel} = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)} = \frac{\tan(-13.64^\circ)}{\tan(39.48^\circ)} = \underline{\underline{-0.2946}}$$

$$\tau_{\parallel} = \frac{2 \cos 26.56^\circ \sin 12.92^\circ}{\sin 39.48^\circ \cos(-13.64^\circ)} = \underline{\underline{0.6474}}$$

(c) $\mathbf{k}_r = -\beta_1 \sin \theta_r \mathbf{a}_y - \beta_1 \cos \theta_r \mathbf{a}_z$. Once k_r is known, $\mathbf{E}_r$ is chosen such that

$\mathbf{k}_r \cdot \mathbf{E}_r = 0$ or $\nabla \cdot \mathbf{E}_r = 0$. Let

$$\mathbf{E}_r = \pm E_{or} (-\cos \theta_r \mathbf{a}_y + \sin \theta_r \mathbf{a}_z) \cos(\omega t + \beta_1 \sin \theta_r y + \beta_1 \cos \theta_r z)$$

Only the positive sign will satisfy the boundary conditions. It is evident that

$$\mathbf{E}_i = E_{oi}(\cos \theta_i \mathbf{a}_y + \sin \theta_i \mathbf{a}_z) \cos(\omega t + 2y - 4z)$$

Since $\theta_r = \theta_i$,

$$E_{or} \cos \theta_r = \Gamma_{//} E_{oi} \cos \theta_i = 10 \Gamma_{//} = -2.946$$

$$E_{or} \sin \theta_r = \Gamma_{//} E_{oi} \sin \theta_i = 5 \Gamma_{//} = -1.473$$

$$\beta_1 \sin \theta_r = 2, \beta_1 \cos \theta_r = 4$$

i.e.

$$\mathbf{E}_r = -(2.946 \mathbf{a}_y - 1.473 \mathbf{a}_z) \cos(\omega t + 2y + 4z)$$

$$\mathbf{E}_1 = \mathbf{E}_i + \mathbf{E}_r = \underline{\underline{(10 \mathbf{a}_y + 5 \mathbf{a}_z) \cos(\omega t + 2y - 4z) + (-2.946 \mathbf{a}_y + 1.473 \mathbf{a}_z) \cos(\omega t + 2y + 4z)}}$$

V/m

(d) $\mathbf{k}_t = -\beta_2 \sin \theta_t \mathbf{a}_y + \beta_2 \cos \theta_t \mathbf{a}_z$. Since $\mathbf{k}_r \bullet \mathbf{E}_r = 0$, let

$$\mathbf{E}_t = E_{ot}(\cos \theta_t \mathbf{a}_y + \sin \theta_t \mathbf{a}_z) \cos(\omega t + \beta_2 y \sin \theta_t - \beta_2 z \cos \theta_t)$$

$$\beta_2 = \omega \sqrt{\mu_2 \epsilon_2} = \beta_1 \sqrt{\epsilon_{r2}} = 2\sqrt{20}$$

$$\sin \theta_t = \frac{1}{2} \sin \theta_i = \frac{1}{2\sqrt{5}}, \quad \cos \theta_t = \frac{\sqrt{9}}{\sqrt{20}}$$

$$\beta_2 \cos \theta_t = 2\sqrt{20} \sqrt{\frac{19}{20}} = 8.718$$

$$E_{ot} \cos \theta_t = \tau_{//} E_{oi} \cos \theta_i = 0.6474 \sqrt{125} \sqrt{\frac{19}{20}} = 7.055$$

$$E_{ot} \sin \theta_t = \tau_{//} E_{oi} \sin \theta_i = 0.6474 \sqrt{125} \sqrt{\frac{1}{20}} = 1.6185$$

Hence

$$\mathbf{E}_2 = \underline{\underline{(7.055 \mathbf{a}_y + 1.6185 \mathbf{a}_z) \cos(\omega t + 2y - 8.718z)}} \quad \text{V/m}$$

$$(d) \tan \theta_{B//} = \sqrt{\frac{\epsilon_2}{\epsilon_1}} = 2 \longrightarrow \underline{\underline{\theta_{B//} = 63.43^\circ}}$$

P.E. 10.13

$$S_i = \frac{1 + 0.4}{1 - 0.4} = \frac{1.4}{0.6} = \underline{\underline{2.333}}$$

$$S_o = \frac{1 + 0.2}{1 - 0.2} = \frac{1.2}{0.8} = \underline{\underline{1.5}}$$

Prob. 10.1 (a) Wave propagates along $+\underline{\underline{a_x}}$.

(b)

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2\pi \times 10^6} = \underline{\underline{1\mu s}}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{6} = \underline{\underline{1.047\text{m}}}$$

$$u = \frac{\omega}{\beta} = \frac{2\pi \times 10^6}{6} = \underline{\underline{1.047 \times 10^6 \text{m/s}}}$$

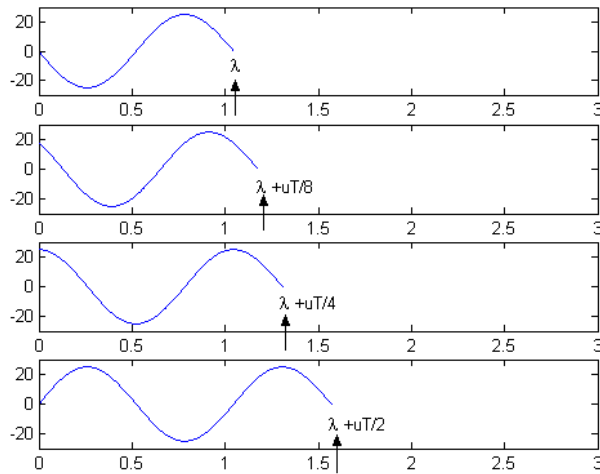
(c) At $t=0$, $E_z = 25 \sin(-6x) = -25 \sin 6x$

$$\text{At } t=T/8, E_z = 25 \sin\left(\frac{2\pi}{T} \frac{T}{8} - 6x\right) = 25 \sin\left(\frac{\pi}{4} - 6x\right)$$

$$\text{At } t=T/4, E_z = 25 \sin\left(\frac{2\pi}{T} \frac{T}{4} - 6x\right) = 25 \sin(-6x + 90^\circ) = 25 \cos 6x$$

$$\text{At } t=T/2, E_z = 25 \sin\left(\frac{2\pi}{T} \frac{T}{2} - 6x\right) = 25 \sin(-6x + \pi) = 25 \sin 6x$$

These are sketched below.

**Prob. 10.2**

$$(a) \lambda = \frac{c}{f} = \frac{3 \times 10^8}{60} = \underline{\underline{5 \times 10^6 \text{ m}}}$$

$$(b) \lambda = \frac{3 \times 10^8}{2 \times 10^6} = \underline{\underline{150 \text{ m}}}$$

$$(c) \lambda = \frac{3 \times 10^8}{120 \times 10^6} = \underline{\underline{2.5 \text{ m}}}$$

$$(d) \lambda = \frac{3 \times 10^8}{2.4 \times 10^9} = \underline{\underline{0.125 \text{ m}}}$$

Prob. 10.3

$$(a) \omega = \underline{\underline{10^8 \text{ rad/s}}}$$

$$(b) \beta = \frac{\omega}{c} = \frac{10^8}{3 \times 10^8} = \underline{\underline{0.333 \text{ rad/m}}}$$

$$(c) \lambda = \frac{2\pi}{\beta} = 6\pi = \underline{\underline{18.85 \text{ m}}}$$

(d) Along $-\mathbf{a}_y$

At $y=1$, $t=10\text{ms}$,

$$(e) H = 0.5 \cos(10^8 t \times 10 \times 10^{-9} + \frac{1}{3} \times 3) = 0.5 \cos(1+1) \\ = \underline{\underline{-0.1665 \text{ A/m}}}$$

Prob. 10.4

(a)

$$\frac{\partial E}{\partial x} = -\sin(x + \omega t) - \sin(x - \omega t)$$

$$\frac{\partial^2 E}{\partial x^2} = -\cos(x + \omega t) - \cos(x - \omega t) = -E$$

$$\frac{\partial E}{\partial t} = -\omega \sin(x + \omega t) - \omega \sin(x - \omega t)$$

$$\frac{\partial^2 E}{\partial t^2} = -\omega^2 \cos(x + \omega t) - \omega^2 \cos(x - \omega t) = -\omega^2 E$$

$$\frac{\partial^2 E}{\partial t^2} - u^2 \frac{\partial^2 E}{\partial x^2} = -\omega^2 E + u^2 E = 0$$

if $\omega^2 = u^2$ and hence, eq. (10.1) is satisfied.

(b) $u = \omega$

Prob. 10.5 If

$$\gamma^2 = j\omega\mu(\sigma + j\omega\varepsilon) = -\omega^2\mu\varepsilon + j\omega\mu\sigma \quad \text{and } \gamma = \alpha + j\beta, \text{ then}$$

$$|\gamma^2| = \sqrt{(\alpha^2 - \beta^2)^2 + 4\alpha^2\beta^2} = \sqrt{(\alpha^2 + \beta^2)^2} = \alpha^2 + \beta^2$$

i.e.

$$\alpha^2 + \beta^2 = \omega\mu\sqrt{(\sigma^2 + \omega^2\varepsilon^2)} \quad (1)$$

$$\text{Re}(\gamma^2) = \alpha^2 - \beta^2 = -\omega^2\mu\varepsilon$$

$$\beta^2 - \alpha^2 = \omega^2\mu\varepsilon \quad (2)$$

Subtracting and adding (1) and (2) lead respectively to

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right]}$$

$$\beta = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1\right]}$$

(b) From eq. (10.25), $\mathbf{E}_s(z) = E_o e^{-\gamma z} \mathbf{a}_x$.

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}_s \longrightarrow \mathbf{H}_s = \frac{j}{\omega\mu} \nabla \times \mathbf{E}_s = \frac{j}{\omega\mu} (-\gamma E_o e^{-\gamma z} \mathbf{a}_y)$$

But $\mathbf{H}_s(z) = H_o e^{-\gamma z} \mathbf{a}_y$, hence $H_o = \frac{E_o}{\eta} = -\frac{j\gamma}{\omega\mu} E_o$

$$\eta = \frac{j\omega\mu}{\gamma}$$

(c) From (b),

$$\eta = \frac{j\omega\mu}{\sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \frac{\sqrt{\mu/\varepsilon}}{\sqrt{1 - j\frac{\sigma}{\omega\varepsilon}}}$$

$$|\eta| = \frac{\sqrt{\mu/\varepsilon}}{\sqrt[4]{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2}}, \tan 2\theta_\eta = \left(\frac{\omega\varepsilon}{\sigma}\right)^{-1} = \frac{\sigma}{\omega\varepsilon}$$

Prob. 10.6 (a)

From eq. (10.18),

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} = \sqrt{j\omega\mu j\omega\varepsilon \left(1 - \frac{j\sigma}{\omega\varepsilon}\right)} = j\omega\sqrt{\mu\varepsilon} \left(1 - \frac{j\sigma}{\omega\varepsilon}\right)^{1/2}$$

Assuming that $\frac{\sigma}{\omega\varepsilon} \ll 1$, we include up to the second power in $\frac{\sigma}{\omega\varepsilon}$ and neglect higher-order terms.

$$\gamma = \frac{\sigma}{2} \sqrt{\frac{\mu}{\varepsilon}} \left(1 - \frac{\sigma^2}{\omega^2\varepsilon^2}\right) + j\omega\sqrt{\mu\varepsilon} \left(1 + \frac{\sigma^2}{8\omega^2\varepsilon^2}\right) = \alpha + j\beta$$

Thus,

$$\beta = \omega\sqrt{\mu\varepsilon} \left(1 + \frac{\sigma^2}{8\omega^2\varepsilon^2}\right)$$

Prob. 10.7

(a)

$$\frac{\sigma}{\omega\varepsilon} = \frac{8 \times 10^{-2}}{2\pi \times 50 \times 10^6 \times 3.6 \times \frac{10^{-9}}{36\pi}} = 8$$

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} - 1 \right]} = \frac{2\pi \times 50 \times 10^6}{3 \times 10^8} \sqrt{\frac{2.1 \times 3.6}{2} [\sqrt{65} - 1]} = 5.41$$

$$\beta = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} + 1 \right]} = 6.129$$

$$\gamma = \alpha + j\beta = \underline{\underline{5.41 + j6.129 \text{ /m}}}$$

$$(b) \quad \lambda = \frac{2\pi}{\beta} = \frac{2\pi}{6.129} = \underline{\underline{1.025 \text{ m}}}$$

$$(c) \quad u = \frac{\omega}{\beta} = \frac{2\pi \times 50 \times 10^6}{6.129} = \underline{\underline{5.125 \times 10^7 \text{ m/s}}}$$

$$(d) \quad |\eta| = \frac{\sqrt{\frac{\mu}{\epsilon}}}{\sqrt[4]{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2}} = \frac{120\pi \sqrt{\frac{2.1}{3.6}}}{\sqrt[4]{65}} = 101.4$$

$$\tan 2\theta_\eta = \frac{\sigma}{\omega\epsilon} = 8 \longrightarrow \theta_\eta = 41.44^\circ$$

$$\eta = \underline{\underline{101.41 \angle 41.44^\circ \Omega}}$$

$$(e) \quad \mathbf{H}_s = \mathbf{a}_k \times \frac{\mathbf{E}_s}{\eta} = \mathbf{a}_x \times \frac{6}{\eta} e^{-\gamma z} \mathbf{a}_z = -\frac{6}{\eta} e^{-\gamma z} \mathbf{a}_y = \underline{\underline{-59.16 e^{-j41.44^\circ} e^{-\gamma z} \mathbf{a}_y \text{ mA/m}}}$$

Prob. 10.8

$$(a) \quad \tan \theta = \frac{\sigma}{\omega\epsilon} = \frac{10^{-2}}{2\pi \times 12 \times 10^6 \times 10 \times \frac{10^{-9}}{36\pi}} = \underline{\underline{1.5}}$$

$$(b) \quad \tan \theta = \frac{10^{-4}}{2\pi \times 12 \times 10^6 \times 4 \times \frac{10^{-9}}{36\pi}} = \underline{\underline{3.75 \times 10^{-2}}}$$

$$(c) \tan \theta = \frac{4}{2\pi \times 12 \times 10^6 \times 81 \times \frac{10^{-9}}{36\pi}} = \underline{\underline{74.07}}$$

Prob. 10.9

(a)

$$\begin{aligned} \alpha &= \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} - 1 \right]} = \frac{2\pi \times 15 \times 10^9}{3 \times 10^8} \sqrt{\frac{1 \times 9.6}{2} \left[\sqrt{1 + 9 \times 10^{-8}} - 1 \right]} \\ &= 100\pi \sqrt{4.8 \left(\frac{1}{2} \times 9 \times 10^{-8} \right)} = 0.146 \end{aligned}$$

$$\delta = \frac{1}{\alpha} = \underline{\underline{6.85 \text{ m}}}$$

$$(b) A = \alpha \ell = 0.146 \times 5 \times 10^{-3} = \underline{\underline{0.73 \times 10^{-3} \text{ Np}}}$$

Prob. 10.10

If $\sigma = \epsilon\omega$, the loss tangent is $\frac{\sigma}{\epsilon\omega} = 1$

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} - 1 \right]} = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + 1^2} - 1 \right]} = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{2} - 1 \right]}$$

$$\text{But } \lambda_o = \frac{2\pi}{\beta} = \frac{2\pi c}{\omega} \rightarrow \omega = \frac{2\pi c}{\lambda_o}$$

$$\alpha = \frac{2\pi c}{\lambda_o} \sqrt{\frac{\mu\epsilon}{2}} \sqrt{\sqrt{2} - 1} = \frac{2\pi \times 3 \times 10^8}{12 \times 10^{-2}} \sqrt{\frac{1}{2} \times 4\pi \times 10^{-7} \times 4 \times \frac{10^{-9}}{36\pi}} (0.6436) = \underline{\underline{47.66 \text{ Np/m}}}$$

$$\begin{aligned} \beta &= \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} + 1 \right]} = \frac{2\pi c}{\lambda_o} \sqrt{\frac{\mu\epsilon}{2}} \sqrt{\sqrt{2} + 1} \\ &= \frac{2\pi \times 3 \times 10^8}{12 \times 10^{-2}} \sqrt{\frac{1}{2} \times 4\pi \times 10^{-7} \times 4 \times \frac{10^{-9}}{36\pi}} (1.5538) \\ &= \underline{\underline{115.06 \text{ rad/m}}} \end{aligned}$$

$$u = \frac{\omega}{\beta} = \frac{2\pi c}{\lambda_o \beta} = \frac{2\pi \times 3 \times 10^8}{12 \times 10^{-2} \times 115.06} = \underline{\underline{1.3652 \times 10^8 \text{ m/s}}}$$

Prob. 10.11

For silver, the loss tangent is

$$\frac{\sigma}{\omega\epsilon} = \frac{6.1 \times 10^7}{2\pi \times 10^8 \times \frac{10^{-9}}{36\pi}} = 6.1 \times 18 \times 10^8 \gg 1$$

Hence, silver is a good conductor

For rubber,

$$\frac{\sigma}{\omega\epsilon} = \frac{10^{-15}}{2\pi \times 10^8 \times 3.1 \times \frac{10^{-9}}{36\pi}} = \frac{18}{3.1} \times 10^{-14} \ll 1$$

Hence, rubber is a poor conductor or a good insulator.

Prob. 10.12

$$\frac{\sigma}{\omega\epsilon} = \frac{4}{2\pi \times 10^5 \times 80 \times 10^{-9} / 36\pi} = 9,000 \gg 1$$

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}} = \sqrt{\frac{2\pi \times 10^5}{2} \times 4\pi \times 10^{-7} \times 4} = 0.4\pi$$

$$(a) \quad u = \omega / \beta = \frac{2\pi \times 10^5}{0.4\pi} = \underline{\underline{5 \times 10^5}} \text{ m/s}$$

$$(b) \quad \lambda = 2\pi / \beta = \frac{2\pi}{0.4\pi} = \underline{\underline{5}} \text{ m}$$

$$(c) \quad \delta = 1 / \alpha = \frac{1}{0.4\pi} = \underline{\underline{0.796}} \text{ m}$$

$$(d) \quad \eta = |\eta| \angle \theta_\eta, \theta_\eta = 45^\circ$$

$$|\eta| = \frac{\sqrt{\frac{\mu}{\epsilon}}}{\sqrt[4]{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}} \cong \sqrt{\frac{\mu}{\epsilon} \frac{\omega\epsilon}{\sigma}} = \sqrt{\frac{4\pi \times 10^{-7} \times 2\pi \times 10^5}{4}} = 0.4443$$

$$\eta = \underline{\underline{0.4443 \angle 45^\circ}} \quad \Omega$$

Prob. 10.13 (a)

$$T = 1/f = 2\pi/\omega = \frac{2\pi}{\pi \times 10^8} = \underline{\underline{20 \text{ ns}}}$$

$$(b) \text{ Let } x = \sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2}$$

$$\frac{\alpha}{\beta} = \left(\frac{x-1}{x+1}\right)^{1/2}$$

$$\text{But } \alpha = \frac{\omega}{c} \sqrt{\frac{\mu_r \epsilon_r}{2}} \sqrt{x-1}$$

$$\sqrt{x-1} = \frac{\alpha c}{\omega \sqrt{\frac{\mu_r \epsilon_r}{2}}} = \frac{0.1 \times 3 \times 10^8}{\pi \times 10^8 \sqrt{2}} = 0.06752 \longrightarrow x = 1.0046$$

$$\beta = \left(\frac{x+1}{x-1}\right)^{1/2} \quad \alpha = \left(\frac{2.0046}{0.0046}\right)^{1/2} \quad 0.1 = 2.088$$

$$\lambda = 2\pi/\beta = \frac{2\pi}{2.088} = \underline{\underline{3 \text{ m}}}$$

$$(c) \quad |\eta| = \frac{\sqrt{\mu/\epsilon}}{\sqrt{x}} = \frac{377}{2\sqrt{1.0046}} = 188.1$$

$$x = \sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} = 1.0046$$

$$\frac{\sigma}{\omega \epsilon} = 0.096 = \tan 2\theta_\eta \longrightarrow \theta_\eta = 274^\circ$$

$$\eta = 188.1 \angle 274^\circ \quad \Omega$$

$$E_o = \eta H_o = 12 \times 188.1 = 2257.2$$

$$\mathbf{a}_E \times \mathbf{a}_H = \mathbf{a}_k \longrightarrow \mathbf{a}_E \times \mathbf{a}_x = \mathbf{a}_y \longrightarrow \mathbf{a}_E = \mathbf{a}_z$$

$$\underline{\underline{\mathbf{E} = 2.257e^{-0.1y} \sin(\pi \times 10^8 t - 2.088y + 2.74^\circ) \mathbf{a}_z \text{ kV/m}}}$$

(d) The phase difference is 2.74° .

Prob. 10.14

This is a lossy medium in which $\mu = \mu_0$.

$$\text{Let } x = \left(\frac{\sigma}{\omega \epsilon} \right)^2$$

$$\eta = \frac{j\omega\mu}{\gamma} = \frac{j2\pi \times 10^9 \times 4\pi}{100 + j200} = 35.31 \angle 26.57^\circ$$

$$E_o = 0.05 \times 35.31 = 1.765$$

$$\mathbf{a}_E = \mathbf{a}_H \times \mathbf{a}_k = -\mathbf{a}_z$$

Thus, we obtain

$$\underline{\underline{\mathbf{E} = -1.765 \cos(2\pi \times 10^9 t - 200x + 26.57^\circ) \mathbf{a}_z \text{ V/m}}}$$

$$\sqrt{\epsilon_r(1/3)} = \frac{\alpha c}{\omega} = \frac{100 \times 3 \times 10^8}{2\pi \times 10^9} = \frac{15}{\pi}$$

$$\epsilon_r \frac{1}{3} = 4.776 \longrightarrow \epsilon_r = 14.32$$

$$\tan 2\theta_\eta = \frac{\sigma}{\omega \epsilon} = \frac{4}{3} \longrightarrow \theta_\eta = 26.57^\circ$$

$$|\eta| = \frac{\sqrt{\mu/\epsilon}}{\sqrt[4]{1+x}} = \frac{\frac{377}{\sqrt{14.32}}}{\sqrt{5/3}} = 77.175$$

$$E_o = |\eta| H_o = 77.175 \times 50 \times 10^{-3} = 3.858$$

$$\mathbf{a}_E = -(\mathbf{a}_k \times \mathbf{a}_H) = -(\mathbf{a}_x \times \mathbf{a}_y) = -\mathbf{a}_z$$

$$\underline{\underline{\mathbf{E} = -3.858 e^{-100x} \cos(2\pi \times 10^9 t - 200x + 26.57^\circ) \mathbf{a}_z \text{ V/m}}}$$

Prob. 10.15

$$\frac{\sigma}{\omega\epsilon} = \frac{1}{2\pi \times 10^9 \times 4 \times \frac{10^{-9}}{36\pi}} = 4.5$$

$$\begin{aligned}\alpha &= \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1 \right]} \\ &= 2\pi \times 10^9 \sqrt{\frac{4\pi}{2} \times 10^{-7} \times 4 \times 9 \times \frac{10^{-9}}{36\pi} \left[\sqrt{1 + 4.5^2} - 1 \right]} \\ &= 20\pi \sqrt{2[\sqrt{21.25} - 1]} = \underline{\underline{168.8 \text{ Np/m}}}\end{aligned}$$

$$\beta = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} + 1 \right]} = 20\pi \sqrt{2[\sqrt{21.25} + 1]} = \underline{\underline{210.5 \text{ rad/m}}}$$

$$\tan 2\theta_\eta = \frac{\sigma}{\omega\epsilon} = 4.5 \quad \longrightarrow \quad \theta_\eta = 38.73^\circ$$

$$|\eta| = \frac{\sqrt{\mu/\epsilon}}{\sqrt[4]{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}} = \frac{120\pi\sqrt{9/4}}{\sqrt[4]{1 + 4.5^2}} = 263.38$$

$$\eta = \underline{\underline{263.38 \angle 38.73^\circ \ \Omega}}$$

$$u = \frac{\omega}{\beta} = \frac{2\pi \times 10^9}{210.5} = \underline{\underline{2.985 \times 10^7 \text{ m/s}}}$$

Prob. 10.16

$$(a) \quad \beta = 6.5 = \omega \sqrt{\mu_o \epsilon_o} = \frac{\omega}{c}$$

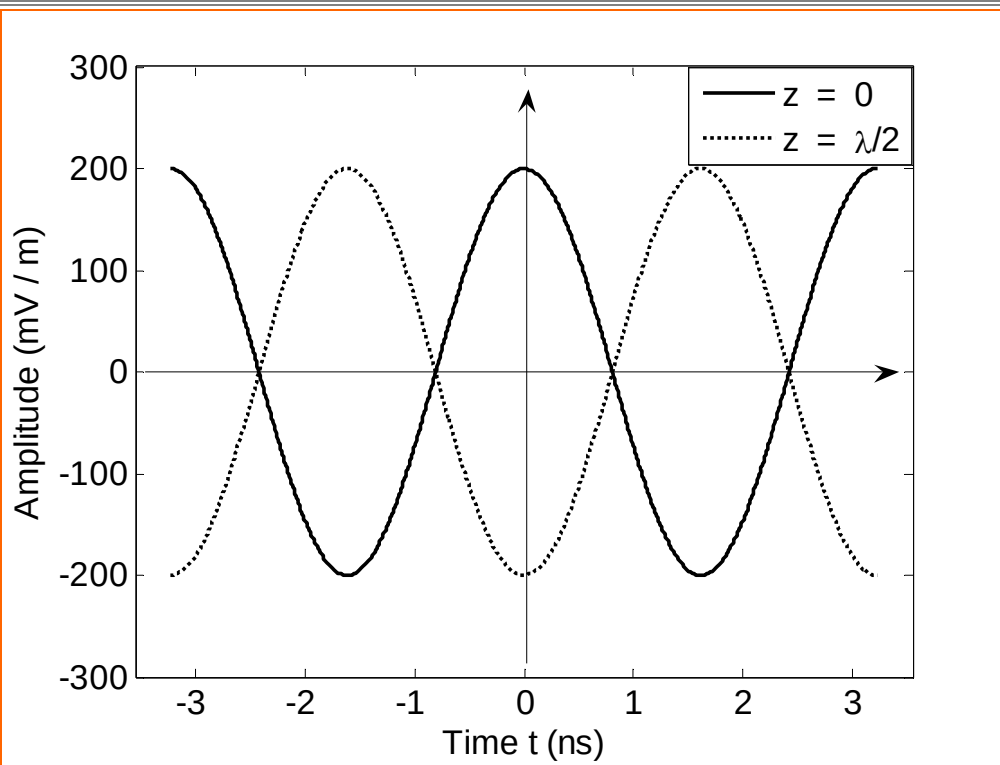
$$\omega = \beta c = 6.5 \times 3 \times 10^8 = \underline{\underline{1.95 \times 10^9 \text{ rad/s}}}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{6.5} = \underline{\underline{0.9666 \text{ m}}}$$

$$(b) \quad \text{For } z=0, \quad E_z = 0.2 \cos \omega t$$

$$\text{For } z=\lambda/2, \quad E_z = 0.2 \cos\left(\omega t - \frac{2\pi}{\lambda} \frac{\lambda}{2}\right) = -0.2 \cos \omega t$$

The two waves are sketched below.



(c) $\mathbf{H} = H_o \cos(\omega t - 6.5z)\mathbf{a}_H$

$$H_o = \frac{E_o}{\eta_o} = \frac{0.2}{377} = 5.305 \times 10^{-4}$$

$$\mathbf{a}_E \times \mathbf{a}_H = \mathbf{a}_k \longrightarrow \mathbf{a}_x \times \mathbf{a}_H = \mathbf{a}_z \longrightarrow \mathbf{a}_H = \mathbf{a}_y$$

$$\underline{\underline{\mathbf{H} = 0.5305 \cos(\omega t - 6.5z)\mathbf{a}_y \text{ mA/m}}}$$

Prob. 10.17

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{\mu\epsilon}}$$

$$\frac{\lambda_o}{\lambda} = \frac{\sqrt{\mu_o\epsilon_o\epsilon_r}}{\sqrt{\mu_o\epsilon_o}} = \sqrt{\epsilon_r}$$

$$\sqrt{\epsilon_r} = \frac{\lambda_o}{\lambda} = \frac{6.4}{2.8} = 2.286 \longrightarrow \underline{\underline{\epsilon_r = 5.224}}$$

Prob. 10.18 (a) Along -x direction.

(b) $\beta = 6, \quad \omega = 2 \times 10^8,$

$$\beta = \omega \sqrt{\mu \epsilon} = \frac{\omega}{c} \sqrt{\mu_r \epsilon_r}$$

$$\sqrt{\epsilon_r} = \beta c / \omega = \frac{6 \times 3 \times 10^8}{2 \times 10^8} = 9 \quad \longrightarrow \quad \epsilon_r = 81$$

$$\epsilon = \epsilon_o \epsilon_r = \frac{10^{-9}}{36\pi} \times 81 = \underline{\underline{7.162 \times 10^{-10} \text{ F/m}}}$$

(c) $\eta = \sqrt{\mu / \epsilon} = \sqrt{\mu_o / \epsilon_o} \sqrt{\mu_r / \epsilon_r} = \frac{120\pi}{9} = 41.89 \Omega$

$$E_o = H_o \eta = 25 \times 10^{-3} \times 41.88 = 1.047$$

$$\mathbf{a}_E \times \mathbf{a}_H = \mathbf{a}_k \longrightarrow \mathbf{a}_E \times \mathbf{a}_y = -\mathbf{a}_x \longrightarrow \mathbf{a}_E = \mathbf{a}_z$$

$$\underline{\underline{E = 1.047 \sin(2 \times 10^8 t + 6x) \mathbf{a}_z \text{ V/m}}}$$

Prob. 10.19 (a) $\frac{\sigma}{\omega \epsilon} = \frac{10^{-6}}{2\pi \times 10^7 \times 5 \times \frac{10^{-9}}{36\pi}} = 3.6 \times 10^{-4} \ll 1$

Thus, the material is lossless at this frequency.

(b) $\beta = \omega \sqrt{\mu \epsilon} = \frac{2\pi \times 10^7}{3 \times 10^8} \sqrt{5 \times 750} = \underline{\underline{12.83 \text{ rad/m}}}$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{12.83} = \underline{\underline{0.49 \text{ m}}}$$

(c) Phase difference = $\beta l = \underline{\underline{25.66 \text{ rad}}}$

(d) $\eta = \sqrt{\mu / \epsilon} = 120\pi \sqrt{\frac{\mu_r}{\epsilon_r}} = 120\pi \sqrt{\frac{750}{5}} = \underline{\underline{4.62 \text{ k}\Omega}}$

Prob. 10.20

(a)

$$\beta = \omega\sqrt{\mu\epsilon} = \frac{\omega}{c} = \frac{10^8\pi}{3\times 10^8} = \frac{\pi}{3} = \underline{\underline{1.0472 \text{ rad/m}}}$$

(b)

$$E = 0 \longrightarrow \sin(10^8\pi t_o - \beta x_o) = 0 = \sin(n\pi), n = 1, 2, 3, \dots$$

$$10^8\pi t_o - \beta x_o = \pi$$

$$10^8\pi \times 5 \times 10^{-3} - \frac{\pi}{3} x_o = \pi \longrightarrow x_o = \underline{\underline{5 \times 10^5 \text{ m}}}$$

(c)

$$\mathbf{H} = H_o \sin(10^8\pi t - \beta x) \mathbf{a}_H$$

$$H_o = \frac{E_o}{\eta} = \frac{50 \times 10^{-3}}{120\pi} = 132.63 \mu\text{A/m}$$

$$\mathbf{a}_H = \mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_x \times \mathbf{a}_z = -\mathbf{a}_y$$

$$\underline{\underline{\mathbf{H} = -132.63 \sin(10^8\pi t - 1.0472x) \mathbf{a}_y \mu\text{A/m}}}$$

Prob. 10.21

This is a lossless material.

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = 377 \sqrt{\frac{\mu_r}{\epsilon_r}} = 105 \quad (1)$$

$$u = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\mu_r\epsilon_r}} = 7.6 \times 10^7 \quad (2)$$

From (1),

$$\sqrt{\frac{\mu_r}{\epsilon_r}} = \frac{105}{377} = 0.2785 \quad (1)a$$

From (2),

$$\frac{1}{\sqrt{\mu_r\epsilon_r}} = \frac{7.6 \times 10^7}{3 \times 10^8} = 0.2533 \quad (2)a$$

Multiplying (1)a by (2)a,

$$\frac{1}{\epsilon_r} = 0.2785 \times 0.2533 = 0.07054 \longrightarrow \epsilon_r = \underline{\underline{14.175}}$$

Dividing (1)a by (2)a,

$$\mu_r = \frac{0.2785}{0.2533} = \underline{\underline{1.0995}}$$

Prob. 10.22

$$\begin{aligned}
 \text{(a)} \quad \nabla \times \mathbf{E} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x(z,t) & E_y(z,t) & 0 \end{vmatrix} = -\frac{\partial E_y}{\partial z} \mathbf{a}_x + \frac{\partial E_x}{\partial z} \mathbf{a}_y \\
 &= -6\beta \cos(\omega t - \beta z) \mathbf{a}_x + 8\beta \sin(\omega t - \beta z) \mathbf{a}_y
 \end{aligned}$$

$$\text{But } \nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \longrightarrow \mathbf{H} = -\frac{1}{\mu} \int \nabla \times \mathbf{E} dt$$

$$\mathbf{H} = \underline{\underline{\frac{6\beta}{\mu\omega} \sin(\omega t - \beta z) \mathbf{a}_x + \frac{8\beta}{\mu\omega} \cos(\omega t - \beta z) \mathbf{a}_y}}}$$

$$\text{(b)} \quad \beta = \omega \sqrt{\mu\epsilon} = \frac{2\pi f}{c} \sqrt{4.5} = \frac{2\pi \times 40 \times 10^6}{3 \times 10^8} \sqrt{4.5} = \underline{\underline{1.777 \text{ rad/m}}}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{1.777} = \underline{\underline{3.536 \text{ m}}}$$

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{120\pi}{\sqrt{4.5}} = \underline{\underline{177.72 \Omega}}$$

$$u = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{4.5}} = \frac{3 \times 10^8}{\sqrt{4.5}} = \underline{\underline{1.4142 \times 10^8 \text{ m/s}}}$$

Prob. 10.23

$$\text{(a)} \quad \mathbf{E} = \text{Re}[\mathbf{E}_s e^{j\omega t}] = (5\mathbf{a}_x + 12\mathbf{a}_y) e^{-0.2z} \cos(\omega t - 3.4z)$$

$$\text{At } z = 4\text{m}, t = T/8, \quad \omega t = \frac{2\pi}{T} \frac{T}{8} = \frac{\pi}{4}$$

$$\mathbf{E} = (5\mathbf{a}_x + 12\mathbf{a}_y) e^{-0.8} \cos(\pi/4 - 13.6)$$

$$|\mathbf{E}| = 13e^{-0.8} |\cos(\pi/4 - 13.6)| = \underline{\underline{5.662 \text{ V/m}}}$$

$$\text{(b)} \quad \text{loss} = \alpha \Delta z = 0.2(3) = 0.6 \text{ Np. Since } 1 \text{ Np} = 8.686 \text{ dB,}$$

$$\text{loss} = 0.6 \times 8.686 = \underline{\underline{5.212 \text{ dB}}}$$

$$(c) \text{ Let } x = \sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2}$$

$$\frac{\alpha}{\beta} = \left(\frac{x-1}{x+1}\right)^{1/2} = 0.2/3.4 = \frac{1}{17}$$

$$\frac{x-1}{x+1} = 1/289 \quad \longrightarrow \quad x = 1.00694$$

$$\alpha = \omega \sqrt{\mu \epsilon / 2} \sqrt{x-1} = \frac{\omega}{c} \sqrt{\epsilon_r / 2} \sqrt{x-1}$$

$$\sqrt{\frac{\epsilon_r}{2}} = \frac{\alpha c}{\omega \sqrt{x-1}} = \frac{0.2 \times 3 \times 10^8}{10^8 \sqrt{0.00694}} = 7.2 \quad \longrightarrow \quad \epsilon_r = 103.68$$

$$|\eta| = \frac{\sqrt{\frac{\mu_o}{\epsilon_o}} \cdot \frac{1}{\sqrt{\epsilon_r}}}{\sqrt{x}} = \frac{120\pi}{\sqrt{103.68 \times 1.00694}} = 36.896$$

$$\tan 2\theta_\eta = \frac{\sigma}{\omega \epsilon} = \sqrt{x^2 - 1} = 0.118 \quad \longrightarrow \quad \theta_\eta = 3.365^\circ$$

$$\eta = \underline{\underline{36.896 \angle 3.365^\circ \Omega}}$$

Prob. 10.24

$$\alpha = \beta = \sqrt{\frac{\omega \mu \sigma}{2}} \quad \longrightarrow \quad \sigma = \frac{2\alpha^2}{\omega \mu} = \frac{2 \times 12^2}{2\pi \times 10^6 \times 4\pi \times 10^{-7}} = \underline{\underline{36.48}}$$

$$\eta = |\eta| \angle \theta_\eta = \sqrt{\frac{\omega \mu}{\sigma}} \angle 45^\circ$$

$$|\eta| = \sqrt{\frac{\omega \mu}{\sigma}} = \sqrt{\frac{2\pi \times 10^6 \times 4\pi \times 10^{-7}}{36.48}} = 0.4652$$

$$E_o = |\eta| H_o = 0.4652 \times 20 \times 10^{-3} = 9.305 \times 10^{-3}$$

$$\mathbf{a}_E = \mathbf{a}_H \times \mathbf{a}_k = \mathbf{a}_y \times (-\mathbf{a}_z) = -\mathbf{a}_x$$

$$\mathbf{E} = E_o e^{-\alpha z} \sin(\omega t + \beta z) \mathbf{a}_E$$

$$= \underline{\underline{-9.305 e^{-12z} \sin(2\pi \times 10^6 t + 12z + 45^\circ) \mathbf{a}_x \text{ mV/m}}}$$

Prob. 10.25 For a good conductor, $\frac{\sigma}{\omega\epsilon} \gg 1$, say $\frac{\sigma}{\omega\epsilon} > 100$

$$(a) \quad \frac{\sigma}{\omega\epsilon} = \frac{10^{-2}}{2\pi \times 8 \times 10^6 \times 15 \times \frac{10^{-9}}{36\pi}} = 1.5 \longrightarrow \text{lossy}$$

No, not conducting.

$$(b) \quad \frac{\sigma}{\omega\epsilon} = \frac{0.025}{2\pi \times 8 \times 10^6 \times 16 \times \frac{10^{-9}}{36\pi}} = 3.515 \longrightarrow \text{lossy}$$

No, not conducting.

$$(c) \quad \frac{\sigma}{\omega\epsilon} = \frac{25}{2\pi \times 8 \times 10^6 \times 81 \times \frac{10^{-9}}{36\pi}} = 694.4 \longrightarrow \text{conducting}$$

Yes, conducting.

Prob. 10.26

$$\alpha = \beta = \frac{1}{\delta}$$

$$\text{But } u = \frac{\omega}{\beta} = \delta\omega = 0.02 \times 10^{-3} \times 2\pi \times 100 \times 10^6 = 4\pi \times 10^3 = \underline{\underline{1.256 \times 10^4 \text{ m/s}}}$$

Prob. 10.27 (a)

$$R_{dc} = \frac{l}{\sigma S} = \frac{l}{\sigma \pi a^2} = \frac{600}{5.8 \times 10^7 \times \pi \times (1.2)^2 \times 10^{-6}} = \underline{\underline{2.287 \Omega}}$$

$$(b) \quad R_{ac} = \frac{l}{\sigma 2\pi a \delta}. \quad \text{At 100 MHz, } \delta = 6.6 \times 10^{-3} \text{ mm} = 6.6 \times 10^{-6} \text{ m mm for copper}$$

(see Table 10.2).

$$R_{ac} = \frac{600}{5.8 \times 10^7 \times 2\pi \times (1.2 \times 10^{-3}) \times 6.6 \times 10^{-6}} = \underline{\underline{207.61 \Omega}}$$

$$(c) \quad \frac{R_{ac}}{R_{dc}} = \frac{a}{2\delta} = 1 \longrightarrow \delta = a/2 = \frac{66.1 \times 10^{-3}}{\sqrt{f}}$$

$$\sqrt{f} = \frac{66.1 \times 2 \times 10^{-3}}{a} = \frac{66.1 \times 2}{1.2} \longrightarrow f = \underline{\underline{12.137 \text{ kHz}}}$$

Prob. 10.28

(a) Copper is a good conductor.

$$\alpha = \sqrt{\pi f \mu \sigma} = \sqrt{\pi \times 10^{10} \times 4\pi \times 10^{-7} \times 5.8 \times 10^7} = 2\pi \times 10^5 \sqrt{5.8}$$

$$\underline{\underline{\alpha = 1.513 \times 10^6 \text{ Np/m}}}$$

$$(b) \quad \delta = \frac{1}{\alpha} = \frac{1}{1.513 \times 10^6} = \underline{\underline{6.609 \times 10^{-7} \text{ m}}}$$

$$(c) \quad \eta = \frac{1+j}{\sigma \delta} = \frac{1+j}{5.8 \times 10^7 \times 6.609 \times 10^{-7}} = \underline{\underline{26.09(1+j) \times 10^{-3} \Omega}}$$

Prob. 10.29

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} \longrightarrow f = \frac{1}{\delta^2 \pi \mu \sigma}$$

$$f = \frac{1}{4 \times 10^{-6} \times \pi \times 4\pi \times 10^{-7} \times 6.1 \times 10^7} = \underline{\underline{1.038 \text{ kHz}}}$$

Prob. 10.30

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = \frac{1}{\sqrt{\pi \times 5.8 \times 10^7 (60) 4\pi \times 10^{-7}}} = \frac{1}{2\pi \sqrt{5.8(60)}} = \underline{\underline{8.531 \text{ mm}}}$$

Prob. 10.31

This is a good conductor.

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ$$

$$24.6 = \sqrt{\frac{2\pi \times 12 \times 10^6 \times 4\pi \times 10^{-7}}{\sigma}} = \sqrt{\frac{16\pi^2(0.6)}{\sigma}}$$

$$\sigma = \frac{16\pi^2(0.6)}{24.6^2} = \underline{\underline{0.1566 \text{ S/m}}}$$

$$\beta = \alpha = \frac{1}{\delta} = \frac{1}{0.12} = \underline{\underline{8.333 \text{ rad/m}}}$$

$$\lambda = \frac{2\pi}{\beta} = 2\pi(0.12) = \underline{\underline{0.754 \text{ m}}}$$

$$u = \frac{\omega}{\beta} = \omega\delta = 2\pi \times 12 \times 10^6 \times 0.12 = \underline{\underline{9.05 \times 10^6 \text{ m/s}}}$$

Prob. 10.32

$$\frac{\sigma}{\omega\epsilon} = \frac{0.12}{2\pi \times 2.42 \times 10^9 \times 5.5 \times \frac{10^{-9}}{36\pi}} = 0.1623$$

This is a lossy material.

$$\begin{aligned} \alpha &= \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} - 1 \right]} \\ &= 2\pi \times 2.42 \times 10^9 \sqrt{\frac{4\pi}{2} \times 10^{-7} \times 5.5 \times \frac{10^{-9}}{36\pi} \left[\sqrt{1 + 0.1623^2} - 1 \right]} \\ &= 15.21(10^9)(10^{-8})\sqrt{0.01308} \\ &= 17.39 \end{aligned}$$

$$\delta = \frac{1}{\alpha} = \underline{\underline{57.5 \text{ mm}}}$$

Prob. 10.33

$$t = 5\delta = \frac{5}{\sqrt{\pi f \mu \sigma}} = \frac{5}{\sqrt{\pi \times 12 \times 10^9 \times 4\pi \times 10^{-7} \times 6.1 \times 10^7}} = \underline{\underline{2.94 \times 10^{-6} \text{ m}}}$$

Prob. 10.34

$$\frac{\sigma}{\omega\epsilon} = \frac{4}{2\pi \times 2 \times 10^9 \times 24 \times \frac{10^{-9}}{36\pi}} = 1.5$$

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} - 1 \right]} = 2\pi \times 2 \times 10^9 \sqrt{\frac{4\pi \times 10^{-7} \times 24 \times \frac{10^{-9}}{36\pi}}{2} \left[\sqrt{1 + (1.5)^2} - 1 \right]}$$

$$= 130.01 \text{ Np/m}$$

$$10^{-5} E_o = E_o e^{-\alpha d}$$

Taking the log of both sides gives

$$-5 \ln 10 = -\alpha d \quad \longrightarrow \quad d = \frac{5 \ln 10}{\alpha} = \frac{5 \ln 10}{130.01} = \underline{\underline{0.0886 \text{ m}}}$$

Prob. 10.35

(a) Linearly polarized along $\mathbf{a}_z$

$$(b) \quad \omega = 2\pi f = 2\pi \times 10^7 \quad \longrightarrow \quad f = 10^7 = \underline{\underline{10 \text{ MHz}}}$$

$$\beta = \omega \sqrt{\mu\epsilon} = \omega \sqrt{\mu_o \epsilon_o} \sqrt{\epsilon_r} = \frac{\omega}{c} \sqrt{\epsilon_r}$$

$$(c) \quad \sqrt{\epsilon_r} = \frac{\beta c}{\omega} = \frac{3 \times 3 \times 10^8}{2\pi \times 10^7} = 14.32 \quad \longrightarrow \quad \epsilon_r = \underline{\underline{205.18}}$$

$$\text{Let } \mathbf{H} = H_o \sin(\omega t - 3y) \mathbf{a}_H$$

$$H_o = \frac{E_o}{\eta}, \quad \eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{\eta_o}{\sqrt{\epsilon_r}} = \frac{120\pi}{14.32} = 26.33$$

$$(d) \quad H_o = \frac{12}{26.33} = 0.456$$

$$\mathbf{a}_H = \mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_y \times \mathbf{a}_z = \mathbf{a}_x$$

$$\underline{\underline{\mathbf{H} = 0.456 \sin(2\pi \times 10^7 t - 3y) \mathbf{a}_x \text{ A/m}}}$$

Prob. 10.36

$$\mathbf{E} = (2\mathbf{a}_y - 5\mathbf{a}_z) \sin(\omega t - \beta x)$$

The ratio E_y / E_z remains the same as t changes. Hence the wave is linearly polarized

Prob. 10.37

(a)

$$E_x = E_o \cos(\omega t + \beta y), \quad E_y = E_o \sin(\omega t + \beta y)$$

$$E_x(0, t) = E_o \cos \omega t \quad \longrightarrow \quad \cos \omega t = \frac{E_x(0, t)}{E_o}$$

$$E_y(0, t) = E_o \sin \omega t \quad \longrightarrow \quad \sin \omega t = \frac{E_y(0, t)}{E_o}$$

$$\cos^2 \omega t + \sin^2 \omega t = 1 \quad \longrightarrow \quad \left(\frac{E_x}{E_o} \right)^2 + \left(\frac{E_y}{E_o} \right)^2 = 1$$

Hence, we have circular polarization.

(b)

$$E_x = E_o \cos(\omega t - \beta y), \quad E_y = -3E_o \sin(\omega t - \beta y)$$

In the $y=0$ plane,

$$E_x(0, t) = E_o \cos \omega t \quad \longrightarrow \quad \cos \omega t = \frac{E_x(0, t)}{E_o}$$

$$E_y(0, t) = E_o \sin \omega t \quad \longrightarrow \quad \sin \omega t = \frac{-E_y(0, t)}{3E_o}$$

$$\cos^2 \omega t + \sin^2 \omega t = 1 \quad \longrightarrow \quad \left(\frac{E_x}{E_o} \right)^2 + \frac{1}{9} \left(\frac{E_y}{E_o} \right)^2 = 1$$

Hence, we have elliptical polarization.

Prob. 10.38

(a) We can write

$$\mathbf{E} = \text{Re}(E_s e^{j\omega t}) = (40\mathbf{a}_x + 60\mathbf{a}_y) \cos(\omega t - 10z)$$

Since E_x / E_y does not change with time, the wave is linearly polarized.

(b) This is elliptically polarized.

Prob. 10.39

(a) The wave is elliptically polarized.

(b)

$$\text{Let } \mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2,$$

$$\text{where } \mathbf{E}_1 = 40 \cos(\omega t - \beta z) \mathbf{a}_x, \quad \mathbf{E}_2 = 60 \sin(\omega t - \beta z) \mathbf{a}_y$$

$$\mathbf{H}_1 = H_{o1} \cos(\omega t - \beta z) \mathbf{a}_{H1}$$

$$H_{o1} = \frac{40}{\eta_o} = \frac{40}{120\pi} = 0.106$$

$$\mathbf{a}_{H1} = \mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y$$

$$\mathbf{H}_1 = 0.106 \cos(\omega t - \beta z) \mathbf{a}_y$$

$$\mathbf{H}_2 = H_{o2} \sin(\omega t - \beta z) \mathbf{a}_{H2}$$

$$H_{o2} = \frac{60}{\eta_o} = \frac{60}{120\pi} = 0.1592$$

$$\mathbf{a}_{H2} = \mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_z \times \mathbf{a}_y = -\mathbf{a}_x$$

$$\mathbf{H}_2 = -0.1592 \sin(\omega t - \beta z) \mathbf{a}_x$$

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 = \underline{\underline{-159.2 \sin(\omega t - \beta z) \mathbf{a}_x + 106 \cos(\omega t - \beta z) \mathbf{a}_y \text{ mA/m}}}$$

Prob. 10.40

We can write $\mathbf{E}_s$ as

$$\mathbf{E}_s = \mathbf{E}_1(z) + \mathbf{E}_2(z)$$

where

$$\mathbf{E}_1(z) = \frac{1}{2} E_o (\mathbf{a}_x - j \mathbf{a}_y) e^{-j\beta z}$$

$$\mathbf{E}_2(z) = \frac{1}{2} E_o (\mathbf{a}_x + j \mathbf{a}_y) e^{-j\beta z}$$

We recognize that $\mathbf{E}_1$ and $\mathbf{E}_2$ are circularly polarized waves. The problem is therefore proved.

Prob. 10.41

(a)

When $\phi = 0$,

$$\mathbf{E}(y, t) = (E_{o1} \mathbf{a}_x + E_{o2} \mathbf{a}_z) \cos(\omega t - \beta y)$$

The two components are in phase and the wave is linearly polarized.

(b)

When $\phi = \pi/2$,

$$\mathbf{E}_z = E_{o2} \cos(\omega t - \beta y + \pi/2) = -E_{o2} \sin(\omega t - \beta y)$$

We can combine E_x and E_z to show that the wave is elliptically polarized.

(c)

When $\phi = \pi$,

$$\begin{aligned}\mathbf{E}(y, t) &= E_{o1} \cos(\omega t - \beta y) \mathbf{a}_x + E_{o2} \cos(\omega t - \beta y + \pi) \mathbf{a}_z \\ &= (E_{o1} \mathbf{a}_x - E_{o2} \mathbf{a}_z) \cos(\omega t - \beta y)\end{aligned}$$

Thus, the wave is linearly polarized.**Prob. 10.42**

$$\text{Let } \mathbf{E}_s = \mathbf{E}_r + j\mathbf{E}_i \quad \text{and} \quad \mathbf{H}_s = \mathbf{H}_r + j\mathbf{H}_i$$

$$\mathbf{E} = \text{Re}(\mathbf{E}_s e^{j\omega t}) = \mathbf{E}_r \cos \omega t - \mathbf{E}_i \sin \omega t$$

Similarly,

$$\mathbf{H} = \mathbf{H}_r \cos \omega t - \mathbf{H}_i \sin \omega t$$

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} = \mathbf{E}_r \times \mathbf{H}_r \cos^2 \omega t + \mathbf{E}_i \times \mathbf{H}_i \sin^2 \omega t - \frac{1}{2}(\mathbf{E}_r \times \mathbf{H}_i + \mathbf{E}_i \times \mathbf{H}_r) \sin 2\omega t$$

$$\begin{aligned}\mathcal{P}_{\text{ave}} &= \frac{1}{T} \int_0^T \mathcal{P} dt = \frac{1}{T} \int_0^T \cos^2 \omega t dt (\mathbf{E}_r \times \mathbf{H}_r) + \frac{1}{T} \int_0^T \sin^2 \omega t dt (\mathbf{E}_i \times \mathbf{H}_i) - \frac{1}{2T} \int_0^T \sin 2\omega t dt (\mathbf{E}_i \times \mathbf{H}_i + \mathbf{E}_i \times \mathbf{H}_r) \\ &= \frac{1}{2}(\mathbf{E}_r \times \mathbf{H}_r + \mathbf{E}_i \times \mathbf{H}_i) = \frac{1}{2} \text{Re}[(\mathbf{E}_r + j\mathbf{E}_i) \times (\mathbf{H}_r - j\mathbf{H}_i)]\end{aligned}$$

$$\mathcal{P}_{\text{ave}} = \frac{1}{2} \text{Re}(\mathbf{E}_s \times \mathbf{H}_s^*)$$

as required.

Prob. 10.43

(a)

$$\beta = \omega \sqrt{\mu \epsilon} = \omega \sqrt{\mu_o \epsilon_o} \sqrt{\epsilon_r} = \frac{\omega}{c} \sqrt{\epsilon_r}$$

$$\sqrt{\epsilon_r} = \frac{\beta c}{\omega} = \frac{8 \times 3 \times 10^8}{10^9} = 2.4$$

$$\epsilon_r = \underline{\underline{5.76}}$$

$$(b) \quad \eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_o}{\epsilon_o}} \frac{1}{\sqrt{\epsilon_r}} = \frac{377}{2.4} = \underline{\underline{157.1 \Omega}}$$

$$(c) \quad u = \frac{\omega}{\beta} = \frac{10^9}{8} = \underline{\underline{1.25 \times 10^8 \text{ m/s}}}$$

(d)

Let $\mathbf{H} = H_o \cos(10^9 t + 8x) \mathbf{a}_H$

$$H_o = \frac{E_o}{\eta} = \frac{150}{157.1} = 0.955$$

$$\mathbf{a}_H = \mathbf{a}_k \times \mathbf{a}_E = -\mathbf{a}_x \times \mathbf{a}_z = \mathbf{a}_y$$

$$\mathbf{H} = \underline{\underline{0.955 \cos(10^9 t + 8x) \mathbf{a}_y \text{ A/m}}}$$

(e)

$$\begin{aligned} \mathcal{P} &= \mathbf{E} \times \mathbf{H} = -150(0.955) \cos^2(10^9 t + 8x) \mathbf{a}_x \\ &= \underline{\underline{-143.25 \cos^2(10^9 t + 8x) \mathbf{a}_x \text{ W/m}^2}} \end{aligned}$$

Prob. 10.44

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} = \frac{E_o^2}{\eta} \cos^2(\omega t - 10z) \mathbf{a}_z$$

$$\mathcal{P}_{ave} = \frac{1}{T} \int_0^T \mathcal{P} dt = \frac{E_o^2}{2\eta} \mathbf{a}_z$$

$$P = \int_S \mathcal{P}_{ave} \cdot d\mathbf{S} = \frac{E_o^2 S}{2\eta} = \frac{(40)^2 \times \pi (1.5)^2}{2 \times 120\pi} = \frac{(60)^2}{240} = \underline{\underline{15 \text{ W}}}$$

Prob. 10.45

(a)

Let $\mathbf{H}_s = \frac{H_o}{r} \sin \theta e^{-j3r} \mathbf{a}_H$

$$H_o = \frac{E_o}{\eta_o} = \frac{10}{120\pi} = \frac{1}{12\pi}$$

$$\mathbf{a}_H = \mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_r \times \mathbf{a}_\theta = \mathbf{a}_\phi$$

$$\mathbf{H}_s = \underline{\underline{\frac{1}{12\pi r} \sin \theta e^{-j3r} \mathbf{a}_\phi \text{ A/m}}}$$

(b)

$$\mathcal{P}_{ave} = \frac{1}{2} \text{Re}(\mathbf{E}_s \times \mathbf{H}_s) = \frac{10}{2 \times 12\pi r^2} \sin^2 \theta \mathbf{a}_r$$

$$P_{ave} = \int_S \mathcal{P}_{ave} \cdot d\mathbf{S}, \quad d\mathbf{S} = r^2 \sin \theta d\theta d\phi \mathbf{a}_r$$

$$\begin{aligned} P_{ave} &= \frac{10}{24\pi} \int_{\phi=0}^{\pi} \int_{\theta=0}^{\pi/6} r^2 \sin^3 \theta d\theta \bigg|_{r=2} = \frac{5}{8} - \frac{5\sqrt{3}}{32} = 0.007145 \\ &= \underline{\underline{7.145 \text{ mW}}} \end{aligned}$$

Prob. 10.46

$$(a) \quad P_{ave} = \frac{1}{2} \operatorname{Re}(E_s H_s^*) = \frac{1}{2} \operatorname{Re}\left(\frac{|E_s|}{|\eta|}\right) = \frac{8^2}{2|\eta|} e^{-0.2z}$$

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1 \right]}$$

$$\beta = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} + 1 \right]}$$

$$\text{Let } x = \sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}$$

$$\frac{\alpha}{\beta} = \frac{\sqrt{x-1}}{\sqrt{x+1}} = 0.1/0.3 = 1/3$$

$$\frac{x-1}{x+1} = \frac{1}{9} \quad \longrightarrow \quad x = 5/4$$

$$\frac{5}{4} = \sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} \quad \longrightarrow \quad \frac{\sigma}{\omega\epsilon} = 3/4$$

$$|\eta| = \frac{\sqrt{\frac{\mu}{\epsilon}}}{\sqrt[4]{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}} = \frac{120\pi / \sqrt{81}}{\sqrt[4]{\frac{5}{4}}} = 37.4657$$

$$P_{ave} = \frac{64}{2(37.4657)} e^{-0.2z} = \underline{\underline{0.8541 e^{-0.2z} \text{ W/m}^2}}$$

$$(b) \quad 20\text{dB} = 10 \log \frac{P_1}{P_2} \quad \longrightarrow \quad \frac{P_1}{P_2} = 100$$

$$\frac{P_2}{P_1} = e^{-0.2z} = \frac{1}{100} \quad \longrightarrow \quad e^{0.2z} = 100$$

$$z = 5 \log 100 = \underline{\underline{23 \text{ m}}}$$

Prob. 10.47

$$(a) \quad u = \omega / \beta \quad \longrightarrow \quad \omega = u\beta = \frac{\beta c}{\sqrt{4.5}} = \frac{2 \times 3 \times 10^8}{\sqrt{4.5}} = \underline{\underline{2.828 \times 10^8}} \text{ rad/s}$$

$$\eta = \frac{120\pi}{\sqrt{4.5}} = 177.7 \Omega$$

$$\mathbf{H} = \mathbf{a}_k \times \frac{\mathbf{E}}{\eta} = \frac{\mathbf{a}_z}{\eta} \times \frac{40}{\rho} \sin(\omega t - 2z) \mathbf{a}_\rho = \frac{0.225}{\rho} \sin(\omega t - 2z) \mathbf{a}_\phi \text{ A/m}$$

$$(b) \quad \mathbf{P} = \mathbf{E} \times \mathbf{H} = \frac{9}{\rho^2} \sin^2(\omega t - 2z) \mathbf{a}_z \text{ W/m}^2$$

$$(c) \quad \mathcal{P}_{\text{ave}} = \frac{4.5}{\rho^2} \mathbf{a}_z, \quad d\mathbf{S} = \rho d\phi d\rho \mathbf{a}_z$$

$$\mathbf{P}_{\text{ave}} = \int \mathbf{P}_{\text{ave}} \cdot d\mathbf{S} = 4.5 \int_{2\text{mm}}^{3\text{mm}} \frac{d\rho}{\rho} \int_0^{2\pi} d\phi = 4.5 \ln(3/2) (2\pi) = \underline{\underline{11.46 \text{ W}}}$$

Prob. 10.48

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} = \frac{E_o^2}{\eta r^2} \sin^2 \theta \sin^2 \omega(t - r/c) \mathbf{a}_r$$

$$\mathbf{P}_{\text{ave}} = \frac{1}{T} \int_0^T \mathcal{P} dt = \frac{E_o^2}{2\eta r^2} \sin^2 \theta \mathbf{a}_r$$

Prob. 10.49

$$\beta = \frac{\omega}{c} \quad \longrightarrow \quad \omega = \beta c = 40(3 \times 10^8) = \underline{\underline{12 \times 10^9}} \text{ rad/s}$$

$$\mathbf{E} = \frac{1}{\epsilon} \int \nabla \times \mathbf{H} dt$$

$$\begin{aligned} \nabla \times \mathbf{H} &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 10 \sin(\omega t - 40x) & -20 \sin(\omega t - 40x) \end{vmatrix} \\ &= -800 \cos(\omega t - 40x) \mathbf{a}_y - 400 \cos(\omega t - 40x) \mathbf{a}_z \end{aligned}$$

$$\begin{aligned}
\mathbf{E} &= \frac{1}{\epsilon} \int \nabla \times \mathbf{H} dt = -\frac{800}{\omega \epsilon} \sin(\omega t - 40x) \mathbf{a}_y - \frac{400}{\omega \epsilon} \sin(\omega t - 40x) \mathbf{a}_z \\
&= -\frac{800}{12 \times 10^9 \times \frac{10^{-9}}{36\pi}} \sin(\omega t - 40x) \mathbf{a}_y - \frac{400}{12 \times 10^9 \times \frac{10^{-9}}{36\pi}} \sin(\omega t - 40x) \mathbf{a}_z \\
&= -7.539 \sin(\omega t - 40x) \mathbf{a}_y - 3.77 \sin(\omega t - 40x) \mathbf{a}_z \text{ kV/m}
\end{aligned}$$

$$\begin{aligned}
\mathbf{P} &= \mathbf{E} \times \mathbf{H} = \begin{vmatrix} 0 & E_y & E_z \\ 0 & H_y & H_z \end{vmatrix} = (E_y H_z - E_z H_y) \mathbf{a}_x \\
&= [20(7.537) \sin^2(\omega t - 40x) + 37.7 \sin^2(\omega t - 40x)] \mathbf{a}_x 10^3 \\
\mathbf{P}_{\text{ave}} &= \frac{1}{2} [20(7.537) + 37.7] \mathbf{a}_x 10^3 = \underline{\underline{94.23 \mathbf{a}_x \text{ kW/m}^2}}
\end{aligned}$$

Prob. 10.50

$$P = \frac{E_o^2}{2\eta_o} \longrightarrow E_o^2 = 2\eta_o P = 2(120\pi)10 \times 10^{-3} = 7.539$$

$$E_o = \underline{\underline{2.746 \text{ V/m}}}$$

Prob. 10.51

Let $T = \omega t - \beta z$.

$$-\frac{\partial \mathbf{B}}{\partial t} = \nabla \times \mathbf{E} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \cos T & \sin T & 0 \end{vmatrix}$$

$$-\mu \frac{\partial \mathbf{H}}{\partial t} = \beta \cos T \mathbf{a}_x + \beta \sin T \mathbf{a}_y$$

$$\mathbf{H} = -\frac{\beta}{\mu} \int [\cos T \mathbf{a}_x + \sin T \mathbf{a}_y] dt = -\frac{\beta}{\mu \omega} \sin T \mathbf{a}_x + \frac{\beta}{\mu \omega} \cos T \mathbf{a}_y$$

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} = \begin{vmatrix} \cos T & \sin T & 0 \\ -\frac{\beta}{\mu \omega} \sin T & \frac{\beta}{\mu \omega} \cos T & 0 \end{vmatrix} = \frac{\beta}{\mu \omega} (\cos^2 T + \sin^2 T) \mathbf{a}_z$$

$$= \frac{\beta}{\mu \omega} \mathbf{a}_z = \sqrt{\frac{\epsilon}{\mu}} \mathbf{a}_z$$

which is constant everywhere.

Prob. 10.52

$$\mathcal{P} = \frac{E_o^2}{2\eta_o}$$

$$P = \mathcal{P}S = \frac{E_o^2 S}{2\eta_o} = \frac{(2.4 \times 10^3)^2 \times 450 \times 10^{-4}}{2 \times 377} = \underline{\underline{343.8 \text{ W}}}$$

Prob. 10.53

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} = \frac{V_o I_o}{2\pi \rho^2 \ln(b/a)} \sin^2(\omega t - \beta z) \mathbf{a}_z$$

$$\begin{aligned} \text{(a)} \quad \mathcal{P}_{ave} &= \frac{1}{T} \int_0^T \mathcal{P} dt = \frac{V_o I_o}{2\pi \rho^2 \ln(b/a)} \frac{1}{T} \int_0^T \sin^2(\omega t - \beta z) dt \mathbf{a}_z = \frac{V_o I_o}{2\pi \rho^2 \ln(b/a)} \frac{1}{2} \mathbf{a}_z \\ &= \underline{\underline{\frac{V_o I_o}{4\pi \rho^2 \ln(b/a)} \mathbf{a}_z}} \end{aligned}$$

(b)

$$\begin{aligned} P_{ave} &= \int_S \mathcal{P}_{ave} \cdot d\mathbf{S}, \quad d\mathbf{S} = \rho d\rho d\phi \mathbf{a}_z \\ &= \frac{V_o I_o}{4\pi \ln(b/a)} \int_{\phi=0}^{2\pi} \int_{\rho=a}^b \frac{1}{\rho^2} \rho d\rho d\phi = \frac{V_o I_o}{4\pi \ln(b/a)} (2\pi) \ln(b/a) \\ &= \underline{\underline{\frac{1}{2} V_o I_o}} \end{aligned}$$

Prob. 10.54

$$\text{(a)} \quad P_{i,ave} = \frac{E_{io}^2}{2\eta_1}, \quad P_{r,ave} = \frac{E_{ro}^2}{2\eta_1}, \quad P_{t,ave} = \frac{E_{to}^2}{2\eta_2}$$

$$R = \frac{P_{r,ave}}{P_{i,ave}} = \frac{E_{ro}^2}{E_{io}^2} = \Gamma^2 = \underline{\underline{\left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \right)^2}}$$

$$R = \left(\frac{\sqrt{\frac{\mu_o}{\epsilon_2}} - \sqrt{\frac{\mu_o}{\epsilon_1}}}{\sqrt{\frac{\mu_o}{\epsilon_2}} + \sqrt{\frac{\mu_o}{\epsilon_1}}} \right)^2 = \left(\frac{\sqrt{\mu_o \epsilon_1} - \sqrt{\mu_o \epsilon_2}}{\sqrt{\mu_o \epsilon_1} + \sqrt{\mu_o \epsilon_2}} \right)^2$$

$$\text{Since } \eta_1 = c\sqrt{\mu_1 \epsilon_1} = c\sqrt{\mu_o \epsilon_1}, \quad \eta_2 = c\sqrt{\mu_o \epsilon_2},$$

$$R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

$$T = \frac{P_{t,ave}}{P_{i,ave}} = \frac{\eta_1}{\eta_2} \frac{E_{to}^2}{E_{io}^2} = \frac{\eta_1}{\eta_2} \tau^2 = \frac{\eta_1}{\eta_2} (1 + \Gamma)^2 = \frac{4n_1 n_2}{(n_1 + n_2)^2}$$

$$(b) \text{ If } P_{r,ave} = P_{t,ave} \longrightarrow RP_{i,ave} = TP_{i,ave} \longrightarrow R = T$$

$$\text{i.e. } (n_1 - n_2)^2 = 4n_1 n_2 \longrightarrow n_1^2 - 6n_1 n_2 + n_2^2 = 0$$

$$\text{or } \left(\frac{n_1}{n_2} \right)^2 - 6 \left(\frac{n_1}{n_2} \right) + 1 = 0, \text{ so}$$

$$\frac{n_1}{n_2} = 3 \pm \sqrt{8} = \underline{\underline{5.828}} \quad \text{or} \quad \underline{\underline{0.1716}}$$

(Note that these values are mutual reciprocals, reflecting the inherent symmetry of the problem.)

Prob. 10.55

$$\eta_1 = \sqrt{\frac{\mu_1}{\epsilon_1}} = \sqrt{\frac{2\mu_o}{8\epsilon_o}} = \frac{\eta_o}{2}, \quad \eta_2 = \sqrt{\frac{\mu_2}{\epsilon_2}} = \sqrt{\frac{\mu_o}{16\epsilon_o}} = \frac{\eta_o}{4}$$

$$\Gamma = \frac{\eta_1 - \eta_2}{\eta_1 + \eta_2} = \frac{\eta_o/4 - \eta_o/2}{\eta_o/4 + \eta_o/2} = -\frac{1}{3}, \quad \tau = \Gamma + 1 = \frac{2}{3}$$

$$\begin{aligned} \mathbf{E}_r &= -\frac{1}{3}(60)\sin(\omega t + 10z)\mathbf{a}_x - \frac{1}{3}(30)\sin(\omega t + 10z + \pi/6)\mathbf{a}_y \\ &= -20\sin(\omega t + 10z)\mathbf{a}_x - 10\sin(\omega t + 10z + \pi/6)\mathbf{a}_y \text{ V/m} \end{aligned}$$

$$\begin{aligned} \mathbf{E}_t &= \frac{2}{3}(60)\sin(\omega t - 10z)\mathbf{a}_x + \frac{2}{3}(30)\sin(\omega t - 10z + \pi/6)\mathbf{a}_y \\ &= 40\sin(\omega t - 10z)\mathbf{a}_x + 20\sin(\omega t - 10z + \pi/6)\mathbf{a}_y \text{ V/m} \end{aligned}$$

Prob. 10.56

$$\eta_1 = \eta_o = 120\pi, \quad \eta_2 = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_o}{9\epsilon_o}} = \frac{\eta_o}{3} = 40\pi$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{\eta_o/3 - \eta_o}{\eta_o/3 + \eta_o} = -\frac{1}{2}, \quad \tau = 1 + \Gamma = \frac{1}{2}$$

$$E_r = \Gamma E_o = -E_o/2, \quad E_t = \tau E_o = E_o/2$$

$$P_{iave} = \frac{|E_o|^2}{2\eta_1} = \frac{E_o^2}{2\eta_o}$$

$$P_{tave} = \frac{|E_t|^2}{2\eta_2} = \frac{\frac{1}{4}E_o^2}{2(\eta_o/3)} = \frac{E_o^2}{2\eta_o} \frac{3}{4}$$

$$\frac{P_{tave}}{P_{iave}} = \frac{3}{4} = \underline{\underline{0.75}}$$

Prob. 10.57

$$(a) \quad \eta_1 = \eta_o$$

$$\mathbf{E}_i = E_{io} \sin(\omega t - 5x) \mathbf{a}_E$$

$$E_{io} = H_{io} \eta_o = 120\pi \times 4 = 480\pi$$

$$\mathbf{a}_E \times \mathbf{a}_H = \mathbf{a}_k \longrightarrow \mathbf{a}_E \times \mathbf{a}_y = \mathbf{a}_x \longrightarrow \mathbf{a}_E = -\mathbf{a}_z$$

$$\mathbf{E}_i = -480\pi \sin(\omega t - 5x) \mathbf{a}_z$$

$$\eta_2 = \sqrt{\frac{\mu_o}{4\epsilon_o}} = \frac{120\pi}{\sqrt{4}} = 60\pi$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{60\pi - 120\pi}{60\pi + 120\pi} = -1/3, \quad \tau = 1 + \Gamma = 2/3$$

$$E_{ro} = \Gamma E_{io} = (-1/3)(-480\pi) = 160\pi$$

$$\mathbf{E}_r = 160\pi \sin(\omega t + 5x) \mathbf{a}_z$$

$$\mathbf{E}_1 = \mathbf{E}_i + \mathbf{E}_r = \underline{\underline{-1.508 \sin(\omega t - 5x) \mathbf{a}_z + 0.503 \sin(\omega t + 5x) \mathbf{a}_z}} \quad \text{kV/m}$$

$$(b) \quad E_{to} = \tau E_{io} = (2/3)(480\pi) = 320\pi$$

$$\mathcal{P} = \frac{E_{to}^2}{2\eta_2} \mathbf{a}_x = \frac{(320\pi)^2}{2(60\pi)} \mathbf{a}_x = \underline{\underline{2.68\mathbf{a}_x \text{ kW/m}^2}}$$

$$(c) \quad s = \frac{1+|\Gamma|}{1-|\Gamma|} = \frac{1+1/3}{1-1/3} = 2$$

$$\text{Prob. 10.58} \quad \eta_1 = \sqrt{\frac{\mu_1}{\epsilon_1}} = \eta_o / 2, \quad \eta_2 = \eta_o$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = 1/3, \quad \tau = 1 + \Gamma = 4/3$$

$$E_{or} = \Gamma E_{io} = (1/3)(5) = 5/3, \quad E_{ot} = \tau E_{io} = 20/3$$

$$\beta = \frac{\omega}{c} \sqrt{\mu_r \epsilon_r} = \frac{10^8}{3 \times 10^8} \sqrt{4} = 2/3$$

$$(a) \quad \mathbf{E}_r = \frac{5}{3} \cos(10^8 t - 2y/3) \mathbf{a}_z$$

$$\underline{\underline{\mathbf{E}_1 = \mathbf{E}_i + \mathbf{E}_r = 5 \cos(10^8 t + \frac{2}{3}y) \mathbf{a}_z + \frac{5}{3} \cos(10^8 t - \frac{2}{3}y) \mathbf{a}_z \text{ V/m}}}$$

$$(b) \quad \mathcal{P}_{\text{ave1}} = \frac{E_{io}^2}{2\eta_1} (-\mathbf{a}_y) + \frac{E_{ro}^2}{2\eta_1} (+\mathbf{a}_y) = \frac{25}{2(60\pi)} (1 - \frac{1}{9}) (-\mathbf{a}_y) = \underline{\underline{-0.0589\mathbf{a}_y \text{ W/m}^2}}$$

$$(c) \quad \mathcal{P}_{\text{ave2}} = \frac{E_{to}^2}{2\eta_2} (-\mathbf{a}_y) = \frac{400}{9(2)(120\pi)} (-\mathbf{a}_y) = \underline{\underline{-0.0589\mathbf{a}_y \text{ W/m}^2}}$$

Prob. 10.59

$$\beta_1 = \omega \sqrt{\mu_1 \epsilon_1} = \omega \sqrt{\mu_o \epsilon_o} \sqrt{\mu_{r1} \epsilon_{r1}} = \frac{\omega}{c} \sqrt{16} = \frac{90 \times 10^9 (4)}{3 \times 10^8} = 300(4) = 1200$$

$$\beta_2 = \omega \sqrt{\mu_o \epsilon_o} = \frac{90 \times 10^9}{3 \times 10^8} = 300$$

$$\eta_1 = \sqrt{\frac{\mu_1}{\epsilon_1}} = \sqrt{\frac{\mu_o}{\epsilon_o}} \sqrt{\frac{\mu_{r1}}{\epsilon_{r1}}} = \frac{\eta_o}{4} = 30\pi, \quad \eta_2 = \eta_o = 120\pi$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{\eta_o - \eta_o/4}{\eta_o + \eta_o/4} = \frac{3}{5}, \quad \tau = 1 + \Gamma = \frac{8}{5}$$

$$\mathcal{P} = \mathbf{E} \times \mathbf{H}$$

$$\mathcal{P}_1 = \mathcal{P}_i + \mathcal{P}_r = \frac{60^2}{\eta_1} \cos^2(\omega t - \beta_1 z) \mathbf{a}_x + \frac{(60 \times \frac{3}{5})^2}{\eta_1} \cos^2(\omega t + \beta_1 z) \mathbf{a}_x$$

$$\frac{60^2}{\eta_1} = \frac{60^2}{30\pi} = 38.197, \quad \frac{(60 \times \frac{3}{5})^2}{\eta_1} = 13.75$$

$$\underline{\underline{\mathcal{P}_1 = 38.197 \cos^2(\omega t - \beta_1 z) \mathbf{a}_x + 13.75 \cos^2(\omega t + \beta_1 z) \mathbf{a}_x \text{ W/m}^2, \text{ where } \beta_1 = 1200}}$$

$$\mathcal{P}_2 = \mathcal{P}_t = \frac{(60 \times \frac{8}{5})^2}{\eta_o} \cos^2(\omega t - \beta_2 z) \mathbf{a}_x$$

$$\frac{(60 \times \frac{8}{5})^2}{\eta_o} = \frac{96^2}{120\pi} = 24.46$$

$$\underline{\underline{\mathcal{P}_2 = 24.46 \cos^2(\omega t - \beta_2 z) \mathbf{a}_x \text{ W/m}^2, \text{ where } \beta_2 = 300}}$$

Prob. 10.60

(a) In air, $\beta_1 = 1, \lambda_1 = 2\pi / \beta_1 = 2\pi = \underline{\underline{6.283 \text{ m}}}$

$$\omega = \beta_1 c = \underline{\underline{3 \times 10^8 \text{ rad/s}}}$$

In the dielectric medium, ω is the same .

$$\omega = \underline{\underline{3 \times 10^8 \text{ rad/s}}}$$

$$\beta_2 = \frac{\omega}{c} \sqrt{\epsilon_{r2}} = \beta_1 \sqrt{\epsilon_{r2}} = \sqrt{3}$$

$$\lambda_2 = \frac{2\pi}{\beta_2} = \frac{2\pi}{\sqrt{3}} = \underline{\underline{3.6276 \text{ m}}}$$

$$(b) \quad H_o = \frac{E_o}{\eta_o} = \frac{10}{120\pi} = 0.0265$$

$$\mathbf{a}_H = \mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_z \times \mathbf{a}_y = -\mathbf{a}_x$$

$$\underline{\underline{\mathbf{H}_i = -26.5 \cos(\omega t - z) \mathbf{a}_x \text{ mA/m}}}$$

$$(c) \quad \eta_1 = \eta_o, \quad \eta_2 = \eta_o / \sqrt{3}$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{(1/\sqrt{3}) - 1}{(1/\sqrt{3}) + 1} = \underline{\underline{-0.268}}, \quad \tau = 1 + \Gamma = \underline{\underline{0.732}}$$

$$(d) \quad E_{to} = \tau E_{io} = 7.32, \quad E_{ro} = \Gamma E_{io} = -2.68$$

$$\underline{\underline{\mathbf{E}_1 = \mathbf{E}_i + \mathbf{E}_r = 10 \cos(\omega t - z) \mathbf{a}_y - 2.68 \cos(\omega t + z) \mathbf{a}_y \text{ V/m}}}$$

$$\underline{\underline{\mathbf{E}_2 = \mathbf{E}_t = 7.32 \cos(\omega t - z) \mathbf{a}_y \text{ V/m}}}$$

$$\mathcal{P}_{\text{ave1}} = \frac{1}{2\eta_1} (\mathbf{a}_z) [E_{io}^2 - E_{ro}^2] = \frac{1}{2(120\pi)} (\mathbf{a}_z) (10^2 - 2.68^2) = \underline{\underline{0.1231 \mathbf{a}_z \text{ W/m}^2}}$$

$$\mathcal{P}_{\text{ave2}} = \frac{E_{to}^2}{2\eta_2} (\mathbf{a}_z) = \frac{\sqrt{3}}{2 \times 120\pi} (7.32)^2 (\mathbf{a}_z) = \underline{\underline{0.1231 \mathbf{a}_z \text{ W/m}^2}}$$

Prob. 10.61

$$\eta_1 = \eta_o = 120\pi$$

For seawater (lossy medium),

$$\eta_2 = \sqrt{\frac{j\omega\mu_o}{\sigma + j\omega\epsilon}} = \sqrt{\frac{j2\pi \times 10^8 \times 4}{4 + j2\pi \times 10^8 \times 81 \times \frac{10^{-9}}{36\pi}}} = 10.44 + j9.333$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = 0.9461 \angle 177.16$$

$$|\Gamma|^2 = 0.8952, \quad 1 - |\Gamma| = 0.1084$$

$$\frac{P_r}{P_i} = \underline{\underline{89.51\%}}, \quad \frac{P_t}{P_i} = \underline{\underline{10.84\%}},$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{7.924 \angle 43.975 - 377}{7.924 \angle 43.975 + 377} = 0.9702 \angle 178.2^\circ$$

The fraction of the incident power reflected is

$$\frac{P_r}{P_i} = |\Gamma|^2 = 0.9702^2 = \underline{\underline{0.9413}}$$

The transmitted fraction is

$$\frac{P_t}{P_i} = 1 - |\Gamma|^2 = 1 - 0.9702^2 = \underline{\underline{0.0587}}$$

Prob. 10.62

(a)

$$\eta_1 = \sqrt{\frac{\mu_1}{\epsilon_1}} = \frac{120\pi}{\sqrt{4}} = 188.5, \quad \eta_2 = \sqrt{\frac{\mu_2}{\epsilon_2}} = \frac{120\pi}{\sqrt{3.2}} = 210.75$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{210.75 - 188.5}{210.75 + 188.5} = 0.0557, \quad \tau = \frac{2\eta_2}{\eta_2 + \eta_1} = \frac{2 \times 210.75}{210.75 + 188.5} = 1.0557$$

$$E_{ro} = \Gamma E_{io} = (0.0557)(12) = 0.6684 \quad E_{to} = \tau E_{io} = 1.0557(12) = 12.668$$

$$\beta_1 = \omega \sqrt{\mu_1 \epsilon_1} = \frac{\omega}{c} \sqrt{4} \quad \longrightarrow \quad \omega = \frac{\beta_1 c}{2} = \frac{40\pi(3 \times 10^8)}{2} = \underline{\underline{6\pi \times 10^9 \text{ rad/s}}}$$

(b)

$$\beta_2 = \omega \sqrt{\mu_2 \epsilon_2} = \frac{\omega}{c} \sqrt{3.2} = \frac{6\pi \times 10^9 \sqrt{3.2}}{3 \times 10^8} = 112.4$$

$$\mathbf{E}_r = E_{ro} \cos(\omega t + 40\pi x) \mathbf{a}_z = \underline{\underline{0.6684 \cos(6\pi \times 10^9 t + 40\pi x) \text{ V/m}}}$$

$$\mathbf{E}_t = E_{to} \cos(\omega t - \beta_2 x) \mathbf{a}_z = \underline{\underline{12.668 \cos(6\pi \times 10^9 t - 112.4x) \text{ V/m}}}$$

Prob. 10.63 (a) $\omega = \beta c = 3 \times 3 \times 10^8 = \underline{\underline{9 \times 10^8 \text{ rad/s}}}$

(b) $\lambda = 2\pi / \beta = 2\pi / 3 = \underline{\underline{2.094 \text{ m}}}$

(c) $\frac{\sigma}{\omega \epsilon} = \frac{4}{9 \times 10^8 \times 80 \times 10^{-9} / 36\pi} = 2\pi = \underline{\underline{6.288}}$

$$\tan 2\theta_\eta = \frac{\sigma}{\omega \epsilon} = 6.288 \quad \longrightarrow \quad \theta_\eta = 40.47^\circ$$

$$|\eta_2| = \frac{\sqrt{\mu_2 / \epsilon_2}}{\sqrt[4]{1 + \left(\frac{\sigma_2}{\omega \epsilon_2}\right)^2}} = \frac{377 / \sqrt{80}}{\sqrt[4]{1 + 4\pi^2}} = 16.71$$

$$\eta_2 = \underline{\underline{16.71 \angle 40.47^\circ \Omega}}$$

$$(d) \quad \Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{16.71 \angle 40.47^\circ - 377}{16.71 \angle 40.47^\circ + 377} = 0.935 \angle 179.7^\circ$$

$$E_{or} = \Gamma E_{oi} = 9.35 \angle 179.7^\circ$$

$$\underline{\underline{E_r = 9.35 \sin(\omega t - 3z + 179.7) \mathbf{a}_x \text{ V/m}}}$$

$$\alpha_2 = \frac{\omega}{c} \sqrt{\frac{\mu_{r2} \epsilon_{r2}}{2} \left[\sqrt{1 + \left(\frac{\sigma_2}{\omega \epsilon_2} \right)^2} - 1 \right]} = \frac{9 \times 10^8}{3 \times 10^8} \sqrt{\frac{80}{2} \left[\sqrt{1 + 4\pi^2} - 1 \right]} = 43.94 \text{ Np/m}$$

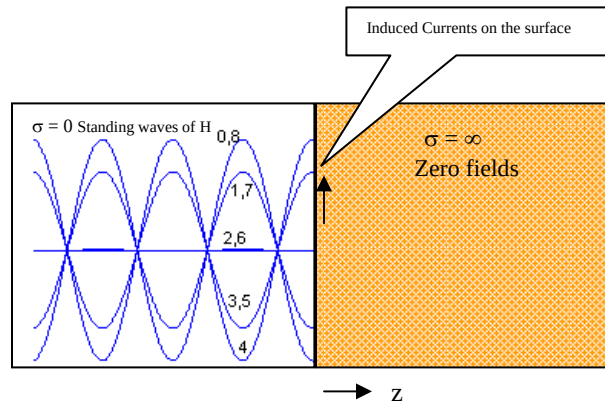
$$\beta_2 = \frac{9 \times 10^8}{3 \times 10^8} \sqrt{\frac{80}{2} \left[\sqrt{1 + 4\pi^2} + 1 \right]} = 51.48 \text{ rad/m}$$

$$\tau = \frac{2\eta_2}{\eta_2 + \eta_1} = \frac{2 \times 16.71 \angle 40.47^\circ}{16.71 \angle 40.47^\circ + 377} = 0.0857 \angle 38.89^\circ$$

$$E_{ot} = \tau E_o = 0.857 \angle 38.89^\circ$$

$$\underline{\underline{E_t = 0.857 e^{43.94z} \sin(9 \times 10^8 t + 51.48z + 38.89^\circ) \mathbf{a}_x \text{ V/m}}}$$

Prob. 10.64



Curve 0 is at $t = 0$; curve 1 is at $t = T/8$; curve 2 is at $t = T/4$; curve 3 is at $t = 3T/8$, etc.

Prob. 10.65 Since $\mu_o = \mu_1 = \mu_2$,

$$\sin \theta_{t1} = \sin \theta_i \sqrt{\frac{\epsilon_o}{\epsilon_1}} = \frac{\sin 45^\circ}{\sqrt{4.5}} = 0.3333 \quad \longrightarrow \quad \underline{\underline{\theta_{t1} = 19.47^\circ}}$$

$$\sin \theta_{t2} = \sin \theta_{t1} \sqrt{\frac{\epsilon_1}{\epsilon_2}} = \frac{1}{3} \sqrt{\frac{4.5}{2.25}} = 0.4714 \quad \longrightarrow \quad \underline{\underline{\theta_{t2} = 28.13^\circ}}$$

Prob. 10.66

$$\begin{aligned} \mathbf{E}_s &= \frac{20(e^{jk_x x} - e^{-jk_x x})}{j2} \frac{(e^{jk_y y} + e^{-jk_y y})}{2} \mathbf{a}_z \\ &= -j5 \left[e^{j(k_x x + k_y y)} + e^{j(k_x x - k_y y)} - e^{-j(k_x x - k_y y)} - e^{-j(k_x x + k_y y)} \right] \mathbf{a}_z \end{aligned}$$

which consists of four plane waves.

$$\nabla \times \mathbf{E}_s = -j\omega\mu_o \mathbf{H}_s \quad \longrightarrow \quad \mathbf{H}_s = \frac{j}{\omega\mu_o} \nabla \times \mathbf{E}_s = \frac{j}{\omega\mu_o} \left(\frac{\partial E_z}{\partial y} \mathbf{a}_x - \frac{\partial E_z}{\partial x} \mathbf{a}_y \right)$$

$$\underline{\underline{\mathbf{H}_s = -\frac{j20}{\omega\mu_o} \left[k_y \sin(k_x x) \sin(k_y y) \mathbf{a}_x + k_x \cos(k_x x) \cos(k_y y) \mathbf{a}_y \right]}}$$

Prob. 10.67

$$\eta_1 = \eta_o = 377\Omega$$

For η_2 ,

$$\frac{\sigma_2}{\omega\epsilon_2} = \frac{4}{2\pi \times 1.2 \times 10^9 \times 50 \times \frac{10^{-9}}{36\pi}} = 1.2$$

$$\tan 2\theta_{\eta_2} = \frac{\sigma_2}{\omega\epsilon_2} = 1.2 \quad \longrightarrow \quad \theta_{\eta_2} = 25.1^\circ$$

$$|\eta_2| = \frac{\sqrt{\mu/\epsilon}}{\sqrt[4]{1 + \left(\frac{\sigma_2}{\omega\epsilon_2}\right)^2}} = \frac{120\pi\sqrt{1/50}}{\sqrt[4]{1 + 1.2^2}} = 42.658$$

$$\eta_2 = 42.658 \angle 25.1^\circ$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{42.658 \angle 25.1^\circ - 377}{42.658 \angle 25.1^\circ + 377} = \underline{\underline{0.8146 \angle 174.4^\circ}}$$

Prob. 10.68

(a)

$$P_t = (1 - |\Gamma|^2) P_i$$

$$s = \frac{1 + |\Gamma|}{1 - |\Gamma|} \longrightarrow |\Gamma| = \frac{s - 1}{s + 1}$$

$$\frac{P_t}{P_i} = 1 - \left(\frac{s - 1}{s + 1} \right)^2 = \frac{4s}{(s + 1)^2}$$

$$(b) \quad P_i = P_r + P_t \longrightarrow \frac{P_r}{P_i} = 1 - \frac{P_t}{P_i} = \left(\frac{s - 1}{s + 1} \right)^2$$

Prob. 10.69

If $\mathbf{A}$ is a uniform vector and $\Phi(r)$ is a scalar,

$$\nabla \times (\Phi \mathbf{A}) = \nabla \Phi \times \mathbf{A} + \Phi (\nabla \times \mathbf{A}) = \nabla \Phi \times \mathbf{A}$$

since $\nabla \times \mathbf{A} = \mathbf{0}$.

$$\begin{aligned} \nabla \times \mathbf{E} &= \left(\frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z \right) \times \mathbf{E}_0 e^{j(k_x x + k_y y + k_z z - \omega t)} = j(k_x \mathbf{a}_x + k_y \mathbf{a}_y + k_z \mathbf{a}_z) e^{j(\mathbf{k} \cdot \mathbf{r} - \omega t)} \times \mathbf{E}_0 \\ &= j\mathbf{k} \times \mathbf{E}_0 e^{j(\mathbf{k} \cdot \mathbf{r} - \omega t)} = j\mathbf{k} \times \mathbf{E} \end{aligned}$$

$$\text{Also, } -\frac{\partial \mathbf{B}}{\partial t} = j\omega\mu\mathbf{H}. \quad \text{Hence } \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \text{ becomes } \mathbf{k} \times \mathbf{E} = \omega\mu\mathbf{H}$$

$$\text{From this, } \underline{\mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_H}$$

Prob. 10.70

$$k = |\mathbf{k}| = \sqrt{124^2 + 124^2 + 263^2} = 316.1$$

$$\lambda = \frac{2\pi}{k} = \underline{\underline{19.88 \text{ mm}}}$$

$$k = \omega\sqrt{\mu\epsilon} = \frac{2\pi f}{c} \longrightarrow f = \frac{kc}{2\pi} = \frac{316.1 \times 3 \times 10^8}{2\pi} = \underline{\underline{15.093 \text{ GHz}}}$$

$$\mathbf{k} \cdot \mathbf{a}_x = k \cos \theta_x \longrightarrow \cos \theta_x = \frac{124}{316.1} \longrightarrow \theta_x = 66.9^\circ = \theta_y$$

$$\theta_z = \cos^{-1} \frac{263}{316.1} = 33.69^\circ$$

Thus,

$$\underline{\underline{\theta_x = \theta_y = 66.9^\circ, \theta_z = 33.69^\circ}}$$

Prob. 10.71

$$\mathbf{k} = -3.4\mathbf{a}_x + 4.2\mathbf{a}_y$$

$$\mathbf{k} \cdot \mathbf{E} = 0 \quad \longrightarrow \quad 0 = -3.4E_o + 4.2$$

$$E_o = \frac{4.2}{3.4} = \underline{\underline{1.235}}$$

$$k = |\mathbf{k}| = \beta = \sqrt{(-3.4)^2 + (4.2)^2} = 5.403$$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{5.403} = 1.162$$

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{1.162} = \underline{\underline{258 \text{ MHz}}}$$

$$\mathbf{H}_s = \frac{1}{\mu\omega} \mathbf{k} \times \mathbf{E}_s = \frac{1}{\mu k c} \mathbf{k} \times \mathbf{E}_s$$

$$= \frac{1}{4\pi \times 10^{-7} \times 5.403 \times 3 \times 10^8} \begin{vmatrix} -3.4 & 4.2 & 0 \\ E_o & 1 & 3 + j4 \end{vmatrix} A_o$$

$$\text{where } A_o = e^{-j3.4x + 4.2y}$$

$$\begin{aligned} \mathbf{H}_s &= 4.91 A_o \times 10^{-4} [4.2(3 + j4)\mathbf{a}_x + 3.4(3 + j4)\mathbf{a}_y + (-3.4 - 4.2E_o)\mathbf{a}_z] \\ &= 0.491 [(12.6 + j16.8)\mathbf{a}_x + (10.2 + j13.6)\mathbf{a}_y - 8.59\mathbf{a}_z] e^{-j3.4x + 4.2y} \text{ mA/m} \end{aligned}$$

Prob. 10.72

$$\nabla \cdot \mathbf{E} = \left(\frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z \right) \cdot \mathbf{E}_o e^{j(k_x x + k_y y + k_z z - \omega t)} = j(k_x \mathbf{a}_x + k_y \mathbf{a}_y + k_z \mathbf{a}_z) e^{j(\mathbf{k} \cdot \mathbf{r} - \omega t)} \cdot \mathbf{E}_o$$

$$= j\mathbf{k} \cdot \mathbf{E}_o e^{j(\mathbf{k} \cdot \mathbf{r} - \omega t)} = j\mathbf{k} \cdot \mathbf{E} = 0 \quad \longrightarrow \quad \mathbf{k} \cdot \mathbf{E} = 0$$

Similarly,

$$\nabla \cdot \mathbf{H} = j\mathbf{k} \cdot \mathbf{H} = 0 \quad \longrightarrow \quad \mathbf{k} \cdot \mathbf{H} = 0$$

It has been shown in the previous problem that

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \longrightarrow \quad \mathbf{k} \times \mathbf{E} = \omega \mu \mathbf{H}$$

Similarly,

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} \quad \longrightarrow \quad \mathbf{k} \times \mathbf{H} = -\epsilon \omega \mathbf{E}$$

From $\mathbf{k} \times \mathbf{E} = \omega \mu \mathbf{H}$, $\mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_H$ and

From $\mathbf{k} \times \mathbf{H} = -\varepsilon \omega \mathbf{E}$, $\mathbf{a}_k \times \mathbf{a}_H = -\mathbf{a}_E$

Prob. 10.73

$$\text{If } \mu_o = \mu_1 = \mu_2, \quad \eta_1 = \frac{\eta_o}{\sqrt{\varepsilon_{r1}}}, \eta_2 = \frac{\eta_o}{\sqrt{\varepsilon_{r2}}}$$

$$\Gamma_{\parallel} = \frac{\frac{1}{\sqrt{\varepsilon_{r2}}} \cos \theta_t - \frac{1}{\sqrt{\varepsilon_{r1}}} \cos \theta_i}{\frac{1}{\sqrt{\varepsilon_{r2}}} \cos \theta_t + \frac{1}{\sqrt{\varepsilon_{r1}}} \cos \theta_i}$$

$$\sqrt{\varepsilon_{r1}} \sin \theta_i = \sqrt{\varepsilon_{r2}} \sin \theta_t \quad \longrightarrow \quad \frac{\sqrt{\varepsilon_{r2}}}{\sqrt{\varepsilon_{r1}}} = \frac{\sin \theta_i}{\sin \theta_t}$$

$$\begin{aligned} \Gamma_{\parallel} &= \frac{\cos \theta_t - \frac{\sin \theta_i}{\sin \theta_t} \cos \theta_i}{\cos \theta_t + \frac{\sin \theta_i}{\sin \theta_t} \cos \theta_i} = \frac{\sin \theta_t \cos \theta_t - \sin \theta_i \cos \theta_i}{\sin \theta_t \cos \theta_t + \sin \theta_i \cos \theta_i} \\ &= \frac{\sin 2\theta_t - \sin 2\theta_i}{\sin 2\theta_t + \sin 2\theta_i} = \frac{\sin(\theta_t - \theta_i) \cos(\theta_t + \theta_i)}{\cos(\theta_t - \theta_i) \sin(\theta_t + \theta_i)} = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)} \end{aligned}$$

Similarly,

$$\begin{aligned} \tau_{\parallel} &= \frac{\frac{2}{\sqrt{\varepsilon_{r2}}} \cos \theta_i}{\frac{1}{\sqrt{\varepsilon_{r2}}} \cos \theta_t + \frac{1}{\sqrt{\varepsilon_{r1}}} \cos \theta_i} = \frac{2 \cos \theta_i}{\cos \theta_t + \frac{\sin \theta_i}{\sin \theta_t} \cos \theta_i} \\ &= \frac{2 \cos \theta_i \sin \theta_t}{\sin \theta_t \cos \theta_t (\sin^2 \theta_i + \cos^2 \theta_i) + \sin \theta_i \cos \theta_i (\sin^2 \theta_t + \cos^2 \theta_t)} \\ &= \frac{2 \cos \theta_i \sin \theta_t}{(\sin \theta_i \cos \theta_t + \sin \theta_t \cos \theta_i)(\cos \theta_i \cos \theta_t + \sin \theta_i \sin \theta_t)} \\ &= \frac{2 \cos \theta_i \sin \theta_t}{\sin(\theta_i + \theta_t) \cos(\theta_i - \theta_t)} \end{aligned}$$

$$\Gamma_{\perp} = \frac{\frac{1}{\sqrt{\epsilon_{r2}}} \cos \theta_i - \frac{1}{\sqrt{\epsilon_{r1}}} \cos \theta_t}{\frac{1}{\sqrt{\epsilon_{r2}}} \cos \theta_i + \frac{1}{\sqrt{\epsilon_{r1}}} \cos \theta_t} = \frac{\cos \theta_i - \frac{\sin \theta_i}{\sin \theta_t} \cos \theta_t}{\cos \theta_i + \frac{\sin \theta_i}{\sin \theta_t} \cos \theta_t} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

$$\tau_{\perp} = \frac{\frac{2}{\sqrt{\epsilon_{r2}}} \cos \theta_i}{\frac{1}{\sqrt{\epsilon_{r2}}} \cos \theta_i + \frac{1}{\sqrt{\epsilon_{r1}}} \cos \theta_t} = \frac{2 \cos \theta_i}{\cos \theta_i + \frac{\sin \theta_i}{\sin \theta_t} \cos \theta_t} = \frac{2 \cos \theta_i \sin \theta_t}{\sin(\theta_t + \theta_i)}$$

Prob. 10.74

(a) $n_1 = 1, \quad n_2 = c\sqrt{\mu_2 \epsilon_2} = c\sqrt{6.4 \epsilon_o \times \mu_o} = \sqrt{6.4} = 2.5298$

$$\sin \theta_t = \frac{n_1}{n_2} \sin \theta_i = \frac{1}{2.5298} \sin 12^\circ = 0.082185 \quad \longrightarrow \quad \theta_t = 4.714^\circ$$

$$\eta_1 = 120\pi, \quad \eta_2 = 120\pi \sqrt{\frac{1}{6.4}} = 47.43\pi$$

$$\begin{aligned} \frac{E_{ro}}{E_{io}} = \Gamma &= \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} = \frac{47.43\pi \cos 4.714^\circ - 120\pi \cos 12^\circ}{47.43\pi \cos 4.714^\circ + 120\pi \cos 12^\circ} \\ &= \frac{47.27 - 117.38}{47.27 + 117.38} = \underline{\underline{-0.4258}} \end{aligned}$$

$$\frac{E_{to}}{E_{io}} = \tau = \frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} = \frac{2 \times 47.43 \cos 12^\circ}{47.27 + 117.33} = \frac{92.787}{164.65} = \underline{\underline{0.5635}}$$

Prob. 10.75

(a) $\mathbf{k}_i = 4\mathbf{a}_y + 3\mathbf{a}_z$

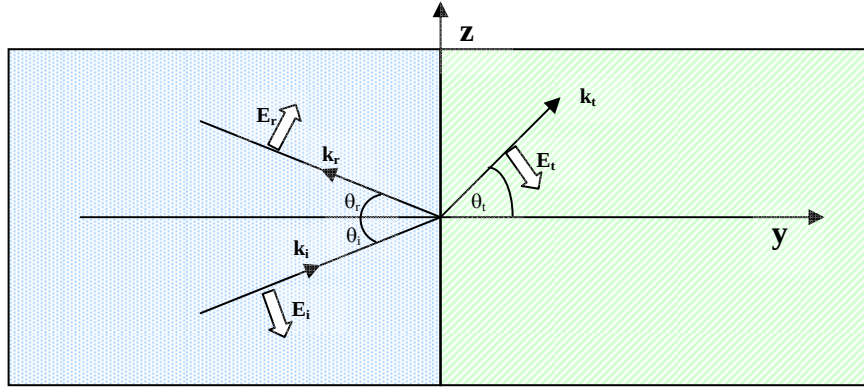
$\mathbf{k}_i \bullet \mathbf{a}_n = k_i \cos \theta_i \quad \longrightarrow \quad \cos \theta_i = 4/5 \quad \longrightarrow \quad \underline{\underline{\theta_i = 36.87^\circ}}$

(b)

$$\mathbf{P}_{\text{ave}} = \frac{1}{2} \operatorname{Re}(\mathbf{E}_s \times \mathbf{H}_s^*) = \frac{E_o^2}{2\eta} \mathbf{a}_k = \frac{(\sqrt{8^2 + 6^2})^2}{2 \times 120\pi} \frac{(4\mathbf{a}_y + 3\mathbf{a}_z)}{5} = \underline{\underline{106.1\mathbf{a}_y + 79.58\mathbf{a}_z \text{ mW/m}^2}}$$

(c) $\theta_r = \theta_i = 36.87^\circ$. Let

$$\mathbf{E}_r = (E_{ry}\mathbf{a}_x + E_{rz}\mathbf{a}_z) \sin(\omega t - \mathbf{k}_r \bullet \mathbf{r})$$



From the figure, $\mathbf{k}_r = k_{rz}\mathbf{a}_z - k_{ry}\mathbf{a}_y$. But $k_r = k_i = 5$

$$k_{rz} = k_r \sin \theta_r = 5(3/5) = 3, \quad k_{ry} = k_r \cos \theta_r = 5(4/5) = 4,$$

Hence, $\mathbf{k}_r = -4\mathbf{a}_y + 3\mathbf{a}_z$

$$\sin \theta_t = \frac{n_1}{n_2} \sin \theta_i = \frac{c\sqrt{\mu_1 \epsilon_1}}{c\sqrt{\mu_2 \epsilon_2}} \sin \theta_i = \frac{3/5}{\sqrt{4}} = 0.3$$

$$\theta_t = 17.46, \quad \cos \theta_t = 0.9539, \quad \eta_1 = \eta_o = 120\pi, \eta_2 = \eta_o / 2 = 60\pi$$

$$\Gamma_{//} = \frac{E_{ro}}{E_{io}} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} = \frac{\frac{\eta_o}{2}(0.9539) - \eta_o(0.8)}{\frac{\eta_o}{2}(0.9539) + \eta_o(0.8)} = -0.253$$

$$E_{ro} = \Gamma_{//} E_{io} = -0.253(10) = -2.53$$

$$\text{But } (E_{ry}\mathbf{a}_y + E_{rz}\mathbf{a}_z) = E_{ro}(\sin \theta_r \mathbf{a}_y + \cos \theta_r \mathbf{a}_z) = -2.53\left(\frac{3}{5}\mathbf{a}_y + \frac{4}{5}\mathbf{a}_z\right)$$

$$\underline{\underline{\mathbf{E}_r = -(1.518\mathbf{a}_y + 2.024\mathbf{a}_z) \sin(\omega t + 4y - 3z) \text{ V/m}}}$$

Similarly, let

$$\mathbf{E}_t = (E_{ty}\mathbf{a}_y + E_{tz}\mathbf{a}_z) \sin(\omega t - \mathbf{k}_t \bullet \mathbf{r})$$

$$k_t = \beta_2 = \omega \sqrt{\mu_2 \epsilon_2} = \omega \sqrt{4\mu_o \epsilon_o}$$

But $k_i = \beta_1 = \omega \sqrt{\mu_o \epsilon_o}$

$$\frac{k_t}{k_i} = 2 \quad \longrightarrow \quad k_t = 2k_i = 10$$

$$k_{ty} = k_t \cos \theta_t = 9.539, \quad k_{tz} = k_t \sin \theta_t = 3,$$

$$k_t = 9.539a_y + 3a_z$$

Note that $k_{iz} = k_{rz} = k_{tz} = 3$

$$\tau_{\parallel} = \frac{E_{to}}{E_{io}} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} = \frac{\eta_o(0.8)}{\frac{\eta_o}{2}(0.9539) + \eta_o(0.8)} = 0.6265$$

$$E_{to} = \tau_{\parallel} E_{io} = 6.265$$

But

$$(E_{ty}a_y + E_{tz}a_z) = E_{to}(\sin \theta_t a_y - \cos \theta_t a_z) = 6.256(0.3a_y - 0.9539a_z)$$

Hence,

$$\underline{\underline{E_t = (1.879a_y - 5.968a_z) \sin(\omega t - 9.539y - 3z) \text{ V/m}}}$$

Prob. 10.76

(a)

$$\tan \theta_i = \frac{k_{ix}}{k_{iz}} = \frac{1}{\sqrt{8}} \quad \longrightarrow \quad \underline{\underline{\theta_i = \theta_r = 19.47^\circ}}$$

$$\sin \theta_t = \sin \theta_i \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \frac{1}{3}(3) = 1 \quad \longrightarrow \quad \underline{\underline{\theta_t = 90^\circ}}$$

$$(b) \quad \beta_1 = \frac{\omega}{c} \sqrt{\epsilon_{r1}} = \frac{10^9}{3 \times 10^8} \times 3 = 10 = k\sqrt{1+8} = 3k \quad \longrightarrow \quad \underline{\underline{k = 3.333}}$$

$$(c) \quad \lambda = 2\pi / \beta, \quad \lambda_1 = 2\pi / \beta_1 = 2\pi / 10 = \underline{\underline{0.6283 \text{ m}}}$$

$$\beta_2 = \omega / c = 10/3, \quad \lambda_2 = 2\pi / \beta_2 = 2\pi \times 3/10 = \underline{\underline{1.885 \text{ m}}}$$

$$(d) \quad \mathbf{E}_i = \eta_1 \mathbf{H}_x \times \mathbf{a}_k = 40\pi(0.2) \cos(\omega t - \mathbf{k} \cdot \mathbf{r}) \mathbf{a}_y \times \frac{(\mathbf{a}_x + \sqrt{8}\mathbf{a}_z)}{3}$$

$$= \underline{\underline{(23.6954\mathbf{a}_x - 8.3776\mathbf{a}_z) \cos(10^9 t - kx - k\sqrt{8}z) \text{ V/m}}}$$

$$(e) \quad \tau_{//} = \frac{2 \cos \theta_i \sin \theta_t}{\sin(\theta_i + \theta_t) \cos(\theta_t - \theta_i)} = \frac{2 \cos 19.47^\circ \sin 90^\circ}{\sin 19.47^\circ \cos 19.47^\circ} = 6$$

$$\Gamma_{//} = -\frac{\cot 19.47^\circ}{\cot 19.47^\circ} = -1$$

$$\text{Let } \mathbf{E}_t = -E_{io} (\cos \theta_t \mathbf{a}_x - \sin \theta_t \mathbf{a}_z) \cos(10^9 t - \beta_2 x \sin \theta_t - \beta_2 z \cos \theta_t)$$

where

$$\mathbf{E}_t = -E_{io} (\cos \theta_t \mathbf{a}_x - \sin \theta_t \mathbf{a}_z) \cos(10^9 t - \beta_1 x \sin \theta_t - \beta_1 z \cos \theta_t)$$

$$\sin \theta_t = 1, \quad \cos \theta_t = 0, \quad \beta_2 \sin \theta_t = 10/3$$

$$E_{io} \sin \theta_t = \tau_{//} E_{io} = 6(24\pi)(3)(1) = 1357.2$$

Hence,

$$\underline{\underline{\mathbf{E}_t = 1357 \cos(10^9 t - 3.333x) \mathbf{a}_z \text{ V/m}}}$$

$$\text{Since } \Gamma = -1, \quad \theta_r = \theta_i$$

$$\underline{\underline{\mathbf{E}_r = (213.3\mathbf{a}_x + 75.4\mathbf{a}_z) \cos(10^9 t - kx + k\sqrt{8}z) \text{ V/m}}}$$

$$(f) \quad \tan \theta_{B//} = \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \sqrt{\frac{\epsilon_o}{9\epsilon_o}} = 1/3 \quad \longrightarrow \quad \underline{\underline{\theta_{B//} = 18.43^\circ}}$$

Prob. 10.77

$$(a) \quad \underline{\underline{\mathbf{E}_{i\perp} = 5 \cos(\omega t - 0.5\pi x - 0.866\pi z) \mathbf{a}_y}}}$$

$$\underline{\underline{\mathbf{E}_{i\parallel} = (4\mathbf{a}_x - 3\mathbf{a}_z) \cos(\omega t - 0.5\pi x - 0.866\pi z)}}$$

(b) Comparing $\mathbf{E}_{i\parallel}$ with eq. (10.115a),

$$4\mathbf{a}_x - 3\mathbf{a}_z = (\cos \theta_i \mathbf{a}_x - \sin \theta_i \mathbf{a}_z) E_{io}$$

$$\tan \theta_i = \frac{\sin \theta_i}{\cos \theta_i} = \frac{3}{4} \quad \longrightarrow \quad \underline{\underline{\theta_i = 36.87^\circ}}$$

Prob. 10.78

$$\tan \theta_B = \sqrt{\frac{\epsilon_1}{\epsilon_2}} = \sqrt{\frac{\epsilon_o \epsilon_r}{\epsilon_o}} = \sqrt{\epsilon_r}$$

$$\epsilon_r = \tan^2 \theta_B = \tan^2 68 = \underline{\underline{6.126}}$$

Prob. 10.79

$$(a) \quad n = \frac{c}{u} = \sqrt{\mu_r \epsilon_r} = \sqrt{2.1 \times 1} = \underline{\underline{1.45}}$$

$$(b) \quad n = \sqrt{\mu_r \epsilon_r} = \sqrt{1 \times 81} = \underline{\underline{9}}$$

$$(c) \quad n = \sqrt{\epsilon_r} = \sqrt{2.7} = \underline{\underline{1.643}}$$

Prob. 10.80

Microwave is used:

- (1) For surveying land with a piece of equipment called the *tellurometer*. This radar system can precisely measure the distance between two points.
- (2) For guidance. The guidance of missiles, the launching and homing guidance of space vehicles, and the control of ships are performed with the aid of microwaves.
- (3) In semiconductor devices. A large number of new microwave semiconductor devices have been developed for the purpose of microwave oscillator, amplification, mixing/detection, frequency multiplication, and switching. Without such achievement, the majority of today's microwave systems could not exist.

Prob. 10.81

(a) In terms of the S-parameters, the T-parameters are given by

$$T_{11} = 1/S_{21}, \quad T_{12} = -S_{22}/S_{21}, \quad T_{21} = S_{11}/S_{21}, \quad T_{22} = S_{12} - S_{11} S_{22}/S_{21}$$

$$(b) \quad T_{11} = 1/0.4 = 2.5, \quad T_{12} = -0.2/0.4,$$

$$T_{21} = 0.2/0.4, \quad T_{22} = 0.4 - 0.2 \times 0.2/0.4 = 0.3$$

Hence,

$$T = \underline{\underline{\begin{bmatrix} 2.5 & -0.5 \\ 0.5 & 0.3 \end{bmatrix}}}$$

Prob. 10.82

Since $Z_L = Z_o$, $\Gamma_L = 0$.

$$\Gamma_i = S_{11} = \underline{0.33 - j0.15}$$

$$\Gamma_g = (Z_g - Z_o) / (Z_g + Z_o) = (2 - 1) / (2 + 1) = 1/3$$

$$\Gamma_o = S_{22} + S_{12}S_{21}\Gamma_g / (1 - S_{11}\Gamma_g)$$

$$= 0.44 - j0.62 + 0.56 \times 0.56 \times (1/3) / [1 - (0.11 - j0.05)]$$

$$= \underline{0.5571 - j0.6266}$$

Prob. 10.83 The microwave wavelengths are of the same magnitude as the circuit components. The wavelength in air at a microwave frequency of 300 GHz, for example, is 1 mm. The physical dimension of the lumped element must be in this range to avoid interference. Also, the leads connecting the lumped element probably have much more inductance and capacitance than is needed.

Prob. 10.84

$$\lambda = c/f = \frac{3 \times 10^8}{8.4 \times 10^9} = \underline{35.71 \text{ mm}}$$

CHAPTER 11

P.E. 11.1 Since Z_o is real and $\alpha \neq 0$, this is a distortionless line.

$$Z_o = \sqrt{\frac{R}{G}} \quad (1)$$

$$\text{or } \frac{L}{R} = \frac{C}{G} \quad (2)$$

$$\alpha = \sqrt{RG} \quad (3)$$

$$\beta = \omega L \sqrt{\frac{G}{R}} = \frac{\omega L}{Z_o} \quad (4)$$

$$(1) \times (3) \rightarrow R = \alpha Z_o = 0.04 \times 80 = \underline{\underline{3.2 \Omega / \text{m}}},$$

$$(3) \div (1) \rightarrow G = \frac{\alpha}{Z_o} = \frac{0.04}{80} = \underline{\underline{5 \times 10^{-4} \text{ S / m}}}$$

$$L = \frac{\beta Z_o}{\omega} = \frac{1.5 \times 80}{2\pi \times 5 \times 10^8} = \underline{\underline{38.2 \text{ nH / m}}}$$

$$C = \frac{LG}{R} = \frac{12}{\pi} \times 10^{-8} \times \frac{0.04}{80} \times \frac{1}{0.04 \times 80} = \underline{\underline{5.97 \text{ pF / m}}}$$

P.E. 11.2

$$\begin{aligned} \text{(a) } Z_o &= \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{0.03 + j2\pi \times 0.1 \times 10^{-3}}{0 + j2\pi \times 0.02 \times 10^{-6}}} \\ &= 70.73 - j1.688 = \underline{\underline{70.75 \angle -1.367^\circ \Omega}} \end{aligned}$$

$$\begin{aligned} \text{(b) } \gamma &= \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{(0.03 + j0.2\pi)(j0.4 \times 10^{-4} \pi)} \\ &= \underline{\underline{2.121 \times 10^{-4} + j8.888 \times 10^{-3} / \text{m}}} \end{aligned}$$

$$\text{(c) } u = \frac{\omega}{\beta} = \frac{2\pi \times 10^3}{8.888 \times 10^{-3}} = \underline{\underline{7.069 \times 10^5 \text{ m/s}}}$$

P.E. 11.3

$$\text{(a) } Z_o = Z_l \rightarrow Z_{in} = Z_o = \underline{\underline{30 + j60 \Omega}}$$

$$(b) V_{in} = V_o = \frac{Z_{in}}{Z_{in} + Z_o} V_g = \frac{V_g}{2} = \underline{\underline{7.5 \angle 0^\circ \text{ V}_{\text{rms}}}}$$

$$I_{in} = I_o = \frac{V_g}{Z_g + Z_{in}} = \frac{V_g}{2Z_o} = \frac{15 \angle 0^\circ}{2(30 + j60)}$$

$$= \underline{\underline{0.2236 \angle -63.43^\circ \text{ A}}}$$

assuming that $Z_g = 0$.

(c) Since $Z_o = Z_r$, $\Gamma = 0 \rightarrow V_o^- = 0, V_o^+ = V_o$

The load voltage is $V_L = V_s(z = l) = V_o^+ e^{-\gamma l}$

$$e^{-\gamma l} = \frac{V_o^+}{V_L} = \frac{7.5 \angle 0^\circ}{5 \angle -48^\circ} = 1.5 \angle 48^\circ$$

$$e^{\alpha l} e^{j\beta l} = 1.5 \angle 48^\circ$$

$$e^{\alpha l} = 1.5 \rightarrow \alpha = \frac{1}{l} \ln(1.5) = \frac{1}{40} \ln(1.5) = 0.0101$$

$$e^{j\beta l} = e^{j48^\circ} \rightarrow \beta = \frac{1}{l} \frac{48^\circ}{180^\circ} \pi \text{ rad} = 0.02094$$

$$\underline{\underline{\gamma = 0.0101 + j0.02094 \text{ /m}}}$$

assuming that $Z_g = 0$.

P.E. 11.4

(a) Using the Smith chart, locate S at $s = 1.6$. Draw a circle of radius OS. Locate P where $\theta_r = 300^\circ$. At P,

$$|\Gamma| = \frac{OP}{OQ} = \frac{2.1 \text{ cm}}{9.2 \text{ cm}} = 0.228$$

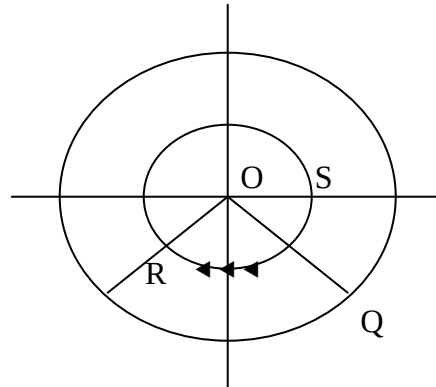
$$\underline{\underline{\Gamma = 0.228 \angle 300^\circ}}$$

Also at P, $\underline{\underline{z_L = 1.15 - j0.48}}$,

$$\underline{\underline{Z_L = Z_o z_L = 70(1.15 - j0.48) = 80.5 - j33.6 \Omega}}$$

$$\ell = 0.6\lambda \rightarrow 0.6 \times 720^\circ = 432^\circ = \underline{\underline{360^\circ + 73^\circ}}$$

From P, move 432° to R. At R, $\underline{\underline{z_{in} = 0.68 - j0.25}}$



$$Z_{in} = Z_o Z_{in} = 70(0.68 - j0.25) = \underline{\underline{47.6 - j17.5\Omega}}$$

(b) The minimum voltage (the only one) occurs at $\theta_r = 180^\circ$; its distance from the

$$\text{load is } \frac{180 - 60}{720} \lambda = \frac{\lambda}{6} = \underline{\underline{0.1667\lambda}}$$

Values obtained using formulas are as follows:

$$\Gamma = \frac{s-1}{s+1} \angle 300^\circ = 0.2308 \angle 300^\circ$$

$$Z_L = 80.5755 - j34.018 \Omega$$

$$Z_{in} = 48.655 - j17.63 \Omega$$

These are pretty close.

P.E. 11.5

$$(a) \Gamma = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{60 + j60 - 60}{60 + j60 + 60} = \frac{j}{2 + j} = \underline{\underline{0.4472 \angle 63.43^\circ}}$$

$$s = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.4472}{1 - 0.4472} = \underline{\underline{2.618}}$$

$$\text{Let } x = \tan(\beta l) = \tan \frac{2\pi l}{\lambda}$$

$$Z_{in} = Z_o \left[\frac{Z_L + jZ_o \tan(\beta l)}{Z_o + jZ_L \tan(\beta l)} \right]$$

$$120 - j60 = 60 \left[\frac{60 + j60 + j60x}{60 + j(60 + j60)x} \right]$$

$$\text{Or } 2 - j = \frac{1 + j(1+x)}{1 - x + jx} \rightarrow 1 - x + j(2x - 2) = 0$$

$$\text{Or } x = 1 = \tan(\beta l)$$

$$\frac{\pi}{4} + n\pi = \frac{2\pi l}{\lambda}$$

$$\text{i.e. } \underline{\underline{l = \frac{\lambda}{8}(1 + 4n), n = 0, 1, 2, 3 \dots}}$$

$$(b) \ z_L = \frac{Z_L}{Z_o} \frac{60 + j60}{60} = 1 + j$$

Locate the load point P on the Smith chart.

$$|\Gamma| = \frac{OP}{OQ} = \frac{4.1cm}{9.2cm} = 0.4457, \theta_\Gamma = 62^\circ$$

$$\underline{\underline{\Gamma = 0.4457 \angle 62^\circ}}$$

Locate the point S on the Smith chart. At S, $r = s = 2.6$

$$Z_{in} = \frac{Z_{in}}{Z_o} = \frac{120 + j60}{60} = 2 - j, \text{ which is located at R on the chart. The angle between OP}$$

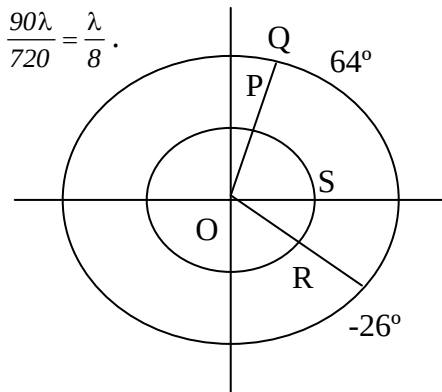
and OR is $64^\circ - (-25^\circ) = 90^\circ$ which is equivalent to $\frac{90\lambda}{720} = \frac{\lambda}{8}$.

$$\text{Hence } l = \frac{\lambda}{8} + n \frac{\lambda}{2} = \frac{\lambda}{8} (1 + 4n), n = 0, 1, 2, \dots$$

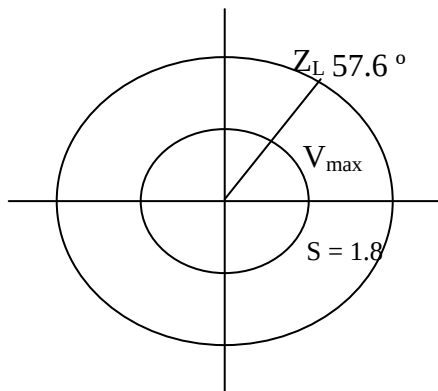
$$(Z_{in})_{\max} = sZ_o = 2.618(60) = 157.08\Omega$$

$$(Z_{in})_{\min} = Z_o / s = 60 / 2.618 = \underline{\underline{22.92 \Omega}}$$

$$l = \frac{62^\circ}{720^\circ} \lambda = 0.0851\lambda$$



P.E. 11.6



$$\frac{\lambda}{2} = 37.5 - 25 = 12.5cm \text{ or } \lambda = 25cm$$

$$l = 37.5 - 35.5 = 2cm = \frac{2\lambda}{25}$$

$$l = 0.08\lambda \rightarrow 57.6^\circ$$

$$z_L = 1.184 + j0.622$$

$$Z_L = Z_o z_L = 50(1.184 - j0.622)$$

$$\underline{\underline{= 59.22 + j31.11\Omega}}$$

P.E. 11.7 See the Smith chart

$$z_L = \frac{100 - j80}{75} = 1.33 - j1.067$$

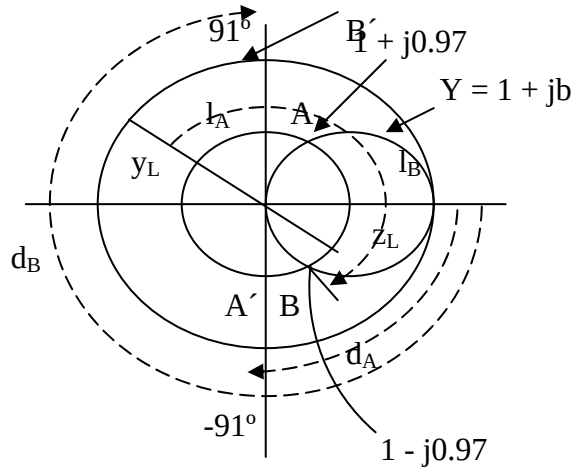
$$l_A = \frac{132^\circ - 65^\circ}{72} \lambda = \underline{\underline{0.093\lambda}}$$

$$l_B = \frac{132^\circ + 64^\circ}{720^\circ} = \underline{\underline{0.272\lambda}}$$

$$d_A = \frac{91}{720} \lambda = 0.126\lambda$$

$$d_B = 0.5\lambda - d_A = \underline{\underline{0.374\lambda}}$$

$$Y_s = \pm \frac{j0.95}{75} = \pm \underline{\underline{j12.67 \text{ mS}}}$$

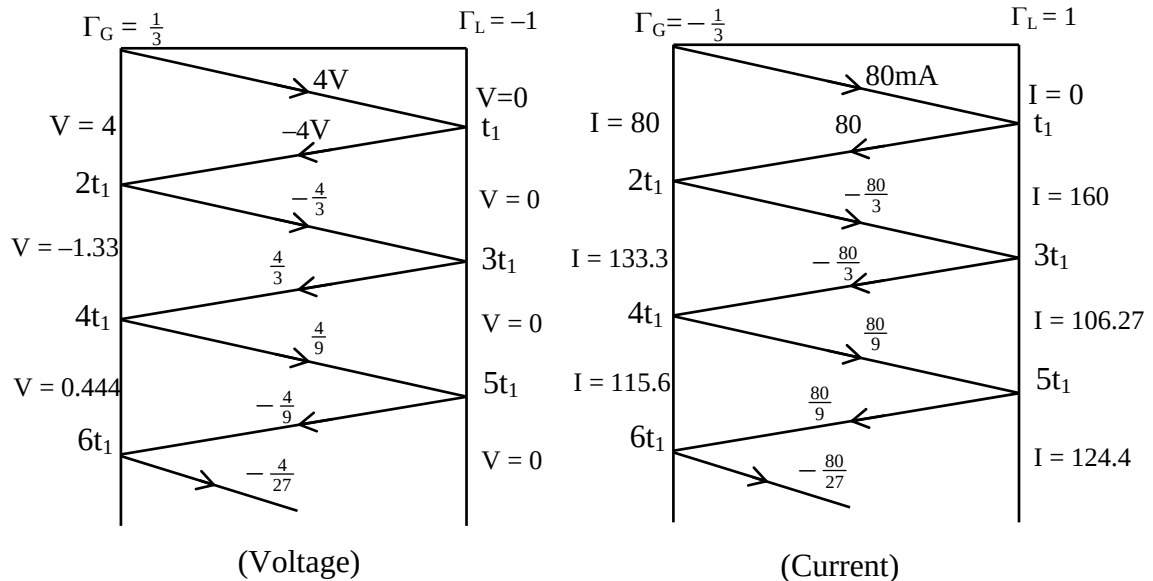


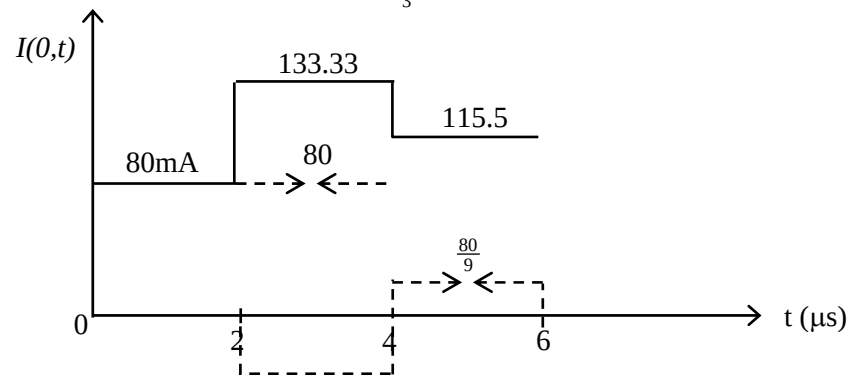
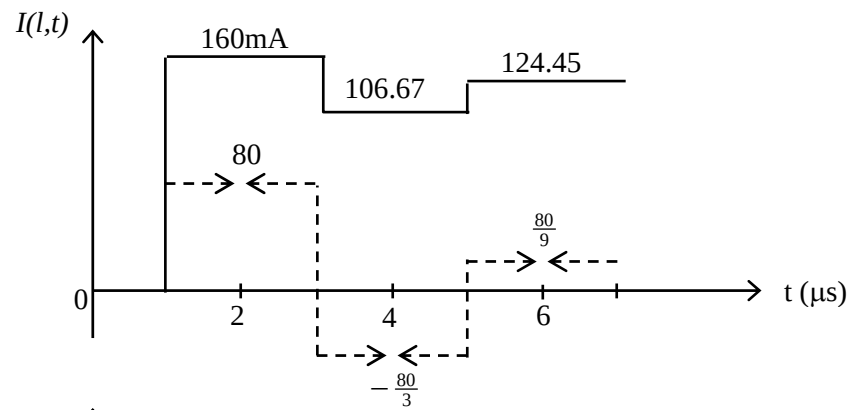
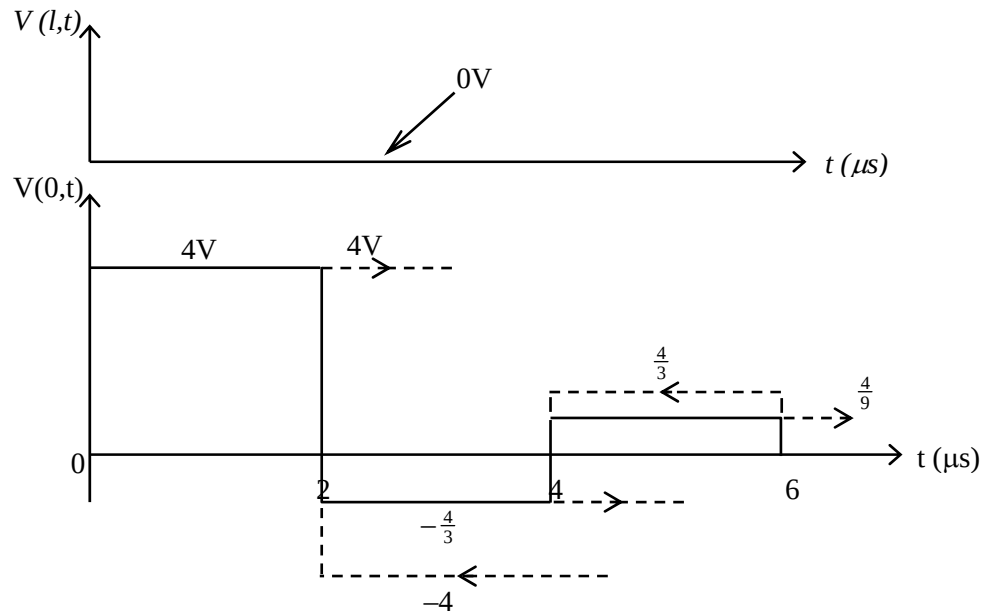
P.E. 11.8

$$(a) \Gamma_G = \frac{1}{3}, \Gamma_L = z_L \xrightarrow{\lim} 0 \frac{Z_L - Z_o}{Z_L + Z_o} = -1$$

$$V_\infty = z_L \xrightarrow{\lim} 0 \frac{Z_L}{Z_L + Z_g} V_g = 0, \quad I_\infty = z_L \xrightarrow{\lim} 0 \frac{V_g}{Z_g + Z_L} = \frac{V_g}{Z_g} = \frac{12}{100} = 120 \text{ mA}$$

Thus the bounce diagrams for current and waves are as shown below.

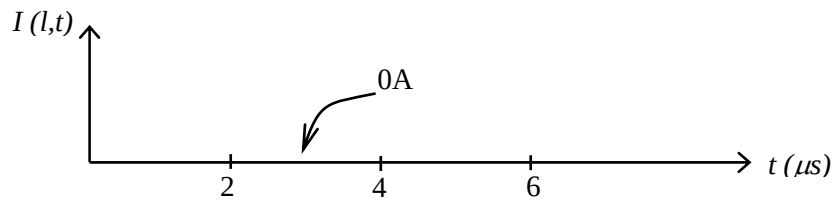
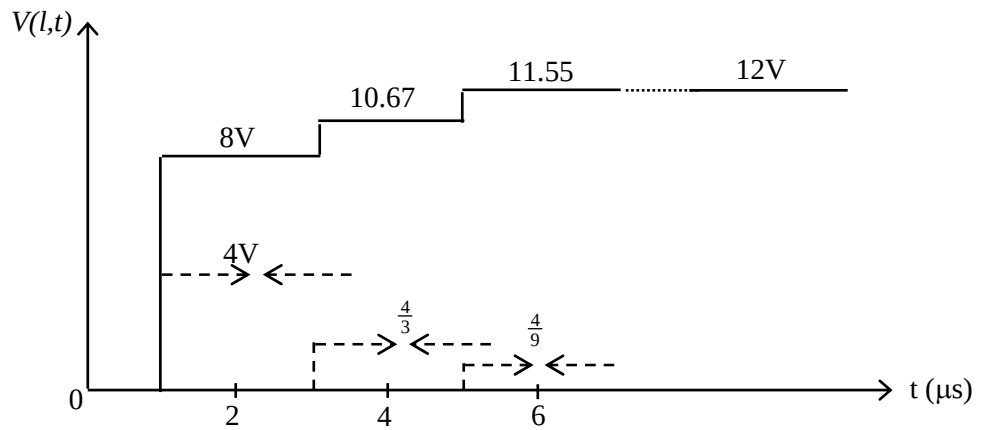
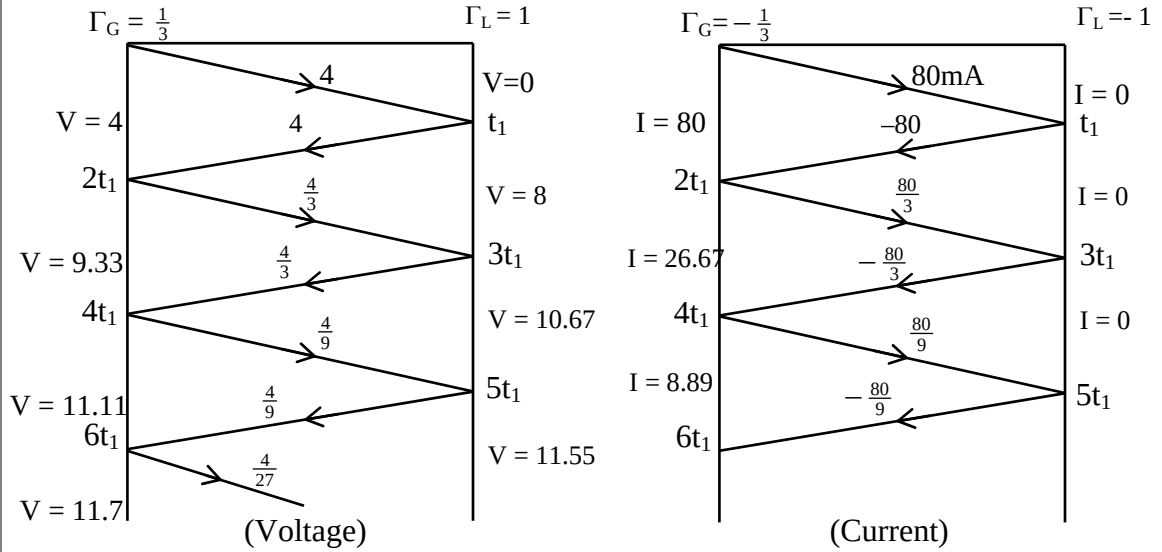


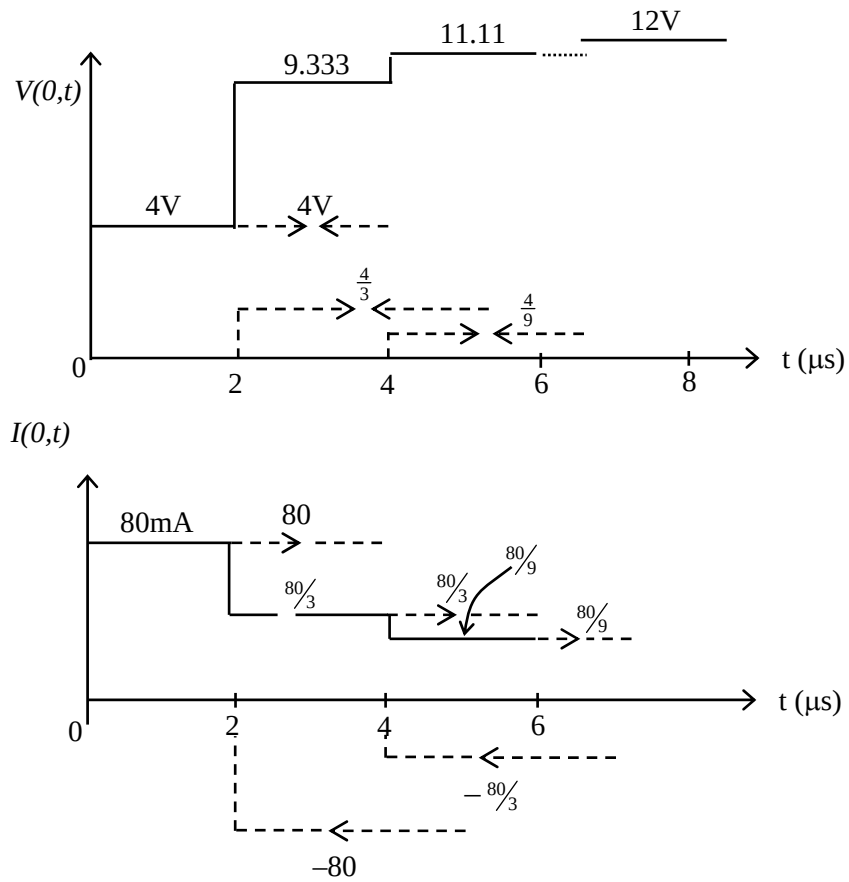


$$(b) \quad \Gamma_G = \frac{1}{3}, \Gamma_L = Z_L \xrightarrow{\lim} \infty \frac{Z_L - Z_o}{Z_L + Z_o} = 1$$

$$V_{\infty} = Z_L \xrightarrow{\lim} \infty \frac{Z_L}{Z_L + Z_g} V_g = V_g = 12V, \quad I_{\infty} = Z_L \xrightarrow{\lim} \infty \frac{V_g}{Z_L + Z_g} = 0$$

The bounce diagrams for current and voltage waves are as shown below.

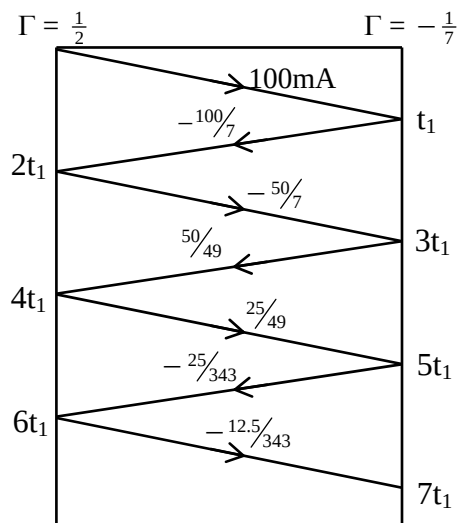


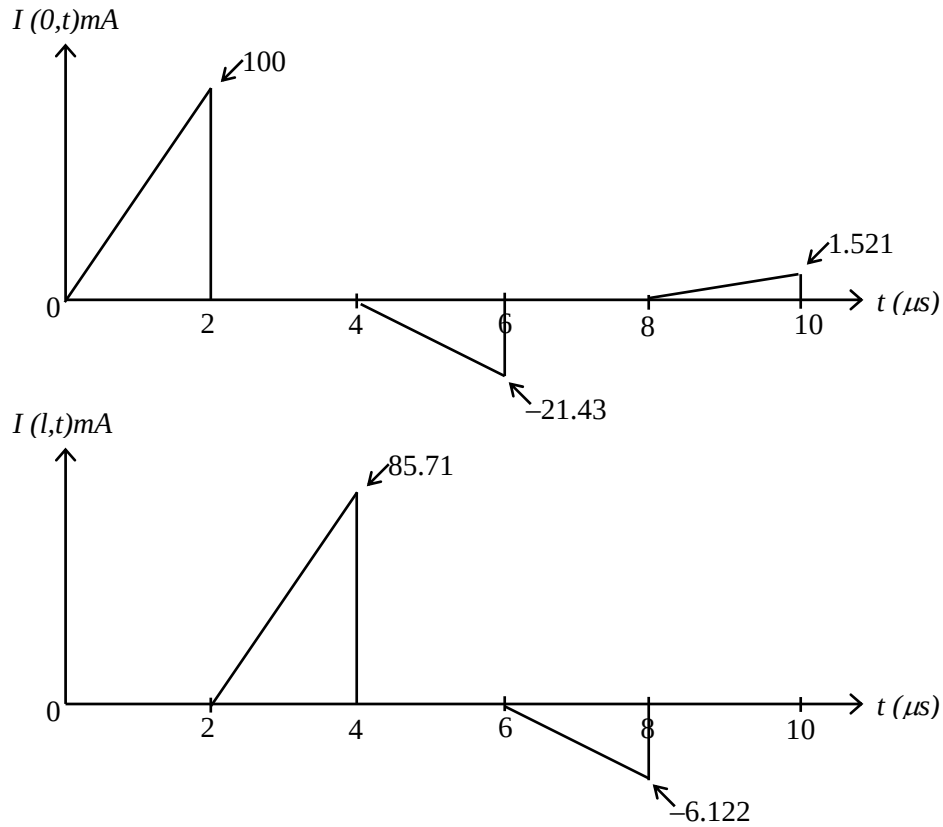
**P.E. 11.9**

$$\Gamma_G = -\frac{1}{2}, \Gamma_L = \frac{1}{7}, t_1 = 2\mu s$$

$$(I_o)_{\max} = \frac{(V_g)_{\max}}{Z_g + Z_o} = \frac{10}{100} = 100\text{mA}$$

The bounce diagrams for maximum current are as shown below.



**P.E. 11.10**

$$(a) \text{ For } w/h = 0.8, \quad \epsilon_{eff} = \frac{4.8}{2} + \frac{2.8}{2} \left[1 + \frac{12}{0.8} \right]^{-1/2} = \underline{\underline{2.75}}$$

$$(b) Z_o = \frac{60}{\sqrt{2.75}} \ln \left(\frac{8}{0.8} + \frac{0.8}{4} \right) = 36.18 \ln 10.2 = \underline{\underline{84.03 \Omega}}$$

$$(c) \lambda = \frac{3 \times 10^8}{10^{10} \sqrt{2.75}} = \underline{\underline{18.09 \text{ mm}}}$$

P.E. 11.11

$$\begin{aligned} R_s &= \sqrt{\frac{\pi f \mu_o}{\sigma_c}} = \sqrt{\frac{\pi \times 20 \times 10^9 \times 4\pi \times 10^{-7}}{5.8 \times 10^7}} \\ &= 3.69 \times 10^{-2} \\ \alpha_c &= 8.685 \frac{R_s}{w Z_o} = \frac{8.686 \times 3.69 \times 10^{-2}}{2.5 \times 10^{-3} \times 50} \\ &= \underline{\underline{2.564 \text{ dB/m}}} \end{aligned}$$

Prob. 11.1

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma_c}} = \frac{1}{\sqrt{\pi \times 500 \times 10^6 \times 4\pi \times 10^{-7} \times 7 \times 10^7}}$$

$$\delta = 2.6902 \times 10^{-6}$$

$$R = \frac{2}{w \delta \sigma_c} = \frac{2}{0.3 \times 2.6902 \times 10^{-6} \times 7 \times 10^7} = \underline{\underline{0.0354 \Omega / \text{m}}}$$

$$L = \frac{\mu_o d}{w} = \frac{4\pi \times 10^{-7} \times 1.2 \times 10^{-2}}{0.3} = \underline{\underline{50.26 \text{ nH} / \text{m}}}$$

$$C = \frac{\epsilon_o w}{d} = \frac{10^{-9}}{36\pi} \times \frac{0.3}{1.2 \times 10^{-2}} = \underline{\underline{221 \text{ pF} / \text{m}}}$$

Since $\sigma = 0$ for air,

$$G = \frac{\sigma w}{d} = 0$$

Prob. 11.2

$$\delta = \frac{1}{\sqrt{\pi f \mu_c \sigma_c}} = \frac{1}{\sqrt{\pi \times 80 \times 10^6 \times 4\pi \times 10^{-7} \times 5.28 \times 10^7}} = 7.744 \times 10^{-6}$$

$$R = \frac{1}{2\pi \delta \sigma_c} \left[\frac{1}{a} + \frac{1}{b} \right] = \frac{\left[\frac{1}{0.8 \times 10^{-3}} + \frac{1}{2.6 \times 10^{-3}} \right]}{2\pi \times 7.744 \times 10^{-6} \times 5.28 \times 10^7} = \frac{10^3(1.25 + 0.3836)}{2569.09} = \underline{\underline{0.6359 \Omega / \text{m}}}$$

$$L = \frac{\mu}{2\pi} \ln \frac{b}{a} = \frac{4\pi \times 10^{-7}}{2\pi} \ln \frac{2.6}{0.8} = \underline{\underline{2.357 \times 10^{-7} \text{ H} / \text{m}}}$$

$$G = \frac{2\pi \sigma}{\ln \frac{b}{a}} = \frac{2\pi \times 10^{-5}}{\ln \frac{2.6}{0.8}} = \underline{\underline{5.33 \times 10^{-5} \text{ S} / \text{m}}}$$

$$C = \frac{2\pi \epsilon}{\ln \frac{b}{a}} = \frac{2\pi \times 3.5 \times \frac{10^{-9}}{36\pi}}{\ln \frac{2.6}{0.8}} = \underline{\underline{1.65 \times 10^{-10} \text{ F} / \text{m}}}$$

Prob. 11.3Method 1:

Assume a charge per unit length Q on the surface of the inner conductor and $-Q$ on the surface of the outer conductor. Using Gauss's law,

$$E_\rho = \frac{Q}{2\pi\epsilon\rho}, \quad a < \rho < b$$

$$V = -\int_a^b \mathbf{E} \cdot d\mathbf{l} = \frac{Q}{2\pi\epsilon} \ln(b/a)$$

$$J = \sigma E = \frac{\sigma Q}{2\pi\epsilon\rho}$$

$$I = \int_S \mathbf{J} \cdot d\mathbf{S} = \int_{\phi=0}^{2\pi} \frac{\sigma Q}{2\pi\epsilon\rho} (1) \rho d\phi = \frac{\sigma Q}{\epsilon}$$

$$G = \frac{I}{V} = \frac{\frac{\sigma Q}{\epsilon}}{\frac{Q}{2\pi\epsilon} \ln(b/a)} = \frac{2\pi\sigma}{\ln(b/a)}$$

Method 2:

Consider a section of unit length, Assume that a total current of I flows from inner conductor to outer conductor. At any radius ρ between a and b ,

$$\mathbf{J} = \frac{I}{2\pi\rho} \mathbf{a}_\rho, \quad a < \rho < b$$

$$\mathbf{E} = \frac{\mathbf{J}}{\sigma} = \frac{I}{2\pi\sigma\rho} \mathbf{a}_\rho$$

$$V = -\int_a^b \mathbf{E} \cdot d\mathbf{l} = \frac{I}{2\pi\sigma} \ln(b/a)$$

$$G = \frac{I}{V} = \frac{2\pi\sigma}{\ln(b/a)}$$

Prob. 11.4

$$(a) \quad R = \frac{2}{w\delta\sigma_c}$$

$$\delta = \frac{1}{\sqrt{\pi f \sigma_c \mu_c}} = \frac{1}{\sqrt{\pi \times 200 \times 10^6 \times 5.8 \times 10^7 \times 4\pi \times 10^{-7}}} = \frac{10^{-3}}{2\pi\sqrt{200}(5.8)} = 4.67 \times 10^{-6}$$

$$R = \frac{2}{30 \times 10^{-3} \times 4.67 \times 10^{-6} \times 5.8 \times 10^7} = \frac{20}{3(4.67)5.8} = \underline{\underline{0.2461 \, \Omega/\text{m}}}$$

$$L = \frac{\mu d}{w} = \frac{4\pi \times 10^{-7} \times 2 \times 10^{-3}}{30 \times 10^{-3}} = \frac{4\pi \times 10^{-7}}{15} = \underline{\underline{83.77 \, \text{nH/m}}}$$

$$G = \frac{\sigma w}{d} = \frac{10^{-3} \times 30 \times 10^{-3}}{2 \times 10^{-3}} = \underline{\underline{15 \, \text{mS/m}}}$$

$$C = \frac{\epsilon w}{d} = \frac{4 \times \frac{10^{-9}}{36\pi} \times 30 \times 10^{-3}}{2 \times 10^{-3}} = \frac{60 \times 10^{-9}}{36\pi} = \underline{\underline{0.5305 \, \text{nF/m}}}$$

$$(b) \quad \gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$\begin{aligned} \gamma^2 &= (0.2461 + j2\pi \times 200 \times 10^6 \times 83.77 \times 10^{-9})(15 \times 10^{-3} + j2\pi \times 200 \times 10^6 \times 0.5305 \times 10^{-9}) \\ &= (0.2461 + j105.3)(15 \times 10^{-3} + j0.6667) \end{aligned}$$

$$\underline{\underline{\gamma = 0.104 + j8.379 \, / \text{m}}}$$

$$Z = \sqrt{\frac{(R + j\omega L)}{(G + j\omega C)}} = \sqrt{\frac{(0.2461 + j105.3)}{(15 \times 10^{-3} + j0.6667)}} = \underline{\underline{12.565 + j0.1266 \, \Omega}}$$

Prob. 11.5

$$C = \frac{\pi \epsilon l}{\cosh^{-1}(d/2a)} \cong \frac{\pi \epsilon l}{\ln(d/a)}$$

since $(d/2a)^2 = 11.11 \gg 1$.

$$C = \frac{\pi \times \frac{10^{-9}}{36\pi} \times 16 \times 10^{-3}}{\ln(2/0.3)} = \underline{\underline{0.2342 \, \text{pF}}}$$

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma_c}} = \frac{1}{\sqrt{\pi \times 10^7 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7}} = 2.09 \times 10^{-5} \, \text{m} \ll a$$

$$R_{ac} = \frac{l}{\pi a \delta \sigma_c} = \frac{16 \times 10^{-3}}{\pi \times 0.3 \times 10^{-3} \times 2.09 \times 10^{-5} \times 5.8 \times 10^7} = \underline{\underline{1.4 \times 10^{-2} \Omega}}$$

Prob. 11.6

We use the two-wire line formulas in Table 11.1

$$\frac{d}{a} = \frac{15 \text{ mm}}{1.2} = 12.5, \quad \left(\frac{d}{2a} \right)^2 = 39.1 \square 1$$

$$\text{Hence, } \cosh^{-1} \frac{d}{2a} \square \ln \frac{d}{a} = \ln 12.5 = 2.526$$

$$L = \frac{\mu}{\pi} \cosh^{-1} \frac{d}{2a} = \frac{\mu}{\pi} \ln \frac{d}{a} = \frac{4\pi \times 10^{-7} \times 2.526}{\pi} = 10.103 \times 10^{-7} = \underline{\underline{1.01 \mu\text{H/m}}}$$

$$C = \frac{\pi \epsilon}{\cosh^{-1} \frac{d}{2a}} = \frac{\pi \epsilon}{\ln \frac{d}{a}} = \frac{\pi}{2.526} 4 \times \frac{10^{-9}}{36\pi} = \underline{\underline{44.105 \text{ pF/m}}}$$

$$Z_o = \sqrt{\frac{L}{C}} = \sqrt{\frac{1.01 \times 10^{-6}}{44.105 \times 10^{-12}}} = \underline{\underline{151.16 \Omega}}$$

Prob. 11.7

$$V(z + \Delta z, t) = V(z, t) - L \Delta z \frac{\partial I(z, t)}{\partial t}$$

or

$$\frac{V(z + \Delta z, t) - V(z, t)}{\Delta z} = -L \frac{\partial I(z, t)}{\partial t} \quad (1)$$

$$I(z + \Delta z, t) = I(z, t) - C \Delta z \frac{\partial V(z + \Delta z, t)}{\partial t}$$

or

$$\frac{I(z + \Delta z, t) - I(z, t)}{\Delta z} = -C \frac{\partial V(z + \Delta z, t)}{\partial t} \quad (2)$$

As $\Delta z \rightarrow 0$, we obtain

$$\frac{\partial V}{\partial z} = -L \frac{\partial I}{\partial t}$$

$$\frac{\partial I}{\partial z} = -C \frac{\partial V}{\partial t}$$

Prob.11.8

(a)

$$\begin{aligned}\gamma &= \sqrt{(R + j\omega L)(G + j\omega C)} = j\omega\sqrt{LC} \sqrt{\left(1 + \frac{R}{j\omega L}\right)\left(1 + \frac{G}{j\omega C}\right)} \\ &= j\omega\sqrt{LC} \sqrt{1 - \frac{RG}{\omega^2 LC} + \frac{R}{j\omega L} + \frac{G}{j\omega C}}\end{aligned}$$

As $R \ll \omega L$ and $G \ll \omega C$, dropping the ω^2 term gives

$$\begin{aligned}\gamma &\cong j\omega\sqrt{LC} \sqrt{1 + \frac{R}{j\omega L} + \frac{G}{j\omega C}} \cong j\omega\sqrt{LC} \left[1 + \frac{R}{2j\omega L} + \frac{G}{2j\omega C}\right] \\ &= \underline{\underline{\left(\frac{R}{2}\sqrt{\frac{C}{L}} + \frac{G}{2}\sqrt{\frac{L}{C}}\right) + j\omega\sqrt{LC}}}\end{aligned}$$

(b)

$$\begin{aligned}Z_o &= \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{L}{C}} \sqrt{\frac{1 + \frac{R}{j\omega L}}{1 + \frac{G}{j\omega C}}} = \sqrt{\frac{L}{C}} \left(1 + \frac{R}{j\omega L}\right)^{1/2} \left(1 + \frac{G}{j\omega C}\right)^{-1/2} \\ &\cong \sqrt{\frac{L}{C}} \left(1 + \frac{R}{2j\omega L} + \dots\right) \left(1 - \frac{G}{2j\omega C} + \dots\right) = \sqrt{\frac{L}{C}} \left(1 - j\frac{R}{2\omega L} + j\frac{G}{2\omega C} + \dots\right) \\ &\cong \underline{\underline{\sqrt{\frac{L}{C}} \left[1 + j\left(\frac{G}{2\omega C} - \frac{R}{2\omega L}\right)\right]}}\end{aligned}$$

Prob. 11.9

$$(a) \quad R + j\omega L = 0.2 + j2\pi \times 12 \times 10^6 \times 40 \times 10^{-6} = 0.2 + j24\pi(40) = 0.2 + j3015.93$$

$$\begin{aligned}G + j\omega C &= 4 \times 10^{-3} + j2\pi \times 12 \times 10^6 \times 25 \times 10^{-6} = 4 \times 10^{-3} + j50\pi(12) \\ &= 4 \times 10^{-3} + j1884.96\end{aligned}$$

$$\begin{aligned}\gamma &= \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{(0.2 + j3015.93)(4 \times 10^{-3} + j1884.96)} \\ &= \underline{\underline{8.159 \times 10^{-2} + j2.384 \times 10^3 \text{ /m}}}\end{aligned}$$

(b) Since $R \ll \omega L$ and $G \ll \omega C$, we may use the result in Problem 11.8(a).

$$\frac{L}{C} = \frac{40 \times 10^{-6}}{25 \times 10^{-6}} = 1.6, \quad LC = 40(25) \times 10^{-12} = 10^{-9}$$

$$\begin{aligned}\gamma &= \left(\frac{R}{2}\sqrt{\frac{C}{L}} + \frac{G}{2}\sqrt{\frac{L}{C}}\right) + j\omega\sqrt{LC} = \left(\frac{0.1}{\sqrt{1.6}} + \frac{4 \times 10^{-3}}{2}\sqrt{1.6}\right) + j2\pi \times 12 \times 10^6 \times \sqrt{40 \times 10^{-6} \times 25 \times 10^{-6}} \\ &= \underline{\underline{8.159 \times 10^{-2} + j2.384 \times 10^3 \text{ /m}}}\end{aligned}$$

Prob. 11.10

(a) $I_1(\omega t - \beta z)$ travels along +z direction.

(b) From eq. (11.18),

$$Z_o = \frac{V_o^+}{I_o^+} = -\frac{V_o^-}{I_o^-}$$

Hence,

$$\underline{V(z, t) = Z_o I_1(\omega t - \beta z) - Z_o I_2(\omega t + \beta z)}$$

Prob. 11.11

$$Z_o = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad (1)$$

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)} \quad (2)$$

$$\alpha = 0.04 \text{ dB/m} = \frac{0.04}{8.686} \text{ Np/m} = 0.00461 \text{ Np/m}$$

Multiplying (1) and (2),

$$Z_o(\alpha + j\beta) = R + j\omega L \quad \longrightarrow \quad 50(0.00461 + j2.5) = R + j\omega L$$

$$R = 50 \times 0.00461 = \underline{\underline{0.2305 \text{ } \Omega/\text{m}}}$$

$$L = \frac{50 \times 2.5}{2\pi \times 60 \times 10^6} = \underline{\underline{0.3316 \text{ } \mu\text{H/m}}}$$

Dividing (2) by (1),

$$\frac{\alpha + j\beta}{Z_o} = G + j\omega C$$

$$G = \frac{\alpha}{Z_o} = \frac{0.00461}{50} = \underline{\underline{92.2 \text{ } \mu\text{S/m}}}$$

$$C = \frac{\beta}{\omega Z_o} = \frac{2.5}{2\pi \times 60 \times 10^6 \times 50} = \underline{\underline{0.1326 \text{ nF/m}}}$$

Prob. 11.12

$$Z_o = \sqrt{\frac{L}{C}} = \sqrt{\frac{\mu d}{w}} \cdot \frac{d}{\epsilon w} = \frac{d}{w} \sqrt{\frac{\mu}{\epsilon}}$$

$$Z_o = \eta_o \frac{d}{w} = 78$$

$$Z_o' = \eta_o \frac{d}{w'} = 75$$

$$\frac{78}{75} = \frac{w'}{w} \rightarrow w' = 1.04w$$

i.e. the width must be increased by 4%.

Prob. 11.13

$$\begin{aligned} \text{(a)} \quad Z_o &= \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{6.8 + j2\pi \times 10^3 \times 3.4 \times 10^{-3}}{0.42 \times 10^{-6} + j2\pi \times 10^3 \times 8.4 \times 10^{-9}}} \\ &= 10^3 \sqrt{\frac{6.8 + j21.36}{0.42 + j52.78}} = \underline{\underline{644.3 - j97 \, \Omega}} \end{aligned}$$

$$\begin{aligned} \gamma &= \sqrt{(R + j\omega L)(G + j\omega C)} = 10^{-3} \sqrt{(6.8 + j21.36)(0.42 - j52.78)} \\ &= \underline{\underline{(5.415 + j33.96) \times 10^{-3} \, \text{/mi}}} \end{aligned}$$

$$\text{(b)} \quad u = \frac{\omega}{\beta} = \frac{2\pi \times 10^3}{33.96 \times 10^{-3}} = \underline{\underline{1.85 \times 10^5 \, \text{mi/s}}}$$

$$\text{(c)} \quad \lambda = \frac{2\pi}{\beta} = \frac{2\pi}{33.96 \times 10^{-3}} = \underline{\underline{185.02 \, \text{mi}}}$$

Prob. 11.14

Using eq. (11.42a),

$$Z_{in} = -jZ_o \cot \beta \ell$$

$$Z_o = 250, \quad \lambda = \frac{c}{f} = \frac{3 \times 10^8}{400 \times 10^6} = 0.75 \, \text{m}$$

$$\beta \ell = \frac{2\pi}{\lambda} \ell = \frac{2\pi \times 0.1}{0.75} = 48^\circ$$

$$Z_{in} = -j(250) \cot 48^\circ = \underline{\underline{-j225.1 \, \Omega}}$$

Prob. 11.15

Assume that the line is lossless.

$$Z_o = \sqrt{\frac{L}{C}}$$

From Table 11.1, $L = \frac{\mu}{2\pi} \ln \frac{b}{a}$, $C = \frac{2\pi\epsilon}{\ln \frac{b}{a}}$

$$\frac{L}{C} = \frac{\mu}{\epsilon} \left(\frac{1}{2\pi} \ln \frac{b}{a} \right)^2$$

$$Z_o = \sqrt{\frac{L}{C}} = \frac{1}{2\pi} \ln \frac{b}{a} \times \sqrt{\frac{\mu}{\epsilon}} = \frac{\eta_o}{2\pi\sqrt{\epsilon_r}} \ln \frac{b}{a}$$

$$\ln \frac{b}{a} = 2\pi\sqrt{\epsilon_r} \frac{Z_o}{\eta_o} = 2\pi\sqrt{2.25} \frac{75}{120\pi} = 1.875$$

$$\frac{b}{a} = e^{1.875} \longrightarrow a = be^{-1.875} = 3e^{-1.875} \text{ mm} = \underline{\underline{0.46 \text{ mm}}}$$

Prob. 11.16

(a) For a lossless line, $R = 0 = G$.

$$\gamma = j\omega\sqrt{LC} \longrightarrow \beta = \omega\sqrt{LC} = \omega\sqrt{\mu_o\epsilon_o} = \frac{\omega}{c}$$

$$u = \frac{\omega}{\beta} = c = \frac{1}{\sqrt{LC}}$$

(b) For lossless line, $R = 0 = G$

$$L = \frac{\mu}{\pi} \cosh^{-1} \frac{d}{2a}, C = \frac{\pi\epsilon}{\cosh^{-1} \frac{d}{2a}}$$

$$Z_o = \sqrt{\frac{L}{C}} = \sqrt{\frac{\mu}{\pi} \cdot \frac{1}{\pi\epsilon}} \cosh^{-1} \frac{d}{2a} = \frac{120\pi}{\pi\sqrt{\epsilon_r}} \cosh^{-1} \frac{d}{2a}$$

$$= \underline{\underline{\frac{120}{\sqrt{\epsilon_r}} \cosh^{-1} \frac{d}{2a}}}$$

Yes, true for other lossless lines.

Prob. 11.17

$$L = \frac{\mu}{\pi} \cosh^{-1} \frac{d}{2a} = 4 \times 10^{-7} \cosh^{-1} \frac{0.32}{0.12}$$

$$\underline{\underline{L = 0.655 \mu\text{H/m}}}$$

$$C = \frac{\pi \epsilon}{\cosh^{-1} \frac{d}{2a}} = \frac{\pi \times \frac{10^{-9}}{36\pi} \times 3.5}{\cosh^{-1} 2.667}$$

$$\underline{\underline{C = 59.4 \text{ pF/m}}}$$

$$Z_o = \sqrt{\frac{L}{C}} = \sqrt{\frac{0.655 \times 10^{-6}}{59.4 \times 10^{-12}}} = \underline{\underline{105 \Omega}}$$

or

$$Z_o = \frac{120}{\sqrt{3.5}} \cosh^{-1} 2.667 = \underline{\underline{105 \Omega}}$$

Prob. 11.18

For a distortionless cable,

$$\frac{R}{L} = \frac{G}{C} \longrightarrow RC = LG \quad (1)$$

$$Z_o = \sqrt{\frac{L}{C}} = 60 \quad (2)$$

$$u = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}} = \frac{\ell}{t_o} = \frac{4}{80 \times 10^{-6}} \quad (3)$$

$$\alpha \ell = 0.24 \text{ dB} = \frac{0.24}{8.686} \text{ Np} = 0.0276$$

$$\alpha = \sqrt{RG} = 0.00069 \quad (4)$$

From (2) and (3),

$$\frac{1}{C} = \frac{60 \times 4}{80 \times 10^{-6}} \longrightarrow C = \frac{8 \times 10^{-5}}{240} = \underline{\underline{333.3 \text{ nF/m}}}$$

From (2),

$$L = (60)^2 C = 3600 \times 333.3 \times 10^{-9} = \underline{\underline{1.20 \text{ mH/m}}}$$

From (1) and (4),

$$\frac{C}{G} = \frac{LG}{0.00069^2} \longrightarrow G^2 = \frac{0.00069^2}{60^2}$$

$$G = \frac{0.00069}{60} = \underline{\underline{11.51 \mu\text{S/m}}}$$

From (4),

$$R = \frac{0.00069^2}{G} = 0.00069 \times 60 = \underline{\underline{0.0414 \Omega/\text{m}}}$$

Prob. 11.19

$$(a) \frac{R}{L} = \frac{G}{C} \rightarrow G = \frac{R}{L} C = \frac{20 \times 63 \times 10^{-12}}{0.3 \times 10^{-6}}$$

$$G = 4.2 \times 10^{-3} \text{ S/m}$$

$$\alpha = \sqrt{RG} = \sqrt{20 \times 4.2 \times 10^{-3}} = 0.2898$$

$$\beta = \omega \sqrt{LC} = 2\pi \times 120 \times 10^6 \sqrt{0.3 \times 10^{-6} \times 63 \times 10^{-12}} = 3.278$$

$$\underline{\underline{\gamma = 0.2898 + j3.278 \text{ /m}}}$$

$$u = \frac{\omega}{\beta} = \frac{2\pi \times 120 \times 10^6}{3.278} = \underline{\underline{2.3 \times 10^8 \text{ m/s}}}$$

$$Z_o = \sqrt{\frac{L}{C}} = \sqrt{\frac{0.3 \times 10^{-6}}{63 \times 10^{-12}}} = \underline{\underline{69 \Omega}}$$

(b) Let V_o be its original magnitude

$$V_o e^{-\alpha z} = 0.2V_o \rightarrow e^{\alpha z} = 5$$

$$z = \frac{1}{\alpha} \ln 5 = \underline{\underline{5.554 \text{ m}}}$$

$$(c) \beta l = 45^\circ = \frac{\pi}{4} \rightarrow l = \frac{\pi}{4\beta} = \frac{4}{4 \times 3.278} = \underline{\underline{0.3051 \text{ m}}}$$

Prob. 11.20

$$(a) \alpha = \underline{\underline{0.0025 \text{ Np/m}}}, \quad \beta = \underline{\underline{2 \text{ rad/m}}},$$

$$u = \frac{\omega}{\beta} = \frac{10^8}{2} = \underline{\underline{5 \times 10^7 \text{ m/s}}}$$

$$(b) \quad \Gamma = \frac{V_o}{V_o^+} = \frac{60}{120} = \frac{1}{2}$$

$$\text{But } \Gamma = \frac{Z_L - Z_o}{Z_L + Z_o} \rightarrow \frac{1}{2} = \frac{300 - Z_o}{300 + Z_o} \rightarrow \underline{\underline{Z_o = 100 \Omega}}$$

$$\begin{aligned} I(l') &= \frac{120}{Z_o} e^{0.0025l'} \cos(10^8 + 2l') - \frac{60}{Z_o} e^{-0.0025l'} \cos(10^8 t - 2l') \\ &= \underline{\underline{1.2e^{0.0025l'} \cos(10^8 + 2l') - 0.6e^{-0.0025l'} \cos(10^8 t - 2l') A}} \end{aligned}$$

Prob. 11.21

$$\alpha = 10^{-3}, \quad \beta = 0.01$$

$$\gamma = \alpha + j\beta = 0.001 + j0.01 = \underline{\underline{(1 + j10) \times 10^{-3} \text{ /m}}}$$

$$u = \frac{\omega}{\beta} = \frac{2\pi \times 10^4}{0.01} = \underline{\underline{6.283 \times 10^6 \text{ m/s}}}$$

Prob. 11.22

$$Z_o = \sqrt{\frac{L}{C}} = \sqrt{\frac{0.6 \times 10^{-6}}{82 \times 10^{-12}}} = \underline{\underline{85.54 \Omega}}$$

$$RC = LG \quad \longrightarrow \quad G = \frac{RC}{L}$$

$$\alpha = \sqrt{RG} = \sqrt{R \frac{RC}{L}} = \frac{R}{Z_o} = \frac{10 \times 10^{-3}}{85.54} = 1.169 \times 10^{-4} \text{ Np/m}$$

$$\begin{aligned} \beta &= \omega \sqrt{LC} = 2\pi \times 80 \times 10^6 \sqrt{0.6 \times 10^{-6} \times 82 \times 10^{-12}} \\ &= 3.5258 \text{ rad/m} \end{aligned}$$

$$\gamma = \underline{\underline{1.169 \times 10^{-4} + j3.5258 \text{ /m}}}$$

Prob. 11.23

$$R + j\omega L = 6.5 + j2\pi \times 2 \times 10^6 \times 3.4 \times 10^{-6} = 6.5 + j4273$$

$$G + j\omega C = 8.4 \times 10^{-3} + j2\pi \times 2 \times 10^6 \times 21.5 \times 10^{-12} = (8.4 + j0.27) \times 10^{-3}$$

$$Z_o = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{6.5 + j4273}{(8.4 + j0.27) \times 10^{-3}}}$$

$$Z_o = 71.71 \angle 39.75^\circ = \underline{\underline{55.12 + j45.85 \Omega}}$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{(43.19 \angle 81.34^\circ)(8.4 \times 10^{-3} \angle 1.84^\circ)}$$

$$= 0.45 + j0.4 \text{ /m}$$

$$\begin{array}{cc} \uparrow & \uparrow \\ \alpha & \beta \end{array}$$

$$t = \frac{l}{u}, \text{ but } u = \frac{\omega}{\beta},$$

$$t = \frac{\beta l}{\omega} = \frac{0.39 \times 5.6}{2\pi \times 2 \times 10^6} = \underline{\underline{0.1783 \mu s}}$$

Prob. 11.24

(a) For a lossy line,

$$Z_{in} = Z_o \left[\frac{Z_L + Z_o \tanh \gamma \ell}{Z_o + Z_L \tanh \gamma \ell} \right]$$

For a short-circuit, $Z_L = 0$.

$$Z_{sc} = Z_{in} \Big|_{Z_L = 0} = Z_o \tanh \gamma \ell$$

$$\tanh \gamma \ell = \frac{Z_{sc}}{Z_o} = \frac{30 - j12}{80 + j60} = 0.168 - j0.276$$

$$\gamma \ell = \alpha \ell + j\beta \ell = \tanh^{-1}(0.168 - j0.276) = 0.1571 - j0.2762$$

$$\alpha = \frac{0.1571}{2.1} = \underline{\underline{0.0748 \text{ Np/m}}}$$

$$\beta = \frac{0.2762}{2.1} = \underline{\underline{0.1316 \text{ rad/m}}}$$

(b)

$$Z_{in} = (80 + j60) \left[\frac{(40 + j30) + (80 + j60)(0.168 - j0.276)}{(80 + j60) + (40 + j30)(0.168 - j0.276)} \right]$$

$$= \underline{\underline{61.46 + j24.43 \Omega}}$$

Prob. 11.25

$$Z_{in} = Z_o \left[\frac{Z_L + Z_o \tanh \gamma \ell}{Z_o + Z_L \tanh \gamma \ell} \right]$$

$$\gamma \ell = \alpha \ell + j\beta \ell = 1.4 \times 0.5 + j2.6 \times 0.5 = 0.7 + j1.3$$

$$\tanh \gamma \ell = 1.4716 + j0.3984$$

$$\begin{aligned} Z_{in} &= (75 + j60) \left[\frac{200 + (75 + j60)(1.4716 + j0.3984)}{(75 + j60) + 200(1.4716 + j0.3984)} \right] \\ &= \underline{\underline{57.44 + j48.82 \, \Omega}} \end{aligned}$$

Prob. 11.26

$$\begin{aligned} \text{(a)} \quad T_L &= \frac{V_L}{V_o^+} = \frac{Z_L I_L}{\frac{1}{2}(V_L + Z_o I_L)} = \frac{2Z_L I_L}{Z_L I_L + Z_o I_L} \\ &= \underline{\underline{\frac{2Z_L}{Z_L + Z_o}}} \end{aligned}$$

$$1 + \Gamma_L = 1 + \frac{Z_L - Z_o}{Z_L + Z_o} = \underline{\underline{\frac{2Z_L}{Z_L + Z_o}}}$$

$$\text{(b)} \quad \text{(i)} \quad \tau_L = \frac{2nZ_o}{nZ_o + Z_o} = \frac{2n}{n+1}$$

$$\text{(ii)} \quad \tau_L =_{Y_L} \xrightarrow{\lim} 0 = \frac{2}{1 + \frac{Z_o}{Z_L}} = 2$$

$$\text{(iii)} \quad \tau_L =_{Z_L} \xrightarrow{\lim} 0 = \frac{2Z_L}{Z_L + Z_o} = 0$$

$$\text{(iv)} \quad \tau_L = \frac{2Z_o}{2Z_o} = 1$$

Prob. 11.27

$$\text{(a)} \quad \Gamma = \frac{Z_L - Z_o}{Z_L + Z_o} \quad \rightarrow \quad \Gamma Z_L + \Gamma Z_o = Z_L - Z_o \quad \rightarrow \quad Z_L(\Gamma - 1) = -(\Gamma + 1)Z_o$$

$$\text{Hence,} \quad Z_L = \frac{1 + \Gamma}{1 - \Gamma} Z_o$$

$$(b) \quad Z_L = \frac{1 + 0.6\angle 45^\circ}{1 - 0.6\angle 45^\circ} (50) = \frac{1.424 + j0.424}{0.5751 - j0.424} (50) = (1.252 + j1.6586)(50) \\ = \underline{\underline{62.6 + j83.03 \, \Omega}}$$

Prob. 11.28

$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{200 - j240 - 120}{200 - j240 + 120} = \frac{80 - j240}{320 - j240} = 0.52 - j0.36 = \underline{\underline{0.6325\angle -34.7^\circ}} \\ s = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = \frac{1 + 0.6325}{1 - 0.6325} = \underline{\underline{4.4415}}$$

Prob. 11.29

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{(3.5 + j2\pi \times 400 \times 10^6 \times 2 \times 10^{-6})(0 + j2\pi \times 400 \times 10^6 \times 120 \times 10^{-12})} \\ = \sqrt{(3.5 + j5026.55)(j0.3016)} = 0.0136 + j38.94$$

$$\alpha = \underline{\underline{0.0136 \text{ Np/m}}}, \quad \beta = \underline{\underline{38.94 \text{ rad/m}}}$$

$$u = \frac{\omega}{\beta} = \frac{2\pi \times 400 \times 10^6}{38.94} = \underline{\underline{6.452 \times 10^7 \text{ m/s}}}$$

$$Z_o = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{3.5 + j5026.55}{j0.3016}} = \underline{\underline{129.1 - j0.045 \, \Omega}}$$

Prob. 11.30

From eq. (11.33)

$$Z_{sc} = Z_{in} \Big|_{Z_L=0} = Z_o \tanh \gamma l$$

$$Z_{oc} = Z_{in} \Big|_{Z_L=\infty} = \frac{Z_o}{\tanh \gamma l} = Z_o \coth(\gamma l)$$

For lossless line, $\gamma = j\beta$, $\tan(\gamma l) = \tanh(j\beta l) = j \tan(\beta l)$

$$Z_{sc} = jZ_o \tan(\beta l), Z_{oc} = -jZ_o \cot(\beta l)$$

$$Z_{sc} = jZ_o \tan(\beta l) = (65 + j38) \tanh[(0.7 + j2.5)0.8] = (65 + j38) \tanh(0.56 + j2) = 75.25\angle 30.3^\circ \frac{\exp(0.56) - \exp(-0.56)}{\exp(0.56) + \exp(-0.56)} \\ = 7.86\angle 60.3^\circ = 3.89 + j6.83$$

Prob. 11.31

$$\Gamma = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{(1 + j2) - 1}{(1 + j2) + 1} = \frac{j}{1 + j} = \frac{1 \angle 90^\circ}{\sqrt{2} \angle 45^\circ} = \underline{\underline{0.7071 \angle -45^\circ}}$$

Prob. 11.32

$$(a) \quad \Gamma = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{120 - 50}{170} = \underline{\underline{0.4118}}$$

$$\text{For resistive load, } s = \frac{Z_L}{Z_o} = \underline{\underline{2.4}}$$

$$(b) \quad Z_{in} = Z_o \frac{Z_L + jZ_o \tan(\beta l)}{Z_o + jZ_L \tan(\beta l)}$$

$$\beta l = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{6} = 60^\circ$$

$$Z_{in} = 50 \left[\frac{120 + j50 \tan(60^\circ)}{50 + j120 \tan(60^\circ)} \right] = \underline{\underline{34.63 \angle -40.65^\circ \Omega}}$$

Prob. 11.33

$$(a) \quad \beta l = \frac{1}{4} \times 100 = 25 \text{ rad} = 1432.4^\circ = 352.4^\circ$$

$$Z_{in} = 60 \left[\frac{j40 + j60 \tan 352.4^\circ}{60 - 40 \tan 352.4^\circ} \right] = \underline{\underline{j29.375 \Omega}}$$

$$\begin{aligned} V(z=0) &= V_o = \frac{Z_{in}}{Z_{in} + Z_g} V_g = \frac{j29.375(10 \angle 0^\circ)}{j29.375 + 50 - j40} \\ &= \frac{293.75 \angle 90^\circ}{51.116 \angle -12^\circ} = \underline{\underline{5.75 \angle 102^\circ}} \end{aligned}$$

$$(b) \quad Z_{in} = Z_L = \underline{\underline{j40 \Omega}}$$

$$V_o^+ = \frac{V_g}{(e^{j\beta l} + \Gamma e^{-j\beta l})} \quad (\text{l is from the load})$$

$$V_L = \frac{V_g(1+\Gamma)}{(e^{j\beta l} + \Gamma e^{-j\beta l})} = \underline{\underline{12.62\angle 0^\circ \text{ V}}}$$

$$(c) \quad \beta l' = \frac{1}{4} \times 4 = 1 \text{ rad} = 57.3^\circ$$

$$Z_{in} = 60 \left[\frac{j40 + j60 \tan 57.3^\circ}{60 - 40 \tan 57.3^\circ} \right] = \underline{\underline{-j3471.88 \Omega}}$$

$$V = \frac{V_g(e^j + \Gamma e^{-j})}{(e^{j25} + \Gamma e^{-j25})} = \underline{\underline{22.74\angle 0^\circ \text{ V}}}$$

(d) 3m from the source is the same as 97m from the load, i.e.

$$l' = 100 - 3 = 97 \text{ m}, \quad \beta l' = \frac{1}{4} \times 97 = 24.25 \text{ rad} = 309.42^\circ$$

$$Z_{in} = 60 \left[\frac{j40 + j60 \tan 309.42^\circ}{60 - 40 \tan 309.42^\circ} \right] = \underline{\underline{-j18.2 \Omega}}$$

$$V = \frac{V_g(e^{j97/4} + \Gamma e^{-j97/4})}{(e^{j25} + \Gamma e^{-j25})} = \underline{\underline{6.607\angle 180^\circ \text{ V}}}$$

Prob. 11.34

$$V_1 = V_s(z=0) = V_o^+ + V_o^- \quad (1)$$

$$V_2 = V_s(z=l) = V_o^+ e^{-\gamma l} + V_o^- e^{\gamma l} \quad (2)$$

$$I_1 = I_s(z=0) = \frac{V_o^+}{Z_o} - \frac{V_o^-}{Z_o} \quad (3)$$

$$I_2 = -I_s(z=l) = -\frac{V_o^+}{Z_o} e^{-\gamma l} + \frac{V_o^-}{Z_o} e^{\gamma l} \quad (4)$$

$$(1) + (3) \rightarrow V_o^+ = \frac{1}{2}(V_1 + Z_o I_1)$$

$$(1) - (3) \rightarrow V_o^- = \frac{1}{2}(V_1 - Z_o I_1)$$

Substituting V_o^+ and V_o^- in (2) gives

$$\begin{aligned} V_2 &= \frac{1}{2}(V_1 + Z_o I_1)e^{-\gamma l} + \frac{1}{2}(V_1 - Z_o I_1)e^{\gamma l} \\ &= \frac{1}{2}(e^{\gamma l} + e^{-\gamma l})V_1 + \frac{1}{2}Z_o(e^{-\gamma l} - e^{\gamma l})I_1 \end{aligned}$$

$$V_2 = \cosh \gamma l V_1 - Z_o \sinh \gamma l I_1 \quad (5)$$

Substituting V_o^+ and V_o^- in (4),

$$\begin{aligned}
 I_2 &= -\frac{1}{2Z_o}(V_1 + Z_o I_1)e^{-\gamma l} + \frac{1}{2Z_o}(V_1 - Z_o I_1)e^{\gamma l} \\
 &= \frac{1}{2Z_o}(e^{\gamma l} - e^{-\gamma l})V_1 + \frac{1}{2}(e^{\gamma l} + e^{-\gamma l})I_1 \\
 I_2 &= \frac{1}{Z_o} \sinh \gamma l V_1 - \cosh \gamma l I_1
 \end{aligned} \tag{6}$$

From (5) and (6)

$$\begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} = \begin{bmatrix} \cosh \gamma l & -Z_o \sinh \gamma l \\ -\frac{1}{Z_o} \sinh \gamma l & \cosh \gamma l \end{bmatrix} \begin{bmatrix} V_1 \\ I_1 \end{bmatrix}$$

But

$$\begin{bmatrix} \cosh \gamma l & -Z_o \sinh \gamma l \\ -\frac{1}{Z_o} \sinh \gamma l & \cosh \gamma l \end{bmatrix}^{-1} = \begin{bmatrix} \cosh \gamma l & Z_o \sinh \gamma l \\ \frac{1}{Z_o} \sinh \gamma l & \cosh \gamma l \end{bmatrix}$$

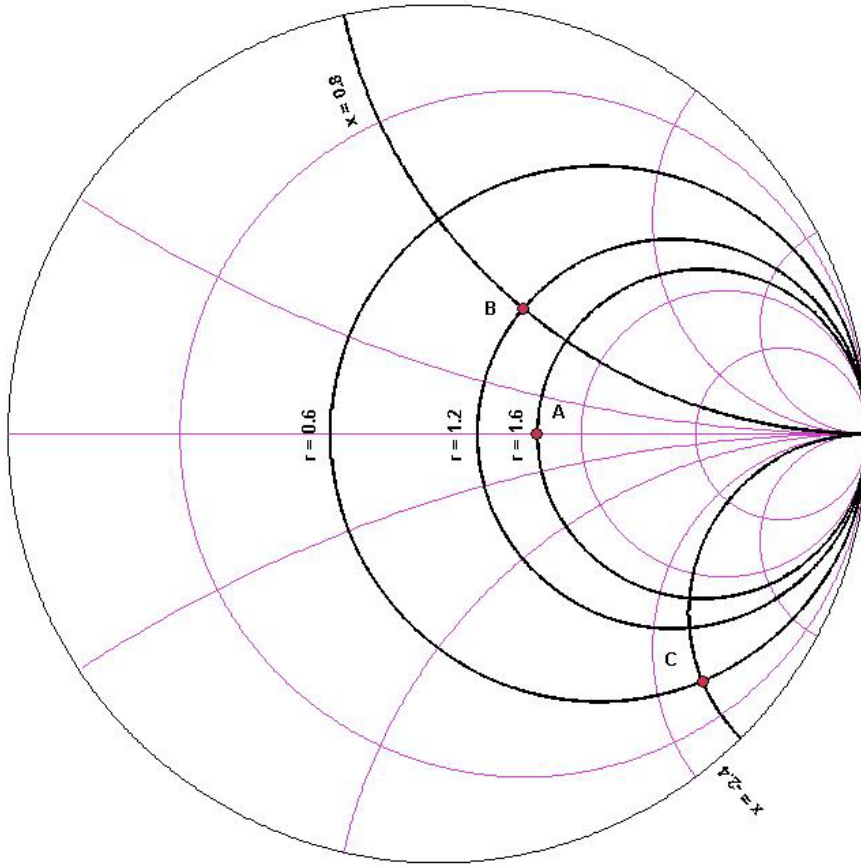
Thus

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} \cosh \gamma l & Z_o \sinh \gamma l \\ \frac{1}{Z_o} \sinh \gamma l & \cosh \gamma l \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

Prob. 11.35

- (a) $z_a = \frac{Z_a}{Z_o} = \frac{80}{50} = 1.6$
- (b) $z_b = \frac{Z_b}{Z_o} = \frac{60 + j40}{50} = 1.2 + j0.8$
- (c) $z_c = \frac{Z_c}{Z_o} = \frac{30 - j120}{50} = 0.6 - j2.4$

The three loads are located on the Smith chart, as A, B, and C as shown next.

**Prob. 11.36**

$$z_L = \frac{Z_L}{Z_0} = \frac{210}{100} = 2.1 = s$$

$$\text{Or } \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{110}{310},$$

$$s = \frac{1 + |\Gamma|}{1 - |\Gamma|} = 2.1$$

$$\text{But } s = \frac{V_{\max}}{V_{\min}} \rightarrow V_{\max} = sV_{\min}$$

$$\text{Since the line is } \frac{\lambda}{4} \text{ long, } \frac{\lambda}{4} \rightarrow \frac{720^\circ}{4} = 120^\circ$$

Hence the sending end will be $V_{\min}$, while the receiving end at $V_{\max}$

$$V_{\min} = V_{\max} / s = 80 / 2.1 = 38.09$$

$$V_{\text{sending}} = \underline{\underline{38.09 \angle 90^\circ}}$$

Prob. 11.37

$$Z_L = (1 + j2)Z_o \longrightarrow z_L = \frac{Z_L}{Z_o} = 1 + j2$$

We locate z_L on the Smith chart.

$$\frac{\lambda}{4} \longrightarrow \frac{720^\circ}{4} = 180^\circ$$

We move 180° toward the generator and locate point Q at which

$$z = 0.2 - j0.4$$

$$Z = zZ_o = \underline{\underline{(0.2 - j0.4)Z_o}}$$

Prob. 11.38

(a) Method 1: At Y,

$$Z_{in} = Z_o \left[\frac{Z_L + jZ_o \tan \beta \ell}{Z_o + jZ_L \tan \beta \ell} \right]$$

$$\beta \ell = \frac{2\pi}{\lambda} \frac{\lambda}{2} = \pi, \quad \tan \pi = 0$$

$$Z_{in} = Z_o \frac{Z_L}{Z_o} = Z_L = 150 \Omega$$

At X, $Z_L' = 150 \Omega$

$$\beta \ell = \frac{2\pi}{\lambda} \frac{\lambda}{4} = \pi / 2, \quad \tan \pi / 2 = \infty$$

$$Z_{in} = \lim_{\tan \beta \ell \rightarrow \infty} Z_o \left[\frac{jZ_o + \frac{Z_L'}{\tan \beta \ell}}{jZ_L' + \frac{Z_o}{\tan \beta \ell}} \right] = \frac{Z_o^2}{Z_L'} = \frac{(75)^2}{150} = 37.5 \Omega$$

Method 2: Using the Smith chart,

$$z_L = \frac{Z_L'}{Z_o} = \frac{150}{50} = 3$$

Since $\ell = \lambda/2$, we must move 360° toward the generator. We arrive at the same point. Hence,

$$Z_{in} = Z_L = 150$$

$$z_L' = \frac{150}{75} = 2$$

$$\ell = \frac{\lambda}{4} \rightarrow 180^\circ$$

We move 180° toward the generator. $z_{in} = 0.5$

$$Z_{in} = 75(0.6) = \underline{\underline{37.5\Omega}}$$

(b) From the Smith chart,

$s = 3$ for section XY

$s = 2$ for section YZ

$$(c) \quad \Gamma = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{150 - 50}{150 + 50} = \underline{\underline{0.5}}$$

Prob. 11.39

$$z_{in} = \frac{Z_{in}}{Z_o} = \frac{100 - j120}{50} = 2 - j2.4$$

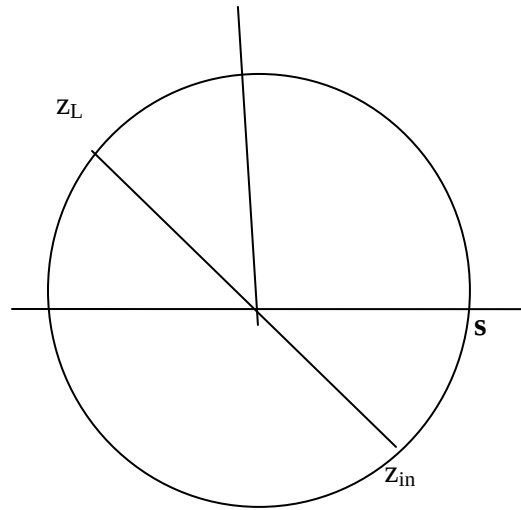
$$\lambda = \frac{u}{f} = \frac{0.8 \times 3 \times 10^8}{6 \times 10^8} = 0.4 \text{ m}, \quad \ell = 0.1 \text{ m} = \frac{\lambda}{4}$$

If $\lambda \rightarrow 720^\circ$, then $\lambda/4 \rightarrow 180^\circ$

We move 180° counterclockwise from z_{in} to get z_L on the Smith chart as shown below.

$$z_L = 0.2 + j0.25 \rightarrow Z_L = Z_o z_L = 50(0.2 + j0.25) = \underline{\underline{10 + j12.5 \Omega}}$$

From the Smith chart, $\underline{\underline{s = 5.19}}$

**Prob. 11.40**

$$z_L = \frac{Z_L}{Z_o} = \frac{40 - j25}{50} = 0.8 - j0.5$$

We locate this at point P on the Smith chart shown below

$$|\Gamma_L| = \frac{OP}{OQ} = \frac{2.4 \text{ cm}}{8 \text{ cm}} = 0.3, \quad \theta_r = -96^\circ$$

$$\underline{\underline{\Gamma_L = 0.3 \angle -96^\circ}}$$

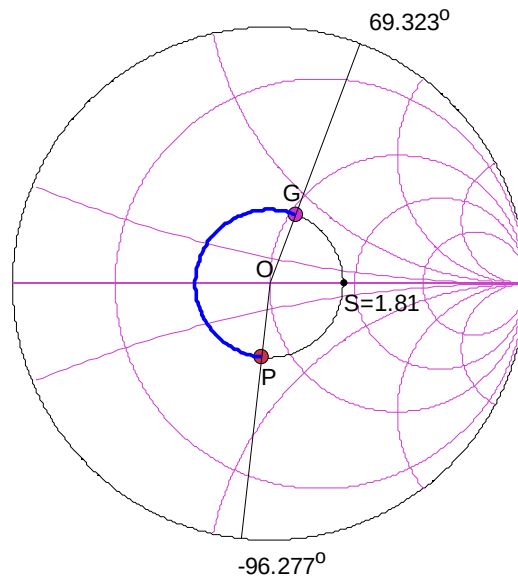
$$\text{At S, } s = r = \underline{\underline{1.81}}$$

$$\ell = 0.27\lambda \longrightarrow 0.27 \times 720^\circ = 194.4^\circ$$

From P, we move 194.4° toward the generator to G. At G,

$$z_{in} = 1.0425 + j0.6133$$

$$\underline{\underline{Z_{in} = Z_o z_{in} = 50(1.0425 + j0.6133) = 52.13 - j30.66 \Omega}}$$

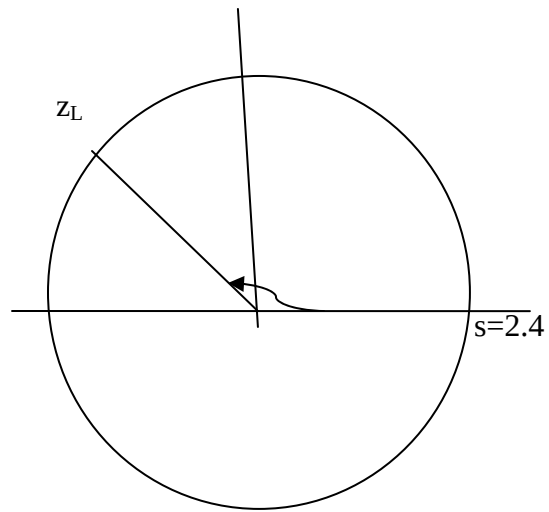
**Prob. 11.41**

$\lambda \rightarrow 720^\circ$ implies that $0.2\lambda \rightarrow 144^\circ$. The minimum voltage is located at s on the Smith chart. We move 144° from there to locate the load z_L as shown on the Smith chart below.

At z_L , we obtain

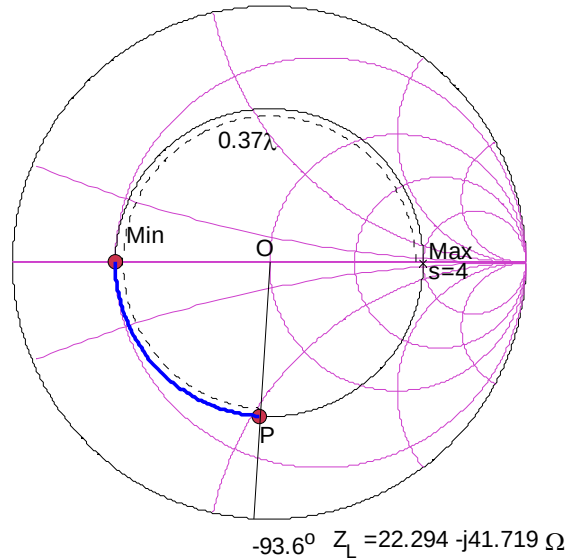
$$z_L = 0.46 + j0.26, \quad \rightarrow \quad Z_L = 100(0.46 + j0.26) = \underline{\underline{46 + j26 \, \Omega}}$$

$$\Gamma = |\Gamma| \angle \theta_\Gamma = \frac{OP}{OQ} \angle 144^\circ = \frac{3.5cm}{8.4cm} \angle 144^\circ = \underline{\underline{0.4167 \angle 144^\circ}}$$

**Prob. 11.42**

(a) $0.12\lambda \longrightarrow 0.12 \times 720^\circ = 86.4^\circ$

We draw the $s=4$ circle and locate $V_{\min}$. We move from that location 86.4° toward the load.



At P, $z_L = 0.45 - j0.83$

$$Z = 50(0.45 - j0.83) = \underline{\underline{22.3 - j41.72 \, \Omega}}$$

(b) the load is capacitive.

(c) $V_{\min}$ and $V_{\max}$ are $\lambda/4$ apart. Hence the first maximum occurs at

$$0.12\lambda + 0.25\lambda = \underline{\underline{0.37\lambda}}$$

Prob. 11.43

$$(a) \quad z_L = \frac{Z_L}{Z_o} = \frac{75 + j60}{50} = 1.5 + j1.2$$

$$|\Gamma| = \frac{OP}{OQ} = \frac{3.8\text{cm}}{8\text{cm}} = 0.475, \quad \theta_r = 42^\circ$$

$$\Gamma = \underline{\underline{0.475 \angle 42^\circ}}$$

(Exact value = $0.4688 \angle 41.76^\circ$)

(b) $s=2.8$
(Exact value = 2.765)

(c) $0.2\lambda \rightarrow 0.2 \times 720^\circ = 144^\circ$

$$z_{in} = 0.55 - j0.65$$

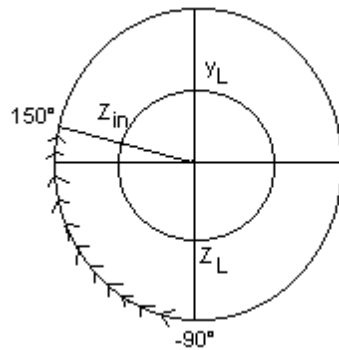
$$Z_{in} = Z_o z_{in} = 50(0.55 + j0.65) = \underline{\underline{27.5 + j32.5 \Omega}}$$

(d) Since $\theta_r = 42^\circ$, V_{min} occurs at

$$\frac{42}{720} \lambda = \underline{\underline{0.05833\lambda}}$$

(e) same as in (d), i.e., $\underline{\underline{0.05833\lambda}}$

Prob. 11.44



If $\lambda \rightarrow 720^\circ$, then $\frac{\lambda}{6} \rightarrow 120^\circ$

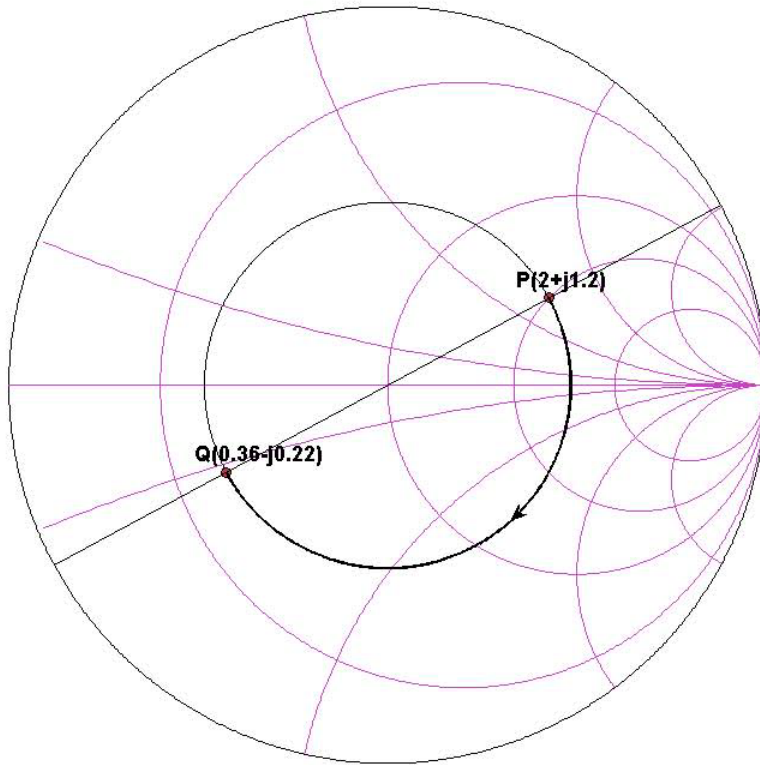
$$z_{in} = \underline{\underline{0.35 + j0.24}}$$

Prob. 11.45

$$z = \frac{Z}{Z_o} = \frac{100 + j60}{50} = 2 + j1.2$$

We locate z on the Smith chart. We move 180° toward the generator to reach point Q. At Q, $y = 0.36 - j0.22$

$$Y = yY_o = \frac{1}{50}(0.36 - j0.22) = \underline{\underline{7.4 - j4.4 \text{ mS}}}$$

**Prob. 11.46**

$$\lambda = \frac{u}{f} = \frac{0.5 \times 3 \times 10^8}{160 \times 10^6} = 0.9375 \text{ m}$$

$$z_L = \frac{Z_L}{Z_o} = \frac{50 + j30}{50} = 1 + j0.6$$

We locate z at P on the Smith chart. We draw a circle that passes through P.

We locate point Q as the point where the circle crosses the Γ_r - axis. At Q,

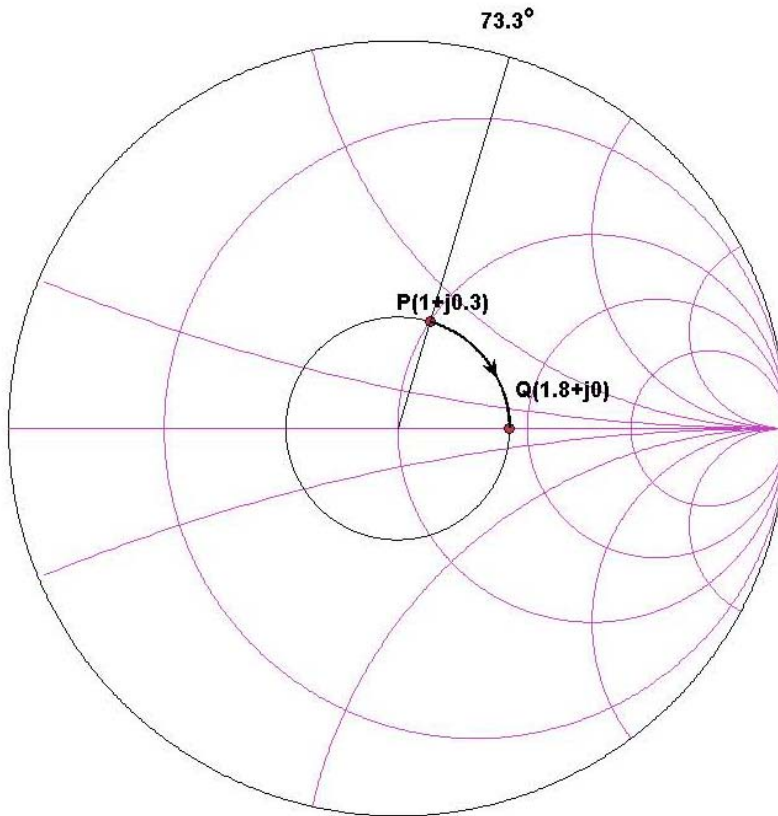
$$z_{in} = 1.8$$

$$Z_{in} = Z_o z_{in} = 50(1.8) = \underline{\underline{90 \Omega}}$$

The angular distance between P and Q is $\theta_r = 73.3^\circ$.

$$\text{If } \lambda \rightarrow 720^\circ, \quad 73.3^\circ \rightarrow \frac{\lambda}{720^\circ} 73.3^\circ$$

$$\ell = \frac{73.3^\circ}{720^\circ} \times 0.9375 = \underline{\underline{0.0954 \text{ m}}}$$

**Prob. 11.47**

$$z_L = \frac{60}{100} = 0.6$$

If $\lambda \rightarrow 720^\circ$, then $\lambda / 2 \rightarrow 360^\circ$. This means that the input impedance is located on the same spot as z_L on the Smith chart.

$$Z'_{in} = Z_o z_L = 60 \rightarrow z'_{in} = 60 / 50 = 1.2$$

$$\lambda / 4 \rightarrow 180^\circ, \quad z_{in} = 0.84$$

$$Z_{in} = 50(0.84) = \underline{\underline{42 \, \Omega}}$$

Prob. 11.48

$$(a) \quad \Gamma = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{0.5 - j - 1}{0.5 - j + 1} = 0.0769 - j0.6154 = \underline{\underline{0.6202 \angle -82.87^\circ}}$$

(b)

$$\Gamma = \frac{z_L - 1}{z_L + 1} \longrightarrow z_L = \frac{1 + \Gamma}{1 - \Gamma} = \frac{1 + 0.4 \angle 25^\circ}{1 - 0.4 \angle 25^\circ}$$

$$z_L = 1.931 + j0.7771$$

$$Z_L = \underline{\underline{(1.931 + j0.7771)Z_o}}$$

Prob. 11.49

$$(a) \quad Z_{in} = \frac{Z_{in}}{Z_o} = \frac{100 - j120}{80} = 1.25 - j1.5$$

$$\lambda = \frac{u}{f} = \frac{0.8 \times 3 \times 10^8}{12 \times 10^6} = 20 \text{ m}$$

$$l_1 = 22 \text{ m} = \frac{22\lambda}{20} = 1.1\lambda \rightarrow 720^\circ + 72^\circ$$

$$l_2 = 28 \text{ m} = \frac{28\lambda}{20} = 1.4\lambda \rightarrow 720^\circ + 72^\circ + 216^\circ$$

To locate P (the load), we move 2 revolutions plus 72° toward the load. At P,

$$|\Gamma_L| = \frac{OP}{OQ} = \frac{5.1 \text{ cm}}{9.2 \text{ cm}} = 0.5543$$

$$\theta_\Gamma = 72^\circ - 47^\circ = 25^\circ$$

$$\Gamma_L = \underline{\underline{0.5543 \angle 25^\circ}}$$

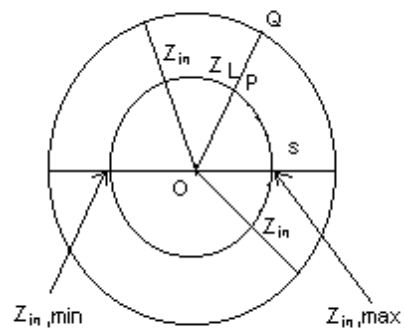
(Exact value = $0.5624 \angle 25.15^\circ$)

$$Z_{in, \max} = sZ_o = 3.7(80) = \underline{\underline{296 \Omega}}$$

(Exact value = 285.59Ω)

$$Z_{in, \min} = \frac{Z_o}{s} = \frac{80}{3.7} = \underline{\underline{21.622 \Omega}}$$

(Exact value = 22.41Ω)



(b) Also, at P, $Z_L = 2.3 + j1.55$

$$Z_L = 80(2.3 + j1.55) = \underline{\underline{184 + j124\Omega}}$$

(Exact value = $183.45 + j128.25 \Omega$)

At S, $s = \underline{\underline{3.7}}$

To Locate Z'_{in} , we move 216° from Z_{in} toward the generator.

At Z'_{in} ,

$$z'_{in} = 0.48 + j0.76$$

$$Z'_{in} = 80(0.48 + j0.76) = \underline{\underline{38.4 + j60.8\Omega}}$$

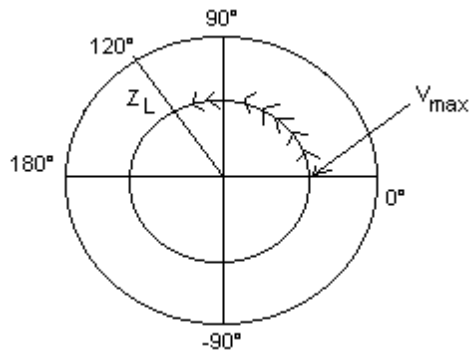
(Exact = $37.56 + j61.304 \Omega$)

(c) Between Z_L and Z_{in} , we move 2 revolutions and 72° . During the movement, we pass through $Z_{in, \max}$ 3 times and $Z_{in, \min}$ twice.

Thus there are:

$$\underline{\underline{3 Z_{in, \max} \text{ and } 2 Z_{in, \min}}}$$

Prob. 11.50



$$(a) \quad \frac{\lambda}{2} = 120\text{cm} \rightarrow \lambda = 2.4\text{m}$$

$$u = f\lambda \rightarrow f = \frac{u}{\lambda} = \frac{3 \times 10^8}{2.4} = \underline{\underline{125\text{MHz}}}$$

$$(b) \quad 40\text{cm} = \frac{40\lambda}{240} = \frac{\lambda}{6} \rightarrow \frac{720^\circ}{6} = 120^\circ$$

$$Z_L = Z_o z_L = 150(0.48 + j0.48) \\ = \underline{\underline{72 + j72 \Omega}}$$

(Exact value = $73.308 + j70.324 \Omega$)

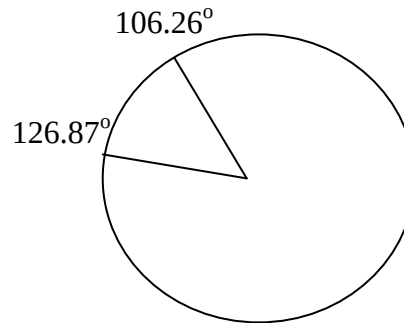
$$(c) \quad |\Gamma| = \frac{s-1}{s+1} = \frac{1.6}{3.9} = 0.444, \\ \Gamma = \underline{\underline{0.444 \angle 120^\circ}}$$

Prob. 11.51

(a)

$$z_L = \frac{Z_L}{Z_o} = \frac{j60}{80} = j0.75, \quad z_{in} = \frac{Z_{in}}{Z_o} = \frac{j40}{80} = j0.5$$

The two loads fall on the $r=0$ circle, the outermost resistance circle. The shortest distance between them is



$$\frac{360^\circ - (126.87^\circ - 106.26^\circ)\lambda}{720^\circ} = \underline{\underline{0.4714\lambda}}$$

$$(b) \quad \underline{\underline{s = \infty}}, \quad \Gamma_L = \underline{\underline{1 \angle 106.26^\circ}}$$

Prob. 11.52

$$z_L = \frac{Z_L}{Z_o} = \frac{40 + j25}{50} = 0.8 + j0.5$$

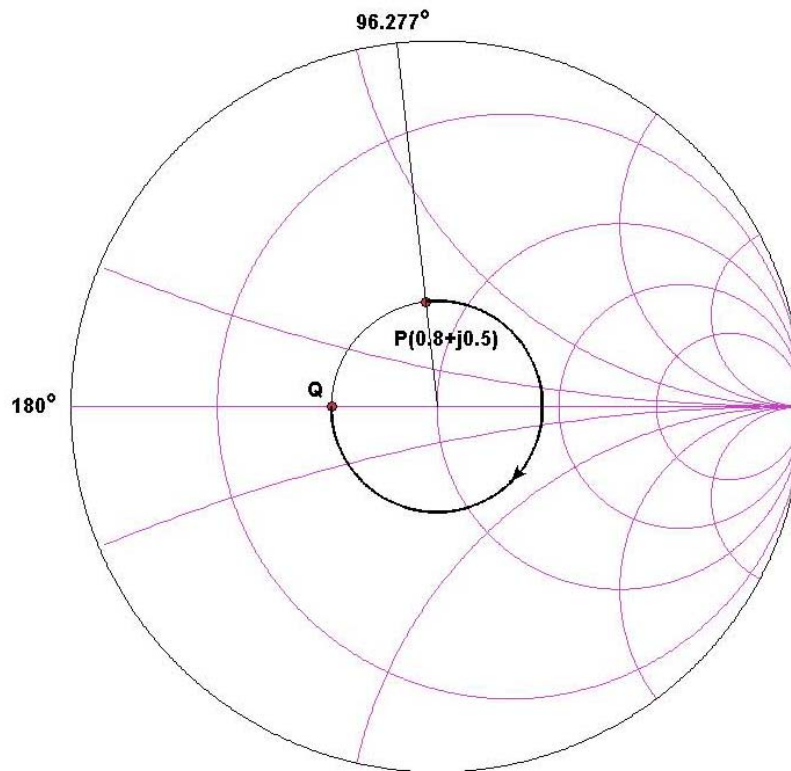
Locate this load at point P on the Smith chart. Draw a circle that passes through P. Locate point Q where the negative Γ_r - axis crosses the circle.

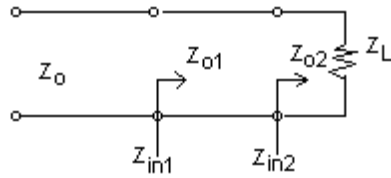
$\theta_p = 97^\circ$. The angular distance between P and Q is

$$\theta = 180^\circ + \theta_p = 277^\circ.$$

$$720^\circ \rightarrow \lambda$$

$$277^\circ \rightarrow \ell = \frac{\lambda}{720^\circ} 277^\circ = \underline{\underline{0.3847\lambda}}$$



Prob. 11.53

(a) From Eq. (11.43), $Z_{in2} = \frac{Z_{o2}^2}{Z_L}$

$$Z_{in1} = \frac{Z_{o1}^2}{Z_{in2}} = Z_o, \text{ i.e. } Z_{in2} = \frac{Z_{o1}^2}{Z_o} = \frac{Z_{o2}^2}{Z_L}$$

$$Z_{o1} = Z_{o2} \sqrt{\frac{Z_o}{Z_L}} = 30 \sqrt{\frac{50}{75}} = \underline{\underline{24.5\Omega}}$$

(b) Also, $\frac{Z_o}{Z_{o1}} = \left(\frac{Z_{o2}}{Z_L} \right) \rightarrow Z_{o2} = \frac{Z_o Z_L}{Z_{o1}}$ (1)

Also, $\frac{Z_{o1}}{Z_{o2}} = \left(\frac{Z_{o2}}{Z_L} \right)^2 \rightarrow (Z_{o2})^3 = Z_{o1} Z_L^2$ (2)

From (1) and (2), $(Z_{o2})^3 = Z_{o1} Z_L^2 = \frac{Z_o^3 Z_L^3}{Z_{o1}^3}$ (3)

or $Z_{o1} = \sqrt[4]{Z_o^3 Z_L} = \sqrt[4]{(50)^3 (75)} = \underline{\underline{55.33\Omega}}$

From (3), $Z_{o2} = \sqrt[3]{Z_{o1} Z_L^2} = \sqrt[3]{(55.33)(75)^2} = \underline{\underline{67.74\Omega}}$

Prob. 11.54

$$\frac{\lambda}{4} \rightarrow 180^\circ, \quad z_L = \frac{74}{50} = 1.48, \quad \frac{1}{z_L} = 0.6756$$

This acts as the load to the left line. But there are two such loads in parallel due to the two lines on the right. Thus

$$Z_L' = 50 \frac{\left(\frac{1}{Z_L} \right)}{2} = 25(0.6756) = 16.892$$

$$z_L' = \frac{16.892}{50} = 0.3378, \quad z_{in} = \frac{1}{z_L'} = 2.96$$

$$Z_{in} = 50(2.96) = \underline{\underline{148\Omega}}$$

Prob. 11.55

From the previous problem, $Z_{in} = 148\Omega$

$$I_{in} = \frac{V_g}{Z_g + Z_{in}} = \frac{120}{80 + 148} = 0.5263A$$

$$P_{ave} = \frac{1}{2} |I_{in}|^2 R_{in} = \frac{1}{2} (0.5263)^2 (148) = 20.5W$$

Since the lines are lossless, the average power delivered to either antenna is 10.25W

Prob. 11.56

$$(a) \quad \beta l = \frac{2\pi}{4} \cdot \frac{\lambda}{4} = \frac{\pi}{2}, \quad \tan \beta l = \infty$$

$$Z_{in} = Z_o \left(\frac{Z_L + jZ_o \tan \beta l}{Z_o + jZ_L \tan \beta l} \right) = Z_o \left(\frac{\frac{Z_L}{\tan \beta l} + jZ_o}{\frac{Z_o}{\tan \beta l} + jZ_L} \right)$$

As $\tan \beta l \rightarrow \infty$,

$$Z_{in} = \frac{Z_o^2}{Z_L} = \frac{(50)^2}{200} = \underline{\underline{12.5\Omega}}$$

(b) If $Z_L = 0$,

$$Z_{in} = \frac{Z_o^2}{0} = \infty \quad (\text{open})$$

$$(c) \quad Z_L = 25 // \infty = \frac{25 \times \infty}{25 + \infty} = \frac{25}{1 + \frac{25}{\infty}} = 25\Omega$$

$$Z_{in} = \frac{(50)^2}{12.5} = \underline{\underline{200\Omega}}$$

Prob. 11.57

$$l_1 = \frac{\lambda}{4} \rightarrow Z_{in1} = \frac{Z_o^2}{Z_L} \text{ or } y_{in1} = \frac{Z_L}{Z_o^2}$$

$$y_{in1} = \frac{200 + j150}{(100)^2} = \underline{\underline{20 + j15 \text{ mS}}}$$

$$l_2 = \frac{\lambda}{8} \rightarrow Z_{in2} = Z_L \lim_{Z_o \rightarrow 0} \left(\frac{Z_L + jZ_o \tan \frac{\pi}{4}}{Z_o + jZ_L \tan \frac{\pi}{4}} \right) = jZ_o$$

$$y_{in2} = \frac{1}{jZ_o} = \frac{1}{j100} = \underline{\underline{-j10 \text{ mS}}}$$

$$l_3 = \frac{7\lambda}{8} \rightarrow Z_{in3} = Z_o \left(\frac{Z_i + jZ_o \tan \frac{7\pi}{4}}{Z_o + jZ_i \tan \frac{7\pi}{4}} \right) = \frac{Z_o (Z_i - jZ_o)}{(Z_o - jZ_i)}$$

But

$$y_i = y_{in1} + y_{in2} = 20 + j5 \text{ mS}$$

$$z_i = \frac{1}{y_i} = \frac{1000}{20 + j5} = 47.06 - j11.76$$

$$y_{in3} = \frac{Z_o - jZ_{in}}{Z_o (Z_{in} - jZ_o)} = \frac{100 - j47.06 - 11.76}{100(47.06 - j11.76 - j100)} \\ = \underline{\underline{6.408 + j5.189 \text{ mS}}}$$

If the shorted section were open,

$$y_{in1} = \underline{\underline{20 + j15 \text{ mS}}}$$

$$y_{in2} = \frac{1}{Z_{in2}} = \frac{j \tan \frac{\pi}{4}}{Z_o} = \frac{j}{100} = \underline{\underline{j10 \text{ mS}}}$$

$$l_3 = \frac{7\lambda}{8} \rightarrow Z_{in3} = Z_o \frac{\left(Z_i + jZ_o \tan \frac{7\pi}{4} \right)}{\left(Z_o + jZ_i \tan \frac{7\pi}{4} \right)} = \frac{Z_o(Z_i - jZ_o)}{(Z_o - jZ_i)}$$

$$y_i = y_{in1} + y_{in2} = 20 + j15 + j10 = 20 + j25 \text{ mS}$$

$$Z_i = \frac{1}{y_i} = \frac{1000}{20 + j25} = 19.51 - j24.39 \Omega$$

$$y_{in3} = \frac{Z_o - jZ_i}{Z_o(Z_i - jZ_o)} = \frac{75.61 - j19.51}{100(19.51 - j124.39)}$$

$$= \underline{\underline{2.461 + j5.691 \text{ mS}}}$$

Prob. 11.58

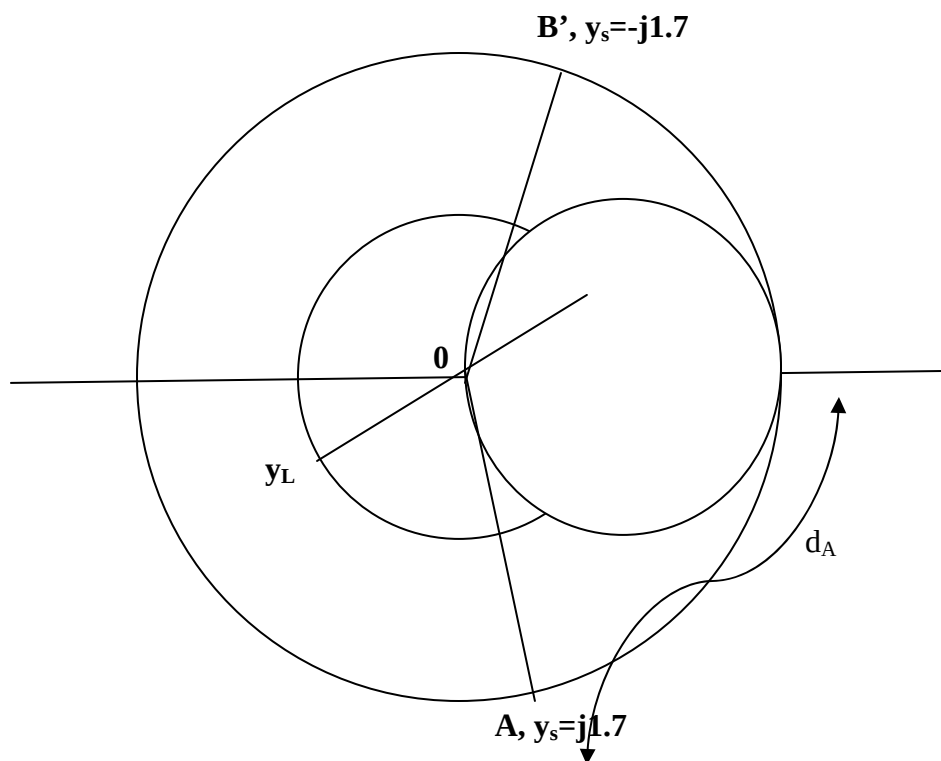
$$z_L = \frac{Z_L}{Z_o} = \frac{75 + j100}{50} = 1.5 + j2$$

Locate z_L on the Smith chart. Draw the s-circle passing through z_L .

Extend the diameter through O to y_L . Locate points A and B where the s-circle intersects the $g=1$ circle. At A, $y_s = -j1.7$ and at B, $y_s = j1.7$.

Locate points A' and B' where the stubs admittance is $j1.7$ and $-j1.7$ respectively. Calculate

$$d_A = \frac{26.5}{720} \lambda = \underline{\underline{0.0368 \lambda}}$$

**Prob. 11.59**

$$d_A = 0.12\lambda \rightarrow 0.12 \times 720^\circ = 86.4^\circ$$

$$l_A = 0.3\lambda \rightarrow 0.3 \times 720^\circ = 216^\circ$$

(a) From the Smith Chart below,

$$z_L = 0.57 + j0.69$$

$$Z_L = 60(0.57 + j0.69)$$

$$= \underline{\underline{34.2 + j41.4\Omega}}$$

$$(b) \quad d_B = \frac{360^\circ - 86.4^\circ}{720^\circ} \lambda = \underline{\underline{0.38\lambda}}$$

$$l_B = \frac{\lambda}{2} - \frac{(-62.4^\circ - -82^\circ)}{720^\circ} \lambda = \underline{\underline{0.473\lambda}}$$

$$(c) \quad \underline{\underline{s = 2.65}}$$

(Exact value = 2.7734)

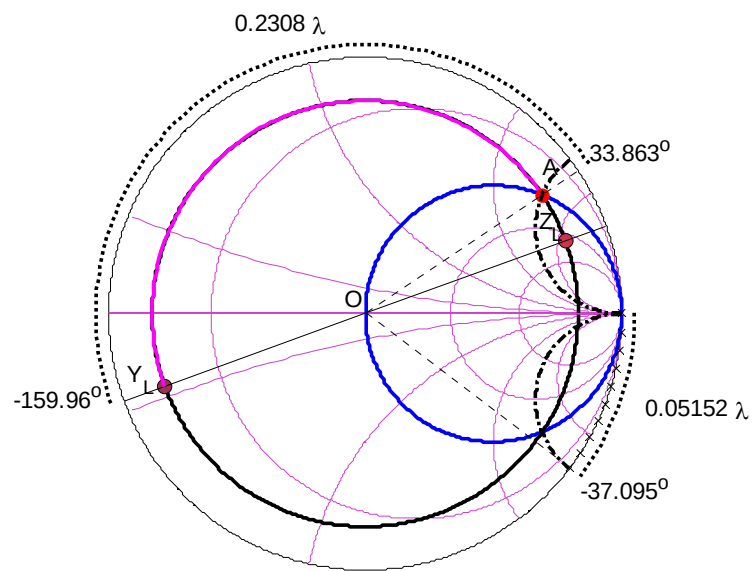
$$Z_L = \frac{Z_L}{Z_o} = \frac{120 + j220}{50} = 2.4 + j4.4$$
$$Y_s = Y_o y_s = \frac{\pm j3}{50} = \pm j0.06 \text{ S}$$

For B,

$$l_B = \frac{180+10+17}{720} \lambda = \underline{\underline{0.2875\lambda}}$$

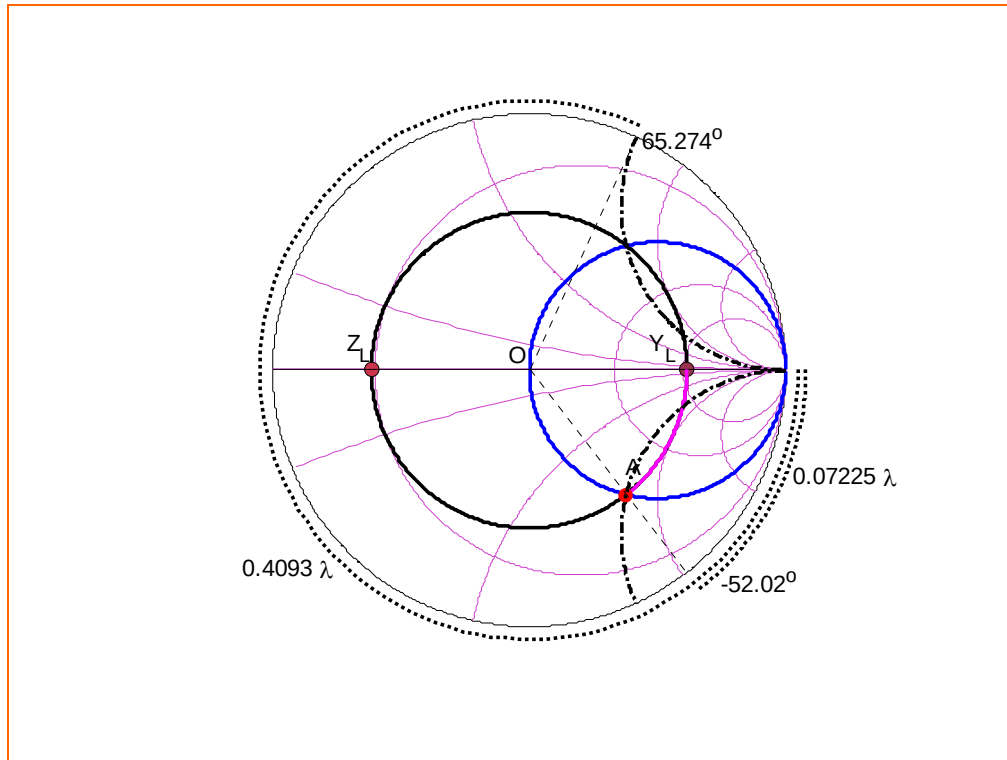
$$d_A = \frac{19}{720} \lambda = \underline{\underline{0.0264 \lambda}}$$
$$d_B = \frac{360-19}{720} \lambda = \underline{\underline{0.4736\lambda}}$$

(Exact value = 0.4485λ)



Prob. 11.61

$$\frac{\lambda}{4} \rightarrow \frac{720^\circ}{4} = 180^\circ$$



At A, $y_{in} = 1 - j1.561$

$$y_{stub} = j1.5614$$

Position of the stub = 0.0723λ

Length of the stub = 0.4093λ

Prob. 11.62

$$s = \frac{V_{\max}}{V_{\min}} = \frac{4V}{1V} = 4$$

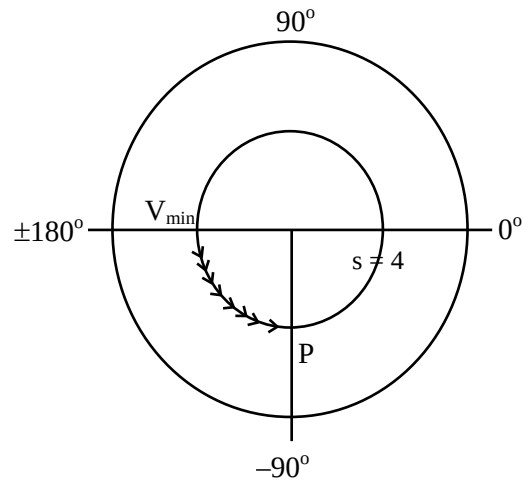
$$|\Gamma| = \frac{s-1}{s+1} = \frac{3}{5} = 0.6$$

$$\frac{\lambda}{2} = 25 \text{ cm} - 5 \text{ cm} = 20 \text{ cm}$$

$$\rightarrow \lambda = 40 \text{ cm}$$

The load is $l=5\text{cm}$ from $V_{\min}$, i.e.

$$l = \frac{5\lambda}{40} = \frac{\lambda}{8} \rightarrow 90^\circ$$



On the $s = 4$ circle, move 90° from $V_{\min}$ towards the load and obtain $Z_L = 0.46 - j0.88$ at

P.

$$Z_L = Z_0 z_L = 60(0.46 - j0.88) = \underline{\underline{27.6 - j52.8 \Omega}}$$

(Exact value = $28.2353 - j52.9412 \Omega$)

$$\theta_\Gamma = 270^\circ \text{ or } -90^\circ$$

$$\Gamma = \underline{\underline{0.6 \angle -90^\circ}}$$

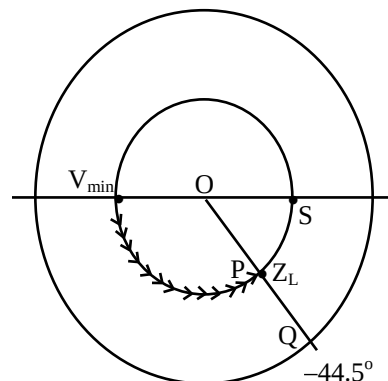
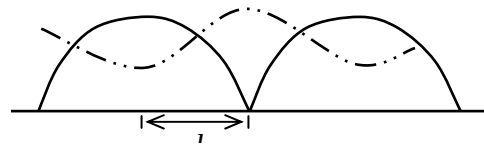
Prob. 11.63

$$s = \frac{V_{\max}}{V_{\min}} = \frac{0.95}{0.45} = \underline{\underline{2.11}}$$

$$\frac{\lambda}{2} = 22.5 - 14 = 8.5 \rightarrow \lambda = 17 \text{ cm}$$

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{0.17} = \underline{\underline{1.764 \text{ GHz}}}$$

$$l = 3.2 \text{ cm} = \frac{3.2}{17} \lambda \rightarrow 135.5^\circ$$



At P, $z_L = 1.4 - j0.8$

$$Z_L = 50(1.4 - j0.8) = \underline{\underline{70 - j40\Omega}}$$

(Exact value = $70.606 - j40.496 \Omega$)

$$|\Gamma| = \frac{s-1}{s+1} = \frac{1.11}{3.11} = 0.357, \quad \theta_r = -44.5^\circ$$

$$\Gamma = \underline{\underline{0.357 \angle -44.5^\circ}}$$

(Exact value = $0.3571 \angle -44.471^\circ$)

Prob. 11.64

$$\Gamma_s = \frac{R_g - R_o}{R_g + R_o} = \frac{0 - 50}{0 + 50} = \underline{\underline{-1}}$$

$$\Gamma_L = \frac{R_L - R_o}{R_L + R_o} = \frac{80 - 50}{80 + 50} = \underline{\underline{0.231}}$$

Prob. 11.65

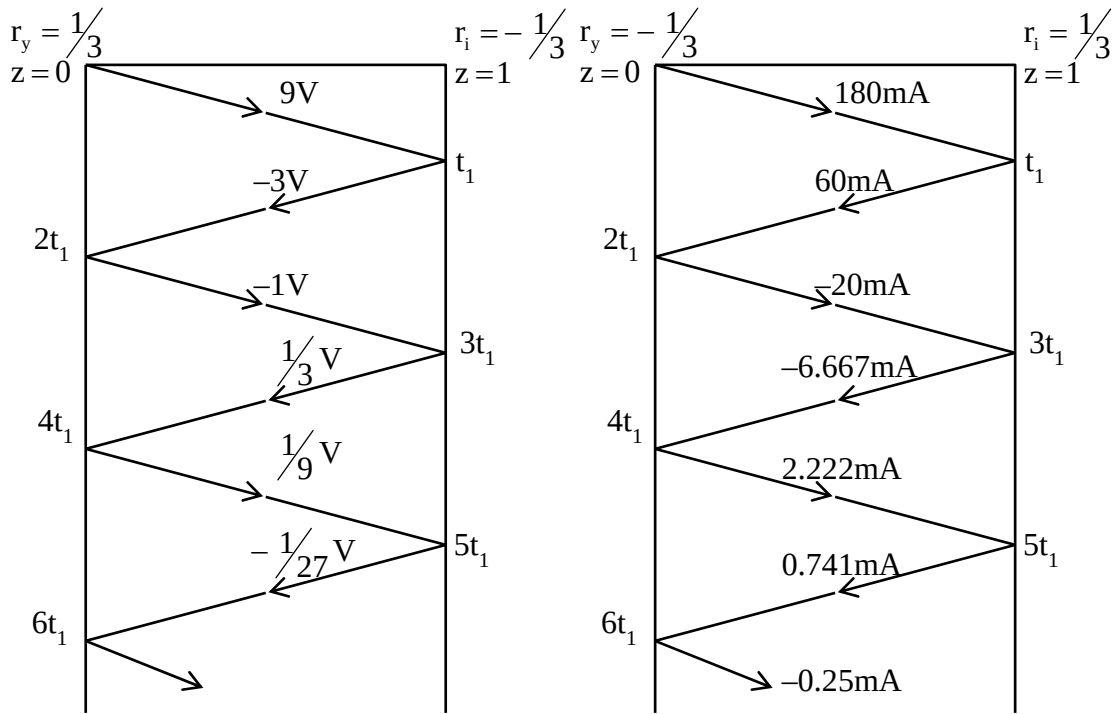
$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{0.5Z_o - Z_o}{1.5Z_o} = -\frac{1}{3}$$

$$\Gamma_g = \frac{Z_g - Z_o}{Z_g + Z_o} = \frac{Z_o}{3Z_o} = \frac{1}{3}$$

$$t_1 = \frac{l}{u} = 2\mu s, \quad V_o = \frac{Z_o}{3Z_o}(27) = 9 \text{ V}, \quad I_o = \frac{V_o}{Z_o} = 180 \text{ mA}$$

$$V_\infty = \frac{Z_L}{Z_g + Z_L} V_g = \frac{0.5}{2.5}(27) = 5.4 \text{ V}, \quad I_\infty = \frac{V_\infty}{Z_L} = 216 \text{ mA}$$

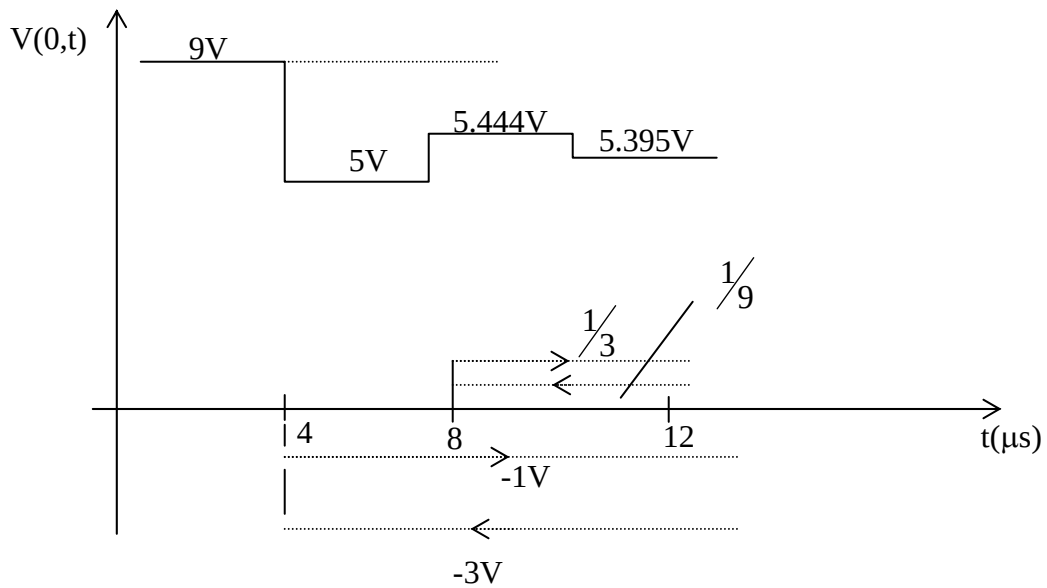
The voltage and current bounce diagrams are shown below

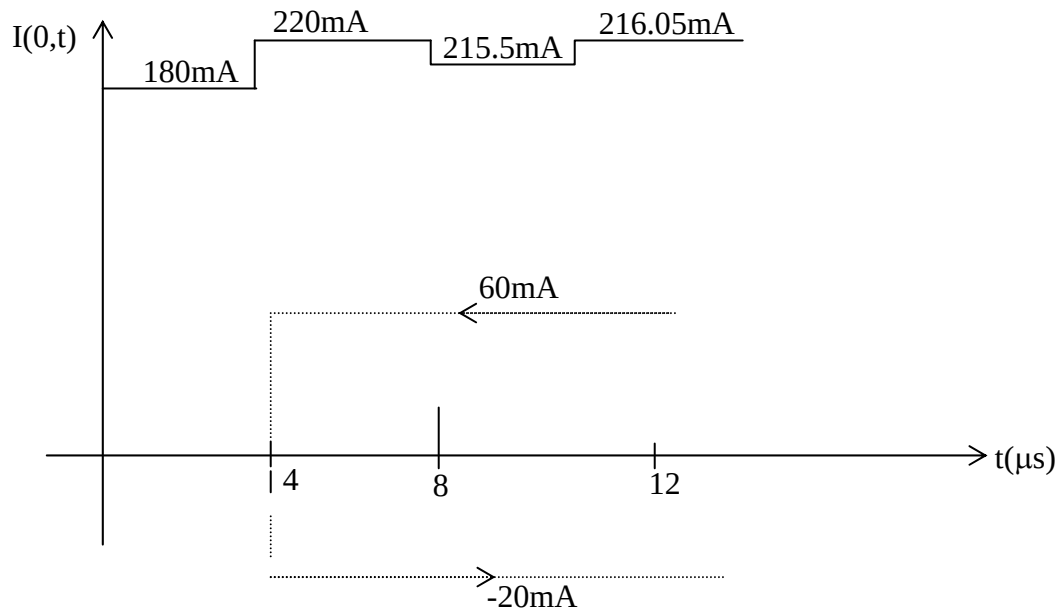


(Voltage bounce diagram)

(Current bounce diagram)

From the bounce diagrams, we obtain $V(0,t)$ and $I(0,t)$ as shown below:



**Prob. 11.66**

Using Thevenin equivalent at $z = 0$ gives

$$R_g = R_s = 4Z_o = 200\Omega$$

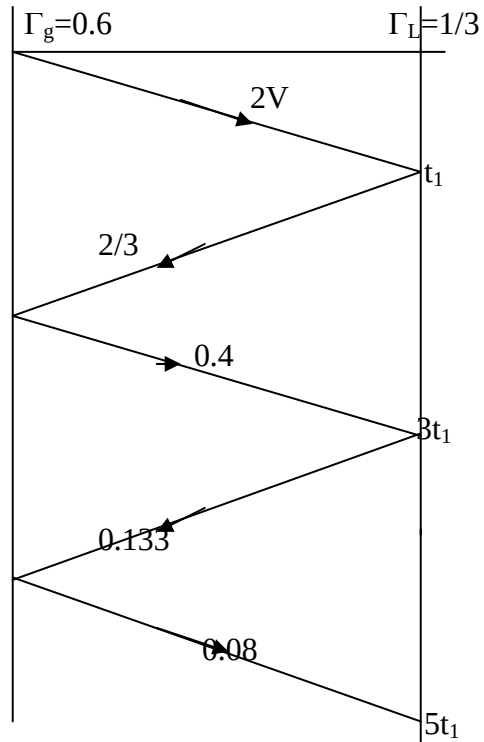
$$V_g = I_s R_s = 10 \times 200 \times 10^{-3} = 2V$$

$$\Gamma_g = \frac{Z_g - Z_o}{Z_g + Z_o} = \frac{4Z_o - Z_o}{4Z_o + Z_o} = \frac{3}{5}$$

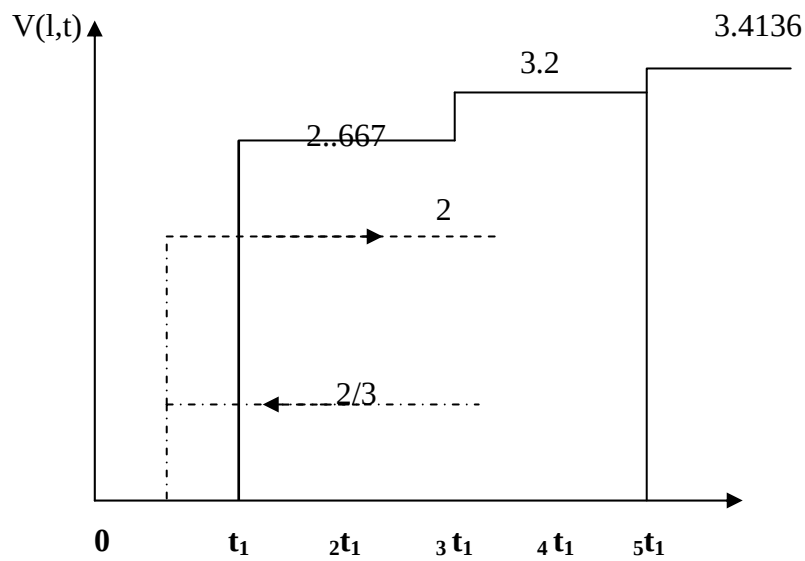
$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{2Z_o - Z_o}{2Z_o + Z_o} = \frac{1}{3}$$

$$t_1 = \frac{\ell}{u} = \frac{10}{2 \times 10^8} = 50 \text{ ns}$$

The bounce diagram is shown below.



The load voltage is sketched below.



$$I(l,t) = \frac{V(l,t)}{Z_L} = \frac{V(l,t)}{100}$$

To get $I(l,t)$, we just scale down $V(l,t)$ by 100.

Prob. 11.67

$$\Gamma_g = \frac{Z_g - Z_o}{Z_g + Z_o} = \frac{32 - 75}{32 + 75} = 0.4019$$

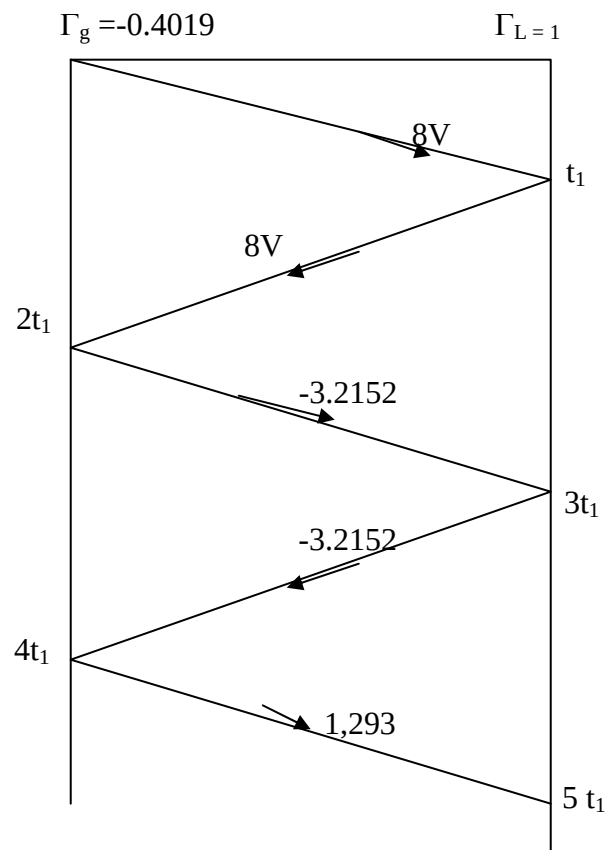
$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{2 \times 10^6 - 75}{2 \times 10^6 + 75} \approx 1$$

$$t_1 = \frac{\ell}{u} = \frac{50 \times 10^{-2}}{2 \times 10^8} = 2.5 \text{ ns}$$

The bounce diagram is shown below.

At $t = 20 \text{ ns} = 4t_1$,

$$V = 8 + 8 - 3.2152 - 3.2152 = \underline{\underline{9.57 \text{ V}}}$$



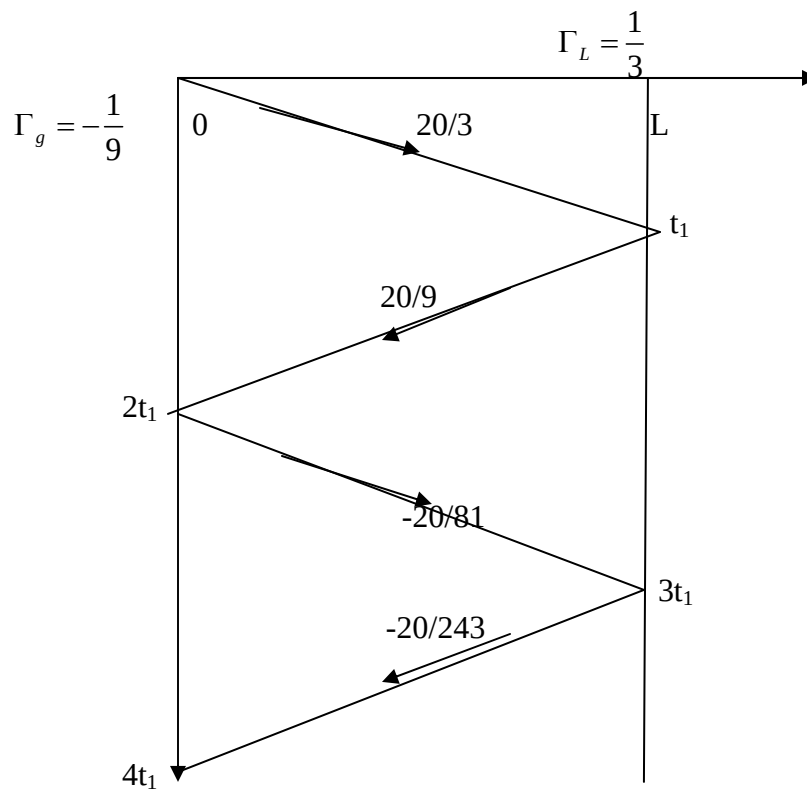
Prob. 11.68

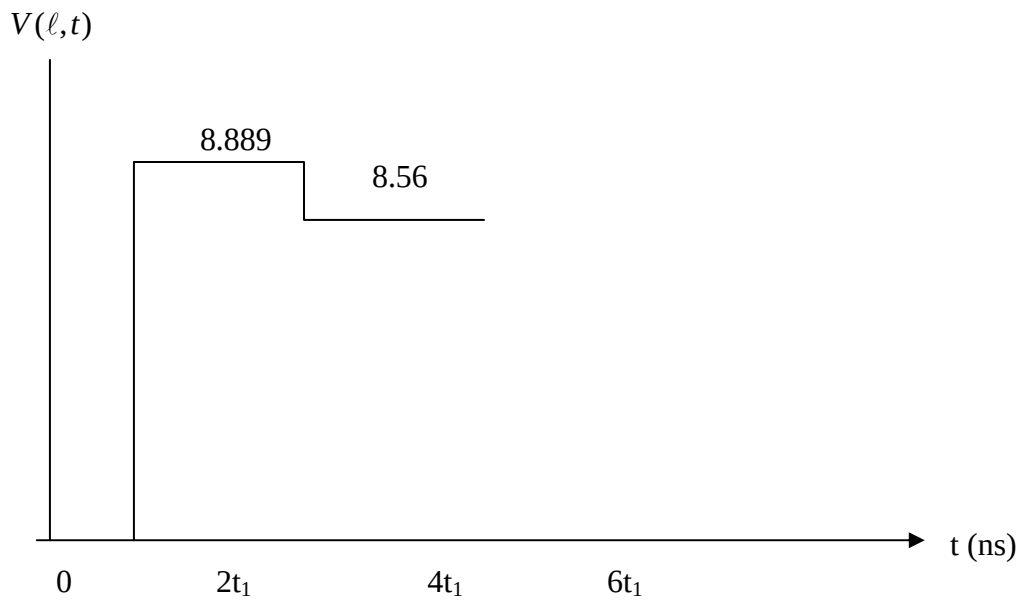
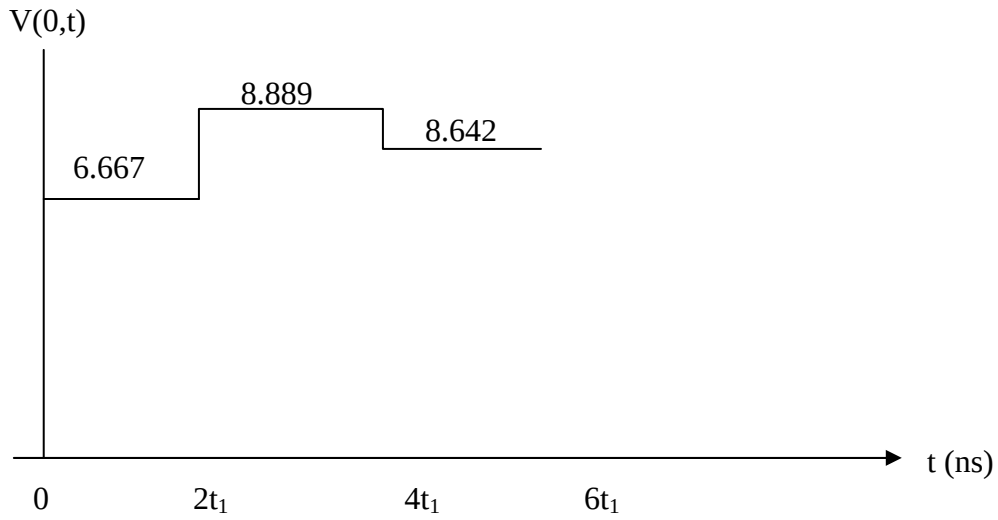
$$t_1 = \frac{l}{u} = \frac{40 \times 10^{-2}}{2.5 \times 10^8} = 01.6ns,$$

$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{100 - 50}{150} = \frac{1}{3}, \quad \Gamma_g = \frac{Z_g - Z_o}{Z_g + Z_o} = \frac{40 - 50}{90} = -\frac{1}{9},$$

$$V_o = \frac{Z_o V_g}{Z_o + Z_g} = \frac{50(12)}{90} = \frac{20}{3} = 6.668V$$

The bounce diagram is sketched below. From it, we obtain $V(0,t)$ and $V(L,t)$.



**Prob. 11.69**

$$V_o = 8V = \frac{Z_o}{Z_o + Z_g} V_g = \frac{50}{50 + 60} V_g \quad \longrightarrow \quad V_g = \frac{8 \times 110}{50} = \underline{\underline{17.6 \text{ V}}}$$

$$2t_1 = 4\mu s \quad \longrightarrow \quad t_1 = 2\mu s = \frac{\ell}{u}$$

$$\ell = ut_1 = 3 \times 10^8 \times 2 \times 10^{-6} = \underline{\underline{600 \text{ m}}}$$

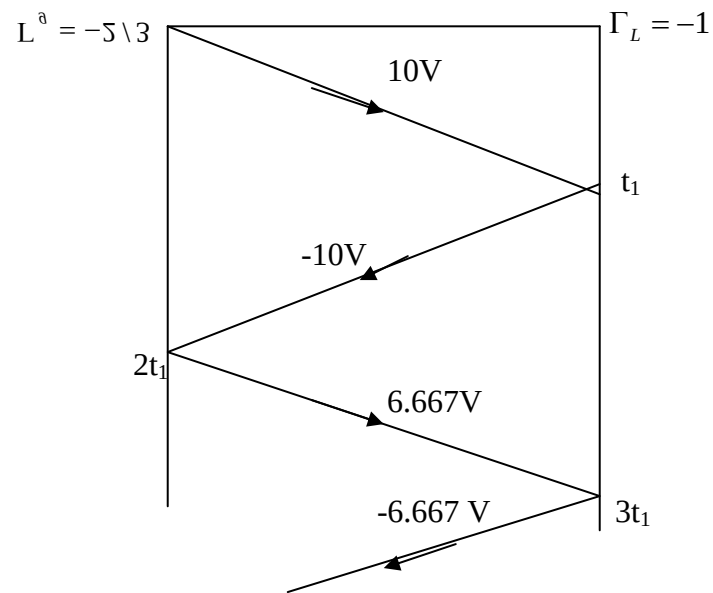
Prob. 11.70

$$t_1 = \frac{20}{2 \times 10^8} = 10^{-7} = 0.1 \mu s, \quad V_o = \frac{Z_o}{Z_o + Z_g} V_g = \frac{50}{60} (12) = 10V$$

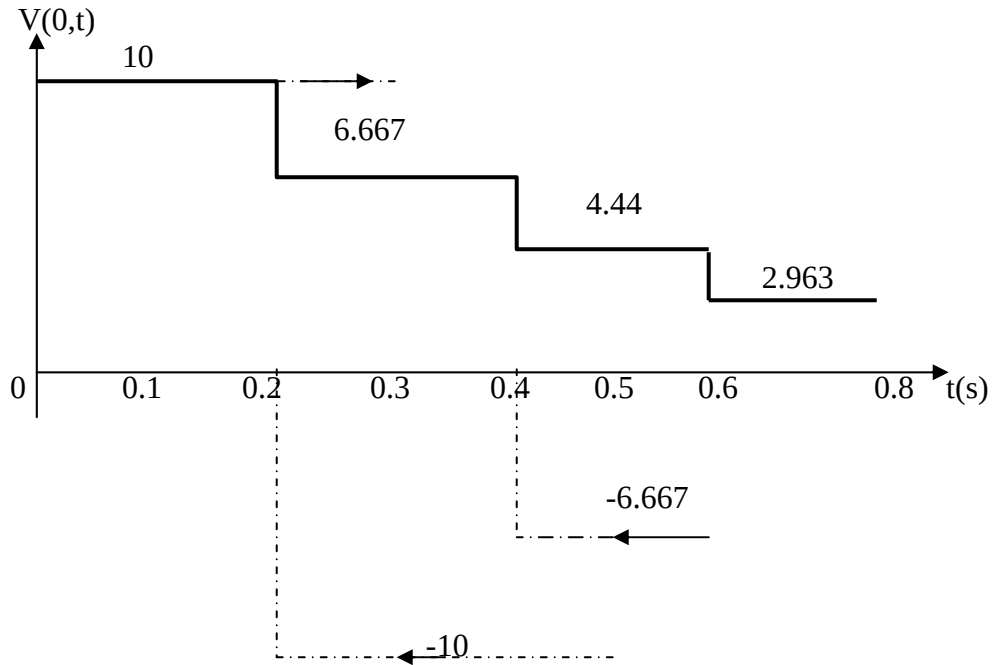
$$\Gamma_g = \frac{Z_g - Z_o}{Z_g + Z_o} = \frac{10 - 50}{10 + 50} = -2/3$$

$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{0 - 50}{0 + 50} = -1$$

The voltage bounce diagram is shown below



From the bounce diagram, we obtain $V(0,t)$ as shown. $V(1,t) = 0$ due to the short circuit.



Prob. 11.71

The initial pulse on the line is

$$V_o = \frac{Z_o}{Z_o + Z_g} V_g \quad (1)$$

The reflection coefficient is

$$\Gamma = \frac{Z_g - Z_o}{Z_g + Z_o} \quad (2)$$

The first reflected wave has amplitude ΓV_o , while the second reflected wave has amplitude $\Gamma^2 V_o$, etc. For a very long time, the total voltage is

$$V_T = V_o + \Gamma V_o + \Gamma^2 V_o + \dots$$

$$= V_o (1 + \Gamma + \Gamma^2 + \Gamma^3 + \dots) = V_o \left(\frac{1}{1 - \Gamma} \right) \quad (3)$$

Substituting (1) and (2) into (3) gives

$$V_T = \frac{Z_o}{Z_o + Z_g} V_g \left(\frac{1}{1 - \frac{Z_g - Z_o}{Z_g + Z_o}} \right) = Z_o V_g \frac{1}{Z_g + Z_o - Z_g + Z_o} = \underline{\underline{\frac{V_g}{2}}}$$

Prob.11.72

$$w = 1.5\text{cm}, \quad h = 1\text{cm}, \quad \frac{w}{h} = 1.$$

$$(a) \quad \epsilon_{\text{eff}} = \left(\frac{6 + 1}{2} \right) + \frac{\epsilon_r - 1}{2\sqrt{1 + 12h/w}} = 1.6 + \frac{0.6}{\sqrt{1 + 12/1.5}} = \underline{\underline{1.8}}$$

$$Z_0 = \frac{377}{\sqrt{1.8} (1.5 + 1.393 + 0.667 \ln(2.944))} = \frac{281}{3.613} = \underline{\underline{77.77\Omega}}$$

$$(b) \quad \alpha_c = 8.686 \frac{R_s}{wZ_0}$$

$$R_s = \frac{1}{\sigma_c \sigma} = \sqrt{\frac{\mu\pi f}{\sigma_c}} = \sqrt{\frac{19 \times 2.5 \times 10^9 \times 4\pi \times 10^{-3}}{1.1 \times 10^7}}$$

$$= 2.995 \times 10^{-2}$$

$$\alpha_c = \frac{8.686 \times 2.995 \times 10^{-2}}{1.5 \times 10^{-2} \times 77.77} = \underline{\underline{0.223\text{dB/m}}}$$

$$u = \frac{c}{\sqrt{\epsilon_{\text{eff}}}} \rightarrow \lambda = \frac{u}{f} = \frac{c}{f\sqrt{\epsilon_{\text{eff}}}} = \frac{3 \times 10^8}{2.5 \times 10^9 \sqrt{1.8}} = 8.944 \times 10^{-2}$$

$$\alpha_d = 27.3 \times \frac{0.8 (2.2)}{1.2 \sqrt{1.8}} \frac{2 \times 10^{-2}}{8.944 \times 10^{-2}} = \frac{96.096}{14.3996}$$

$$\alpha_d = \underline{\underline{6.6735 \text{ dB/m}}}$$

$$(c) \quad \alpha = \alpha_c + \alpha_d = 6.8965 \text{ dB/m}$$

$$\alpha \ell = 20\text{dB} \rightarrow \ell = \frac{20}{\alpha} = \frac{20}{6.8965} = \underline{\underline{2.9 \text{ m}}}$$

Prob. 11.73

(a) Let $x = w/h$. If $x < 1$,

$$50 = \frac{60}{\sqrt{4.6}} \ln \left(\frac{8}{x} + x \right)$$

$$5\sqrt{4.6} - 6 \ln \left(\frac{8}{x} + x \right) = 0$$

we solve for x (e.g using Maple) and get $x = 2.027$ or 3.945

which contradicts our assumption that $x < 1$. If $x > 1$,

$$50 = \frac{120\pi}{\sqrt{4.6} [x + 1.393 + 0.667 \ln(x + 1.444)]}$$

We solve this iteratively and obtain:

$$x = 1.8628, w = xh = 14.9024 \text{ mm}$$

For this w and h ,

$$\epsilon_{\text{eff}} = 3.4598$$

$$(b) \quad \beta = \frac{\omega \sqrt{\epsilon_{\text{eff}}}}{c}$$

$$\beta \ell = 45^\circ = \frac{\pi}{4} = \frac{\omega \ell \sqrt{\epsilon_{\text{eff}}}}{c}$$

$$\ell = \frac{\pi c}{4 \sqrt{\epsilon_{\text{eff}}} 2\pi f} = \frac{3 \times 10^8}{8 \times \sqrt{3.4598} \times 8 \times 10^9}$$

$$\underline{\underline{\ell = 0.00252 \text{ m}}}$$

Prob. 11.74

$$\text{For } w = 0.4 \text{ mm}, \quad \frac{w}{h} = \frac{0.4 \text{ mm}}{2 \text{ m}} = 0.2 \rightarrow \text{narrow strip}$$

$$\text{For } \frac{w}{h} = 0.2, \quad \varepsilon_{\text{eff}} = 5.851, \quad Z_o = 91.53\Omega$$

$$\text{For } \frac{w}{h} = 0.4, \quad \varepsilon_{\text{eff}} = 6.072, \quad Z_o = 73.24\Omega$$

Hence,

$$\underline{\underline{73.24\Omega < Z_o < 91.53\Omega}}$$

Prob. 11.75

$$w' = 0.5 + \frac{0.1}{3.2} \ln \left(\frac{5 \times 1.2}{0.1} \right) = 0.5 + 0.1279 = 0.6279$$

$$\begin{aligned} Z_o &= \frac{377}{2\pi\sqrt{2}} \ln \left\{ 1 + \frac{4 \times 1.2}{\pi \times 0.6177} \left[\frac{8 \times 1.2}{4 \times 0.6279} + 3.354 \right] \right\} \\ &= 42.43 \ln \{ 1 + 2.49(3.822 + 3.354) \} = 42.43 \ln(20.255) \\ &= \underline{\underline{127.64 \Omega}} \end{aligned}$$

Prob. 11.76

Suppose we guess that $w/h < 2$

$$A = \frac{75}{60} \sqrt{\frac{3.3}{2}} + \frac{1.3}{3.3} \left(0.23 + \frac{0.11}{2.3} \right) = 1.117$$

$$\frac{w}{h} = \frac{8e^A}{e^{2A} - 2} = \frac{24.44}{7.337} = 3.331 \rightarrow w = 3.331h = \underline{\underline{4\text{mm}}}$$

If we guess that $w/h > 2$,

$$B = \frac{60\pi^2}{Z_o\sqrt{\varepsilon_r}} = \frac{60\pi^2}{75\sqrt{2.3}} = 5.206$$

$$\frac{w}{h} = \frac{2}{\pi} \left[4.266 - \ln 9.412 + \frac{1.3}{4.6} \left(\ln 4.206 + 0.39 - \frac{0.61}{2.3} \right) \right]$$

$$= 1.665 < 2$$

$$\text{Thus } \frac{w}{h} = 3.331 > 2$$

$$\epsilon_{\text{eff}} = \frac{3.3}{2} + \frac{1.3}{2\sqrt{1 + \frac{12}{3.331}}} = 1.953$$

$$u = \frac{3 \times 10^8}{\sqrt{1.953}} = \underline{\underline{2.1467 \times 10^8 \text{ m/s}}}$$

Prob. 11.77

$$\Gamma = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{100 - 150}{250} = -0.2$$

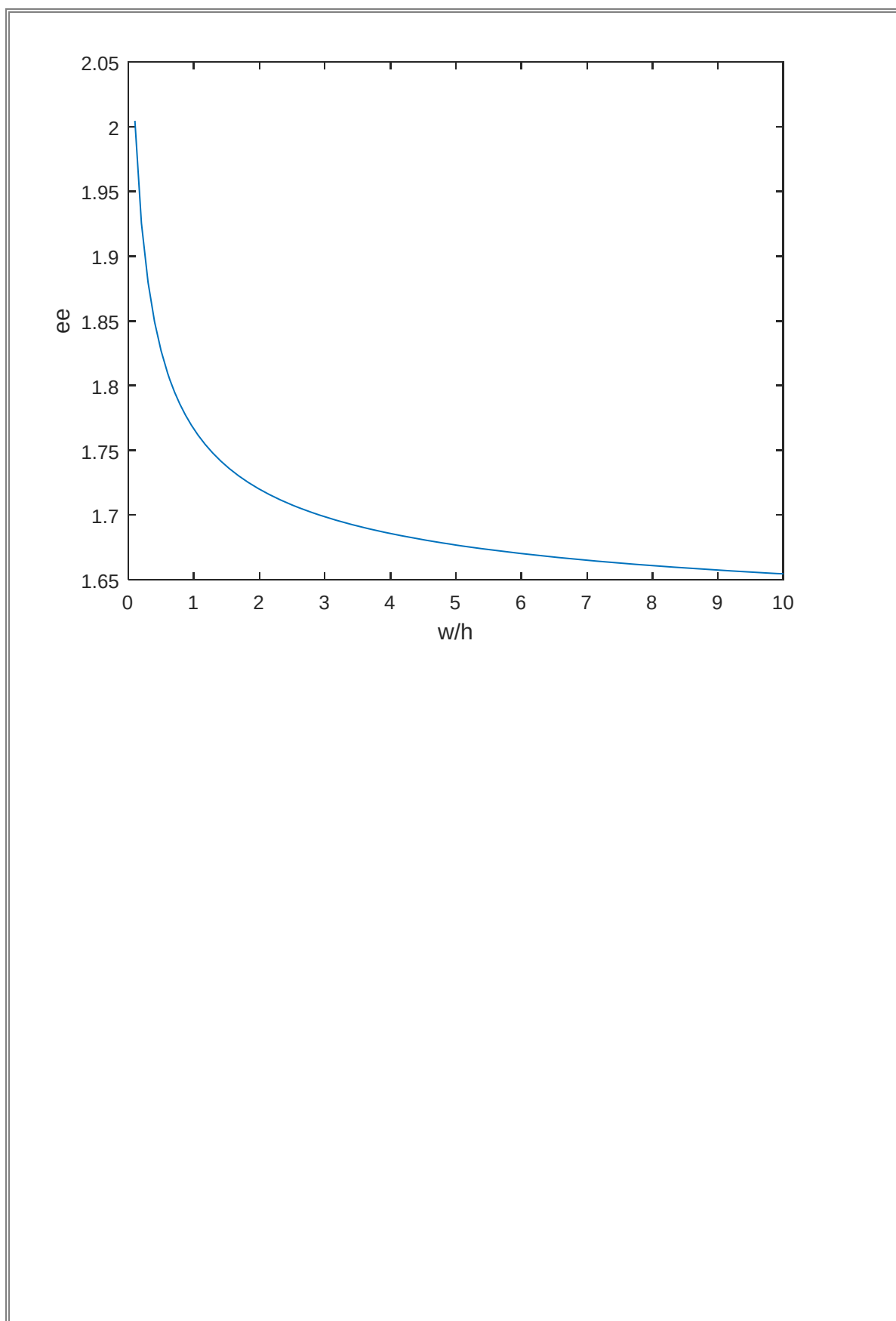
$$RL = -20 \log |\Gamma| = \underline{\underline{13.98 \text{ dB}}}$$

Prob. 11.78

The MATLAB code and the plot of the effective relative permittivity are presented below.

```
% Plot effective permittivity versus x = h/w
```

```
er = 2.2
e1 = (er + 1)/2; e2 = (er - 1)/2;
% x = 0.1*0.1*100
for k=1:100
    x(k)=0.1*k
    fac=x(k);
    ee(k) = e1 + e2/sqrt(1 + 12*fac);
end
plot(x,ee)
xlabel('w/h')
ylabel('ee')
```



CHAPTER 12

P. E. 12.1 (a) For TE_{10} , $f_c = 3$ GHz,

$$\sqrt{1 - (f_c / f)^2} = \sqrt{1 - (3 / 15)^2} = \sqrt{0.96}, \quad \beta_o = \omega / u_o = 4\pi f / c$$

$$\beta = \frac{4\pi f}{c} \sqrt{0.96} = \frac{4\pi \times 15 \times 10^9}{3 \times 10^8} \sqrt{0.96} = \underline{\underline{615.6}} \text{ rad/m}$$

$$u = \frac{\omega}{\beta} = \frac{2\pi \times 15 \times 10^9}{615.6} = \underline{\underline{1.531 \times 10^8}} \text{ m/s}$$

$$\eta' = \sqrt{\frac{\mu}{\epsilon}} = 60\pi, \quad \eta_{TE} = \frac{60\pi}{\sqrt{0.96}} = \underline{\underline{192.4\Omega}}$$

(b) For TM_{11} , $f_c = 3\sqrt{7.25}$ GHz, $\sqrt{1 - (f_c / f)^2} = 0.8426$

$$\beta = \frac{4\pi f}{c} (0.8426) = \frac{4\pi \times 15 \times 10^9 (0.8426)}{3 \times 10^8} = \underline{\underline{529.4}} \text{ rad/m}$$

$$u = \frac{\omega}{\beta} = \frac{2\pi \times 15 \times 10^9}{529.4} = \underline{\underline{1.78 \times 10^8}} \text{ m/s}$$

$$\eta_{TM} = 60\pi (0.8426) = \underline{\underline{158.8\Omega}}$$

P. E. 12.2 (a) Since $E_z \neq 0$, this is a TM mode

$$E_{zs} = E_o \sin(m\pi x / a) \sin(n\pi y / b) e^{-j\beta z}$$

$$E_o = 20, \quad \frac{m\pi}{a} = 40\pi \quad \longrightarrow \quad m=2, \quad \frac{n\pi}{b} = 50\pi \quad \longrightarrow \quad n=1$$

i.e. TM₂₁ mode.

$$(b) \quad f_c = \frac{u'}{2} \sqrt{(m/a)^2 + (n/b)^2} = \frac{3 \times 10^8}{2} \sqrt{40^2 + 50^2} = 1.5\sqrt{41} \text{ GHz}$$

$$\beta = \omega \sqrt{\mu\epsilon} \sqrt{1 - (f_c / f)^2} = \frac{2\pi f}{c} \sqrt{f^2 - f_c^2} = \frac{2\pi \times 10^9}{3 \times 10^8} \sqrt{225 - 92.25} = \underline{\underline{241.3 \text{ rad/m}}}$$

(c)

$$E_{xs} = \frac{-j\beta}{h^2} (40\pi) 20 \cos 40\pi x \sin 50\pi y e^{-j\beta z}$$

$$E_{ys} = \frac{-j\beta}{h^2} (50\pi) 20 \sin 40\pi x \cos 50\pi y e^{-j\beta z}$$

$$\frac{E_y}{E_x} = \underline{\underline{1.25 \tan 40\pi x \cot 50\pi y}}$$

P. E. 12.3 If TE₁₃ mode is assumed, f_c and β remain the same.

$$\underline{\underline{f_c = 28.57 \text{ GHz}, \beta = 1718.81 \text{ rad/m}, \gamma = j\beta}}$$

$$\eta_{TE13} = \frac{377 / 2}{\sqrt{1 - (28.57 / 50)^2}} = \underline{\underline{229.69 \Omega}}$$

For $m=1, n=3$, the field components are:

$$E_z = 0$$

$$H_z = H_o \cos(\pi x / a) \cos(3\pi y / b) \cos(\omega t - \beta z)$$

$$E_x = -\frac{\omega\mu}{h^2} \left(\frac{3\pi}{b} \right) H_o \cos(\pi x / a) \sin(3\pi y / b) \sin(\omega t - \beta z)$$

$$E_y = \frac{\omega\mu}{h^2} \left(\frac{\pi}{a} \right) H_o \sin(\pi x / a) \cos(3\pi y / b) \sin(\omega t - \beta z)$$

$$H_x = -\frac{\beta}{h^2} \left(\frac{\pi}{a} \right) H_o \sin(\pi x / a) \cos(3\pi y / b) \sin(\omega t - \beta z)$$

$$H_y = -\frac{\beta}{h^2} \left(\frac{3\pi}{b} \right) H_o \cos(\pi x / a) \sin(3\pi y / b) \sin(\omega t - \beta z)$$

$$\text{Given that } H_{ox} = 2 = -\frac{\beta}{h^2} (\pi / a) H_o,$$

$$H_{oy} = -\frac{\beta}{h^2} (3\pi / b) H_o = 6a / b = 6(15) / 8 = 11.25$$

$$H_{oz} = H_o = -\frac{2h^2 a}{\beta\pi} = \frac{-2 \times 14.51\pi^2 \times 10^4 \times 1.5 \times 10^{-2}}{1718.81\pi} = -7.96$$

$$E_{oy} = \frac{\omega\mu}{h^2} \left(\frac{\pi}{a} \right) H_o = -\frac{2\omega\mu}{\beta} = 2\eta_{TE} = -459.4$$

$$E_{ox} = -E_{oy} \frac{3a}{b} = 459.4(4.5 / 0.8) = 2584.1$$

$$E_x = 2584.1 \cos(\pi x / a) \sin(3\pi y / b) \sin(\omega t - \beta z) \text{ V/m,}$$

$$E_y = -459.4 \sin(\pi x / a) \cos(3\pi y / b) \sin(\omega t - \beta z) \text{ V/m,}$$

$$E_z = 0,$$

$$H_y = 11.25 \cos(\pi x / a) \sin(3\pi y / b) \sin(\omega t - \beta z) \text{ A/m},$$

$$H_z = -7.96 \cos(\pi x / a) \cos(3\pi y / b) \cos(\omega t - \beta z) \text{ A/m}$$

P. E. 12.4

$$f_{c11} = \frac{u'}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = \frac{3 \times 10^8 \times 10^2}{2} \sqrt{1/8.636^2 + 1/4.318^2} = 3.883 \text{ GHz}$$

$$u_p = \frac{3 \times 10^8}{\sqrt{1 - (3.883/4)^2}} = \underline{\underline{12.5 \times 10^8}} \text{ m/s},$$

$$u_g = \frac{9 \times 10^{16}}{12.5 \times 10^8} = \underline{\underline{7.2 \times 10^7}} \text{ m/s}$$

P. E. 12.5 The dominant mode becomes TE₀₁ mode

$$f_{c01} = \frac{c}{2b} = 3.75 \text{ GHz}, \quad \eta_{TE} = 406.7\Omega$$

From Example 12.2,

$$E_x = -E_o \sin(3\pi y / b) \sin(\omega t - \beta z), \quad \text{where } E_o = \frac{\omega \mu b}{\pi} H_o.$$

$$\mathcal{P}_{ave} = \int_{x=0}^a \int_{y=0}^b \frac{|E_{xs}|^2}{2\eta} dx dy = \frac{E_o^2 ab}{4\eta}$$

Hence $E_o = 63.77 \text{ V/m}$ as in Example 12.5.

$$H_o = \frac{\pi E_o}{\omega \mu b} = \frac{\pi \times 63.77}{2\pi \times 10^{10} \times 4\pi \times 10^{-7} \times 4 \times 10^{-2}} = \underline{\underline{63.34}} \text{ mA/m}$$

P. E. 12.6 (a) For $m=1, n=0$, $f_c = u'/(2a)$

$$\frac{\sigma}{\omega \epsilon} = \frac{10^{-15}}{2\pi \times 9 \times 10^9 \times 2.6 \times 10^{-9} / (36\pi)} = \frac{10^{-15}}{1.3} \ll 1$$

Hence,

$$u' \cong \frac{1}{\sqrt{\mu\epsilon}} = c / \sqrt{2.6}, \quad f_c = \frac{3 \times 10^8}{2 \times 2.4 \times 10^{-2} \sqrt{2.6}} = 2.2149 \text{ GHz}$$

$$\alpha_d = \frac{\sigma \eta'}{2 \sqrt{1 - (f_c / f)^2}} = \frac{10^{-15} \times 377 / \sqrt{2.6}}{2 \sqrt{1 - (2.2149 / 9)^2}} = 1.205 \times 10^{-13} \text{ Np/m}$$

For $n = 0, m=1$,

$$\begin{aligned} \alpha_c &= \frac{2R_s}{b\eta' \sqrt{1 - (f_c / f)^2}} \left[\frac{1}{2} + \frac{b}{a} (f_c / f)^2 \right] \\ &= \frac{2\sqrt{2.6} \sqrt{\pi \times 9 \times 10^9 \times 1.1 \times 10^7 \times 4\pi \times 10^{-7}}}{377 \times 1.5 \times 10^{-2} \times 1.1 \times 10^7 \sqrt{1 - (2.2149 / 9)^2}} [0.5 + (2.4 / 1.5)(2.2148 / 9)^2] = \underline{\underline{2 \times 10^{-2}}} \text{ Np/m} \end{aligned}$$

(b) Since $\alpha_c \gg \alpha_d, \alpha = \alpha_c + \alpha_d \cong \alpha_c = 2 \times 10^{-2}$

$$\text{loss} = \alpha l = 2 \times 10^{-2} \times 0.4 = 0.8 \times 10^{-2} \text{ Np} = \underline{\underline{0.06945 \text{ dB}}}$$

P. E. 12.7 For TE_{11} , $m = 1 = n$,

$$H_{zs} = H_o \cos(\pi x / a) \cos(\pi y / b) e^{-\gamma z}$$

$$E_{xs} = \frac{j\omega}{h^2} (\pi / b) H_o \cos(\pi x / a) \sin(\pi y / b) e^{-\gamma z}$$

$$E_{ys} = -\frac{j\omega\mu}{h^2} (\pi / a) H_o \sin(\pi x / a) \cos(\pi y / b) e^{-\gamma z}$$

$$H_{xs} = \frac{j\beta}{h^2} (\pi / a) H_o \sin(\pi x / a) \cos(\pi y / b) e^{-\gamma z}$$

$$H_{ys} = \frac{j\beta}{h^2} (\pi / b) H_o \cos(\pi x / a) \sin(\pi y / b) e^{-\gamma z}$$

$$E_{zs} = 0$$

For the electric field lines,

$$\frac{dy}{dx} = \frac{E_y}{E_x} = (a / b) \tan(\pi x / a) \cot(\pi y / b)$$

For the magnetic field lines

$$\frac{dy}{dx} = \frac{H_y}{H_x} = -(a/b) \cot(\pi x/a) \tan(\pi y/b)$$

Notice that $\left(\frac{E_y}{E_x}\right)\left(\frac{H_y}{H_x}\right) = -1$

showing that the electric and magnetic field lines are mutually orthogonal. The field lines are as shown in Fig. 12.14.

P. E. 12.8

$$u' = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\epsilon_r}}$$

$$f_{TE101} = \frac{1.5 \times 10^{10}}{\sqrt{3}} \sqrt{1/25 + 0 + 1/100} = \underline{\underline{1.936 \text{ GHz}}}$$

$$Q_{TE101} = \frac{1}{61\delta}, \text{ where}$$

$$\delta = \frac{1}{\sqrt{\pi f_{101} \mu \sigma_c}} = \frac{1}{\sqrt{\pi \times 1.936 \times 10^9 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7}} = 1.5 \times 10^{-6}$$

$$Q_{TE101} = \frac{10^6}{61 \times 1.5} = \underline{\underline{10,929}}$$

P. E. 12.9

(a) By Snell's law, $n_1 \sin \theta_1 = n_2 \sin \theta_2$. Thus
 $\theta_2 = 90^\circ \longrightarrow \sin \theta_2 = 1$

$$\sin \theta_1 = n_2/n_1, \quad \theta_1 = \sin^{-1} n_2/n_1 = \sin^{-1} 1.465/1.48 = \underline{\underline{81.83^\circ}}$$

$$(b) \text{ NA} = \sqrt{n_1^2 - n_2^2} = \sqrt{1.48^2 - 1.465^2} = \underline{\underline{0.21}}$$

P. E. 12.10

$$\alpha_l = 10 \log P(0)/P(l) = 0.2 \times 10 = 2$$

$$P(0)/P(l) = 10^{0.2}, \text{ i.e. } P(l) = P(0) 10^{-0.2} = 0.631 P(0)$$

$$\text{i.e. } \underline{\underline{63.1\%}}$$

Prob. 12.1

(a) For TE_{10} mode, $f_c = \frac{u'}{2a} = \frac{3 \times 10^8}{2 \times 6 \times 10^{-2}} = \underline{\underline{2.5 \text{ GHz}}}$

(b) $f = 3f_c = 7.5 \text{ GHz}$

$$f_{cmn} = \frac{u'}{2} \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} = \frac{3 \times 10^8 \times 10^2}{2} \sqrt{\left(\frac{m}{6}\right)^2 + \left(\frac{n}{4}\right)^2}$$

$$= 15 \sqrt{\left(\frac{m}{6}\right)^2 + \left(\frac{n}{4}\right)^2} \text{ GHz}$$

$$f_{c20} = 15 \times \frac{2}{6} = 5 \text{ GHz}$$

$$f_{c01} = 3.75 \text{ GHz}, \quad f_{c02} = 7.5 \text{ GHz}$$

$$f_{c10} = 2.5 \text{ GHz}, \quad f_{c20} = 5.0 \text{ GHz}$$

$$f_{c21} = 6.25 \text{ GHz}, \quad f_{c30} = 7.5 \text{ GHz}$$

$$f_{12} = 15 \sqrt{\left(\frac{1}{6}\right)^2 + \left(\frac{2}{4}\right)^2} = 7.91 \text{ GHz}$$

$$f_{11} = 15 \sqrt{\left(\frac{1}{6}\right)^2 + \left(\frac{1}{4}\right)^2} = 4.507 \text{ GHz}$$

The following modes are transmitted

$TE_{01}, TE_{02}, TE_{10}, TE_{11}, TE_{20}, TE_{21}, TE_{30}$

TM_{11}, TM_{21}

i.e. 7 TE modes and 2 TM modes

Prob. 12.2

$$f_{c10} = \frac{u'}{2a}, \quad f_{c11} = \frac{u'}{2} \sqrt{\frac{1}{a^2} + \frac{1}{a^2}} = \frac{u'}{2a} \sqrt{2}$$

Since the guide can only propagate TE_{10} mode,

$$f_{c10} < f < f_{c11} \rightarrow \frac{u'}{2a} < f < \frac{u'}{2a} \sqrt{2} \rightarrow u' < 2af < u' \sqrt{2}$$

$$\frac{u'}{f} < 2a < \frac{u'}{f} \sqrt{2} \rightarrow \lambda < 2a < \lambda \sqrt{2}$$

$$\underline{\underline{\frac{\lambda}{2} < a < \frac{\lambda}{\sqrt{2}}}}}$$

Prob. 12.3

$$f_c = \frac{u'}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{3 \times 10^8}{2 \sqrt{2.25} \times 10^{-2}} \left[\left(\frac{m}{2.28}\right)^2 + \left(\frac{n}{1.01}\right)^2 \right]^{1/2}$$

$$= \frac{15}{\sqrt{2.25}} \left[\left(\frac{m}{2.28}\right)^2 + \left(\frac{n}{1.01}\right)^2 \right]^{1/2} \text{ GHz}$$

Using this formula, we obtain the cutoff frequencies for the given modes as shown below.

Mode	f_c (GHz)
TE ₀₁	9.901
TE ₁₀	4.386
TE ₁₁	10.829
TE ₀₂	19.802
TE ₂₂	21.658
TM ₁₁	10.829
TM ₁₂	20.282
TM ₂₁	13.228

Prob. 12.4

(a)

For TE₁₀ mode,

$$f_c = \frac{u'}{2} \sqrt{\left(\frac{1}{a}\right)^2} = \frac{3 \times 10^8}{2 \times 2.4 \times 10^{-2}} = \underline{\underline{6.25 \text{ GHz}}}$$

For TE₀₁ mode,

$$f_c = \frac{u'}{2} \sqrt{\left(\frac{1}{b}\right)^2} = \frac{3 \times 10^8}{2 \times 1.2 \times 10^{-2}} = \underline{\underline{12.5 \text{ GHz}}}$$

For TE₂₀ mode,

$$f_c = \frac{u'}{2} \sqrt{\left(\frac{2}{a}\right)^2} = 2 \times 6.25 = \underline{\underline{12.5 \text{ GHz}}}$$

For TE₀₂ mode,

$$f_c = \frac{u'}{2} \sqrt{\left(\frac{2}{b}\right)^2} = 2 \times 12.5 = \underline{\underline{25 \text{ GHz}}}$$

(b) Since $f = 12 \text{ GHz}$, only TE₁₀ mode will propagate.

Prob. 12.5 $a/b = 3 \longrightarrow a = 3b$

$$f_{c10} = \frac{u'}{2a} \longrightarrow a = \frac{u'}{2f_{c10}} = \frac{3 \times 10^8}{2 \times 18 \times 10^9} \text{ m} = 0.833 \text{ cm}$$

A design could be $a = 9 \text{ mm}$, $b = 3 \text{ mm}$.

Prob. 12.6 For the dominant mode,

$$f_c = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 8} = 18.75 \text{ MHz}$$

(a) It will not pass the AM signal, (b) it will pass the FM signal.

Prob. 12.7 (a) For TE_{10} mode, $f_c = \frac{u'}{2a}$

$$\text{Or } a = \frac{u'}{2f_c} = \frac{3 \times 10^8}{2 \times 5 \times 10^9} = \underline{\underline{3 \text{ cm}}}$$

For TE_{01} mode, $f_c = \frac{u'}{2b}$

$$\text{Or } b = \frac{u'}{2f_c} = \frac{3 \times 10^8}{2 \times 12 \times 10^9} = \underline{\underline{1.25 \text{ cm}}}$$

(b) Since $a > b$, $1/a < 1/b$, the next higher modes are calculated as shown below.

Mode	f_c (GHz)
TE_{10}	5
* TE_{20}	10
TE_{30}	15
TE_{40}	20
* TE_{01}	12
TE_{02}	24
* TE_{11}	13
TE_{21}	15.62

The next three higher modes are starred ones, i.e. TE_{20} , TE_{01} , TE_{11}

$$(c) \ u' = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{2.25}} = 2 \times 10^8 \text{ m/s}$$

For TE_{11} modes,

$$f_c = \frac{3 \times 10^8}{2 \times 10^{-2} \sqrt{2.25}} \sqrt{\frac{1}{3^2} + \frac{1}{1.25^2}} = \underline{\underline{8.67 \text{ GHz}}}$$

Prob. 12.8

$$\text{Let } F_{12} = \sqrt{1 - \left(\frac{f_{c12}}{f} \right)^2} = \sqrt{1 - \left(\frac{25}{40} \right)^2} = 0.7806$$

$$\lambda' = \frac{c}{f} = \frac{3 \times 10^8}{40 \times 10^9} = 0.0075 \text{ m} = 7.5 \times 10^{-3} \text{ m}$$

$$\lambda_{12} = \frac{\lambda'}{F_{12}} = \frac{7.5 \times 10^{-3} \text{ m}}{0.7806} = \underline{\underline{9.608 \times 10^{-3} \text{ m}}}$$

$$u_{12} = \frac{u'}{F_{12}} = \frac{3 \times 10^8}{0.7806} = \underline{\underline{3.843 \times 10^8 \text{ m/s}}}$$

$$\beta_{12} = \frac{2\pi}{\lambda_{12}} = \frac{2\pi}{9.608 \times 10^{-3}} = \underline{\underline{653.95 \text{ rad/m}}}$$

$$\eta_{TE12} = \frac{\eta'}{F_{12}} = \frac{120\pi}{0.7806} = \underline{\underline{482.95 \Omega}}$$

Prob. 12.9

$$f_c = \frac{u'}{2} \sqrt{\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2}$$

For TE₁₀ mode, m=1, n=0,

$$f_c = \frac{u'}{2a} = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 5 \times 10^{-2}} = 3 \text{ GHz}$$

$$\beta = \omega \sqrt{\mu \epsilon} \sqrt{1 - \left(\frac{f_c}{f} \right)^2} = 2\pi \times 12.5 \times 10^9 \sqrt{4\pi \times 10^{-7} \times \frac{10^{-9}}{36\pi}} \sqrt{1 - \left(\frac{3}{12.5} \right)^2} = \frac{785.4}{3} (0.9708)$$

$$\underline{\underline{\beta = 254.15 \text{ rad/m}}}$$

$$u = \frac{\omega}{\beta} = \frac{2\pi \times 12.5 \times 10^9}{254.15} = \underline{\underline{3.09 \times 10^8 \text{ m/s}}}$$

$$\eta_{TE} = \frac{\eta'}{\sqrt{1 - \left(\frac{f_c}{f} \right)^2}} = \frac{120\pi}{0.9708} = \underline{\underline{388.3 \Omega}}$$

Prob. 12.10

$$f_c = \frac{u'}{2a} = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 4 \times 10^{-2}} = \underline{\underline{3.75 \text{ GHz}}}$$

$$\beta = \omega \sqrt{\mu \epsilon} \sqrt{1 - \left(\frac{f_c}{f} \right)^2} = \frac{2\pi \times 24 \times 10^9}{3 \times 10^8} \sqrt{1 - \left(\frac{3.75}{24} \right)^2} = \frac{480\pi}{3} (0.9877)$$

$$\underline{\underline{\beta = 496.48 \text{ rad/m}}}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{496.48} = \underline{\underline{0.0127 \text{ m}}}$$

Prob. 12.11

$$u = \frac{\omega}{\beta} = \frac{u'}{\sqrt{1 - (f_c / f)^2}} = \frac{3 \times 10^8}{\sqrt{1 - (6.5 / 7.2)^2}} = 6.975 \times 10^8 \text{ m/s}$$

$$u_g = \frac{9 \times 10^{16}}{u} = 1.2903 \times 10^8 \text{ m/s}$$

$$t = \frac{2l}{u_g} = \frac{300}{1.2903 \times 10^8} = \underline{\underline{2.325 \mu\text{s}}}$$

Prob. 12.12

$$f_c = \frac{u'}{2a} = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 2.4 \times 10^{-2}} = \underline{\underline{6.25 \text{ GHz}}}$$

$$f = 1.25 f_c = \underline{\underline{7.813 \text{ GHz}}}$$

$$\eta = \frac{\eta'}{\sqrt{1 - \left(\frac{f_c}{f} \right)^2}} = \frac{377}{\sqrt{1 - \left(\frac{1}{1.25} \right)^2}} = \frac{377}{0.6} = \underline{\underline{628.32 \Omega}}$$

Prob. 12.13

$$\begin{aligned} (a) f_{c10} &= \frac{u'}{2a} = \frac{c}{2a\sqrt{\epsilon_r}} \\ &= \frac{3 \times 10^8}{2 \times 1.067 \times 10^{-2} \sqrt{6.8}} \\ &= \frac{30}{2 \times 1.067 \sqrt{6.8}} \text{ GHz} \\ &= \underline{\underline{5.391 \text{ GHz}}} \end{aligned}$$

(b)

$$\begin{aligned}
 F &= \sqrt{1 - \left(\frac{f_c}{f}\right)^2} \\
 &= \sqrt{1 - \left(\frac{5.391}{6}\right)^2} \\
 &= 0.439
 \end{aligned}$$

$$u_\rho = \frac{u'}{F} = \frac{c}{F\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{0.439 \times \sqrt{6.8}} = \underline{\underline{2.62 \times 10^8 \text{ m/s}}}$$

(c)

$$\begin{aligned}
 \lambda &= \frac{\lambda'}{F} = \frac{u'/f}{F} = \frac{c}{fF\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{0.439 \times 6 \times 10^9 \times \sqrt{6.8}} \\
 &= \frac{10^{-1}}{2 \times 0.439 \sqrt{6.8}} = 0.04368 \text{ m} = \underline{\underline{4.368 \text{ cm}}}
 \end{aligned}$$

Prob.12.14

In evanescent mode,

$$\begin{aligned}
 k^2 &= \omega^2 \mu \epsilon < \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \\
 \beta &= 0, \quad \gamma = \alpha = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - k^2} = \sqrt{4\pi^2 \mu \epsilon f_c^2 - \omega^2 \mu \epsilon} \\
 \alpha &= \sqrt{\mu \epsilon} \sqrt{4\pi^2 f_c^2 - 4\pi^2 f^2} = \underline{\underline{2\pi \sqrt{\mu \epsilon} f_c \sqrt{1 - \left(\frac{f}{f_c}\right)^2}}}
 \end{aligned}$$

Prob. 12.15 $E_z \neq 0$. This must be TM₂₃ mode (m=2, n=3). Since a= 2b,

$$f_c = \frac{c}{4b} \sqrt{m^2 + 4n^2} = \frac{3 \times 10^8}{4 \times 3 \times 10^{-2}} \sqrt{4 + 36} = 15.81 \text{ GHz}, \quad f = \frac{\omega}{2\pi} = \frac{10^{12}}{2\pi} = 159.2 \text{ GHz}$$

$$\eta_{TM} = 377 \sqrt{1 - (15.81/159.2)^2} = \underline{\underline{375.1 \Omega}}$$

$$\mathcal{P}_{\text{ave}} = \frac{|E_{xs}|^2 + |E_{ys}|^2}{2\eta_{TM}} \mathbf{a}_z$$

$$= \frac{\beta^2 E_o^2}{2h^4 \eta_{TM}} \left[(2\pi/a)^2 \cos^2(2\pi x/a) \sin^2(3\pi y/b) + (3\pi/b)^2 \sin^2(2\pi x/a) \cos^2(3\pi y/b) \right] \mathbf{a}_z$$

$$P_{ave} = \int \mathcal{P}_{ave} \cdot d\mathbf{S} = \int_{x=0}^a \int_{y=0}^b \mathcal{P}_{ave} \cdot dxdy \mathbf{a}_z$$

$$= \frac{\beta^2 E_o^2}{2h^4 \eta_{TM}} \frac{ab}{4} \left[\frac{4\pi^2}{a^2} + \frac{9\pi^2}{b^2} \right] = \frac{\beta^2 E_o^2 ab}{8h^2 \eta_{TM}}$$

But

$$\beta = \frac{\omega}{c} \sqrt{1 - (f_c/f)^2} = \frac{10^{12}}{3 \times 10^8} \sqrt{1 - (15.81/159.2)^2} = 3.317 \times 10^3$$

$$h^2 = \frac{4\pi^2}{a^2} + \frac{9\pi^2}{b^2} = \frac{10\pi^2}{b^2} = 1.097 \times 10^5$$

$$P_{ave} = \frac{(3.317)^2 \times 10^6 \times 5^2 \times 18 \times 10^{-4}}{8 \times (1.098 \times 10^5) \times 375.1} = \underline{\underline{1.5 \text{ mW}}}$$

Prob. 12.16 (a) Since $m=2$ and $n=1$, we have TE₂₁ mode

$$(b) \beta = \beta' \sqrt{1 - (f_c/f)^2} = \omega \sqrt{\mu_o \epsilon_o} \sqrt{1 - (\omega_c/\omega)^2}$$

$$\beta c = \sqrt{\omega^2 - \omega_c^2} \longrightarrow \omega_c^2 = \sqrt{\omega^2 - \beta^2 c^2}$$

$$f_c = \frac{\omega_c}{2\pi} = \sqrt{f^2 - \frac{\beta^2 c^2}{4\pi^2}} = \sqrt{36 \times 10^{18} - \frac{144 \times 9 \times 10^{16}}{4\pi^2}} = \underline{\underline{5.973 \text{ GHz}}}$$

$$(c) \eta_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} = \frac{377}{\sqrt{1 - (5.973/6)^2}} = \underline{\underline{3978 \Omega}}$$

(d) For TE mode,

$$E_y = \frac{\omega \mu}{h^2} (m\pi/a) H_o \sin(m\pi x/a) \cos(n\pi y/b) \sin(\omega t - \beta z)$$

$$H_x = \frac{-\beta}{h^2} (m\pi/a) H_o \sin(m\pi x/a) \cos(n\pi y/b) \sin(\omega t - \beta z)$$

$$\beta = 12, m = 2, n = 1$$

$$E_{oy} = \frac{\omega \mu}{h^2} (m\pi / a) H_o, \quad H_{ox} = \frac{\beta}{h^2} (m\pi / a) H_o$$

$$\eta_{TE} = \frac{E_{oy}}{H_{ox}} = \frac{\omega \mu}{\beta} = \frac{2\pi \times 6 \times 10^9 \times 4\pi \times 10^{-7}}{12} = 4\pi^2 \times 100$$

$$H_{ox} = \frac{E_{oy}}{\eta_{TE}} = \frac{5}{4\pi^2 \times 100} = 1.267 \text{ mA/m}$$

$$\underline{H_x = -1.267 \sin(m\pi x / a) \cos(n\pi y / b) \sin(\omega t - \beta z) \text{ mA/m}}$$

Prob. 12.17 (a) Since $m=2, n=3$, the mode is TE₂₃.

$$(b) \quad \beta = \beta' \sqrt{1 - (f_c / f)^2} = \frac{2\pi f}{c} \sqrt{1 - (f_c / f)^2}$$

But

$$f_c = \frac{u'}{2} \sqrt{(m/a)^2 + (n/b)^2} = \frac{3 \times 10^8}{2 \times 10^{-2}} \sqrt{(2/2.86)^2 + (3/1.016)^2} = 46.19 \text{ GHz}, \quad f = 50 \text{ GHz}$$

$$\beta = \frac{2\pi \times 50 \times 10^9}{3 \times 10^8} \sqrt{1 - (46.19/50)^2} = 400.68 \text{ rad/m}$$

$$\gamma = j\beta = \underline{\underline{j400.7 \text{ /m}}}$$

$$(c) \quad \eta = \frac{\eta'}{\sqrt{1 - (f_c / f)^2}} = \frac{377}{\sqrt{1 - (46.19/50)^2}} = \underline{\underline{985.3 \Omega}}$$

Prob. 12.18 In free space,

$$\eta_1 = \frac{\eta_o}{\sqrt{1 - (f_c / f)^2}}, \quad f_c = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 5 \times 10^{-2}} = 3 \text{ GHz}$$

$$\eta_1 = \frac{377}{\sqrt{1 - (3/8)^2}} = 406.7 \Omega$$

$$\eta_2 = \frac{\eta'_1}{\sqrt{1 - (f_c / f)^2}}, \quad \eta'_1 = \frac{120\pi}{\sqrt{2.25}} = 80\pi, \quad f_c = \frac{u'}{2a}, \quad u' = \frac{c}{\sqrt{\epsilon_r}}$$

$$f_c = \frac{3 \times 10^8}{2 \times 5 \times 10^{-2} \sqrt{2.25}} = 2 \text{ GHz}, \quad \eta_2 = \frac{80\pi}{\sqrt{1 - (2/8)^2}} = 259.57\Omega$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = -0.2208$$

$$s = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \underline{\underline{1.5667}}$$

Prob. 12.19 Substituting $E_z = R\Phi Z$ into the wave equation,

$$\frac{\Phi Z}{\rho} \frac{d}{d\rho}(\rho R') + \frac{RZ}{\rho^2} \Phi'' + R\Phi Z'' + k^2 R\Phi Z = 0$$

Dividing by $R\Phi Z$,

$$\frac{1}{R\rho} \frac{d}{d\rho}(\rho R') + \frac{\Phi''}{\Phi\rho^2} + k^2 = -\frac{Z''}{Z} = -k_z^2$$

$$\text{i.e.} \quad \underline{\underline{Z'' - k_z^2 Z = 0}}$$

$$\frac{1}{R\rho} \frac{d}{d\rho}(\rho R') + \frac{\Phi''}{\Phi\rho^2} + (k^2 + k_z^2) = 0$$

$$\frac{\rho}{R} \frac{d}{d\rho}(\rho R') + (k^2 + k_z^2)\rho^2 = -\frac{\Phi''}{\Phi} = k_\phi^2$$

or

$$\underline{\underline{\Phi'' + k_\phi^2 \Phi = 0}}$$

$$\rho \frac{d}{d\rho}(\rho R') + (k_\rho^2 \rho^2 - k_\phi^2)R = 0, \text{ where } k_\rho^2 = k^2 + k_z^2. \text{ Hence}$$

$$\underline{\underline{\rho^2 R'' + \rho R' + (k_\rho^2 \rho^2 - k_\phi^2)R = 0}}$$

Prob. 12.20

The MATLAB code and the plot of the phase and group velocities are presented below.

```
% Plot U_p and U_g versus frequency f (10<f<100) in GHz
```

```
c=3*10^8;
```

```
for k=1:91
```

```
    f(k)=9+k
```

```
    fac = sqrt( 1 - (8/f(k))^2 );
```

```
    up(k) = c/fac;
```

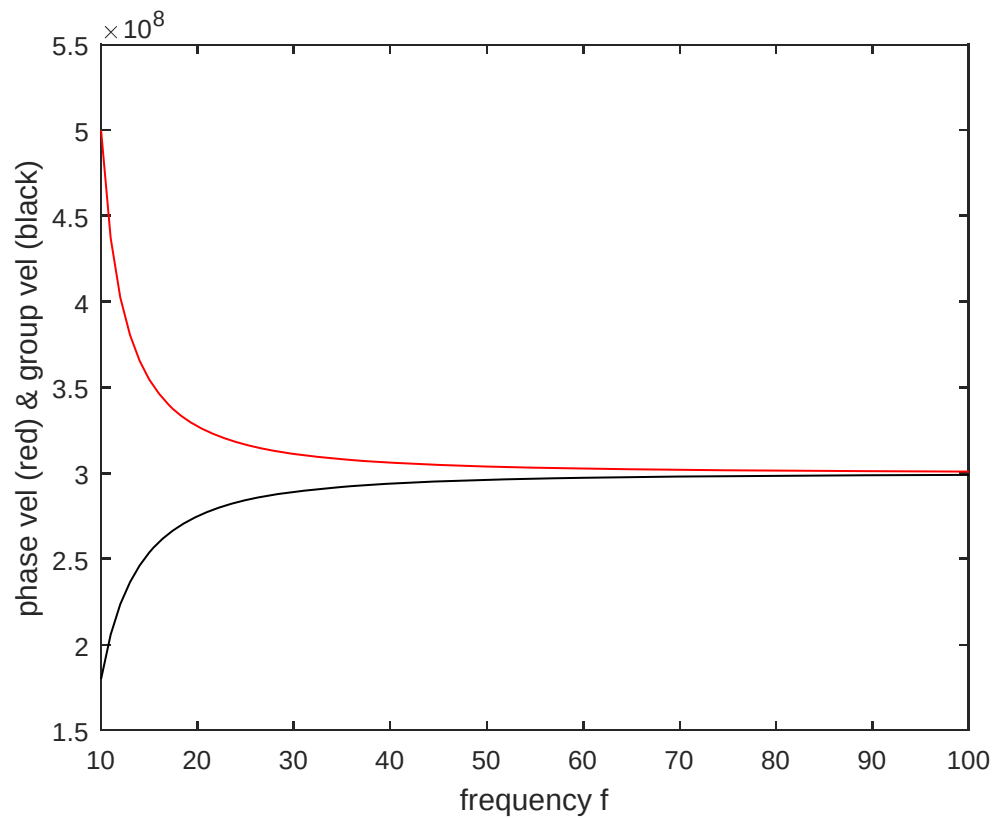
```
    ug(k) = c*fac;
```

```
end
```

```
plot(f,up, 'r', f, ug, 'k')
```

```
xlabel('frequency f')
```

```
ylabel('phase vel (red) & group vel (black)')
```



Prob. 12.21

(a)

For TE_{10} mode,

$$f_c = \frac{u'}{2a} = \frac{3 \times 10^8}{2 \times 7.2 \times 10^{-2}} = 2.083 \text{ GHz}$$

$$\text{Let } F = \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \sqrt{1 - \left(\frac{2.083}{6.2}\right)^2} = 0.942$$

$$\beta = \omega \sqrt{\mu \epsilon} F = \frac{\omega F}{c} = \frac{2\pi \times 6.2 \times 10^9 \times 0.942}{3 \times 10^8} = \underline{\underline{122.32 \text{ rad/m}}}$$

$$u_p = \frac{\omega}{\beta} = \frac{c}{F} = \frac{3 \times 10^8}{0.942} = \underline{\underline{3.185 \times 10^8 \text{ m/s}}}$$

$$u_g = u' F = 3 \times 10^8 (0.942) = \underline{\underline{2.826 \times 10^8 \text{ m/s}}}$$

$$\eta_{TE} = \frac{\eta'}{F} = \frac{377}{0.942} = \underline{\underline{400.21 \Omega}}$$

(b)

$$u' = \frac{1}{\sqrt{\mu \epsilon}} = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.25}} = 2 \times 10^8$$

$$f_c = \frac{u'}{2a} = \frac{2 \times 10^8}{2 \times 7.2 \times 10^{-2}} = 1.389 \text{ GHz}$$

$$\text{Let } F = \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \sqrt{1 - \left(\frac{1.389}{6.2}\right)^2} = 0.9746$$

$$\beta = \omega \sqrt{\mu \epsilon} F = \frac{\omega F \sqrt{\epsilon_r}}{c} = \frac{2\pi \times 6.2 \times 10^9 \times 0.9746 \times 1.5}{3 \times 10^8} = \underline{\underline{189.83 \text{ rad/m}}}$$

$$u_p = \frac{\omega}{\beta} = \frac{2\pi \times 6.2 \times 10^9}{189.83} = \underline{\underline{2.052 \times 10^8 \text{ m/s}}}$$

$$u_g = u' F = 2 \times 10^8 (0.9746) = \underline{\underline{1.949 \times 10^8 \text{ m/s}}}$$

$$\eta_{TE} = \frac{\eta'}{F} = \frac{377}{1.5 \times 0.9746} = \underline{\underline{257.88 \Omega}}$$

Prob. 12.22

$$f_{c10} = \frac{u'}{2a} = \frac{1}{2a\sqrt{\mu\epsilon}} = \frac{c}{2a\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{2 \times 7.214 \times 10^{-2} \sqrt{2.5}} = 1.315 \text{ GHz}$$

$$\text{Let } F = \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \sqrt{1 - \left(\frac{1.315}{4}\right)^2} = 0.9444$$

$$\beta = \omega\sqrt{\mu\epsilon}F = \frac{\omega\sqrt{\epsilon_r}F}{c}$$

$$u_p = \frac{\omega}{\beta} = \frac{c}{F\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{0.9444 \times \sqrt{2.5}} = \underline{\underline{2.009 \times 10^8 \text{ m/s}}}$$

$$u_g = u'F = \frac{cF}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8 \times 0.9444}{\sqrt{2.5}} = \underline{\underline{1.792 \times 10^8 \text{ m/s}}}$$

Prob. 12.23

$$u_p = \frac{\omega}{\beta} = \frac{\omega\lambda}{2\pi} \rightarrow \omega = \beta u_p = \beta c \frac{\lambda_o^2}{\lambda^2} = \frac{c\lambda_o^2\beta^3}{4\pi^2}, \quad (\lambda = 2\pi / \beta)$$

$$u_g = \frac{d\omega}{d\beta} = 3 \left(\frac{c\lambda_o^2}{4\pi^2} \right) \beta^2 = 3c \left(\frac{\lambda_o}{\lambda} \right)^2 = \underline{\underline{3u_p}}$$

Prob. 12.24

$$u_g = \frac{1}{4}c = u' \sqrt{1 - \left(\frac{f_c}{f_1}\right)^2} \quad (1)$$

$$u_g = \frac{1}{3}c = u' \sqrt{1 - \left(\frac{f_c}{f_2}\right)^2} \quad (2)$$

Dividing (1) by (2),

$$\frac{1/4}{1/3} = \frac{\sqrt{1 - \left(\frac{f_c}{f_1}\right)^2}}{\sqrt{1 - \left(\frac{f_c}{f_2}\right)^2}} \rightarrow \left(\frac{3}{4}\right)^2 = 0.5625 = \frac{1 - \left(\frac{f_c}{f_1}\right)^2}{1 - \left(\frac{f_c}{f_2}\right)^2}$$

$$1 - \left(\frac{f_c}{f_1}\right)^2 = 0.5625 \left[1 - \left(\frac{f_c}{f_2}\right)^2 \right]$$

Assuming f_c is in GHz,

$$1 - \frac{f_c^2}{144} = 0.5625 - \frac{0.5625 f_c^2}{225} \rightarrow f_c^2 = 98.44 \rightarrow \underline{\underline{f_c = 9.9216 \text{ GHz}}}$$

From (1),

$$u' = \frac{0.25c}{\sqrt{1 - \left(\frac{f_c}{f_1}\right)^2}} = \frac{0.25c}{\sqrt{1 - \left(\frac{9.9216}{12}\right)^2}} = 0.4444c$$

$$\text{But } u' = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\epsilon_r}}$$

$$\frac{c}{\sqrt{\epsilon_r}} = 0.4444c \rightarrow \epsilon_r = \left(\frac{1}{0.4444}\right)^2 = \underline{\underline{5.0625}}$$

Prob. 12.25

$$f_c = \frac{u'}{2a}$$

$$u_g = u' \sqrt{1 - \left(\frac{f_c}{f}\right)^2} \longrightarrow \left(\frac{f_c}{f}\right)^2 = 1 - \left(\frac{u_g}{u'}\right)^2 = 1 - \left(\frac{\frac{1.8 \times 10^8}{3 \times 10^8}}{\frac{\sqrt{2.2}}{2.2}}\right)^2 = 0.208$$

$$f_c = \sqrt{0.208} f = 2.0523 \text{ GHz}$$

$$a = \frac{u'}{2f_c} = \frac{3 \times 10^8}{2\sqrt{2.2} \times 2.053 \times 10^9} = \underline{\underline{4.927 \text{ cm}}}$$

Prob. 12.26

$$\text{Let } F = \sqrt{1 - (f_c / f)^2} = \sqrt{1 - (16 / 24)^2} = 0.7453$$

$$u' = \frac{1}{\sqrt{\mu\epsilon}} = \frac{3 \times 10^8}{\sqrt{2.25}} = 2 \times 10^8, \quad u_p = \frac{u'}{F}, \quad u_g = u' F = 2 \times 10^8 \times 0.7453 = \underline{\underline{1.491 \times 10^8}}$$

m/s

$$\eta_{TE} = \eta' / F = \frac{377}{1.5 \times 0.7453} = \underline{\underline{337.2 \Omega}}$$

Prob. 12.27For the TE₁₀ mode,

$$H_{zs} = H_o \cos\left(\frac{\pi x}{a}\right) e^{-j\beta z}$$

$$H_{xs} = \frac{j\beta a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) e^{-j\beta z}$$

$$E_{ys} = -\frac{j\omega\mu a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) e^{-j\beta z}$$

$$E_{xs} = 0 = E_{zs} = H_{ys}$$

$$\begin{aligned} \mathbf{E}_s \times \mathbf{H}_s^* &= \begin{vmatrix} 0 & E_{ys} & 0 \\ H_{xs}^* & 0 & H_{zs}^* \end{vmatrix} = E_{ys} H_{zs}^* \mathbf{a}_x - E_{ys} H_{xs}^* \mathbf{a}_z \\ &= -\frac{j\omega\mu a}{\pi} H_o^2 \cos\left(\frac{\pi x}{a}\right) \sin\left(\frac{\pi x}{a}\right) \mathbf{a}_x + \frac{\omega\mu\beta a^2}{\pi^2} H_o^2 \sin^2\left(\frac{\pi x}{a}\right) \mathbf{a}_z \\ \mathcal{P}_{ave} &= \frac{1}{2} \text{Re}[\mathbf{E}_s \times \mathbf{H}_s^*] = \underline{\underline{\frac{\omega\mu\beta a^2}{2\pi^2} H_o^2 \sin^2\left(\frac{\pi x}{a}\right) \mathbf{a}_z}} \end{aligned}$$

Prob. 12.28

$$\mathbf{P}_{ave} = \frac{|E_{xs}|^2 + |E_{ys}|^2}{2\eta} \mathbf{a}_z = \underline{\underline{\frac{\omega^2 \mu^2 \pi^2}{2\eta b^2 h^4} H_o^2 \sin^2 \pi y / b \mathbf{a}_z}}$$

where $\eta = \eta_{TE10}$.

$$\mathbf{P}_{ave} = \int \mathbf{P}_{ave} \cdot d\mathbf{S} = \frac{\omega^2 \mu^2 \pi^2}{2\eta b^2 h^4} H_o^2 \int_{x=0}^a \int_{y=0}^b \sin^2 \pi y / b dx dy$$

$$P_{\text{ave}} = \frac{\omega^2 \mu^2 \pi^2}{2\eta b^2 h^4} H_o^2 ab / 2$$

$$\text{But } h^2 = (m\pi / a)^2 + (n\pi / b)^2 = \frac{\pi^2}{b^2},$$

$$P_{\text{ave}} = \frac{\omega^2 \mu^2 ab^3 H_o^2}{4\pi^2 \eta}$$

Prob. 12.29

$$R_s = \sqrt{\frac{\pi \mu f}{\sigma_c}} = \sqrt{\frac{\pi \times 12 \times 10^9 \times 4\pi \times 10^{-7}}{5.8 \times 10^7}} = 2.858 \times 10^{-2}$$

$$f_{c10} = \frac{u'}{2a} = \frac{3 \times 10^8}{2\sqrt{2.6} \times 2 \times 10^{-2}} = 4.651 \text{ GHz}$$

$$f_{c11} = \frac{u'}{2} \left[\frac{1}{a^2} + \frac{1}{b^2} \right]^{1/2} = 10.4 \text{ GHz}$$

$$\eta' = \sqrt{\frac{\mu}{\epsilon}} = \frac{377}{\sqrt{2.6}} = 233.81 \Omega$$

(a) For TE₁₀ mode, eq.(12.57) gives

$$\begin{aligned} \alpha_d + j\beta_d &= \sqrt{-\omega^2 \mu \epsilon + k_x^2 + k_y^2 + j\omega \mu \sigma_d} \\ &= \sqrt{-\omega^2 / u^2 + \frac{\pi^2}{a^2} + j\omega \mu \sigma_d} \\ &= \sqrt{-\left(\frac{2\pi \times 12 \times 10^9}{3 \times 10^8} \right)^2 (2.6) + \frac{\pi^2}{(2 \times 10^{-2})^2} + j2\pi \times 12 \times 10^9 \times 4\pi \times 10^{-7} \times 10^{-4}} \\ &= 0.012682 + j373.57 \end{aligned}$$

$$\underline{\alpha_d = 0.012682 \text{ Np/m}}$$

$$\alpha_c = \frac{2R_s}{b\eta' \sqrt{1 - (f_c / f)^2}} \left[\frac{1}{2} + \frac{b}{a} \left(\frac{f_c}{f} \right)^2 \right]$$

$$= \frac{2 \times 2.858 \times 10^{-2}}{10^{-2} (233.81) \sqrt{1 - (4.651/12)^2}} \left[\frac{1}{2} + \frac{1}{2} \left(\frac{4.651}{12} \right)^2 \right] = \underline{\underline{0.0153}} \text{ Np/m}$$

(b) For TE₁₁ mode,

$$\begin{aligned} \alpha_d + j\beta_d &= \sqrt{-\omega^2 / u^2 + 1/a^2 + 1/b^2 + j\omega\mu\sigma_d} \\ &= \sqrt{-139556.21 + \frac{\pi^2}{(10^{-2})^2} + j9.4748} = 0.02344 + j202.14 \end{aligned}$$

$$\underline{\underline{\alpha_d = 0.02344 \text{ Np/m}}}$$

$$\alpha_c = \frac{2R_s}{b\eta' \sqrt{1 - (f_c/f)^2}} \left[\frac{(b/a)^3 + 1}{(b/a)^2 + 1} \right] = \frac{2 \times 2.858 \times 10^{-2}}{10^{-2} (233.81) \sqrt{1 - (10.4/12)^2}} \left[\frac{(1/8) + 1}{(1/4) + 1} \right]$$

$$\underline{\underline{\alpha_c = 0.0441 \text{ Np/m}}}$$

Prob. 12.30 $\epsilon_c = \epsilon' - j\epsilon'' = \epsilon - j \frac{\sigma}{\omega}$

Comparing this with

$$\epsilon_c = 16\epsilon_o(1 - j10^{-4}) = 16\epsilon_o - j16\epsilon_o \times 10^{-4}$$

$$\epsilon = 16\epsilon_o, \quad \frac{\sigma}{\omega} = 16\epsilon_o \times 10^{-4}$$

For TM₂₁ mode,

$$f_c = \frac{u'}{2} \left[\frac{m^2}{a^2} + \frac{n^2}{b^2} \right]^{1/2} = 2.0963 \text{ GHz}, \quad f = 1.1f_c = 2.3059 \text{ GHz}$$

$$\sigma = 16\epsilon_o \omega \times 10^{-4} = 16 \times 2\pi \times 2.3059 \times 10^9 \times \frac{10^{-9}}{36\pi} \times 10^{-4} = 2.0525 \times 10^{-4}$$

$$\eta' = \sqrt{\frac{\mu}{\epsilon}} = 30\pi \Omega$$

$$\alpha_d = \frac{\sigma\eta'}{2\sqrt{1 - (f_c/f)^2}} = \frac{4.1 \times 10^{-4} \times 30\pi}{2\sqrt{1 - 1/1.12}} = \underline{\underline{0.0231}} \text{ Np/m}$$

$$E_o e^{-\alpha_d z} = 0.8E_o \quad \longrightarrow \quad z = \frac{1}{\alpha_d} \ln(1/0.8) = \underline{\underline{9.66 \text{ m}}}$$

Prob. 12.31For TM₂₁ mode,

$$\alpha_c = \frac{2R_s}{b\eta' \sqrt{1 - (f_c / f)^2}}$$

$$R_s = \frac{1}{\sigma_c \delta} = \sqrt{\frac{\pi f \mu}{\sigma_c}} = \sqrt{\frac{\pi \times 2.3059 \times 10^9 \times 4\pi \times 10^{-7}}{1.5 \times 10^7}} = 0.0246 \, \Omega$$

$$\alpha_c = \frac{2 \times 0.0246}{4\pi \times 10^{-2} \times 30\pi \times 0.4166} = 0.0314 \, \text{Np/m}$$

$$E_o e^{-(\alpha_c + \alpha_d)z} = 0.7 E_o \quad \longrightarrow \quad z = \frac{1}{\alpha_c + \alpha_d} \ln(1/0.7) = \underline{\underline{6.5445 \, \text{m}}}$$

Prob. 12.32

$$f_{c10} = \frac{u'}{2a} = \frac{3 \times 10^8}{2 \times 6 \times 10^{-2}} = 2.5 \, \text{GHz}$$

$$\eta_{TE} = \frac{\eta'}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{377}{\sqrt{1 - \left(\frac{2.5}{4}\right)^2}} = 483 \, \Omega$$

From Example 12.5,

$$P_{ave} = \frac{E_o^2 ab}{2\eta} = \frac{(2.2)^2 \times 10^6 \times 6 \times 3 \times 10^{-4}}{2 \times 483} = \underline{\underline{9.0196 \, \text{mW}}}$$

Prob. 12.33For TE₁₀ mode,

$$f_c = \frac{u'}{2a} = \frac{3 \times 10^8}{2\sqrt{2.11 \times 4.8 \times 10^{-2}}} = 2.151 \, \text{GHz}$$

$$(a) \text{ loss tangent} = \frac{\sigma}{\omega \epsilon} = d$$

$$\sigma = d\omega\epsilon = 3 \times 10^{-4} \times 2\pi \times 4 \times 10^9 \times 2.11 \times \frac{10^{-9}}{36\pi} = 1.4086 \times 10^{-4}$$

$$\eta' = \frac{120\pi}{\sqrt{2.11}} = 259.53 \, \Omega$$

$$\alpha_d = \frac{\sigma \eta'}{2\sqrt{1-(f_c/f)^2}} = \frac{1.4067 \times 10^{-4} \times 259.53}{2\sqrt{1-(2.151/4)^2}} = \underline{\underline{2.165 \times 10^{-2} \text{ Np/m}}}$$

$$(b) R_s = \sqrt{\frac{\mu f \pi}{\sigma_c}} = \sqrt{\frac{\pi \times 4 \times 10^9 \times 4\pi \times 10^{-7}}{4.1 \times 10^7}} = 1.9625 \times 10^{-2} \Omega$$

$$\alpha_c = \frac{2R_s}{b\eta' \sqrt{1-(f_c/f)^2}} \left[\frac{1}{2} + \frac{b}{a} (f_c/f)^2 \right] = \frac{3.925 \times 10^{-2} (0.5 + 0.5 \times 0.2892)}{2.4 \times 10^{-2} \times 259.53 \times 0.8431}$$

$$= \underline{\underline{4.818 \times 10^{-3} \text{ Np/m}}}$$

Prob.12.34

$$\alpha_c = \frac{2R_s}{b\eta' \sqrt{1-\left(\frac{f_c}{f}\right)^2}} \left[\frac{1}{2} + \frac{b}{a} \left(\frac{f_c}{f}\right)^2 \right] = \frac{2\sqrt{\frac{\pi f \mu}{\sigma_c}}}{b\eta' \sqrt{1-\left(\frac{f_c}{f}\right)^2}} \left[\frac{1}{2} + \frac{1}{2} \left(\frac{f_c}{f}\right)^2 \right]$$

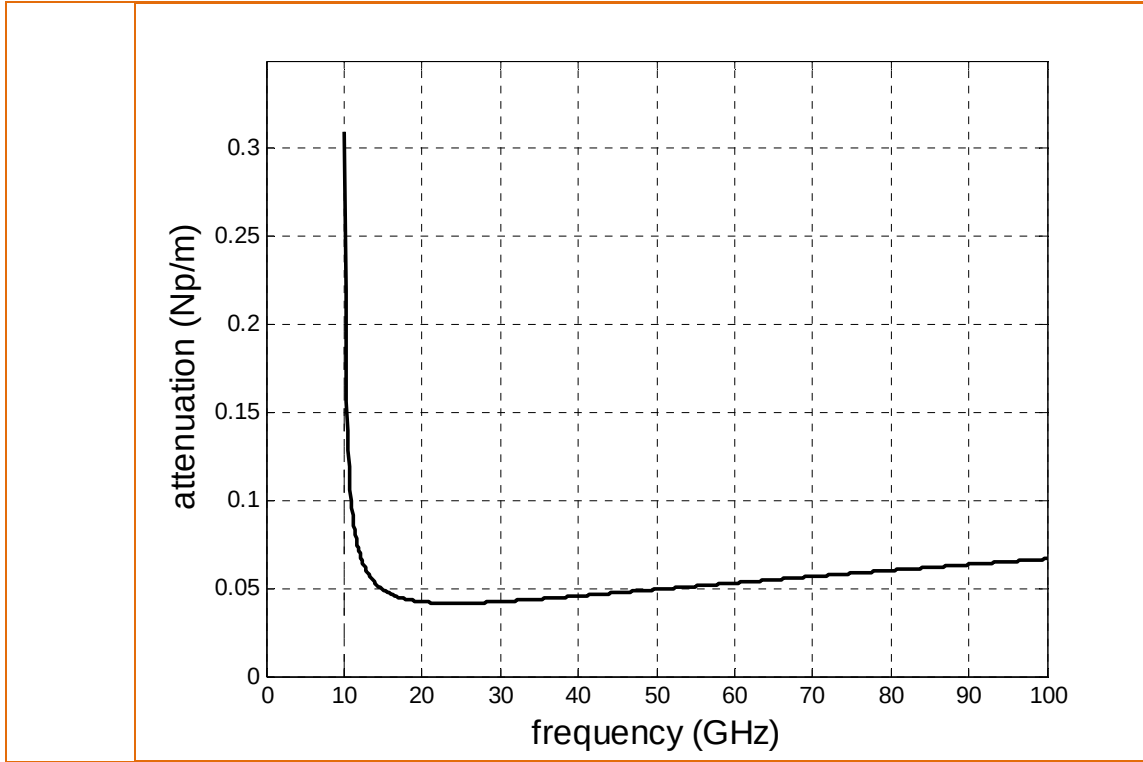
$$= \frac{2\sqrt{4\pi \times 10^{-7} \times \pi} \sqrt{f} \times \frac{1}{2}}{0.5 \times 10^{-2} \times (120\pi / \sqrt{2.25}) \sqrt{5.8 \times 10^7} \sqrt{1-\left(\frac{f_c}{f}\right)^2}} \left[1 + \left(\frac{f_c}{f}\right)^2 \right]$$

$$= \frac{10^{-5} \sqrt{f}}{30\sqrt{(5.8/2.25)} \sqrt{1-\left(\frac{f_c}{f}\right)^2}} \left[1 + \left(\frac{f_c}{f}\right)^2 \right]$$

The MATLAB code is shown below

```
k=10^(-5)/(30*sqrt(5.8/2.25));
fc=10^10;
for n=1:1000
    f(n)=fc*(n/100+1);
    fn=f(n);
    num=sqrt(fn)*(1+(fc/fn)^2);
    den=sqrt(1-(fc/fn)^2);
    alpha(n)=k*num/den;
end
plot(f/10^9,alpha)
xlabel('frequency (GHz)')
ylabel('attenuation')
grid
```

The plot of attenuation versus frequency is shown below.



Prob. 12.35

The cutoff frequency of the dominant mode is

$$f_{c10} = \frac{u}{2a} = \frac{3 \times 10^8}{4.576 \times 10^{-2}} = 6.56 \text{ GHz}$$

The surface resistance is

$$R_s = \sqrt{\frac{\pi f \mu}{\sigma_c}} = \sqrt{\frac{\pi \times 8.4 \times 10^9 \times 4\pi \times 10^{-7}}{5.8 \times 10^7}} = 23.91 \times 10^{-3} \Omega$$

For TE₁₀ mode,

$$\alpha_c = \frac{2R_s}{b\eta' \sqrt{1 - \left(\frac{f_c}{f}\right)^2}} \left[0.5 + \frac{b}{a} \left(\frac{f_c}{f}\right)^2 \right]$$

$$\frac{f_c}{f} = \frac{6.56}{8.4} = 0.781, \quad \eta' = \eta_o = 377$$

$$\begin{aligned}
 \alpha_c &= \frac{2 \times 23.91 \times 10^{-3}}{1.016 \times 10^{-2} \times 377 \sqrt{1 - 0.781^2}} \left[0.5 + \frac{1.016}{2.286} (0.781)^2 \right] \\
 &= \frac{47.82 \times 10^{-3} (0.5 + 0.2711)}{3.83 \times 0.6245} = 15.42 \times 10^{-3} \text{ Np/m} \\
 &= 15.42 \times 10^{-3} \times 8.686 \text{ dB/m} = \underline{\underline{0.1339 \text{ dB/m}}}
 \end{aligned}$$

Prob. 12.36

$$(a) \quad f_{c10} = \frac{u'}{2a} = \frac{3 \times 10^8}{2 \times 3.8 \times 10^{-2}} = 3.947 \text{ GHz}$$

$$u_g = u' \sqrt{1 - \left(\frac{f_c}{f} \right)^2} = 3 \times 10^8 \sqrt{1 - (0.3947)^2} = \underline{\underline{2.756 \times 10^8 \text{ m/s}}}$$

$$(b) \quad \alpha = \alpha_d + \alpha_c$$

$\alpha_d = 0$ since the guide is air-filled.

$$R_s = \sqrt{\frac{\pi f \mu}{\sigma_c}} = \sqrt{\frac{\pi \times 10^{10} \times 4\pi \times 10^{-7}}{5.8 \times 10^7}} = 2.609 \times 10^{-2} \Omega$$

$$\begin{aligned}
 \alpha_c &= \frac{2R_s}{b\eta' \sqrt{1 - \left(\frac{f_c}{f} \right)^2}} \left[0.5 + \frac{b}{a} \left(\frac{f_c}{f} \right)^2 \right] \\
 &= \frac{2 \times 2.609 \times 10^{-2}}{1.6 \times 10^{-2} (377) \sqrt{1 - (0.3947)^2}} \left[0.5 + \frac{1.6}{3.8} (0.3947)^2 \right] = \frac{5.218 \times 0.5656}{554.23} \\
 &= 5.325 \times 10^{-3} \text{ Np/m}
 \end{aligned}$$

$$\alpha_c (\text{dB}) = 8.686 \times 5.325 \times 10^{-3} = 0.04626 \text{ dB/m}$$

Prob. 12.37

$$f_{c10} = \frac{u'}{2a} = \frac{c}{2a\sqrt{\epsilon_r \mu_r}} = \frac{3 \times 10^8}{2 \times 2.5 \times 10^{-2} \sqrt{2.26}} = 3.991 \text{ GHz}$$

$$\beta' = \frac{\omega}{u'} = \frac{2\pi f \sqrt{\epsilon_r}}{c}$$

$$F = \sqrt{1 - \left(\frac{f_c}{f} \right)^2} = \sqrt{1 - \left(\frac{3.991}{7.5} \right)^2} = 0.8467$$

$$\beta = \beta' F = \frac{2\pi \times 7.5 \times 10^9 \sqrt{2.26}}{3 \times 10^8} \times 0.8467 = \underline{\underline{199.94 \text{ rad/m}}}$$

$$\alpha_d = \frac{\sigma \eta'}{2F} = \frac{\sigma \eta_o}{2F \sqrt{\epsilon_r}} = \frac{10^{-4}(377)}{2 \times 0.8467 \sqrt{2.26}} = \underline{\underline{1.481 \times 10^{-2} \text{ Np/m}}}$$

$$\alpha_c = \frac{2R_s}{b\eta' \sqrt{1 - \left(\frac{f_c}{f}\right)^2}} \left[0.5 + \frac{b}{a} \left(\frac{f_c}{f}\right)^2 \right]$$

$$R_s = \sqrt{\frac{\pi f \mu}{\sigma_c}} = \sqrt{\frac{\pi \times 7.5 \times 10^9 \times 4\pi \times 10^{-7}}{1.1 \times 10^7}} = 0.0519 \Omega$$

$$\alpha_c = \frac{2 \times 0.0519 \left[0.5 + \frac{1.5}{2.5} \left(\frac{3.991}{7.5}\right)^2 \right]}{1.5 \times 10^{-2} \times \frac{377}{\sqrt{2.66}} \times 0.8467} = \frac{0.1038 \times 0.6698}{3.1848}$$

$$= \underline{\underline{0.02183 \text{ Np/m}}}$$

$$u_p = \frac{u'}{F} = \frac{c}{F \sqrt{\epsilon_r}} = \frac{3 \times 10^8}{0.8467 \sqrt{2.26}} = \underline{\underline{2.357 \times 10^8 \text{ m/s}}}$$

$$u_g = u' F = \frac{3 \times 10^8 \times 0.8467}{\sqrt{2.26}} = \underline{\underline{1.689 \times 10^8 \text{ m/s}}}$$

$$\lambda_c = \frac{u'}{f_c} = \frac{c}{f_c \sqrt{\epsilon_r}} = \frac{3 \times 10^8}{3.991 \times 10^9 \sqrt{2.26}} = 0.05 \text{ m} = \underline{\underline{5 \text{ cm}}} (= 2a, \text{ as expected})$$

Prob. 12.38 (a) For TE₁₀ mode,

$$f_c = \frac{u'}{2a}, \quad u' = \frac{c}{\sqrt{2.11}}$$

$$f_c = \frac{3 \times 10^8}{\sqrt{2.11}(2 \times 2.25 \times 10^{-2})} = \underline{\underline{4.589 \text{ GHz}}}$$

$$(b) \quad \alpha_{cTE10} = \frac{2R_s}{b\eta' \sqrt{1 - (f_c / f)^2}} \left[\frac{1}{2} + \frac{b}{a} (f_c / f)^2 \right]$$

$$R_s = \sqrt{\frac{\pi f \mu}{\sigma_c}} = \sqrt{\frac{\pi \times 5 \times 10^9 \times 4\pi \times 10^{-7}}{1.37 \times 10^7}} = 3.796 \times 10^{-2} \Omega$$

$$\eta' = \frac{377}{\sqrt{2.11}} = 259.54 \Omega$$

$$\alpha_c = \frac{2 \times 3.796 \times 10^{-2} [0.5 + \frac{1.5}{2.25} (4.589/5)^2]}{1.5 \times 10^{-4} (259.54) \sqrt{1 - (4.589/5)^2}} = \underline{\underline{0.05217}} \text{ Np/m}$$

Prob. 12.39 For TE₁₀ mode,

$$\alpha_c = \frac{2R_s}{b\eta' \sqrt{1 - (f_c/f)^2}} \left[\frac{1}{2} + \frac{b}{a} \left(\frac{f_c}{f} \right)^2 \right]$$

But $a = b$, $R_s = \frac{1}{\sigma_c \delta} = \sqrt{\frac{\pi f \mu}{\sigma_c}}$

$$\alpha_c = \frac{2 \sqrt{\frac{\pi f \mu}{\sigma_c}}}{a\eta' \sqrt{1 - (f_c/f)^2}} \left[\frac{1}{2} + \left(\frac{f_c}{f} \right)^2 \right] = \frac{k \sqrt{f} \left[\frac{1}{2} + \left(\frac{f_c}{f} \right)^2 \right]}{\sqrt{1 - (f_c/f)^2}}$$

where k is a constant.

$$\frac{d\alpha_c}{df} = \frac{k[1 - (\frac{f_c}{f})^2]^{1/2} [\frac{1}{4} f^{-1/2} - \frac{3}{2} f_c^2 f^{-5/2}] - \frac{k}{2} [\frac{1}{2} f^{1/2} + f_c^2 f^{-3/2}] (2 f_c^2 f^{-3}) [1 - (\frac{f_c}{f})^2]^{-1/2}}{1 - (f_c/f)^2}$$

For minimum value, $\frac{d\alpha_c}{df} = 0$. This leads to $\underline{\underline{f = 2.962 f_c}}$.

Prob. 12.40 For the TE mode to z ,

$$E_{zs} = 0, H_{zs} = H_o \cos(m\pi x/a) \cos(n\pi y/b) \sin(p\pi z/c)$$

$$E_{ys} = -\frac{\gamma}{h^2} \frac{\partial E_{zs}}{\partial y} + \frac{j\omega\mu}{h^2} \frac{\partial H_{zs}}{\partial x} = -\frac{j\omega\mu}{h^2} (m\pi/a) H_o \sin(m\pi x/a) \cos(n\pi y/b) \sin(p\pi z/c)$$

as required.

$$E_{xs} = -\frac{\gamma}{h^2} \frac{\partial E_{zs}}{\partial x} - \frac{j\omega\mu}{h^2} \frac{\partial H_{zs}}{\partial y} = \frac{j\omega\mu}{h^2} (n\pi/b) H_o \cos(m\pi x/a) \sin(n\pi y/b) \sin(p\pi z/c)$$

From Maxwell's equation,

$$-j\omega\mu\mathbf{H}_s = \nabla \times \mathbf{E}_s = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_{xs} & E_{ys} & 0 \end{vmatrix}$$

$$H_{xs} = \frac{1}{j\omega\mu} \frac{\partial E_{ys}}{\partial z} = -\frac{1}{h^2} (m\pi/a)(p\pi/c) H_o \sin(m\pi x/a) \cos(n\pi y/b) \cos(p\pi z/c)$$

Prob. 12.41 Maxwell's equation can be written as

$$H_{xs} = \frac{j\omega\epsilon}{h^2} \frac{\partial E_{zs}}{\partial y} - \frac{\gamma}{h^2} \frac{\partial H_{zs}}{\partial x}$$

For a rectangular cavity,

$$h^2 = k_x^2 + k_y^2 = (m\pi/a)^2 + (n\pi/b)^2$$

For TM mode, $H_{zs} = 0$ and

$$E_{zs} = E_o \sin(m\pi x/a) \sin(n\pi y/b) \cos(p\pi z/c)$$

Thus

$$H_{xs} = \frac{j\omega\epsilon}{h^2} \frac{\partial E_{zs}}{\partial y} = \frac{j\omega\epsilon}{h^2} (n\pi/b) E_o \sin(m\pi x/a) \cos(n\pi y/b) \cos(p\pi z/c)$$

as required.

$$\begin{aligned} H_{xs} &= -\frac{j\omega\epsilon}{h^2} \frac{\partial E_{zs}}{\partial x} - \frac{\gamma}{h^2} \frac{\partial H_{zs}}{\partial y} \\ &= -\frac{j\omega\epsilon}{h^2} (m\pi/a) E_o \cos(m\pi x/a) \sin(n\pi y/b) \cos(p\pi z/c) \end{aligned}$$

From Maxwell's equation,

$$j\omega\epsilon\mathbf{E}_s = \nabla \times \mathbf{H}_s = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_{xs} & H_{ys} & 0 \end{vmatrix}$$

$$E_{ys} = \frac{1}{j\omega\epsilon} \frac{\partial H_{xs}}{\partial z} = \frac{1}{h^2} (n\pi/b)(p\pi/c) E_o \sin(m\pi x/a) \cos(n\pi y/b) \cos(p\pi z/c)$$

Prob. 12.42

$$f_r = \frac{u'}{2} \sqrt{(m/a)^2 + (n/b)^2 + (p/c)^2}$$

where for TM mode to z, $m = 1, 2, 3, \dots$, $n = 1, 2, 3, \dots$, $p = 0, 1, 2, \dots$

and for TE mode to z, $m = 0, 1, 2, 3, \dots$, $n = 0, 1, 2, 3, \dots$, $p = 1, 2, 3, \dots$, $(m+n) \neq 0$.

(a) If $a < b < c$, $1/a > 1/b > 1/c$,

The lowest TM mode is TM_{110} with $f_r = \frac{u'}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$

The lowest TE mode is TE_{011} with $f_r = \frac{u'}{2} \sqrt{\frac{1}{b^2} + \frac{1}{c^2}} < \frac{u'}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$

Hence the dominant mode is TE_{011} .

(b) If $a > b > c$, $1/a < 1/b < 1/c$,

The lowest TM mode is TM_{110} with $f_r = \frac{u'}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$

The lowest TE mode is TE_{101} with $f_r = \frac{u'}{2} \sqrt{\frac{1}{a^2} + \frac{1}{c^2}} > \frac{u'}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$

Hence the dominant mode is TM_{110} .

(c) If $a = c > b$, $1/a = 1/c < 1/b$,

The lowest TM mode is TM_{110} with $f_r = \frac{u'}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$

The lowest TE mode is TE_{101} with $f_r = \frac{u'}{2} \sqrt{\frac{1}{a^2} + \frac{1}{c^2}} < \frac{u'}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$

Hence the dominant mode is TE_{101} .

Prob. 12.43

$$(a) \quad u' = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{4.6}}$$

$$f_c = \frac{u'}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}$$

For the dominant mode, $m = 1, n=0, p=1$

$$f_c = \frac{u'}{2} \sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{c}\right)^2} = \frac{3 \times 10^8}{2\sqrt{4.6}} \sqrt{\frac{1}{9 \times 10^{-4}} + \frac{1}{36 \times 10^{-4}}} = \frac{3 \times 10^{10}}{2(2.1447)} (0.37267) = \underline{\underline{2.606 \text{ GHz}}}$$

$$(b) \quad Q = \frac{(a^2 + c^2)abc}{\delta [2b(a^3 + c^3) + ac(a^2 + c^2)]}$$

$$\delta = \frac{1}{\sqrt{\pi f_{r101} \sigma \mu_o}} = \frac{1}{\sqrt{\pi \times 2.606 \times 10^9 \times 1.57 \times 10^7 \times 4\pi \times 10^{-7}}} = 2.49 \times 10^{-6} \text{ m}$$

$$Q = \frac{(9+36)(72) \times 10^{-2}}{\delta [8(27+216) + 18(9+36)]} = \frac{32.42}{2.49 \times 10^{-6} (2754)} = \underline{\underline{4727.7}}$$

Prob. 12.44

(a)

$$f_r = \frac{u'}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}$$

$$f_{rTE101} = 1.5 \times 10^{10} \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{1}{4}\right)^2} = 1.5 \times 10^{10} \sqrt{0.1736} = \underline{\underline{6.25 \text{ GHz}}}$$

$$f_{rTE011} = 1.5 \times 10^{10} \sqrt{\frac{1}{6.25} + \frac{1}{16}} = 1.5 \times 10^{10} \sqrt{0.2225} = \underline{\underline{7.075 \text{ GHz}}}$$

$$f_{rTE110} = 1.5 \times 10^{10} \sqrt{\frac{1}{9} + \frac{1}{6.25}} = 1.5 \times 10^{10} \sqrt{0.2711} = \underline{\underline{7.81 \text{ GHz}}}$$

Prob. 12.45

$$u' = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{2.5}} = \frac{3 \times 10^8}{\sqrt{2.5}} = 1.897 \times 10^8$$

$$f_c = \frac{u'}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2} = \frac{1.897 \times 10^8 \times 10^2}{2} \sqrt{\left(\frac{m}{1}\right)^2 + \left(\frac{n}{2}\right)^2 + \left(\frac{p}{3}\right)^2}$$

$$= 9.485 \sqrt{m^2 + 0.25n^2 + 0.111p^2} \text{ GHz}$$

$$f_{r101} = 9.485 \sqrt{1 + 0 + 0.111} = 10 \text{ GHz}$$

$$f_{r011} = 9.485 \sqrt{0 + 0.25 + 0.111} = 5.701 \text{ GHz}$$

$$f_{r012} = 9.485 \sqrt{0 + 0.25 + 0.444} = 7.906 \text{ GHz}$$

$$f_{r013} = 9.485 \sqrt{0 + 0.25 + 0.999} = 10.61 \text{ GHz}$$

$$f_{r021} = 9.485 \sqrt{0 + 1 + 0.111} = 10 \text{ GHz}$$

Thus, the first five resonant frequencies are:

$$5.701 \text{ GHz (TE}_{011}\text{)}$$

$$7.906 \text{ GHz (TE}_{012}\text{)}$$

$$10 \text{ GHz (TE}_{101}\text{ and TE}_{021}\text{)}$$

$$10.61 \text{ GHz (TE}_{013}\text{ or TM}_{110}\text{)}$$

$$\underline{\underline{11.07 \text{ GHz (TE}_{111}\text{ or TM}_{111}\text{)}}}$$

Prob. 12.46

$$Q = \frac{(a^2 + c^2)abc}{\delta [2b(a^3 + c^3) + ac(a^2 + c^2)]}$$

When $a = b = c$,

$$Q = \frac{2a^2a^3}{\delta [2a \times 2a^3 + a^2 \times 2a^2]} = \frac{2a^5}{6\delta a^4} = \underline{\underline{\frac{a}{3\delta}}}$$

Prob. 12.47

(a) Since $a > b < c$, the dominant mode is TE_{101}

$$f_{r101} = \frac{u'}{2} \sqrt{\frac{1}{a^2} + 0 + \frac{1}{c^2}} = \frac{3 \times 10^8 \times 10^2}{2} \sqrt{\frac{1}{2^2} + \frac{1}{1^2}} = \underline{\underline{16.77 \text{ GHz}}}$$

$$\begin{aligned}
 \text{(b)} \quad Q_{TE_{101}} &= \frac{(a^2 + c^2)abc}{\delta [2b(a^3 + c^3) + ac(a^2 + c^2)]} \\
 &= \frac{(400 + 100)20 \times 8 \times 10 \times 10^{-3}}{\delta [16(8000 + 1000) + 200(400 + 100)]} = \frac{3.279 \times 10^{-3}}{\delta}
 \end{aligned}$$

$$\text{But } \delta = \frac{1}{\sqrt{\pi f_{r101} \mu_o \sigma}} = \frac{1}{\sqrt{\pi 16.77 \times 10^9 \times 4\pi \times 10^{-7} \times 6.1 \times 10^7}} = \frac{10^{-4}}{200.961} \text{ m}$$

$$Q_{TE_{101}} = 3.279 \times 10^{-3} \frac{200.961}{10^{-4}} = \underline{\underline{6589.51}}$$

Prob. 12.48

$$f_r = \frac{c}{2a} \sqrt{m^2 + n^2 + p^2}$$

The lowest possible modes are TE_{101} , TE_{011} , and TM_{110} . Hence

$$f_r = \frac{c}{2a} \sqrt{2} \longrightarrow a = \frac{c}{f_r \sqrt{2}} = \frac{3 \times 10^8}{\sqrt{2} \times 3 \times 10^9} = 7.071 \text{ cm}$$

$$\underline{\underline{a = b = c = 7.071 \text{ cm}}}$$

Prob. 12.49

$$\text{(a)} \quad a = b = c$$

$$f_r = \frac{u'}{2a} \sqrt{m^2 + n^2 + p^2}$$

For the dominant mode TE_{101} ,

$$f_r = \frac{u'}{2a} \sqrt{1+1} = \frac{c}{2a} \sqrt{2}$$

$$a = \frac{c\sqrt{2}}{2f_r} = \frac{3 \times 10^8 \sqrt{2}}{2 \times 5.6 \times 10^9} = 0.03788 \text{ m}$$

$$\underline{\underline{a = b = c = 3.788 \text{ cm}}}$$

(b)

$$\text{For } \epsilon_r = 2.05, \quad u' = \frac{c}{\sqrt{\epsilon_r}}$$

$$a = \frac{c\sqrt{2}}{2f_r\sqrt{\epsilon_r}} = \frac{0.03788}{\sqrt{2.05}} = 0.02646$$

$$\underline{\underline{a = b = c = 2.646 \text{ cm}}}$$

Prob. 12.50

(a)

This is a TM mode to z. From Maxwell's equations,

$$\nabla \times \mathbf{E}_s = -j\omega\mu\mathbf{H}_s$$

$$\mathbf{H}_s = -\frac{1}{j\omega\mu} \nabla \times \mathbf{E}_s = \frac{j}{\omega\mu} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & E_{zs}(x, y) \end{vmatrix} = \frac{j}{\omega\mu} \left(\frac{\partial E_{zs}}{\partial y} \mathbf{a}_x - \frac{\partial E_{zs}}{\partial x} \mathbf{a}_y \right)$$

But

$$E_{zs} = 200 \sin 30\pi x \sin 30\pi y, \quad \frac{1}{\omega\mu} = \frac{1}{6 \times 10^9 \times 4\pi \times 10^{-7}} = \frac{10^{-2}}{24\pi}$$

$$\mathbf{H}_s = \frac{j10^{-2}}{24\pi} \times 200 \times 30\pi \left\{ \sin 30\pi x \cos 30\pi y \mathbf{a}_x - \cos 30\pi x \sin 30\pi y \mathbf{a}_y \right\}$$

$$\mathbf{H} = \text{Re} (\mathbf{H}_s e^{j\omega t})$$

$$\mathbf{H} = 2.5 \left\{ -\sin 30\pi x \cos 30\pi y \mathbf{a}_x + \cos 30\pi x \sin 30\pi y \mathbf{a}_y \right\} \sin 6 \times 10^9 \pi t \quad \text{A/m}$$

(b)

$$\mathbf{E} = E_z \mathbf{a}_z, \quad \mathbf{H} = H_x \mathbf{a}_x + H_y \mathbf{a}_y$$

$$\mathbf{E} \cdot \mathbf{H} = 0$$

Prob. 12.51

$$(a) \quad a = b = c \quad \longrightarrow \quad f_{r101} = \frac{3 \times 10^8}{a\sqrt{2}} = 12 \times 10^9$$

$$a = \frac{3 \times 10^8}{\sqrt{2} \times 12 \times 10^9} = \underline{\underline{1.77 \text{ cm}}}$$

$$(b) \quad Q_{TE101} = \frac{a}{3\delta} = \frac{a\sqrt{\pi f_{r101}\mu\sigma}}{3}$$

$$= \frac{1.77 \times 10^{-2} \sqrt{\pi \times 12 \times 10^9 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7}}{3} = \underline{\underline{9767.61}}$$

Prob. 12.52

$$f_r = \frac{u'}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}$$

$$f_{r101} = \frac{3 \times 10^8}{2} \sqrt{\frac{1}{(10.2)^2} + \frac{1}{(3.6)^2}} = 44.186 \text{ MHz}$$

$$f_{r011} = 150 \sqrt{\frac{1}{(8.7)^2} + \frac{1}{(3.6)^2}} \text{ MHz} = 45.093 \text{ MHz}$$

$$f_{r111} = 150 \sqrt{\frac{1}{(10.2)^2} + \frac{1}{(8.7)^2} + \frac{1}{(3.6)^2}} \text{ MHz} = 47.43 \text{ MHz}$$

$$f_{r110} = 150 \sqrt{\frac{1}{(10.2)^2} + \frac{1}{(8.7)^2}} \text{ MHz} = 22.66 \text{ MHz}$$

$$f_{r102} = 150 \sqrt{\frac{1}{(10.2)^2} + \frac{4}{(3.6)^2}} \text{ MHz} = 84.62 \text{ MHz}$$

$$f_{r201} = 150 \sqrt{\frac{4}{(10.2)^2} + \frac{1}{(3.6)^2}} \text{ MHz} = 51 \text{ MHz}$$

Thus, the resonant frequencies below 50 MHz are

$$\underline{\underline{f_{r110}, f_{r101}, f_{r011}, \text{ and } f_{r111}}}$$

Prob. 12.53

$$n = c/u_m = \frac{3 \times 10^8}{2.1 \times 10^8} = \underline{\underline{1.4286}}$$

Prob. 12.54

$$NA = \sqrt{n_1^2 - n_2^2} = \sqrt{1.51^2 - 1.45^2} = \sqrt{0.1776} = 0.421$$

Prob. 12.55

$$(a) NA = \sqrt{n_1^2 - n_2^2} = \sqrt{1.62^2 - 1.604^2} = \underline{\underline{0.2271}}$$

$$(b) NA = \sin \theta_a = 0.2271 \text{ or } \theta_a = \sin^{-1} 0.2271 = \underline{\underline{13.13^\circ}}$$

$$(c) V = \frac{\pi d}{\lambda} NA = \frac{\pi \times 50 \times 10^{-6} \times 0.2271}{1300 \times 10^{-9}} = 27.441$$

$$N = V^2/2 \quad \underline{\underline{6 \text{ modes}}}$$

Prob. 12.56

$$V = \frac{\pi d}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{\pi \times 2 \times 5 \times 10^{-6}}{1300 \times 10^{-9}} \sqrt{1.48^2 - 1.46^2} = 5.86$$

$$N = \frac{V^2}{2} = 17.17 \text{ or } \underline{\underline{17 \text{ modes}}}$$

Prob. 12.57

$$(a) \text{ NA} = \sin \theta_a = \sqrt{n_1^2 - n_2^2} = \sqrt{1.53^2 - 1.45^2} = 0.4883$$

$$\theta_a = \sin^{-1} 0.4883 = \underline{\underline{29.23^\circ}}$$

$$(b) P(l)/P(0) = 10^{-\alpha l / 10} = 10^{-0.4 \times 5 / 10} = 0.631$$

$$\text{i.e. } \underline{\underline{63.1 \%}}$$

Prob. 12.58

$$P(\ell) = P(0) 10^{-\alpha \ell / 10} = 10 \times 10^{-0.5 \times 0.85 / 10} = \underline{\underline{9.0678 \text{ mW}}}$$

Prob. 12.59

As shown in Eq. (10.35), $\log_{10} P_1/P_2 = 0.434 \ln P_1/P_2$,

$$1 \text{ Np} = 20 \log_{10} e = 8.686 \text{ dB} \text{ or } 1 \text{ Np/km} = 8.686 \text{ dB/km},$$

or $1 \text{ Np/m} = 8686 \text{ dB/km}$. Thus,

$$\alpha_{12} = \underline{\underline{8686 \alpha_{10}}}$$

Prob. 12.60

$$\alpha \ell = 10 \log_{10} \frac{P_{in}}{P_{out}} = 10 \log_{10} \frac{1.2 \times 10^{-3}}{1 \times 10^{-6}} = 30.792$$

$$\alpha = 0.4 \text{ dB/km} = \frac{0.4}{8.686} \text{ Np/km}$$

$$\ell = \frac{30.792}{\alpha} = \frac{30.392 \text{ dB}}{0.4 \text{ dB/km}} = \underline{\underline{76.98 \text{ km}}}$$

Prob. 12.61

$$P(0) = P(l) 10^{\alpha l / 10} = 0.2 \times 10^{0.4 \times 30 / 10} \text{ mW} = \underline{\underline{3.1698 \text{ mW}}}$$

Prob. 12.62 See text.

CHAPTER 13

P. E. 13.1

(a) For this case, r is at near field.

$$H_{\phi s} = \frac{I_o dl \sin \theta}{4\pi} \left(\frac{j\beta}{r} + \frac{1}{r^2} \right) e^{-j\beta r}, \quad \beta r = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{5} = 72^\circ$$

$$\lambda = \frac{2\pi c}{\omega} = \frac{2\pi \times 3 \times 10^8}{10^8} = 6\pi, \quad \beta = \frac{2\pi}{\lambda} = \frac{1}{3}$$

$$H_{\phi s} = \frac{(0.25) \frac{6\pi}{100} \sin 30^\circ}{4\pi} \left(\frac{j1/3}{6\pi/5} + \frac{1}{(6\pi/5)^2} \right) e^{-j72^\circ} = 0.2119 \angle -20.511^\circ \text{ mA/m}$$

$$\mathbf{H} = \text{Im} (H_{\phi s} e^{j\omega t} \mathbf{a}_\phi) \quad \text{Im is used since } I = I_o \sin \omega t$$

$$= \underline{\underline{0.2119 \sin(10^8 - 20.5^\circ) \mathbf{a}_\phi \text{ mA/m}}}$$

(b) For this case, r is at far field. $\beta = \frac{2\pi}{\lambda} \times 200\lambda = 0^\circ$

$$H_{\phi s} = \frac{j(0.25) \left(\frac{2\pi}{\lambda} \right) \frac{\lambda}{100} \sin 60^\circ e^{-j0^\circ}}{4\pi(6\pi \times 200)} = 0.2871 e^{j90^\circ} \mu\text{A/m}$$

$$\mathbf{H} = \text{Im} (H_{\phi s} \mathbf{a}_\phi e^{j\omega t}) = \underline{\underline{0.2871 \sin(10^8 + 90^\circ) \mathbf{a}_\phi \mu\text{A/m}}}$$

P. E. 13.2

(a) $l = \frac{\lambda}{4} = \underline{\underline{1.5\text{m}}},$

(b) $I_o = \underline{\underline{83.3\text{mA}}}$

(c) $R_{\text{rad}} = 36.56 \Omega, P_{\text{rad}} = \frac{1}{2} (0.0833)^2 36.56$
 $= \underline{\underline{126.8 \text{ mW}}}.$

(d) $Z_L = 36.5 + j21.25,$

$$\Gamma = \frac{36.5 + j21.25 - 75}{36.5 + j21.25 + 75} = 0.3874 \angle 140.3^\circ$$

$$s = \frac{1 + 0.3874}{1 - 0.3874} = \underline{\underline{2.265}}$$

P. E. 13.3

$$D = \frac{4\pi U_{\max}}{P_{\text{rad}}}$$

(a) For the Hertzian monopole

$$U(\theta, \phi) = \sin^2 \theta, \quad 0 < \theta < \pi/2, \quad 0 < \phi < 2\pi, \quad U_{\max} = 1$$

$$P_{\text{rad}} = \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} \sin^2 \theta \sin \theta d\theta d\phi = \frac{4\pi}{3}$$

$$D = \frac{4\pi(1)}{4\pi/3} = \underline{\underline{3}}$$

(b) For the $\frac{\lambda}{4}$ monopole,

$$U(\theta, \phi) = \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}, \quad U_{\max} = 1$$

$$P_{\text{rad}} = \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta} \sin \theta d\theta d\phi = 2\pi(0.609)$$

$$D = \frac{4\pi(1)}{2\pi(0.609)} = \underline{\underline{3.28}}$$

P. E. 13.4

(a) $P_{\text{rad}} = \eta_r P_{\text{in}} = 0.95(0.4)$

$$D = \frac{4\pi U_{\max}}{P_{\text{rad}}} = \frac{4\pi(0.5)}{0.4 \times 0.95} = \underline{\underline{16.53}}$$

(b) $D = \frac{4\pi(0.5)}{0.3} = \underline{\underline{20.94}}$

P. E. 13. 5

$$P_{rad} = \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} \sin\theta \sin\theta d\theta d\phi = \frac{\pi^2}{2}, U_{\max} = 1$$

$$D = \frac{4\pi(1)}{\pi^2/2} = \underline{\underline{2.546}}$$

P. E. 13. 6

$$(a) f(\theta) = |\cos\theta| \cos\left[\frac{1}{2}(\beta d \cos\theta + \alpha)\right]$$

$$\text{where } \alpha = \pi, \beta d = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} = \pi$$

$$f(\theta) = |\cos\theta| \cos\left[\frac{1}{2}(\pi \cos\theta + \pi)\right]$$



unit pattern

group pattern

For the group pattern, we have nulls at

$$\frac{\pi}{2}(\cos\theta + 1) = \pm \frac{\pi}{2} \longrightarrow \theta = \pm \frac{\pi}{2}$$

and maxima at

$$\frac{\pi}{2}(\cos\theta + 1) = 0, \pi \longrightarrow \cos\theta = -1, 1 \longrightarrow \theta = 0, \pi$$

Thus the group pattern and the resultant patterns are as shown in Fig.13.15(a)

$$(b) f(\theta) = |\cos\theta| \cos\left[\frac{1}{2}(\beta d \cos\theta + \alpha)\right]$$

$$\text{where } \alpha = -\pi/2, \beta d = \pi/2$$

$$f(\theta) = |\cos\theta| \cos\left[\frac{1}{2}\left(\frac{\pi}{2}\cos\theta - \pi/2\right)\right]$$



unit pattern

group pattern

For the group pattern, the nulls are at

$$\frac{\pi}{4}(\cos\theta - 1) = -\frac{\pi}{2} \longrightarrow \theta = 180^\circ$$

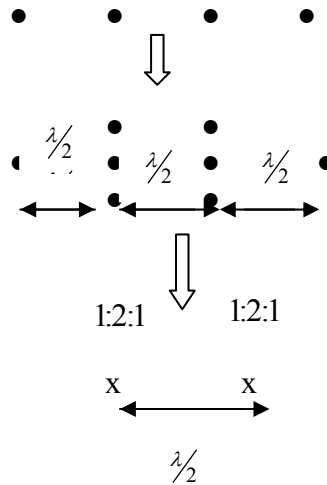
and maxima at

$$\cos\theta - 1 = 0 \quad \longrightarrow \quad \theta = 0$$

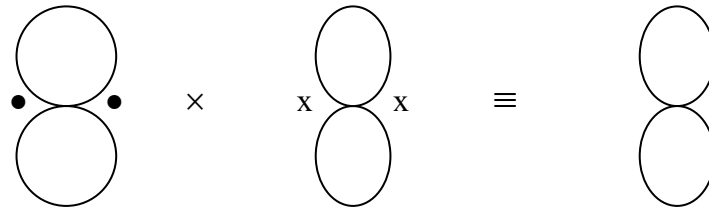
Thus the group pattern and the resultant patterns are as shown in Fig.13.15(b)

P. E. 13.7

(a)



Thus, we take a pair at a time and multiply the patterns as shown below.



(b) The group pattern is the normalized array factor, i.e.

$$(AF)_n = \frac{1}{2^{N-1}} \left| 1 + Ne^{i\psi} + \frac{N(N-1)}{2!} e^{i2\psi} + \frac{N(N-1)(N-2)}{3!} e^{i3\psi} + \dots + e^{i(N-1)\psi} \right|$$

$$\text{where } \sum = \sum_{i=1}^{N-1} \binom{N}{i} = 1 + N + \frac{N-1}{2!} + \frac{N(N-1)(N-2)}{3!} + \dots$$

$$= (1+1)^{N-1} = 2^{N-1}$$

$$(AF)_n = \frac{1}{2^{N-1}} |1 + e^{j\psi}|^{N-1} = \frac{1}{2^{N-1}} \left| e^{j\psi/2} \left(e^{-j\psi/2} + e^{j\psi/2} \right) \right|^{N-1}$$

$$= \frac{1}{2^{N-1}} \left| 2 \cos \frac{\psi}{2} \right|^{N-1} = \underline{\underline{\left| \cos \frac{\psi}{2} \right|^{N-1}}}$$

P. E. 13.8

$$A_e = \frac{\lambda^2}{4\pi} G_d, \quad \lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3 \text{ m}$$

For the Hertzian dipole,

$$G_d = 1.5 \sin^2 \theta$$

$$A_e = \frac{\lambda^2}{4\pi} (1.5 \sin^2 \theta)$$

$$A_{e,\max} = \frac{1.5\lambda^2}{4\pi} = \frac{1.5 \times 9}{4\pi} = \underline{\underline{1.074 \text{ m}^2}}$$

By definition,

$$P_r = A_e P_{ave} \longrightarrow P_{ave} = \frac{P_r}{A_e} = \frac{3 \times 10^{-6}}{1.074} = \underline{\underline{2793 \text{ } \mu\text{W} / \text{m}^2}}$$

P. E. 13.9

$$\begin{aligned} \text{(a)} \quad G_d &= \frac{4\pi r^2 P_{ave}}{P_{rad}} = \frac{4\pi r^2 \frac{1}{2} \frac{E^2}{\eta}}{P_{rad}} = \frac{2\pi r^2 E^2}{\eta P_{rad}} \\ &= \frac{2\pi \times 400 \times 10^6 \times 144 \times 10^{-6}}{120\pi \times 100 \times 10^3} = 0.0096 \end{aligned}$$

$$G = 10 \log_{10} G_d = \underline{\underline{-20.18 \text{ dB}}}$$

$$\text{(b)} \quad G = \eta_r G_d = 0.98 \times 0.0096 = \underline{\underline{9.408 \times 10^{-3}}}$$

P. E. 13.10

$$r = \left[\frac{\lambda^2 G_d^2 \sigma}{(4\pi)^3} \frac{P_{rad}}{P_r} \right]^{1/4}$$

$$\text{where } \lambda = \frac{c}{f} = \frac{3 \times 10^8}{6 \times 10^9} = 0.05 \text{ m}$$

$$A_e = 0.7\pi a^2 = 0.7\pi (1.8)^2 = 7.125 \text{ m}^2$$

$$G_d = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi (7.125)}{25 \times 10^{-4}} = 3.581 \times 10^4$$

$$r = \left[\frac{25 \times 10^{-4} \times (3.581)^2 \times 10^8 \times 5 \times 60 \times 10^3}{(4\pi)^3 \times 0.26 \times 10^{-3}} \right]^{1/4}$$

$$= 1168.4 \text{ m} = \underline{\underline{0.631 \text{ nm}}}$$

$$\text{At } r = \frac{r_{\max}}{2} = 584.2 \text{ m},$$

$$P = \frac{G_d P_{\text{rad}}}{4\pi r^2} = \frac{3.581 \times 10^4 \times 60 \times 10^3}{4\pi (584.2)^2} = \underline{\underline{501 \text{ W/m}^2}}$$

Prob. 13.1

Using vector transformation,

$$A_{rs} = A_{xs} \sin \theta \cos \phi, \quad A_{\theta s} = A_{xs} \cos \theta \cos \phi, \quad A_{\phi s} = -A_{xs} \sin \phi$$

$$\mathbf{A}_s = \frac{50e^{-j\beta r}}{r} (\sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi)$$

$$\begin{aligned} \frac{\nabla \times \mathbf{A}_s}{\mu} = \mathbf{H}_s &= \frac{100 \cos \theta \sin \phi}{\mu r^2 \sin \theta} e^{-j\beta r} \mathbf{a}_r - \frac{50}{\mu r^2} (1 - j\beta r) \sin \phi e^{-j\beta r} \mathbf{a}_\theta \\ &- \frac{50}{\mu r^2} \cos \theta \cos \phi (1 + j\beta r) e^{-j\beta r} \mathbf{a}_\phi \end{aligned}$$

At far field, only $\frac{1}{r}$ term remains. Hence

$$\mathbf{H}_s = \frac{j50}{\mu r} \beta e^{-j\beta r} (\sin \phi \mathbf{a}_\theta - \cos \theta \cos \phi \mathbf{a}_\phi)$$

$$\mathbf{E}_s = -\eta \mathbf{a}_r \times \mathbf{H}_s = \frac{-j50\beta\eta e^{-j\beta r}}{\mu r} (\sin \phi \mathbf{a}_\phi + \cos \theta \cos \phi \mathbf{a}_\theta)$$

$$\mathbf{E} = \text{Re}[\mathbf{E}_s e^{j\omega t}] = \underline{\underline{\frac{50\eta\beta}{\mu r} \sin(\omega t - \beta r) (\sin \phi \mathbf{a}_\phi + \cos \theta \cos \phi \mathbf{a}_\theta) \text{ V/m}}}$$

$$\mathbf{H} = \text{Re}[\mathbf{H}_s e^{j\omega t}] = \underline{\underline{\frac{-50}{\mu r} \beta \sin(\omega t - \beta r) (\sin \phi \mathbf{a}_\theta - \cos \theta \cos \phi \mathbf{a}_\phi) \text{ A/m}}}$$

Prob. 13.2

$$(a) \quad \lambda = \frac{c}{f} = \frac{3 \times 10^8}{400 \times 10^6} = 0.75 \text{ m}$$

$$r_{\min} = \frac{2d^2}{\lambda} = \frac{2(0.02\lambda)^2}{\lambda} \quad r$$

i.e. r is in the far field.

$$H_{\phi s} = \frac{jI_o \beta dl}{4\pi r} \sin \theta e^{-j\beta r}$$

$$|H_{\phi s}| = \frac{I_o \beta dl}{4\pi r} \sin \theta = \frac{3 \times \frac{2\pi}{\lambda} \times 0.02\lambda \times \sin 90^\circ}{4\pi(60)} = 5 \times 10^{-4} = 0.5 \text{ mA/m}$$

$$|E_{\theta s}| = \eta_o |H_{\phi s}| = \underline{\underline{0.1885 \text{ V/m}}}$$

$$(b) \quad |H_{\phi s}| = 0.5 \text{ mA/m}$$

$$(c) \quad R_{\text{rad}} = 80\pi^2 \left(\frac{dl}{\lambda} \right)^2 = 80\pi^2 (0.02)^2 = \underline{\underline{0.3158 \Omega}}$$

$$(d) \quad P_{\text{rad}} = \frac{1}{2} |I_o|^2 R_{\text{rad}} = \frac{1}{2} (9)(0.3158) = \underline{\underline{1.421 \text{ W}}}$$

Prob. 13.3

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{50 \times 10^6} = 6 \text{ m}$$

$$H_{\phi s} = \frac{jI_o \beta dl}{4\pi r} \sin \theta e^{-j\beta r}, \quad E_{\theta s} = \eta H_{\phi s}$$

$$dl = 10 \text{ cm} \ll \lambda$$

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{6} = \pi / 3$$

$$\frac{I_o \beta dl}{4\pi} = \frac{20 \times \frac{\pi}{3} \times 0.1}{4\pi} = 0.1667$$

$$\underline{\underline{H_s = \frac{j0.1667}{r} \sin \theta e^{-j\pi r/3} \mathbf{a}_\phi \text{ A/m}}}$$

$$\underline{\underline{E_s = \frac{j0.1667 \times 377}{r} \sin \theta e^{-j\pi r/3} \mathbf{a}_\theta = \frac{j62.83}{r} \sin \theta e^{-j\pi r/3} \mathbf{a}_\theta \text{ V/m}}}$$

Prob. 13.4

$$\lambda = c / f, \quad R_{\text{rad}} = 80\pi^2 \left(\frac{dl}{\lambda} \right)^2$$

$$P_{\text{rad}} = \frac{1}{2} I_o^2 R_{\text{rad}} \quad \rightarrow \quad I_o^2 = \frac{2P_{\text{rad}}}{R_{\text{rad}}}$$

$$I_o^2 = 2P_{\text{rad}} \left(\frac{\lambda}{dl} \right)^2 \frac{1}{80\pi^2} = 2P_{\text{rad}} \left(\frac{c}{fdl} \right)^2 \frac{1}{80\pi^2} = 2 \times 12 \left(\frac{3 \times 10^8}{140 \times 10^6 \times 2 \times 10^{-2}} \right)^2 \frac{1}{80\pi^2} = 348.93$$

$$\underline{\underline{I_o = 18.68 \text{ A}}}$$

Prob. 13.5

$$(a) \quad A_{zs} = \frac{e^{-j\beta r}}{4\pi r} \int_{-l/2}^{l/2} I_o \left(1 - \frac{2|z|}{l} \right) e^{j\beta z \cos \theta} dz$$

$$= \frac{e^{-j\beta r}}{4\pi r} I_o \left[\int_{-l/2}^{l/2} \left(1 - \frac{2|z|}{l} \right) \cos(\beta z \cos \theta) dz + j \int_{-l/2}^{l/2} \left(1 - \frac{2|z|}{l} \right) \sin(\beta z \cos \theta) dz \right]$$

$$= \frac{e^{-j\beta r}}{4\pi r} 2I_o \int_0^{l/2} \left(1 - \frac{2z}{l} \right) \cos(\beta z \cos \theta) dz$$

$$= \frac{I_o e^{-j\beta r}}{2\pi r \beta^2 \cos^2 \theta} \cdot \frac{2}{l} \left[1 - \cos\left(\frac{\beta l}{2} \cos \theta\right) \right]$$

$$\underline{\underline{E_s = -j\omega\mu A_s \longrightarrow E_{\theta s} = j\omega\mu \sin \theta A_{zs} = j\beta\eta \sin \theta A_{zs}}}$$

$$E_{\theta s} = \frac{j\eta I_o e^{-j\beta r} \sin \theta \left[1 - \cos\left(\frac{\beta l}{2} \cos \theta\right) \right]}{\pi r l \beta \cos^2 \theta}$$

$$\text{If } \frac{\beta l}{2} \ll 1, \quad \cos\left(\frac{\beta l}{2} \cos \theta\right) = 1 - \frac{\left(\frac{\beta l}{2} \cos \theta\right)^2}{2!}.$$

Hence

$$E_{\theta s} = \frac{j\eta I_o}{8\pi r} \beta l e^{-j\beta r} \sin \theta, \quad H_{\phi s} = E_{\theta s} / \eta$$

$$P_{\text{ave}} = \frac{|E_{\theta s}|^2}{2\eta}, \quad P_{\text{rad}} = \int P_{\text{ave}} dS$$

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^\pi \frac{n}{2} \left(\frac{I_o \beta l}{8\pi} \right)^2 \frac{1}{r^2} \sin^2 \theta r^2 \sin \theta d\theta d\phi$$

$$= 10\pi^2 I_o^2 \left(\frac{l}{\lambda}\right)^2 = \frac{1}{2} I_o^2 R_{rad}$$

$$\text{or } R_{rad} = 20\pi^2 \left(\frac{l}{\lambda}\right)^2$$

$$(b) \quad 0.5 = 20\pi^2 \left(\frac{l}{\lambda}\right)^2 \longrightarrow \underline{\underline{l = 0.05\lambda}}$$

Prob. 13.6

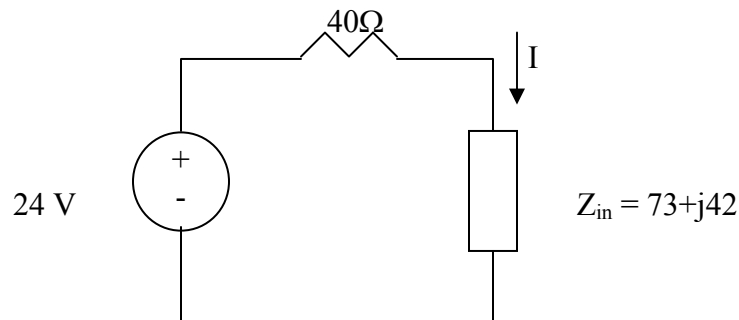
$$R_{rad} = 80\pi^2 \left(\frac{dl}{\lambda}\right)^2$$

$$\lambda = c / f = \frac{3 \times 10^8}{1.2 \times 10^6} = 250m$$

$$\left(\frac{dl}{\lambda}\right)^2 = \frac{R_{rad}}{80\pi^2} = \frac{0.5}{80\pi^2} = 6.33 \times 10^{-4}$$

$$\frac{dl}{\lambda} = 2.516 \times 10^{-2} \quad \rightarrow \quad dl = 2.516 \times 10^{-2} \times 250 = \underline{\underline{6.29 \text{ m}}}$$

Prob. 13.7



$$I = \frac{V}{R_s + Z_{in}} = \frac{24}{40 + 73 + j42} = 0.1866 - j0.0694$$

$$P_{rad} = \frac{1}{2} |I|^2 R_{rad}, \quad R_{rad} = 73$$

$$P_{rad} = \frac{1}{2} (0.1991)^2 \times 73 = \underline{\underline{1.447 \text{ W}}}$$

Prob. 13.8

Let us model this as a short Hertzian dipole.

$$R_{rad} = 80\pi^2 \left(\frac{dl}{\lambda} \right)^2 = 80\pi^2 (1/8)^2 = 12.34\Omega$$

$$P_{rad} = \frac{1}{2} I_o^2 R_{rad} = 4 \quad \longrightarrow \quad I_o = \underline{\underline{0.8052 \text{ A}}}$$

Prob. 13.9

Change the limits in Eq. (13.16) to $\pm l/2$ i.e.

$$\begin{aligned} A_s &= \frac{\mu I_o e^{j\beta z \cos\theta}}{4\pi r} \left(\frac{j\beta \cos\theta \cos\beta t + \beta \sin\beta t}{-\beta^2 \cos^2\theta + \beta^2} \right) \Big|_{-l/2}^{l/2} \\ &= \frac{\mu I_o e^{j\beta r}}{2\pi r} \frac{1}{\beta \sin^2\theta} \left[\sin \frac{\beta l}{2} \cos \left(\frac{\beta l}{2} \cos\theta \right) - \cos\theta \cos \frac{\beta l}{2} \sin \left(\frac{\beta l}{2} \cos\theta \right) \right] \end{aligned}$$

But $\mathbf{B} = \mu \mathbf{H} = \nabla \times \mathbf{A}$

$$H_{\phi s} = \frac{1}{\mu r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right],$$

where $A_\theta = -A_z \sin\theta$, $A_r = A_z \cos\theta$

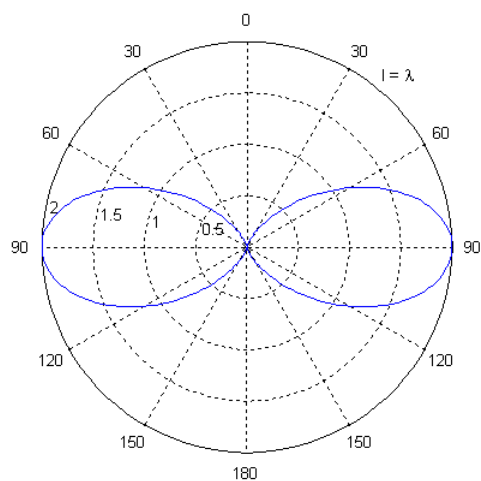
$$H_{\phi s} = \frac{I_o}{2\pi r} \frac{e^{-j\beta r}}{\beta} \left(\frac{j\beta}{\sin\theta} \right) \left[\sin \frac{\beta l}{2} \cos \left(\frac{\beta l}{2} \cos\theta \right) - \cos\theta \cos \frac{\beta l}{2} \sin \left(\frac{\beta l}{2} \cos\theta \right) \right] + \frac{I_o}{2\pi r^2} e^{-j\beta r} (\dots)$$

For far field, only the $\frac{1}{r}$ -term remains. Hence

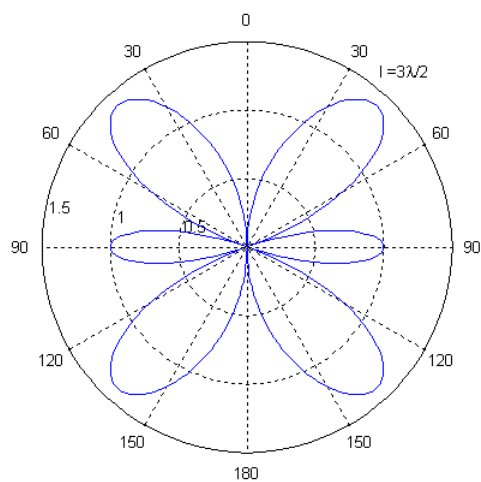
$$H_{\phi s} = \frac{j I_o}{2\pi r} e^{-j\beta r} \frac{\left[\sin \frac{\beta l}{2} \cos \left(\frac{\beta l}{2} \cos\theta \right) - \cos\theta \cos \frac{\beta l}{2} \sin \left(\frac{\beta l}{2} \cos\theta \right) \right]}{\sin\theta}$$

$$(b) \quad f(\theta) = \frac{\cos \left(\frac{\beta l}{2} \cos\theta \right) - \cos \frac{\beta l}{2}}{\sin\theta}$$

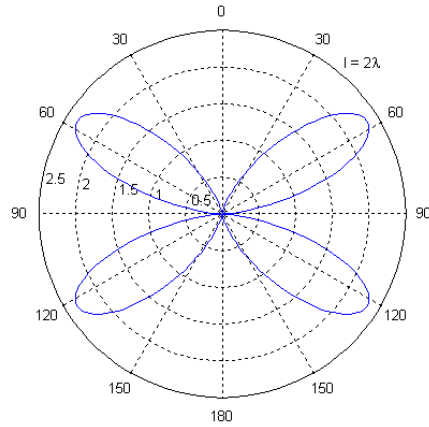
$$\text{For } l = \lambda, f(\theta) = \frac{\cos(\pi \cos\theta) + 1}{\sin\theta}$$



For $l = \frac{3\lambda}{2}$, $f(\theta) = \frac{\cos\left(\frac{3\pi}{2} \cos\theta\right)}{\sin\theta}$



For $l = 2\lambda$, $f(\theta) = \frac{\cos\theta \sin(2\pi \cos\theta)}{\sin\theta}$

**Prob. 13.10**

(a)

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{450 \times 10^6} = 0.6667 \text{ m}$$

$$\ell = \frac{\lambda}{2} = \underline{\underline{0.333 \text{ m}}}$$

(b)

$$\frac{\sigma}{\omega \epsilon} = \frac{4}{2\pi \times 450 \times 10^6 \times 81 \times \frac{10^{-9}}{36\pi}} = 1.975$$

$$\begin{aligned} \beta &= \omega \sqrt{\frac{\mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon} \right)^2} + 1 \right]} = \frac{2\pi \times 460 \times 10^6}{c} \sqrt{\frac{81}{2} \left[\sqrt{1 + (1.975)^2} + 1 \right]} \\ &= \frac{2\pi \times 460 \times 10^6}{3 \times 10^8} \times 11.4086 = 109.91 \end{aligned}$$

$$\lambda = \frac{2\pi}{\beta} = 0.0572$$

$$\ell = \frac{\lambda}{2} = \underline{\underline{28.58 \text{ mm}}}$$

Prob. 13.11

(a)

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1.150 \times 10^6} = 260.8 \text{ m}$$

$$\ell = \frac{\lambda}{4} = \underline{\underline{65.22 \text{ m}}}$$

(b)

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{90 \times 10^6} = 3.333 \text{ m}$$

$$\ell = \frac{\lambda}{4} = \underline{\underline{0.8333 \text{ m}}}$$

(c)

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{80 \times 10^6} = 3.75 \text{ m}$$

$$\ell = \frac{\lambda}{4} = \underline{\underline{0.9375 \text{ m}}}$$

(d)

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{600 \times 10^6} = 0.5 \text{ m}$$

$$\ell = \frac{\lambda}{4} = \underline{\underline{0.125 \text{ m}}}$$

Prob. 13.12

$\ell \ll \lambda$, hence it is a Hertzian monopole.

$$R_{rad} = 80\pi^2 \left(\frac{dl}{\lambda} \right)^2 = 80\pi^2 \left(\frac{1}{8} \right)^2 = \underline{\underline{12.34 \Omega}}$$

Prob. 13.13

$$(a) \quad \lambda = \frac{c}{f} = \frac{3 \times 10^8}{10 \times 10^6} = 30 \text{ m}$$

$$E_{\max} = \frac{\eta \pi I_o S}{r \lambda^2} \longrightarrow I_o = \frac{E_{\max} r \lambda^2}{\eta \pi S}$$

$$I_o = \frac{50 \times 10^{-3} \times 3 \times 30^2}{120 \pi^2 \pi (0.2)^2 100} = \underline{\underline{9.071 \text{ mA}}} \quad (S = N \pi r^2)$$

$$(b) \quad R_{rad} = \frac{320 \pi^4 S^2}{\lambda^4} = \frac{320 \pi^4 \pi^2 (0.2)^4 \times 10^4}{30^4} = 6.077 \Omega$$

$$P_{rad} = \frac{1}{2} I_o^2 R_{rad} = \frac{1}{2} (9.071)^2 \times 10^{-6} \times 6.077$$

$$= \underline{0.25 \text{ mW}}$$

Prob. 13.14

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{80 \times 10^6} = 3.75 \text{ m}$$

$$S = N\pi\rho_o^2$$

$$R_{rad} = \frac{320\pi^4 S^2}{\lambda^4} = \frac{320\pi^4 N^2 \pi^2 \rho_o^4}{\lambda^4} \longrightarrow N^2 = \frac{\lambda^4 R_{rad}}{320\pi^6 (1.2 \times 10^{-2})^4}$$

$$N^2 = \frac{(3.75)^4 \times 8}{320\pi^6 (1.2 \times 10^{-2})^4} = 248006 \longrightarrow N \approx \underline{498}$$

Prob. 13.15

$$(a) \quad R_{rad} = \frac{320\pi^4 S^2}{\lambda^4}$$

$$S = \pi\rho_o^2 = \pi(0.4)^2 = 0.5027 \text{ m}^2$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{6 \times 10^6} = 50 \text{ m}$$

$$R_{rad} = \frac{320\pi^4 (0.5027)^2}{(50)^4} = \underline{1.26 \text{ m}\Omega}$$

$$(b) \quad P_{rad} = \frac{1}{2} I_o^2 R_{rad} = \frac{1}{2} (50)^2 \times 1.26 \times 10^{-3} = \underline{1.575 \text{ W}}$$

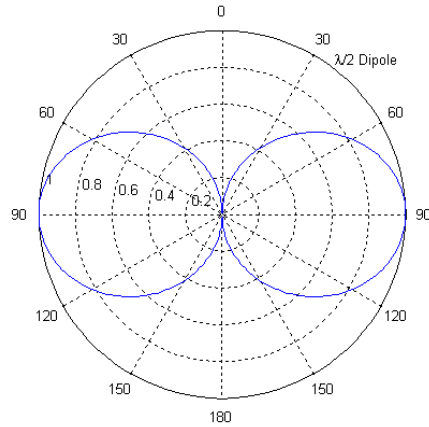
$$(c) \quad R_\ell = \frac{a}{2\delta} R_{dc} = \frac{a}{2\delta} \frac{\ell}{\sigma S} = \frac{a\ell}{2\sigma\pi a^2} \sqrt{\pi f \mu \sigma} = \frac{2\pi R}{2\pi a} \sqrt{\frac{\mu f \pi}{\sigma}}$$

$$R_\ell = \frac{R}{a} \sqrt{\frac{\mu f \pi}{\sigma}} = \frac{0.4}{4 \times 10^{-3}} \sqrt{\frac{4\pi \times 10^{-7} \times 6 \times 10^6 \times \pi}{5.8 \times 10^7}} = 63.91 \text{ m}\Omega$$

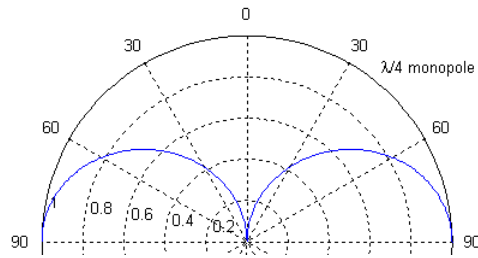
$$\eta = \frac{R_{rad}}{R_{rad} + R_\ell} = \frac{1.26}{1.26 + 63.91} \times 100\% = \underline{1.933\%}$$

Prob. 13.16

$$(a) \quad f(\theta) = \left| \frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \right|$$



(b) The same as for $\frac{\lambda}{2}$ dipole except that the fields are zero for $\theta = \frac{\pi}{2}$ as shown.



Prob. 13.17

Let P_{rad1} and P_{rad2} be the old and new radiated powers respectively.

Let P_{ohm1} and P_{ohm2} be the old and new ohmic powers respectively.

$$\eta_{r1} = 20\% = \frac{P_{rad1}}{P_{rad1} + P_{ohm1}} = \frac{1}{5} \longrightarrow 4P_{rad1} = P_{ohm1} \quad (1)$$

$$\text{But } P_{ohm1} = \frac{1}{2} I^2 R_s \Delta z$$

$$P_{ohm2} = \frac{1}{2} I^2 R_s 2\Delta z = 2P_{ohm1} \quad (2)$$

$$P_{rad1} = \frac{1}{2} I_o^2 R_{rad} = \frac{1}{2} I_o^2 \times 80\pi^2 \left(\frac{\Delta z}{\lambda} \right)^2$$

$$P_{rad2} = \frac{1}{2} I_o^2 R_{rad} = \frac{1}{2} I_o^2 \times 80\pi^2 \left(\frac{2\Delta z}{\lambda} \right)^2 = 4P_{rad1} \quad (3)$$

From (1) to (3),

$$\eta_{r2} = \frac{P_{rad2}}{P_{rad2} + P_{ohm2}} = \frac{4P_{rad1}}{4P_{rad1} + 2P_{ohm1}} = \frac{P_{ohm1}}{3P_{ohm1}} = \underline{\underline{33.3\%}}$$

Prob. 13.18

$$\eta_r = \frac{P_{rad}}{P_{in}} = \frac{R_{rad}}{R_{rad} + R_\ell}$$

$$R_{rad} = 73\Omega, \quad R_\ell = \frac{\ell}{\sigma_c S} = \frac{\ell}{\sigma_c \pi a^2}$$

$$\ell = \frac{\lambda}{2}, \quad \lambda = c / f = \frac{3 \times 10^8}{6 \times 10^6} = 50m, \quad \ell = 25m$$

$$R_\ell = \frac{25}{58 \times 10^6 \times \pi (1.2)^2 \times 10^{-6}} = \frac{25}{262.4} = 0.09528\Omega$$

$$\eta_r = \frac{73}{73 + 0.09528} = 0.9987 = \underline{\underline{99.87\%}}$$

Prob. 13.19

$$(a) \text{ Let } \mathbf{H}_s = \frac{\cos 2\theta}{\eta_o r} e^{-j\beta r} \mathbf{a}_H$$

$$\mathbf{a}_E \times \mathbf{a}_H = \mathbf{a}_k \longrightarrow \mathbf{a}_\theta \times \mathbf{a}_H = \mathbf{a}_r \longrightarrow \mathbf{a}_H = \mathbf{a}_\phi$$

$$\mathbf{H}_s = \frac{\cos 2\theta}{120\pi r} e^{-j\beta r} \mathbf{a}_\phi$$

$$(b) \mathbf{P}_{ave} = \frac{|E_s|^2}{2\eta} \mathbf{a}_r = \frac{\cos^2(2\theta)}{2\eta r^2} \mathbf{a}_r$$

$$P_{rad} = \frac{1}{2\eta} \iint \frac{\cos^2 2\theta}{r^2} r^2 \sin \theta d\theta d\phi = \frac{1}{240\pi} (2\pi) \int_0^\pi \cos^2 2\theta \sin \theta d\theta$$

$$\text{But } \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2\cos^2 \theta - 1$$

$$\begin{aligned} P_{rad} &= -\frac{1}{120} \int_0^\pi (2\cos^2 \theta - 1)^2 d(\cos \theta) \\ &= -\frac{1}{120} \int_0^\pi (4\cos^4 \theta - 4\cos^2 \theta + 1) d(\cos \theta) \\ &= -\frac{1}{120} \left(\frac{4\cos^5 \theta}{5} - \frac{4\cos^3 \theta}{3} + \cos \theta \right) \Big|_0^\pi \\ &= -\frac{1}{120} \left[-\frac{4}{5} + \frac{4}{3} - 1 - \left(-\frac{4}{5} + \frac{4}{3} - 1 \right) \right] = \frac{1}{120} \left(\frac{14}{15} \right) \\ &= \underline{\underline{7.778 \text{ mW}}} \end{aligned}$$

(c)

$$\begin{aligned}
 P_{rad} &= -\frac{1}{120} \int_{60^\circ}^{120^\circ} (2 \cos^2 \theta - 1)^2 d(\cos \theta) \\
 &= -\frac{1}{120} \left(\frac{4 \cos^5 \theta}{5} - \frac{4 \cos^3 \theta}{3} + \cos \theta \right) \bigg|_{60^\circ}^{120^\circ} \\
 &= -\frac{1}{120} \left[\frac{4}{5} \left(-\frac{1}{32} \right) - \frac{4}{3} \left(-\frac{1}{8} \right) - \frac{1}{2} - \frac{4}{5} \left(\frac{1}{32} \right) + \frac{4}{3} \left(\frac{1}{8} \right) - \frac{1}{2} \right] = \frac{1}{60} \left[\frac{1}{40} + \frac{1}{2} - \frac{1}{6} \right] \\
 &= 5.972 \text{ mW}
 \end{aligned}$$

which is $\frac{5.972}{7.778} = 0.7678$ or 76.78%

Prob. 13.20

$$\mathcal{P}_{ave} = \frac{1}{2} \text{Re}(\mathbf{E}_s \times \mathbf{H}_s^*) = \frac{1}{2} \eta |H_{\phi_s}|^2 \mathbf{a}_r$$

$$\begin{aligned}
 P_{rad} &= \int \mathcal{P}_{ave} \cdot d\mathbf{S} = \frac{1}{2} \eta \iint \frac{\beta^2 I_o^2}{16\pi^2 r^2} \sin^2 \theta \cos^2 \phi r^2 \sin \theta d\theta d\phi \\
 &= \frac{\eta \beta^2 I_o^2}{32\pi^2} \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} \cos^2 \phi d\phi = \frac{\eta \beta^2 I_o^2}{32\pi^2} \left(\frac{4}{3} \right) \frac{1}{2} \int_0^{2\pi} (1 + \cos 2\phi) d\phi \\
 &= \frac{\eta \beta^2 I_o^2}{32\pi^2} \left(\frac{4}{3} \right) \pi = \frac{\eta \beta^2 I_o^2}{24\pi}
 \end{aligned}$$

$$R_{rad} = \frac{2P_{rad}}{I_o^2} = \frac{\eta \beta^2}{12\pi^2}$$

Assuming free space, $\eta = 120\pi$,

$$\underline{\underline{R_{rad} = \frac{10\beta^2}{\pi}}}$$

Prob. 13.21

(a) $P_{rad} = \int \mathbf{P}_{rad} \cdot d\mathbf{S} = P_{ave} \cdot 2\pi r^2$ (hemisphere)

$$P_{ave} = \frac{P_{rad}}{2\pi r^2} = \frac{200 \times 10^3}{2\pi (2500 \times 10^6)} = 12.73 \mu\text{W} / \text{m}^2$$

$$\underline{\underline{\mathbf{P}_{ave} = 12.73 \mathbf{a}_r \mu\text{W}/\text{m}^2}}$$

(b) $P_{ave} = \frac{(E_{\max})^2}{2\eta}$

$$E_{\max} = \sqrt{2\eta P_{\text{ave}}} = \sqrt{240\pi \times 1273 \times 10^{-6}}$$

$$= \underline{\underline{0.098 \text{ V/m}}}$$

Prob. 13.22

$$U(\theta, \phi) = r^2 P_{\text{ave}} = k \sin^2 \theta \sin^3 \phi$$

$$G_d(\theta, \phi) = \frac{U(\theta, \phi)}{U_{\text{ave}}}$$

$$U_{\text{ave}} = \frac{k}{4\pi} \iint U(\theta, \phi) \sin \theta d\theta d\phi = \frac{k}{4\pi} \int_0^\pi \sin^3 \phi d\phi \int_0^\pi \sin^3 \theta d\theta = \frac{k}{4\pi} \left[\int_0^\pi (1 - \cos^2 \phi) d(-\cos \phi) \right]^2$$

$$= \frac{k}{4\pi} \left[\frac{4}{3} \right]^2 = \frac{4k}{9\pi}$$

$$G_d(\theta, \phi) = \frac{9\pi}{4k} k \sin^2 \theta \sin^3 \phi = \underline{\underline{7.069 \sin^2 \theta \sin^3 \phi}}$$

$$D = G_{d,\max} = \underline{\underline{7.069}}$$

Prob. 13.23

From Prob. 13.11, set $\ell = \frac{3\lambda}{2}$, $\beta\ell = \frac{2\pi}{\lambda} \frac{3\lambda}{2} = 3\pi$

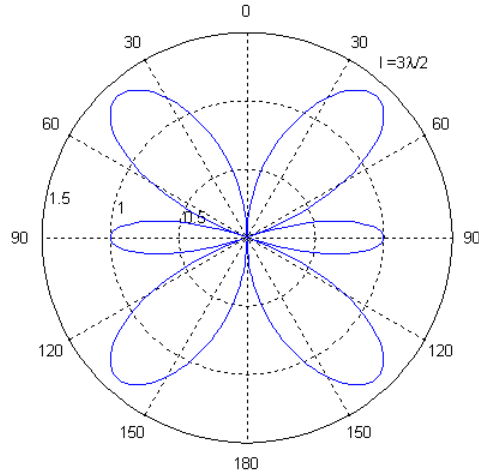
$$H_{\phi s} = \frac{jI_o e^{-\beta r}}{2\pi r} \frac{\left[\cos\left(\frac{2\pi}{\lambda} \frac{1}{2} \frac{3\lambda}{2} \cos \theta\right) - \cos\left(\frac{2\pi}{\lambda} \frac{1}{2} \frac{3\lambda}{2}\right) \right]}{\sin \theta}$$

$$= \frac{jI_o e^{-\beta r}}{2\pi r} \frac{\left[\cos\left(\frac{3\pi}{2} \cos \theta\right) - \cos\left(\frac{3\pi}{2}\right) \right]}{\sin \theta} = \frac{jI_o e^{-\beta r}}{2\pi r} \frac{\cos(1.5\pi \cos \theta)}{\sin \theta}$$

Hence, the normalized radiated field pattern is

$$f(\theta) = \frac{\cos(1.5\pi \cos \theta)}{\sin \theta}$$

which is plotted below.

**Prob. 13.24**

The MATLAB code is shown below

```
N=20;
del= 2*pi/N;
sum=0;
for k=1:N
    theta = del*k;
    term = (1 - cos(theta))/theta;
    sum = sum + term;
end
int = del*sum
```

When the program is run, it gives the value of 2.4335. The accuracy may be increased by increasing N.

Prob. 13.25

$$(a) \quad E_{\theta s} = \frac{j\eta I_o \beta dl}{4\pi r} \sin\theta e^{-j\beta r}$$

$$R_{rad} = 80\pi^2 \left(\frac{dl}{\lambda} \right)^2$$

$$G_d = \frac{4\pi r^2 P_{ave}}{P_{rad}} = \frac{4\pi r^2 \cdot \frac{1}{2\eta} |E_{\theta s}|^2}{\frac{1}{2} I_o^2 R_{rad}}$$

$$= \frac{4\pi r^2}{I_o^2} \cdot \frac{1}{80\pi^2} \left(\frac{\lambda}{dl} \right)^2 \cdot \frac{1}{\eta} \frac{\eta^2 I_o^2 \beta^2 (dl)^2 \sin^2 \theta}{16\pi^2 r^2}$$

$$G_d = \underline{\underline{1.5 \sin^2 \theta}}$$

$$(b) \quad D = G_{d, \max} = \underline{\underline{1.5}}$$

$$(c) \quad A_e = \frac{\lambda^2}{4\pi} G_d = \frac{1.5 \lambda^2 \sin^2 \theta}{\underline{\underline{4\pi}}}$$

$$(d) \quad R_{rad} = 80\pi^2 \left(\frac{1}{16} \right)^2 = \underline{\underline{3.084 \, \Omega}}$$

Prob. 13.26

$$G_d(\theta, \phi) = \frac{4\pi U(\theta, \phi)}{P_{rad}} = \frac{4\pi f^2(\theta)}{\iint f^2(\theta) d\Omega}$$

$$f(\theta) = \sin \theta$$

$$G_d(\theta, \phi) = \frac{4\pi \sin^2 \theta}{\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \sin^3 \theta d\theta} = \frac{4\pi \sin^2 \theta}{2\pi(4/3)} = \underline{\underline{1.5 \sin^2 \theta}}$$

$$D = G_{d, \max} = \underline{\underline{1.5}}$$

Prob. 13.27

(a)

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1.2 \times 10^6} = 250$$

$$\ell = \frac{\lambda}{4} = \underline{\underline{62.5 \, \text{m}}}$$

$$(b) \quad \text{From eq. (13.30), } R_{rad} = \underline{\underline{36.5 \, \Omega}}$$

(c)

For $\lambda/4$ -monopole,

$$f(\theta) = \frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta}, \quad 0 < \theta < \pi/2$$

$$G_d(\theta, \phi) = \frac{4\pi f^2(\theta)}{\int f^2(\theta) d\Omega} = \frac{\frac{4\pi \cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}}{\int_0^{2\pi} \int_0^{\pi/2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin \theta} d\theta d\phi}$$

$$= \frac{4\pi \cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta} \frac{1}{2\pi(0.6094)} = \frac{3.282 \cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$$

$$D = G_{d, \max} = \underline{\underline{3.282}}$$

Prob. 13.28(a) $U_{\max} = 1$

$$U_{ave} = \frac{P_{rad}}{4\pi} = \frac{\int U d\Omega}{4\pi}$$

$$= \frac{1}{4\pi} \int \int \sin^2 2\theta \sin \theta d\theta d\phi$$

$$= \frac{1}{4\pi} (2\pi) \int_0^\pi (2 \sin \theta \cos \theta)^2 d(-\cos \theta)$$

$$= 2 \int_0^\pi (\cos^4 \theta - \cos^2 \theta) d(\cos \theta)$$

$$= 2 \left[\frac{\cos^5 \theta}{5} - \frac{\cos^3 \theta}{3} \right] \bigg|_0^\pi$$

$$= 2 \left[-\frac{2}{5} + \frac{2}{3} \right] = \frac{8}{15}$$

$$U_{ave} = \underline{\underline{0.5333}}$$

$$D = \frac{U_{\max}}{U_{ave}} = \underline{\underline{1.875}}$$

$$(b) \quad U_{\max} = 4$$

$$U_{ave} = \frac{1}{4\pi} \int U d\Omega = \frac{4}{4\pi} \int \int \frac{\sin \theta}{\sin^2 \theta} d\theta d\phi$$

$$= \frac{1}{\pi} \int_0^\pi d\phi \int_{\pi/3}^{\pi/2} \operatorname{cosec} \theta d\theta = \frac{\pi}{\pi} \ln \sqrt{3}$$

$$U_{ave} = \underline{\underline{0.5493}}$$

$$D = \frac{U_{\max}}{U_{ave}} = \frac{16}{3 \ln \sqrt{3}} = \underline{\underline{9.7092}}$$

$$(c) \quad U_{\max} = 2$$

$$U_{ave} = \frac{1}{4\pi} \int U d\Omega = \frac{1}{4\pi} \int \int 2 \sin^2 \theta \sin^2 \phi \sin \theta d\theta d\phi$$

$$= \frac{1}{2\pi} \int_0^\pi \sin^2 \phi d\phi \int_0^\pi (1 - \cos^2 \theta) d(-\cos \theta)$$

$$= \frac{1}{2\pi} \cdot \frac{\pi}{2} \left(\frac{\cos^3 \theta}{3} - \cos \theta \right) \Big|_0^\pi = \frac{1}{4} \left[-\frac{2}{3} + 2 \right] = \frac{1}{3}$$

$$U_{ave} = \underline{\underline{0.333}}$$

$$D = \frac{U_{\max}}{U_{ave}} = \underline{\underline{6}}$$

Prob. 13.29

$$(a) \quad G_d(\theta, \phi) = \frac{U(\theta, \phi)}{U_{ave}}$$

$$\begin{aligned} U_{ave} &= \frac{1}{4\pi} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} 10 \sin \theta \sin^2 \phi \times \sin \theta d\theta d\phi \\ &= \frac{10}{4\pi} \int_0^{2\pi} \sin^2 \phi d\phi \int_0^{\pi} \sin^2 \theta d\theta \\ &= \frac{10}{4\pi} \frac{1}{2} \left(\theta - \frac{\sin 2\theta}{2} \right) \bigg|_0^{\pi} \frac{2\pi}{2} \left(\phi - \frac{\sin 2\phi}{2} \right) \bigg|_0^{\pi} \\ &= \frac{10}{16\pi} (2\pi - 0)(\pi - 0) = \frac{5\pi}{4} \end{aligned}$$

$$G_d(\theta, \phi) = \frac{40 \sin \theta \sin^2 \phi}{5\pi} = \underline{\underline{2.546 \sin \theta \sin^2 \phi}}$$

$$D = G_{d, \max} = \underline{\underline{2.546}}$$

$$\begin{aligned} (b) \quad U_{ave} &= \frac{1}{4\pi} \int_{\phi=0}^{\pi} \int_{\theta=0}^{\pi} 2 \sin^2 \theta \sin^3 \phi \times \sin \theta d\theta d\phi \\ &= \frac{2}{4\pi} \int_0^{\pi} \sin^3 \phi d\phi \int_0^{\pi} \sin^3 \theta d\theta = \frac{2}{4\pi} \left[\int_0^{\pi} (1 - \cos^2 \phi) d(-\cos \phi) \right]^2 \\ &= \frac{1}{2\pi} \left[\left(\frac{\cos^3 \phi}{3} - \cos \phi \right) \bigg|_0^{\pi} \right] = \frac{1}{2\pi} \left(\frac{4}{3} \right)^2 = \frac{16}{18\pi} \end{aligned}$$

$$G_d(\theta, \phi) = \frac{18\pi}{16} 2 \sin^2 \theta \sin^3 \phi = \underline{\underline{2.25\pi \sin^2 \theta \sin^3 \phi}}$$

$$D = G_{d, \max} = \underline{\underline{7.069}}$$

$$\begin{aligned}
 \text{(c)} \quad U_{ave} &= \frac{1}{4\pi} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} 5(1 + \sin^2 \theta \sin^2 \phi) \times \sin \theta d\theta d\phi \\
 &= \frac{5}{4\pi} \left[\int_0^{\pi} \sin \theta d\theta \int_0^{2\pi} d\phi + \int_0^{\pi} \sin^3 \theta d\theta \int_0^{2\pi} \sin^2 \phi d\phi \right] \\
 &= \frac{5}{4\pi} \left[2\pi(-\cos \theta) \Big|_0^{\pi} + \frac{4}{3} \left(\frac{\phi}{2} - \frac{\sin 2\phi}{4} \right) \Big|_0^{2\pi} \right] \\
 &= \frac{5}{4\pi} \left[4\pi + \frac{4}{3}\pi \right] = \frac{20}{3}
 \end{aligned}$$

$$G_d(\theta, \phi) = \frac{3}{20} 5(1 + \sin^2 \theta \sin^2 \phi) = \underline{\underline{0.75(1 + \sin^2 \theta \sin^2 \phi)}}$$

$$D = G_{d,\max} = \underline{\underline{1.5}}$$

Prob. 13.30

$$U_{\max} = 4$$

$$\begin{aligned}
 U_{ave} &= \frac{1}{4\pi} \int U d\Omega = \frac{1}{4\pi} \iint 4 \sin^2 \theta \sin \frac{\phi}{2} \sin \theta d\theta d\phi \\
 &= \frac{1}{\pi} \int_0^{\pi} \sin^3 \theta d\theta \int_0^{\pi} \sin \frac{\phi}{2} d\phi = \frac{1}{\pi} \int_0^{\pi} (1 - \cos^2 \theta) d(-\cos \theta) (-2 \cos \frac{\phi}{2}) \Big|_0^{\pi} \\
 &= \frac{1}{\pi} \left(\frac{4}{3} \right) (2) = \frac{8}{3\pi}
 \end{aligned}$$

$$D = \frac{U_{\max}}{U_{ave}} = 4 \times \frac{3\pi}{8} = \underline{\underline{4.712}}$$

Prob. 13.31

$$\mathbf{P}_{ave} = \frac{|E_r|^2}{2\eta} \mathbf{a}_r = \frac{I_o^2 \sin^2 \theta}{2\eta r^2} \mathbf{a}_r$$

$$\begin{aligned}
 P_{rad} &= \frac{I_o^2}{2\eta} \iint \frac{\sin^2 \theta}{r^2} r^2 \sin \theta d\theta d\phi = \frac{I_o^2}{240\pi} (2\pi) \int_0^{\pi} (1 - \cos^2 \theta) d(-\cos \theta) \\
 &= \frac{I_o^2}{120} \left(\frac{\cos^3 \theta}{3} - \cos \theta \right) \Big|_0^{\pi} = \frac{I_o^2}{120} (-1/3 + 1 - 1/3 + 1) = \frac{I_o^2}{90}
 \end{aligned}$$

$$I_o^2 = 90 P_{ave} = 90 \times 50 \times 10^{-3} \longrightarrow I_o = \underline{\underline{2.121 \text{ A}}}$$

Prob. 13.32

$$U(\theta, \phi) = r^2 P_{ave} = r^2 \frac{|E_\phi|^2}{2\eta} = \frac{r^2}{2\eta} \frac{1400\pi^4 I_o^2 S^2 \sin^2 \theta}{r^2 \lambda^4}, \quad \text{where } \eta = 120\pi$$

$$U(\theta, \phi) = \frac{120\pi^3 I_o^2 S^2 \sin^2 \theta}{2\lambda^4} = \underline{\underline{\frac{60\pi^3 I_o^2 S^2 \sin^2 \theta}{\lambda^4}}}$$

Prob. 13.33

$$P_{ave} = \frac{E^2}{2\eta} = \frac{\alpha^2 I^2}{2\eta r^2}$$

$$R_{rad} = \frac{2P_{rad}}{I^2} = \frac{2P_{ave}}{I^2} 4\pi r^2 = \frac{8\pi r^2}{I^2} \frac{\alpha^2 I^2}{2\eta r^2} = \frac{4\pi \alpha^2}{\eta} = \frac{4\pi \alpha^2}{120\pi} = \underline{\underline{\frac{\alpha^2}{30}}}$$

Prob. 13.34

According to eq. (13.10),

$$P_{rad} = k \int_0^\pi \sin^3 \theta d\theta = \frac{4k}{3}, \quad \text{where } k \text{ is a constant.}$$

$$P'_{rad} = k \int_0^{\pi/3} \sin^3 \theta d\theta = k \left[\frac{\cos^3 \theta}{3} - \cos \theta \right]_0^{\pi/3} = k \left[\frac{1}{3 \times 8} - \frac{1}{2} - \frac{1}{3} + 1 \right] = \frac{5}{24} k$$

$$\text{Fraction} = \frac{\frac{5}{24} k}{\frac{4k}{3}} = \frac{5}{32} = \underline{\underline{0.1562}}$$

Prob. 13.35

This is similar to Fig. 13.10 except that the elements are z-directed.

$$\mathbf{E}_s = \mathbf{E}_{s1} + \mathbf{E}_{s2} = \frac{j\eta\beta I_o dl}{4\pi} \left[\sin \theta_1 \frac{e^{-j\beta r_1}}{r_1} \mathbf{a}_{\theta 1} + \sin \theta_2 \frac{e^{-j\beta r_2}}{r_2} \mathbf{a}_{\theta 2} \right]$$

$$\text{where } r_1 \cong r - \frac{d}{2} \cos \theta, \quad r_2 \cong r + \frac{d}{2} \cos \theta, \quad \theta_1 \cong \theta_2 \cong \theta, \quad \mathbf{a}_{\theta 1} \cong \mathbf{a}_{\theta 2} = \mathbf{a}_\theta$$

$$\mathbf{E}_s = \frac{j\eta\beta I_o dl}{4\pi} \sin \theta \mathbf{a}_\theta \left[e^{j\beta d \cos \theta / 2} + e^{-j\beta d \cos \theta / 2} \right]$$

$$\underline{\underline{E_s = \frac{j\eta\beta I_o dl}{2\pi} \sin\theta \cos\left(\frac{1}{2}\beta d \cos\theta\right) \mathbf{a}_\theta}}$$

Prob. 13.36

$$(a) \quad AF = 2 \cos\left[\frac{1}{2}(\beta d \cos\theta + \alpha)\right], \quad \alpha = 0, \quad \beta d = \frac{2\pi}{\lambda} \lambda = 2\pi$$

$$AF = \underline{\underline{2 \cos(\pi \cos\theta)}}$$

(b) Nulls occur when

$$\cos(\pi \cos\theta) = 0 \quad \longrightarrow \quad \pi \cos\theta = \pm\pi/2, \pm3\pi/2, \dots$$

or

$$\theta = \underline{\underline{60^\circ, 120^\circ}}$$

(c) Maxima and minima occur when

$$\frac{df}{d\theta} = 0 \quad \longrightarrow \quad \sin(\pi \cos\theta) \pi \sin\theta = 0$$

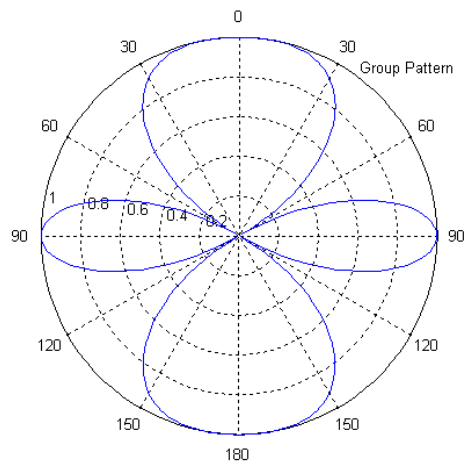
$$\text{i.e. } \sin\theta = 0 \quad \longrightarrow \quad \theta = 0^\circ, 180^\circ$$

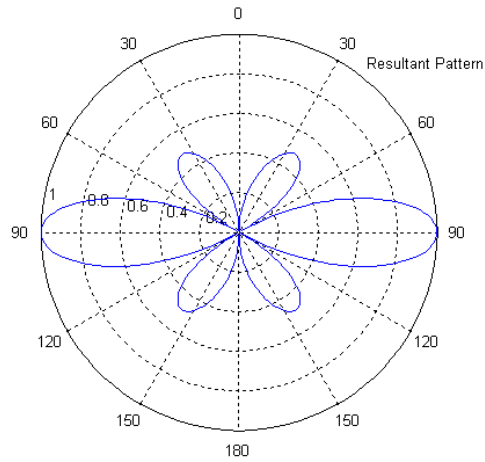
$$\cos\theta = 0 \quad \longrightarrow \quad \theta = 90^\circ$$

or

$$\theta = \underline{\underline{0^\circ, 90^\circ, 180^\circ}}$$

(d) The group pattern is sketched below.



**Prob. 13.37**

$$f(\theta) = \cos \left[\frac{1}{2} (\beta d \cos \theta + \alpha) \right]$$

$$(a) \quad \alpha = \pi/2, \beta d = \frac{2\pi}{\lambda} \cdot \lambda = 2\pi$$

$$f(\theta) = \cos \left(\pi \cos \theta + \frac{\pi}{4} \right)$$

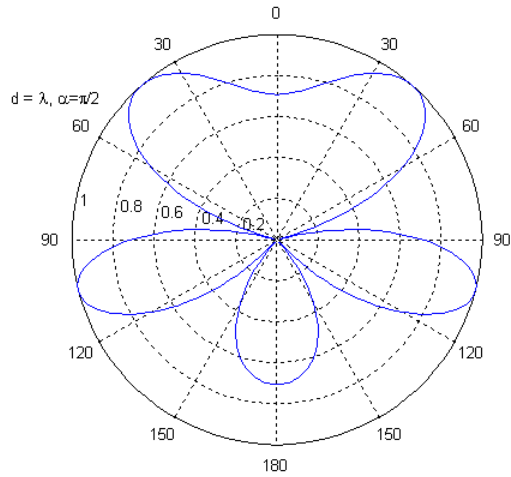
$$\text{Nulls occur at } \pi \cos \theta + \frac{\pi}{4} = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots \text{ or } \theta = 75.5^\circ, 138.6^\circ$$

$$\text{Maxima occur at } \frac{\partial f}{\partial \theta} = 0 \longrightarrow \sin \theta = 0 \longrightarrow \theta = 0^\circ, 180^\circ$$

$$\text{Or } \sin \left(\pi \cos \theta + \frac{\pi}{4} \right) \equiv 0 \longrightarrow \theta = 41.4^\circ, 104.5^\circ$$

$$\text{With } f_{\max} = 0.71, 1.$$

Hence the group pattern is sketched below.



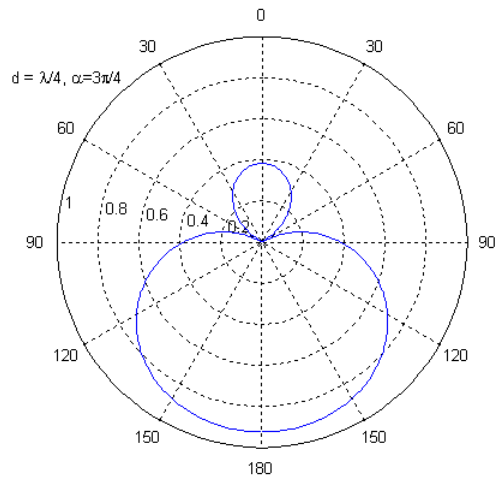
$$(b) \quad \alpha = \frac{3\pi}{4}, \beta d = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$$

$$f(\theta) = \left| \cos\left(\frac{\pi}{4} \cos\theta + \frac{3\pi}{8}\right) \right|$$

$$\text{Nulls occur at } \frac{\pi}{4} \cos\theta + \frac{3\pi}{8} = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots \longrightarrow \theta = 60^\circ$$

$$\text{Minima and maxima occur at } \sin\theta \cos\left(\frac{\pi}{4} \cos\theta + \frac{3\pi}{8}\right) = 0$$

$$\text{i.e. } \theta = 0^\circ, 180^\circ \rightarrow f(\theta) = 0.383, 0.924$$



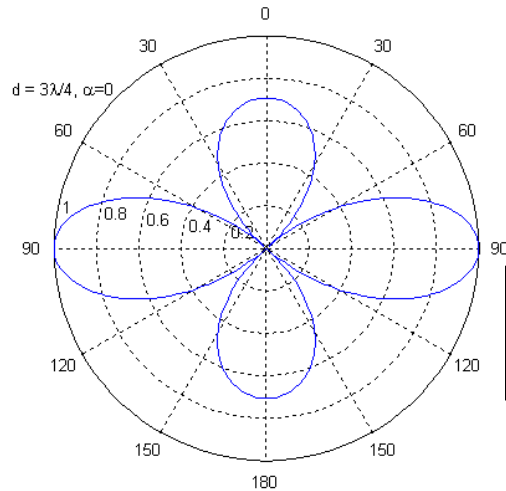
$$(c) \quad \alpha = 0, \beta d = \frac{2\pi}{\lambda} \cdot \frac{3\lambda}{4} = \frac{3\pi}{2}$$

$$f(\theta) = \left| \cos\left(\frac{3\pi}{4} \cos\theta\right) \right|$$

It has nulls at $\frac{3\pi}{4} \cos\theta = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots \rightarrow \theta = 48.2^\circ, 131.8^\circ$

It has maxima and minima at $\frac{df}{d\theta} = 0 \rightarrow \sin\theta \sin\left(\frac{3\pi}{4} \cos\theta\right) = 0$

i.e. $\theta = 0^\circ, 180^\circ \rightarrow f(\theta) = 0.707, 1$, $\theta = \pm 90^\circ \rightarrow f(\theta) = 1$



$\theta = 0^\circ$,	$\text{abs}(f) = 1/\sqrt{2} = 0.707$
$\theta = 90^\circ$,	$\text{abs}(f) = 1$,
$\theta = 180^\circ$.	$\text{abs}(f) = 1/\sqrt{2} = 0.707$

Prob. 13.38

(a) For $N = 2$, $f(\theta) = \cos\left[\frac{1}{2}(\beta d \cos\theta + \alpha)\right]$

$$\alpha = 0, d = \frac{\lambda}{4}$$

$$f(\theta) = \cos\left[\frac{1}{2}\left(\frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} \cos\theta + 0\right)\right] = \cos\left(\frac{\pi}{4} \cos\theta\right)$$

Maxima and minima occur at

$$\frac{d}{d\theta} \left[\cos\left(\frac{\pi}{4} \cos\theta\right) \right] = 0$$

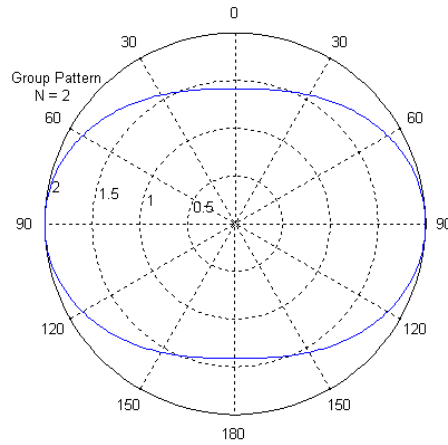
$$\sin\theta \sin\left(\frac{\pi}{4} \cos\theta\right) = 0$$

$$\sin\theta = 0 \rightarrow \theta = \pi, 0 \text{ and } f(\theta) = 0.707$$

$$\sin\left(\frac{\pi}{4} \cos\theta\right) \rightarrow \cos = 0 \rightarrow \theta = 90^\circ, f(\theta) = 1$$

Nulls occur as $\frac{\pi}{4} \cos \theta = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$ (No Solution)

The group pattern is sketched below.



(b) For $N = 4$,

$$AF = \frac{\sin 2(\beta d \cos \theta + 0)}{\sin \frac{1}{2}(\beta d \cos \theta + 0)}$$

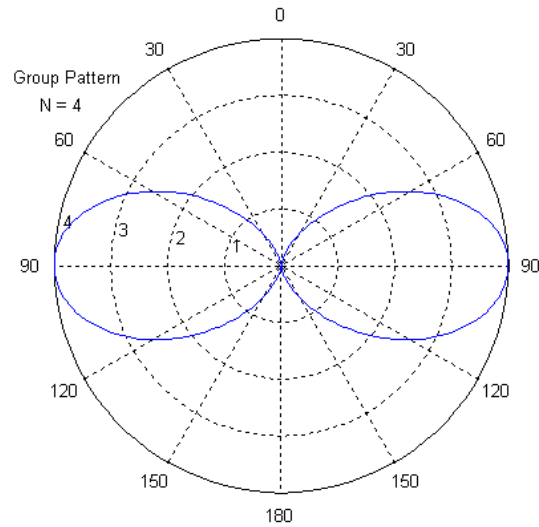
$$\text{Now, } \frac{\sin 4\theta}{\sin \theta} = \frac{2 \sin 2\theta \cos 2\theta}{\sin \theta} = 4 \cos 2\theta \cos \theta$$

$$AF = 4 \cos(\beta d \cos \theta) \cos\left(\frac{1}{2} \beta d \cos \theta\right)$$

$$f(\theta) = \cos\left(\frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} \cos \theta\right) \cos\left(\frac{1}{2} \frac{2\pi}{\lambda} \frac{\lambda}{4}\right) \cos \theta$$

$$= \cos\left(\frac{\pi}{2} \cos \theta\right) \cos\left(\frac{\pi}{4} \cos \theta\right)$$

The plot is shown below.

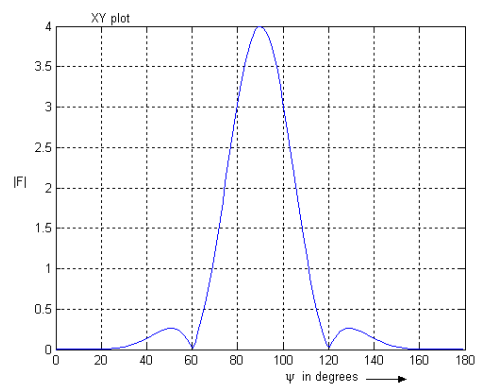
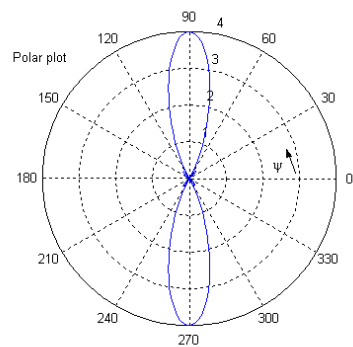


Prob. 13.39

The MATLAB code is shown below.

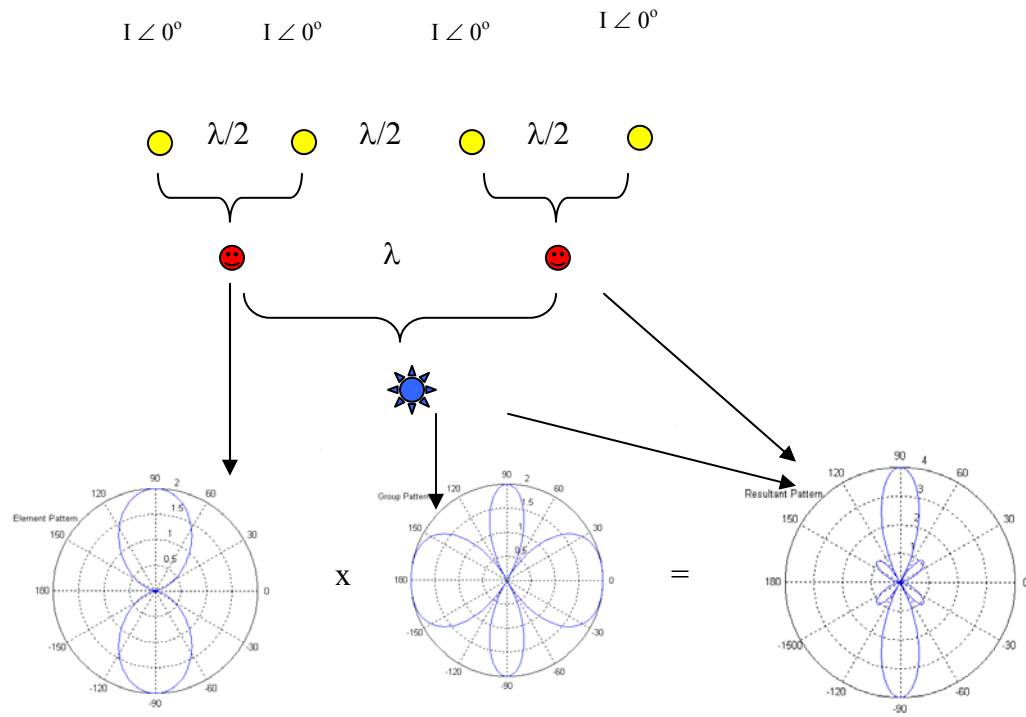
```
for n=1:180
    phi=n*pi/180;
    p(n)=n;
    sn=sin(2*pi*cos(phi));
    cn=cos(0.5*pi*cos(phi));
    sd=sin(0.5*pi*cos(phi));
    fun=sn*cn*cn/sd;
    f(n)=abs(fun);
end
polar(p,f)
```

The polar plot and the xy plot are shown below.



Prob. 13.40

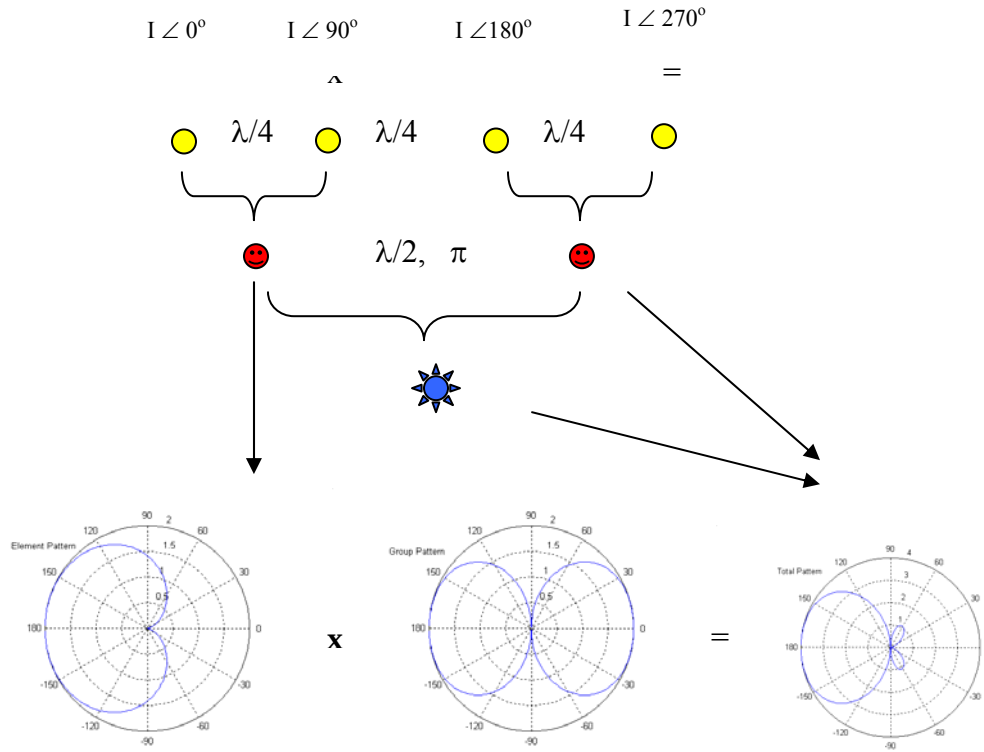
(a) The resultant pattern is obtained as follows.



(b) The array is replaced by by $\begin{matrix} + & & + \\ \angle 0^\circ & \frac{\lambda}{4} & \angle \pi/2 \end{matrix}$

where + stands for $\begin{matrix} \cdot & \longleftrightarrow & \cdot \\ & \angle 0^\circ & \angle \pi \end{matrix}$

Thus the resultant pattern is obtained as shown.



Prob. 13.41

$$G_d(\text{dB}) = 20\text{dB} = 10 \log_{10} G_d \longrightarrow G_d = 10^2 = 100$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10 \times 10^9} = 3 \times 10^{-2}$$

$$A_e = \frac{\lambda^2}{4\pi} G_d = \frac{9 \times 10^{-4}}{4\pi} 100 = \underline{\underline{7.162 \times 10^{-2} \text{ m}^2}}$$

Prob. 13.42

$$A_e = \frac{P_r}{P_{ave}} = \frac{P_r}{\frac{|E_r|^2}{2\eta}} = \frac{2\eta P_r}{|E_r|^2}$$

$$= \frac{2 \times 120\pi \times 2 \times 10^{-6}}{25 \times 10^2 \times 10^{-6}} = \frac{48\pi}{250} = \underline{\underline{0.6031 \text{ m}^2}}$$

Prob. 13.43

Friis equation states that

$$\frac{P_r}{P_t} = G_r G_t \left(\frac{\lambda}{4\pi r} \right)^2$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{200 \times 10^6} = 1.5 \text{ m}, \quad r = 238,857 \times 1.609 \times 10^3 = 3.843 \times 10^8$$

$$G_t(\text{dB}) = 15 \text{ dB} = 10 \log_{10} G_t \quad \longrightarrow \quad G_t = 10^{15/10} = 31.623$$

$$G_r = \left(\frac{4\pi r}{\lambda} \right)^2 \frac{P_r}{P_t} = \left(\frac{4\pi \times 3.843 \times 10^8}{1.5} \right)^2 \frac{4 \times 10^{-9}}{120 \times 10^{-3}} = 34.55 \times 10^{10}$$

$$G_r(\text{dB}) = 10 \log_{10} G_r = 10 \log_{10} 34.55 \times 10^{10} = \underline{\underline{115.384 \text{ dB}}}$$

Prob. 13.44

Using Frii's equation,

$$P_r = G_r G_t \left[\frac{\lambda}{4\pi r} \right]^2 P_t$$

$$P_t = \left[\frac{4\pi r}{\lambda} \right]^2 \frac{P_r}{G_r G_t}$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^9} = 0.1, \quad r = 42 \text{ km}$$

$$G_t(\text{dB}) = 10 \log_{10} G_t = 25 \quad \longrightarrow \quad G_t = 10^{2.5} = 316.23$$

$$G_r(\text{dB}) = 10 \log_{10} G_r = 20 \quad \longrightarrow \quad G_r = 10^2 = 100$$

$$P_t = \left(\frac{4\pi \times 42 \times 10^3}{0.1} \right)^2 \frac{3 \times 10^{-6}}{31623} = \underline{\underline{2.642 \text{ kW}}}$$

Prob. 13.45

$$G_{dt} = 10^4, G_{dr} = 10^{3.2} = 1585$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{15 \times 10^9} = 0.02 \text{ m} = \frac{1}{50}$$

$$P_r = G_{dr} G_{dt} \left(\frac{\lambda}{4\pi r} \right)^2 P_t = 10^4 (1585) \left(\frac{0.02}{4\pi \times 2.456741 \times 10^7} \right)^2 320$$

$$= 2.129 \times 10^{-11} \text{ W} = \underline{\underline{21.29 \text{ pW}}}$$

Prob. 13.46

$$P_r = G_{dt} G_{dr} \left(\frac{\lambda}{4\pi r} \right)^2 P_t$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{20 \times 10^6} = 15 \text{ m}, \quad G_{dt} = G_{dr} = 1.64$$

$$P_t = \frac{P_r}{G_{dt} G_{dr}} \left(\frac{4\pi r}{\lambda} \right)^2 = 0.5 \times 10^{-6} \left(\frac{4\pi \times 80 \times 10^3}{1.64 \times 15} \right)^2 = 835.025 \text{ W}$$

$$\text{But } R_{rad} = \frac{2P_{rad}}{I_o^2} = 73 \quad \rightarrow \quad I_o^2 = \frac{2P_{rad}}{73} = \frac{2 \times 835.025}{73} = 22.8774$$

$$I_o = \underline{\underline{4.783 \text{ A}}}$$

Prob. 13.47

$$30 \text{ dB} = \log \frac{P_t}{P_r} \rightarrow \frac{P_t}{P_r} = 10^3 = 1000$$

$$\text{But } P_r = (G_d)^2 \left(\frac{3}{50 \times 4\pi \times 12} \right)^2 P_t = P_t \left(\frac{G_d}{800\pi} \right)^2$$

$$\left(\frac{G_d}{800\pi} \right)^2 = \frac{P_r}{P_t} = \frac{1}{1000} = \left(\frac{1}{10\sqrt{10}} \right)^2$$

$$\text{or } G_d = \frac{800\pi}{10\sqrt{10}} = 79.476$$

$$G_d = 10 \log 79.476 = \underline{\underline{19 \text{ dB}}}$$

Prob. 13.48

$$(a) \quad P_i = \frac{|E|^2}{2\eta_o} = \frac{P_{rad} G_d}{4\pi r^2} \rightarrow |E_i| = \sqrt{\frac{240\pi P_{rad} G_d}{4\pi r^2}}$$

$$|E_i| = \frac{1}{r} \sqrt{60 P_{rad} G_d} = \frac{1}{120 \times 10^3} \sqrt{60 \times 200 \times 10^3 \times 3500}$$

$$= \underline{\underline{1708 \text{ V/m}}}$$

$$(b) \quad |E_s| = \sqrt{\frac{|E_i|^2 \sigma}{4\pi r^2}} = \sqrt{\frac{1708^2 \times 8}{4\pi \times 14400 \times 10^6}} = \underline{\underline{11.36 \text{ } \mu\text{V/m}}}$$

$$(c) \quad P_c = P_i \sigma = \frac{1708^2}{240\pi} (8) = \underline{\underline{30.95 \text{ mW}}}$$

$$(d) \quad P_i = \frac{|E|^2}{2\eta_o} = \frac{(11.36)^2 \times 10^{-12}}{240\pi} = 1712 \times 10^{-13} \text{ W/m}^2$$

$$\lambda = \frac{3 \times 10^8}{15 \times 10^8} = 0.2 \text{ m}, A_{2r} = \frac{\lambda^2 G}{4\pi} = \frac{0.04 \times 3500}{4\pi}$$

$$P_r = P_a A_{er} = 1712 \times 10^{-13} \times 11.14 = 1.907 \times 10^{-12}$$

$$\text{or } P_r = \frac{(\lambda G_d)^2 \sigma P_{rad}}{(4\pi)^3 r^4} = \frac{(0.2 \times 3500)^2 \times 8 \times 2 \times 10^5}{(4\pi)^3 \times 12^4 \times 10^{16}}$$

$$= \underline{\underline{1.91 \times 10^{-12} \text{ W}}}$$

Prob. 13.49

$$P_r = \frac{(\lambda G_d)^2 \sigma P_{rad}}{(4\pi)^3 r^4}$$

$$G_d(\text{dB}) = 30 \text{ dB} = 10 \log_{10} G_d \quad \longrightarrow \quad G_d = 10^3$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{4 \times 10^9} = 0.075 \text{ m}$$

$$P_r = \frac{(0.075 \times 10^3)^2 \times 12 \times 80 \times 10^3}{(4\pi)^3 (10 \times 10^3)^4} = \underline{\underline{272.1 \text{ pW}}}$$

Prob. 13.50

$$(a) \quad t_r = \frac{2d}{u} = \frac{2 \times 160 \times 10^3}{3 \times 10^8} = \frac{32}{3} \times 10^{-4} = \underline{\underline{1.067 \text{ ms}}}$$

$$(b) \quad \lambda = c / f = \frac{3 \times 10^8}{4 \times 10^9} = 0.075 \text{ m}, \quad P_r = \frac{A_e \sigma G_d P_{rad}}{(4\pi r^2)^2}$$

$$\text{But } A_e = \frac{\lambda^2 G_d}{4\pi} \rightarrow G_d = \frac{4\pi A_e}{\lambda^2}$$

$$P_r = 4\pi\sigma \left(\frac{A_e}{4\pi r^2 \lambda} \right)^2 P_{rad} = 4\pi \times 5 \left(\frac{2}{0.075 \times 4\pi \times 160^2 \times 10^6} \right)^2 (60 \times 10^3) = \underline{\underline{2.59 \times 10^{-14} \text{ W}}}$$

$$(c) \quad r = \left[\frac{\lambda^2 G_d^2 \sigma P_{rad}}{(4\pi)^3 P_r} \right]^{1/4} = \left[\frac{\lambda^2 \sigma (4\pi)^2 A_e^2 P_{rad}}{(4\pi)^3 \lambda^4 P_r} \right]^{1/4} = \left[\frac{A_e^2 \sigma P_{rad}}{4\pi \lambda^2 P_r} \right]^{1/4}$$

$$= \left[\frac{5 \times 4}{4\pi (0.075)^2} \frac{60 \times 10^3}{8 \times 10^{-12}} \right]^{1/4} = \underline{\underline{38.167 \text{ km}}}$$

Prob. 13.51

$$P_r = \frac{k P_{rad}}{r^4} \rightarrow P_{rad} = \frac{r^4}{k} P_r$$

$$\text{If } R = 2r, \quad P_{rad} = \frac{(2r)^4}{k} P_r = 16 \frac{r^4}{k} P_r = 16 P_{rad}$$

i.e. the transmitted power must be increased 16 times.

Prob. 13.52

$$P_{rad} = \frac{4\pi}{G_{dt} G_{dr}} \left(\frac{4\pi r_1 r_2}{\lambda} \right)^2 \frac{P_r}{\sigma}$$

$$\text{But } G_{dt} = 36 \text{ dB} = 10^{3.6} = 3981.1$$

$$G_{dr} = 20 \text{ dB} = 10^2 = 100$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{5 \times 10^9} = 0.06$$

$$r_1 = 3 \text{ km}, \quad r_2 = 5 \text{ km}$$

$$P_{rad} = \frac{4\pi}{3981.1 \times 100} \left(\frac{4\pi \times 15 \times 10^6}{6 \times 10^{-2}} \right)^2 \frac{8 \times 10^{-12}}{2.4}$$

$$= \underline{\underline{1.038 \text{ kW}}}$$

Prob. 13.53

(a)

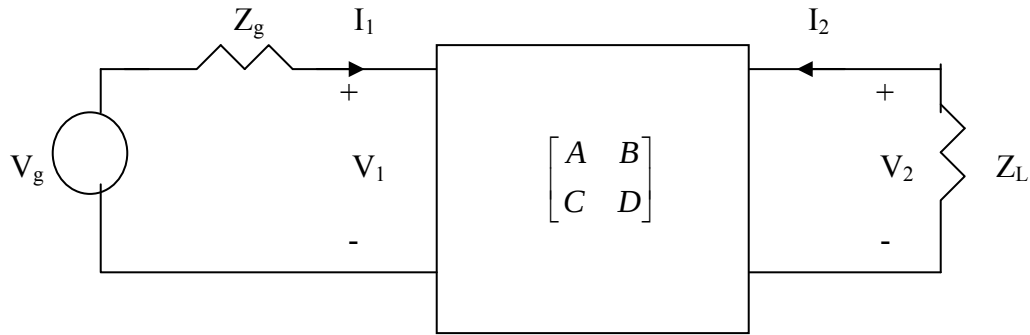
$$F = \frac{\pi fL}{R} = \frac{\pi \times 300 \times 10^6 \times 50 \times 10^{-9}}{20} = 2.356$$

$$IL = 10 \log_{10}(1 + F^2) = 10 \log_{10}(1 + 2.356^2) = \underline{\underline{8.164 \text{ dB}}}$$

(b)

$$F = \pi fRC = \pi \times 300 \times 10^6 \times 10 \times 10^3 \times 60 \times 10^{-12} = 180\pi = 565.5$$

$$IL = 10 \log_{10}(1 + F^2) = 10 \log_{10}(1 + 565.5^2) = \underline{\underline{55.05 \text{ dB}}}$$

Prob. 13.54

By definition,

$$V_1 = AV_2 - BI_2 \quad (1)$$

$$I_1 = CV_2 - DI_2 \quad (2)$$

Let V_2 and $\bar{V}_2$ be respectively the load voltages when the filter circuit is present and when it is absent.

$$V_2 = -I_2 Z_L = \frac{I_1 Z_L}{CZ_L + D}$$

$$= \frac{V_g Z_L}{\left(Z_g + \frac{V_1}{I_1} \right) (CZ_L + D)} = \frac{V_g Z_L}{\left(Z_g + \frac{AV_2 - BI_2}{CV_2 - DI_2} \right) (CZ_L + D)}$$

$$= \frac{V_g Z_L}{\left(Z_g + \frac{AZ_L + B}{CZ_L + D} \right) (CZ_L + D)}$$

$$= \frac{V_g Z_L}{(Z_g (CZ_L + D) + AZ_L + B)}$$

$$\bar{V}_2 = \frac{V_g Z_L}{(Z_g + Z_L)}$$

Ratio and modulus give

$$\left| \frac{\bar{V}_2}{V_2} \right| = \frac{(Z_g (CZ_L + D) + AZ_L + B)}{Z_g + Z_L}$$

Insertion loss =

$$\text{IL} = 20 \log_{10} \left| \frac{\bar{V}_2}{V_2} \right| = 20 \log_{10} \left| \frac{(Z_g (CZ_L + D) + AZ_L + B)}{Z_g + Z_L} \right|$$

which is the required result

Prob. 13.55

$$(a) R_{dc} = \frac{l}{\sigma S} = \frac{10^3}{0.96 \times 10^{-4} \times 6.1 \times 10^7} = \underline{\underline{17.1 \text{ m}\Omega / \text{km}}}$$

$$(b) R_{ac} = \frac{l}{\delta w \sigma}, \quad \pi a^2 = 0.8 \times 1.2 = 0.96 \quad \text{or} \quad a = 0.5528$$

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = \frac{1}{\sqrt{\pi \times 6 \times 10^6 \times 4\pi \times 10^{-7} \times 6.1 \times 10^7}} = \frac{1}{12.1 \times \pi \times 10^3}$$

$$R_{ac} = \frac{1000 \times 12.1 \times \pi \times 10^3}{1.2 \times 10^{-2} \times 6.1 \times 10^7} = \underline{\underline{51.93 \Omega}}$$

Prob. 13.56

$$\begin{aligned} SE &= 20 \log_{10} \frac{E_i}{E_o} = 20 \log_{10} \frac{6}{20 \times 10^{-6}} = 20 \log_{10} (3 \times 10^5) \\ &= \underline{\underline{109.54 \text{ dB}}} \end{aligned}$$

CHAPTER 14

P. E. 14.1 The program in Fig. 14.3 was used to obtain the plot in Fig. 14.5.

P. E. 14.2 For the exact solution,

$$(D^2 + 1)y = 0 \quad \longrightarrow \quad y = A \cos x + B \sin x$$

$$y(0) = 0 \quad \longrightarrow \quad A = 0$$

$$y(1) = 1 \quad \longrightarrow \quad 1 = B \sin 1 \text{ or } B = 1/\sin 1$$

Thus, $y = \sin x / \sin 1$

For the finite difference solution,

$$y'' + y = 0 \quad \longrightarrow \quad \frac{y(x + \Delta) - 2y(x) + y(x - \Delta))}{\Delta^2} + y = 0$$

or

$$y(x) = \frac{y(x + \Delta) + y(x - \Delta))}{2 - \Delta^2}, y(0) = 0, y(1) = 1, \Delta = 1/4$$

With the MATLAB program shown below, we obtain the exact result y_e and FD result y .

```

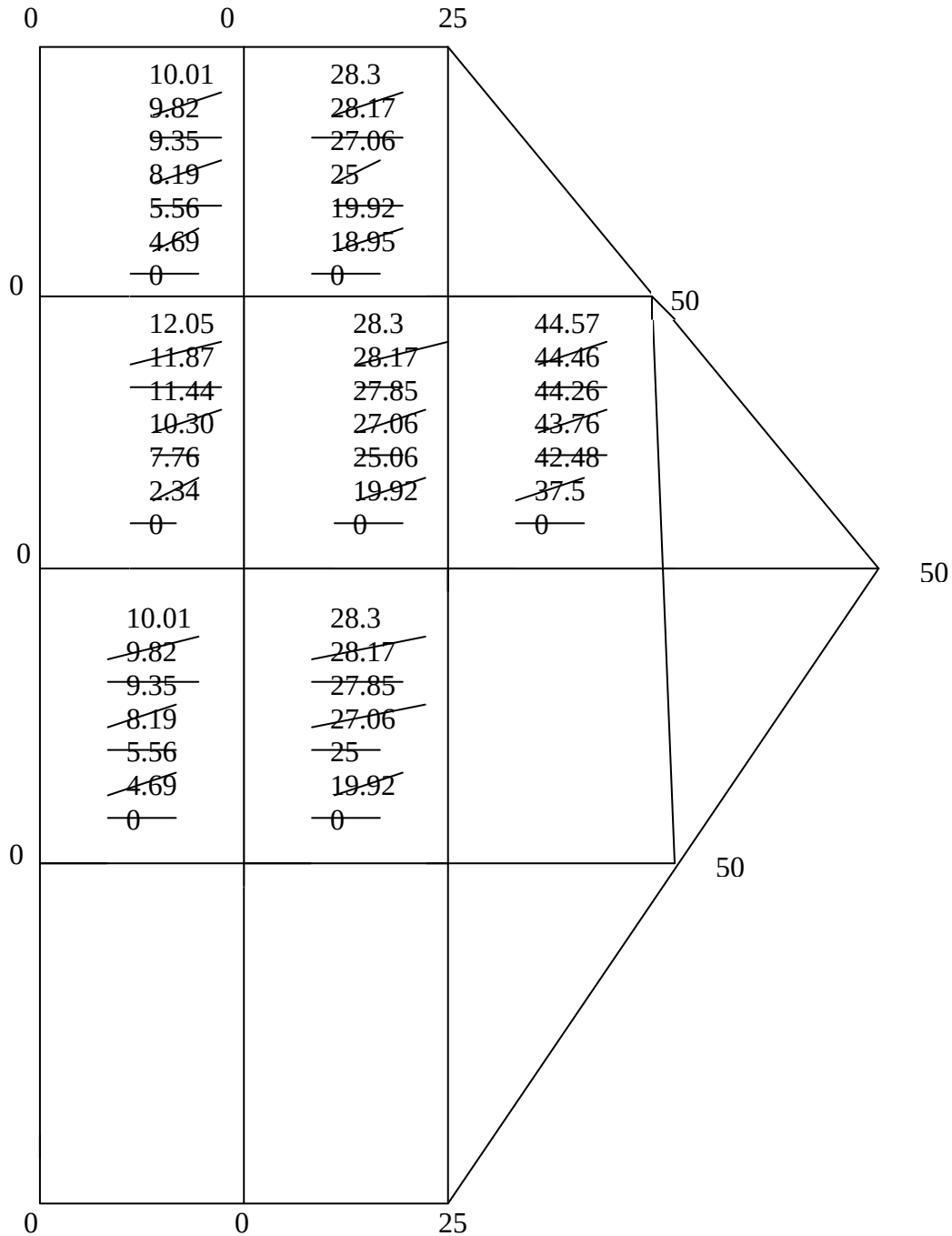
y(1)=0.0;
y(5)=1.0;
del=0.25;
for n=1:20
    for k=2:4
        y(k)=( y(k+1) +y(k-1) )/(2-del*del)
        x=(k-1)*del;
        ye=sin(x)/sin(1.0)
    end
end
end

```

The results are listed below.

y(x)	N=5	N=10	N=15	N=20	Exact $y_e(x)$
y(0.25)	0.2498	0.2924	0.2942	0.2943	0.2940
y(0.5)	0.5242	0.5682	0.5701	0.5702	0.5697
y(0.75)	0.7867	0.8094	0.8104	0.8104	0.8101

P. E. 14.3 By applying eq. (14.16) to each node as shown below, we obtain the following results after 5 iterations.



P. E. 14.4 (a) Using the program in Fig. 14.16 with $n_x = 4+1=5$ and $n_y = 8+1=9$, we obtain the potential at center as

$$V(3,5) = \underline{\underline{23.796 \text{ V}}}$$

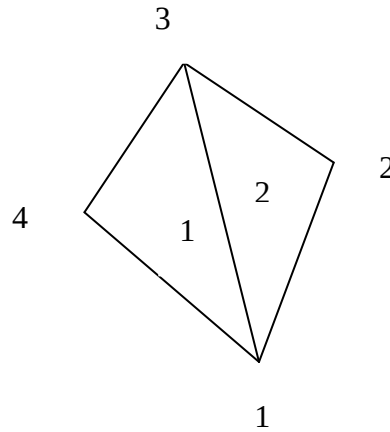
(b) Using the same program with $n_x = 12+1=13$ and $n_y = 24+1=25$, the potential at the center is

$$V(7,13) = \underline{23.883 \text{ V}}$$

P. E. 14.5 By combining the ideas in Figs. 14.20 and 14.24, and dividing each wire into N segments, the results listed in Table 14.2 is obtained.

P. E. 14.6

(a)



For element 1, local 1-2-3 corresponds with global 1-3-4 so that $A_1 = 0.35$,

$$P_1 = 0.8, P_2 = 0.6, P_3 = -1.4, Q_1 = -0.5, Q_2 = 0.5, Q_3 = 0$$

$$C^{(1)} = \begin{bmatrix} 0.6357 & 0.1643 & -0.8 \\ 0.1643 & 0.4357 & -0.6 \\ -0.8 & -0.6 & 1.4 \end{bmatrix}$$

For element 2, local 1-2-3 corresponds with global 1-2-3 so that $A_2 = 0.7$,

$$P_1 = 0.1, P_2 = 1.4, P_3 = -1.5, Q_1 = -1, Q_2 = 0, Q_3 = 1$$

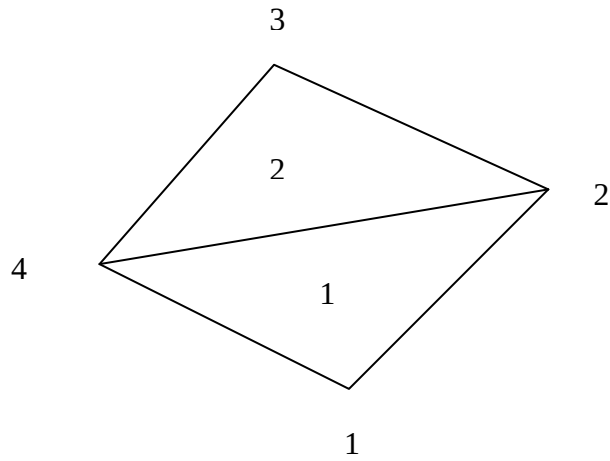
$$C^{(2)} = \begin{bmatrix} 0.3607 & 0.05 & -0.4107 \\ 0.05 & 0.7 & -0.75 \\ -0.4107 & -0.75 & 1.1607 \end{bmatrix}$$

The global coefficient matrix is given by

$$C = \begin{bmatrix} C_{11}^{(1)} + C_{11}^{(2)} & C_{12}^{(2)} & C_{12}^{(1)} + C_{13}^{(2)} & C_{13}^{(1)} \\ C_{21}^{(2)} & C_{22}^{(2)} & C_{23}^{(2)} & 0 \\ C_{21}^{(1)} + C_{31}^{(2)} & C_{32}^{(2)} & C_{22}^{(1)} + C_{33}^{(2)} & C_{23}^{(1)} \\ C_{31}^{(1)} & 0 & C_{32}^{(2)} & C_{33}^{(1)} \end{bmatrix}$$

$$= \begin{bmatrix} 0.9964 & 0.05 & -0.2464 & -0.8 \\ 0.05 & 0.7 & -0.75 & 0 \\ -0.2464 & -0.75 & 1.596 & -0.6 \\ -0.8 & 0 & -0.75 & 1.4 \end{bmatrix}$$

(b)



For element 1, local 1-2-3 corresponds with global 1-2-4 .

$$P_1 = 0.9000; \quad P_2 = 0.6000; \quad P_3 = -1.5000$$

$$Q_1 = -1.5000; \quad Q_2 = 0.5000; \quad Q_3 = 1;$$

$$A_1 = 0.6750;$$

$$C^{(1)} = \begin{bmatrix} 1.1333 & -0.0778 & -1.0556 \\ -0.0778 & 0.2259 & -0.1481 \\ -1.0556 & -0.1481 & 1.2037 \end{bmatrix}$$

For element 2, local numbering 1-2-3 corresponds with global numbering 2-3-4.

$$P_1 = 0.8000; \quad P_2 = -0.9000; \quad P_3 = 0.1000;$$

$$Q_1 = -0.5000; \quad Q_2 = 1.5000; \quad Q_3 = -1;$$

$$A_2 = 0.3750;$$

$$C^{(2)} = \begin{bmatrix} 0.5933 & -0.9800 & 0.3867 \\ -0.9800 & 2.0400 & -1.0600 \\ 0.3867 & -1.0600 & 0.6733 \end{bmatrix}$$

The global coefficient matrix is

$$C = \begin{bmatrix} C_{11}^{(1)} & C_{12}^{(1)} & 0 & C_{13}^{(1)} \\ C_{21}^{(1)} & C_{22}^{(1)} + C_{11}^{(2)} & C_{12}^{(2)} & C_{23}^{(1)} + C_{13}^{(2)} \\ 0 & C_{12}^{(2)} & C_{22}^{(2)} & C_{23}^{(2)} \\ C_{31}^{(1)} & C_{32}^{(1)} + C_{31}^{(2)} & C_{32}^{(2)} & C_{33}^{(1)} + C_{33}^{(2)} \end{bmatrix}$$

$$C = \begin{bmatrix} 1.1333 & -0.0778 & 0 & -1.0556 \\ -0.0778 & 0.8193 & -0.9800 & 0.2385 \\ 0 & -0.9800 & 2.0400 & -1.0600 \\ -1.0556 & 0.2385 & -1.0600 & 1.8770 \end{bmatrix}$$

P. E. 14.7 We use the MATLAB program in Fig. 14.33. The input data for the region in Fig. 14.34 is as follows:

NE = 32; ND = 26; NP = 18;

NL = [1 2 4

2 5 4

2 3 5

3 6 5

4 5 9

5 10 9

5 6 10

6 11 10

7 8 12

8 13 12

8 9 13

9 14 13

9 10 14

10 15 14

10 11 15

11 16 15

12 13 17

13 18 17

13 14 18


```

14 19 18
14 15 19
15 20 19
15 16 20
16 21 20
17 18 22
18 23 22
18 19 23
19 24 23
19 20 24
20 25 24
20 21 25
21 26 25];
X = [ 1.0 1.5 2.0 1.0 1.5 2.0 0.0 0.5 1.0 1.5 2.0 0.0 0.5 1.0 1.5 2.0 0.0 0.5 1.0 1.5
      2 0.0 0.5 1.0 1.5 2.0];
Y = [ 0.0 0.0 0.0 0.5 0.5 0.5 1.0 1.0 1.0 1.0 1.0 1.5 1.5 1.5 1.5 1.5 2.0 2.0 2.0
      2.0 2.0 2.5 2.5 2.5 2.5 2.5 ];
NDP = [ 1 2 3 6 11 16 21 26 25 24 23 22 17 12 7 8 9 4];
VAL = [0.0 0.0 15.0 30.0 30.0 30.0 30.0 25.0 20.0 20.0 20.0 10.0 0.0 0.0 0.0
       0.0 0.0 0.0];

```

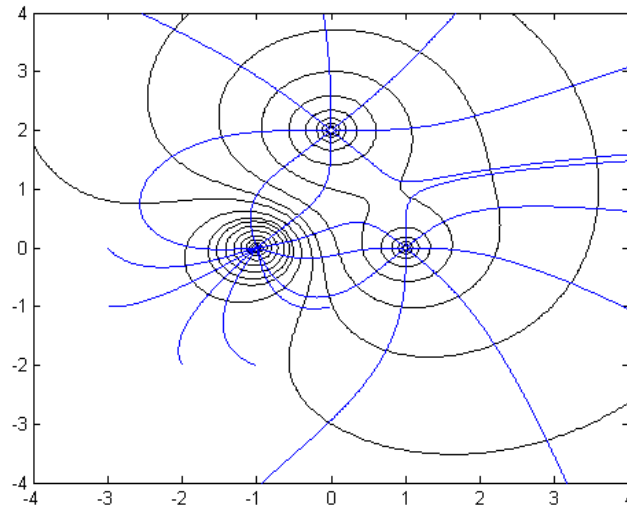
With this data, the finite element (FEM) solution is compared with the finite difference (FD) solution as shown in the table below.

Node #	X	Y	FEM	FD
5	1.5	0.5	11.265	11.25
10	1.5	1.0	15.06	15.02
13	0.5	1.5	4.958	4.705
14	1.0	1.5	9.788	9.545
15	1.0	1.5	18.97	18.84
18	0.5	2.0	10.04	9.659
19	1.0	2.0	15.32	15.85
20	1.5	2.0	21.05	20.87

Prob. 14.1 (a) Using the Matlab code in Fig. 14.3, we input the data as:

```
>> plotit( [-1 2 1], [-1 0; 0 2; 1 0], 1, 1, 0.01, 0.01, 8, 2, 5 )
```

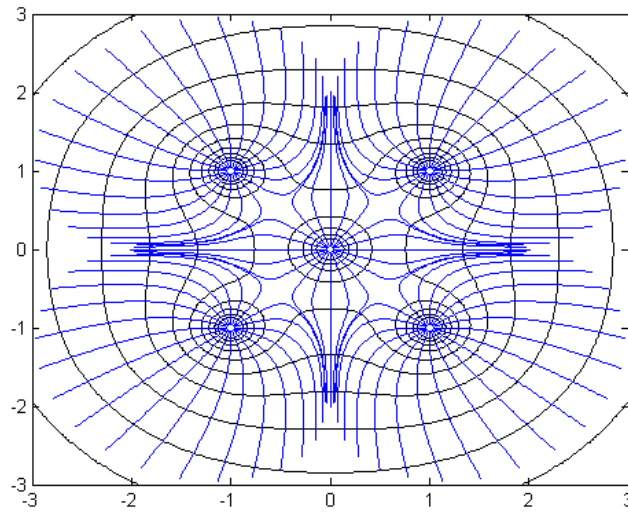
and the plot is shown below.



(b) Using the MATLAB code in Fig. 14.3, we input the required data as:

```
>> plotit( [1 1 1 1 1], [-1 -1; -1 1; 1 -1; 1 1; 0 0], 1, 1, 0.02, 0.01, 6, 2, 5 )
```

and obtain the plot shown below.

**Prob.14.2**

The exact solution is

$$V(x) = \frac{1}{6}x^3 + \frac{1}{2}x^2 + \frac{1}{3}x$$

so that $V(0.5) = 0.3125$. For the finite difference solution,

$$\frac{V(x+\Delta) - 2V(x) + V(x-\Delta)}{\Delta^2} = x+1$$

which leads to

$$V(x) = \frac{1}{2} [V(x+\Delta) + V(x-\Delta) - \Delta^2(x+1)]$$

We apply this at $x = 0.25, 0.5, 0.75$ for 5 iterations as tabulated below.

No. of iterations	V(0)	V(0.25)	V(0.5)	V(0.75)	V(1.0)
0	0	0	0	0	1
1	0	-0.03916	-0.0664	0.4121	1
2	0	-0.07226	0.1232	0.5069	1
3	0	0.02254	0.2178	0.5542	1
4	0	0.06984	0.2651	0.5779	1
5	0	0.09349	0.2888	0.5897	1

From the table, $V(0.5) = 0.2888$ which is smaller than the exact value due to the fact that the number of iterations is not sufficiently large and also $\Delta = 0.25$ is large.

Prob. 14.3 (a)

$$\frac{dV}{dx} = \frac{V(x+\Delta x) - V(x-\Delta x)}{2\Delta x}$$

For $\Delta x = 0.05$ and at $x = 0.15$,

$$\frac{dV}{dx} = \frac{2.0134 - 1.00}{0.05 \times 2} = \underline{\underline{10.117}}$$

$$\frac{d^2V}{dx^2} = \frac{V(x + \Delta x) - 2V(x) + V(x - \Delta x)}{(\Delta x)^2} = \frac{2.0134 + 1.0017 - 2 \times 1.5056}{(0.05)^2} = \underline{\underline{1.56}}$$

(b) $V = 10 \sinh x$, $dV/dx = 10 \cosh x$. At $x = 0.15$, $dV/dx = \underline{\underline{10.113}}$

which is close to the numerical estimate.

$d^2V/dx^2 = 10 \sinh x$. At $x = 0.15$, $d^2V/dx^2 = \underline{\underline{1.5056}}$

which is slightly lower than the numerical value.

Prob. 14.4

$$\nabla^2 V = \frac{\partial^2 V}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial V}{\partial \rho} + \frac{\partial^2 V}{\partial z^2} = 0$$

The equivalent finite difference expression is

$$\frac{V(\rho_o + \Delta\rho, z_o) - 2V(\rho_o, z_o) + V(\rho_o - \Delta\rho, z_o)}{(\Delta\rho)^2} + \frac{1}{\rho_o} \frac{V(\rho_o + \Delta\rho, z_o) - V(\rho_o - \Delta\rho, z_o)}{2\Delta\rho} + \frac{V(\rho_o, z_o + \Delta z) - 2V(\rho_o, z_o) + V(\rho_o, z_o - \Delta z)}{(\Delta z)^2} = 0$$

If $\Delta z = \Delta\rho = h$, rearranging terms gives

$$V(\rho_o, z_o) = \frac{1}{4}V(\rho_o, z_o + h) + \frac{1}{4}V(\rho_o, z_o - h) + \left(1 + \frac{h}{2\rho_o}\right)V(\rho_o + h, z_o) + \left(1 - \frac{h}{2\rho_o}\right)V(\rho_o - h, z_o)$$

as expected.

Prob. 14.5

$$\nabla^2 V = \frac{\partial^2 V}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial V}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} = 0, \quad (1)$$

$$\frac{\partial^2 V}{\partial \rho^2} = \frac{V_{m+1}^n - 2V_m^n + V_{m-1}^n}{(\Delta \rho)^2}, \quad (2)$$

$$\frac{\partial^2 V}{\partial \phi^2} = \frac{V_m^{n+1} - 2V_m^n + V_m^{n-1}}{(\Delta \phi)^2}, \quad (3)$$

$$\left. \frac{\partial V}{\partial \rho} \right|_{m,n} = \frac{V_{m+1}^n - V_{m-1}^n}{2\Delta \rho}. \quad (4)$$

Substituting (2) to (4) into (1) gives

$$\begin{aligned} \nabla^2 V &= \frac{V_{m+1}^n - V_{m-1}^n}{m\Delta \rho(2\Delta \rho)} + \frac{V_{m+1}^n - 2V_m^n + V_{m-1}^n}{(\Delta \rho)^2} + \frac{V_m^{n+1} - 2V_m^n + V_m^{n-1}}{(m\Delta \rho \Delta \phi)^2} \\ &= \frac{1}{(\Delta \rho)^2} \left[\left(1 - \frac{1}{2m}\right) V_{m-1}^n - 2V_m^n + \left(1 + \frac{1}{2m}\right) V_{m+1}^n + \frac{1}{(m\Delta \phi)^2} (V_m^{n+1} - 2V_m^n + V_m^{n-1}) \right] \end{aligned}$$

as required.

Prob. 14.6

$$V_o = \frac{V_1 + V_2 + V_3 + V_4}{4} = \frac{10 - 40 + 50 + 80}{4} = \underline{\underline{25V}}$$

Prob. 14.7

Iteration →	0	1	2	3	4	5
V_1	0.0000	25.0000	35.6250	38.9063	39.7266	39.9316
V_2	0.0000	26.2500	32.8125	34.4531	34.8633	34.9658
V_3	0.0000	16.2500	22.8125	24.4531	24.8633	24.9658
V_4	0.0000	15.6250	18.9063	19.7266	19.9316	19.9829

Prob. 14.8

$$V_a = \frac{1}{4}[0 + 100 + 100 + V_b] = \frac{1}{4}(V_b + 200) \quad (1)$$

$$V_b = \frac{1}{4}[0 + 0 + V_a + V_c] = \frac{1}{4}(V_a + V_c) \quad (2)$$

$$V_c = \frac{1}{4}[V_b + 100 + 100 + 0] = \frac{1}{4}(200 + V_b) \quad (3)$$

Using these relationships, we obtain the data in the table below.

Iteration	1st	2nd	3rd	4th	5th
V_a	50	53.115	56.641	57.08	57.135
V_b	12.5	26.56	28.32	28.54	28.57
V_c	53.125	56.64	57.08	57.135	57.142

Alternatively, we can solve (1) to (3) simultaneously.

From (1) and (3), $V_a = V_c$

From (2), $V_b = \frac{V_a}{2}$

Thus (1) becomes $V_a = \frac{1}{4}\left(\frac{V_a}{2} + 200\right) \rightarrow V_a = 400/7 = 57.143 = V_c$

$$V_b = \frac{V_a}{2} = 28.57$$

Prob. 14.9

(a) We follow Example 6.5 with $a=b$.

$$V = V_1 + V_2 = \frac{4V_o}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{\sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right)}{n \sinh(n\pi)} + \frac{4V_o}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{\sin\left(\frac{n\pi y}{a}\right) \sinh\left(\frac{n\pi x}{a}\right)}{n \sinh(n\pi)}$$

(b) At the center of the region, finite difference gives

$$V(a/2, a/2) = \frac{1}{4}(0 + 0 + V_o + V_o) = \frac{V_o}{2} = \underline{\underline{25 \text{ V}}}$$

Prob. 14.10

$$k = \frac{h^2 \rho_s}{\varepsilon} = 10^{-4} \times \frac{50 \times 10^{-9}}{\frac{10^{-9}}{36\pi}} = 0.18\pi = 0.5655$$

At node 1,

$$V_1 = \frac{1}{4}[0 + V_2 + V_3 + k] \longrightarrow 4V_1 - V_2 - V_3 = k \quad (1)$$

At node 2,

$$V_2 = \frac{1}{4}[0 + V_1 + V_4 + k] \longrightarrow 4V_2 - V_1 - V_4 = k \quad (2)$$

At node 3,

$$V_3 = \frac{1}{4}[0 + 2V_1 + V_4 + k] \longrightarrow 4V_3 - 2V_1 - V_4 = k \quad (3)$$

At node 4,

$$V_4 = \frac{1}{4}[0 + 2V_2 + V_3 + k] \longrightarrow 4V_4 - 2V_2 - V_3 = k \quad (4)$$

Putting (1) to (4) in matrix form,

$$\begin{bmatrix} 4 & -1 & -1 & 0 \\ -1 & 4 & 0 & -1 \\ -2 & 0 & 4 & -1 \\ 0 & -2 & -1 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0.5655 \\ 0.5655 \\ 0.5655 \\ 0.5655 \end{bmatrix}$$

Using a calculator or MATLAB, we obtain

$$\underline{\underline{V_1 = V_2 = 0.3231 \text{ V}, \quad V_3 = V_4 = 0.4039 \text{ V}}}$$

Prob. 14.11

(a)

$$\begin{bmatrix} -4 & 1 & 0 & 1 & 0 & 0 \\ 1 & -4 & 1 & 0 & 1 & 0 \\ 0 & 1 & -4 & 0 & 0 & 1 \\ 1 & 0 & 0 & -4 & 1 & 0 \\ 0 & 1 & 0 & 1 & -4 & 1 \\ 0 & 0 & 1 & 0 & 1 & -4 \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \\ V_d \\ V_e \\ V_f \end{bmatrix} = \begin{bmatrix} -200 \\ -100 \\ -100 \\ -100 \\ 0 \\ 0 \end{bmatrix}$$

[A] [B]

(b)

$$\begin{array}{cccccccc}
 \begin{bmatrix} -4 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & -4 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -4 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & -4 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & -4 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & -4 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & -4 \end{bmatrix} &
 \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \\ V_6 \\ V_7 \\ V_8 \end{bmatrix} &
 = &
 \begin{bmatrix} -30 \\ -15 \\ -30 \\ -7.5 \\ 0 \\ -7.5 \\ 0 \\ 0 \end{bmatrix} \\
 [A] & & & [B]
 \end{array}$$

Prob. 14.12 (a) Matrix [A] remains the same. To each term of matrix [B], we add

$$-h^2 \rho_v / \varepsilon.$$

(b) Let $\Delta x = \Delta y = h = 0.25$ so that $n_x = 5 = n_y$.

$$\frac{\rho_v}{\varepsilon} = \frac{x(y-1)10^{-9}}{10^{-9}/36\pi} = 36\pi x(y-1)$$

Modify the program in Fig. 14.16 as follows.

```

H=0.25;
for I=1:nx-1
    for J=1:ny-1
        X = H*I;
        Y=H*J;
        RO = 36.0*pi*X*(Y-1);
        V(I,J) = 0.25*( V(I+1,J) + V(I-1,J) + V(I,J+1) + V(I,J-1) + H*H*RO );
    end
end

```

This is the major change. However, in addition to this, we must set

```

v1 = 0.0;
v2 = 10.0;
v3 = 20.0;
v4 = -10.0;
nx = 5;
ny = 5;

```

The results are:

$$\begin{aligned}
 V_a &= 4.6095 & V_b &= 9.9440 & V_c &= 11.6577 \\
 V_d &= -1.5061 & V_e &= 3.5090 & V_f &= 6.6867 \\
 \underline{V_g} &= \underline{-3.2592} & \underline{V_h} &= \underline{0.2366} & \underline{V_i} &= \underline{3.3472}
 \end{aligned}$$

Prob. 14.13

$$V_1 = \frac{1}{4}(0 + 0 + V_2 + V_4) = \frac{1}{4}(V_2 + V_4)$$

$$V_2 = \frac{1}{4}(0 + 50 + V_1 + V_3) = \frac{1}{4}(50 + V_1 + V_3)$$

$$V_3 = \frac{1}{4}(0 + 100 + 50 + V_2) = \frac{1}{4}(150 + V_2)$$

$$V_4 = \frac{1}{4}(0 + 50 + V_1 + V_5) = \frac{1}{4}(50 + V_1 + V_5)$$

$$V_5 = \frac{1}{4}(0 + 0 + V_4 + V_6) = \frac{1}{4}(V_4 + V_6)$$

$$V_6 = \frac{1}{4}(0 + 50 + V_5 + V_7) = \frac{1}{4}(50 + V_5 + V_7)$$

$$V_7 = \frac{1}{4}(0 + 100 + V_6 + 50) = \frac{1}{4}(150 + V_6)$$

Initially set all free potentials equal to zero. Apply the seven formulas above iteratively and obtain the results shown below.

n	1	2	3	4	5
V_1	0	6.25	9.77	10.63	10.97
V_2	12.5	24.22	25.83	26.15	26.25
V_3	40.625	43.55	43.96	44.04	44.06
V_4	12.5	14.84	16.70	17.73	17.97
V_5	3.12	7.03	10.29	10.93	11.05
V_6	13.281	24.46	25.98	26.23	26.28
V_7	40.82	43.62	43.99	44.06	44.07

Prob. 14.14

$$\begin{aligned}
 \frac{1}{c^2} \frac{\Phi^{j+1}_{m,n} + \Phi^{j-1}_{m,n} - 2\Phi^j_{m,n}}{(\Delta t)^2} &= \frac{\Phi^j_{m+1,n} + \Phi^j_{m-1,n} - 2\Phi^j_{m,n}}{(\Delta x)^2} \\
 &+ \frac{\Phi^j_{m,n+1} + \Phi^j_{m,n-1} - 2\Phi^j_{m,n}}{(\Delta z)^2}
 \end{aligned}$$

If $h = \Delta x = \Delta z$, then after rearranging we obtain

$$\Phi^{j+1}_{m,n} = 2\Phi^j_{m,n} - \Phi^{j-1}_{m,n} + \alpha(\Phi^j_{m+1,n} + \Phi^j_{m-1,n} - 2\Phi^j_{m,n}) \\ + \alpha(\Phi^j_{m,n+1} + \Phi^j_{m,n-1} - 2\Phi^j_{m,n})$$

where $\alpha = (c\Delta t / h)^2$.

Prob. 14.15

$$\frac{\partial^2 V}{\partial x^2} = \frac{\partial^2 V}{\partial t^2} \longrightarrow \frac{V(x + \Delta x, t) - 2V(x, t) + V(x - \Delta x, t)}{(\Delta x)^2} = \\ \frac{V(x, t + \Delta t) - 2V(x, t) + V(x, t - \Delta t)}{(\Delta t)^2}$$

$$V(x, t + \Delta t) = \left(\frac{\Delta t}{\Delta x} \right)^2 [V(x + \Delta x, t) - 2V(x, t) + V(x - \Delta x, t)] + 2V(x, t) - V(x, t - \Delta t)$$

or

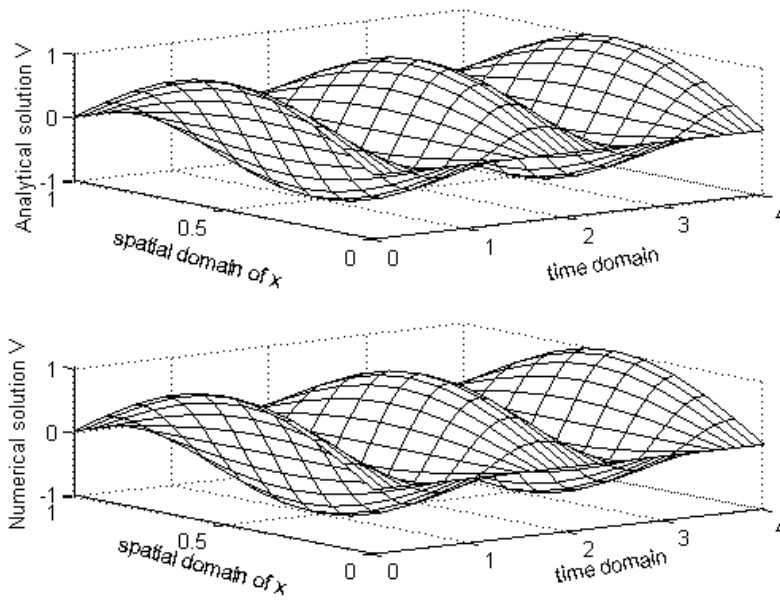
$$V(i, j + 1) = \alpha [V(i + 1, j) + V(i - 1, j)] + 2(1 - \alpha)V(i, j) - V(i, j - 1)$$

where $\alpha = \left(\frac{\Delta t}{\Delta x} \right)^2$. Applying the finite difference formula derived above, the following

programs was developed.

```
xd=0:.1:1;td=0:.1:4;
[t,x]=meshgrid(td,xd);
Va=sin(pi*x).*cos(pi*t);%Analytical result
subplot(211);mesh(td,xd,Va);colormap([0 0 0])
% Numerical result
N=length(xd);M=length(td);
v(:,1)=sin(pi*xd');
v(2:N-1,2)=(v(1:N-2,1)+v(3:N,1))/2;
for k=2:M-1
    v(2:N-1,k+1)=-v(2:N-1,k-1)+v(1:N-2,k)+v(3:N,k);
end
subplot(212);mesh(td,xd,v);colormap([0 0 0])
```

The results of the finite difference algorithm agree perfectly with the exact solution as shown below.

**Prob. 14.16**

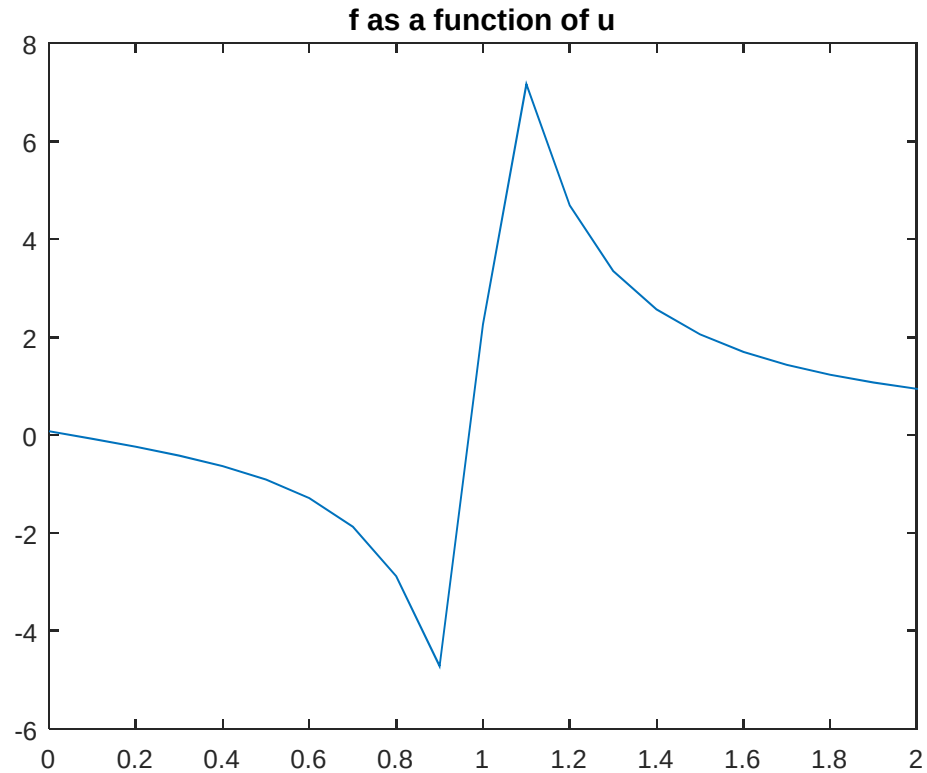
The MATLAB code and the plot of $F(u)$ are presented below.

% Integration using MATLAB

```

N=40;
del=pi/N;
for k=1:21
    u(k)=0.1*(k-1);
    sum=0.0;
    for n=1:N
        theta=del*n;
        num= u(k)- cos(theta)
        den=( 1 +u(k)^2 - 2*u(k)*cos(theta) )^1.5;
        term = num/den;
        sum=sum + term;
    end
    f(k) = sum*del;
end
plot(u,f)
title('f as a function of u')

```



Prob. 14.17 Combining the ideas in the programs in Figs. 14.20 and 14.24, we develop a MATLAB code which gives

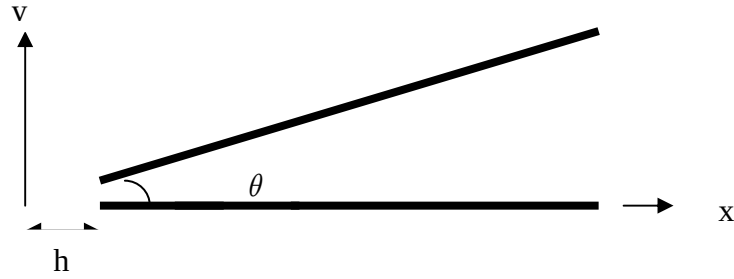
$$N = 20 \longrightarrow C = 19.4 \text{ pF/m}$$

$$N = 40 \longrightarrow C = 13.55 \text{ pF/m}$$

$$N = 100 \longrightarrow \underline{C = 12.77 \text{ pF/m}}$$

For the exact value, $d/2a = 50/10 = 5$

$$C = \frac{\pi\epsilon}{\cosh^{-1} \frac{d}{2a}} = \frac{\pi \times 10^{-9} / 36\pi}{\cosh^{-1} 5} = \underline{\underline{12.12 \text{ pF/m}}}$$

Prob. 14.18

To find C , take the following steps:

(1) Divide each line into N equal segments. Number the segments in the lower conductor as $1, 2, \dots, N$ and segments in the upper conductor as $N+1, N+2, \dots, 2N$,

(2) Determine the coordinate (x_k, y_k) for the center of each segment.

For the lower conductor, $y_k = 0, k=1, \dots, N, x_k = h + \Delta (k-1/2), k = 1, 2, \dots, N$

For the upper conductor, $y_k = [h + \Delta (k-1/2)] \sin \theta, k=N+1, N+2, \dots, 2N$,

$x_k = [h + \Delta (k-1/2)] \cos \theta, k = N+1, N+2, \dots, 2N$

where h is determined from the gap g as

$$h = \frac{g}{2 \sin \theta / 2}$$

(3) Calculate the matrices $[V]$ and $[A]$ with the following elements

$$V_k = \begin{cases} V_o, k = 1, \dots, N \\ -V_o, k = N + 1, \dots, 2N \end{cases}$$

$$A_{ij} = \begin{cases} \frac{\Delta}{4\pi\epsilon R_{ij}}, i \neq j \\ 2 \ln \Delta / a, i = j \end{cases}$$

where $R_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$

(4) Invert matrix $[A]$ and find $[\rho] = [A]^{-1} [V]$.

(5) Find the charge Q on one conductor

$$Q = \sum \rho_k \Delta = \Delta \sum_{k=1}^N \rho_k$$

(6) Find $C = |Q|/2V_o$

Taking $N = 10$, $V_o = 1.0$, a program was developed to obtain the following result.

θ	C (in pF)
10	8.5483
20	9.0677
30	8.893
40	8.606
50	13.004
60	8.5505
70	9.3711
80	8.7762
90	8.665
100	8.665
110	10.179
120	8.544
130	9.892
140	8.7449
150	9.5106
160	8.5488
170	11.32
180	8.6278

Prob. 14.19 We may modify the program in Fig. 14.24 and obtain the result in the table below. $Z_o \cong \underline{\underline{100 \Omega}}$.

N	Z_o , in Ω
10	97.2351
20	97.8277
30	98.0515
40	98.1739
50	98.2524

Prob. 14.20

We make use of the formulas in Problem 14.19.

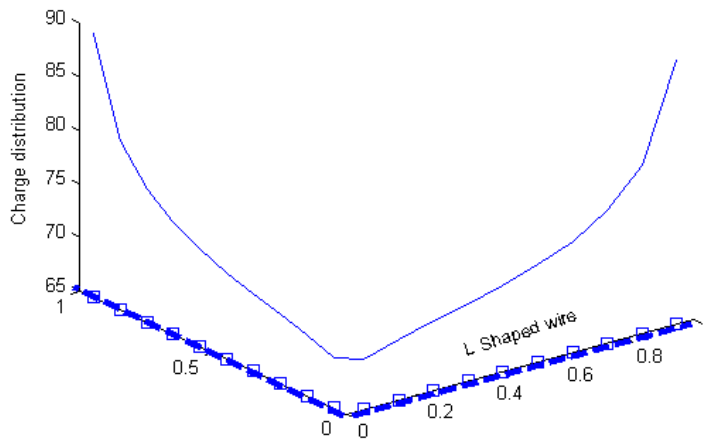
$$V_i = \sum_{j=1}^{2N} A_{ij} \rho_j$$

where N is the number of divisions on each arm of the conductor.

The MATLAB code is as follows:

```
aa=0.001;
vo=10;
eo = (10^(-9))/(36*pi);
L=2.0;
N=10; %no.of divisions on each arm
NT=N*2;
delta=L/(NT);
x=zeros(NT,1);
y=zeros(NT,1);
%Second calculate the elements of the coefficient matrix
for i=1:N-1
    y(i)=0;
    x(i)=delta*(i-0.5)
end
for i=N+1:NT
    x(i)=0;
    y(i)=delta*(i-N-0.5);
end
for i=1:NT
    for j=1:NT
        if (i ~=j)
            R=sqrt( (x(i)-x(j))^2 + (y(i)-y(j))^2 )
            A(i,j)=-delta*R;
        else
            A(i,j)=-delta*(log(delta)-1.5);
        end
    end
end
end
%Determine the matrix of constant vector B and find rho
B=2*pi*eo*vo*ones(NT,1);
rho=inv(A)*B;
```

The result is presented below.



Segment	x	y	ρ in pC/m
1	0.9500		89.6711
2	0.8500	0	80.7171
3	0.7500	0	77.3794
4	0.6500	0	75.4209
5	0.5500	0	74.0605
6	0.4500	0	73.0192
7	0.3500	0	72.1641
8	0.2500	0	71.4150
9	0.1500	0	70.6816
10	0.0500	0	69.6949
11	0	0	69.6949
12	0	0.0500	70.6816
13	0	0.1500	71.4150
14	0	0.2500	72.1641
15	0	0.3500	73.0192
16	0	0.4500	74.0605
17	0	0.5500	75.4209
18	0	0.6500	77.3794
19	0	0.7500	80.7171
20	0	0.8500	89.6711

Prob. 14.21(a) Exact solution yields

$$C = 2\pi\epsilon / \ln(\Delta / a) = 8.02607 \times 10^{-11} \text{ F/m and } \underline{\underline{Z_o = 41.559\Omega}}$$

where $a = 1\text{cm}$ and $\Delta = 2\text{cm}$. The numerical solution is shown below.

N	C (pF/m)	$Z_o (\Omega)$
10	82.386	40.486
20	80.966	41.197
40	80.438	41.467
100	80.025	41.562

(b) For this case, the numerical solution is shown below.

N	C (pF/m)	$Z_o (\Omega)$
10	109.51	30.458
20	108.71	30.681
40	108.27	30.807
100	107.93	30.905

Prob. 14.22 We modify the MATLAB code in Fig. 14.24 (for Example 14.5) by changing the input data and matrices [A] and [B]. We let

$$x_i = h + \Delta (i-1/2), \quad i = 1, 2, \dots, N, \quad \Delta = L/N$$

$$y_i = h/2, \quad j = 1, 2, \dots, N, \quad z_k = t/2, \quad k = 1, 2, \dots, N$$

and calculate

$$R_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}$$

We obtain matrices [A] and [B]. Inverting [A] gives

$$[q] = [A]^{-1} [B], \quad [\rho_v] = [q]/(ht\Delta), \quad C = \frac{\sum_{i=1}^N q_i}{10}$$

The computed values of $[\rho_v]$ and C are shown below.

I	$\rho_{vi}(\times 10^{-6})C/m^3$
1, 20	0.5104
2, 19	0.4524
3, 18	0.4324
4, 17	0.4215
5, 16	0.4144
6, 15	0.4096
7, 14	0.4063
8, 13	0.4041
9, 12	0.4027
10, 11	0.4020

C = 17.02 pF

Prob. 14.23 From the given figure, we obtain

$$\alpha_1 = \frac{A_1}{A} = \frac{1}{2A} \begin{vmatrix} 1 & x & y \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix} = \frac{1}{2A} [(x_2 y_3 - x_3 y_2) + (y_2 - y_3)x + (x_3 - x_2)y]$$

as expected. The same applies for α_2 and α_3 .

Prob. 14.24

(a) $P_1 = 1.5, P_2 = 0.5, P_3 = -2, Q_1 = -1, Q_2 = 1.5, Q_3 = -0.5$

$$A = \frac{1}{2}(P_2 Q_3 - P_3 Q_2) = 1.375$$

$$C_{ij} = \frac{1}{4A} [P_i P_j + Q_i Q_j]$$

$$C = \begin{bmatrix} 0.5909 & -0.1364 & -0.4545 \\ -0.1364 & 0.4545 & -0.3182 \\ -0.4545 & -0.3182 & 0.7727 \end{bmatrix}$$

(b)

$$P_1 = -4, P_2 = 4, P_3 = 0, Q_1 = 0, Q_2 = -3, Q_3 = 3$$

$$A = \frac{1}{2}(P_2 Q_3 - P_3 Q_2) = 6$$

$$C = \begin{bmatrix} 0.6667 & -0.6667 & 0 \\ -0.6667 & 1.042 & -0.375 \\ 0 & -0.375 & 0.375 \end{bmatrix}$$

Prob. 14.25 (a)

$$2A = \begin{vmatrix} 1 & 1/2 & 1/2 \\ 1 & 3 & 1/2 \\ 1 & 2 & 2 \end{vmatrix} = 15/4$$

$$\alpha_1 = \frac{4}{15}[(6-1) + (-1\frac{1}{2})x + (-1)y] = \frac{4}{15}(5 - 1.5x - y)$$

$$\alpha_2 = \frac{4}{15}[(1-1) + \frac{3}{2}x - \frac{3}{2}y] = \frac{4}{15}(1.5x - 1.5y)$$

$$\alpha_3 = \frac{4}{15}[(1/4 - 3/2) + 0x + \frac{5}{2}y] = \frac{4}{15}(-1.25 + 2.5y)$$

$$V = \alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3$$

Substituting $V=80$, $V_1 = 100$, $V_2 = 50$, $V_3 = 30$, α_1 , α_2 , and α_3 leads to

$$20 = 7.5x + 10y + 3.75$$

Along side 12, $y=1/2$ so that

$$20 = 15x/2 + 5 + 15/4 \longrightarrow x=3/2, \text{ i.e. } (1.5, 0.5)$$

Along side 13, $x=y$

$$20 = 15x/2 + 10x + 15/4 \longrightarrow x=13/4, \text{ i.e. } (13/4, 13/4)$$

$$\text{Along side 23, } y = -3x/2 + 5$$

$$20 = 15x/2 - 15 + 50 + 15/4 \longrightarrow x=-5/2 \text{ (not possible)}$$

Hence intersection occurs at

(1.5, 0.5) along 12 and (0.9286, 0.9286) along 13

(b) At (2,1),

$$\alpha_1 = \frac{4}{15}, \alpha_2 = \frac{6}{15}, \alpha_3 = \frac{5}{15}$$

$$V(2,1) = \alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3 = (400 + 300 + 150)/15 = \underline{\underline{56.67 \text{ V}}}$$

Prob. 14.26

$$2A = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 1 & 4 \end{vmatrix} = 9$$

$$\alpha_1 = \frac{1}{9}[(0-0) + (4-0)x + (0-1)y] = \frac{1}{9}(4x - y)$$

$$\alpha_2 = \frac{1}{9}[(0-0) + (0+1)x + (2-0)y] = \frac{1}{9}(x + 2y)$$

$$\alpha_3 = \frac{1}{9}[(8+1) + (-1-4)x + (1-2)y] = \frac{1}{9}(9 - 5x - y)$$

$$V_e = \alpha_1 V_{e1} + \alpha_2 V_{e2} + \alpha_3 V_{e3}$$

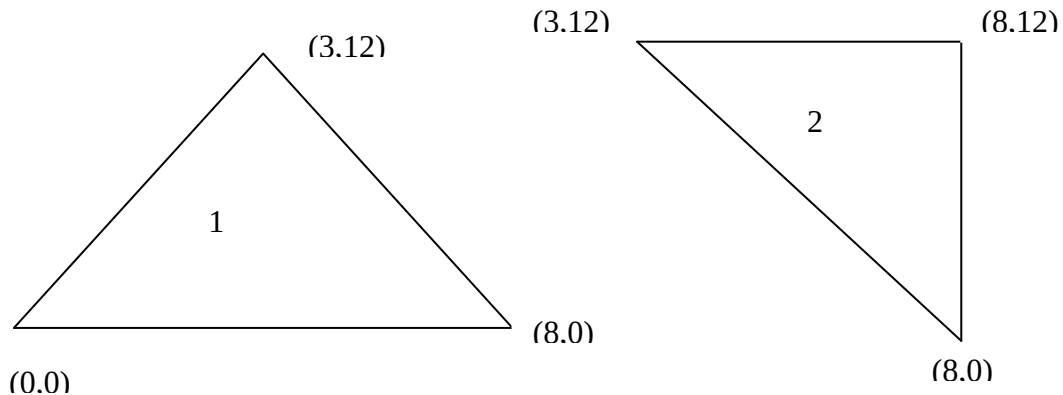
$$V(1,2) = 8(4-2)/9 + 12(1+4)/9 + 10(9-5-1)/9 = 96/9 = \underline{10.667 \text{ V}}$$

At the center $\alpha_1 = \alpha_2 = \alpha_3 = 1/3$ so that

$$V(\text{center}) = (8 + 12 + 10)/3 = 10$$

Or at the center, $(x, y) = (0 + 1 + 2, 0 + 4 - 1)/3 = (1, 1)$

$$V(1,1) = 8(3)/9 + 12(3)/9 + 10(3)/9 = \underline{10 \text{ V}}$$

Prob. 14.27

For element 1, local numbering 1-2-3 corresponds to global numbering 4-2-1.

$$P_1 = 12, P_2 = 0, P_3 = -12, Q_1 = -3, Q_2 = 8, Q_3 = -5,$$

$$A = (0 + 12 \times 8)/2 = 48$$

$$C_{ij} = \frac{1}{4 \times 48} [P_j P_i + Q_j Q_i]$$

$$C^{(1)} = \begin{bmatrix} 0.7969 & -0.125 & -0.6719 \\ -0.125 & 0.3333 & -0.2083 \\ -0.6719 & -0.2083 & 0.8802 \end{bmatrix}$$

For element 2, local numbering 1-2-3 corresponds to global numbering 2-4-3.

$$P_1 = -12, P_2 = 0, P_3 = 12, Q_1 = 0, Q_2 = -5, Q_3 = 5,$$

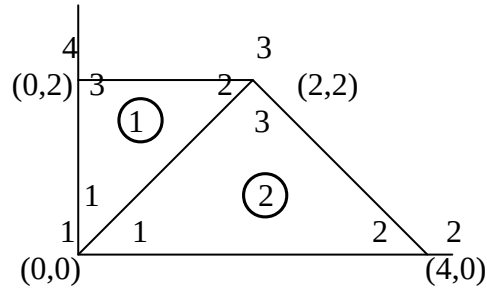
$$A = (0 + 60)/2 = 30$$

$$C_{ij} = \frac{1}{4 \times 48} [P_j P_i + Q_j Q_i]$$

$$C^{(2)} = \begin{bmatrix} 1.2 & 0 & -1.2 \\ 0 & 0.208 & -0.208 \\ -1.2 & -0.208 & 1.408 \end{bmatrix}$$

$$C = \begin{bmatrix} C_{33}^{(1)} & C_{23}^{(1)} & 0 & C_{31}^{(1)} \\ C_{23}^{(1)} & C_{22}^{(1)} + C_{11}^{(2)} & C_{13}^{(2)} & C_{21}^{(1)} + C_{12}^{(2)} \\ 0 & C_{31}^{(2)} & C_{33}^{(2)} & C_{32}^{(2)} \\ C_{13}^{(1)} & C_{21}^{(1)} + C_{21}^{(2)} & C_{23}^{(2)} & C_{22}^{(2)} + C_{11}^{(1)} \end{bmatrix}$$

$$= \begin{bmatrix} 0.8802 & -0.2083 & 0 & -0.6719 \\ -0.2083 & 1.533 & -1.2 & -0.125 \\ 0 & -1.2 & 1.4083 & -0.2083 \\ -0.6719 & -0.125 & -0.2083 & 1.0052 \end{bmatrix}$$

Prob. 14.28

For element 1,

$$P_1 = 0, P_2 = 2, P_3 = -2, Q_1 = -2, Q_2 = 0, Q_3 = 2$$

$$A = \frac{1}{2}(4 - 0) = 2, \quad 4A = 8$$

$$C^{(1)} = \frac{1}{8} \begin{bmatrix} 4 & 0 & -4 \\ 0 & 4 & -4 \\ -4 & -4 & 8 \end{bmatrix} = \begin{bmatrix} 0.5 & 0 & -0.5 \\ 0 & 0.5 & -0.5 \\ -0.5 & -0.5 & 1 \end{bmatrix}$$

For element 2,

$$P_1 = -2, P_2 = 2, P_3 = 0, Q_1 = -2, Q_2 = -2, Q_3 = 4$$

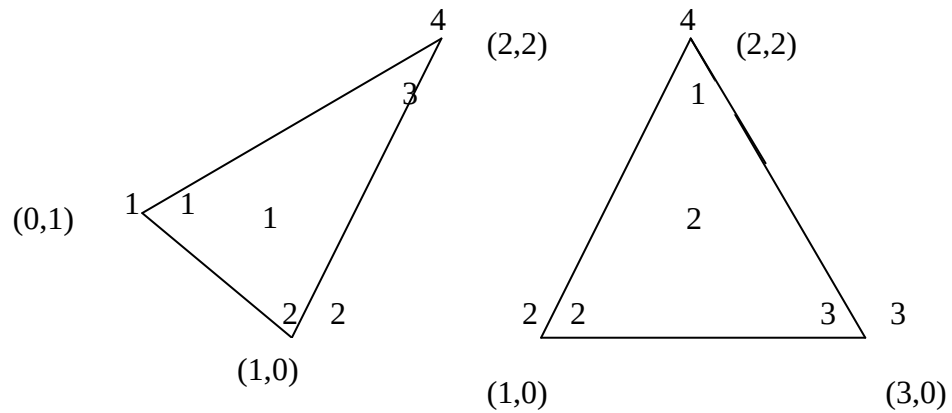
$$A = \frac{1}{2}(8 - 0) = 4, \quad 4A = 16$$

$$C^{(2)} = \frac{1}{16} \begin{bmatrix} 8 & 0 & -8 \\ 0 & 8 & -8 \\ -8 & -8 & 16 \end{bmatrix} = \begin{bmatrix} 0.5 & 0 & -0.5 \\ 0 & 0.5 & -0.5 \\ -0.5 & -0.5 & 1 \end{bmatrix}$$

The global coefficient matrix is

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{bmatrix} = \begin{bmatrix} C_{11}^{(1)} + C_{11}^{(2)} & C_{12}^{(2)} & C_{12}^{(1)} + C_{13}^{(2)} & C_{13}^{(1)} \\ C_{21}^{(2)} & C_{22}^{(2)} & C_{23}^{(2)} & 0 \\ C_{21}^{(1)} + C_{31}^{(2)} & C_{32}^{(2)} & C_{22}^{(1)} + C_{33}^{(2)} & C_{23}^{(1)} \\ C_{31}^{(1)} & 0 & C_{32}^{(1)} & C_{33}^{(1)} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & -0.5 & -0.5 \\ 0 & 0.5 & -0.5 & 0 \\ -0.5 & -0.5 & 1.5 & -0.5 \\ -0.5 & 0 & -0.5 & 1 \end{bmatrix}$$

Prob. 14.29

For element 1, local numbering 1-2-3 corresponds to global numbering 1-2-4.

$$P_1 = -2, P_2 = 1, P_3 = 1, Q_1 = 1, Q_2 = -2, Q_3 = 1,$$

$$A = (P_2 Q_3 - P_3 Q_2) / 2 = 3/2, \text{ i.e. } 4A = 6$$

$$C_{ij} = \frac{1}{4A} [P_j P_i + Q_j Q_i]$$

$$C^{(1)} = \frac{1}{6} \begin{bmatrix} 5 & -4 & -1 \\ -4 & 5 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

For element 2, local numbering 1-2-3 corresponds to global numbering 4-2-3.

$$P_1 = 0, P_2 = -2, P_3 = 2, Q_1 = 2, Q_2 = -1, Q_3 = -1,$$

$$A = 2, 4A = 8$$

$$C^{(2)} = \frac{1}{8} \begin{bmatrix} 4 & -2 & -2 \\ -2 & 5 & -3 \\ -2 & -3 & 5 \end{bmatrix}$$

The global coefficient matrix is

$$C = \begin{bmatrix} C_{11}^{(1)} & C_{12}^{(1)} & 0 & C_{13}^{(1)} \\ C_{12}^{(1)} & C_{22}^{(1)} + C_{22}^{(2)} & C_{23}^{(2)} & C_{23}^{(1)} + C_{21}^{(2)} \\ 0 & C_{23}^{(2)} & C_{33}^{(2)} & C_{31}^{(2)} \\ C_{13}^{(1)} & C_{23}^{(1)} + C_{21}^{(2)} & C_{31}^{(2)} & C_{33}^{(1)} + C_{11}^{(2)} \end{bmatrix}$$

$$= \begin{bmatrix} 0.8333 & -0.667 & 0 & -0.1667 \\ -0.6667 & 1.4583 & -0.375 & -0.4167 \\ 0 & -0.375 & 0.625 & -0.25 \\ -0.1667 & -0.4167 & -0.25 & 0.833 \end{bmatrix}$$

Prob. 14.30 We can do it by hand as in Example 14.6. However, it is easier to prepare an input file and use the program in Fig. 14.54. The MATLAB input data is

```
NE = 2;
ND = 4;
NP = 2;
NL = [1 2 4
      4 2 3];
X = [ 0.0  1.0  3.0  2.0];
Y = [ 1.0  0.0  0.0  2.0];
NDP= [ 1 3 ];
VAL = [ 10.0  30.0]
```

The result is $V = \begin{bmatrix} 10 \\ 18 \\ 30 \\ 20 \end{bmatrix}$

From this,

$$\underline{V_2 = 18 \text{ V}}, \quad \underline{V_4 = 20 \text{ V}}$$

Prob. 14.31

Compare your finite element solution to the exact or finite difference solution:

$$\underline{V_5 = 25 \text{ V}}$$

Prob. 14.32 As in P. E. 14.7, we use the program in Fig. 14.33. The input data based on Fig. 14.64 is as follows.

NE =50; ND= 36; NP= 20;

NL = [1	8	7
1	2	8
2	9	8
2	3	9
3	10	9
3	4	10
4	11	10
4	5	11
5	12	11
5	6	12
7	14	13
7	8	14
8	15	14
8	9	15
9	16	15
9	10	16
10	17	16
10	11	17
11	18	17
11	12	18
13	20	19
13	14	20
14	21	20
14	15	21
15	22	21
15	16	22
16	23	22
16	17	23
17	24	23
17	18	24
19	26	25
19	20	26
20	27	26
20	21	27

```

21    28    27
21    22    28
22    29    28
22    23    29
23    30    29
23    24    30
25    32    31
25    26    32
26    33    32
26    27    33
27    34    33
27    28    34
28    35    34
28    29    35
29    36    35
29    30    36];

```

```

X = [0.0 0.2 0.4 0.6 0.8 1.0 0.0 0.2 0.4 0.6 0.8 1.0 0.0 0.2 0.4 0.6 0.8 1.0
      0.0 0.2 0.4 0.6 0.8 1.0 0.0 0.2 0.4 0.6 0.8 1.0 0.2 0.4 0.6 0.8 1.0];
Y = [0.0 0.0 0.0 0.0 0.0 0.0 0.2 0.2 0.2 0.2 0.2 0.2 0.2 0.4 0.4 0.4 0.4 0.4
      0.4 0.6 0.6 0.6 0.6 0.6 0.6 0.8 0.8 0.8 0.8 0.8 0.8 1.0 1.0 1.0 1.0 1.0];
NDP = [ 1  2  3  4  5  6 12 18 24 30 36 35 34 33 32 31 25 19 13 7];
VAL = [ 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 50.0 100.0 100.0 100.0
        100.0 50.0 0.0 0.0 0.0 0.0];

```

With this data, the potentials at the free nodes are compared with the exact values as shown below.

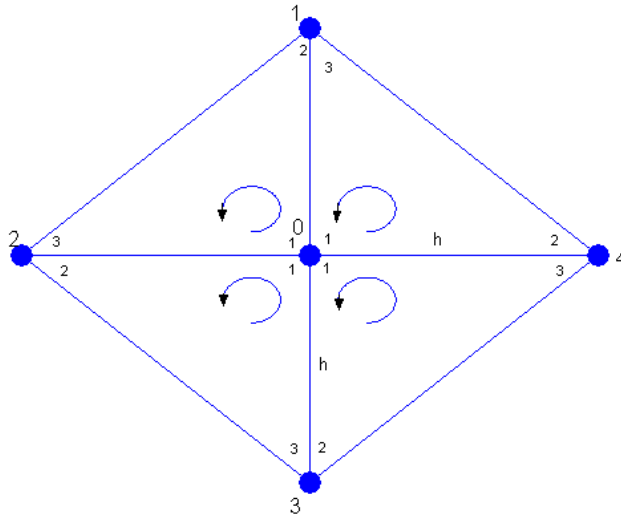
Node no.	FEM Solution	Exact Solution
8	4.546	4.366
9	7.197	7.017
10	7.197	7.017
11	4.546	4.366
14	10.98	10.60
15	17.05	16.84
16	17.05	16.84
17	10.98	10.60
20	22.35	21.78
21	32.95	33.16
22	32.95	33.16
23	22.35	21.78
26	45.45	45.63
27	59.49	60.60
28	59.49	60.60
29	45.45	45.63

Prob. 14.33 We use exactly the same input data as in the previous problem except that the last few lines are replaced by the following lines.

VAL = [0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 29.4 58.8 95.1 95.1
58.8 29.4 0.0 0.0 0.0 0.0];

The potential at the free nodes obtained with the input data are compared with the exact solution as shown below.

Node no.	FEM Solution	Exact Solution
8	3.635	3.412
9	5.882	5.521
10	5.882	5.521
11	3.635	3.412
14	8.659	8.217
15	14.01	13.30
16	14.01	13.30
17	8.659	8.217
20	16.99	16.37
21	27.49	26.49
22	27.49	26.49
23	16.99	16.37
26	31.81	31.21
27	51.47	50.5
28	51.47	50.5
29	31.81	31.21

Prob. 14.34

For element 1, the local numbering 1-2-3 corresponds with nodes with V_1 , V_2 , and V_3 .

$$V_o = -\frac{1}{C_{oo}} \sum_{i=1}^4 V_i C_{io}$$

$$C_{oo} = \sum_{j=1}^4 C_{oj}^{(e)} = \frac{1}{4h^2/2} (hh + hh) \times 2 + \frac{1}{4h^2/2} (hh + 0) \times 4 = 4$$

$$C_{o1} = \frac{2 \times 1}{2h^2} [P_3 P_1 + Q_3 Q_1] = \frac{2}{2h^2} [-hh - 0] = -1$$

$$C_{o2} = \frac{2 \times 1}{2h^2} [P_1 P_2 + Q_1 Q_2] = \frac{2}{2h^2} [-h \times 0 + h \times (-h)] = -1$$

Similarly, $C_{o3} = -1 = C_{o4}$. Thus

$$V_o = (V_1 + V_2 + V_3 + V_4)/4$$

which is the same result obtained using FDM.

Prob. 4.35

$$V_1 = \frac{1}{4}(0 + 0 + 100 + V_2) = 25 + \frac{V_2}{4}$$

$$V_2 = \frac{1}{4}(0 + 100 + V_1 + V_3) = 25 + \frac{V_1 + V_3}{4}$$

$$V_3 = \frac{1}{4}(0 + 0 + 100 + V_2) = 25 + \frac{V_2}{4}$$

$$V_4 = \frac{1}{4}(0 + 0 + 100 + V_5) = 25 + \frac{V_5}{4}$$

$$V_5 = \frac{1}{4}(0 + 0 + 100 + V_4 + V_6) = 25 + \frac{(V_4 + V_6)}{4}$$

$$V_6 = \frac{1}{4}(0 + 0 + 100 + V_5) = 25 + \frac{V_5}{4}$$

We initially set $V_1 = V_2 = V_3 = V_4 = V_5 = V_6 = 0$ and then apply above formulas iteratively. The solutions are presented in the table below.

iteration	1st	2nd	3rd	4th	5th
V_1	25	32.81	35.35	35.67	35.71
V_2	31.25	41.41	42.68	42.83	42.85
V_3	32.81	35.35	35.67	35.71	35.71
V_4	25	23.81	35.35	35.67	35.71
V_5	31.25	41.41	42.68	42.83	42.85
V_6	32.81	35.35	35.67	35.71	35.71

$$\underline{\underline{V_1 = V_4 = 35.71 \text{ V}, \quad V_2 = V_5 = 42.85 \text{ V}, \quad V_3 = V_6 = 35.71 \text{ V}}}$$

Alternatively, if we take advantage of the symmetry, $V_1 = V_3 = V_4 = V_6$ and $V_2 = V_5$. We need to find solve two equations, namely,

$$V_1 = 25 + V_2 / 4$$

$$V_2 = 25 + V_1 / 2$$

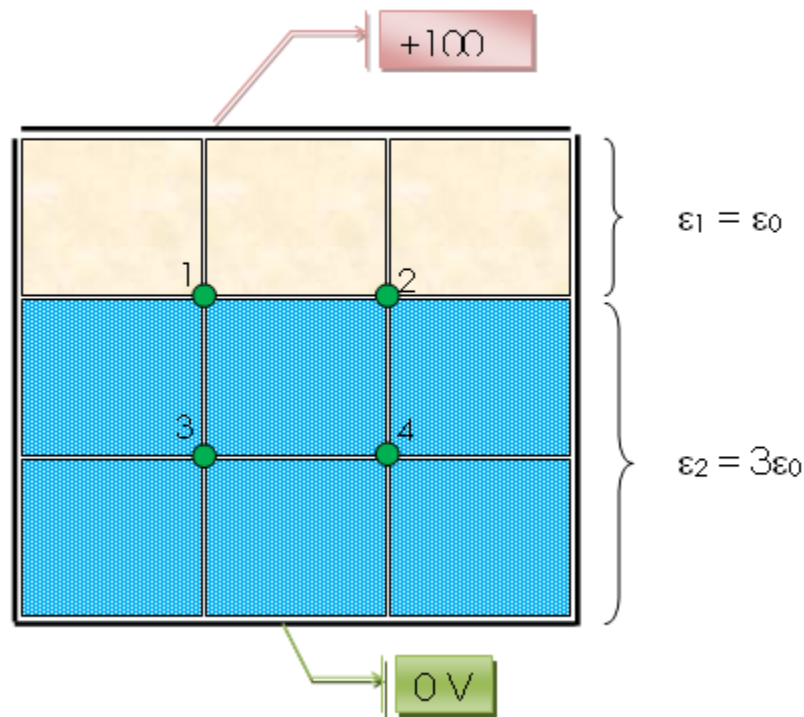
Solving these gives

$$V_1 = 35.714$$

$$V_2 = 42.857$$

Other node voltages follow.

Prob. 14.36



On the interface,

$$\frac{\varepsilon_1}{2(\varepsilon_1 + \varepsilon_2)} = \frac{1}{8}, \quad \frac{\varepsilon_2}{2(\varepsilon_1 + \varepsilon_2)} = \frac{3}{8}$$

$$V_1 = \frac{V_2}{4} + \frac{3V_3}{8} + 12.5$$

$$V_2 = 12.5 + \frac{3V_4}{8} + \frac{V_1}{4}$$

$$V_3 = \frac{1}{4}(V_1 + V_4)$$

$$V_4 = \frac{1}{4}(V_2 + V_3)$$

Applying this iteratively, we obtain the results shown in the table below.

No. of iterations	0	1	2	3	4	5...	100
V_1	0	12.5	17.57	19.25	19.77	19.93	20
V_2	0	15.62	18.65	19.58	19.87	19.96	20
V_3	0	3.125	5.566	6.33	6.56	6.6634	6.667
V_4	0	4.688	6.055	6.477	6.608	6.649	6.667

Prob. 14.37

The MATLAB code is similar to the one in Fig.14.40. When the program is run, it gives $Z_o = \underline{\underline{40.587 \Omega}}$.

Prob. 14.38

The finite difference solution is obtained by following the same steps as in Example 14.8. We obtain $\underline{\underline{Z_o = 43 \Omega}}$

Prob.14.39

$$V_1 = \frac{1}{4}(V_2 + 100 + 100 + 100) = \frac{1}{4}V_2 + 75$$

$$V_2 = \frac{1}{4}(V_1 + V_4 + 2V_3)$$

$$V_3 = \frac{1}{4}(V_2 + V_5 + 200) = \frac{1}{4}(V_2 + V_5) + 50$$

$$V_4 = \frac{1}{4}(V_2 + V_7 + 2V_5)$$

$$V_5 = \frac{1}{4}(V_3 + V_4 + V_6 + V_8)$$

$$V_6 = \frac{1}{4}(V_5 + V_9 + 200) = \frac{1}{4}(V_5 + V_9) + 50$$

$$V_7 = \frac{1}{4}(V_4 + 2V_8 + 0) = \frac{1}{4}(V_4 + 2V_8)$$

$$V_8 = \frac{1}{4}(V_5 + V_7 + V_9)$$

$$V_9 = \frac{1}{4}(V_6 + V_8 + 100 + 0) = \frac{1}{4}(V_6 + V_8) + 25$$

Using these equations, we apply iterative method and obtain the results shown below.

	1 st	2 nd	3 rd	4 th	5 th
V_1	75	79.687	87.11	89.91	92.01
V_2	18.75	48.437	59.64	68.06	74.31
V_3	54.69	65.82	73.87	79.38	82.89
V_4	4.687	19.824	34.57	46.47	53.72
V_5	14.687	35.14	49.45	57.24	61.78
V_6	53.71	68.82	74.2	77.01	78.6

V_7	1.172	6.958	18.92	26.08	30.194
V_8	4.003	20.557	28.93	33.53	36.153
V_9	39.43	47.34	50.78	52.63	53.69

Prob. 14.40

Applying the difference method,

$$V_1 = \frac{V_3}{4} + \frac{V_2}{2} + 25$$

$$V_2 = \frac{1}{4}(V_1 + V_4) + 50$$

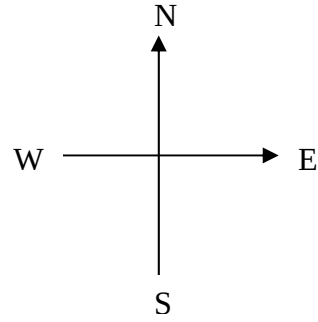
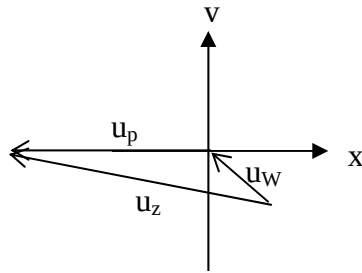
$$V_3 = \frac{1}{4}(V_1 + 2V_4)$$

$$V_4 = \frac{1}{4}(V_2 + V_3 + V_5)$$

$$V_5 = \frac{V_4}{4} + 50$$

Applying these equations iteratively, we obtain the results below.

Iterations	0	1	2	3	4	5...	100
V_1	0	25.0	54.68	64.16	70.97	73.79	74.68
V_2	0	56.25	67.58	74.96	78.23	79.54	80.41
V_3	0	6.25	17.58	33.91	38.72	40.63	41.89
V_4	0	15.63	35.74	41.96	44.86	45.31	45.95
V_5	0	53.91	58.74	60.49	61.09	61.37	51.49

**P. E. 1.4**

Using the dot product,

$$\cos \theta_{AB} = \frac{\mathbf{A} \cdot \mathbf{B}}{AB} = \frac{-13}{\sqrt{10}\sqrt{65}} = -\sqrt{\frac{13}{50}}$$

$$\underline{\underline{\theta_{AB} = 120.66^\circ}}$$

P. E. 1.5

$$\begin{aligned} \text{(a) } \mathbf{E}_F &= (\mathbf{E} \cdot \mathbf{a}_F) \mathbf{a}_F = \frac{(\mathbf{E} \cdot \mathbf{F}) \mathbf{F}}{|\mathbf{F}|^2} = \frac{-10(4, -10, 5)}{141} \\ &= \underline{\underline{-0.2837\mathbf{a}_x + 0.7092\mathbf{a}_y - 0.3546\mathbf{a}_z}} \end{aligned}$$

$$\begin{aligned} \text{(b) } \mathbf{E} \times \mathbf{F} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 0 & 3 & 4 \\ 4 & -10 & 5 \end{vmatrix} = (55, 16, -12) \\ \mathbf{a}_{E \times F} &= \underline{\underline{\pm (0.9398, 0.2734, -0.205)}} \end{aligned}$$

P. E. 1.6 $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$ showing that $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ form the sides of a triangle.

$$\mathbf{a} \cdot \mathbf{b} = 0,$$

hence it is a right angle triangle.

$$\begin{aligned} \text{Area} &= \frac{1}{2} |\mathbf{a} \times \mathbf{b}| = \frac{1}{2} |\mathbf{b} \times \mathbf{c}| = \frac{1}{2} |\mathbf{c} \times \mathbf{a}| \\ \frac{1}{2} |\mathbf{a} \times \mathbf{b}| &= \frac{1}{2} \begin{vmatrix} 4 & 0 & -1 \\ 1 & 3 & 4 \end{vmatrix} = \frac{1}{2} |(3, -17, 12)| \\ \text{Area} &= \frac{1}{2} \sqrt{9 + 289 + 144} = \underline{\underline{10.51}} \end{aligned}$$

P. E. 1.7

$$(a) P_1P_2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$= \sqrt{25 + 4 + 64} = \underline{\underline{9.644}}$$

$$(b) \mathbf{r}_P = \mathbf{r}_{P_1} + \lambda(\mathbf{r}_{P_2} - \mathbf{r}_{P_1})$$

$$= (1, 2, -3) + \lambda(-5, -2, 8)$$

$$= \underline{\underline{(1 - 5\lambda, 2 - 2\lambda, -3 + 8\lambda)}}$$

(c) The shortest distance is

$$d = P_1P_3 \sin \theta = |\mathbf{P}_1\mathbf{P}_3 \times \mathbf{a}_{P_1P_2}|$$

$$= \frac{1}{\sqrt{93}} \begin{vmatrix} 6 & -3 & 5 \\ -5 & -2 & 8 \end{vmatrix}$$

$$= \frac{1}{\sqrt{93}} |(-14, -73, -27)| = \underline{\underline{8.2}}$$

Prob.1.1

$$\mathbf{r}_{OP} = 4\mathbf{a}_x - 5\mathbf{a}_y + \mathbf{a}_z$$

$$\mathbf{a}_{r_{OP}} = \frac{\mathbf{r}_{OP}}{|\mathbf{r}_{OP}|} = \frac{(4, -5, 1)}{\sqrt{(16 + 25 + 1)}} = \underline{\underline{0.6172\mathbf{a}_x - 0.7715\mathbf{a}_y + 0.1543\mathbf{a}_z}}$$

Prob. 1.2

Method 1:

$$\mathbf{r}_{AB} = \mathbf{r}_B - \mathbf{r}_A, \quad \mathbf{r}_{BC} = \mathbf{r}_C - \mathbf{r}_B, \quad \mathbf{r}_{CA} = \mathbf{r}_A - \mathbf{r}_C$$

$$\mathbf{r}_{AB} + \mathbf{r}_{BC} + \mathbf{r}_{CA} = \mathbf{r}_B - \mathbf{r}_A + \mathbf{r}_C - \mathbf{r}_B + \mathbf{r}_A - \mathbf{r}_C = \mathbf{0}$$

Method 2

$$\mathbf{r}_{AB} = \mathbf{r}_B - \mathbf{r}_A = (-2, 0, 3) - (4, -6, 2) = (-6, 6, 1)$$

$$\mathbf{r}_{BC} = \mathbf{r}_C - \mathbf{r}_B = (10, 1, -7) - (-2, 0, 3) = (12, 1, -10)$$

$$\mathbf{r}_{CA} = \mathbf{r}_A - \mathbf{r}_C = (4, -6, 2) - (10, 1, -7) = (-6, -7, 9)$$

$$\mathbf{r}_{AB} + \mathbf{r}_{BC} + \mathbf{r}_{CA} = (0, 0, 0) = \mathbf{0}$$

Prob. 1.3

(a)

$$\begin{aligned} \mathbf{A} - 3\mathbf{B} &= (4, -2, 6) - 3(12, 18, -8) = (4, -2, 6) - (36, 54, -24) \\ &= \underline{\underline{(-32, -56, -30)}} \end{aligned}$$

(b)

$$2\mathbf{A} + 5\mathbf{B} = 2(4, -2, 6) + 5(12, 18, -8) = (68, 86, -28)$$

$$|\mathbf{B}| = \sqrt{12^2 + 18^2 + 8^2} = \sqrt{532} = 23.065$$

$$(2\mathbf{A} + 5\mathbf{B}) / |\mathbf{B}| = (68, 86, -28) / 23.065 = 2.948\mathbf{a}_x + 3.728\mathbf{a}_y - 1.214\mathbf{a}_z$$

(c)

$$\mathbf{a}_x \times \mathbf{A} = \begin{vmatrix} 1 & 0 & 0 \\ 4 & -2 & 6 \end{vmatrix} = \underline{\underline{-6\mathbf{a}_y - 2\mathbf{a}_z}}$$

(d)

$$\mathbf{B} \times \mathbf{a}_x = \begin{vmatrix} 12 & 18 & -8 \\ 1 & 0 & 0 \end{vmatrix} = -8\mathbf{a}_y - 18\mathbf{a}_z$$

$$(\mathbf{B} \times \mathbf{a}_x) \bullet \mathbf{a}_y = -8$$

Prob. 1.4

$$(a) \mathbf{A} \cdot \mathbf{B} = (10, -6, 8) \cdot (1, 0, 2) = 10 + 16 = \underline{\underline{26}}$$

$$\begin{aligned} (b) \quad \mathbf{A} \times \mathbf{B} &= \begin{vmatrix} 10 & -6 & 8 \\ 1 & 0 & 2 \end{vmatrix} = (-12 - 0)\mathbf{a}_x + (8 - 20)\mathbf{a}_y + (0 + 6)\mathbf{a}_z \\ &= \underline{\underline{-12\mathbf{a}_x - 12\mathbf{a}_y + 6\mathbf{a}_z}} \end{aligned}$$

$$(c) \quad 2\mathbf{A} - 3\mathbf{B} = (20, -12, 16) - (3, 0, 6) = \underline{\underline{17\mathbf{a}_x - 12\mathbf{a}_y + 10\mathbf{a}_z}}$$

Prob. 1.5

$$(a) \quad \mathbf{A} - \mathbf{B} + \mathbf{C} = (-2, 5, 1) + (-1, 0, -3) + (4, -6, 10) = \underline{\underline{(1, -1, 8)}}$$

$$(b) \quad \mathbf{B} \times \mathbf{C} = \begin{vmatrix} 1 & 0 & 3 \\ 4 & -6 & 10 \end{vmatrix} = (18, 2, -6)$$

$$\mathbf{A} \bullet (\mathbf{B} \times \mathbf{C}) = (-2, 5, 1) \bullet (18, 2, -6) = -36 + 10 - 6 = \underline{\underline{-32}}$$

$$(c) \quad \cos \theta_{AB} = \frac{\mathbf{A} \bullet \mathbf{B}}{AB} = \frac{-2 + 0 + 3}{\sqrt{4 + 25 + 1}\sqrt{1 + 9}} = 0.05773 \quad \rightarrow \quad \theta_{AB} = \underline{\underline{86.69^\circ}}$$

Prob. 1.6

$$(a) \quad \mathbf{B} \times \mathbf{C} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{vmatrix} = \mathbf{a}_x - 2\mathbf{a}_y + \mathbf{a}_z$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (1, 0, -1) \cdot (1, -2, 1) = 1 + 0 - 1 = \underline{\underline{0}}$$

$$(b) \quad \mathbf{A} \times \mathbf{B} = \begin{vmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{vmatrix} = \mathbf{a}_x - 2\mathbf{a}_y + \mathbf{a}_z$$

$$(\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C} = (1, -2, 1) \cdot (0, 1, 2) = 0 - 2 + 2 = \underline{\underline{0}}$$

$$(c) \quad \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} 1 & 0 & -1 \\ 1 & -2 & 1 \end{vmatrix} = \underline{\underline{-2\mathbf{a}_x - 2\mathbf{a}_y - 2\mathbf{a}_z}}$$

$$(d) \quad (\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \begin{vmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \end{vmatrix} = \underline{\underline{-5\mathbf{a}_x - 2\mathbf{a}_y + \mathbf{a}_z}}$$

Prob.1.7

$$(a) \quad \mathbf{T} = \underline{(4, 6, -1)} \text{ and } \mathbf{S} = \underline{(10, 12, 8)}$$

$$(b) \quad r_{TS} = r_S - r_T = (10, 12, 8) - (4, 6, -1) = \underline{\underline{6\mathbf{a}_x + 6\mathbf{a}_y + 9\mathbf{a}_z}}$$

$$(c) \quad TS = |r_{TS}| = \sqrt{36 + 36 + 81} = \underline{\underline{12.37}}$$

Prob. 1.8

(a) If $\mathbf{A}$ and $\mathbf{B}$ are parallel, $\mathbf{B} = k\mathbf{A}$, where k is a constant.

$$B_x = kA_x, \quad B_y = kA_y, \quad B_z = kA_z$$

$$\text{For } B_z, \quad 3 = k(-1) \rightarrow k = -3$$

$$B_x = \alpha = kA_x = (-3)(4) = -12$$

$$B_y = \beta = kA_y = (-3)(2) = -6$$

$$\text{Hence, } \underline{\underline{\alpha = -12, \beta = -6}}$$

(b) If $\mathbf{A}$ and $\mathbf{B}$ are perpendicular to each other,

$$\mathbf{A} \cdot \mathbf{B} = 0 \rightarrow \underline{\underline{4\alpha + 2\beta - 3 = 0}}$$

Prob. 1.9

$$(a) \quad \mathbf{A} \times \mathbf{a}_y = \begin{vmatrix} 10 & 5 & -2 \\ 0 & 1 & 0 \end{vmatrix} = \underline{\underline{2\mathbf{a}_x + 10\mathbf{a}_z}}$$

$$(b) \quad \mathbf{A} \cdot \mathbf{a}_z = \underline{\underline{-2}}$$

$$(c) \quad \cos \theta_z = \frac{\mathbf{A} \cdot \mathbf{a}_z}{\sqrt{100 + 25 + 4}} = \frac{-2}{11.358} \rightarrow \theta_z = \underline{\underline{100.14^\circ}}$$

Prob. 1.10

$$(a) A \cdot B = AB \cos \theta_{AB}$$

$$A \times B = AB \sin \theta_{AB} \mathbf{a}_n$$

$$(A \cdot B)^2 + |A \times B|^2 = (AB)^2 (\cos^2 \theta_{AB} + \sin^2 \theta_{AB}) = (AB)^2$$

$$(b) \mathbf{a}_x \cdot (\mathbf{a}_y \times \mathbf{a}_z) = \mathbf{a}_x \cdot \mathbf{a}_x = 1. \text{ Hence,}$$

$$\frac{\mathbf{a}_y \times \mathbf{a}_z}{\mathbf{a}_x \cdot \mathbf{a}_y \times \mathbf{a}_z} = \frac{\mathbf{a}_x}{1} = \mathbf{a}_x$$

$$\frac{\mathbf{a}_z \times \mathbf{a}_x}{\mathbf{a}_x \cdot \mathbf{a}_y \times \mathbf{a}_z} = \frac{\mathbf{a}_y}{1} = \mathbf{a}_y$$

$$\frac{\mathbf{a}_x \times \mathbf{a}_y}{\mathbf{a}_x \cdot \mathbf{a}_y \times \mathbf{a}_z} = \frac{\mathbf{a}_z}{1} = \mathbf{a}_z$$

Prob. 1.11

$$(a) \mathbf{P} + \mathbf{Q} = (6, 2, 0), \mathbf{P} + \mathbf{Q} - \mathbf{R} = (7, 1, -2)$$

$$|\mathbf{P} + \mathbf{Q} - \mathbf{R}| = \sqrt{49 + 1 + 4} = \sqrt{54} = \underline{\underline{7.3485}}$$

$$(b) \mathbf{P} \cdot \mathbf{Q} \times \mathbf{R} = \begin{vmatrix} 2 & -1 & -2 \\ 4 & 3 & 2 \\ -1 & 1 & 2 \end{vmatrix} = 2(6 - 2) + (8 + 2) - 2(4 + 3) = 8 + 10 - 14 = \underline{\underline{4}}$$

$$\mathbf{Q} \times \mathbf{R} = \begin{vmatrix} 4 & 3 & 2 \\ -1 & 1 & 2 \end{vmatrix} = (4, -10, 7)$$

$$\mathbf{P} \cdot \mathbf{Q} \times \mathbf{R} = (2, -1, -2) \cdot (4, -10, 7) = 8 + 10 - 14 = \underline{\underline{4}}$$

$$(c) \mathbf{Q} \times \mathbf{P} = \begin{vmatrix} 4 & 3 & 2 \\ 2 & -1 & -2 \end{vmatrix} = (-4, 12, -10)$$

$$\mathbf{Q} \times \mathbf{P} \cdot \mathbf{R} = (-4, 12, -10) \cdot (-1, 1, 2) = 4 + 12 - 20 = \underline{\underline{-4}}$$

$$\text{or } \mathbf{Q} \times \mathbf{P} \cdot \mathbf{R} = \mathbf{R} \cdot \mathbf{Q} \times \mathbf{P} = \begin{vmatrix} -1 & 1 & 2 \\ 4 & 3 & 2 \\ 2 & -1 & -2 \end{vmatrix} = -(-6 + 2) - (-8 - 4) + 2(-4 - 6) = \underline{\underline{-4}}$$

$$(d) (\mathbf{P} \times \mathbf{Q}) \cdot (\mathbf{Q} \times \mathbf{R}) = (4, -12, 10) \cdot (4, -10, 7) = 16 + 120 + 70 = \underline{\underline{206}}$$

$$(e) (\mathbf{P} \times \mathbf{Q}) \times (\mathbf{Q} \times \mathbf{R}) = \begin{vmatrix} 4 & -12 & 10 \\ 4 & -10 & 7 \end{vmatrix} = \underline{\underline{16\mathbf{a}_x + 12\mathbf{a}_y + 8\mathbf{a}_z}}$$

$$(f) \cos \theta_{PR} = \frac{\mathbf{P} \cdot \mathbf{R}}{|\mathbf{P}| |\mathbf{R}|} = \frac{(-2-1-4)}{\sqrt{4+1+4}\sqrt{1+1+4}} = \frac{-7}{3\sqrt{6}} = -0.9526$$

$$\theta_{PR} = \underline{\underline{162.3^\circ}}$$

$$(g) \sin \theta_{PQ} = \frac{|\mathbf{P} \times \mathbf{Q}|}{|\mathbf{P}| |\mathbf{Q}|} = \frac{\sqrt{16+144+100}}{3\sqrt{16+9+4}} = \frac{\sqrt{260}}{3\sqrt{29}} = 0.998$$

$$\theta_{PQ} = \underline{\underline{86.45^\circ}}$$

Prob. 1.12

$$\mathbf{A} \cdot \mathbf{B} = (4, -6, 1) \cdot (2, 0, 5) = 8 - 0 + 5 = 13$$

$$(a) |\mathbf{B}|^2 = 2^2 + 5^2 = 29$$

$$\mathbf{A} \cdot \mathbf{B} + 2 |\mathbf{B}|^2 = 13 + 2 \times 29 = \underline{\underline{71}}$$

(b)

$$\mathbf{a}_\perp = \pm \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|}$$

$$\text{Let } \mathbf{C} = \mathbf{A} \times \mathbf{B} = \begin{vmatrix} 4 & -6 & 1 \\ 2 & 0 & 5 \end{vmatrix} = (-30, -18, 12)$$

$$\mathbf{a}_\perp = \pm \frac{\mathbf{C}}{|\mathbf{C}|} = \pm \frac{(-30, -18, 12)}{\sqrt{30^2 + 18^2 + 12^2}} = \pm (-0.8111\mathbf{a}_x - 0.4867\mathbf{a}_y + 0.3244\mathbf{a}_z)$$

Prob. 1.13

$$\mathbf{P} \cdot \mathbf{Q} = (2, -6, 5) \cdot (0, 3, 1) = 0 - 18 + 5 = \underline{\underline{-13}}$$

$$\mathbf{P} \times \mathbf{Q} = \begin{vmatrix} 2 & -6 & 5 \\ 0 & 3 & 1 \end{vmatrix} = \underline{\underline{-21\mathbf{a}_x - 2\mathbf{a}_y + 6\mathbf{a}_z}}$$

$$\cos \theta_{PQ} = \frac{\mathbf{P} \cdot \mathbf{Q}}{PQ} = \frac{-13}{\sqrt{10}\sqrt{65}} = -0.51 \longrightarrow \theta_{PQ} = \underline{\underline{120.66^\circ}}$$

Prob. 1.14

$\mathbf{P}$ and $\mathbf{Q}$ are orthogonal if the angle between them is 90° . Hence

$$\mathbf{P} \cdot \mathbf{Q} = PQ \cos \theta = 0$$

$$\mathbf{P} \cdot \mathbf{Q} = (2, 4, -6) \cdot (5, 2, 3) = 10 + 8 - 18 = 0$$

showing that they are perpendicular or orthogonal.

Prob. 1.15

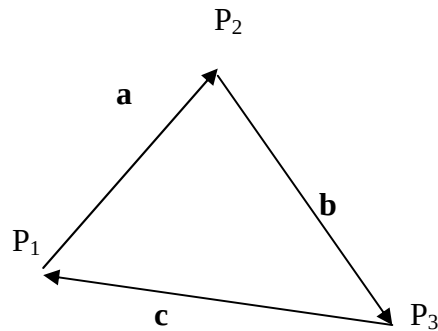
(a) Using the fact that

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{B} \cdot \mathbf{C})\mathbf{A},$$

we get

$$\mathbf{A} \times (\mathbf{A} \times \mathbf{B}) = -(\mathbf{A} \times \mathbf{B}) \times \mathbf{A} = \underline{\underline{(\mathbf{B} \cdot \mathbf{A})\mathbf{A} - (\mathbf{A} \cdot \mathbf{A})\mathbf{B}}}$$

$$\begin{aligned} \text{(b) } \mathbf{A} \times (\mathbf{A} \times (\mathbf{A} \times \mathbf{B})) &= \mathbf{A} \times [(\mathbf{A} \cdot \mathbf{B})\mathbf{A} - (\mathbf{A} \cdot \mathbf{A})\mathbf{B}] \\ &= (\mathbf{A} \cdot \mathbf{B}) (\mathbf{A} \times \mathbf{A}) - (\mathbf{A} \cdot \mathbf{A}) (\mathbf{A} \times \mathbf{B}) \\ &= \underline{\underline{-\mathbf{A}^2 (\mathbf{A} \times \mathbf{B})}} \end{aligned}$$

since $\mathbf{A} \times \mathbf{A} = \mathbf{0}$ **Prob. 1.16**

$$\mathbf{a} = \mathbf{r}_{p_2} - \mathbf{r}_{p_1} = (1, -2, 4) - (5, -3, 1) = (-4, 1, 3)$$

$$\text{(a) } \mathbf{b} = \mathbf{r}_{p_3} - \mathbf{r}_{p_2} = (3, 3, 5) - (1, -2, 4) = (2, 5, 1)$$

$$\mathbf{c} = \mathbf{r}_{p_1} - \mathbf{r}_{p_3} = (5, -3, 1) - (3, 3, 5) = (2, -6, -4)$$

Note that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$

$$\mathbf{a} \cdot \mathbf{b} = -8 + 5 + 3 = 0 \quad \longrightarrow \quad \text{perpendicular}$$

$$\mathbf{b} \cdot \mathbf{c} = 4 - 30 - 4 \neq 0$$

$$\mathbf{c} \cdot \mathbf{a} = -8 - 6 - 12 \neq 0$$

Hence $\underline{\underline{P_2}}$ is a right angle.

$$\begin{aligned} \text{(b) } \text{Area} &= \frac{1}{2} |\mathbf{a} \times \mathbf{b}| = \frac{1}{2} \begin{vmatrix} -4 & 1 & 3 \\ 2 & 5 & 1 \end{vmatrix} = \frac{1}{2} |(1-15)\mathbf{a}_x + (6+4)\mathbf{a}_y + (-20-2)\mathbf{a}_z| \\ &= \frac{1}{2} |(-14, 10, -22)| = \frac{1}{2} \sqrt{196 + 100 + 484} = \underline{\underline{13.96}} \end{aligned}$$

Prob. 1.17

Given $\mathbf{r}_P = (-1, 4, 8)$, $\mathbf{r}_Q = (2, -1, 3)$, $\mathbf{r}_R = (-1, 2, 3)$

(a) $|PQ| = \sqrt{9 + 25 + 25} = \underline{\underline{7.6811}}$

(b) $\underline{\underline{\mathbf{PR} = -2\mathbf{a}_y - 5\mathbf{a}_z}}$

(c)

$$\mathbf{QP} = (-1, 4, 8) - (2, -1, 3) = (-3, 5, 5)$$

$$\mathbf{QR} = (-1, 2, 3) - (2, -1, 3) = (-3, 3, 0)$$

$$\frac{\mathbf{QP} \cdot \mathbf{QR}}{|\mathbf{QP}| |\mathbf{QR}|} = \frac{9 + 15 + 0}{\sqrt{59} \sqrt{18}} = 0.7365$$

$$\angle PQR = \cos^{-1}(0.7365) = \underline{\underline{42.64^\circ}}$$

(d) Area = $\frac{1}{2} |\mathbf{QP} \times \mathbf{QR}| = \frac{1}{2} \begin{vmatrix} -3 & 5 & 5 \\ -3 & 3 & 0 \end{vmatrix} = 0.5 |(-15, -15, 8)| = 0.5 \sqrt{225 + 225 + 36} = \underline{\underline{10.677}}$

(e) Perimeter = $PQ + QR + RP = \sqrt{59} + \sqrt{18} + \sqrt{29} = \underline{\underline{17.31}}$

Prob. 1.18

Let R be the midpoint of PQ.

$$\mathbf{r}_R = \frac{1}{2} \{ (2, 4, -1) + (12, 16, 9) \} = (7, 10, 4)$$

$$OR = \sqrt{49 + 100 + 16} = \sqrt{165} = 12.845$$

$$t = \frac{OR}{v} = \frac{12.845}{300} = \underline{\underline{42.82 \text{ ms}}}$$

Prob. 1.19

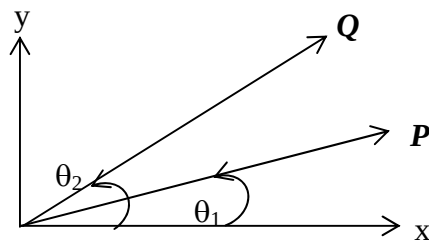
Area = Twice the area of a triangle

$$= |\mathbf{D} \times \mathbf{E}| = \begin{vmatrix} 4 & 1 & 5 \\ -1 & 2 & 3 \end{vmatrix} = (3 - 10)\mathbf{a}_x + (-5 - 12)\mathbf{a}_y + (8 + 1)\mathbf{a}_z$$

$$= |(-7, -19, 9)| = \sqrt{49 + 361 + 81} = \underline{\underline{22.16}}$$

Prob. 1.20

(a) Let $\mathbf{P}$ and $\mathbf{Q}$ be as shown below:



$$|\mathbf{P}| = \cos^2 \theta_1 + \sin^2 \theta_1 = 1, |\mathbf{Q}| = \cos^2 \theta_2 + \sin^2 \theta_2 = 1,$$

Hence $\mathbf{P}$ and $\mathbf{Q}$ are unit vectors.

$$(b) \mathbf{P} \cdot \mathbf{Q} = (1)(1)\cos(\theta_2 - \theta_1)$$

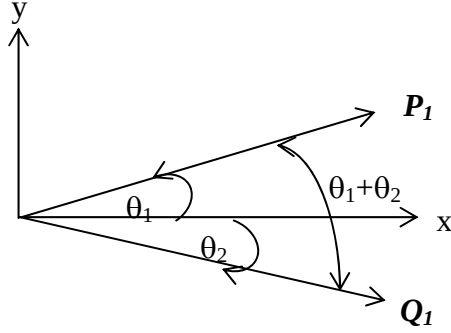
But $\mathbf{P} \cdot \mathbf{Q} = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2$. Thus,

$$\underline{\underline{\cos(\theta_2 - \theta_1) = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2}}$$

Let $\mathbf{P}_1 = \mathbf{P} = \cos \theta_1 \mathbf{a}_x + \sin \theta_1 \mathbf{a}_y$ and

$$\mathbf{Q}_1 = \cos \theta_2 \mathbf{a}_x - \sin \theta_2 \mathbf{a}_y.$$

$\mathbf{P}_1$ and $\mathbf{Q}_1$ are unit vectors as shown below:



$$\mathbf{P}_1 \cdot \mathbf{Q}_1 = (1)(1)\cos(\theta_1 + \theta_2)$$

But $\mathbf{P}_1 \cdot \mathbf{Q}_1 = \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2$,

$$\underline{\underline{\cos(\theta_2 + \theta_1) = \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2}}$$

Alternatively, we can obtain this formula from the previous one by replacing θ_2 by $-\theta_2$ in $\mathbf{Q}$.

(c)

$$\frac{1}{2} |\mathbf{P} - \mathbf{Q}| = \frac{1}{2} |(\cos \theta_1 - \cos \theta_2) \mathbf{a}_x + (\sin \theta_1 - \sin \theta_2) \mathbf{a}_y|$$

$$= \frac{1}{2} \sqrt{\cos^2 \theta_1 + \sin^2 \theta_1 + \cos^2 \theta_2 + \sin^2 \theta_2 - 2 \cos \theta_1 \cos \theta_2 - 2 \sin \theta_1 \sin \theta_2}$$

$$= \frac{1}{2} \sqrt{2 - 2(\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2)} = \frac{1}{2} \sqrt{2 - 2 \cos(\theta_2 - \theta_1)}$$

Let $\theta_2 - \theta_1 = \theta$, the angle between $\mathbf{P}$ and $\mathbf{Q}$.

$$\frac{1}{2} |\mathbf{P} - \mathbf{Q}| = \frac{1}{2} \sqrt{2 - 2 \cos \theta}$$

But $\cos 2A = 1 - 2 \sin^2 A$.

$$\frac{1}{2} |\mathbf{P} - \mathbf{Q}| = \frac{1}{2} \sqrt{2 - 2 + 4 \sin^2 \theta / 2} = \sin \theta / 2$$

Thus,

$$\frac{1}{2} |\mathbf{P} - \mathbf{Q}| = \left| \sin \frac{\theta_2 - \theta_1}{2} \right|$$

Prob. 1.21

$$\boldsymbol{\omega} = \frac{\omega(1, -2, 2)}{3} = (1, -2, 2), \quad r = r_p - r_o = (1, 3, 4) - (2, -3, 1) = (-1, 6, 3)$$

$$u = \boldsymbol{\omega} \times r = \begin{vmatrix} 1 & -2 & 2 \\ -1 & 6 & 3 \end{vmatrix} = (-18, -5, 4)$$

$$\underline{\underline{u = -18a_x - 5a_y + 4a_z}}$$

Prob. 1.22

$$\mathbf{r}_1 = (1, 1, 1), \quad \mathbf{r}_2 = (1, 0, 1) - (0, 1, 0) = (1, -1, 1)$$

$$\cos \theta = \frac{\mathbf{r}_1 \cdot \mathbf{r}_2}{r_1 r_2} = \frac{(1-1+1)}{\sqrt{3}\sqrt{3}} = \frac{1}{3} \quad \longrightarrow \quad \underline{\underline{\theta = 70.53^\circ}}$$

Prob. 1.23

$$(a) T_s = T \cdot a_s = \frac{T \cdot S}{|S|} = \frac{(2, -6, 3) \cdot (1, 2, 1)}{\sqrt{6}} = \frac{-7}{\sqrt{6}} = \underline{\underline{-2.8577}}$$

$$(b) S_T = (\mathbf{S} \cdot \mathbf{a}_T) \mathbf{a}_T = \frac{(\mathbf{S} \cdot \mathbf{T}) \mathbf{T}}{T^2} = \frac{-7(2, -6, 3)}{7^2}$$

$$\underline{\underline{= -0.2857\mathbf{a}_x + 0.8571\mathbf{a}_y - 0.4286\mathbf{a}_z}}$$

$$(c) \sin \theta_{TS} = \frac{|\mathbf{T} \times \mathbf{S}|}{|\mathbf{T}| |\mathbf{S}|} = \frac{\begin{vmatrix} 2 & -6 & 3 \\ 1 & 2 & 1 \end{vmatrix}}{7\sqrt{6}} = \frac{|(-12, 1, 10)|}{7\sqrt{6}} = \frac{\sqrt{245}}{7\sqrt{6}} = 0.9129$$

$$\Rightarrow \theta_{TS} = \underline{\underline{65.91^\circ}}$$

Prob. 1.24

Let $A = A_{B\parallel} + A_{B\perp}$

$$A_{B\parallel} = (A \cdot a_B) a_B = \frac{A \cdot B}{B \cdot B} B$$

Hence,

$$A_{B\perp} = A - A_{B\parallel} = A - \frac{A \cdot B}{B \cdot B} B$$

Prob. 1.25

$$(a) \quad A \cdot B = 20 + 0 - 10 = \underline{\underline{10}}$$

$$(b) \quad A \times B = \begin{vmatrix} 20 & 15 & -10 \\ 1 & 0 & 1 \end{vmatrix} = \underline{\underline{15a_x - 30a_y - 15a_z}}$$

$$(c) \quad A_B = (A \cdot a_B) a_B = \frac{(A \cdot B) B}{B^2} = \frac{10(a_x + a_z)}{2} = \underline{\underline{5a_x + 5a_z}}$$

Prob. 1.26

$$A \cdot a_x = A_x = A \cos \alpha \quad \rightarrow \quad \cos \alpha = \frac{A_x}{A} = \frac{2}{\sqrt{4+16+36}} = 0.2673 \quad \rightarrow \quad \underline{\underline{\alpha = 74.5^\circ}}$$

$$\cos \beta = \frac{A_y}{A} = \frac{-4}{\sqrt{56}} = -0.5345 \quad \rightarrow \quad \underline{\underline{\beta = 122.31^\circ}}$$

$$\cos \gamma = \frac{A_z}{A} = \frac{6}{\sqrt{56}} = 0.8018 \quad \rightarrow \quad \underline{\underline{\gamma = 36.7^\circ}}$$

Prob. 1.27

$$(a) \quad H(1, 3, -2) = 6a_x + a_y + 4a_z$$

$$a_H = \frac{(6, 1, 4)}{\sqrt{36+1+16}} = \underline{\underline{0.8242a_x + 0.1374a_y + 0.5494a_z}}$$

$$(b) \quad |H| = 10 = \sqrt{4x^2y^2 + (x+z)^2 + z^4}$$

or

$$\underline{\underline{100 = 4x^2y^2 + x^2 + 2xz + z^2 + z^4}}$$

Prob. 1.28

$$\mathbf{R} = R\mathbf{a}_R, \quad R = 4$$

$$\mathbf{a}_R = \frac{\mathbf{P} \times \mathbf{Q}}{|\mathbf{P} \times \mathbf{Q}|}$$

$$\mathbf{P} \times \mathbf{Q} = \begin{vmatrix} 2 & -4 & 1 \\ 1 & 2 & 0 \end{vmatrix} = -2\mathbf{a}_x + \mathbf{a}_y + 8\mathbf{a}_z$$

$$\mathbf{a}_R = \frac{-2\mathbf{a}_x + \mathbf{a}_y + 8\mathbf{a}_z}{\sqrt{4+1+64}} = -0.2408\mathbf{a}_x + 0.1204\mathbf{a}_y + 0.9631\mathbf{a}_z$$

$$\mathbf{R} = R\mathbf{a}_R = 4(-0.2408\mathbf{a}_x + 0.1204\mathbf{a}_y + 0.9631\mathbf{a}_z) = \underline{\underline{-0.9631\mathbf{a}_x + 0.4815\mathbf{a}_y + 3.852\mathbf{a}_z}}$$

An alternate choice of $\mathbf{R}$ is $-0.9631\mathbf{a}_x + 0.4815\mathbf{a}_y + 3.852\mathbf{a}_z$

Prob. 1.29

(a) At (1, -2, 3), $x = 1, y = -2, z = 3$.

$$\mathbf{G} = \mathbf{a}_x + 2\mathbf{a}_y + 6\mathbf{a}_z, \quad \mathbf{H} = -6\mathbf{a}_x + 3\mathbf{a}_y - 3\mathbf{a}_z$$

$$G = \sqrt{1+4+36} = \underline{\underline{6.403}}$$

$$H = \sqrt{36+9+9} = \underline{\underline{7.348}}$$

$$(b) \quad \mathbf{G} \cdot \mathbf{H} = -6 + 6 - 18 = \underline{\underline{-18}}$$

$$(c) \quad \cos \theta_{GH} = \frac{\mathbf{G} \cdot \mathbf{H}}{GH} = \frac{-18}{6.403 \times 7.348} = -0.3826$$

$$\underline{\underline{\theta_{GH} = 112.5^\circ}}$$

Prob. 1.30

$$(a) \quad \mathbf{H} = 10(2)(16)\mathbf{a}_x - 8(-8)\mathbf{a}_y + 12(4)\mathbf{a}_z = 320\mathbf{a}_x + 64\mathbf{a}_y + 48\mathbf{a}_z$$

$$\text{Let } \mathbf{F} = \mathbf{a}_x - \mathbf{a}_y$$

$$(b) \quad \mathbf{H}_F = (\mathbf{H} \cdot \mathbf{a}_F)\mathbf{a}_F = \frac{(\mathbf{H} \cdot \mathbf{F})\mathbf{F}}{F^2} = \frac{(320 - 64)(1, -1, 0)}{1+1} = 128\mathbf{a}_x - 128\mathbf{a}_y$$

Prob. 1.31

(a) At (1,2,3), $\mathbf{E} = (2,1,6)$

$$|\mathbf{E}| = \sqrt{4+1+36} = \sqrt{41} = \underline{\underline{6.403}}$$

(b) At (1,2,3), $\mathbf{F} = (2,-4,6)$

$$\begin{aligned} \mathbf{E}_F &= (\mathbf{E} \cdot \mathbf{a}_F) \mathbf{a}_F = \frac{(\mathbf{E} \cdot \mathbf{F}) \mathbf{F}}{|\mathbf{F}|^2} = \frac{36}{56} (2, -4, 6) \\ &= \underline{\underline{1.286 \mathbf{a}_x - 2.571 \mathbf{a}_y + 3.857 \mathbf{a}_z}} \end{aligned}$$

(c) At (0,1,-3), $\mathbf{E} = (0,1,-3)$, $\mathbf{F} = (0,-1,0)$

$$\mathbf{E} \times \mathbf{F} = \begin{vmatrix} 0 & 1 & -3 \\ 0 & -1 & 0 \end{vmatrix} = (-3, 0, 0)$$

$$\mathbf{a}_{\mathbf{E} \times \mathbf{F}} = \pm \frac{\mathbf{E} \times \mathbf{F}}{|\mathbf{E} \times \mathbf{F}|} = \underline{\underline{\pm \mathbf{a}_x}}$$

Prob. 1.32

(a) At P, $x = -1$, $y = 2$, $z = 4$

$$\mathbf{D} = 8\mathbf{a}_x - 4\mathbf{a}_y - 2\mathbf{a}_z, \quad \mathbf{E} = -10\mathbf{a}_x + 24\mathbf{a}_y + 128\mathbf{a}_z$$

$$\mathbf{C} = \mathbf{D} + \mathbf{E} = -2\mathbf{a}_x + 20\mathbf{a}_y + 126\mathbf{a}_z$$

$$\mathbf{C} \cdot \mathbf{a}_x = C \cos \theta_x \quad \longrightarrow \quad \cos \theta_x = \frac{\mathbf{C} \cdot \mathbf{a}_x}{C} = \frac{-2}{\sqrt{2^2 + 20^2 + 126^2}} = -0.01575$$

(b)

$$\underline{\underline{\theta_x = 90.9^\circ}}$$